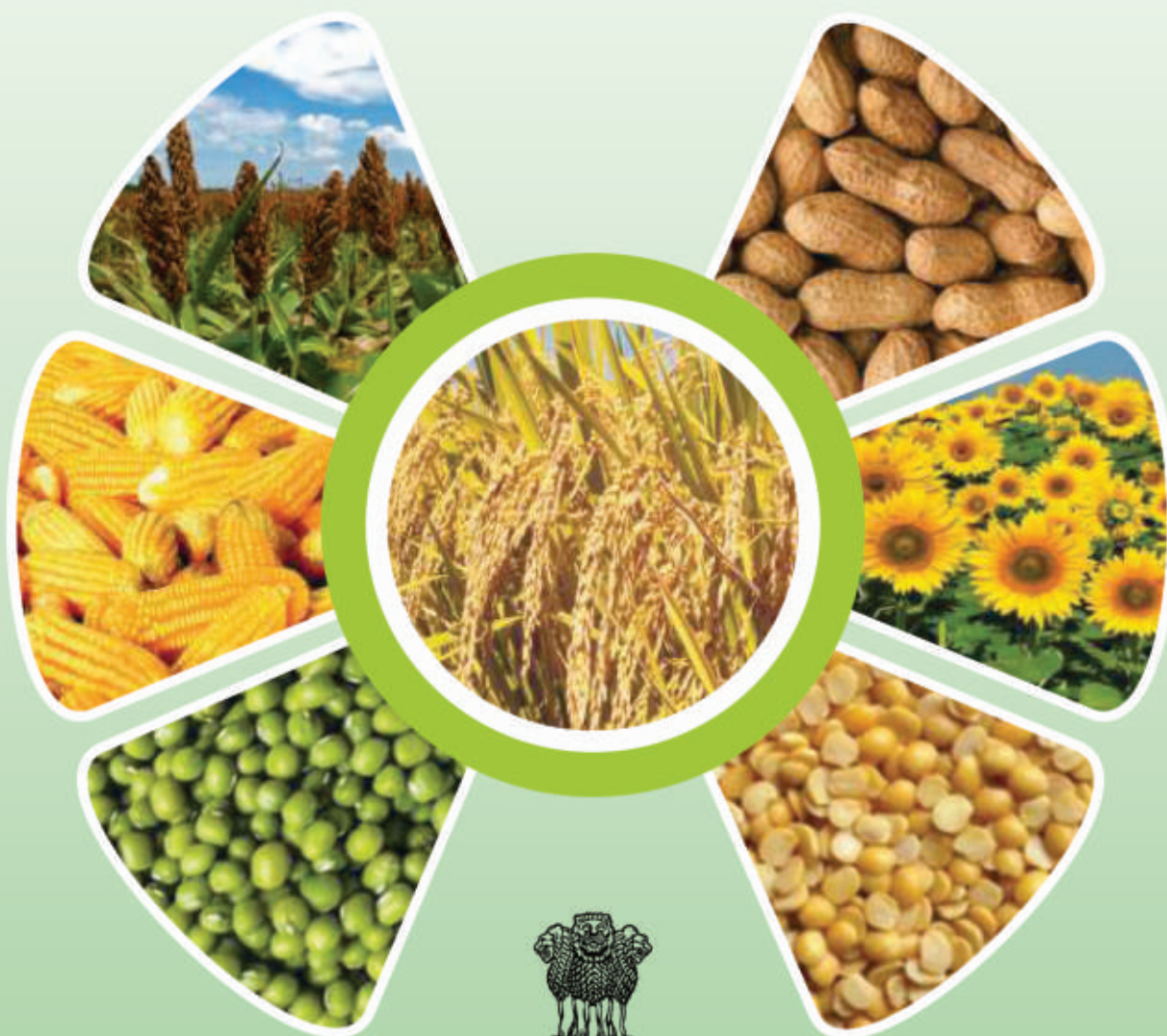


Price Policy for
Kharif Crops
The Marketing Season 2025-26



Commission for Agricultural Costs & Prices

Department of Agriculture & Farmers Welfare

Ministry of Agriculture & Farmers Welfare

Government of India, New Delhi

March 2025

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Preface & Acknowledgments

It is an immense honour and privilege to submit the report of “**Price Policy for Kharif Crops: Marketing Season 2025-26**”. The report contains recommendations on Minimum Support Prices (MSP) for the mandated crops, namely, **paddy, maize, jowar, bajra, ragi, arhar (tur), moong, urad, groundnut, soybean, sunflower, sesamum, nigerseed** and **cotton** and a set of non-price policy measures. I hope that these recommendations will incentivise and encourage farmers to adopt new technologies, reorient production pattern in the light of emerging demand opportunities, ensure remunerative prices to farmers and improve global competitiveness of Indian agriculture.

Summary of Recommendations is followed by an overview of Indian agriculture in Chapter 1. Chapter 2 of the report discusses demand-supply situation, price trends and procurement operations. Productivity trends and patterns and drivers of productivity growth are analysed in Chapter 3. Chapter 4 discusses trade patterns and composition, global prices and brief review of agricultural trade policy. Costs and returns and cost projections for *Kharif* Marketing Season 2025-26 are discussed in Chapter 5. The MSP and non-price policy recommendations are presented in the Chapter 6.

Many people have assisted in the preparation of this report. First and foremost, I would like to express my sincere thanks to farmers, representatives of farmers’ associations/unions, officers from Central and State Governments, various agencies/organizations/companies involved in procurement, post-harvest management, value-addition and marketing of agricultural commodities, and other stakeholders for providing valuable insights and suggestions during the meetings and preparation of this report. Special thanks to the Economics, Statistics & Evaluation Division, Ministry of Agriculture & Farmers Welfare, Government of India for providing data on cost estimates for this report.

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Last but not least, credit is due to the officers and staff of the Commission, who contributed to this report. Sincere gratitude goes to Sh. Ratan Lal Daga, Member (Non-Official) and Dr. Prem Chand, Member (Official) for their support in preparation of the report. Special thanks to Sh. Vivek Shukla, Advisor, for his excellent contribution in coordination of meetings with various stakeholders and report writing. The report would not



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15th March 2025

(Vijay Paul Sharma)



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Acronyms

Acronym	Description
A_2	Paid out cost
$A_2 + FL$	Paid out cost plus imputed value of family labour
AGMARKNET	Agricultural Marketing Information Network
AGRTOT	Agricultural Terms of Trade
AIBP	Accelerated Irrigation Benefit Program
AICRP	All India Coordinated Research Project
AIDC	Agriculture Infrastructure and Development Cess
AIF	Agriculture Infrastructure Fund
AMI	Agricultural Marketing Infrastructure
APEDA	Agricultural and Processed Food Products Export Development Authority
BCD	Basic Custom Duty
BCM	Billion Cubic Meter
BE	Budget Estimates
Bt	Bacillus thuringiensis
C_2	Paid out cost plus imputed value of family labour plus rental value of owned land (net of land revenue) and interest on fixed capital excluding land
CACP	Commission for Agricultural Costs & Prices
CAGR	Compound Annual Growth Rate
CBIC	Central Board of Indirect Taxes and Customs
CCI	Cotton Corporation of India
CFPI	Consumer Food Price Index
CHC	Custom Hiring Centres
CIMMYT	International Maize and Wheat Improvement Center
CIP	Central Issue Price
CIPI	Composite Input Price Index
CoC	Cost of Cultivation
COCPC	Committee on Cotton Production and Consumption
CoP	Cost of Production



Acronym	Description
CPI	Consumer Price Index
CRM	Crop Residue Management
CS	Closer Spacing
CS	Comprehensive Scheme
CWC	Central Water Commission
DCP	Decentralized Procurement
DCS	Digital Crop Survey
DGCES	Digital General Crop Estimation Survey
DGCIS	Directorate General of Commercial Intelligence & Statistics
DGFT	Directorate General of Foreign Trade
DIPP	Department of Industrial Policy & Promotion
DMI	Directorate of Marketing & Inspection
DPDP	Digital Personal Data Protection
DPI	Digital Public Infrastructure
ELS	Extra-Long Staple
e-NAM	National Agriculture Market
ES&E	Economics, Statistics and Evaluation
EU	European Union
FAI	Fertilizers Association of India
FAO	Food and Agricultural Organization
FAQ	Fair Average Quality
FCI	Food Corporation of India
FFPI	FAO Food Price Index
FIAT	Fund for innovation & technology
FLD	Front Line Demonstration
FOB	Freight on Board
FOR	Freight on Road
FPMA	Food Price Monitoring and Analysis
FPO	Farmer Producer Organizations



Acronym	Description
FSSAI	Food Safety and Standards Authority of India
FTA	Free Trade Agreement
FToT	Farmers Terms of Trade
GAP	Good Agricultural Practice
GCA	Gross Cropped Area
GDP	Gross Domestic Product
GIA	Gross Irrigated Area
GM	Genetically Modified
GMP	Good Manufacturing Practice
GVA	Gross Value Added
GVO	Gross Value of Output
HACCP	Hazard Analysis Critical Control Points
HDPS	High Density Planting System
HKKP	<i>Har Khet Ko Pani</i>
HSD	High Speed Diesel
HYV	High Yielding Varieties
ICAC	International Cotton Advisory Committee
ICAR-CICR	Indian Council of Agricultural Research-Central Institute for Cotton Research
ICAR-IIMR	Indian Council of Agricultural Research-Indian Institute of Maize Research
IGC	International Grains Council
ISAM	Integrated Scheme for Agricultural Marketing
KCC	<i>Kisan</i> Credit Card
Krishi DSS	<i>Krishi</i> Decision Support System
LS	Long Staple
MEP	Minimum Export Price
MISS	Modified Interest Subvention Scheme
MS	Marketed Surplus
MSP	Minimum Support Price
MSR	Marketed Surplus Ratio



Acronym	Description
MT	Metric Tonne
NAFED	National Agricultural Cooperative Marketing Federation of India Limited
NCCF	National Cooperative Consumers' Federation of India Limited
NFHS	National Family Health Survey
NFSM	National Food Security Mission
NMEO-Oilseeds	National Mission on Edible Oils – Oilseeds
NMEO-OP	National Mission on Edible Oils – Oil Palm
NMNF	National Mission on Natural Farming
Non-DCP	Centralized Procurement
NSS	National Small Saving
OECD	Organization for Economic Co-operation and Development
OMSS	Open Market Sale Scheme
OWS	Other Welfare Schemes
PACS	Primary Agricultural Credit Societies
PDMC	<i>Per Drop More Crop</i>
PMFBY	<i>Pradhan Mantri Fasal Bima Yojana</i>
PMGKAY	<i>Pradhan Mantri Garib Kalyan Anna Yojana</i>
PM-KISAN	<i>Pradhan Mantri Kisan Samman Nidhi</i>
PMKSY	<i>Pradhan Mantri Krishi Sinchayee Yojana</i>
PM-PRANAM	PM Programme for Restoration, Awareness Generation, Nourishment, and Amelioration of Mother Earth
PRI	Prompt Repayment Incentive
R&M	Rapeseed and Mustard
RBI	Reserve Bank of India
RCAC	Registration-cum-Allocation Certificate
RE	Revised Estimates
RKVY	<i>Rashtriya Krishi Vikas Yojana</i>
ROO	Rules of Origin
SAFTA	South Asian Free Trade Area



Acronym	Description
SAUs	State Agricultural Universities
SHC	Soil Health Card
SHGs	Self Help Groups
SMAM	Sub Mission on Agricultural Mechanisation
SOP	Standard Operating Procedure
SRR	Seed Replacement Rate
STLs	Soil Testing Labs
SUR	Stock to Use Ratio
SWC	Social Welfare Cess
TE	Triennium Ending
ToT	Terms of Trade
TRQ	Tariff Rate Quota
UAE	United Arab Emirates
UPAg	Unified Portal for Agricultural Statistics
USA	United States of America
USDA	United States Department of Agriculture
VRR	Varietal Replacement Rate
w.e.f.	With Effect From
WINDS	Weather Information and Network Data System
WMO	World Meteorological Organization
WPI	Wholesale Price Index
WTO	World Trade Organization
YES-TECH	Yield Estimation System Using Technology



Summary of Recommendations

S.1. The Commission has considered multiple factors including the cost of production, demand-supply situation, domestic and international prices, inter-crop parity, terms of trade between agriculture and non-agriculture sector, likely impact of the price policy on the economy and a minimum 50 percent margin over the cost of production while recommending the Minimum Support Prices (MSP). Considering the aforementioned factors and deliberations with various stakeholders, the Commission makes the following price and non-price policy recommendations.

Price Policy Recommendations

S.2 The Commission recommends the MSPs of mandated *kharif* crops for *Kharif* Marketing Season (KMS) 2025-26 as given in the Table S.1.

Table S.1: Recommended MSPs of Mandated Crops: KMS2025-26

(₹/qtl)				
Crops	Projected A ₂ +FL Cost for KMS2025-26	MSP for KMS2024-25	Recommended MSP for KMS2025-26	MSP as percent of A ₂ +FL
Paddy-Common	1579	2300	2369 (3.0)	150
Paddy-Grade A	-	2320	2389 (3.0)	-
Jowar-Hybrid	2466	3371	3699 (9.7)	150
Jowar-Maldandi	-	3421	3749 (9.6)	-
Bajra	1703	2625	2775 (5.7)	163
Ragi	3257	4290	4886 (13.9)	150
Maize	1508	2225	2400 (7.9)	159
Tur/Arhar	5038	7550	8000 (6.0)	159
Moong	5845	8682	8768 (1.0)	150
Urad	5114	7400	7800 (5.4)	153
Groundnut	4842	6783	7263 (7.1)	150
Sunflower Seed	5147	7280	7721 (6.1)	150
Soybean (Yellow)	3552	4892	5328 (8.9)	150
Sesamum	6564	9267	9846 (6.2)	150
Nigerseed	6358	8717	9537 (9.4)	150
Cotton (Medium Staple)	5140	7121	7710 (8.3)	150
Cotton (Long Staple)	-	7521	8110 (7.8)	-

Note: Figures in parenthesis represent increase in MSP over the previous year



Non-Price Recommendations

Promote Balanced Production Pattern

- S3. With the growing imbalance between domestic production and demand for different agricultural commodities in the country and sustainability challenges associated with paddy-wheat monoculture, particularly, in the northwest Indo-Gangetic Plains, a strategic shift towards pulses and oilseeds has become necessary to achieve *Atmanirbharta* and address these challenges. Over the last few years, the Commission has tried to realign price incentive in favour of pulses, oilseeds and *Shree Anna* to encourage farmers to diversify crop choices and adopt modern technologies but yield levels of these crops are so low that their relative profitability is substantially lower than the competing crops. Therefore, the Commission recommends that a comprehensive diversification strategy and action plan should be prepared with a special focus on technological innovations and remunerative prices to pulses and oilseeds farmers.

Improve Productivity and Bridge Yield Gaps

- S4. Some crops have experienced stagnation or decline in yield in the recent years and a large gap exists between actual and the potential yield of many crops mainly caused by socio-economic and institutional factors. Therefore, identifying technologies and best practices that can address the issue of yield stagnation/decline and socio-economic factors that can effectively address yield gaps is crucial. The Commission recommends that investment in agricultural Research and Development (R&D) should be substantially enhanced and timely availability of climate resilient high yielding variety seeds, quality inputs and services, and improvement in Seed Replacement Rate (SRR) and Varietal Replacement Rate (VRR) should be accorded the highest priority.

Improve Soil Health and Productivity

- S5. There are wide regional differences in fertiliser use in the country and fertiliser consumption is high in some areas. Imbalanced and inefficient use of nutrients, primarily due to heavy subsidy on urea compared to P & K fertilizers and poor extension services, has become a major threat to soil health, with adverse environmental, social and economic effects. Government has launched various initiatives to address these challenges and promote sustainable and balanced use of fertilisers as well as alternative fertilisers. The Commission recommends that massive awareness programmes and campaigns highlighting the harmful effects of imbalanced use should be launched, and the issue of lopsided fertiliser subsidies, which substantially lower the prices of urea relative to P and K and complex fertilisers, needs to be addressed.

Irrigation Development and Management

- S6. Sustained efforts by the Government under '*Pradhan Mantri Krishi Sinchayee Yojana*' have led to significant increase in area under irrigation and also helped in improving water use efficiency at farm level. However, over-exploitation of groundwater resources and low water use efficiency remain



major challenges. The Commission recommends that concerted efforts should be made to increase area under micro-irrigation through awareness and training programmes for farmers and ensuring supply of good quality systems/equipments along with post installation maintenance services. In addition, special emphasis should be given on repair, renovation and restoration of traditional water bodies including tanks.

Promote Farm Mechanization

- S7. Farm mechanization has become a pressing necessity not only due to rising labour costs and non-availability of labour but also to bring efficiency in the agricultural operations. The Sub-Mission on Agricultural Mechanization (SMAM) has played an important role in promoting mechanization but the level of mechanization in some crops and States is still low. The Commission recommends significant increase in budget allocation under SMAM and expansion of Custom Hiring Centres (CHC) network in underserved areas. The issue of high GST rates on farm machinery and implements requires favourable consideration to encourage mechanization.

Ensure Easy and Equitable Access to Credit

- S8. Various policy initiatives taken by the Government have resulted in significant increase in the flow of institutional credit to agriculture over the years and agricultural credit has witnessed a double-digit growth during the last 10 years. However, there is a significant disparity in disbursement of agricultural credit across regions and majority of small and marginal farmers are still dependent on non-institutional sources of credit. The Commission recommends that coordinated efforts should be made by all stakeholders, including Government and the lending agencies, to achieve the objective of equitable and universal access to agricultural credit.

Reduce Cost of Production

- S9. Concerted efforts are needed to improve crop yields and promote adoption of new technologies, scientific practices and effective extension services to reduce large inter-State variations in cost of production caused by differences in productivity, access to technology, use of inputs and services, and prices of inputs and outputs. In addition, encouraging and incentivizing farmers to shift from less profitable crops to alternative crops, which are more suitable and profitable, will help in improving farm profitability.

Address Market Distortions in Agriculture

- S10. State interventions in the form of high market fees, taxes and other incidental charges and direct or indirect bonus over and above the MSP by some States severely distort markets, restrict participation of private sector, and adversely affect global competitiveness. The Commission recommends rationalisation of market fees and other charges to create efficient and competitive national market and discontinuation of crop-specific incentives. The farmers should be encouraged and incentivised for crop diversification, particularly, towards pulses, oilseeds and millets.



Strengthen Procurement Operations at State Level

- S11. The level of rice procurement is low in some major producing States like West Bengal, Uttar Pradesh, Tamil Nadu, Karnataka, Bihar and Assam. In addition, actual procurement of pulses and oilseeds under the Price Support Scheme (PSS) has remained less than sanctioned quantity due to non-availability of adequate and proper storage and warehousing facilities, inadequate procurement centres, delay in start of procurement operations, quality issues, etc. Since PSS is implemented on the request of the State Governments/UTs, the Commission recommends that States should prepare a comprehensive plan for strengthening procurement operations, creating sufficient storage and warehousing facilities and training of farmers to meet FAQ norms as well as provide dryers and other facilities.

Underestimation of MSP Procurement by the National Statistical Office (NSO)

- S12. The underestimates of paddy procurement and number of farmers benefitted from MSP procurement operations in NSO's "Situation Assessment of Agricultural Households and Land and Livestock Holdings of Households in Rural India, 2019 (77th Round)" create widespread misperceptions about the effectiveness of procurement operations and adversely affect farmers. Therefore, the Commission recommends that the issue of divergence between actual procurement and beneficiary farmers, and NSO estimates should be addressed.

Review Import Duty Structure of Pulses and Edible Oils

- S13. Ensuring remunerative prices to farmers is one of the important pathways to achieve *Aatmanirbharta* in pulses and edible oils, and pro-farmer trade policy mechanism will help in ensuring better prices to farmers. However, unrestricted and cheap imports of pulses and edible oils at low/zero import duty have resulted in domestic prices of soybean, groundnut, urad, moong and tur falling below the MSP during KMS2024-25. Aligning import duty structure with MSP will ensure remunerative prices to farmers and encourage them to increase area and production of oilseeds and pulses. The Commission recommends that imports of yellow peas should be banned and import duty on other pulses, particularly, tur/arhar, lentil and urad should be increased.

Increase Coverage and Sample Size under the Comprehensive Scheme

- S14. Inadequate representation of certain crops with significant share in both all-India and State-level area and production under the Comprehensive Scheme and thin sample size for some crops can potentially undermine the reliability and robustness of cost estimates. Therefore, the Commission recommends that sample size for jowar in Andhra Pradesh and Madhya Pradesh, bajra in Tamil Nadu, maize in Jharkhand and Punjab, tur and moong in Madhya Pradesh, and all-India sample size of ragi, sunflower, and nigerseed should be increased and coverage of Comprehensive Scheme be extended to Jammu & Kashmir, Tripura and Uttarakhand.



CHAPTER 1



Overview

Introduction

- 1.1 India's agricultural sector experienced robust growth in the recent years, with gross value added (GVA) from agriculture, livestock, forestry and fishing growing at an average annual growth rate of five percent since 2015-16. This growth has been contributed largely by higher growth in livestock (7.1%) and fishing and aquaculture (8.8%) and a moderate growth of 3.5 percent in crop sector during 2015-16 to 2023-24. As per the Second Advance Estimates of Annual Gross Domestic Product (GDP) for 2024-25, real GVA from agriculture and allied sectors including forestry and logging is estimated to register a growth rate of 4.6 percent in 2024-25 as compared to 2.7 percent in 2023-24. Agriculture sector is an important component of the Indian economy, contributing about 18 percent of national GVA at current prices and employing 59.8 percent of the rural workforce in 2023-24.
- 1.2 India's foodgrains production has witnessed a steady increase since 2016-17 and touched a record 332.3 million tonnes in 2023-24 despite adverse *El Niño* conditions and climate change challenges. Foodgrains production increased from 206.7 million tonnes in Triennium Ending (TE) 2005-06 to 325.9 million tonnes in TE2023-24, an increase of 57.7 percent (Table 1.1). Most crops/crop groups, except *Shree Anna* and cotton, recorded significant improvement in production. Rice production grew at compound annual growth rate (CAGR) of 3.3 percent during last 10 years from 2014-15 to 2023-24, much higher (1.7%) compared with 2005-06 to 2014-15 period. Maize production recorded the highest CAGR of 5.8 percent, followed by oilseeds (5.1%), pulses (4.5%) and the lowest in cotton (0.3%) during the recent decade. All crops/crop groups, except cotton, recorded much higher growth rate in production during the recent period compared with earlier period. It is worth noting that growth rate in productivity of almost all crops/crop groups, except cotton, was higher during the last 10 years compared with 2005-06 to 2014-15 period. During 2014-15 to 2023-24, about 78 percent of growth in foodgrains production came from higher yield per hectare and about 22 percent from area increase, while in case of pulses and rice, yield contributed about two-third of growth. This shows that yield increase has played a major role in growth of foodgrains and pulses production, and policies to sustain yield growth will be crucial for the future. In case of maize and oilseeds, both area expansion and yield improvement contributed to the growth in production, while despite significant improvement in yield of *Shree Anna*, area under cultivation recorded negative growth rate. Negative growth rate in cotton



yield is a matter of concern and needs to be addressed.

- 1.3 Eight successive years of record foodgrains production from 2016-17 to 2023-24 were underpinned by increased yield and strong commodity prices on average across agricultural commodities. Despite delayed and poor monsoons caused by *El Niño*, rice production reached a record 137.8 million tonnes and wheat production at 113.3 million tonnes in 2023-24. *Shree Anna* production increased to 17.6 million tonnes, growth of 1.7 percent in 2023-24, while production of pulses registered a decline of seven percent in 2023-24 compared to the previous year due to fall in production of urad, moong and gram, despite higher output of lentil and tur/arhar. Total oilseeds production marginally declined to 39.7 million tonnes in 2023-24 after reaching a peak of 41.4 million tonnes in 2022-23. The all-India area, production and yield of mandated *kharif* crops during last 10 years are given in Annex Tables 1.1 to 1.3 and share of major producing States in total production during the TE2023-24 are given in Annex Table 1.4.
- 1.4 As per Second Advance Estimates for 2024-25 by Ministry of Agriculture and Farmers Welfare, *kharif* foodgrains production in the country is estimated to increase by 6.8 percent and *kharif* rice by 6.6 percent, while *kharif* pulses production is likely to increase by 2.5 percent. *Kharif* oilseeds production is estimated to increase by 14.4 percent in 2024-25, primarily driven by substantial growth in groundnut production. Maize, jowar and moong production during *kharif* 2024-25 are also estimated to record a strong growth. Tur/arhar production is expected to increase by 2.8 percent, while *kharif* urad production is estimated to fall by 20.1 percent. Production of cotton, one of the most important fiber and cash crop of India, is estimated to decline by 9.5 percent during 2024-25.

Table 1.1: Trends and Compound Annual Growth Rate (CAGR) of Production and Yield of Major Crops in India: 2003-04 to 2023-24

Crop/Crop Group	TE2005-06	TE2014-15	TE2023-24	2005-06 to 2014-15	2014-15 to 2023-24
	Production (million tonnes)			CAGR (%)	
Foodgrains	206.7	258.1	325.9	2.5	3.3
Rice	87.8	105.8	134.4	1.7	3.3
Maize	14.6	23.6	36.5	5.7	5.8
Shree Anna	19.2	16.8	17.0	-1.0	1.2
Jowar	7.2	5.4	4.2	-4.4	-1.3
Bajra	9.2	9.1	10.6	1.5	2.8
Ragi	2.3	1.9	1.7	-0.3	-0.6
Pulses	13.8	18.3	25.9	3.8	4.5
Tur/Arhar	2.5	3.0	3.6	1.8	1.9



Crop/Crop Group	TE2005-06	TE2014-15	TE2023-24	2005-06 to 2014-15	2014-15 to 2023-24
Urad	1.3	1.9	2.6	5.2	1.1
Moong	1.2	1.4	3.3	5.0	9.9
Oilseeds	25.8	30.4	39.7	1.6	5.1
Soybean	7.7	12.3	13.7	4.0	3.8
Groundnut	7.6	7.3	10.2	0.7	5.0
Cotton*	16.2	35.0	32.4	7.5	0.3
	Yield (qtl/ha)			CAGR (%)	
Foodgrains	17.0	20.9	24.8	2.3	2.6
Rice	20.5	24.2	28.4	1.8	2.2
Maize	19.6	26.2	34.3	3.6	3.5
Shree Anna	8.9	10.9	13.3	2.8	3.3
Jowar	8.0	9.0	11.1	0.1	4.7
Bajra	9.3	12.1	14.6	4.7	2.7
Ragi	14.3	15.9	14.1	1.8	-1.3
Pulses	6.0	7.6	8.9	2.9	2.9
Tur/Arhar	7.0	7.7	8.3	0.4	1.5
Urad	4.2	5.9	6.4	4.3	0.7
Moong	3.7	4.7	6.1	5.3	3.9
Oilseeds	9.8	11.4	13.3	1.9	2.8
Soybean	10.6	11.1	10.7	-0.2	1.5
Groundnut	11.9	14.4	20.1	3.8	4.1
Cotton	3.3	4.9	4.4	2.8	-0.4

Note : * Million bales of 170 kg each

Source: Economics, Statistics & Evaluation Division, Ministry of Agriculture and Farmers Welfare

Growth Drivers of Agricultural Sector

- 1.5 The value of output from agriculture and allied sectors excluding forestry and logging grew at CAGR of 4.4 percent during 2015-16 to 2022-23 and growth in high-value segment such as horticulture, livestock including dairy, meat and eggs and fisheries has been a major driver of overall growth in Indian agriculture. Fisheries sector recorded the highest CAGR (8.6%), followed by condiments and spices (7.9%), livestock sector (5.7%), while crop sector grew at 3.3 percent. However, there are large inter-State and inter-sectoral variations in growth in agriculture and allied sectors.



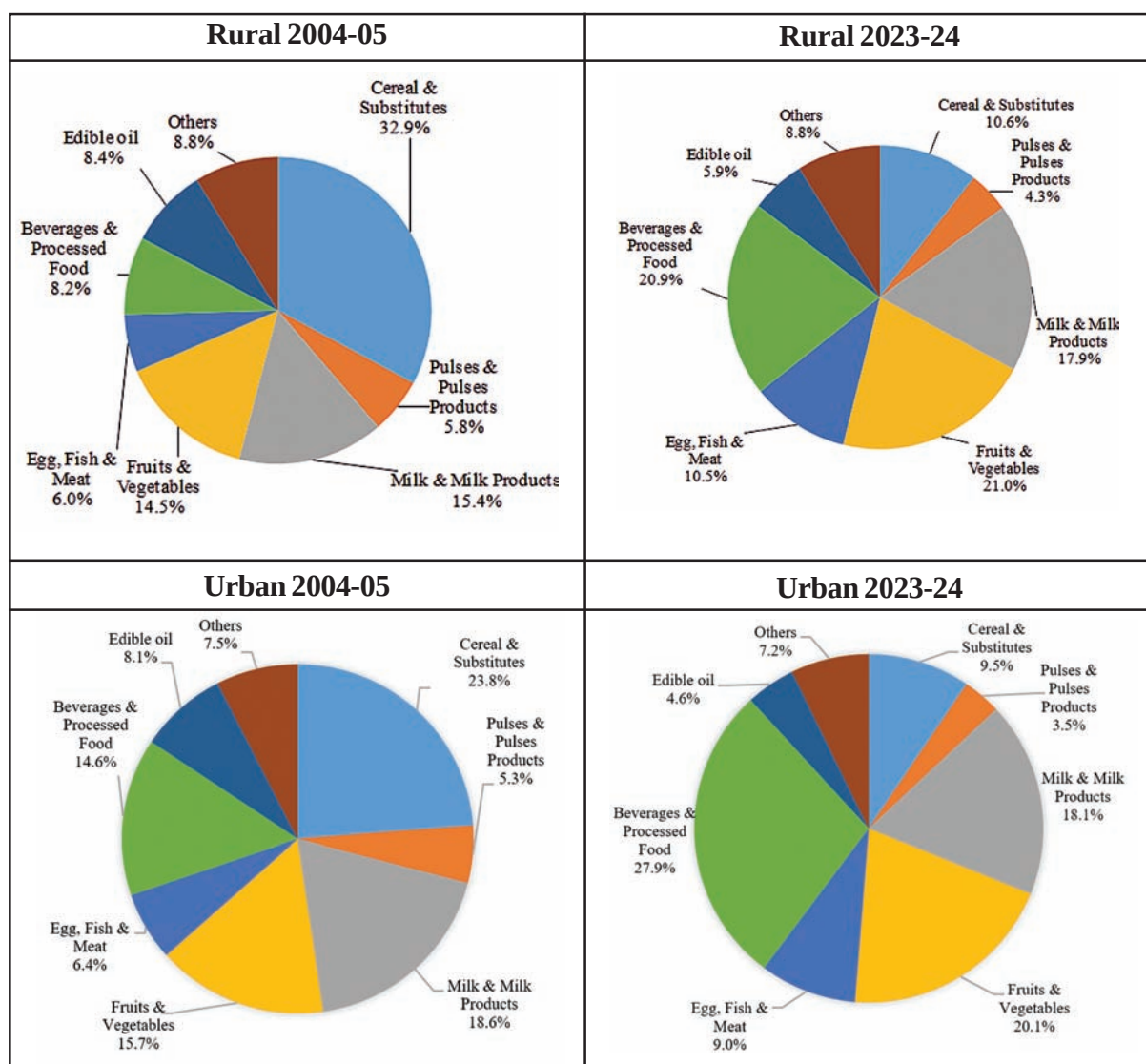
- 1.6 The CAGR of major crops and sub-sectors during the period 2015-16 to 2022-23 for major States are given in Annex Table 1.5. Karnataka recorded the highest growth rate (8.7%) in agriculture and allied sectors (excluding forestry and logging), followed by Telangana (8.5%), Andhra Pradesh (7.2%) and Madhya Pradesh (5.5%) among major States. Telangana registered the highest growth in crop sector (9.3%), while Rajasthan witnessed highest growth rate (7%) in fruits and vegetables. Karnataka had the highest growth rate of 11.5 percent in total livestock products and milk group and 16.5 percent in meat and eggs group. Six States experienced double digit CAGR in fishing and aquaculture, with highest growth in Madhya Pradesh (17%), followed by Andhra Pradesh (15%), Jharkhand (12.3%), Odisha (11.6%), Telangana (10.8%) and Karnataka (10.7%).

Changing Consumer Preferences and Food Consumption Patterns

- 1.7 Consumption patterns and preferences of consumers are changing in both rural and urban areas in the country due to rising income, urbanization, changing demographics and lifestyles. The dietary shift towards high value commodities like fruits, vegetables, milk and dairy products, meat, eggs, fish, beverages, processed foods, etc. is evident from Chart 1.1. As per the latest Household Consumption Expenditure Survey (HCES) 2023-24, share of expenditure on cereals and cereal products in total food expenditure witnessed steepest decline during the last two decades, from 32.9 percent in 2004-05 to 10.6 percent in 2023-24 in rural areas and from 23.8 percent to 9.5 percent in urban areas during the corresponding period. The share of livestock products including milk and dairy products, eggs, meat, fish, and fruits and vegetables witnessed a steady increase, but beverages and processed foods showed a remarkable increase, from 8.2 percent to 20.9 percent in rural and from 14.6 percent to 27.9 percent in urban areas between 2004-05 and 2023-24. The dietary pattern shifts towards energy-dense high-fat and ultra-processed foods are associated with overweight, obesity and other lifestyle diseases and caution should be taken to address these concerns. The National Family Health Survey (NFHS-5) results show that about 22.9 percent of adult men and 24 percent of adult women are overweight or obese, 15.6 percent men and 13.5 percent women have high or very high blood sugar and 21.3 percent women and 24 percent men have elevated blood pressure. However, rising importance of milk and dairy products, egg, meat, fish, fruits and vegetables in Indian diets has enormous demand potential and provides opportunities for crop diversification.
- 1.8 According to OECD-FAO Agricultural Outlook 2024-2033, average daily per capita calorie intake will increase by seven percent by 2033 in the middle-income countries, driven by greater consumption of staples, livestock products and fats. The resulting shifts in agricultural production and consumption will also lead to changes in international agricultural trade patterns.



Chart 1.1: Changing Consumption Pattern in India: 2004-05 and 2023-24



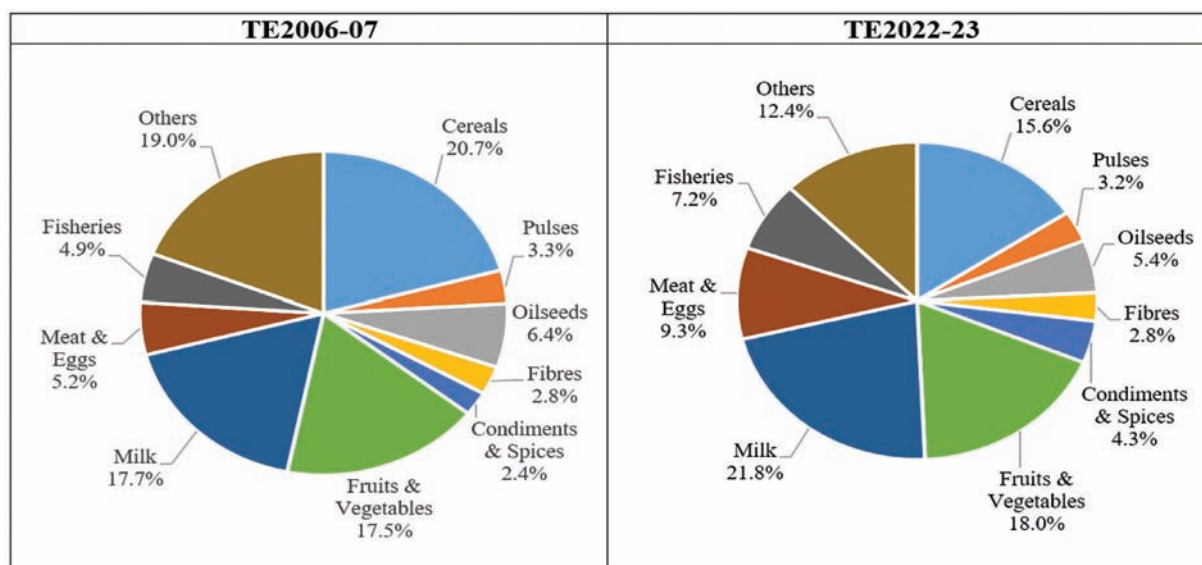
Source: Household Consumption Expenditure Survey (HCES) 2022-23 and 2023-24, MOSPI (2024)

Changing Production Patterns

- 1.9 India has a diverse agricultural, livestock and fisheries sector, producing a range of crops and livestock products. Changes in Indian agriculture can be seen through compositional shifts in its output (Chart 1.2). Over the last two decades, the share of horticulture, livestock and fisheries sector in the total value of output from agriculture and allied sectors has increased, , while share of grains has significantly reduced. There has been a decline in the contribution of cereals from 20.7 percent to 15.6 percent, while the share of livestock sector, mainly, milk increased from 17.7 percent to 21.8 percent and meat and eggs from 5.2 percent to 9.3 percent between TE2006-07 to TE2022-23. The share of fisheries increased from 4.9 percent to 7.2 percent during the period.



Chart 1.2: Changing Shares of Major Groups in Total Value of Output from Agriculture and Allied Sectors*



Note: *Gross Value of Output at current prices excluding Forestry & Logging

Source: Computed from Statistical reports on Value of Output from Agriculture and Allied Sectors, National Statistical Office, MOSPI (2024 & 2010)

Imbalance in Agricultural Production Pattern

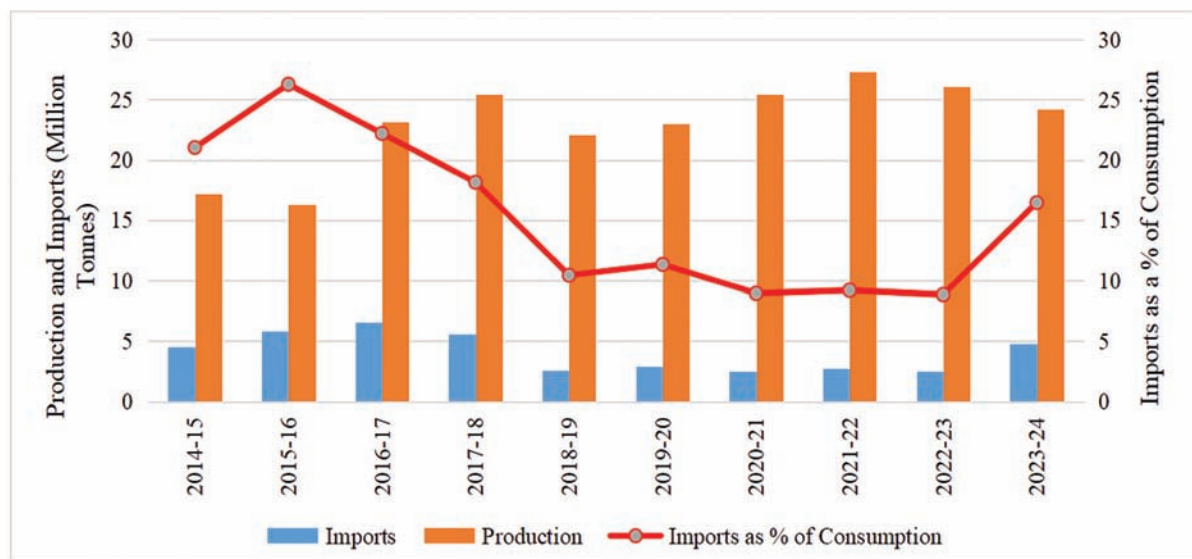
- 1.10 Imbalances between domestic production and demand for different agricultural commodities have been growing in the country over the years. India has large surplus production of rice, while country's deficit in edible oils and pulses is rising year after year resulting in higher import dependency. The country meets about 55-60 percent of domestic demand for vegetable oils through imports. In case of pulses, despite significant increase in production since 2015-16, India's import dependency is quite high in lentil, tur/arhar and urad. Thus, there is a great scope and need to increase domestic oilseeds and pulses production.

Pulses

- 1.11 Domestic production of pulses increased from 16.3 million tonnes in 2015-16 to 27.3 million tonnes in 2021-22, while imports declined from 6.6 million tonnes in 2016-17 to 2.5 million tonnes in 2022-23. However, during the last two years, pulses production declined and imports increased from 2.5 million tonnes in 2022-23 to 4.8 million tonnes in 2023-24. India has already imported about 4.2 million tonnes of pulses during first eight months of 2024-25. The dependence on imports, which declined from about 26.3 percent in 2015-16 to 8.8 percent in 2022-23, reached 16.5 percent in 2023-24 (Chart 1.3). Lentils accounted for 35.1 percent of total pulses imports, followed by peas (24.6%), tur/arhar (16.1%) and urad (13.1%) in 2023-24, which is primarily driven by imbalance in production pattern of pulses in the country (Chart 1.4). Gram production increased by 55 percent and moong production by 89.2 percent during the last 10 years, while production of tur/arhar increased by mere 6.9 percent and urad by 14.7 percent.



Chart 1.3: Trends in Production and Import of Pulses



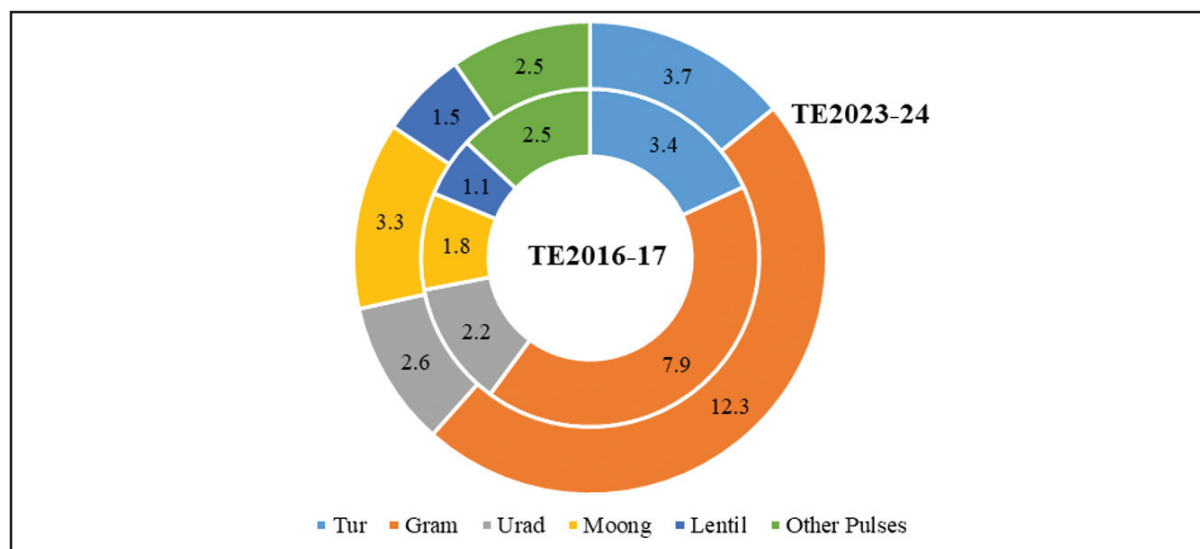
Note: Production plus imports used as a proxy for consumption

Source :- 1. Directorate General of Commercial Intelligence and Statistics

2. Economics, Statistics & Evaluation Division, Ministry of Agriculture & Farmers Welfare

Chart 1.4: Changing Production Pattern of Pulses, TE2016-17 and TE2023-24

(million tonnes)



Source: Economics, Statistics & Evaluation Division, Ministry of Agriculture and Farmers Welfare

- 1.12 In order to reduce dependence on imports and boost production of pulses with special focus on tur/arhar, urad and lentil, the Government announced **“Mission for Aatmanirbharta in Pulses”** in the Union Budget 2025-26 with a special focus on development and commercial availability of climate resilient seeds, enhancing protein content, increasing productivity, improving post-harvest storage and management, and assuring remunerative prices to farmers. This initiative will help in boosting domestic production and reduce dependence on imports for pulses.



Oilseeds

- 1.13 Although production of oilseeds in the country has increased significantly during the last few years, from 27.5 million tonnes in 2014-15 to 41.4 million tonnes in 2022-23, rising demand has significantly outpaced domestic production, leading to a heavy dependence on imports. Vegetable oils is the single largest commodity in India's total agricultural imports with an estimated share of 42.3 percent in 2023-24. During the last 10 years, imports of vegetable oils have increased from 10.5 million tonnes in 2015-16 to 15.7 million tonnes in 2022-23, while value of imports increased by almost three times from about ₹58 thousand crore to ₹167.4 thousand crore during the period.
- 1.14 In order to boost domestic production and achieving *Aatmanirbharta* in edible oils, Central Government has launched National Mission on Edible Oils – Oil Palm (NMEO-OP) to increase area under oil palm from 3.5 lakh hectares in 2019-20 to 10 lakh hectares by 2025-26 and production of Crude Palm Oil from 0.27 lakh tonnes to 11.2 lakh tonnes during the period. The National Mission on Edible Oils – Oilseeds (NMEO-Oilseeds) was approved by the Government in October 2024 with a financial outlay of ₹10,103 crore to increase oilseed production from 39 million tonnes in 2022-23 to 69.7 million tonnes by 2030-31. The Mission (NMEO-OP & Oilseeds) targets to increase domestic edible oil production to about 25.5 million tonnes by 2030-31 and reduce import dependence to less than 30 percent of projected domestic requirement.

Cotton

- 1.15 Bt cotton technology was the key driver of cotton production growth in the country since its introduction in 2002-03, but India's cotton sector is facing a crisis of stagnant or declining yield and area during the last 10 years. However, as the second largest cotton producing and mill use country in the world, India needs to focus on increasing both area and yield of cotton. India is a leading exporter of raw cotton in the world, but highly dependent on imports for meeting domestic demand for extra-long staple (ELS) and long staple (LS) cotton. Therefore, in order to give an impetus to the ELS cotton and improve cotton productivity, Government announced "Mission for Cotton Productivity" in the Union Budget 2025-26 with an allocation of ₹500 crore.

India's Position in World Agriculture

- 1.16 Table 1.2 provides international comparison of selected crops/groups highlighting India's performance. India accounts for 12 percent of the world's total cereal output, with 103.3 million hectares of area producing 374.6 million tonnes at an average yield of 3,626 kilogram per hectare in 2023. India's share in global pulses production is even more significant at 27.3 percent. Notably, India's contribution to world unmilled or rough rice (paddy) production stands at 25.8 percent, with 47.8 million hectare area under cultivation and 206.7 million tonnes of production at yield of 4,322 kilogram per hectare. India's share in world production of groundnut, sesame seed, and sorghum was 19 percent, 11.8 percent, and 6.7 percent, respectively. These data underscore India's important role and contribution in global agriculture. While the country contributes substantially to world production, there is significant scope for improving productivity and narrowing the yield gap compared to the global average yields.

**Table 1.2: International Comparison for Selected Crops, 2023***(Area in million hectares, production in million tonnes and yield in kg/ha)*

Crops	India			World			Share of India in World (%)	
	A	P	Y	A	P	Y	A	P
Total Cereals*	103.3	374.6	3626	738.8	3134.3	4243	14.0	12.0
Total Pulses**	34.8	25.7	740	96.6	94.3	976	36.0	27.3
Rice***	47.8	206.7	4322	168.4	800.0	4752	28.4	25.8
Maize	10.7	38.1	3545	208.2	1241.6	5962	5.2	3.1
Soyabean	13.1	15.0	1145	136.9	371.2	2711	9.6	4.0
Groundnut	5.0	10.3	2076	30.9	54.3	1755	16.0	19.0
Sorghum	3.5	3.8	1079	39.9	57.3	1438	8.9	6.7
Sesame seed	1.5	0.8	527	13.1	6.8	520	11.7	11.8

Note: *Total Cereals include paddy production;

**Pulses include imputed value of dry beans;

***Unmilled or rough rice (paddy)

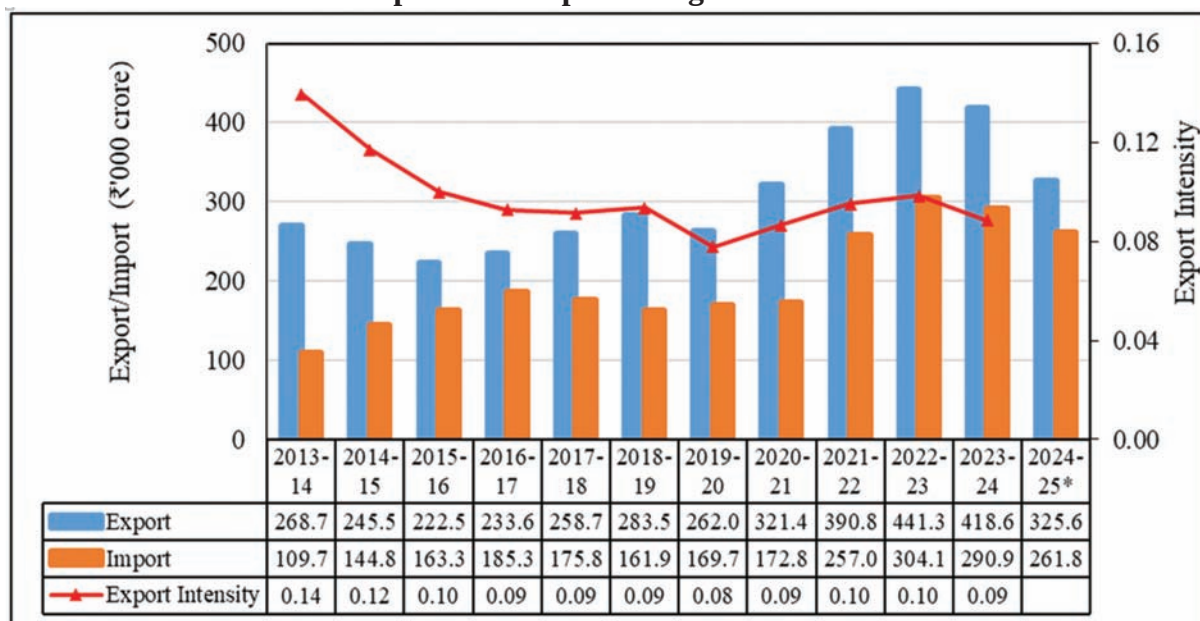
Source: Food and Agriculture Organization, (Data accessed from <http://www.fao.org/faostat/en/#data> on 15th January 2025)

Agricultural Trade

- 1.17 India has emerged as a major agricultural exporter and is among the top ten exporters of agricultural products in the world. India's agricultural exports almost doubled from ₹2.22 lakh crore in 2015-16 to a record of more than ₹4.4 lakh crore in 2022-23 (Chart 1.5). However, due to various export restrictions, agricultural exports registered a decline of about five percent during 2023-24. The trade surplus ranged between ₹48.3 thousand crore in 2016-17 to ₹148.6 thousand crore in 2020-21. The export intensity, measured as the ratio of agricultural exports over GVA from agriculture, declined from 0.12 in 2014-15 to 0.08 in 2019-20 with a slight improvement in 2018-19, showed an increasing trend during 2019-20 to 2022-23, but marginally declined again in 2023-24. The instability, measured as the ratio of standard deviation to mean value for the period 2014-15 to 2023-24, was lower in agricultural exports compared with imports, which suggests that exports are relatively stable when subjected to external policy interventions and economic fluctuations. India's agricultural exports and imports for the period April-December 2024 were ₹325.6 thousand crore and ₹261.8 thousand crore, an increase of 7.9 percent and 20 percent, respectively, over corresponding period last year.
- 1.18 Rice continued to be the largest commodity in Indian agricultural export basket with a share of 20.6 percent despite export restrictions, followed by marine products (14.6%), spices (8.4%) and meat & processed meat (7.6%) during 2023-24. However, value of sugar exports declined by 49.5 percent primarily due to ban on exports. Vegetable oils, with a share of 42.3 percent, were the largest commodity in India's import basket, followed by pulses (10.7%), fresh fruits (7.8%) and sugar (5.7%).



Chart 1.5: India's Exports and Imports of Agricultural Commodities



Note: * indicates April to December period; Export Intensity is the ratio of Exports to Agri-GVA

Source: Directorate General of Commercial Intelligence and Statistics, Ministry of Commerce and Industry

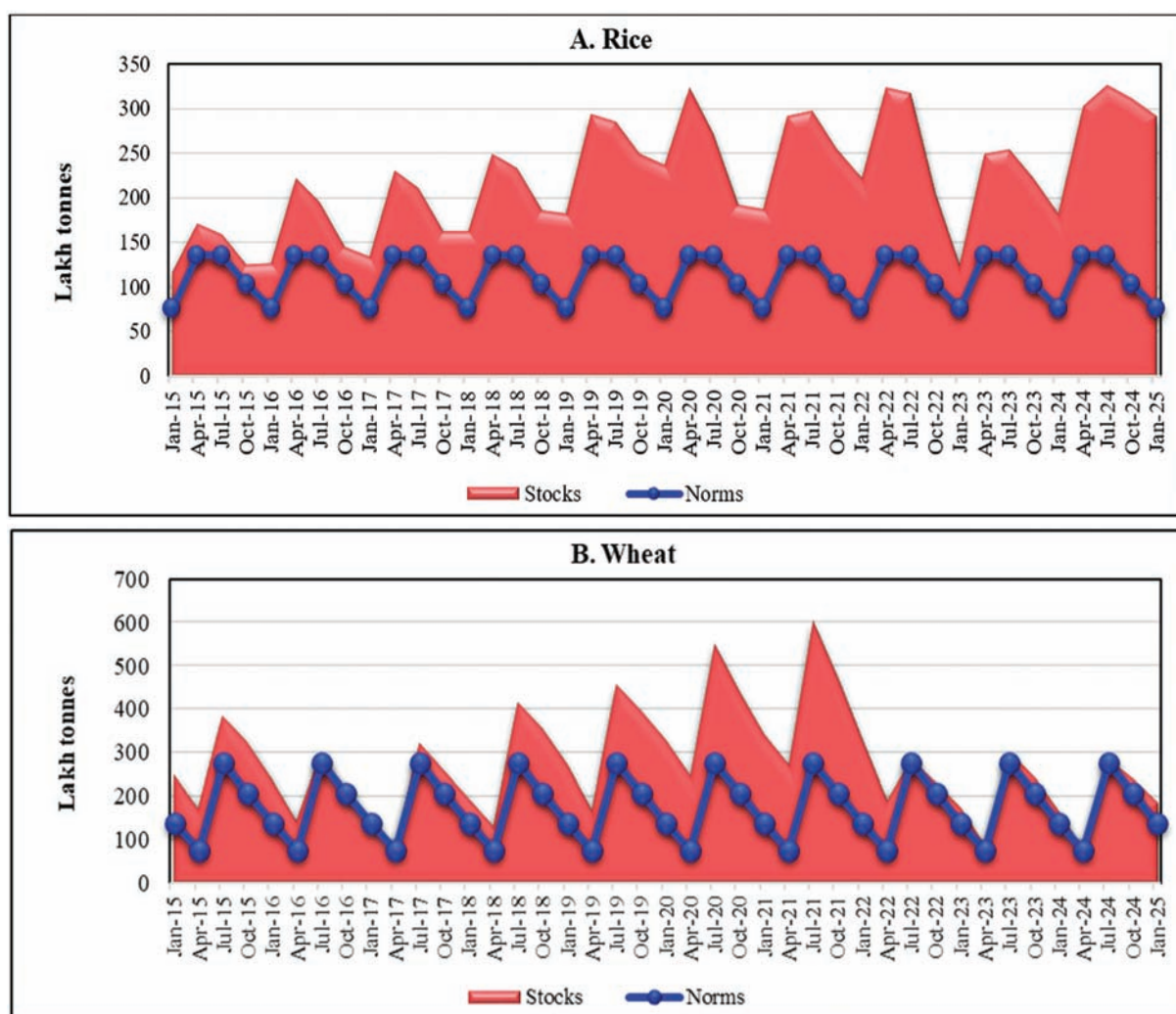
Excess Stocks of Rice in Central Pool

- 1.19 Since January 2015, Central Pool stocks of rice (29.9 million tonnes) and wheat (60.3 million tonnes) were at the record high of 90.2 million tonnes in June 2021, followed by a steady decline to a lowest level of 29.7 million tonnes in January 2023 (Chart 1.6). During the last five years, the lowest stocks of rice (11.5 million tonnes) were in December 2022 and that of wheat in April 2023 at about 8.3 million tonnes due to low procurement. Wheat stocks have been marginally higher than the stocking norms but rice stocks remained significantly higher than norms during last two years. The total stocks of rice and wheat for the quarter beginning 1st January 2025 were 47.5 million tonnes, more than double the stocking norms for the Central Pool. The total stocks of rice and wheat on 31st January 2025 (closing stock) were 50.2 million tonnes comprising 34 million tonnes of rice and 16.2 million tonnes of wheat. The large stocks of grains in the Central Pool result in higher economic cost, which in turn raises the food subsidy. Therefore, there is a need to liquidate excess rice stocks.
- 1.20 Despite excessive stocks of rice, the annual rate of inflation of paddy based on all-India Wholesale Price Index (WPI) number has remained high during the last one year, although it has declined from 12.03 percent in April 2024 (over April 2023) to 6.22 percent (provisional) in January 2025. Similarly, wheat inflation has also remained high since March 2024 and witnessed a steady increase from June 2024 to November 2024, declined marginally in December 2024 but reached the highest level (9.75%) in January 2025.
- 1.21 In order to moderate prices in open market and liquidate excess stocks, Government announced Open Market Sale Scheme (Domestic) [OMSS(D)] Policy for rice and wheat in July 2024 for 2024-



25, which was revised for sale of rice on 7th January and 17th January 2025. During the last six months (August 2024 to January 2025), about 10.4 lakh tonnes of wheat and 11.7 lakh tonnes of rice have been sold under the OMSS(D). Besides, 78.4 thousand tonnes of *Bharat atta* and 34.3 thousand tonnes of *Bharat rice* have been sold since April 2024 under *Bharat* brand, which was launched on 6th November 2023 and 6th June 2024, respectively. In addition, 2.4 million tonnes of rice is also being offered for sale to ethanol distilleries for production of ethanol at a fixed price of ₹2,250 per quintal under OMSS(D) policy. In view of low wheat stocks and relatively high domestic prices, Government of India imposed stock limits on wheat under the Removal of Licensing Requirements, Stock Limits and Movement Restrictions on Specified Foodstuffs (Amendment) Order, 2024 to prevent hoarding and unscrupulous speculation on 24th June 2024, which was revised on 9th September 2024 and again on 11th December 2024. The Government further reduced the wheat stock limit for traders/wholesalers, retailers, and big chain retailers applicable until 31st March 2025 on 20th February 2025.

Chart 1.6: Trends in Stock Position and Stocking Norms of Rice and Wheat in the Central Pool, 2015 to 2025



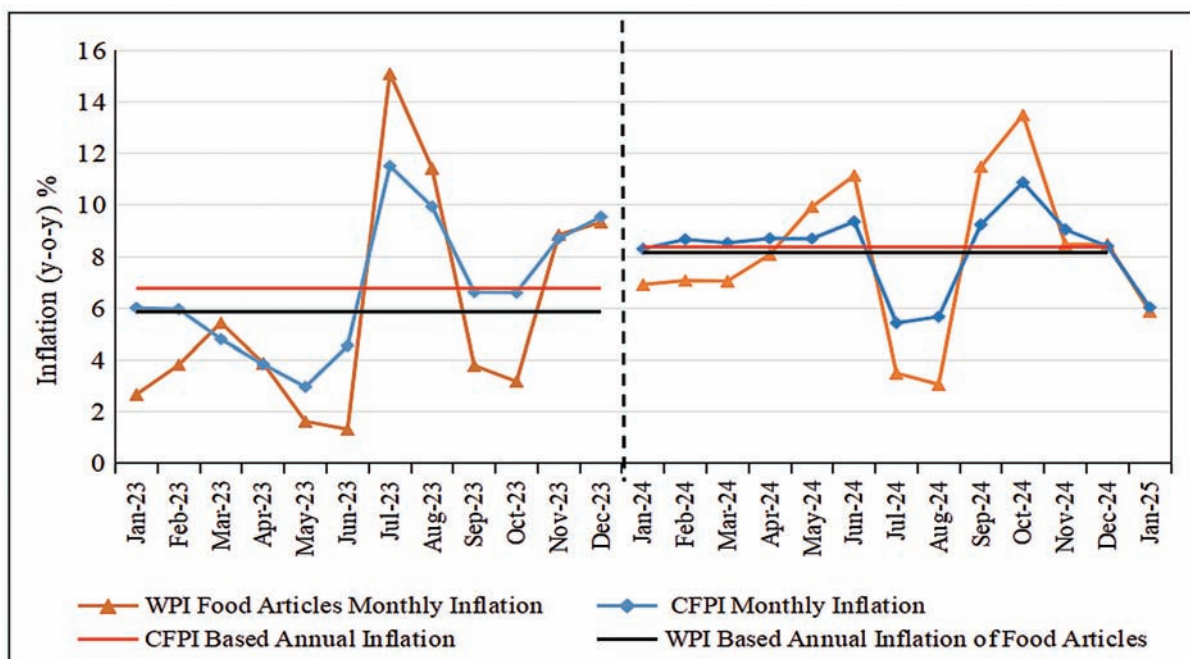
Source: Department of Food and Public Distribution, Ministry of Consumer Affairs, Food and Public Distribution



Food Inflation

- 1.22 The average food inflation based on Consumer Food Price Index (CFPI) in India was higher (8.4%) in 2024 compared to 2023 (6.8%) and was mainly driven by fruits & vegetables, pulses and products, cereals and products and eggs (Chart 1.8). Fruits and vegetables, pulses and products, eggs, meat and fish witnessed high inflation in 2024, while inflation in spices, milk and products, and cereals and products declined in 2024 compared with 2023. Food inflation based on CFPI varied from 2.96 percent in May 2023 to 11.51 percent in July 2023 and showed a declining trend during the last three months (November 2024 to January 2025).
- 1.23 The annual rate of inflation based on all India Wholesale Price Index (WPI) Food Articles increased from 5.87 percent in 2023 to 8.17 percent in 2024. Vegetables, pulses, paddy, fruits and wheat were major drivers of inflation during 2024, though inflation in wheat and milk declined during 2024. Both vegetables and pulses recorded double digit inflation during 2024 and inflation rate for oilseeds was negative. The inflation rate for vegetables, pulses and paddy remained elevated during 2023 and 2024, but it is surprising to note that despite excess stocks of rice in the Central Pool, paddy inflation was high and even rose during 2024. The annual rate of inflation based on all India WPI of Food Articles declined during the last three months (November 2024 to January 2025) from a peak of 13.49 percent in October 2024.

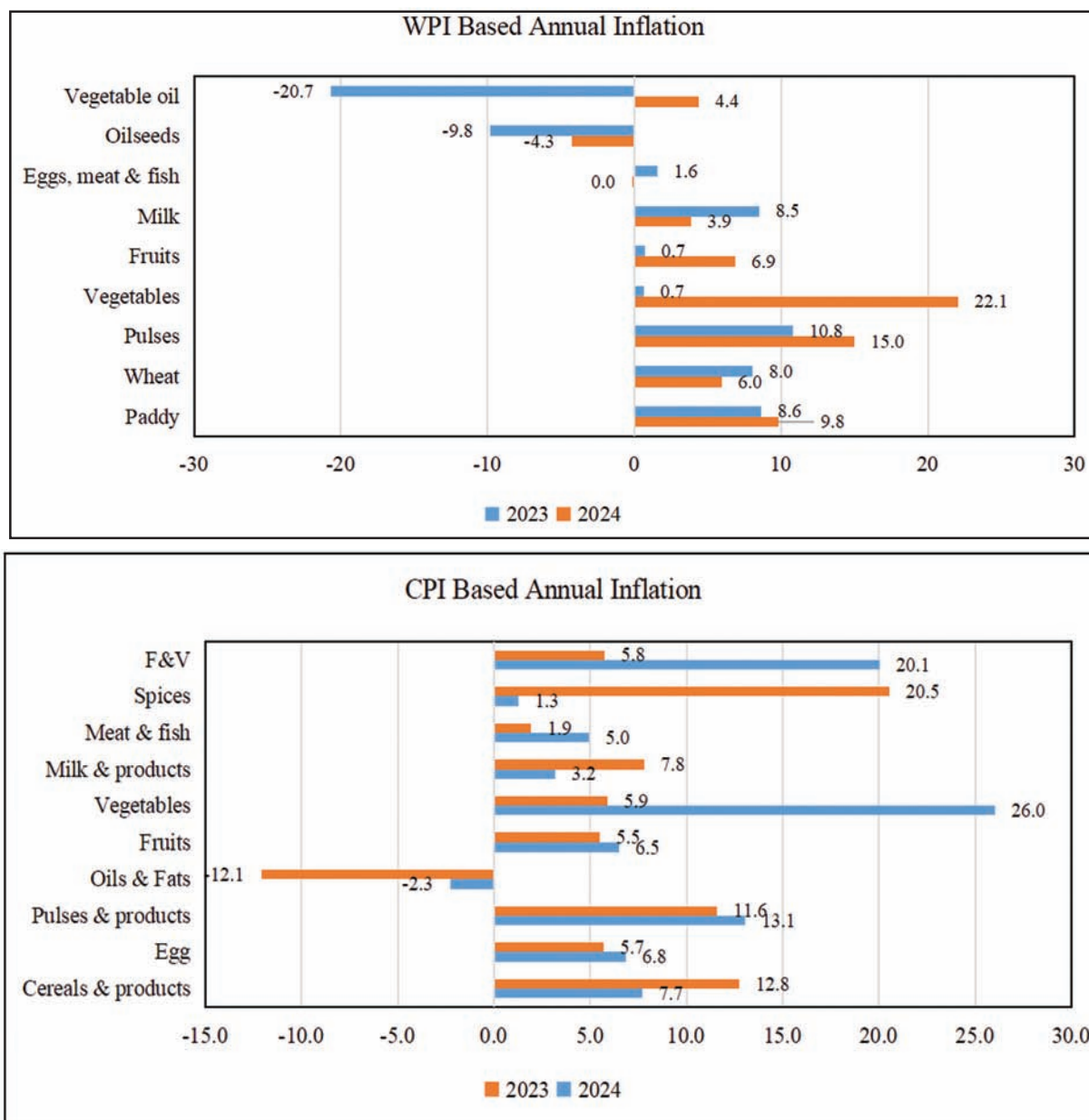
**Chart 1.7: Trends in Food Inflation based on CPI and WPI,
January 2023 to January 2025**



Source: 1. Ministry of Statistics & Programme Implementation
2. Office of Economic Adviser, Ministry of Commerce & Industry



Chart 1.8: Rate of Inflation (%) based on Wholesale Price Index (2011-12=100) and Consumer Price Index (2012=100) for Major Food Commodity Groups



Note: F&V- Fruits and Vegetables

Source: 1. Ministry of Statistics & Programme Implementation

2. Office of the Economic Adviser, Ministry of Commerce & Industry

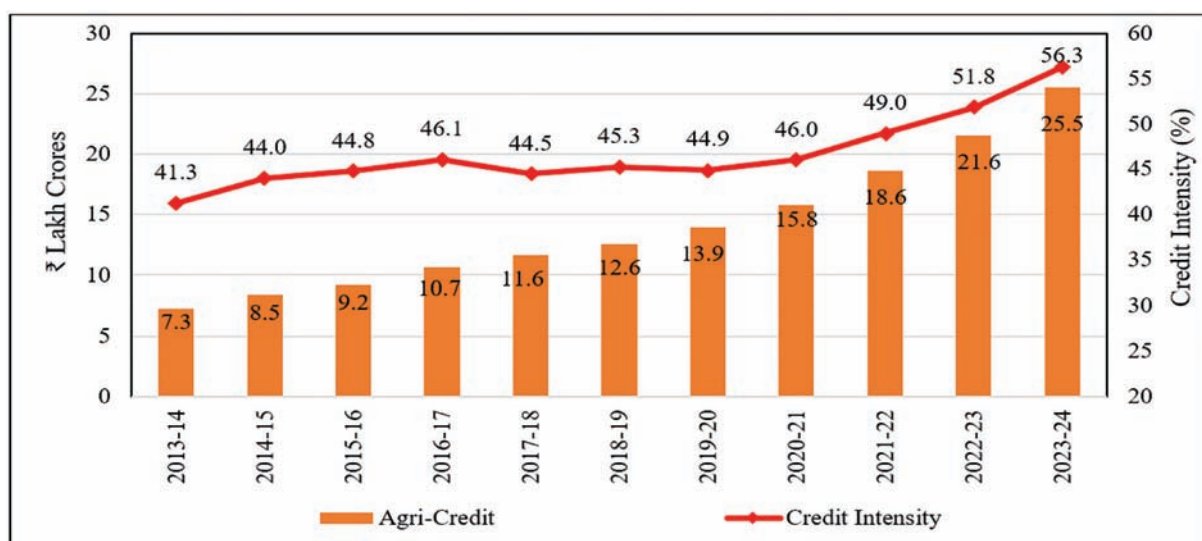
Agriculture Credit: Efficiency and Equity

- 1.24 Indian agricultural credit system has made commendable progress in terms of the scale and expansion of institutional credit. The ground level credit has risen from about ₹8.5 lakh crore in 2014-15 to about ₹25.5 lakh crore in 2023-24. Within this, the share of small and marginal farmers significantly increased from ₹3.46 lakh crore (41%) to ₹14.39 lakh crore (57%) between 2014-15 and 2023-24. Agricultural credit, on an average, grew at a compound annual growth rate of about 12.7 percent (in nominal



terms) during 2014-15 to 2023-24, which was significantly higher than the overall growth in bank credit during this period. This shows an increase in the allocation of credit to agricultural sector as compared to other sectors. The credit intensity, measured by agricultural credit as a ratio of nominal agriculture GVA (excluding forestry and logging), has risen consistently during the last 10 years and stood at 56.3 percent in 2023-24 (Chart 1.9). There has been a continuous decline in share of cooperatives in total credit, from 19.3 percent in 2011-12 to 9.5 percent in 2023-24, while share of Scheduled Commercial Banks (SCBs) has increased from 68.7 percent to 79.4 percent during the period. It is worth noting that share of long-term credit, which witnessed a declining trend during the early-2000s, increased during the last 10 years, from 24.8 percent in 2014-15 to 43.3 percent in 2020-21, but declined during 2021-22 and 2022-23 and marginally improved during 2023-24. However, there are large regional and farm size disparities in disbursement of agricultural credit. The eastern and north-eastern regions and small and marginal farmers have poor access to institutional credit. Therefore, the issue of regional disparity in credit distribution and poor access of small and marginal farmers needs to be addressed.

Chart 1.9: Trends and Intensity of Institutional Credit in Agriculture in India



Note: Credit intensity is the ratio of agricultural credit to Agri-GVA (at current prices) excluding forestry and logging

Source: Ministry of Agriculture and Farmers Welfare

- 1.25 Government has taken various policy initiatives for increasing flow of institutional credit and reduce dependence on non-institutional sources of lending. In 1998, *Kisan* Credit card (KCC) was introduced to provide timely credit to farmers and in 2006-07, Interest Subvention Scheme, a Central Sector Scheme 100 percent funded by Government of India, was launched to provide short-term agricultural credit to farmers through KCC at concessional rate of interest. The allocation under Interest Subvention/ Modified Interest Subvention Scheme (MISS) has increased from ₹6,000 crore in 2014-15 to ₹22,600 crore in 2025-26(BE). The KCC scheme has played a pivotal role in providing timely credit



to farmers, with over 7.72 crore operative KCC accounts involving a total outstanding credit of ₹10.05 lakh crore as on 31st December 2024. The share of institutional credit, which was about 10 percent in 1951, increased to 65 percent in 1991 and hovered around it until 2013. However, due to various initiatives taken by the Government, the share increased significantly during the last 10 years and reached 75 percent in 2022. The KCC saturation drive for PM KISAN beneficiaries, which was launched in February 2020 to cover 2.5 crore farmers, including fishermen and dairy farmers, with a credit boost of ₹2 lakh crore, has made significant progress. As on 17th January 2025, 5.69 crore farmers have been issued KCCs as part of saturation drive. In December 2024, Reserve Bank of India (RBI) enhanced the collateral-free loan limit for farmers from ₹1.6 lakh to ₹2 lakh with effect from January 1, 2025. In addition, the Government increased the KCC loan limit from ₹3 lakh to ₹5 lakh under the Modified Interest Subvention Scheme in the recent budget. These interventions would further help in improving access to institutional credit, particularly to small and marginal farmers and in eastern and north-eastern region, where the institutional credit penetration is low.

High-Yielding and Climate-Resilient Variety Seeds

- 1.26 Inadequate access to good quality seeds of new improved varieties is one of the major constraints faced by farmers. Despite large number of improved varieties released by research institutes and State Agricultural Universities and potential benefits of these varieties, widespread adoption remains limited due to various socio-economic, institutional and infrastructure constraints as well as less developed seed value chains. Seed Replacement Rate (SRR) and Varietal Replacement Rate (VRR), instrumental in improving crop productivity, have improved over the years, but are still quite low in some crops, making introduction of new varieties difficult. It is worth noting that 109 varieties of 61 crops, including 34 field crops of cereals, millets, oilseeds, pulses, sugarcane, cotton, etc. and 27 horticultural crops were released by the Hon'ble Prime Minister in August 2024. However, it is important to ensure production and multiplication of high yielding certified/quality seeds of these varieties in sufficient quantities and make available to farmers. In order to enhance availability of quality seed and disseminate information about new varieties to the farmers, active participation of farmers/farmers' organizations and coordination among research institutions, State Governments, seed industry and other stakeholders in seed production programme is needed. The Commission believes that recently announced '*National Mission on High Yielding Variety/Hybrid Seeds*' with an allocation of ₹100 crore aimed at strengthening the research ecosystem, targeted development and propagation of seeds with high yield, pest resistance and climate resilience, and enhancing commercial availability of more than 100 varieties released recently would help in making available quality seed of these varieties at affordable prices to farmers. This would also help in improving SRR and VRR and enhancing productivity of the crops.

Balanced Fertilizer Nutrient Management and Soil Health

- 1.27 Balanced application of fertilizer nutrients is essential for enhancing their use efficiency and improving crop yields by improving overall health of the soil. Various studies have shown that majority of Indian soils are deficient in macro and micronutrients and have low organic carbon. In order to provide



information to farmers about nutrient status of their soil and recommendation on appropriate dosage of nutrients to be applied for improving soil health and fertility, Soil Health Cards (SHCs) Scheme was launched in 2014-15. Under the Scheme a massive programme of soil sampling, testing and providing soil health cards to all farmers as well as strengthening of soil testing facilities in the country was implemented. The Scheme has made significant progress as 24.74 crore soil health cards have been distributed till February 2025 and 8,242 soil testing laboratories have been set up in the country. However, N:P:K use ratio has remained distorted at 10.9:4.4:1 during 2023-24 against ideal ratio of 4:2:1. The highly imbalanced use of N, P and K along with deficiency of micronutrients result in adverse impact on crop productivity, farmers' profitability and environment. High subsidy for urea, which substantially lowers the price of nitrogen relative to other nutrients, promotes inefficient and ineffective use. Therefore, there is a need to review fertiliser subsidy policy and promote use of phosphatic, potassic and complex fertilisers.

- 1.28 Recognizing the problem of excessive and imbalanced use of fertilisers as well as deficiency of micro-nutrients, the Government has launched various initiatives such as PM Programme for Restoration, Awareness Generation, Nourishment, and Amelioration of Mother Earth (PM-PRANAM), Sulphur-coated urea (Urea Gold), zincated urea, neem coated urea, promotion of drones and fertigation techniques, Nano-fertilizers, National Mission on Natural Farming (NMNF), etc. to improve soil health, fertility and quality, promote balanced use of chemical fertilizers and sustainable agricultural practices.

Irrigation Development and Sustainable Water Management

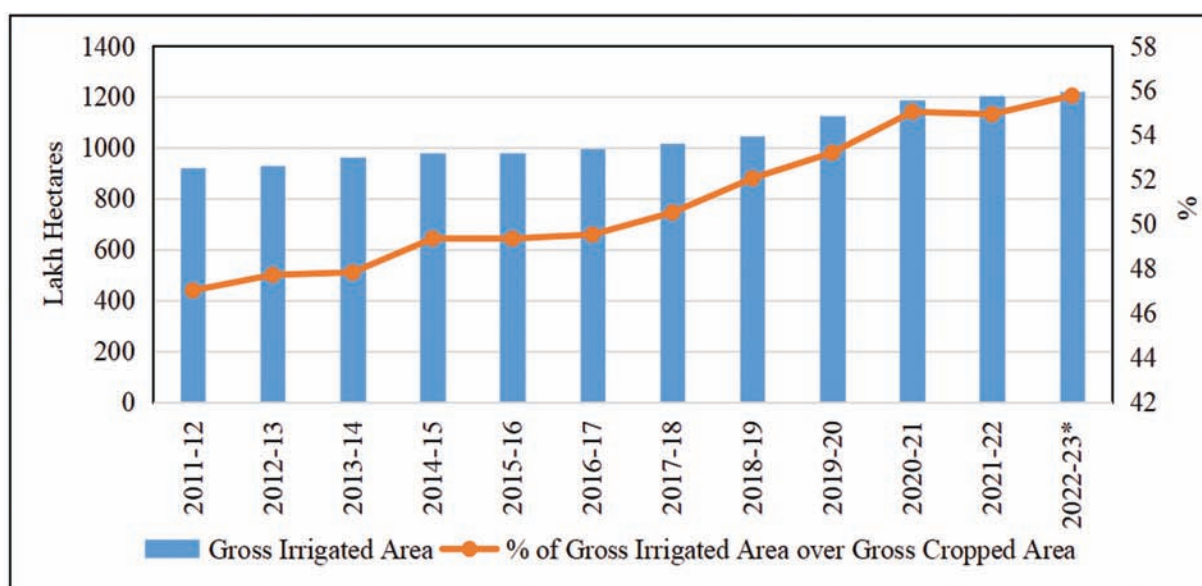
- 1.29 Irrigated agriculture, which accounts for the largest share in water use, will continue to play a key role in agricultural production growth. However, increasing competition between agriculture and other users is becoming a growing concern in many countries, including India. Therefore, there is urgent need for development of efficient irrigation systems and promotion of innovative irrigation practices that enhance water use efficiency. In order to increase coverage of irrigation ('*Har Khet Ko Pani*') and improve water use efficiency ('*Per Drop More Crop*'), Government of India launched *Pradhan Mantri Krishi Sinchayee Yojana* (PMKSY) in July 2015. In addition, Micro Irrigation Fund (MIF) was operationalized by NABARD in 2019-20 to facilitate State Governments for expanding coverage under micro-irrigation and incentivizing its adoption. As on 31st December 2024, 14 projects have been sanctioned to eight States and ₹4,719 crore loan has been sanctioned, against which ₹3,639 crore has been disbursed.
- 1.30 Significant progress has been made in increasing irrigation coverage and intensity and expansion of micro-irrigation during the last few years. Gross Irrigated Area (GIA) has increased from 91.9 million hectares in 2011-12 to 122.3 million hectares during 2022-23 and percentage of GIA over Gross Cropped Area (GCA) has increased from about 47 percent to 55.8 percent during the period (Chart 1.10). The GIA rose by 23.9 percent between TE2015-16 and TE2022-23 compared to about 9.5 percent increase in GCA. However, there are large disparities in irrigation coverage across States and



GIA as percent of GCA is as high as 98 percent in Punjab, 94 percent in Haryana, 86 percent in Telangana and 84 percent in Uttar Pradesh but less than 25 percent in States like Jharkhand, Assam, Kerala and Maharashtra.

- 1.31 Per Drop More Crop (PDMC), a component of the PMKSY, has made good progress and impact on improving water use efficiency. From 2015-16 till December 2024, Central assistance of ₹22,265 crore has been released to the States for implementation of PDMC and about 95.6 lakh hectare area has been covered under micro-irrigation. However, about 75 percent of area under micro-irrigation is concentrated in six States, namely, Karnataka, Rajasthan, Maharashtra, Gujarat, Tamil Nadu and Andhra Pradesh.

Chart 1.10: Trends in Gross Irrigated Area in India



Note: * Provisional

Source: Economics, Statistics & Evaluation Division, Ministry of Agriculture and Farmers Welfare

Direct Income Support to Farmers

- 1.32 The *Pradhan Mantri Kisan Samman Nidhi* (PM-KISAN) Scheme, a Central Sector Scheme launched in February 2019, is a flagship programme of the Government to supplement financial needs of farmers. Under the Scheme, a financial assistance of ₹6,000 per year is transferred in three equal installments to bank accounts of farmers through Direct Benefit Transfer (DBT). Government of India has disbursed over ₹3.68 lakh crore to about 9.8 crore farmers in 19 instalments since inception of the Scheme. Various studies including International Food Policy Research Institute (2019), Agro-Economic Research Centre (2022) and independent researchers found that PM-KISAN eased farmers' credit constraints, increased agricultural input investments, as well as helped in adoption of modern techniques and increase in yield and income. In addition to PM-KISAN, various States provide additional assistance to farmers under State Schemes.



Farm Mechanization

- 1.33 Human labour remains a major cost factor contributing a significant share (nearly 50%) of total operational costs in mandated *kharif* crops. Labour shortages push farm wages, which increase the cost of production. To address this problem, traditional farming practices and methods need to be replaced by widespread adoption of farm mechanization and role of farm mechanization should go beyond tractorization, with emphasis on mechanization of various pre- and post-harvest operations. However, small and fragmented land holdings, high cost of machinery, non-availability of suitable machines/ implements and low capacity of farmers to make initial capital investment are major impediments to the growth of farm mechanization in the country. Level of farm mechanisation is also uneven across the States/regions as farm power availability ranges from 6 kW/ha in Punjab to 0.5 kW/ha in the north-eastern States.
- 1.34 In order to increase reach of farm mechanization to small and marginal farmers and to the regions where availability of farm power is low, Government has been implementing Centrally Sponsored Scheme ‘Sub-Mission on Agricultural Mechanization (SMAM)’ since April 2014. Under the Scheme, ‘Custom Hiring Centres’ (CHC) are being promoted to offset the adverse economies of scale due to small farm size and high cost of individual ownership, and demonstration and capacity building programmes are organized to create awareness among stakeholders. As on 31st January 2025, 52,173 CHCs/Hi-Tech Hubs/Farm Machinery banks have been established under the SMAM and 19.5 lakh machines have been distributed to the farmers. In order to make effective use of machines available with the CHCs and farmers, a multi-lingual mobile app-based aggregator platform ‘FARMS’ App (Farm Machinery Solutions App) has been launched, which connects farmers with CHCs, like cab aggregators. More than 75 thousand CHCs/Service Providers have been registered on this Mobile App to rent out about than 1.76 lakh agricultural machines benefitting 5.2 crore farmers.
- 1.35 To protect environment from air pollution and prevent loss of nutrients and soil micro-organisms due to burning of crop residue, a Central Sector Scheme ‘Promotion of Agricultural Mechanization for *In-Situ* Management of Crop Residue’ in the States of Punjab, Haryana, Uttar Pradesh and NCT of Delhi has been implemented from April 2018. Under the Scheme, 41,968 CHCs/Farm Machinery banks have been established and about 3.23 lakh machines have been distributed to farmers. The Scheme has been merged under SMAM and implemented as Centrally Sponsored component under *Rashtriya Krishi Vikas Yojana* (RKVY).
- 1.36 Given the role that farm mechanization can play in making Indian agriculture more efficient, productive, and competitive, the budget allocation under SMAM is inadequate and needs to be increased substantially.

Agri-Market Infrastructure Development Initiatives

- 1.37 Access to affordable physical infrastructure supporting on-farm production, efficient marketing, storage and warehousing, and value-addition is a major determinant of competitiveness in agriculture.



Therefore, Government has accorded highest priority for development of market infrastructure and launched two major Schemes, namely, Integrated Scheme for Agricultural Marketing (ISAM) and Central Sector Scheme for Formation & Promotion of new 10,000 FPOs. Agricultural Marketing Infrastructure (AMI) aimed to creating storage and other infrastructure, and National Agriculture Market (e-NAM) are two important sub-schemes of ISAM. The e-NAM has made significant progress and as on 31st January 2025 over 1.82 crore farmers, 4,362 FPOs and 2.63 lakh traders have been registered on the e-NAM portal. Currently, 1,466 APMC mandis from 23 States and four Union Territories are linked to the e-NAM network. Under the AMI, 48,611 storage infrastructure projects have been sanctioned and ₹4,795 crore subsidy has been disbursed. In addition, more than 21 thousand projects related to other types of infrastructure under the AMI Scheme with a subsidy of about ₹2,126 crore have been sanctioned.

- 1.38 Government of India launched “Agriculture Infrastructure Fund” (AIF) in 2020-21 to strengthen infrastructure by creating farm gate storage and logistics infrastructure. As on 26th January 2025, 92,393 projects worth ₹56,334 crore have been sanctioned, which have mobilized an investment of ₹91,856 crore. Major projects sanctioned under AIF include warehouses (14,727), sorting and grading units (3,430), cold store projects (2,190), custom hiring centres (24,477), primary processing units (19,030) and about 29 thousand other kinds of post-harvest management projects.
- 1.39 In addition, the Government has approved the “World’s Largest Grain Storage Plan in Cooperative Sector” on 31st May 2023 for creation of various agri-infrastructure including godowns and processing units at Primary Agricultural Credit Societies (PACS) level through convergence of various Schemes such as AIF, AMI, SMAM, *Pradhan Mantri* Formalization of Micro Food Processing Enterprises Scheme (PMFME), etc.

Outlook for Indian Agriculture

- 1.40 In the Union Budget 2025-26, among proposed development measures focusing on *Garib*, Youth, *Annadata* and *Nari*, agricultural growth and productivity has been accorded the top priority and the sector as the first engine of development. The key initiatives announced in the budget for accelerating growth in agricultural sector are given in Box 1.1.
- 1.41 India’s foodgrains production hit a record 332.3 million tonnes in the 2023-24, driven by record rice (137.8 million tonnes) and wheat (113.3 million tonnes) output. However, maize, pulses and oilseeds production in 2023-24 was lower than 2022-23. As per the Second Advance Estimates of Production for 2024-25, production of most of mandated *kharif* crops except bajra, urad and cotton is estimated to increase in 2024-25 over 2023-24. *Kharif* rice production is estimated at 120.7 million tonnes, 6.6 percent higher than 2023-24, while *kharif* oilseeds production is estimated to increase by 14.4 percent. *Kharif* maize production is expected to rise by 11.5 percent, while *kharif Shree Anna* is estimated to increase by 3.5 percent. Cotton production is estimated to



decline by 9.5 percent, from 32.5 million bales in 2023-24 to 29.4 million bales in 2024-25.

- 1.42 *Rabi* foodgrains production in the country is estimated at 164.5 million tonnes in 2024-25, an increase of 2.8 percent over 2023-24. *Rabi* rice is also set to rise by 7.9 percent to 15.76 million tonnes, while *rabi* pulses production is expected to increase by 4.5 percent to 15.9 million tonnes. *Rabi* oilseeds production is estimated to fall by 1.7 percent to 14 million tonnes. Urad, groundnut and moong production during *rabi* 2024-25 are also estimated to register a strong positive growth.
- 1.43 As per Central Water Commission (CWC) reservoir storage bulletin dated 27th February 2025, the total live storage available in 155 reservoirs of the country is 93.53 Billion Cubic Meter (BCM), which is 52 percent of total live storage capacity of these reservoirs. The current year's storage is nearly 119 percent of the live storage of corresponding period of last year and 115 percent of normal storage, which indicates favorable water availability for agricultural sector.

Box 1.1: Key initiatives announced in Budget 2025-26 for Agriculture

Prime Minister Dhan-Dhaanya Krishi Yojana - Developing Agri Districts Programme

- Will cover 100 districts with low productivity, moderate crop intensity and below-average credit parameters. It aims to enhance agricultural productivity, adopt crop diversification and sustainable agriculture practices, augment post-harvest storage at the panchayat and block level, improve irrigation facilities and facilitate availability of long-term and short-term credit. The Scheme is likely to benefit 1.7 crore farmers and will be implemented in partnership with States.

'Aatmanirbharta' in Pulses

- Government will launch a 6-year "Mission for Aatmanirbharta in Pulses" with focus on tur, urad and masoor. NAFED and NCCF to procure these pulses from farmers who register with these agencies during the next 4 years to assure remunerative prices to farmers.

National Mission on High Yielding Seeds

- A National Mission on High Yielding Seeds to be launched aiming at strengthening the research ecosystem, targeted development and propagation of seeds with climate resilient high yield varieties and commercial availability of more than 100 new seed varieties.

Mission for Cotton Productivity

- A 5-year mission announced to facilitate significant improvements in productivity and sustainability of cotton farming, and promote extra-long staple cotton varieties. The mission will be aligned with the 5F (Farmer, Fiber, Factor, Fashion and Foreign) vision for the textile sector to increase incomes of the farmers and ensure a steady supply of quality cotton for rejuvenating India's traditional textile sector.

Enhanced Credit through KCC

- The loan limit under the Modified Interest Subvention Scheme to be enhanced from ₹3 lakh to ₹5 lakh for loans taken through the KCC.



Global Outlook

- 1.44 According to the FAO, USDA and IGC forecasts, world rice production, supply and utilization are expected to increase in 2024-25, while rice trade is forecast to remain at the same level as in 2023-24. World rice stocks are predicted to rise and reach a fresh peak in 2024-25 but little change is estimated in Stock to Use Ratio (SUR) due to record consumption in 2024-25.
- 1.45 Global production and supply of maize is forecast to be lower in 2024-25 than 2023-24, while consumption is estimated to rise due to stronger feed and industrial use. Global trade is projected to drop by about 6-7 percent and stocks are forecast to fall below the opening level in 2024-25, resulting in lower SUR.
- 1.46 As per USDA estimates, global production, supply and utilization of oilseeds complex (oilseeds, oil meal and vegetable oils) is forecast to increase by about three percent in 2024-25, while trade is estimated to marginally increase (about 1%). World stock of oilseeds (5.7%) and oil meals (11.4%) are expected to have a sizeable increase but vegetable oil stocks are projected to fall by about eight percent in 2024-25. World production, supply, and utilization of soybean are forecast to expand by 6-8 percent, while trade is predicted to increase marginally (1%) in 2024-25. World stocks and SUR of soybean are estimated to increase substantially in 2024-25.
- 1.47 According to USDA and International Cotton Advisory Committee (ICAC) forecasts, world cotton production, supply and utilization are predicted to increase in 2024-25 while trade is forecast to be lower than 2023-24. Both USDA and ICAC estimates forecast an increase in stocks but in case of SUR, USDA forecast indicates an increase, while ICAC expects a marginal decline.
- 1.48 As per the latest World Meteorological Organization (WMO) update (6th March 2025), the weak *La Niña* event that emerged in December 2024 is likely to be short-lived and there is a 60 percent probability that conditions will shift back to ENSO-neutral conditions during March-May 2025 and probability increasing to 70 percent for April-June 2025. The probability of developing either *El Niño* or *La Niña* during March-June is negligible.
- 1.49 Global food commodity prices increased in February 2025, due to sharp rise in vegetable oils and sugar. The FAO Food Price Index (FFPI) was 1.6 percent higher than January 2025 but 8.2 percent higher than February 2024 and 21 percent below March 2022 peak. FAO All Rice Price Index declined by 6.8 percent in February, 2025. The FAO Vegetable Oil Price Index in February 2025 was two percent higher than January 2025, but 29.1 percent above its level in February 2024.

Structure of the Report

- 1.50 The report is organized into six chapters. Chapter 2 presents the demand-supply trends and outlook, comparison of market prices with MSP and progress of procurement operations of mandated crops.



Chapter 3 analyses trends in crop productivity, yield gap analysis and discusses major drivers of productivity. Chapter 4 presents an overview of world trade trends and patterns, domestic and world prices, brief review of trade policies and trade outlook. Chapter 5 analyses the cost of production, returns and cost projections of mandated crops. Chapter 6 presents a summary of considerations along with MSP and non-price recommendations.



CHAPTER - 2



Demand-Supply Outlook, Price Trends and Procurement Operations

World Trends and Outlook

- 2.1 During the last two decades, there has been an increase in the consumption of agricultural commodities, driven by population and income growth especially in low and middle-income economies. These countries have also witnessed rapid increase in production through technological advancements, as well as increasing use of natural resources. Consequently, the shifts in agricultural production and consumption locations have led to changes in international agricultural trade patterns. Global food production is perceptibly vulnerable to weather and climate-related shocks, geopolitical tensions, trade disruptions, etc. These factors disrupt the balance between supply and demand as well as global supply chains, and hinder trade that result in volatility in the prices of agricultural commodities, thereby impinging on global food security. Being prone to several vulnerabilities, noticeable shifts in dietary habits and demographic changes impacting global food demand, necessitate informed policy decisions as well as efficient implementation to manage global agricultural situation. This is contingent on reliable demand and supply estimates of agricultural commodities. In this backdrop, the global demand supply trends and outlook for three major *kharif* crops viz. rice, maize and soybean as estimated by the Food and Agricultural Organization (FAO), the United States Department of Agriculture (USDA) and the International Grains Council (IGC) have been given in Table 2.1.

Rice

- 2.2 Global production of rice is projected to increase by about 2 percent in 2024-25 as per USDA and IGC and about one percent as per FAO estimates. World rice supply is also forecast to increase by about 1.5 percent in 2024-25 as per FAO and IGC estimates and 1.3 percent by USDA. Both global utilisation and trade in rice are forecast to increase in 2024-25 as per FAO and IGC, while USDA estimates an increase in utilisation but decline in trade. The stocks are projected to increase as per all three agencies, however, Stocks-to-Use Ratio (SUR) is projected to witness a marginal increase in 2024-25 by FAO but USDA and IGC estimates expect marginal decline in SUR.

Maize

- 2.3 With regards to maize, the global production is forecast to decline by about 2 percent as per FAO in



2024-25, 1.3 percent by USDA and one percent by IGC forecast. Global supply and trade of maize are projected to decrease in 2024-25 as per FAO, USDA and IGC, while utilisation is forecast to marginally increase. These forecasts indicate a decline in maize stocks and SUR in 2024-25 owing to reduced output and increased utilization.

Soybean

- 2.4 Global production as well as supply of soybean is forecast to rise in 2024-25 as per FAO, USDA and IGC. World utilisation and trade are also projected to increase as per USDA and IGC, while FAO forecasts show a marginal decline in soybean trade. FAO, USDA and IGC forecast higher stocks and SUR in 2024-25.

Table 2.1: Global Supply and Demand Outlook for Rice, Maize and Soybean

(Million Tonnes)

Crops	FAO-AMIS				USDA				IGC			
	2021-22	2022-23	2023-24*	2024-25**	2021-22	2022-23	2023-24*	2024-25 [#]	2021-22	2022-23	2023-24*	2024-25 ^{\$}
Rice												
Production	525.5	525.8	534.8	539.4	515.0	514.4	522.6	532.9	516.1	514.5	524.2	534.6
Supply	719.3	721.9	728.8	739.4	703.2	697.6	703.3	712.4	698.5	690.8	696.9	708.1
Utilization	521.4	526.1	527.4	537.2	520.1	526.6	516.6	525.2	519.6	519.4	523.5	532.0
Trade	55.5	52.9	58.4	59.1	56.4	52.7	59.2	58.1	53.6	51.7	55.9	56.3
Stocks	196.9	196.0	200.0	204.0	183.1	178.0	179.5	182.1	178.8	171.4	173.4	176.1
SUR ^α	37.8	37.3	37.9	38.0	35.2	33.8	34.7	34.7	34.4	33.0	33.1	33.1
Maize												
Production	1211.8	1171.0	1241.3	1217.0	1214.9	1157.7	1230.0	1214.3	1219.5	1163.0	1231.2	1218.6
Supply	1497.8	1476.9	1528.3	1523.1	1507.4	1468.5	1534.7	1531.8	1498.5	1458.7	1518.4	1510.8
Utilization	1199.0	1187.4	1219.0	1236.1	1201.5	1159.2	1219.8	1230.0	1217.2	1180.8	1226.2	1239.0
Trade	181.9	183.4	198.4	185.2	193.8	180.7	197.9	189.7	179.4	178.3	198.8	181.6
Stocks	305.6	286.0	306.1	292.9	306.0	302.2	317.5	293.3	281.3	278.0	292.2	272.1
SUR ^α	25.5	24.1	25.1	23.7	25.5	26.1	26.0	23.8	23.1	23.5	23.8	22.0
Soybean												
Production	357.1	377.8	396.0	421.2	358.1	378.2	395.0	424.3	355.9	374.7	395.6	419.6
Supply	407.7	422.9	447.0	485.9	458.1	471.3	496.2	536.6	411.3	427.4	458.0	492.8
Utilization	368.8	368.3	389.8	412.6	363.2	365.8	384.3	405.5	366.4	369.0	384.8	408.4
Trade	154.7	171.5	179.2	179.1	153.9	172.1	177.6	182.0	155.3	171.6	178.7	180.0
Stocks	40.6	47.7	64.8	71.3	98.2	101.3	112.4	128.4	44.8	58.5	73.2	84.4
SUR ^α	11.0	13.0	16.6	17.3	27.0	27.7	29.2	31.7	12.2	15.9	19.0	20.7

Note: *: estimated, **: forecast as on 7th February 2025, #: forecast as on 10th January 2025; \$: forecast as on 16th January 2025; α: Stocks-to-use ratio (SUR) = Stocks/Utilization*100

Source: 1. AMIS-FAO Market Monitor,
2. United States Department of Agriculture (USDA) and
3. International Grains Council (IGC)



Oilseeds Complex

- 2.5 Table 2.2 presents the global demand-supply trends and outlook for oilseeds complex. As per USDA, global oilseeds production and supply in 2024-25 are forecast at 677.7 million tonnes and 810.2 million tonnes, respectively, a growth of 3.1 percent and 3.9 percent, respectively. Global utilisation of oilseeds is also forecast to increase by 2.8 percent at 561.3 million tonnes, while trade is forecast to increase marginally by 0.8 percent. Higher supply compared to utilisation and trade in 2024-25 is expected to result in improved stocks and SUR.
- 2.6 As regards oilmeals, global production, supply, utilisation and trade are projected to increase in 2024-25. Stocks and SUR for oilmeals are expected to increase considerably in 2024-25 due to higher increase in production and supply combined with lower growth in utilisation and trade.
- 2.7 Vegetable oils production, supply and utilization are forecast to increase in 2024-25. However, due to higher growth in utilization compared to production and supply, coupled with marginal increase in trade, stocks and SUR of vegetable oils are forecast to decline in 2024-25.

Table 2.2: Global Supply and Demand Outlook for Oilseeds Complex

(Million Tonnes)

Particulars	2021-22	2022-23	2023-24*	2024-25 [#]
Oilseeds				
Production	611.8	638.4	657.1	677.7
Supply	725.7	752.6	779.5	810.2
Utilization	512.4	526.5	545.9	561.3
Trade	179.4	201.4	205.6	207.3
Stocks	113.1	122.4	132.5	140.1
SUR ^{\$}	22.1	60.8	64.4	67.6
Oilmeals				
Production	351.2	358.1	372.6	385.7
Supply	371.1	378.3	390.0	404.7
Utilization	348.4	356.3	366.2	379.4
Trade	96.4	97.7	106.2	108.1
Stocks	20.5	17.4	18.9	21.1
SUR ^{\$}	5.9	17.8	17.8	19.5
Vegetable Oils				
Production	208.2	217.3	221.7	227.4
Supply	237.2	247.4	252.9	258.2
Utilization	202.8	211.1	217.5	224.7
Trade	79.7	88.9	86.0	86.2
Stocks	29.3	31.2	30.8	28.4
SUR ^{\$}	14.4	35.1	35.9	32.9

Note: * Estimated, # Projected as on February 2025, \$ Stocks-to-use ratio (SUR) = Stocks/Utilization*100

Source: United States Department of Agriculture, WASDE Report February 2025



Coarse Grains

- 2.8 As per USDA report, global production and supply of coarse grains is anticipated to have a marginal decline in 2024-25 to 1492.5 million tonnes and 1839 million tonnes, respectively (Table 2.3). Global utilisation is forecast to increase by 1.7 percent, while trade is projected to decrease by 4.7 percent in 2024-25. Global stocks and SUR of coarse grains are projected to decline as a result of higher utilisation and reduced output. FAO also projects lower production and supply of coarse grains in 2024-25, while global utilisation is forecast to increase by one percent. World trade is projected to decrease significantly by 6.8 percent in 2024-25 while coarse grains stocks are forecast to decline by 3.8 percent and SUR from 24 in 2023-24 to 22.7 in 2024-25.

Table 2.3: Global Supply and Demand Outlook for Coarse Grains

(Million Tonnes)

Particulars	USDA @				FAO ##			
	2021-22	2022-23	2023-24*	2024-25 [#]	2021-22	2022-23	2023-24*	2024-25 ^{\$}
Production	1500.5	1455.4	1506.5	1492.5	1503.3	1480.9	1533.3	1512.7
Supply	1824.1	1797.2	1843.1	1839.0	1853.7	1848.0	1889.9	1881.0
Utilization	1485.9	1460.6	1496.6	1521.8	1496.4	1483.0	1519.8	1534.5
Trade	253.6	220.4	237.0	225.9	230.4	224.5	244.2	227.7
Stocks	338.3	336.6	346.5	317.1	367.1	356.6	368.3	354.2
SUR	22.8	23.0	23.1	20.8	24.8	23.5	24.0	22.7

Note: * Estimate, # Projected, \$ Forecast

Source: @ United States Department of Agriculture, WASDE report February 2025

FAO cereal supply and demand brief released date 07/02/2025

Cotton

- 2.9 USDA estimates indicate higher world cotton production, supply and utilization but 4.7 percent decline in trade. Global production is projected at 120.5 million bales, supply at 194.2 million bales and utilisation at 116 million bales in 2024-25 (Table 2.4). Cotton stocks are also forecast to grow from 73.7 million bales in 2023-24 to 78.4 million bales in 2024-25 and SUR from 64.2 in 2023-24 to 67.6 in 2024-25. The International Cotton Advisory Committee (ICAC) forecasts also indicate higher world cotton production, supply and utilization but lower trade in 2024-25. Due to higher growth in utilisation compared to supply in 2024-25, cotton stocks are expected to increase marginally but SUR is forecast to decline in 2024-25.



Table 2.4: Global Supply and Demand Outlook for Cotton

Particulars	USDA				ICAC			
	Million Bales of 480 lbs each				Million Tonnes			
	2021-22	2022-23	2023-24*	2024-25 [#]	2021-22	2022-23	2023-24	2024-25 ^s
Production	114.5	116.3	113.0	120.5	25.1	24.4	24.1	25.5
Supply	192.2	187.2	188.9	194.2	44.5	43.1	43.6	44.1
Utilization	116.1	112.6	114.8	116.0	25.8	23.6	25.0	25.4
Trade	43.2	36.7	44.6	42.5	9.6	8.3	9.9	9.6
Stocks	76.4	75.9	73.7	78.4	18.7	19.5	18.6	18.7
SUR	65.8	67.4	64.2	67.6	72.4	82.4	74.3	73.5

Note: * Estimated, # Projected as on January 2025, \$ as on 1.11.2024

Source: 1. United States Department of Agriculture, WASDE report February 2025

2. International Cotton Advisory Committee (ICAC)

Domestic Scenario

Rice

2.10 India is the second largest producer as well as the largest exporter of rice in the world. Table 2.5 presents the domestic production, stocks and export situation of rice in India. Domestic rice production increased from 135.8 million tonnes in 2022-23 to 137.8 million tonnes in 2023-24. Rice stocks in Central Pool (as on 1st March) declined in 2022-23 over 2021-22, but increased in 2023-24 and were 34 million tonnes in 2024-25 (as on 31st January, 2025). Rice exports witnessed a steep fall in 2023-24 at 16.4 million tonnes on account of the ban on exports and are estimated at 11 million tonnes in 2024-25 (April-November). As per Second Advance Estimates, rice production (*kharif* and *rabi*) is estimated at 136.4 million tonnes in 2024-25, 6.7 percent higher than 2023-24 *kharif* and *rabi* rice production. In view of higher Central Pool Stocks and increased production in 2024-25, domestic rice supply is expected to rise significantly, which might result in weakening of rice prices in the next season.

Table 2.5: Domestic Supply of Rice in India

(Million Tonnes)

Particulars	2021-22	2022-23	2023-24	2024-25
Production	129.5	135.8	137.8	136.4*
Stocks in Central Pool	29.6 [#]	21.1 [#]	26.3 [#]	34.0 ^{##}
Exports	21.2	22.4	16.4	11.0 ^{**}

Notes: *: as per 2nd Advance Estimates 2024-25 released on 10.03.2025 (only *kharif* and *rabi*, does not include summer rice production),

#: as on 1st March 2022, 2023 and 2024

##: as on 31st January 2025

** April – November, 2024

Source: 1. Economics, Statistics and Evaluation Division, Ministry of Agriculture and Farmers Welfare

2. Department of Food and Public Distribution, Ministry of Consumer Affairs, Food and Public Distribution

3. Directorate General of Commercial Intelligence and Statistics



Pulses

- 2.11 India's pulses production has steadily increased from 17.1 million tonnes in 2014-15 to a record 27.3 million tonnes in 2021-22. However, pulses production in the country witnessed a decline of 4.6 percent in 2022-23 and 7 percent in 2023-24 at 24.2 million tonnes. Among major *kharif* pulses, *tur/arhar* production declined by 21.5 percent in 2022-23 but increased by 3.2 percent in 2023-24 at 3.4 million tonnes. As per Second Advance Estimates, *tur/arhar* production is estimated to further improve in 2024-25, at 3.5 million tonnes, an increase of 2.5 percent. Production of *urad* was 2.3 million tonnes in 2023-24, 11.9 percent lower than 2022-23 while *moong* production is estimated at 3.1 million tonnes in 2023-24, a decline of 15.6 percent over 2022-23. *Kharif* and *rabi moong* production is estimated to be 16.1 lakh tonnes in 2024-25, 28 percent higher than 2023-24 but significantly lower than 2020-21 peak of about 20 lakh tonnes. In case of *kharif* and *rabi urad*, production is estimated to be about 1.8 million tonnes, down from 2.1 million tonnes during last *kharif* and *rabi* seasons and less than half of peak production of 2.75 million tonnes in 2017-18.

Cotton

- 2.12 Cotton production in the country witnessed a significant decline during the last two years from 336.6 lakh bales in 2022-23 to 325.2 lakh bales in 2023-24 to 299.3 lakh bales in 2024-25 (Table 2.6). Lower opening stocks and lower production are expected to result in higher imports in 2024-25, while exports are estimated to fall by 36.5 percent in 2024-25. Total supply and demand are estimated to decline by 7.5 percent and 3 percent, respectively, in 2024-25 but closing stocks are estimated to fall by a whopping 42 percent.

Table 2.6: Demand Supply Estimates of Cotton in India

(Lakh Bales of 170 kg each)

Particulars	2022-23	2023-24	2024-25 (P)
Opening Stocks	39.5	61.2	47.1
Crop (Production)	336.6	325.2	299.3
Imports	14.6	15.2	25.0
Total Supply	390.7	401.6	371.4
Total consumption*	313.6	326.1	326.0
Exports	15.9	28.4	18.0
Total Demand	329.5	354.5	344.0
Closing Stocks	61.2	47.1	27.4

Note: 1. P- Provisional (as per COCPC meeting held on 28.11.2024)

2. * includes MSME, non-MSME & Non-textile

Source: Committee on Cotton Production and Consumption (COCPC), Ministry of Textiles



Inflation

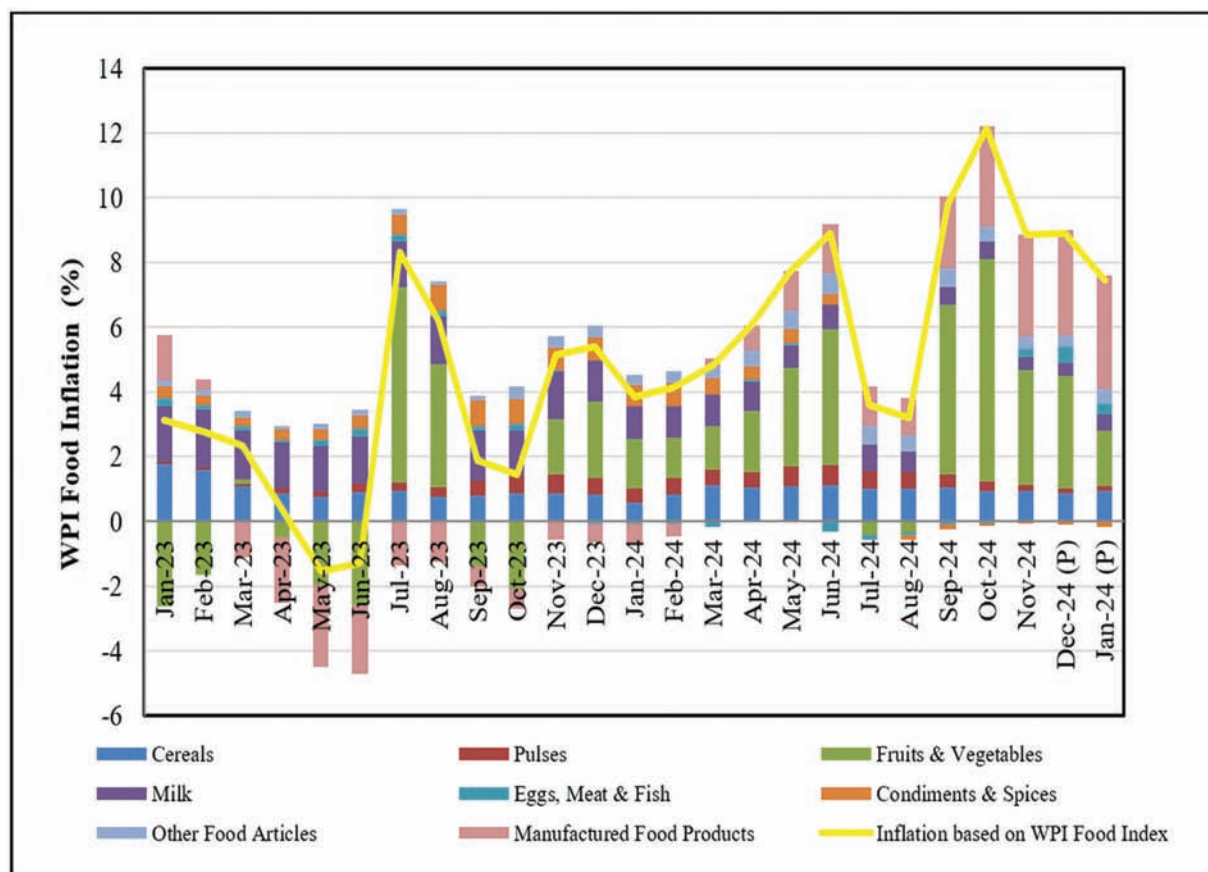
- 2.13 As per the World Bank's 'Food Security Update' (14th February, 2025) between October, 2024 and January 2025, 73.7 percent of low-income countries and 52.5 percent of lower-middle-income countries had high inflation (higher than 5%). In real terms, food price inflation exceeded overall inflation in about 56 percent of the 164 countries for which food and overall CPI are available. Agri-food systems globally have navigated multiple disruptions in recent years, including the COVID-19 pandemic, the impact of Russia-Ukraine conflict, weather related supply fluctuations in several regions, high energy prices and spiralling inflation. The sharp rise in food prices has impacted the cost and affordability of food as well as food security in several regions.

Food Inflation

- 2.14 The Consumer Food Price Index (CFPI) has a significant weightage (39.06%) in the overall Consumer Price Index (CPI), while Food Index consisting of 'Food Articles' from Primary Articles and 'Food Products' from Manufactured Products has 24.38 percent weightage in Wholesale Price Index (WPI) for 'All Commodities'. It is, thus, of vital importance to analyse the trends in inflation based on WPI and CFPIs. The trend in food inflation based on WPI during January 2023 to January 2025 is given in Chart 2.1. Inflation based on Wholesale Food Price Index exhibited a downward trend from 3.1 percent in January 2023 to (-)1.3 percent in June 2023, driven by fruits & vegetables and manufactured food products. However, WPI food inflation saw a sharp jump to 8.3 percent in July 2023 on account of rise in inflation of fruits & vegetables, milk, cereals and condiments & spices. From August 2023 onwards, WPI food inflation declined till October 2023 but rose in November-December 2023. After falling to 3.8 percent in January 2024, WPI food inflation rose and was 8.9 percent in June 2024, but declined to 3.6 percent in July 2024 and 3.2 percent in August 2024, driven by decline in inflation of fruits & vegetables, eggs, meat & fish and condiments & spices. WPI food inflation reached 9.8 percent in September 2024 due to fruits & vegetables, peaked at 12.1 percent in October 2024, declined to 8.9 percent in November-December 2024, and was 7.5 percent in January 2025.



**Chart 2.1: Contribution of Food Subgroups to Inflation based on WPI
Food Index (Base year 2011-12)**

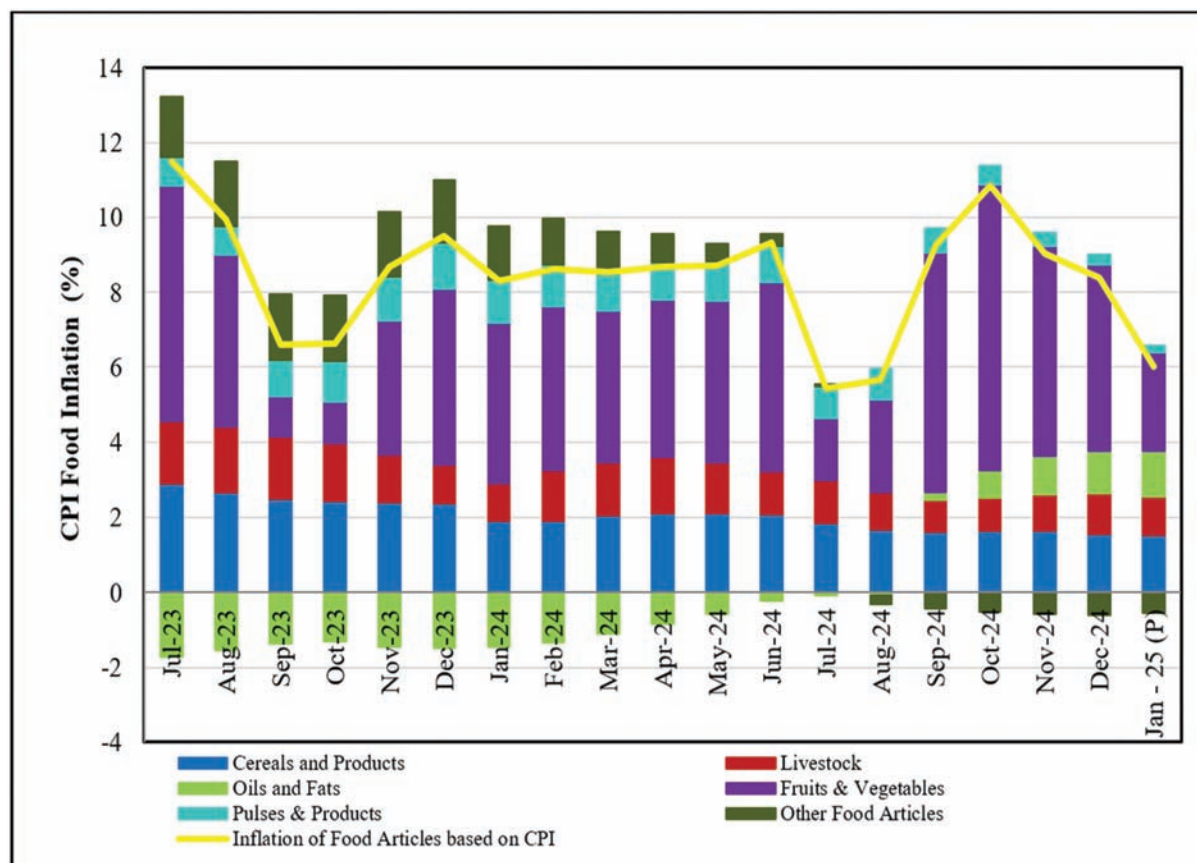


Source: Office of the Economic Adviser, Ministry of Commerce and Industry

- 2.15 The trends in food inflation based on CPI from January 2023 to January 2025 are depicted in Chart 2.2. Food inflation, which was 6 percent in January 2023, started declining from March 2023 and reached 3 percent in May 2023, due to deceleration in prices of oils & fats and fruits & vegetables. CPI based food inflation rose to 11.5 percent in July 2023 driven by a jump of 6.3 percent in fruits & vegetables, after which it declined for sometime before reaching 9.5 percent in December 2023. For the period January-May 2024, CPI food inflation fluctuated around 8.6 percent and was 9.4 percent in June 2024. Driven by fruits & vegetables, food inflation declined to 5.4 percent in July 2024 and from August 2024 onwards, it witnessed an increase and reached 10.9 percent in October 2024, after which food inflation showed a declining trend and was 6 percent in January 2025.



Chart 2.2: Contribution of Food Subgroups to Inflation based on CPI Food Index (Base year 2012)



Source: Ministry of Statistics and Programme Implementation

Market Prices vis-à-vis MSP

- 2.16 Agricultural commodity markets and prices are dependent on the supply and demand dynamics of the commodity. Trends in minimum support price (MSP) and the domestic market prices for mandated *kharif* crops are discussed in this section. Market price data compiled from Unified Portal for Agricultural Statistics (UPAg), Ministry of Agriculture and Farmers Welfare has been used for the analysis. It is pertinent to mention here that the MSP is for the Fair Average Quality (FAQ), while market prices of crops may or may not conform to the FAQ. Thus, though not precisely comparable owing to differences in quality parameters and other market dynamics, but to examine broad trends, comparison of domestic market prices and MSP for various *kharif* crops has been done.

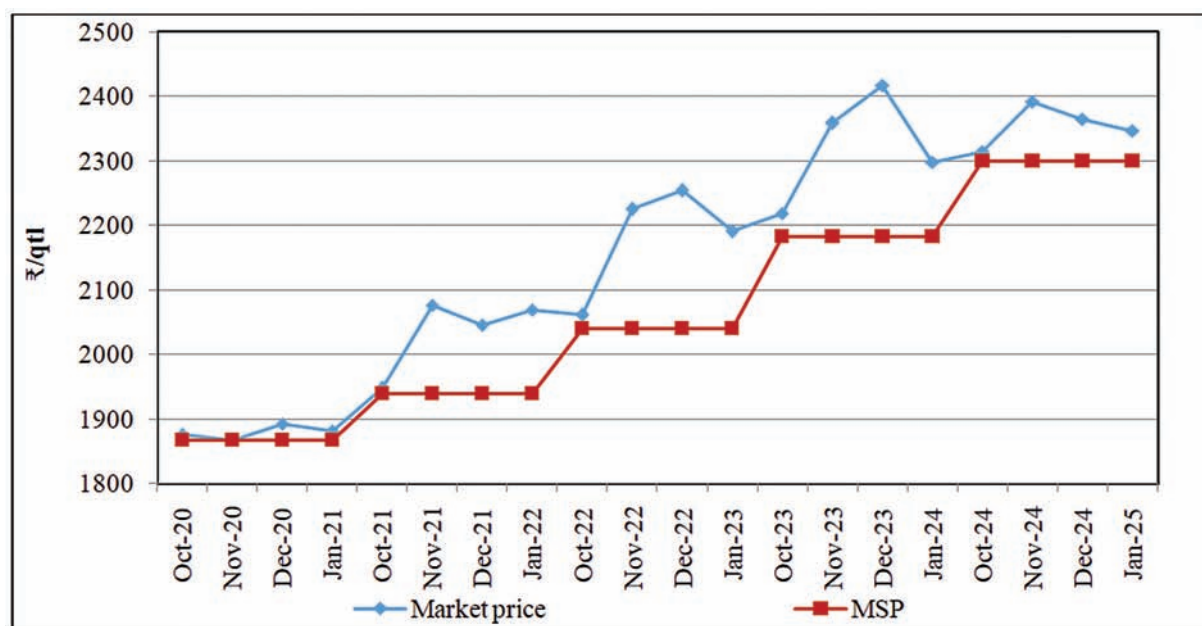
Paddy

- 2.17 Chart 2.3 presents the trends in average market price vis-à-vis MSP of paddy during the last five marketing seasons. During KMS2020-21, market prices stayed above the MSP with a small margin, except November 2020 when market price was equal to MSP. Market prices stayed above MSP in KMS2021-22, with a rising trend in October-November 2021, dipped in December 2021 followed by a rise in January 2022. Further, during KMS2022-23 and KMS2023-24, market prices were



above the MSP and followed a rising trend, with prices increasing during the months of October-December and falling in January. For KMS2024-25, market prices remained above MSP and rose during October-November 2024 but declined in December 2024 and January 2025. These trends clearly show that average market prices have generally remained above the MSP during the last five marketing seasons. Also, it is interesting to note that during the last five marketing seasons, market prices generally witness a rising trend in the months of October-December due to major procurement operations but declined in January.

Chart 2.3: Trends in Domestic Market Prices vis-à-vis MSP of Paddy (KMS2020-21 to KMS2024-25)



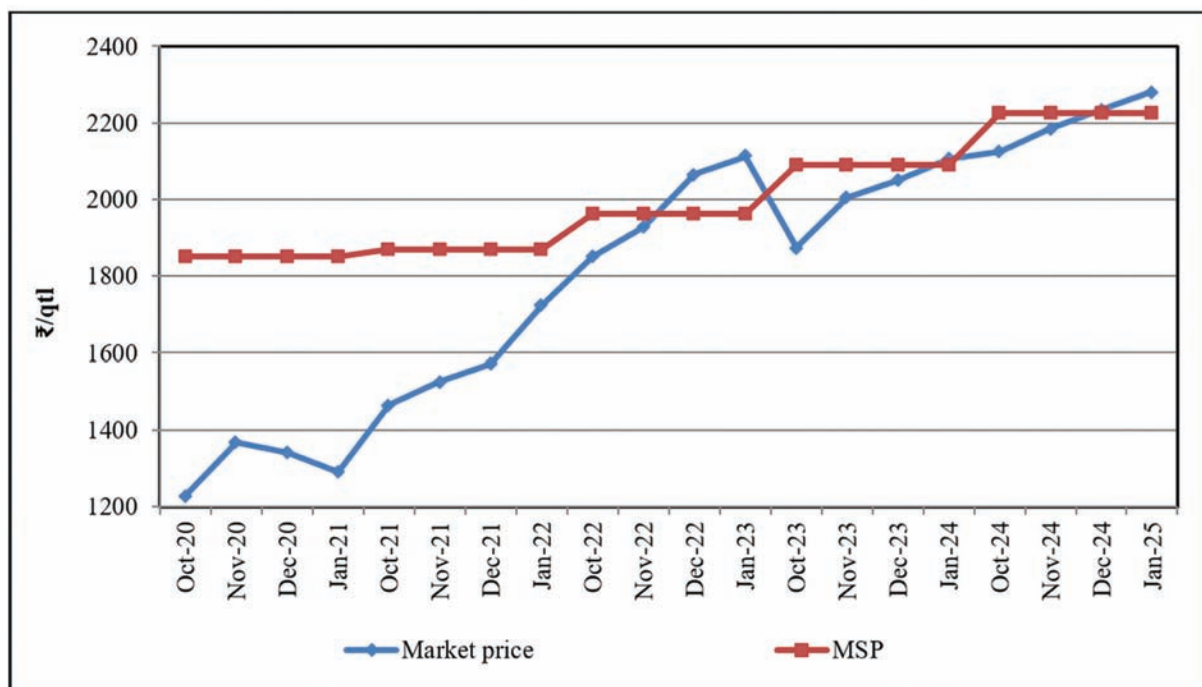
Source: Ministry of Agriculture and Farmers Welfare's UPaG portal, accessed on 24th February, 2025

Maize

- 2.18 As is quite evident from Chart 2.4, that market prices of maize showed a rising trend but stayed below MSP during KMS2020-21 and KMS2021-22, with the gap between the two decreasing. During KMS2022-23, market prices were below the MSP during October-November 2022, but moved above the MSP during December 2022 and January 2023. However, the market prices fell and stayed below MSP in KMS2023-24, except January 2024. During KMS2024-25, market prices were below the MSP in October-November 2024, but surpassed MSP in December 2024 and January 2025. There has been a significant improvement in maize prices during the last five seasons due to growing demand for maize from ethanol production sector.



**Chart 2.4: Trends in Domestic Market Prices vis-à-vis MSP of Maize
(KMS2020-21 to KMS2024-25)**



Source: Ministry of Agriculture and Farmers Welfare's UPAg portal, accessed 24th February, 2025

Pulses

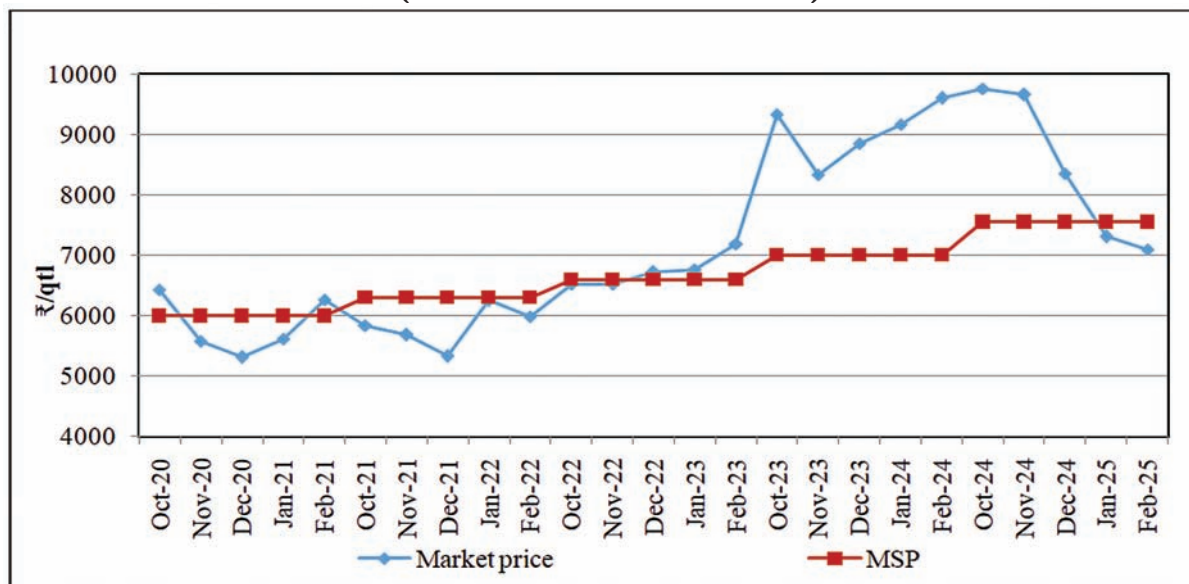
- 2.19 The trends in domestic market prices and MSP for kharif pulses, namely *tur/arhar*, *urad* and *moong* have been presented in Charts 2.5, 2.6 and 2.7.

Tur/Arhar

- 2.20 As is seen in Chart 2.5, market prices of *tur/arhar* declined and stayed below the MSP in KMS2020-21 and KMS2021-22, except October 2020 and February 2021. During KMS2022-23, market prices moved closer to MSP and surpassed MSP in December 2022 and continued the rising trend. The rise in *tur/arhar* prices continued in KMS2023-24 and KMS2024-25, and market prices remained significantly higher than MSP. However, market prices showed a steep decline during December 2024 to February 2025 and even fell below the MSP during the last two months. Steep decline in market prices and even falling below the MSP is a matter of concern and needs urgent attention.



**Chart 2.5: Trends in Domestic Market Prices vis-à-vis MSP of Tur/Arhar
(KMS2020-21 to KMS2024-25)**

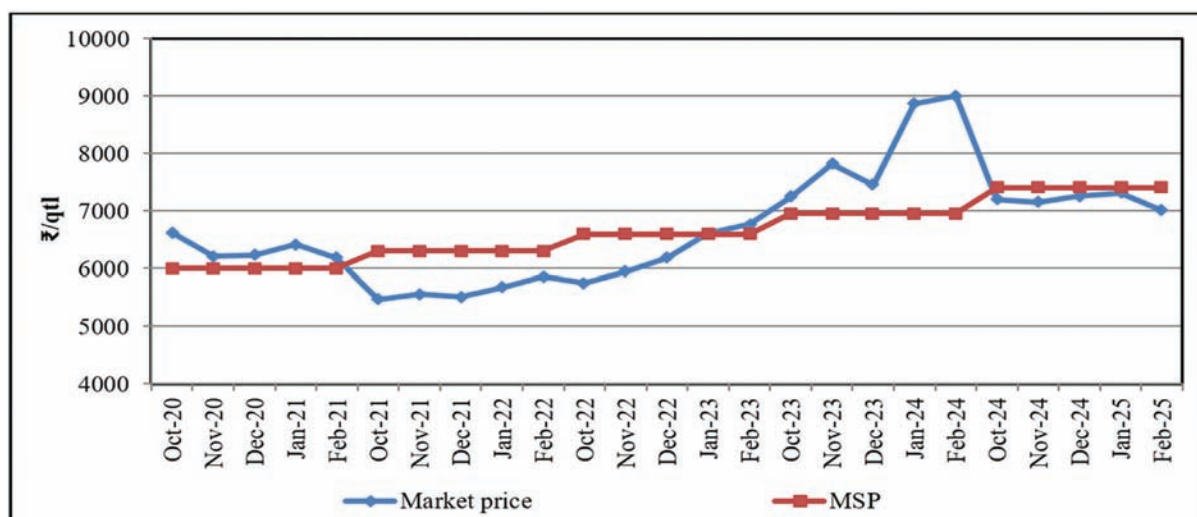


Source: Ministry of Agriculture and Farmers Welfare's UPAg portal, accessed on 24th February, 2025

Urad

- 2.21 The movement of market prices of *urad* and MSP in Chart 2.6 depicts that during KMS2020-21, average market price stayed above the MSP while it fell below the MSP during KMS2021-22. However, during KMS2022-23, though market prices were below MSP in the months of October-December 2022, they surpassed MSP in January-February 2023. For KMS2023-24, market prices remained significantly above the MSP due to lower domestic production. However, market prices fell below the MSP during the current marketing season (KMS2024-25), which might force farmers to shift from *urad* to other crops.

**Chart 2.6: Trends in Domestic Market Prices vis-à-vis MSP of Urad
(KMS2020-21 to KMS2024-25)**



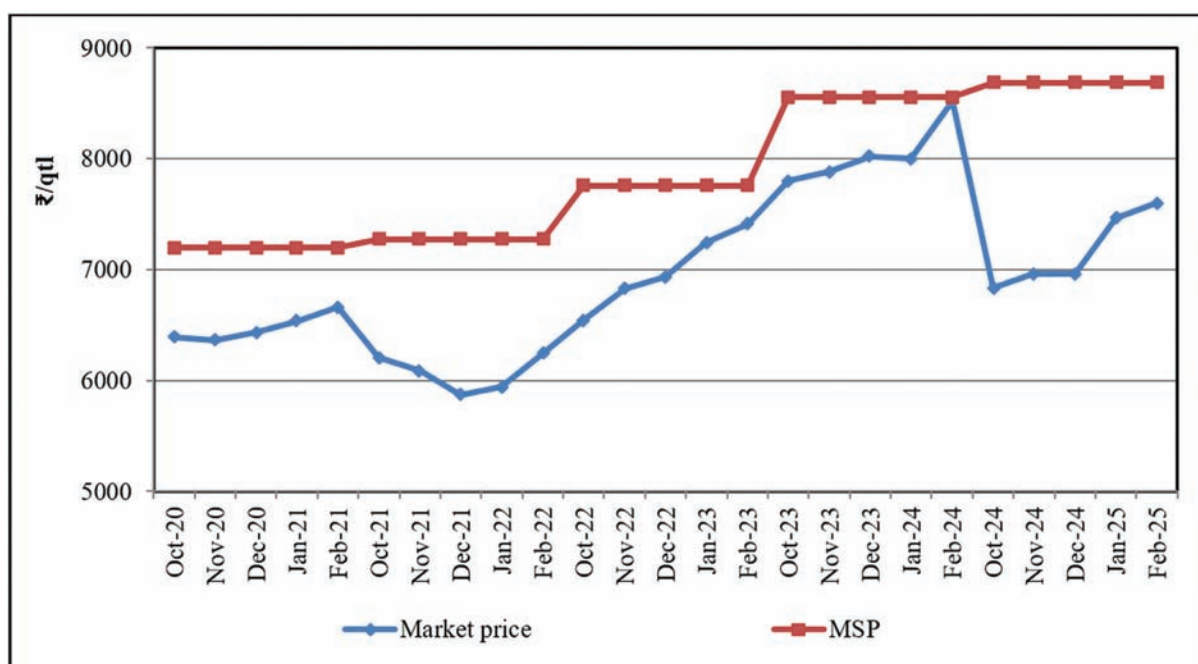
Source: Ministry of Agriculture and Farmers Welfare's UPAg portal, accessed on 24th February, 2025



Moong

- 2.22 Chart 2.7 shows the movement of market prices of *moong* and MSP. The market prices stayed significantly below the MSP during the period KMS2020-21 to KMS2023-24 with prices witnessing an increasing trend from December 2021 to February 2024 and the gap between MSP and market prices decreasing. During KMS2024-25, the market prices stayed much below the MSP but some improvement in prices were recorded during the last three months.
- 2.23 The depressed market prices of pulses can adversely affect Government efforts of achieving self-sufficiency in pulses as farmers might shift to more remunerative crops. Therefore, serious efforts are needed to ensure remunerative prices to pulses growers.

**Chart 2.7: Trends in Domestic Market Prices vis-à-vis MSP of Moong
(KMS2020-21 to KMS2024-25)**



Source: Ministry of Agriculture and Farmers Welfare's UPAd portal, accessed on 24th February, 2025

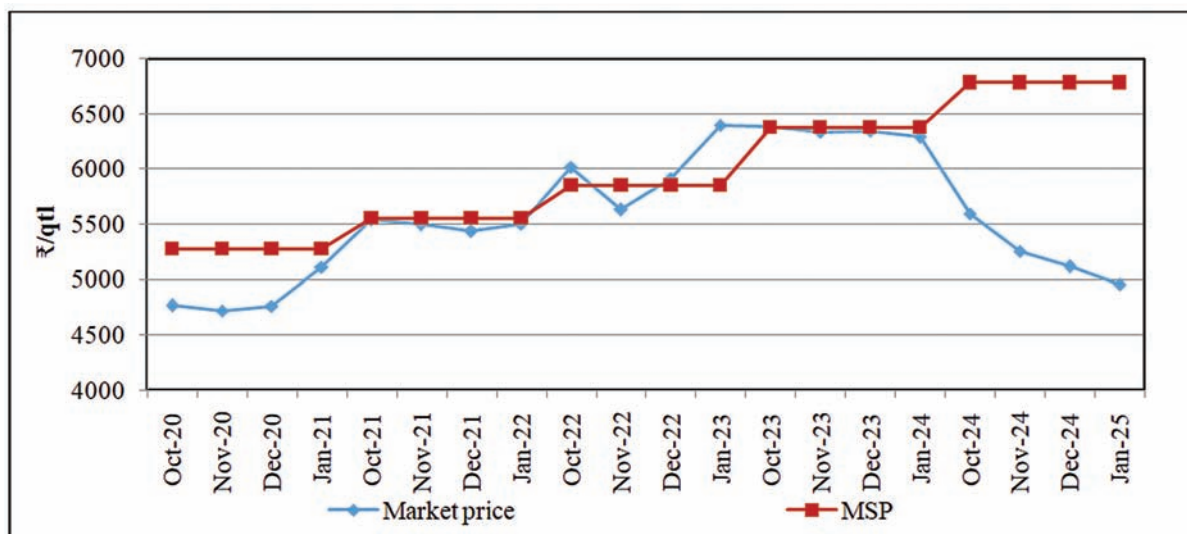
Oilseeds

Groundnut

- 2.24 Market prices of groundnut stayed below the MSP during KMS2020-21 with the gap decreasing due to improvement in market prices (Chart 2.8). During KMS2021-22, market prices were marginally below the MSP but moved closely with it. In KMS2022-23, market prices fluctuated around the MSP and were higher than MSP, except for November 2022. However, market prices fell below the MSP in KMS2023-24 and the trend continued during KMS2024-25. Market prices recorded a very steep decline during the current marketing season and the gap between the MSP and market price widened from (-)17.5 percent in October 2024 to (-)26.9 percent in January 2025.



**Chart 2.8: Trends in Domestic Market Prices vis-à-vis MSP of Groundnut
(KMS2020-21 to KMS2024-25)**

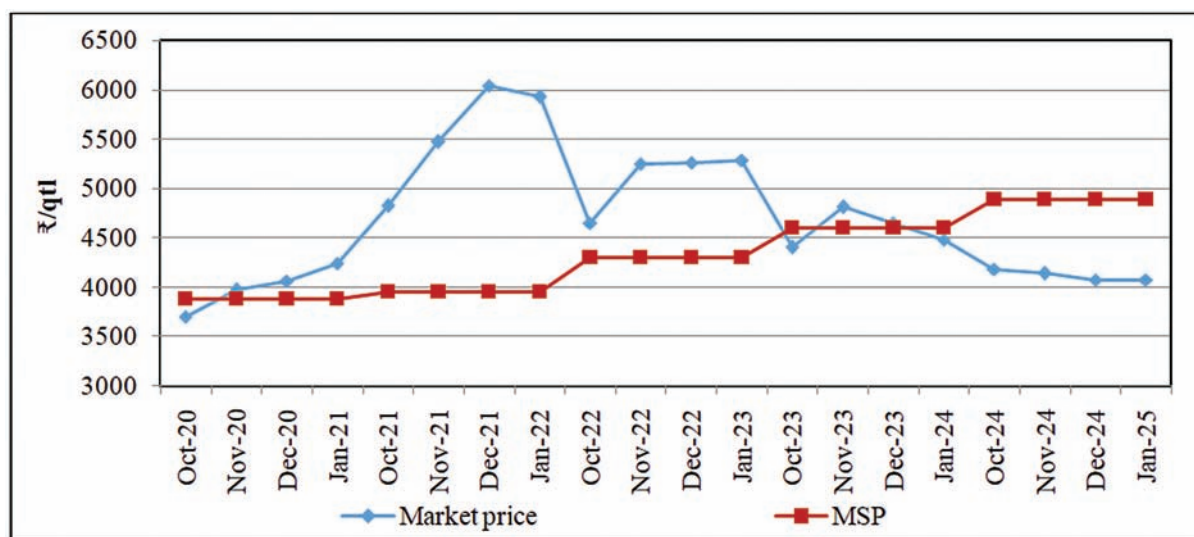


Source: Ministry of Agriculture and Farmers Welfare's UPAg portal, accessed on 24th February, 2025

Soybean

- 2.25 As is seen from Chart 2.9, except for October 2020, market prices of soybean stayed above the MSP for KMS2020-21 and KMS2021-22 and prices showed a steep increase during KMS2021-22. However, market prices were lower in KMS2022-23 compared with KMS2021-22 but stayed above the MSP. Market prices fluctuated around the MSP during KMS2023-24 and were higher than MSP in November-December 2023. During KMS2024-25, market prices recorded a steady decline and were significantly below (-15.8%) the MSP during the season.

**Chart 2.9: Trends in Domestic Market Prices vis-à-vis MSP of Soybean
(KMS2020-21 to KMS2024-25)**



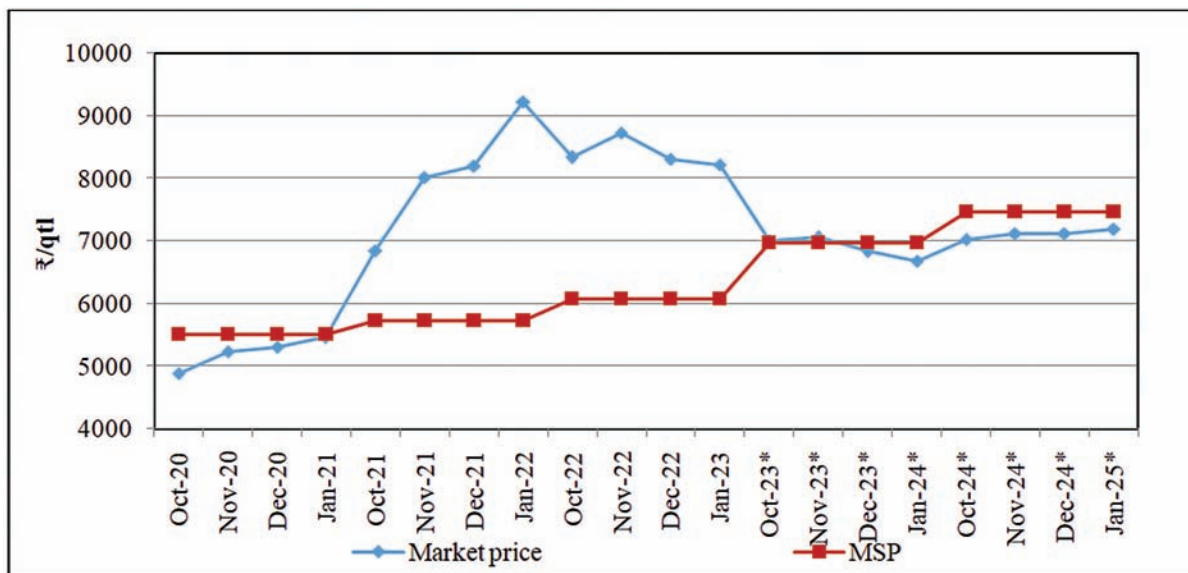
Source: Ministry of Agriculture and Farmers Welfare's UPAg portal, accessed on 24th February, 2025



Cotton

- 2.26 Market prices of cotton were below the MSP in KMS2020-21 and moved above the MSP during KMS2021-22 with steep increase in market prices (Chart 2.10). Market prices showed a declining trend during KMS2022-23 but remained above the MSP. During KMS2023-24, the prices fell below the MSP in December 2023 and remained below the MSP, which continued in KMS2024-25.

**Chart 2.10: Trends in Domestic Market Prices vis-à-vis MSP of Cotton
(KMS2020-21 to KMS2024-25)**



Note: 1. Weighted wholesale modal prices of Gujarat and Maharashtra

2. *:MSP for KMS2023-24 and KMS2024-25 are for staple length 27.5 to 29 mm

Sources: 1. AGMARKNET, Directorate of Marketing & Inspection (DMI), Department of Agriculture and Farmers Welfare, Ministry of Agriculture and Farmers Welfare

2. Economics, Statistics and Evaluation Division, Ministry of Agriculture and Farmers Welfare

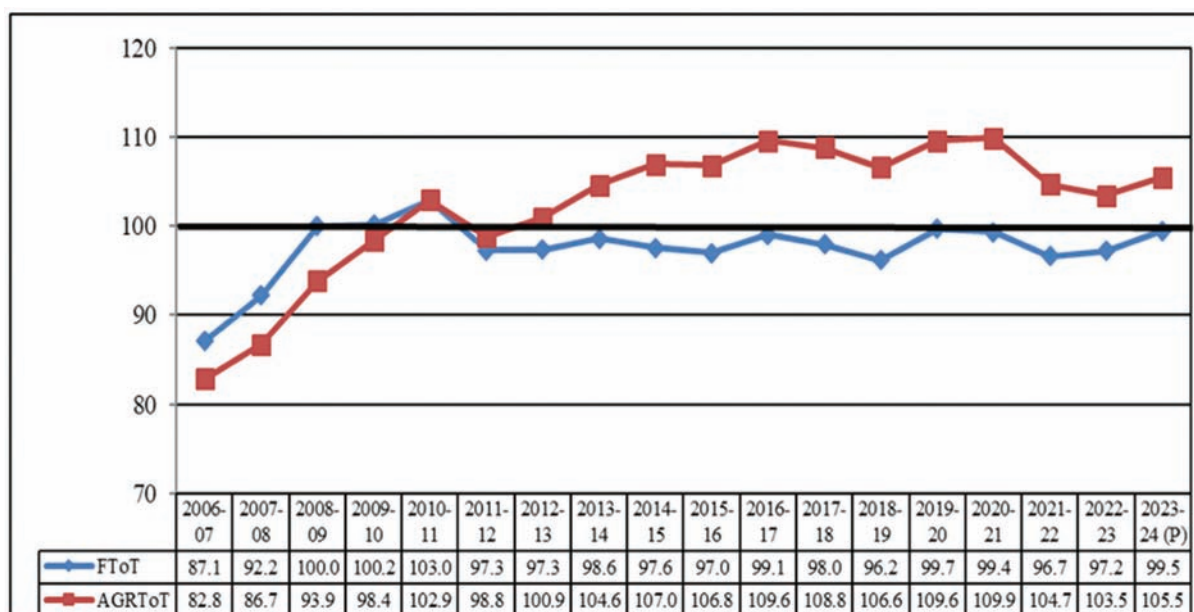
Trends in Terms of Trade

- 2.27 Terms of Trade (ToT) is a factor considered by the Commission while formulating its recommendations on the price policy for agricultural commodities. The index is considered as a measure of relative performance of sectors and crucial for growth, income distribution and consequently demand, savings and investment patterns.
- 2.28 Chart 2.11 depicts the farmers' terms of trade (FToT) and agricultural terms of trade (AGRTToT) during 2006-07 to 2023-24(P). The FToT measures index of prices that farmers receive for their products and the prices paid by the farmers. As may be inferred from the Chart, FToT index has exhibited an increasing trend from 87.1 in 2006-07 to 103 in 2010-11 and improvement in FToT is attributed to factors such as an increase in minimum support prices, a rise in global agricultural prices and high food inflation. However, FToT index stagnated and has remained below 100 since 2011-12 but showed improvement during the last 3 years.



- 2.29 The agricultural terms of trade (AGRToT) includes both farmers and agricultural labourers. The AGRToT improved between 2006-07 and 2010-11 but declined in 2011-12. The index of AGRToT remained above 100 during the last 12 years and reached the highest level (109.9) in 2020-21. The AGRToT rose much faster than FToT and remained higher than the FToT. Improved Terms of Trade along with improvement in productivity are essential for raising farmer's income. In order to improve the FToT and AGRToT and contribute to the economic well-being of farmers, appropriate policy interventions are needed to ensure better prices to farmers for their produce and reduce the unit cost of production. The effective extension services, adoption of new innovative technologies and improved practices can help in reducing unit cost of production, thereby improving farmers' income. In addition, improved infrastructure, efficient and competitive markets, policy reforms realizing economies of scale, and improved bargaining power of farmers by organizing them into groups like Farmer Producer Organizations (FPOs), Cooperatives, SHGs etc. will help in addressing the challenges faced by agricultural sector.

Chart 2.11: Trends in Index of Farmers' Terms of Trade (FToT) and Agricultural Terms of Trade (AGRToT)



Note: P – Provisional

Source: Economics, Statistics and Evaluation Division, Ministry of Agriculture and Farmers Welfare

Procurement Policy and Operations

- 2.30 The Government policy of procurement of foodgrains ensures Minimum Support Price (MSP) to farmers and makes foodgrains available to vulnerable sections of society at affordable prices. This strategy aims to maintain a balance between supporting agricultural producers and ensuring food affordability and accessibility for consumers. The policy also ensures effective market intervention through open market sales to control prices and contributes to overall food security in the country.



Designated procurement agencies, such as the Food Corporation of India (FCI) alongwith other State agencies for wheat and paddy, National Agricultural Cooperative Marketing Federation of India Limited (NAFED) and National Cooperative Consumer's Federation of India Limited (NCCF) for pulses and oilseeds and Cotton Corporation of India (CCI) for cotton, undertake procurement operations. Coarse grains are procured by State Government Agencies for Central Pool as per the directions issued by Government of India from time to time.

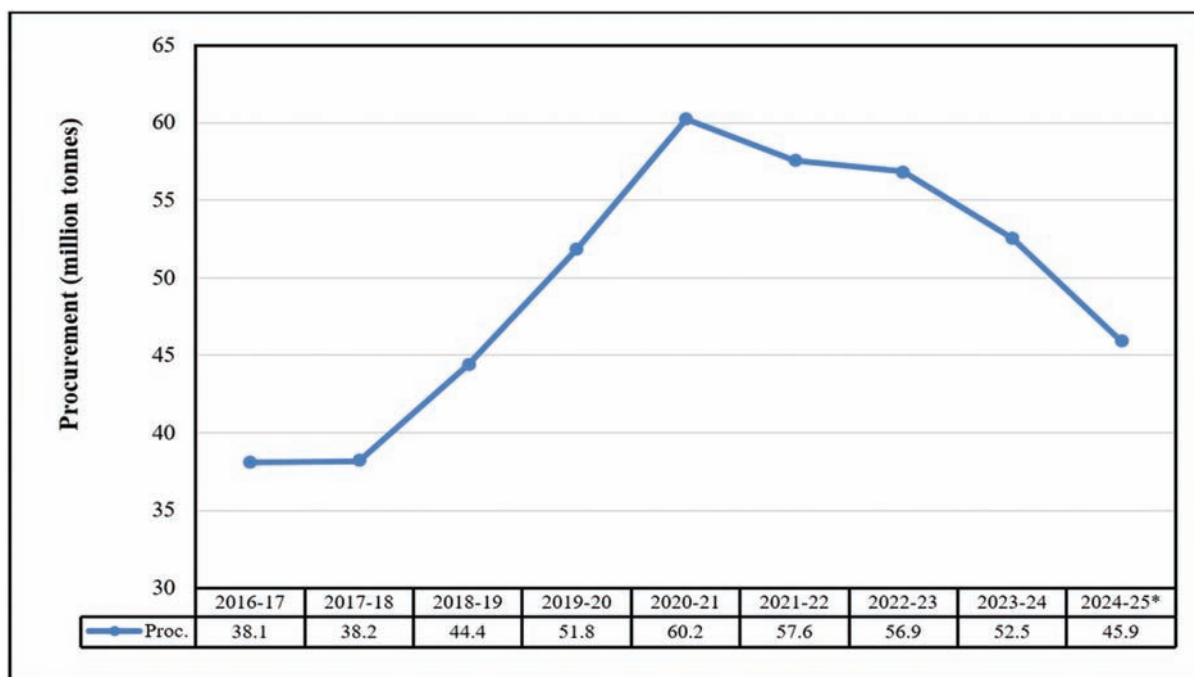
- 2.31 Procurement of paddy is open-ended and whatever quantity is offered by the farmers during the procurement season in conformity to quality standards as per FAQ, is purchased at MSP by the Government agencies for Central Pool. Paddy is procured under Decentralised Procurement System (DCP) and Centralised Procurement System (non-DCP). Under Centralized Procurement System (non-DCP), foodgrains are procured by FCI directly or by the State agencies and stocks are handed over to the FCI for storage and issue against allocation in the same State or surplus stocks are moved to other States, while under Decentralized Procurement (DCP) system, the State Governments procure, store and distribute rice against allocation under the National Food Security Act (NFSA) and Other Welfare Schemes (OWS) in the State and hand over the excess stocks to FCI for Central Pool. Government of India reimburses the expenditure incurred by the State Governments on procurement, storage and distribution under the DCP system based on the laid down principles. At present, 16 States for rice are under the DCP system and 7 States/Union Territories are under non-DCP system of procurement. The list of DCP and non-DCP States is given at Annex Table 2.1.

Procurement Trends

- 2.32 There has been significant increase in rice procurement during the last few years. The level of procurement showed a rising trend from 38.1 million tonnes in 2016-17 to 60.2 million tonnes in 2020-21 (Chart 2.12). However, rice procurement witnessed a declining trend during the last 3 seasons, from 57.6 million tonnes in 2021-22 to 56.9 million tonnes in 2022-23 and was 52.5 million tonnes in 2023-24 season. The decline in rice procurement is attributed to relatively higher market prices. Punjab has the highest share in total procurement and accounted for 22.3 percent in TE2023-24 followed by Telangana (13.5%), Chhattisgarh (12.2%) and Odisha (9%) (Table 2.7). The share of Telangana, Chhattisgarh and Uttar Pradesh in total procurement increased between TE2018-19 and TE2023-24, while other States like Punjab, Odisha, Haryana, Andhra Pradesh and West Bengal lost their share in total procurement.



Chart 2.12: Rice Procurement Trends from KMS2016-17 to KMS2024-25



Note: *upto 02.03.2025

Source: Food Corporation of India

Table 2.7 Rice Procurement Trends in Major States from TE2018-19 to TE2023-24

States	Procurement (million tonnes)			Share %	
	TE2018-19	TE2023-24	2024-25*	TE2018-19	TE2023-24
Punjab	11.4	12.4	11.6	28.4	22.3
Telangana	4.1	7.5	3.6	10.3	13.5
Chhattisgarh	3.7	6.8	7.0	9.3	12.2
Odisha	3.8	5.0	4.4	9.4	9.0
Uttar Pradesh	2.8	4.1	3.9	7.0	7.4
Haryana	3.8	3.9	3.6	9.5	7.0
Madhya Pradesh	1.3	3.0	2.9	3.2	5.4
Bihar	1.0	2.6	2.6	2.5	4.7
Andhra Pradesh	4.2	3.1	1.5	10.4	5.5
Tamil Nadu	0.8	2.2	1.4	2.0	3.9
West Bengal	1.9	2.1	1.9	4.6	3.7
Maharashtra	0.4	1.1	0.6	0.9	1.9
Others	1.0	1.9	0.8	2.5	3.4
All India	40.2	55.7	45.9	100.0	100.0

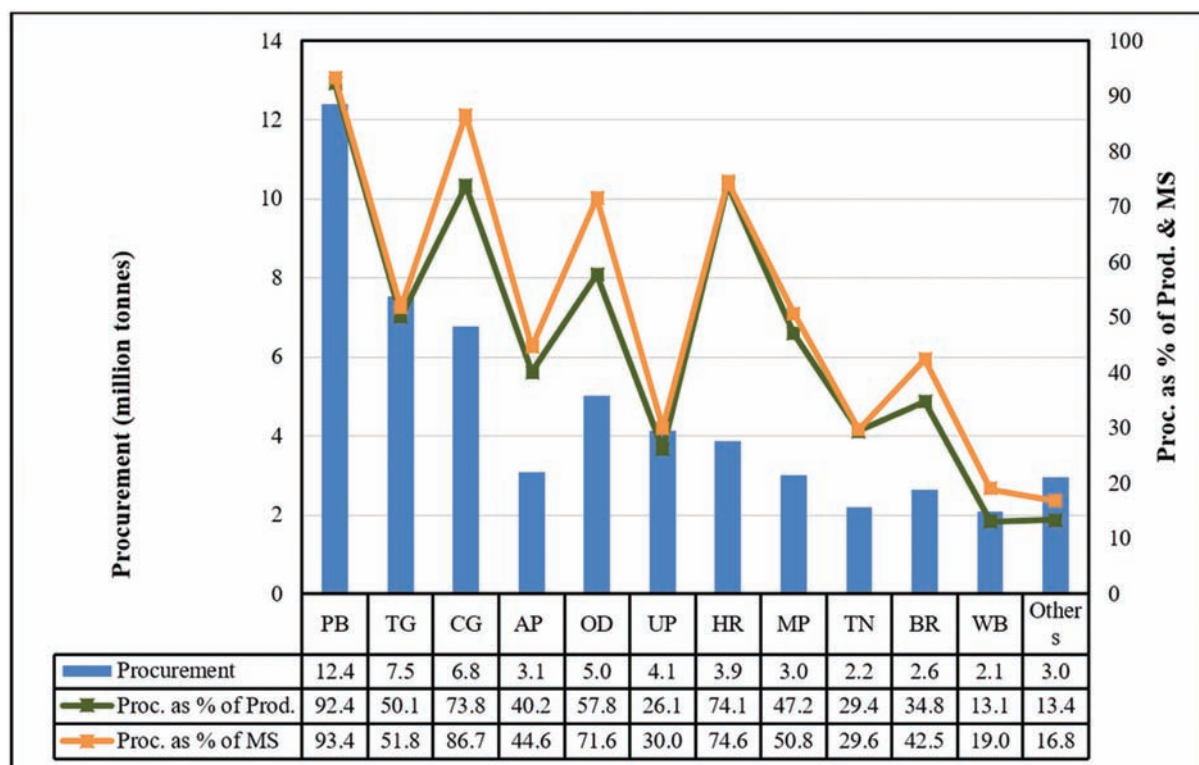
Note: *upto 02.03.2025

Source: Food Corporation of India



- 2.33 Total procurement and procurement as a percent of production and marketed surplus in major rice producing States during TE2023-24 are shown in Chart 2.13. Punjab procured 92.4 percent of total production in TE2023-24, followed by Haryana (74.1%), Chhattisgarh (73.8%), Odisha (57.8%) and Telangana (50.1%). However, West Bengal and Uttar Pradesh, top two producers contributing 23.7 percent of total rice production, procured only 13.1 percent and 26.1 percent of production, respectively, in the State. As regards procurement as a percent of marketed surplus, 93.4 percent of the marketed surplus of rice in Punjab was procured during TE2023-24. Chhattisgarh (86.7%), Haryana (74.6%), Odisha (71.6%), Telangana (51.8%) and Madhya Pradesh (50.8%) had high share of procurement as a percent of marketed surplus in TE2023-24. However, in West Bengal, the largest producer for rice, only 19 percent of marketed surplus was procured, while Uttar Pradesh procured 30 percent of marketed surplus. This calls for concerted efforts to strengthen procurement operations in States like Bihar, West Bengal, Uttar Pradesh and Tamil Nadu.

Chart 2.13: Procurement of Rice in Major Producing States during TE2023-24



Note : MS refers to marketed surplus

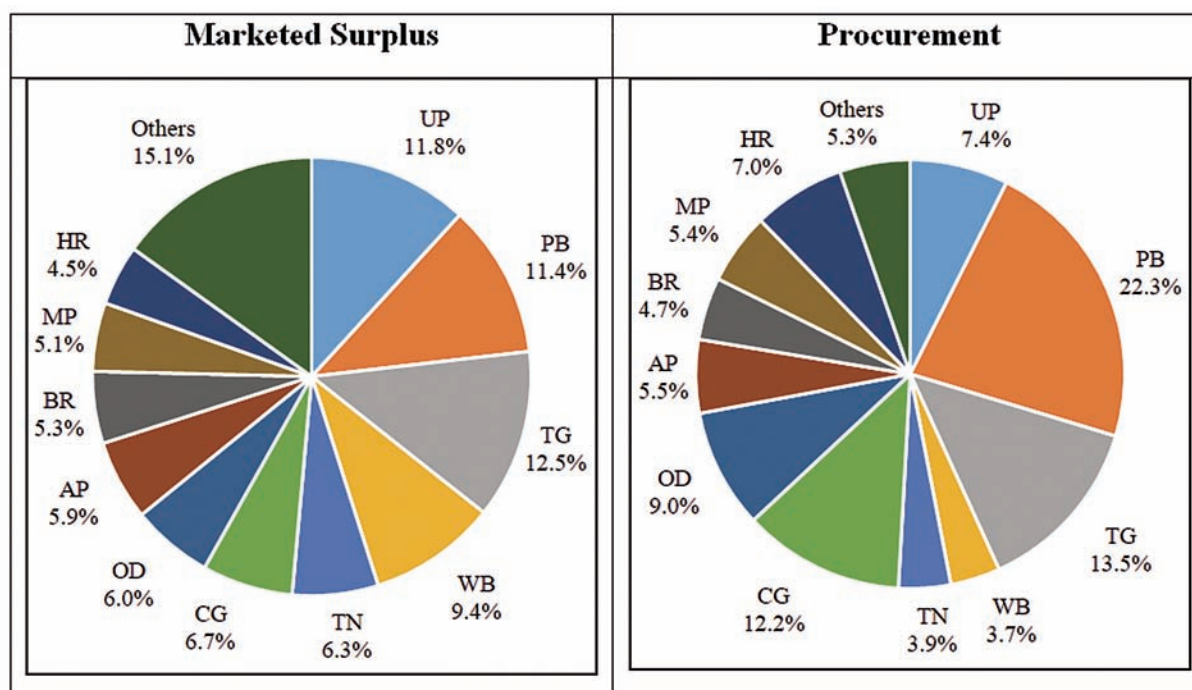
Source: 1. Economics, Statistics & Evaluation Division, Ministry of Agriculture and Farmers Welfare
2. Food Corporation of India

- 2.34 Share of major States in total marketed surplus and procurement of rice during TE2023-24 is given in Chart 2.14 to analyse equity issues in procurement operations. As may be inferred from the Chart, Telangana has the highest share (12.5%) in marketed surplus of rice, followed by Uttar Pradesh (11.8%), Punjab (11.4%), West Bengal (9.4%), Chhattisgarh (6.7%), Tamil Nadu (6.3%) and Odisha (6%). In contrast, Punjab accounted for 22.3 percent of total rice procurement, followed by Telangana



(13.5%), Chhattisgarh (12.2%), Odisha (9%), Uttar Pradesh (7.4%), and Haryana (7%) in TE2023-24. States like Punjab, Chhattisgarh, Odisha and Haryana have significantly higher share in rice procurement vis-à-vis their share in marketed surplus, while West Bengal, Uttar Pradesh and Tamil Nadu have much lower share in procurement compared with their share in marketed surplus. Though West Bengal and Uttar Pradesh are two top producers of rice and accounted for 9.4 percent and 11.8 percent of total marketed surplus of rice, their share in procurement was much lower at 3.7 percent and 7.4 percent, respectively. Therefore, efforts should be made to strengthen procurement operations and improve storage and warehousing infrastructure as well as institutional mechanisms to increase procurement in these States.

Chart 2.14: Share of Major States in Marketed Surplus and Procurement of Rice in TE2023-24



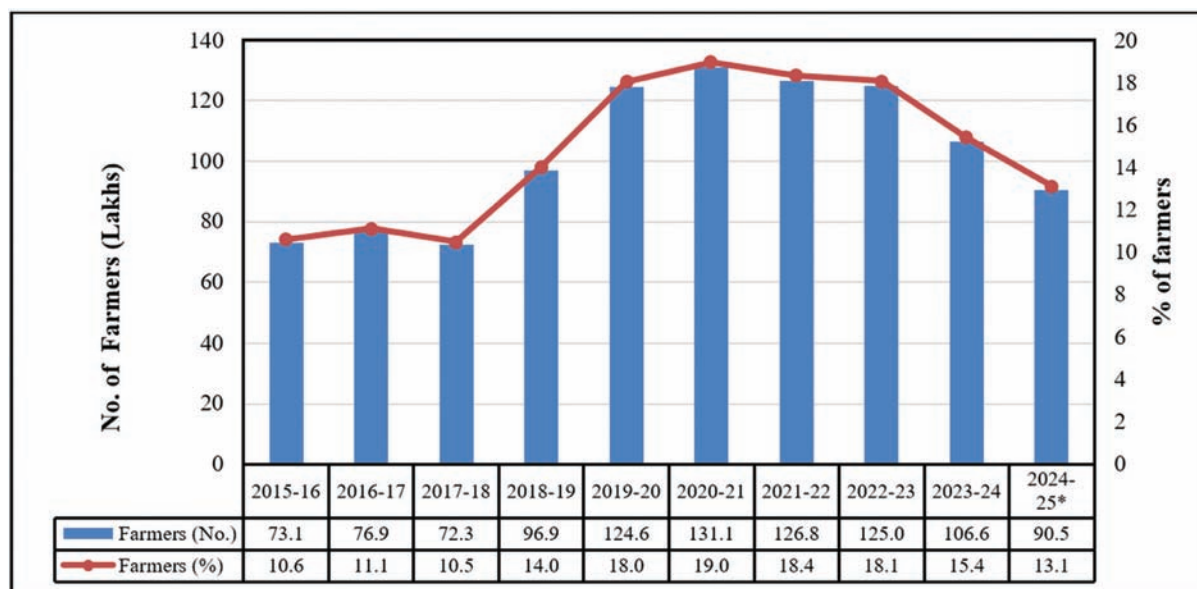
Source: Food Corporation of India

Coverage of Farmers

- 2.35 There has been a continuous expansion in procurement operations, as a result of Government's proactive initiatives and policies. It is evident from Chart 2.15 that the total number of beneficiary farmers under paddy procurement witnessed an increasing trend between 2016-17 and 2020-21, and reached a record level of 1.31 crore in 2020-21. However, a steady decline was observed from 2021-22 onwards and total number of beneficiary farmers was about 1.1 crore in 2023-24, a decline of around 19 percent from 2021-22 and is attributed to high market prices. The percentage of farmers benefitted from paddy procurement increased from 10.5% in 2017-18 to 19% in 2020-21, which declined thereafter due to market prices higher than MSP. Detailed information on State-wise paddy farmers benefitted from procurement operations is given in Annex Table 2.2.



Chart 2.15: Number of Paddy Farmers Benefitting from Procurement



Note: 1. Number of Paddy Operational Holdings as per 2015-16 Agriculture Census has been taken as a proxy to number of paddy farmers.

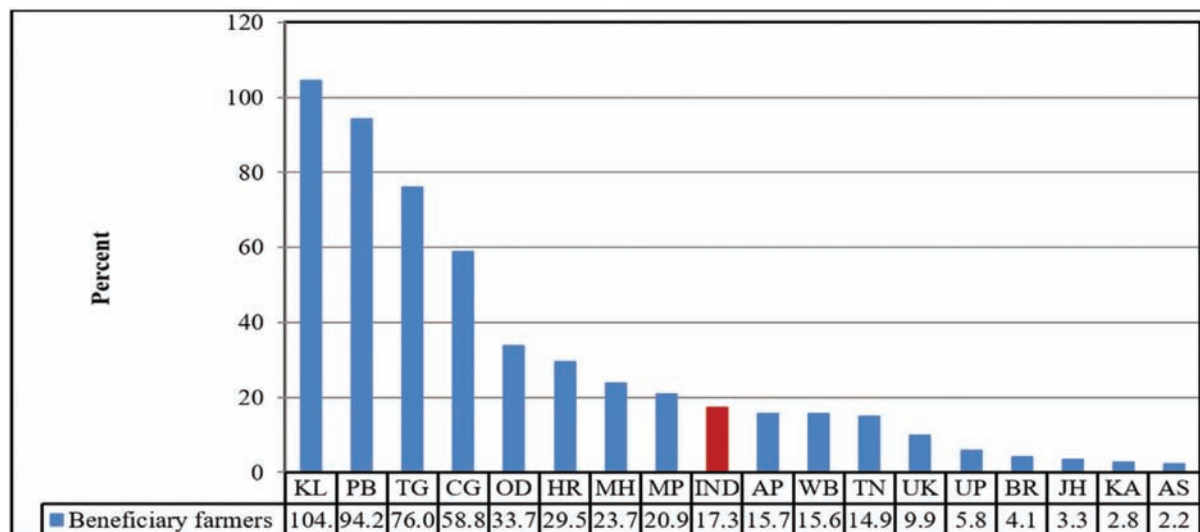
2. *upto 02.03.2025

Source: Food Corporation of India

- 2.36 At the all-India level, 17.3 percent of India's paddy farmers benefitted from procurement operations during TE2023-24 with large variations across States (Chart 2.16). More than 94 percent paddy farmers in Punjab, 76 percent in Telangana and 58.8 percent in Chhattisgarh benefitted from procurement operations. While in other major rice producing States like Uttar Pradesh (5.8%), West Bengal (15.6%), Odisha (33.7%) and Bihar (4.1%) coverage of farmers was low. In Assam, Karnataka, Jharkhand and Bihar less than 5 percent paddy farmers benefitted from procurement.



Chart 2.16: Paddy Beneficiary Farmers as Percent of Total Farmers in Selected States (TE2023-24)

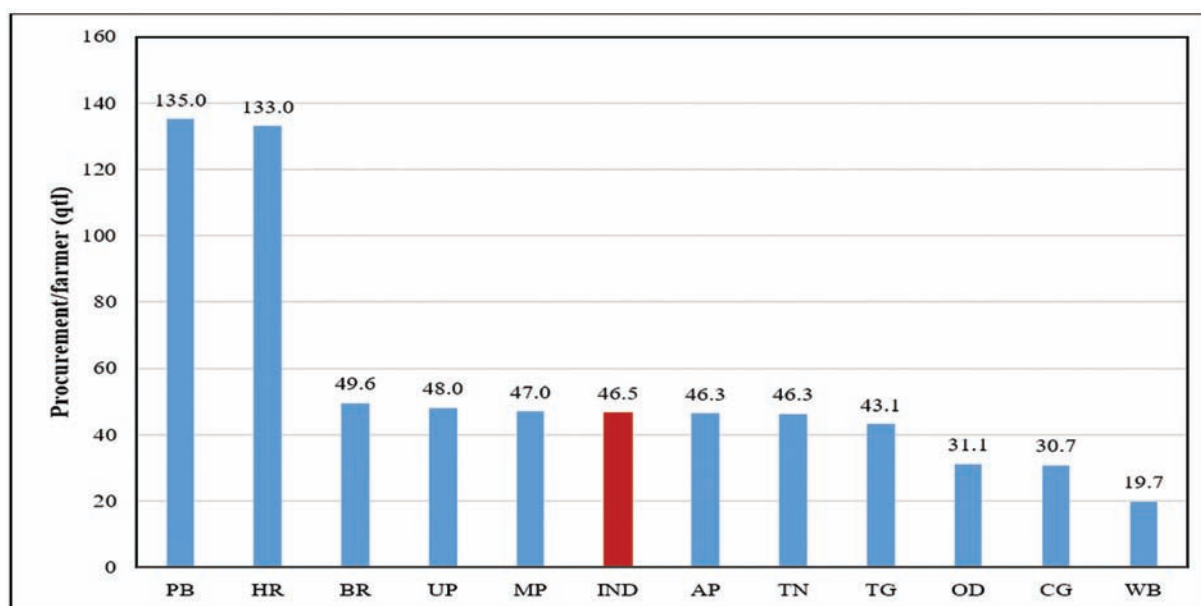


Note: 1. Beneficiary Farmers relate to figures for TE2023-24
 2. Number of Paddy Operational Holdings as per 2015-16 Agriculture Census has been taken as a proxy for number of paddy farmers.

Source: 1. Economics, Statistics and Evaluation Division, Ministry of Agriculture and Farmers Welfare
 2. Food Corporation of India

2.37 Chart 2.17 presents average procurement of rice per farmer in major States during TE2023-24. The all-India average procurement per farmer was 46.5 qtl, there are large variations across States. Punjab has the highest procurement per farmer at 135 qtl followed by Haryana (133 qtl) and Bihar (49.6 qtl). Other States with high procurement per farmer are Uttar Pradesh (48 qtl), Madhya Pradesh (47 qtl), Andhra Pradesh (46.3 qtl), Tamil Nadu (46.3 qtl) and Telangana (43.1 qtl), while West Bengal (19.7 qtl), Chhattisgarh (30.7 qtl) and Odisha (31.1 qtl) had lower procurement per farmer in TE2023-24.

Chart 2.17: Average Procurement of Rice per Farmer in Major States (TE2023-24)



Source: Food Corporation of India

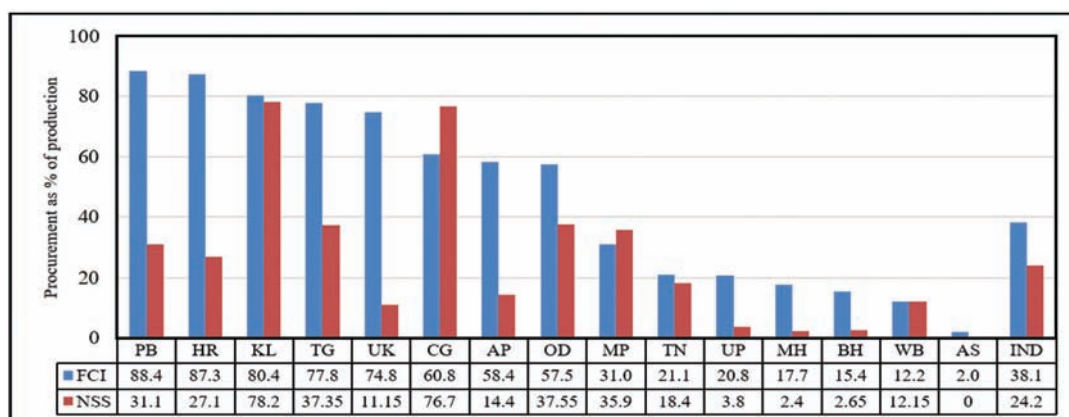


Box 2.1: Divergence in Actual Procurement of Paddy and NSS Estimates

Situation Assessment of Agricultural Households and Land and Livestock Holdings of Households in Rural India, 2019 (NSS 77th Round) estimated various aspects related to MSP including awareness about MSP and procurement agency, and output sold under MSP. However, closer examination of data indicates large divergence between FCI's actual procurement of paddy and beneficiary farmers and NSS estimates.

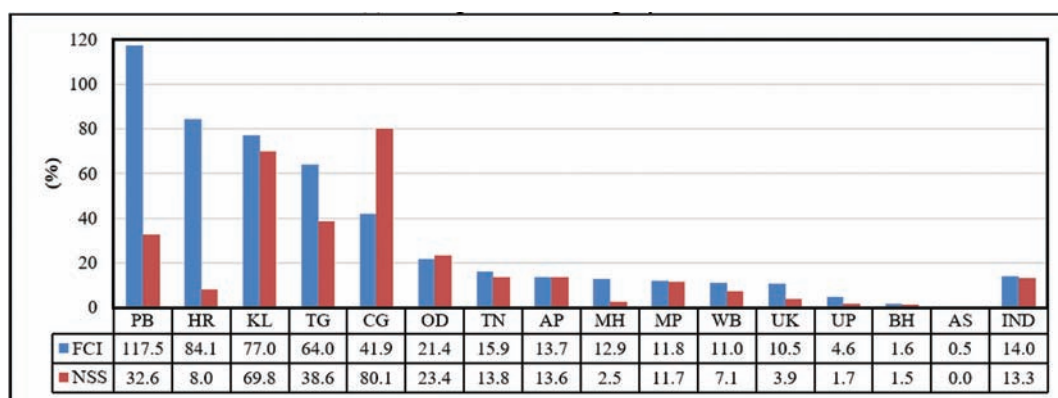
Divergence in Procurement: As is evident from the Chart below, actual procurement as percent of total production during KMS2018-19 is significantly higher in all major States compared with NSS estimates. The divergence is very large in Punjab, Haryana, Andhra Pradesh and Telangana. In case of Punjab, actual procurement was 88.4 percent of production, while as per NSS estimates it was 31.1 percent. Similarly, in Haryana, the difference between two datasets is around 60 percentage points. Therefore, the Commission recommends that NSS data should be validated with actual procurement before publication of the report as it creates negative perception.

Chart Box 2.1(a): Divergence in Procurement in 2018-19



Divergence in Coverage of Farmers: The percentage of paddy farmers who sold their crop to procurement agency as per NSS estimates was also significantly lower than actual number of farmers who benefitted from procurement. The divergence is significant in Punjab, Haryana and Telangana, although at all-India level, the difference is insignificant.

Chart Box 2.1(b): Divergence in Coverage of Farmers in 2018-19



Note: Number of paddy operational holdings as per 2015-16 Agriculture Census has been taken as a proxy for number of paddy farmers

The differences between actual procurement levels and beneficiary farmers as per FCI data and the NSS estimates are significantly large and must be examined. Such underestimation by NSS can undermine the trust and effectiveness of procurement operations. Therefore, the issue needs to be addressed.

Source: 1. Food Corporation of India
2. Situation Assessment of Agricultural Households and Land and Livestock Holdings of Households in Rural India, 2019 (NSS 77th Round)



Participation of Small and Marginal Farmers in Procurement

2.38 Table 2.8 presents procurement of paddy by farm size in different States on the basis of information provided by State Governments. As per the Table, majority of farmers benefitting from paddy procurement in Andhra Pradesh, Telangana and Odisha belong to the marginal and small category. For Andhra Pradesh and Odisha, share of marginal and small farmers benefitting from paddy procurement increased in 2022-23 but declined in 2023-24, while for Telangana share of marginal farmers decreased in 2022-23 followed by an increase in 2023-24. For Uttar Pradesh, marginal farmers declined from 2021-22 to 2023-24. For KMS2023-24, Telangana has the highest share of marginal and small farmers at 92.9 percent, followed by Andhra Pradesh (89.1%) and Odisha (86%), with almost same pattern in KMS2021-22 and KMS2022-23. However, in Uttar Pradesh share of marginal and small farmers has seen a continuous decline from 70.7 percent in KMS2021-22 to less than 50 percent in KMS2023-24 while share of semi-medium farmers has seen a continuous increase over the said period.

Table 2.8: Procurement of Paddy by Farm-Size in selected States during KMS2021-22, KMS2022-23 and KMS2023-24

(in percent)

Farmers benefitted				
Year	Marginal Farmer (<1 ha)	Small Farmer (1-2 ha)	Semi- medium Farmer (2-4 ha)	Medium & Large (>4 ha)
Andhra Pradesh				
2021-22	37.6	33.6	21.6	7.2
2022-23	70.9	20.3	7.3	1.5
2023-24	66.8	22.3	9.0	1.9
Odisha				
2021-22	50.0	35.0	14.0	2.0
2022-23	51.0	35.0	13.0	1.0
2023-24	50.0	36.0	13.0	1.0
Telangana				
2021-22	72.6	20.4	5.8	1.2
2022-23	67.6	24.6	6.6	1.2
2023-24	68.4	24.5	6.1	1.0
Uttar Pradesh				
2021-22	22.2	48.5	24.1	5.1
2022-23	15.9	33.8	42.0	8.3
2023-24	13.7	31.7	54.3	0.3

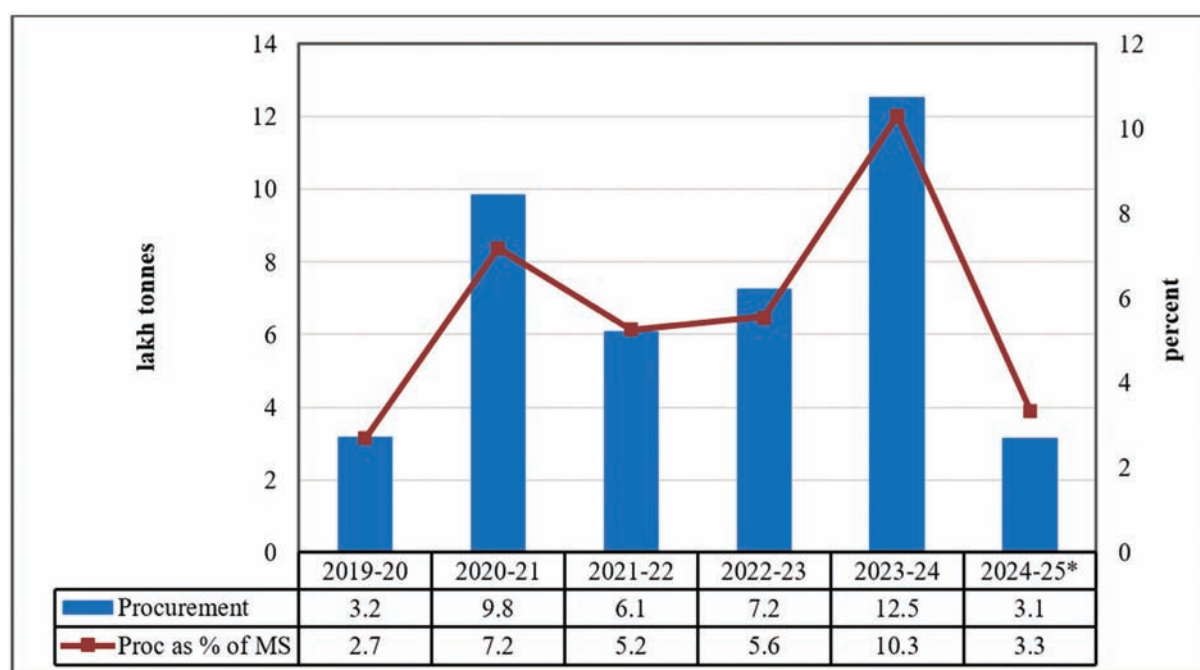
Source: Replies from respective State Governments



Nutri-cereals

- 2.39 Procurement of nutri-cereals increased in KMS2020-21 and was 9.8 lakh tonnes, as against 3.2 lakh tonnes in KMS2019-20 (Chart 2.18). However, procurement declined in KMS2021-22 and was 6.1 lakh tonnes, which increased to 7.2 lakh tonnes in KMS2022-23. However, during KMS2023-24, a significant jump of around 73 percent was recorded in procurement and reached 12.5 lakh tonnes. The procurement as a percent of marketed surplus of nutri-cereals was 7.2 percent in KMS2020-21, declined in KMS2021-22 but increased during the last two seasons and was estimated at 10.3 percent in KMS2023-24.

Chart 2.18: Trends in Procurement and Share of Procurement in the Marketed Surplus (MS) of Nutri-Cereals during KMS2019-20 to KMS2024-25



Note: 1. *KMS 2024-25(Kharif) is under progress. Data upto 03.02.2025

2. Marketed Surplus Ratio (MSR) for year 2023-24 is taken as the proxy for year 2024-25

Source: 1. Economics, Statistics & Evaluation Division, Ministry of Agriculture and Farmers Welfare

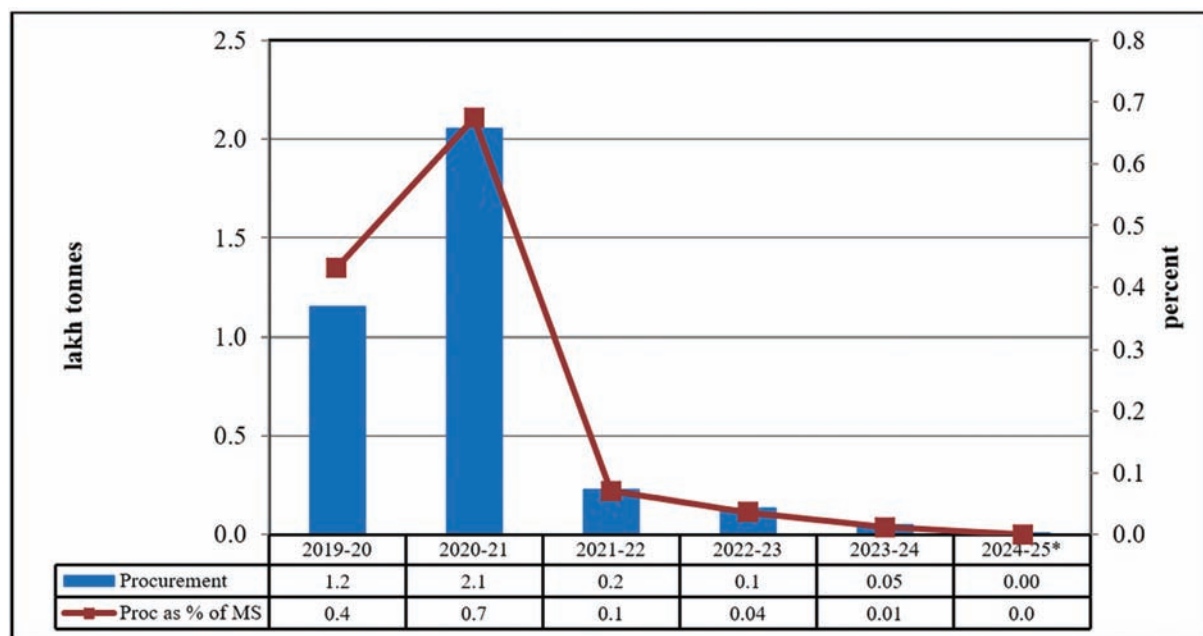
2. Food Corporation of India

Maize

- 2.40 Procurement of maize, which witnessed a sharp increase in KMS2020-21, declined during the last three seasons and negligible quantity was procured as market prices remained higher than MSP (Chart 2.19). Detailed information on State-wise procurement of nutri-cereals and maize during KMS2023-24 and KMS2024-25 is given in Annex Table 2.3.



Chart 2.19: Trends in Procurement and Share of Procurement in the Marketed Surplus (MS) of Maize during KMS2019-20 to KMS2024-25



Note: 1. *KMS 2024-25(Kharif) is under progress. Data upto 03.02.2025

2. Marketed Surplus Ratio (MSR) for year 2023-24 is taken as the proxy for year 2024-25

Source: 1. Economics, Statistics & Evaluation Division, Ministry of Agriculture and Farmers Welfare

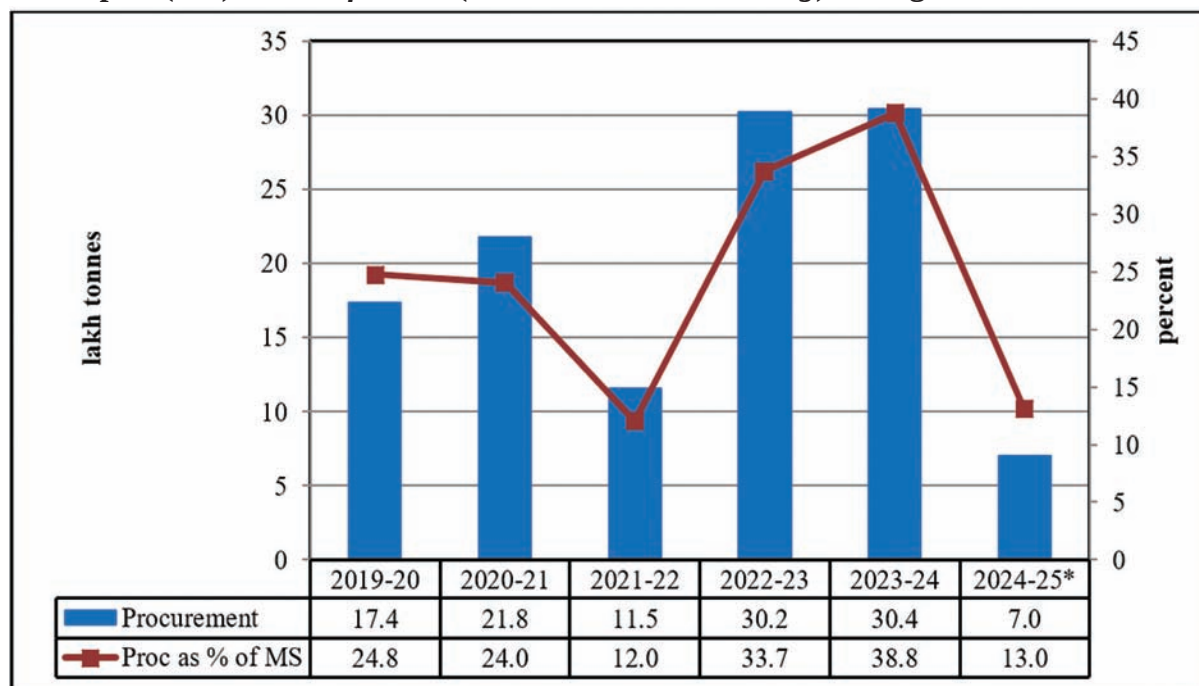
2. Food Corporation of India

Pulses

- 2.41 The trends in procurement of pulses during 2019-20 to 2024-25 are presented in Chart 2.20. A record procurement of 40.5 lakh tonnes was done in 2018-19 due to depressed market prices. The procurement declined in 2019-20 but rose during 2020-21. However, procurement of pulses rose during the last two seasons and was estimated around 30 lakh tonnes in 2023-24. Detailed information on State-wise procurement of *Kharif* pulses, *moong*, *tur/arhar* and *urad*, during KMS2022-23 to KMS2024-25 is provided in Annex Table 2.4.



Chart 2.20: Trends in Procurement and Share of Procurement in the Marketed Surplus (MS) of Kharif Pulses (Tur/Arhar, Urad & Moong) during 2019-20 to 2024-25



Note: 1. *KMS 2024-25 (Kharif) is under progress. Data upto 03.02.2025

2. Marketed Surplus Ratio (MSR) for year 2023-24 is taken as the proxy for year 2024-25

Source: 1. Economics, Statistics & Evaluation Division, Ministry of Agriculture and Farmers Welfare

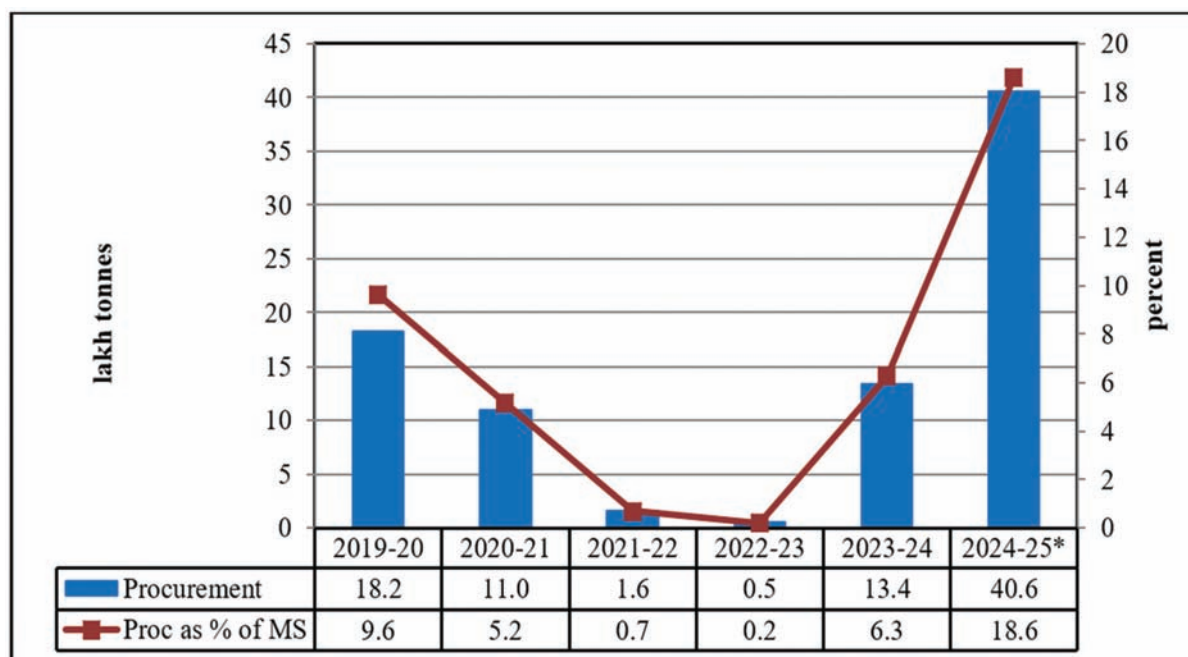
2. National Agricultural Cooperative Marketing Federation of India (NAFED)

Oilseeds

- 2.42 The procurement of oilseeds recorded a steady decline during 2020-21 to 2022-23 due to higher market prices. However, procurement increased significantly during 2023-24 and 2024-25 as market prices of two major *kharif* oilseeds, soybean and groundnut, were lower than MSP (Chart 2.21). During last five marketing seasons, a record procurement of 40.6 lakh tonnes was done in 2024-25 (as on 3rd February, 2025). The detailed information on State-wise procurement of groundnut and soybean during KMS2022-23 to KMS2024-25 is given in Annex Table 2.5.



Chart 2.21: Trends in Procurement and Share of Procurement in the Marketed Surplus (MS) of Oilseeds (Groundnut & Soybean) during 2019-20 to 2024-25



Note: 1. *KMS 2024-25(Kharif) is under progress. Data upto 03.02.2025

2. Marketed Surplus Ratio (MSR) for year 2023-24 is taken as the proxy for year 2024-25

Source: 1. Economics, Statistics & Evaluation Division, Ministry of Agriculture and Farmers Welfare

2. National Agricultural Cooperative Marketing Federation of India (NAFED)



Box 2.2: Need to Strengthen Procurement Operations at State-level

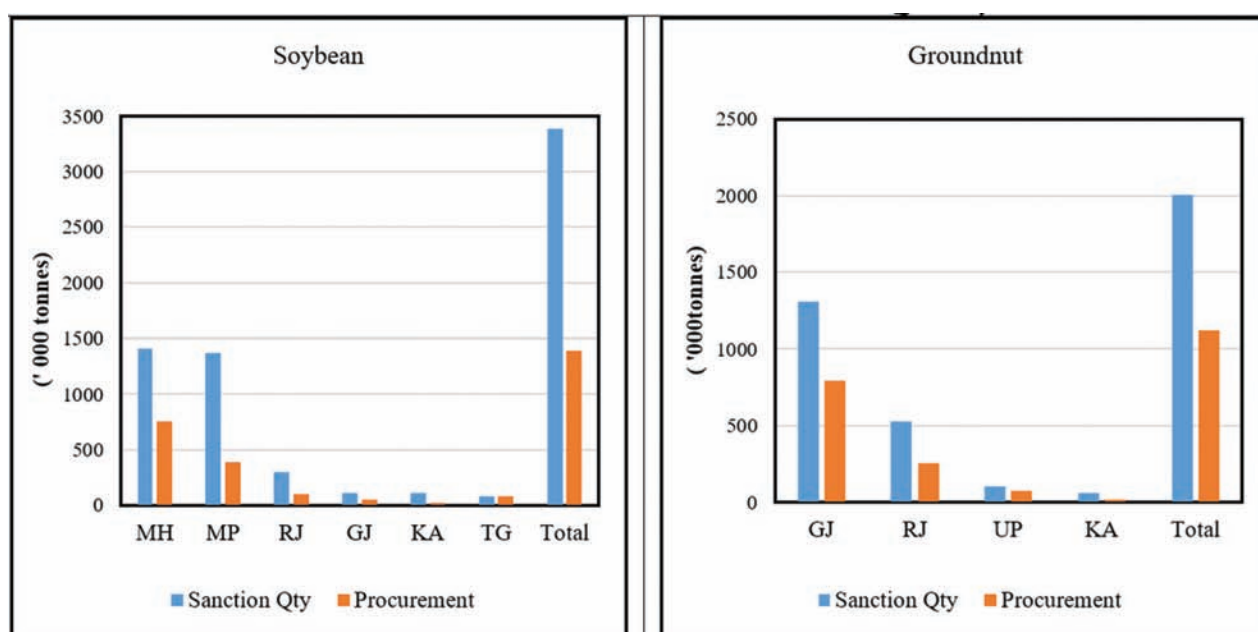
For procurement of pulses and oilseeds, Price Support Scheme (PSS) is implemented on the request of the State Governments/UTs. From 2024-25 procurement season onwards, the sanction for the procurement of notified pulses and oilseeds under PSS is given to the State/UT initially up to a maximum of 25 percent of the production of the State for that particular season and if the State exhausts its limit of 25 percent, the sanction for additional procurement under PSS is given to the State/UT up to a maximum of 25 percent of the national production after necessary approvals. In order to achieve Atmanirbharta in pulses, procurement ceiling of 25 percent has been removed in respect of tur/arhar, urad and lentil/masoor for the year 2024-25.

During Commission's interactions with various stakeholders including farmers and State Governments, most State Governments and farmers' representatives demanded that 25 percent procurement ceiling should be lifted. However, empirical evidence shows that actual procurement of oilseeds during KMS2024-25 in various States has remained much lower than the sanctioned quantity. For example, in case of soybean, as against the total sanctioned quantity of about 33.9 lakh tonnes at all-India level, only 13.9 lakh tonnes was procured as on 3rd February, 2025. In Madhya Pradesh, the second largest producer of soybean in the country, less than 30 percent of total sanctioned quantity was procured, while in Maharashtra, top producer, 53.2 percent of sanctioned quantity was procured. Telangana was the only exception where 98.3 percent of sanctioned quantity was procured.

Similarly, in groundnut, about 56 percent of sanctioned quantity was procured by States, which varied from about 1.1 percent in Karnataka to 66.6 percent in Uttar Pradesh. In Gujarat, the largest producer of groundnut, 60.7 percent of sanctioned quantity was procured, while in Rajasthan, less than 50 percent of sanctioned quantity was procured.

The issue of actual procurement significantly lower than the sanctioned quantity indicates lack of procurement facilities and infrastructure such as procurement centres and storage facilities, Non-FAQ produce brought to procurement centres by farmers, delay in start of procurement operations, etc. Therefore, there is a need to strengthen procurement facilities including setting up procurement centres, scientific storage and warehousing facilities, awareness among farmers, timely start of procurement operations by States, provision of dryers and other facilities to farmers, etc. In this context, effective participation of States is needed.

Chart Box 2.2: Actual Procurement v/s Sanctioned Quantity



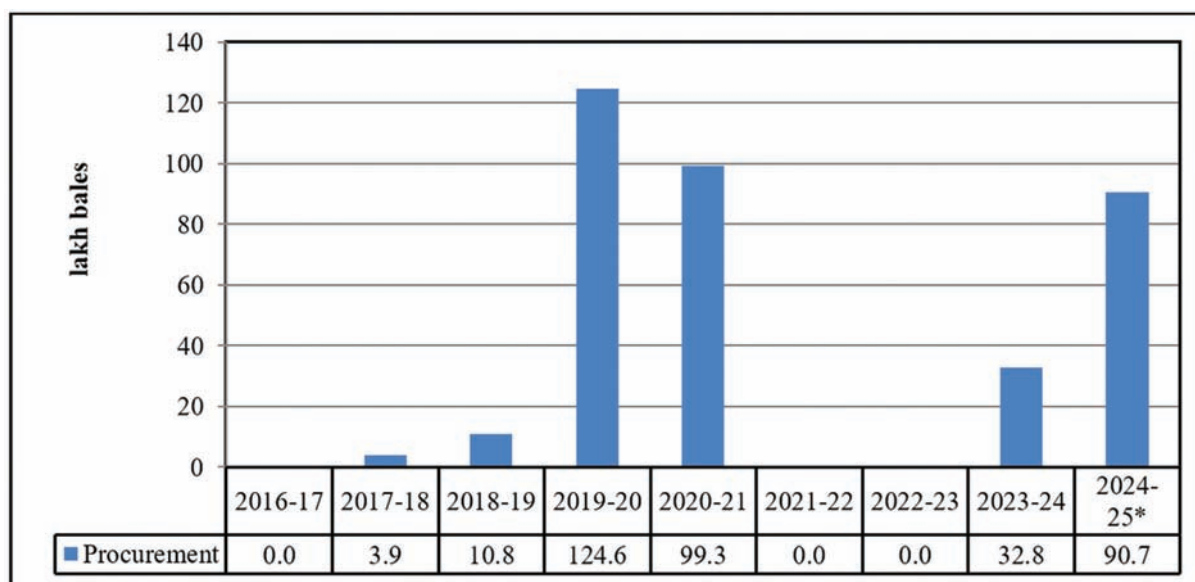
Source: National Agricultural Cooperative Marketing Federation of India (NAFED)



Cotton

- 2.43 The trends in MSP procurement of cotton by Cotton Corporation of India (CCI) are presented in Chart 2.22. A record procurement (124.6 lakh bales) of cotton was done by CCI in 2019-20 due to depressed market prices and Covid-19 induced market disruptions. Procurement, though declined in 2020-21, was quite high (99.3 lakh bales). There were no procurement operations in 2021-22 and 2022-23 by CCI, due to unprecedented high market prices. However, CCI had to intervene in the market due to low market prices and procured 32.8 lakh bales in 2023-24 and 90.7 lakh bales in 2024-25 (as on 24th February, 2025). Out of the total procurement in 2024-25, 43.8 percent was procured in Telangana, followed by Maharashtra (29.3 %), Gujarat (10.6%) and Karnataka (5.3%).

Chart 2.22: MSP Procurement of Cotton during 2016-17 to 2024-25



Note: * Provisional (upto 24.02.2025)

Source: Cotton Corporation of India

Market Distortions: Bonus on Paddy

- 2.44 Some State Governments pay direct or indirect bonus over and above the MSP, which creates distortions in the market. This limits private trade participation and potentially discourages competition, as well as adversely affects global competitiveness. Kerala, a deficit DCP State, announced the highest bonus (₹520/qttl) for paddy in KMS2024-25 but lower than 2023-24 season. Other deficit States, Tamil Nadu, Jharkhand and West Bengal, also declared bonus on paddy procurement during KMS2024-25 (Table 2.9). However, some surplus States like Chhattisgarh and Odisha announced significantly high bonus on paddy procurement during KMS2024-25 but was paid under *Krishak Unnati Yojana* and *Smrudh Krushak Niti*, respectively, on per acre basis. Since, such assistance is not crop-neutral, it would encourage farmers to produce more paddy and thereby, adversely affect crop diversification efforts in the States. The Commission is of the view that such crop-specific incentive/bonus should be discontinued and farmers can be incentivised for diversification towards pulses, oilseeds and millets.



Table 2.9: Bonus Declared by Selected States for Paddy

(₹/qtl)

States	KMS2022-23	KMS2023-24	KMS2024-25
Chhattisgarh	* *	* *	**
Jharkhand	10	117	100
Kerala	C=780	C=637	520
	A=760	A=617	
Odisha			# #
Tamil Nadu	C=75	C=82	C=105
	A= 100	A= 107	A=130
West Bengal	20	20	20

Note: 1. C=Common, A= Grade –A

2. **Govt of Chhattisgarh paid ₹9,000/acre (ceiling of 15qtl/acre) under Rajiv Gandhi Kisan Nyay Yojna during KMS2022-23 and ₹19,257/acre (ceiling of 21qtl/acre) under Krishak Unnati Yojna during KMS2023-24 and KMS2024-25.

3. # # Odisha Government provided input subsidy of ₹800/qtl over and above MSP of ₹2,300/qtl under Samrudh Krushak Yojna.

Source: 1. Food Corporation of India

2. Department of Food and Public Distribution, Ministry of Consumer Affairs, Food and Public Distribution

Market Fees and Other Incidental Charges

- 2.45 Various States levy fees, taxes and other incidental charges on procurement, which result in high procurement incidentals and economic cost of grains. The fees, taxes and other charges levied on rice procurement during KMS2024-25 in major rice growing States are presented in Table 2.10. The incidence of such charges is highest in Punjab and Haryana at ₹91.38 per qtl, followed by Andhra Pradesh, Odisha and Chhattisgarh (₹77.25/qtl); Uttar Pradesh (₹65.75/qtl); Assam, Madhya Pradesh, and Telangana (₹54.25/qtl); Maharashtra (₹48.5/qtl) and Karnataka and West Bengal (₹42.75/qtl). Kerala is the only State, which does not levy any charges. These charges create market inefficiencies and restrict inter-State trade, thereby affecting the overall efficiency of procurement. Such disparities in fees and charges among States need to be addressed and rationalized so as to create competitive and efficient markets. States charging high fee should be dis-incentivised to facilitate private sector participation and also reduce procurement incidentals.

**Table 2.10: State-wise Fees/Taxes/Charges levied on Rice Procurement (KMS2024-25)**

States	Market Fee/Mandi Charges (%)	Mandi Development Cess (%)	Krishak Kalyan Cess (%)	Dami/ Arhatia Charges (₹/qtl)	Commission to society (₹/qtl)
Haryana	2	-	-	45.38	0
Punjab	2	-	-	45.38	0
Andhra Pradesh	2	-	-	0	31.25
Chhattisgarh	1.5	-	0.5	0	31.25
Odisha	2	-	-	0	31.25
Assam	1	-	-	0	31.25
Madhya Pradesh	1	-	-	0	31.25
Telangana	1	-	-	0	31.25
Uttar Pradesh	1	0.5	-	0	31.25
Maharashtra	0.75	-	-	0	31.25
Karnataka	0.5	-	-	0	31.25
West Bengal	0.5	-	-	0	31.25
Kerala	0	-	-	0	0

Source: Department of Food and Public Distribution, Ministry of Consumer Affairs, Food and Public Distribution

Food Subsidy and Economic Cost

- 2.46 Food subsidy has played an important role in achieving food security by ensuring access to food at affordable price, particularly for poor households. However, food subsidy has increased significantly over the years due to rising economic cost of grains and non-revision of Central Issue Price (CIP) of grains. *Pradhan Mantri Garib Kalyan Anna Yojna* (PMGKAY), which was launched in April 2020 to provide free foodgrains to about 81.35 crore beneficiaries, has been extended by the Central Government for a period of 5 years from 1st January, 2024 and about ₹11.8 lakh crore will be spent on food subsidy during these five years.
- 2.47 Food subsidy comprises three components: (i) Consumer Subsidy (difference between economic cost and Central Issue Price (CIP)), (ii) Cost of maintaining buffer stock and (iii) Operational losses of FCI. Economic cost is the major component of food subsidy and it consists of (i) pooled grains cost, (ii) procurement incidentals consisting of statutory charges, gunny cost, labour charges, transportation, storage, interest etc., and (iii) distribution cost consisting of freight, handling, storage, interest, shortages and administrative overheads. The trends in different components of economic cost are given in Table 2.11. It is evident from the Table that there has been a steady increase in economic cost of rice from ₹3,562.5 per qtl in 2021-22 to ₹4,042.2 per qtl in 2024-25(RE). However, it is important to note that



the share of pooled cost of grains has shown a rising trend over the years and accounted for around 80 per cent of economic cost. On the other hand, the share of procurement incidentals and distribution cost has declined. The share of procurement incidentals declined from 19.3 percent in TE2015-16 to 12.8 percent in 2024-25(RE), while share of distribution cost declined from 16 percent in TE2015-16 to 7.1 percent in 2024-25(RE). The decline in distribution cost share was driven by reduction in freight, handling charges, storages and interest cost, with lower interest cost contributing most, as National Small Savings Fund (NSSF) loan to FCI for food subsidy was discontinued in 2021-22 and ₹4,22,618 crore was provided to FCI. The above trends clearly indicate improvement in rice procurement and distribution efficiency.

Table 2.11: Trends in different Components of Economic Cost of Rice

(₹/qtl)

Particulars	TE2015-16	2021-22	2022-23	2023-24	2024-25 (RE)	2025-26 (BE)
(A) Pooled Cost of Grain	1872.5	2731.9	2854.9	3081.6	3238.2	3375.3
(B) Procurement Incidentals	558.8	516.4	475.3	477.3	517.0	526.6
a) Statutory /Obligatory Cost	395.5	310.8	312.9	285.8	290.5	296.3
b) Labour & Transport Charges	29.3	69.1	66.5	69.8	74.1	74.7
c) Storage & Interest charges paid to State Agencies	98.1	69.6	15.2	36.5	49.7	51.0
d) Others charges	35.9	67.0	80.8	85.3	102.7	104.6
(C) Distribution Cost	463.5	314.2	392.2	325.0	287.0	271.4
Freight	170.9	156.5	157.6	128.8	112.7	122.9
Handling Charges	73.3	50.7	60.6	66.6	64.6	57.8
Storage Charges	35.8	39.6	55.5	48.2	45.5	35.3
Interest	138.0	26.3	62.2	24.8	0.0	-
Shortages	18.5	13.4	19.0	7.4	7.1	7.3
Administrative Overheads	27.1	27.7	37.3	49.1	57.1	48.2
Economic Cost (A+B+C)	2894.9	3562.5	3722.4	3883.8	4042.2	4173.3

Source: Food Corporation of India

Awareness about MSP and FAQ Norms

- 2.48 As per NSS 77th Round, Situation Assessment of Agricultural Households and Land and Livestock Holdings of Households in Rural India, 2019, level of awareness about MSP was relatively higher in case of paddy farmers (46.8%), cotton (33.3%), tur/arhar (32.6%) and bajra (Table 2.12). However, awareness among farmers growing ragi, jowar, maize, pulses and oilseeds was low. It is pertinent to note that awareness about procurement agencies was significantly lower than awareness about MSP.



There were also large variations across States about awareness about MSP and procurement agencies. Therefore, concerted efforts are needed to create awareness among farmers about MSP and procurement agencies to expand the coverage of MSP and procurement operations.

- 2.49 Many a times, grains brought by farmers to procurement centres fail to meet FAQ standards due to high moisture content, broken/damaged grains, etc. and are rejected by procurement agencies. Therefore, there is a need to create awareness as well as train farmers to bring FAQ produce to procurement centres. Basic infrastructure for drying, cleaning, etc. should be provided at village level. The Central and State Governments should organise extensive publicity campaigns related to MSP and procurement operations in regional languages using electronic and print media, social media, dissemination through panchayats, farmer organisations/FPOs, PACS, APMC *mandis*, etc. before the start of procurement season.

Table 2.12: Percentage of Agricultural Households reported Awareness about MSP and Procurement Agencies

Crop	Awareness about					
	MSP (%)			Procurement Agency (%)		
	Jul-Dec 2018	Jan-June 2019	All	Jul-Dec 2018	Jan-June 2019	All
Paddy	40.7	52.8	46.8	30.3	40.3	35.3
Jowar	27.4	14.3	20.9	16.6	7.3	12.0
Bajra	31.7	-	31.7	19.7	-	19.7
Maize	21.3	22.7	22.0	15.3	17.7	16.5
Wheat	-	37.1	37.1	-	27.2	27.2
Ragi	4.3	-	4.3	3.1	-	3.1
Gram	-	29	29.0	-	18.8	18.8
Arhar (Tur)	24	41.2	32.6	13	27.9	20.5
Urad	20.9	-	20.9	15.7	-	15.7
Moong	22.7	27.2	25.0	13.7	20.4	17.1
Masoor	-	24.2	24.2	-	138	138.0
Sugarcane	39.8	56.9	48.4	32.7	51	41.9
Groundnut	22.2	-	22.2	10.6	-	10.6
Rapeseed	-	39.2	39.2	-	22.2	22.2
Coconut	11	12	11.5	6.5	6.2	6.4
Soybean	28.3	-	28.3	19.6	-	19.6
Cotton	30.7	35.8	33.3	18	24.7	21.4

Source: NSS Report No. 587- Situation Assessment of Agricultural Households and Land and Livestock Holdings of Households in Rural India, 2019



Recapitulation

- 2.50 Global production of rice and soybean are projected to increase in 2024-25 while maize production is forecast to fall as per FAO, USDA and IGC estimates. Domestic rice production increased from 135.8 million tonnes in 2022-23 to 137.8 million tonnes in 2023-24 and as per Second Advance Estimates, rice production (*kharif* and *rabi*) is estimated at 136.4 million tonnes, 6.7 percent higher than 2023-24 *kharif* and *rabi* rice production. In view of higher Central Pool Stocks and production in 2024-25, domestic rice supply is expected to rise significantly, which might result in weakening of rice prices in the next season.
- 2.51 During KMS2024-25, market prices of paddy remained above MSP while for maize, market prices were below MSP in October-November 2024, but surpassed MSP in December 2024 and January 2025. With respect to pulses, market prices of *tur/arhar* were above the MSP but fell below MSP in January 2025, whereas market prices of *urad* and *moong* stayed below the MSP during KMS2024-25. In case of oilseeds, market prices remained much below the MSP for groundnut and soybean, while for cotton fell below the MSP in December 2023 and remained below the MSP, which continued in KMS2024-25.
- 2.52 Rice procurement, which registered significant increase during 2019-20 and 2020-21, showed a steady decline during the last three seasons and was 52.5 million tonnes in 2023-24. During KMS2024-25, rice procurement was 45.9 million tonnes as on 2nd March, 2025 which was higher than the last year corresponding procurement of 43.9 million tonnes. Procurement was significantly higher in Punjab, Haryana, Chhattisgarh, Odisha and Telangana, while procurement was low in major producers like West Bengal and Uttar Pradesh. The coverage of procurement operations has also increased as is evident from increase in beneficiary farmers during 2016-17 to 2020-21. However, due to better market prices, there is some decline in rice procurement as well as beneficiary farmers. Targeted policy interventions are required to address regional disparities in procurement operations and ensure equitable access to MSP benefits for all farmers, especially small and marginal ones.
- 2.53 Some State Governments pay direct or indirect bonus over and above the MSP, which creates distortions in the market and restricts private trade participation and potentially discourages competition, as well as adversely affects global competitiveness. In addition, some States levy various fees, taxes and other incidental charges on procurement, which result in high procurement incidentals and economic cost of grains and also create market inefficiencies and restrict inter-State trade, thereby affecting the overall efficiency of procurement. Such distortionary practices should be discontinued to create efficient and competitive markets.
- 2.54 Food subsidy, which has contributed towards achievement of food security by ensuring access to food at affordable price especially for poor households, has increased significantly over the years on account of rising economic cost of grains and non-revision of Central Issue Price (CIP) of grains. Due to extension of the *Pradhan Mantri Garib Kalyan Anna Yojna* (PMGKAY) by the Central



Government for a period of 5 years from 1st January, 2024, about ₹11.8 lakh crore will be spent on food subsidy during these five years. There has been a steady increase in economic cost of rice from 2021-22 to 2024-25(RE) due to an increasing trend showed by share of pooled cost of grains while the share of procurement incidentals and distribution cost has declined indicating improvement in rice procurement and distribution efficiency.



CHAPTER 3



Crop Productivity Trends and Drivers

- 3.1 During the last few decades, Indian agricultural sector has undergone significant transformation and productivity was major driver of growth in agriculture. Several factors, including adoption of improved technologies and farming practices has increased productivity. However, growth in agricultural output and productivity has slowed during the recent decades. Given the limited land resources and climate change induced extreme weather shocks, it is important to address climate change challenges and improve efficiency and productivity of agricultural sector.
- 3.2 This chapter analyzes the yield trends and gap between current and potential yields of mandated *kharif* crops at all-India level as well as in major States with an objective to understand how productivity has changed in the recent period and to identify regions/crops of particular concern. Further, the yield of selected *kharif* crops in the country has been compared with the yield levels of major producing countries with a view to assess performance at the international level. This chapter also highlights several initiatives taken by the Government to improve crop yields and reduce yield gaps.

Growth Trends of Area, Production and Yield at All-India Level

- 3.3 Table 3.1 presents the average growth rates in area, production and yield of mandated *kharif* crops for TE2013-14, TE2018-19 and TE2023-24.

Cereals

- 3.4 During TE2023-24, area under rice recorded higher growth but growth in yield was much lower than the previous period, resulting in lower growth in production. As far as maize is concerned, an impressive growth of 6.1 percent was recorded in production in TE2023-24 but lower than the previous period (7.4%) and was driven by a significant growth in area (4.4%). There was significant deceleration in yield growth (1.7%) in the recent period compared with 6.4 percent in TE2018-19.

Shree Anna/Nutri-Cereals

- 3.5 In case of *kharif* Shree Anna/nutri-cereals, jowar and bajra registered positive growth in yield during TE2023-24 but significantly lower than TE2018-19, which resulted in positive growth in production



despite negative growth in area. However, ragi, which registered positive growth in area after continuous negative growth in the previous periods, witnessed negative growth in production due to sharp decline in yield (-6.8%) during TE2023-24. The growth in area, production and yield of *kharif Shree Anna* cereals during the current period was negative, while *rabi* nutri-cereals recorded quite impressive growth in area, production and productivity. Performance of *rabi* cereals in terms of growth in area and production during the current period was found to be higher than that of *kharif* cereals, while yield performance of *kharif* cereals was better than that of *rabi* cereals.

Pulses

- 3.6 Among *kharif* pulses, urad registered positive growth in production (2.5%) on account of impressive growth in yield (7%) during current period, which outweighed the adverse impact of negative growth in area (-4.5%). On the other hand, growth in both area and yield of tur was recorded negative during TE2023-24, resulting in sharp decline (-6.9%) in production. In case of moong, growth in area, production and yield was positive in the recent period, however much lower than TE2018-19. It is evident from the Table that performance of *rabi* pulses was better than *kharif* pulses during the recent period. However, growth performance of pulses, both *kharif* and *rabi*, was highest during TE2018-19. Despite significant increase in production of pulses due to area expansion and yield improvement during the last 10 years, imports of pulses, which declined between 2017-18 and 2022-23, have increased during the last two years. In order to achieve self-sufficiency in production of pulses, with a special focus on tur, urad and lentil, Government has announced a six year “Mission for *Aatmanirbharta* in Pulses” in the Budget 2025-26. Assured procurement of pulses by Central Nodal agencies, improved productivity, climate resilient HYV seeds, improved post-harvest management and storage under the mission would have positive impact on the production of pulses in the country.

Oilseeds

- 3.7 It is encouraging to note that positive growth in yield was registered for all mandated *kharif* oilseeds during the current period. Groundnut registered the highest growth in yield (8.5%) among *kharif* oilseeds during TE2023-24, but a sharp decline in area (-7.8%) resulted in negative growth in production. In case of soybean, positive growth in both area and yield contributed in achieving positive growth in production during TE2023-24. High positive growth in yield of sesamum neutralized the adverse impact of decline in area, and led to positive growth in its production during current period. Almost similar trend was observed in case of sunflower during TE2023-24. For nigerseed, high negative growth in area adversely affected the growth in production during recent period. As far as *kharif* oilseeds are concerned, despite 3.3 percent growth in yield, moderate growth of 0.8 percent in production could be achieved due to negative growth in area (-2.3%) in the recent period. In case of *rabi* oilseeds, higher growth in area (10.8%) contributed to positive growth in production despite negative growth in yield (-2.1%) during TE2023-24.



Table 3.1: Triennial Trends in Growth Rate of Area, Production and Yield of Major Kharif Crops

(Percent)

Crop/Group	Area			Production			Yield		
	TE2013-14	TE2018-19	TE2023-24	TE2013-14	TE2018-19	TE2023-24	TE2013-14	TE2018-19	TE2023-24
Rice	1.0	0.5	1.5	3.7	3.7	3.5	2.6	3.2	2.0
Maize	2.0	1.0	4.4	3.8	7.4	6.1	1.8	6.4	1.7
Shree Anna /Nutri-Cereals									
Jowar	-7.6	-12.2	-1.6	-7.1	-4.9	0.8	0.8	7.7	2.0
Bajra	-6.2	0.0	-0.8	-3.3	3.1	0.2	3.2	2.9	0.8
Ragi	-2.3	-6.2	1.7	-1.5	-6.1	-5.6	0.1	-3.1	-6.8
Kharif	-5.6	-2.5	-2.9	-3.4	0.2	-4.2	2.5	2.6	-1.4
Rabi*	-9.4	-15.8	6.6	-6.9	-8.8	16.5	2.7	7.6	9.5
Total	-6.6	-5.7	-1.0	-4.5	-1.2	-0.5	2.1	4.4	0.4
Cereals									
Kharif	-0.5	-0.2	0.0	2.9	3.8	1.6	3.4	4.0	1.6
Rabi*	0.4	-1.9	3.4	2.8	3.9	3.7	2.4	5.9	1.0
Total	-0.1	-0.9	1.2	2.9	3.8	2.6	3.0	4.8	1.4
Pulses									
Tur	-3.6	6.8	-3.9	3.9	18.5	-6.9	7.7	7.5	-3.2
Moong	0.4	7.7	0.5	-0.9	16.9	1.0	-2.7	7.9	0.5
Urad	-2.1	15.9	-4.5	-1.0	18.8	2.5	1.1	1.6	7.0
Kharif	-5.5	10.1	-4.4	-5.3	19.1	-6.8	0.3	6.1	-2.0
Rabi*	2.1	2.1	1.3	6.2	10.4	1.2	4.3	7.4	-0.2
Total	-1.3	5.7	-1.4	2.0	12.8	-1.4	3.6	5.8	-0.1
Foodgrains									
Kharif	-1.3	1.5	-0.8	2.4	4.3	1.1	3.9	2.7	2.0
Rabi*	0.8	-0.9	2.4	3.1	4.4	3.4	2.4	5.4	0.9
Total	-0.4	0.5	0.6	2.8	4.3	2.3	3.2	3.9	1.7
Oilseeds									
Groundnut	-1.3	1.5	-7.8	19.5	2.5	-0.2	15.4	2.0	8.5
Soybean	6.9	-1.2	1.0	-1.0	19.3	1.8	-7.4	20.7	0.9
Sesamum	-6.9	-10.0	-3.8	-6.8	-6.6	1.3	-0.1	3.6	5.3
Sunflower	-8.8	-18.3	-1.4	-7.3	-9.8	0.7	2.9	10.8	4.7
Nigerseed	-6.8	-13.8	-14.0	-3.0	-12.7	-13.8	4.8	-0.1	0.7
Kharif	2.6	-2.0	-2.3	1.2	8.4	0.8	-1.4	10.7	3.3
Rabi*	-2.0	-0.6	10.8	-0.8	8.2	8.4	1.0	8.9	-2.1
Total	1.1	-1.6	1.6	0.5	8.2	3.5	-0.6	10.0	1.8
Cotton	2.2	1.5	-1.4	2.9	-1.7	-2.3	0.8	-1.6	-1.1

Note: 1. Average growth is average of last three-year annual growth rate, for eg. TE2013-14 is average of annual growth between 2010-11 and 2011-12, 2011-12 and 2012-13 & 2012-13 and 2013-14

2. * Rabi includes summer season.

Source: Economics, Statistics & Evaluation Division, Ministry of Agriculture and Farmers Welfare

Cotton

- 3.8 Growth rate in area, production and yield was recorded negative during current period, while both production and yield growth rates were negative during last two periods. Negative growth rate in



cotton yield and almost stagnant production during the last 7-8 years is a matter of concern and needs immediate attention to reverse the trend. In the Budget 2025-26, the Government has announced a Five-year Mission for improving cotton productivity and promote extra-long staple varieties.

Comparison of Yield in Major Producing States

- 3.9 In this section, yield levels in major States have been compared to understand yield variations across States. For this analysis, average yield in TE2023-24 and average annual growth in yield between 2019-20 and 2023-24 have been compared. Classification of States based on growth in yield of mandated crops during last five years is presented in Table 3.2.
- 3.10 Average annual growth rate in yield at the all-India level has been taken as benchmark for classification of States under two categories i.e. $\geq$ all-India average and $<$ all-India average. Despite low growth rate (1.8%) in rice yield, nine out of 16 States recorded lower growth than all-India average. Even major producers like West Bengal, Uttar Pradesh, Telangana, Odisha, Andhra Pradesh and Tamil Nadu had lower growth than all-India growth rate. In case of maize, ragi and soybean, the number of States having higher than all-India growth rate was higher, while in other crops bajra, jowar, urad, moong, groundnut, sunflower, nigerseed and cotton, number of States having lower growth rate than all-India average was high. Therefore, concerted efforts are needed to improve productivity in low performing States through promotion of modern technologies, good quality seed, best agricultural practices, quality inputs and services to farmers.

Table 3.2: Performance of the States on the basis of Average Y-o-Y Growth in Yield at All-India Level during Last Five Years

Crop	All-India Growth (%)	Classification of States			
		Count	$\geq$ All-India Average States	Count	$<$ All-India Average States
Rice	1.8	7	CG, BR, HR, AS, MP, PB, GJ	9	AP, KA, TG, MH, OD, WB, UP, TN, JH
Maize	1.9	9	BR, TG, RJ, CG, MH, JH, UP, GJ, AP	7	PB, KA, WB, HP, J&K, MP, TN
Bajra	3.7	3	MH, KA, MP	5	RJ, GJ, HR, UP, TN
Jowar	6.7	3	AP, MH, TG	6	UP, KA, GJ, RJ, MP, TN
Ragi	0.8	6	MH, UK, JH, OD, KA, TN	2	AR, AP
Tur	2.9	5	AP, TG, MH, MP, KA	5	JH, OD, GJ, UP, TN
Urad	4.4	5	MP, RJ, TG, UP, OD	8	GJ, AP, MH, WB, JH, AS, KA, TN
Moong	3.4	5	WB, GJ, MP, AP, OD	8	PB, BR, HR, MH, UP, KA, RJ, TN
Groundnut	10.6	2	AP, GJ	8	MH, UP, KA, WB, MP, RJ, TN, TG
Soybean	(-)2.8	6	RJ, GJ, KA, MH, TG, MP	-	-
Sunflower	7.1	3	MH, TN, AP	7	KA, WB, HR, TG, OD, UP, BR
Sesamum	3.3	5	GJ, UP, AP, WB, KA	5	RJ, TG, MP, AS, TN
Nigerseed	2.2	2	MH, AP	7	CG, AS, WB, OD, MP, JH, GJ
Cotton	3.3	4	KA, TG, GJ, MH	6	OD, HR, AP, PB, RJ, MP

Note: States with production share (TE2023-24) of 1 percent or more have been considered for the analysis

Source: Economics, Statistics & Evaluation Division, Ministry of Agriculture and Farmers Welfare

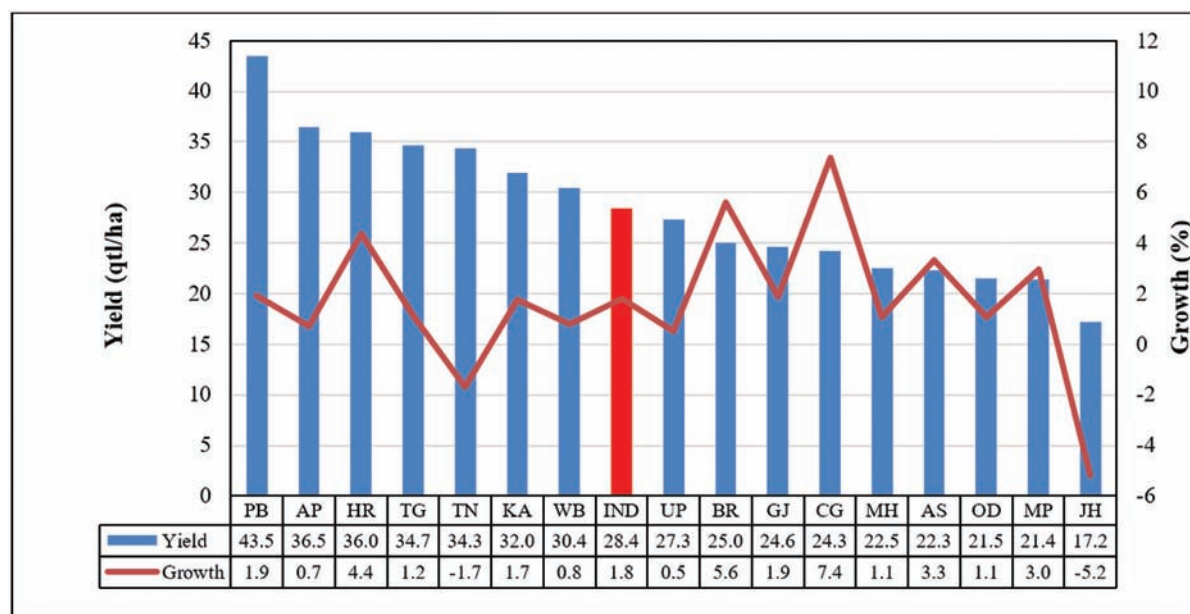


Cereals

Rice

- 3.11 Chart 3.1 exhibits average yield and growth in yield of rice in major producing States. For this analysis, average yield in TE2023-24 and average annual growth in yield between 2019-20 and 2023-24 have been computed. It is evident from the Chart that the highest average yield was recorded in Punjab, followed by Andhra Pradesh and Haryana during TE2023-24, while the highest growth in yield was observed in Chhattisgarh, followed by Bihar and Haryana. Tamil Nadu and Jharkhand witnessed negative growth in yield during the given period. Among the major States, average yield of only 7 States was recorded higher than the national average. The average yield of Uttar Pradesh, second largest producer, and eight other States was lower than the all-India average.

Chart 3.1: Average Yield and Growth in Yield of Rice in Major Producing States, TE2023-24



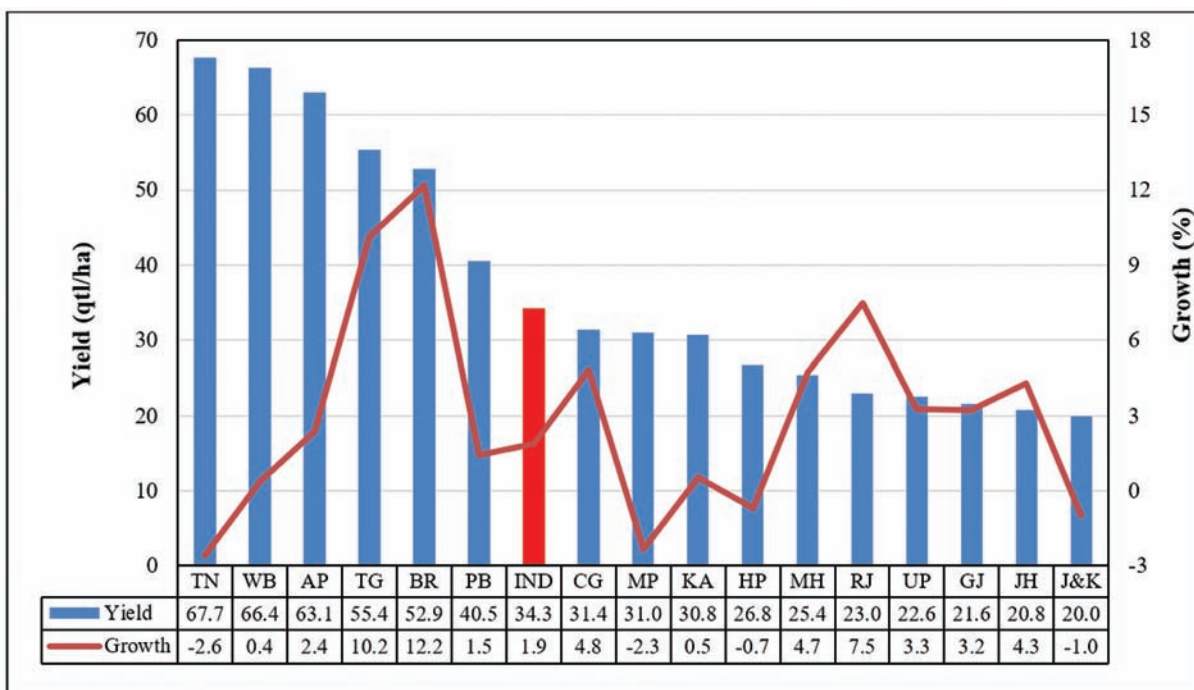
Source: Economics, Statistics & Evaluation Division, Ministry of Agriculture and Farmers Welfare

Maize

- 3.12 In case of maize, average yield varied from 67.7 quintals per hectare in Tamil Nadu to 20 quintals per hectare in Jammu & Kashmir with all-India average of 34.3 quintals per hectare. Majority of States (10) had lower yield than all-India average, while six States recorded higher yield. In yield, large inter-State variations have been observed. Bihar recorded the highest growth rate (12.2%), followed by Telangana (10.2%), while Tamil Nadu (-2.6%), Madhya Pradesh (-2.3%), Jammu & Kashmir (-1.0%) and Himachal Pradesh (-0.7%) showed negative growth rate. Karnataka, the largest producer, registered 0.5 percent growth and Madhya Pradesh, the second largest producer, witnessed a negative growth (-2.3%).



Chart 3.2: Average Yield and Growth in Yield of Maize in Major Producing States, TE2023-24

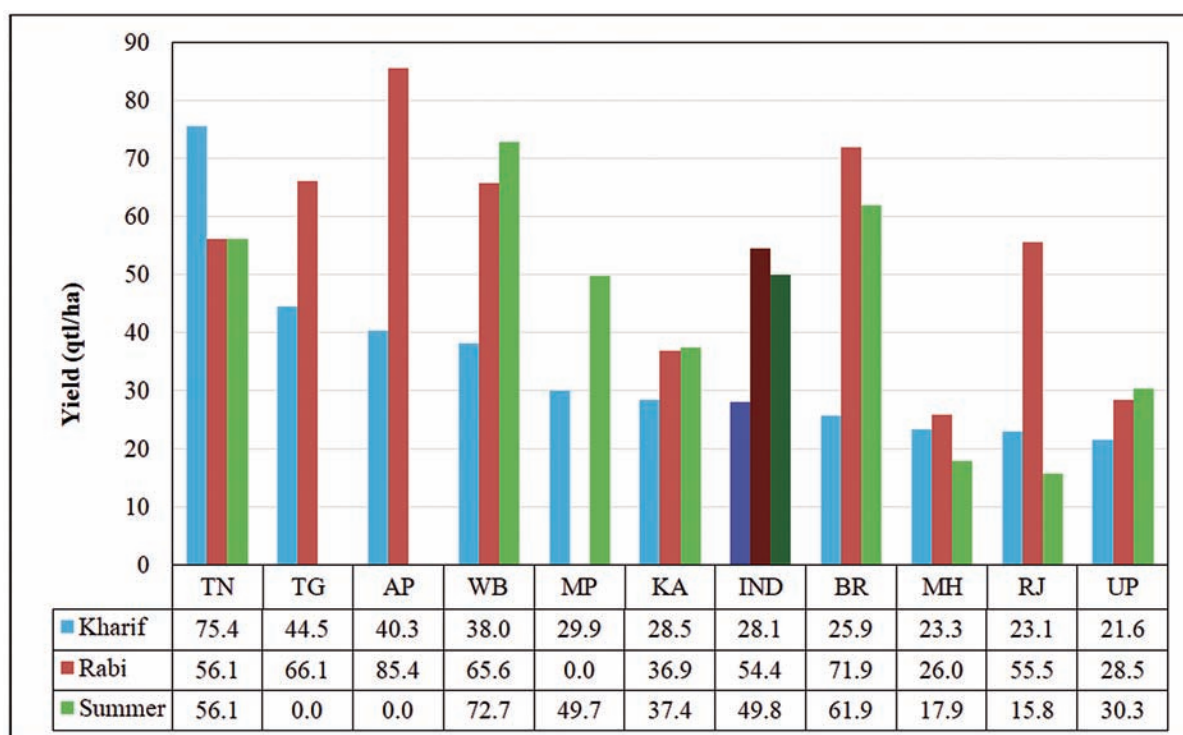


Source: Economics, Statistics & Evaluation Division, Ministry of Agriculture and Farmers Welfare

- 3.13 Maize is grown in all three seasons, *kharif*, *rabi* and *summer*. The share of *kharif* maize in total production has significantly declined from 82.9 percent in TE2005-06 to 62.7 percent in TE2023-24, while share of *rabi/summer* maize has increased over the years. *Rabi/summer* maize is an important crop in Karnataka, Bihar, Maharashtra, Tamil Nadu, Telangana, West Bengal and Andhra Pradesh. It is evident from Chart 3.3 that there are large variations in maize yield across States and seasons. At all-India level, maize productivity was highest (54.4 qtl/ha) in *rabi* season, followed by *summer* (49.8 qtl/ha) and the lowest (28.1qtl/ha) in *kharif* season. Tamil Nadu recorded the highest yield during *kharif* season (75.4 qtl/ha), while Andhra Pradesh had the highest yield during *rabi* (85.4 qtl/ha) and West Bengal during *summer* season (72.7 qtl/ha). Average yield in all major States except Tamil Nadu was significantly higher during *rabi* and *summer* season, compared to *kharif*. Since about two-third of maize production comes from *kharif* season, efforts should be made to improve productivity of *kharif* maize.



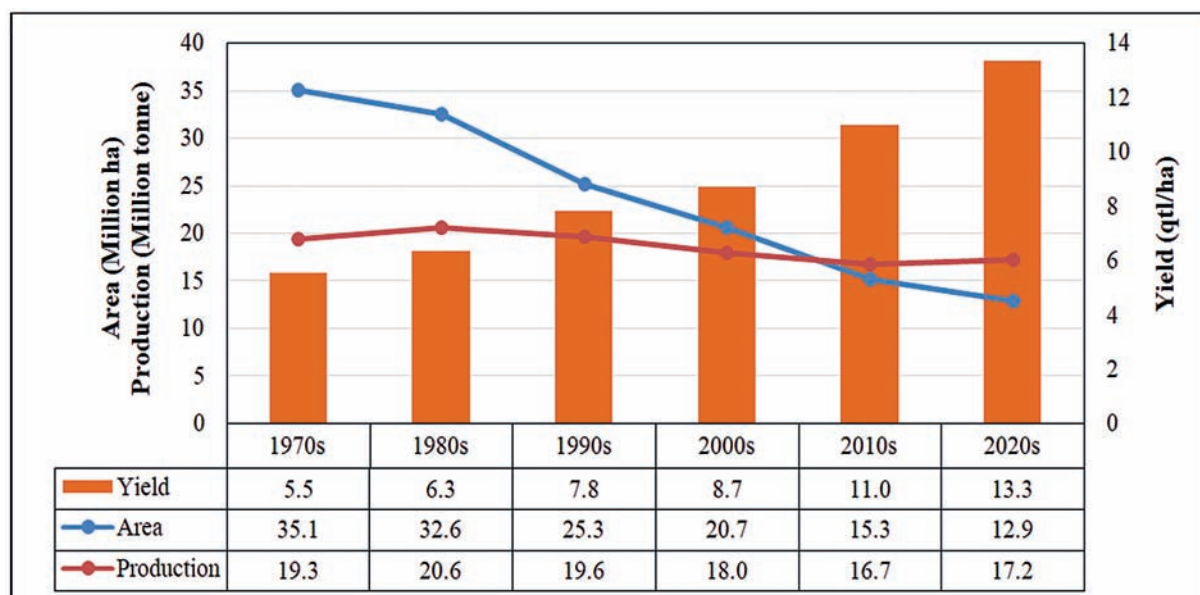
**Chart 3.3: Season-wise Yield of Maize in Major Producing States
(Average of 2022-23 and 2023-24)**



Source: Economics, Statistics & Evaluation Division, Ministry of Agriculture and Farmers Welfare

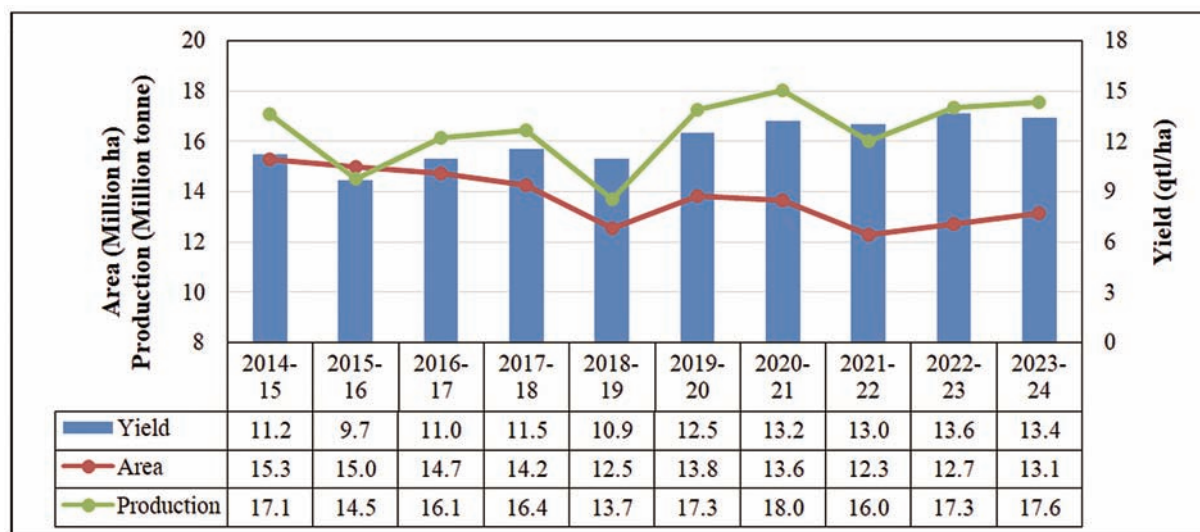
Nutri-Cereals/Shree Anna

- 3.14 *Shree Anna* cereals, drought tolerant crops, are increasingly recognised as sustainable solution for ensuring food and nutritional security, particularly in the face of climate change. Therefore, the Government has given priority for increasing production through area expansion and yield improvements in these crops. Recently, several initiatives have been taken in this regard, including organization of millet (*Shree Anna*) *Mahotsavs* in different parts of the country as part of International Year of Millets 2023, National Year of Millets in India in 2018, identifying specific districts with *Shree Anna* products as “One District One Product”, millet missions launched by various States, etc.
- 3.15 Decadal trends in area, production and yield of *Shree Anna* from 1970s to 2020s have been presented in Chart 3.4. It is evident from the Chart that yield of *Shree Anna* registered a sustained increase over the decades but area witnessed a steep decline. The production, which increased during the 1980s, recorded a continuous decline from 1990s to 2010s and marginally increased during the 2020s due to improvement in yield, which is encouraging. However, more concerted efforts are needed to motivate farmers to grow these crops by providing necessary technological and market support.

Chart 3.4: Decadal Trend in Area, Production and Yield of *Shree Anna*

Source: Economics, Statistics & Evaluation Division, Ministry of Agriculture and Farmers Welfare

- 3.16 Chart 3.5 shows trends in area, production and yield of *Shree Anna* during last 10 years. It is evident from the Chart that fluctuating trends have been observed in area and yield of *Shree Anna*, which resulted in instability in production during the period under consideration. Therefore, there is a need to improve yield and reduce fluctuations in yield to encourage farmers for millet cultivation. Research institutions should intensify their efforts in this regard and farmers should be made aware about improved technologies and best farm practices by organising field level demonstrations and out-reach activities.

Chart 3.5: Area, Production and Yield Trend in *Shree Anna* during Last 10 Years

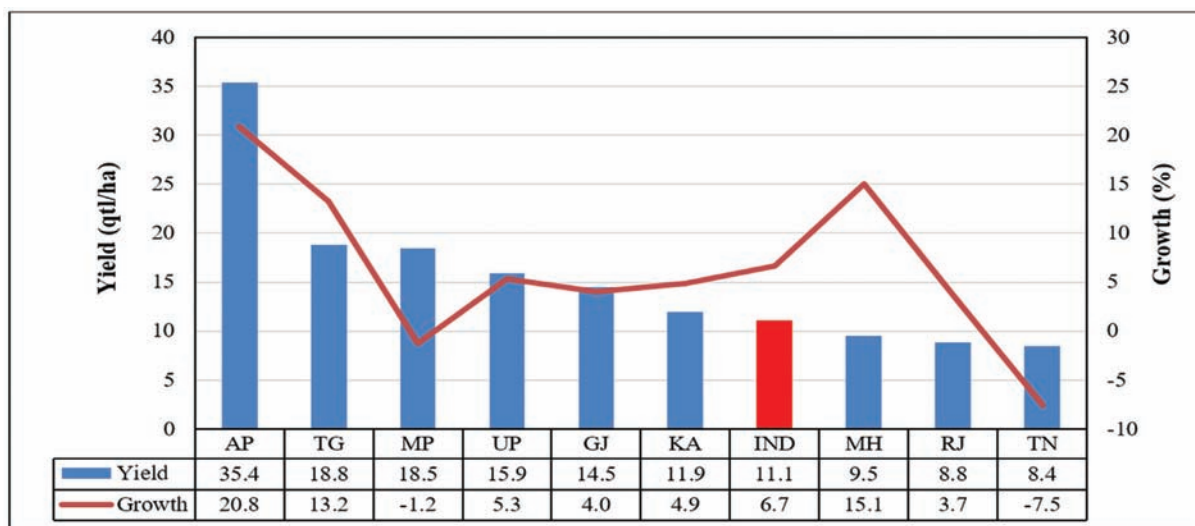
Source: Economics, Statistics & Evaluation Division, Ministry of Agriculture and Farmers Welfare



Jowar

- 3.17 As far as jowar is concerned, the highest average yield was recorded in Andhra Pradesh, followed by Telangana and Madhya Pradesh, while the lowest yield was in Tamil Nadu (Chart 3.6). In Maharashtra, the largest producing State, average yield was 9.5 quintals per hectare, which was much lower than the average yield of 35.4 quintals per hectare in Andhra Pradesh and even all-India average. It is encouraging to note that growth in yield was found to be highest in Andhra Pradesh (20.8%) followed by Maharashtra (15.1%), while Madhya Pradesh (-1.2%) and Tamil Nadu (-7.5%) recorded negative growth.

Chart 3.6: Average Yield and Growth in Yield of Jowar in Major Producing States, TE2023-24



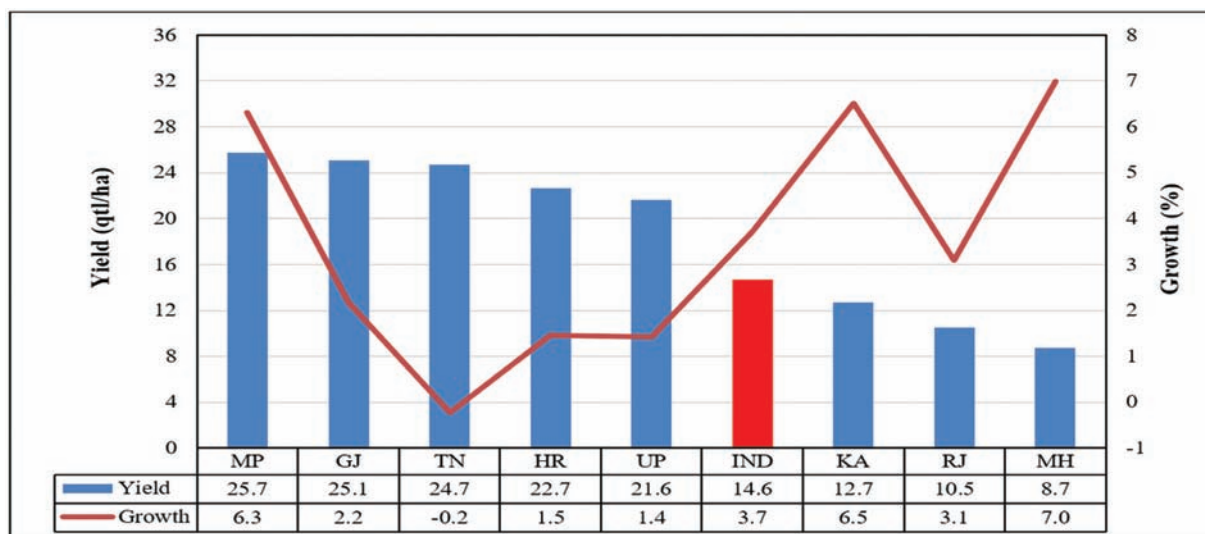
Source: Economics, Statistics & Evaluation Division, Ministry of Agriculture and Farmers Welfare

Bajra

- 3.18 For bajra, the average yield in Madhya Pradesh, Gujarat, Tamil Nadu, Haryana and Uttar Pradesh was found to be higher than all-India average yield, while productivity in Karnataka, Rajasthan and Maharashtra was lower than all-India average (Chart 3.7). Among the major States, Madhya Pradesh had the highest average yield followed by Gujarat during TE2023-24. The average yield in Rajasthan, the largest bajra producing State, was 10.5 quintals per hectare, much lower than all-India average. Growth in yield was highest in Maharashtra, followed by Karnataka and Madhya Pradesh, while Tamil Nadu showed negative growth rate.



Chart 3.7: Average Yield and Growth in Yield of Bajra in Major Producing States, TE2023-24

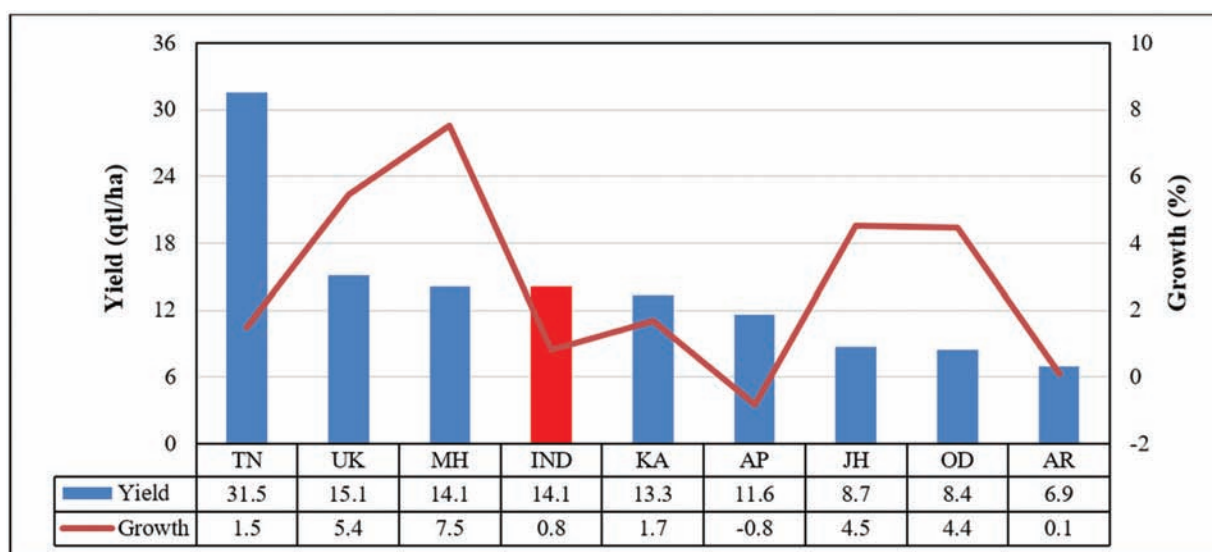


Source: Economics, Statistics & Evaluation Division, Ministry of Agriculture and Farmers Welfare

Ragi

- 3.19 Karnataka, the largest producer with 65.3 percent share, recorded average yield of 13.3 quintals per hectare during TE2023-24 and a moderate growth of 1.7 percent during the last five years. Tamil Nadu, the second largest producer, had the highest average yield (31.5 qtl/ha), while Maharashtra registered the highest growth (7.5%) in yield among major producers. The all-India average yield was recorded at 14.1 quintals per hectare, which was less than half of Tamil Nadu's average yield but marginally higher than Karnataka during TE2023-24 (Chart 3.8).

Chart 3.8: Average Yield and Growth in Yield of Ragi in Major Producing States, TE2023-24



Source: Economics, Statistics & Evaluation Division, Ministry of Agriculture and Farmers Welfare

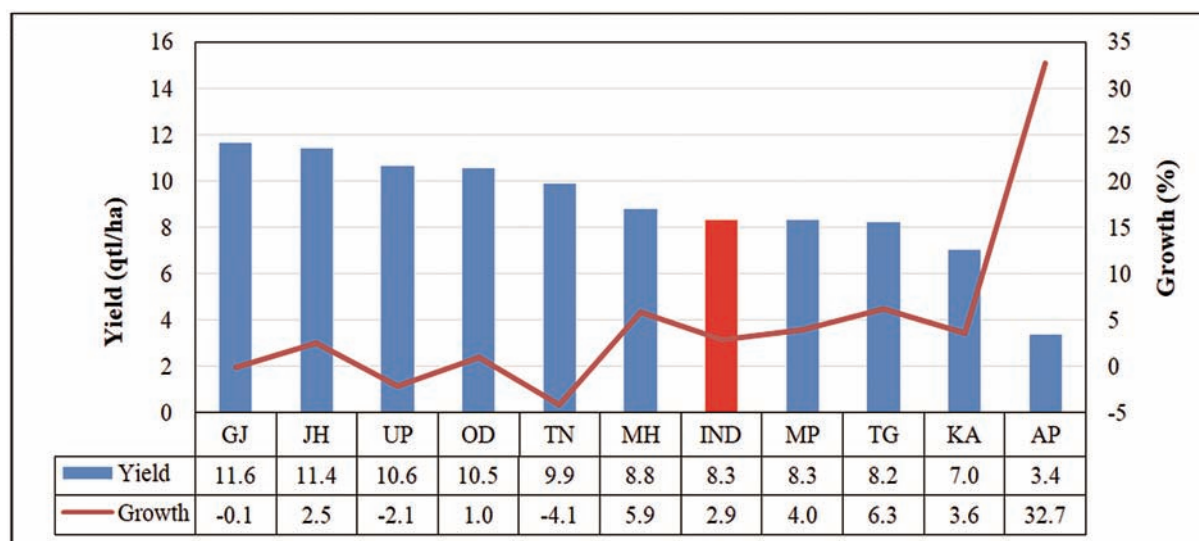


Pulses

Tur

- 3.20 Chart 3.9 exhibits average yield of tur during TE2023-24 and growth in yield between 2019-20 and 2023-24 in major States. Among the major tur producing States, Gujarat had the highest yield and Andhra Pradesh had the lowest. Maharashtra, the largest producer, had marginally higher than all-India average yield, while the second largest producer, Karnataka, recorded lower than all-India average. The highest growth in yield was found to be in Andhra Pradesh, State with the lowest yield, followed by Telangana. Maharashtra and Karnataka, top two producers, also registered positive growth of 5.9 percent and 3.6 percent, respectively. On the other hand, Uttar Pradesh and Gujarat, which also have significant share in production, witnessed negative growth during last five years.

Chart 3.9: Average Yield and Growth in Yield of Tur in Major Producing States, TE2023-24



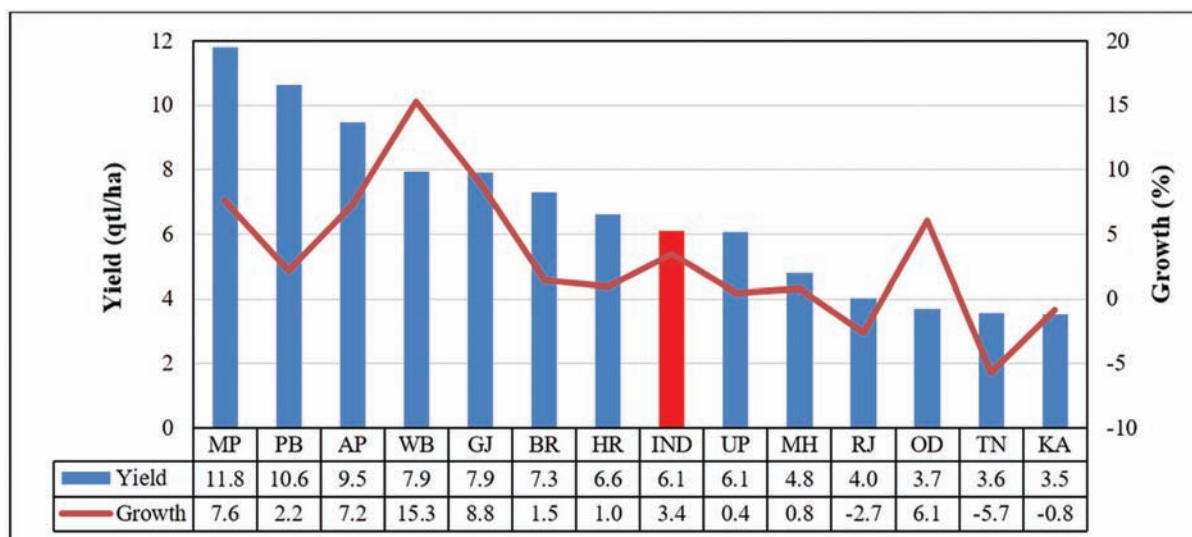
Source: Economics, Statistics & Evaluation Division, Ministry of Agriculture and Farmers Welfare

Moong

- 3.21 Chart 3.10 shows average yield of moong during TE2023-24 and growth in yield in major States. It is encouraging to note that Madhya Pradesh, which has the largest production share, recorded a significant growth in yield as well as the highest average yield. On the other hand, average yield of Rajasthan, the second largest producer, was almost one-third of the average yield of Madhya Pradesh and two-third of all-India average and growth in yield was negative in Rajasthan. Among major States, the highest growth in yield was witnessed in West Bengal (15.3%), while Tamil Nadu, Rajasthan and Karnataka had negative growth in yield.



Chart 3.10: Average Yield and Growth in Yield of Moong in Major Producing States, TE2023-24

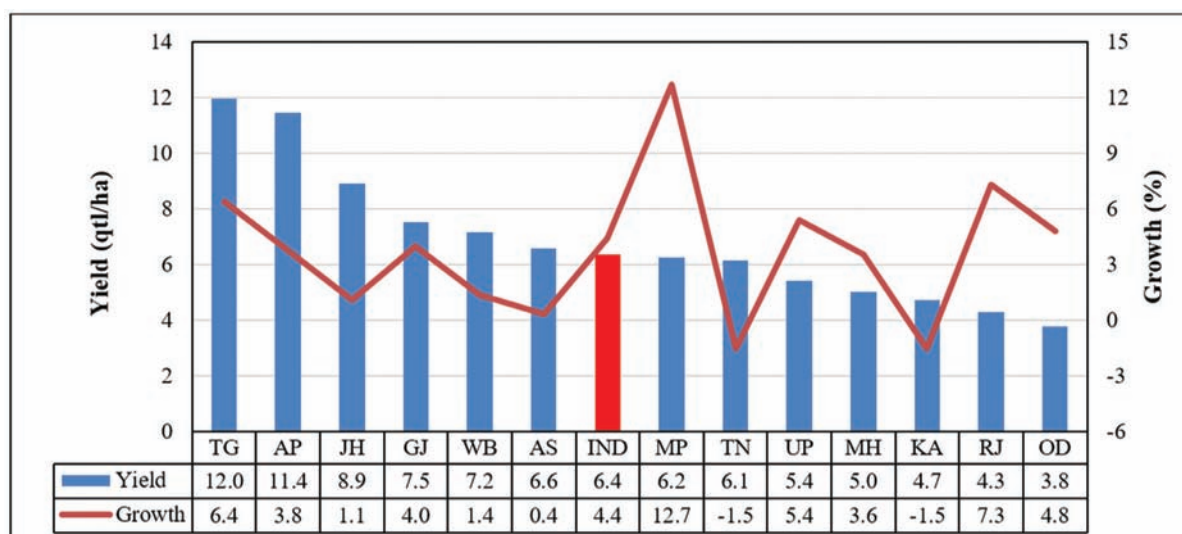


Source: Economics, Statistics & Evaluation Division, Ministry of Agriculture and Farmers Welfare

Urad

- 3.22 Madhya Pradesh, the largest producer of urad registered the highest growth in yield between 2019-20 and 2023-24, however average yield was nearly half of Telangana and marginally lower than the national average during TE2023-24 (Chart 3.11). On the other hand, Andhra Pradesh, the second largest producer, recorded much higher average yield (11.4 qtl/ha) with a significant high growth of 3.8 percent. Telangana registered the highest average yield as well as an impressive growth in yield. Rajasthan, Uttar Pradesh and Odisha also registered significant growth in yield during the reference period. Out of 13 States, seven States had lower yield than all-India average.

Chart 3.11: Average Yield and Growth in Yield of Urad in Major Producing States, TE2023-24



Source: Economics, Statistics & Evaluation Division, Ministry of Agriculture and Farmers Welfare

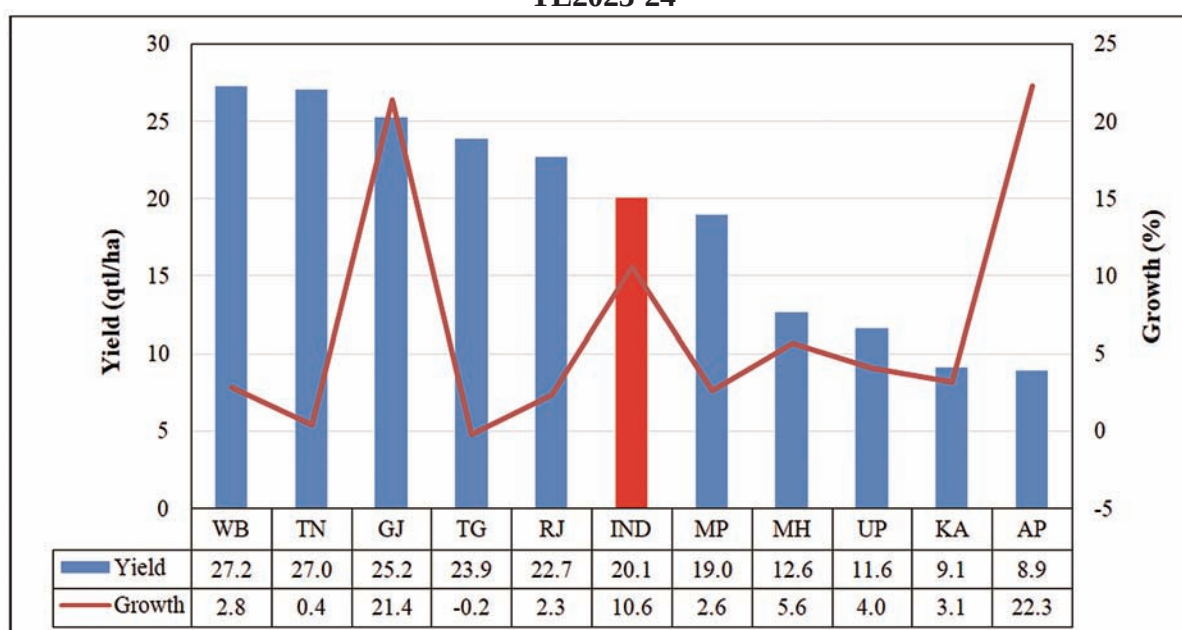


Oilseeds

Groundnut

- 3.23 Chart 3.12 exhibits average yield of groundnut during TE2023-24 and growth in yield between 2019-20 and 2023-24 in major States. Gujarat, the top producer, had recorded yield higher (25.2 qtl/ha) than the national average with a spectacular growth of 21.4 percent. Rajasthan, the second largest producer, also recorded the average yield higher than national average but with a slower growth of 2.3 percent. Top three producers, Gujarat, Rajasthan and Tamil Nadu, had higher productivity compared with all-India average yield.

Chart 3.12: Average Yield and Growth in Yield of Groundnut in Major Producing States, TE2023-24



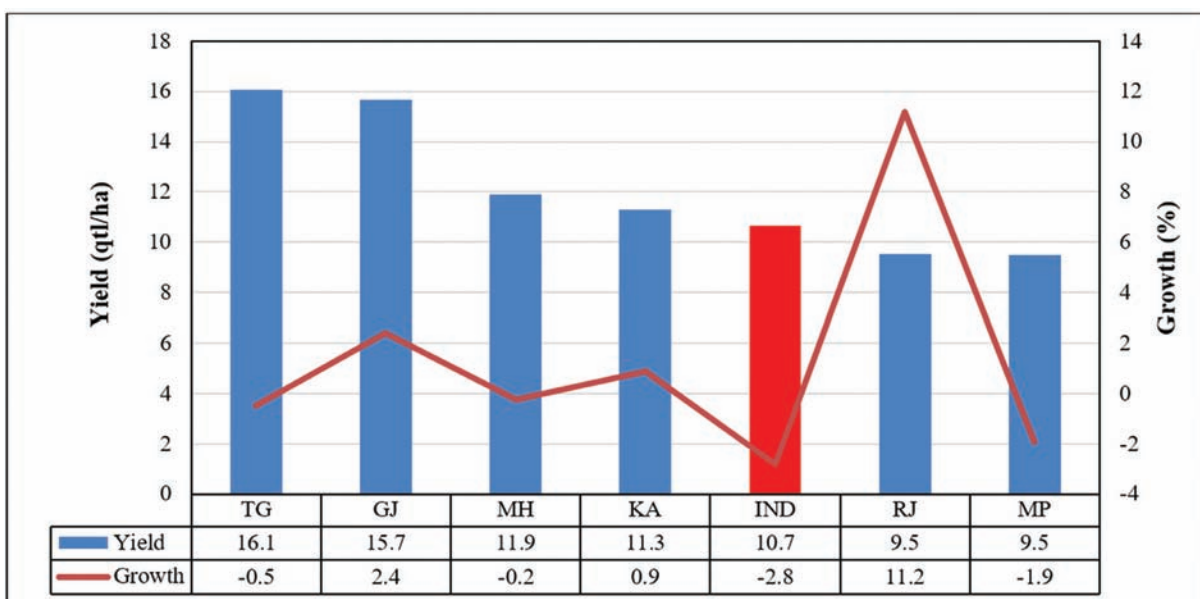
Source: Economics, Statistics & Evaluation Division, Ministry of Agriculture and Farmers Welfare

Soybean

- 3.24 Soybean yield in India has remained low compared with other major producers. The highest average yield was recorded in Telangana, followed by Gujarat, while the lowest yield was recorded in Madhya Pradesh, the second largest producer of soybean. Rajasthan, another major producer of soybean, though, had yield lower than the national average, registered highest growth in yield. Top two producers, Maharashtra and Madhya Pradesh, experienced negative growth during the last 5 years. On account of negative growth in major producing States, growth in soybean yield at the all-India level was recorded negative.



Chart 3.13: Average Yield and Growth in Yield of Soybean in Major Producing States, TE2023-24



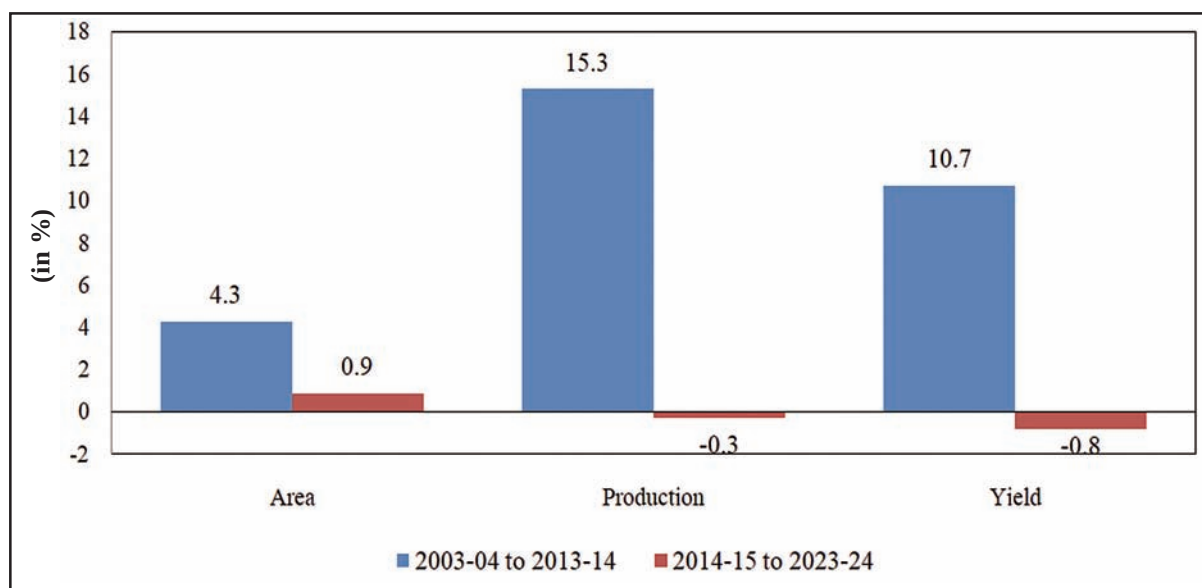
Source: Economics, Statistics & Evaluation Division, Ministry of Agriculture and Farmers Welfare

Cotton

- 3.25 Cotton production in India more than quadrupled due to expansion in area and improvement in yield after introduction of Bt (*Bacillus thuringiensis*) cotton in 2002, from 8.6 million bales in 2002-03 to 35.9 million bales in 2013-14. Cotton productivity increased from 189 kilogram per hectare in TE2002-03 to 496 kilogram per hectare in TE2013-14, while area increased from 8.4 million hectare to over 12 million hectare during the period. However, both production and yield have remained stagnant for the past decade and average yield in India (4.4 qtl/ha) is significantly lower than the world average (7.7 qtl/ha).
- 3.26 Cotton production in the country increased at an annual growth rate of 15.3 percent during 2003-04 to 2013-14, which declined to (-)0.3 percent during the last 10 years, while yield growth rate declined from 10.7 percent to (-)0.8 percent during this period (Chart 3.14). The problem of yield stagnation/decline has an impact on production as well as on farmers' income, signaling the need for new technological interventions and improved practices.



Chart 3.14: Trends in Growth Rate of Cotton Area, Production and Yield in India



Source: Economics, Statistics & Evaluation Division, Ministry of Agriculture and Farmers Welfare

- 3.27 Chart 3.15 compares the average yield and growth in yield of cotton in major producing States. During TE2023-24, Gujarat, top producer, registered the highest average yield and growth in yield was also quite high during last five years. Maharashtra, the second largest producer, had the lowest yield (3.3 qtl/ha). The growth in yield was highest in Karnataka, followed by Telangana, Gujarat and Maharashtra, while it was negative in Rajasthan, Madhya Pradesh, Punjab and Andhra Pradesh. Since there has been a significant deceleration in productivity growth in the recent period, there is a strong need for comprehensive strategy for improving yield in major producing States.

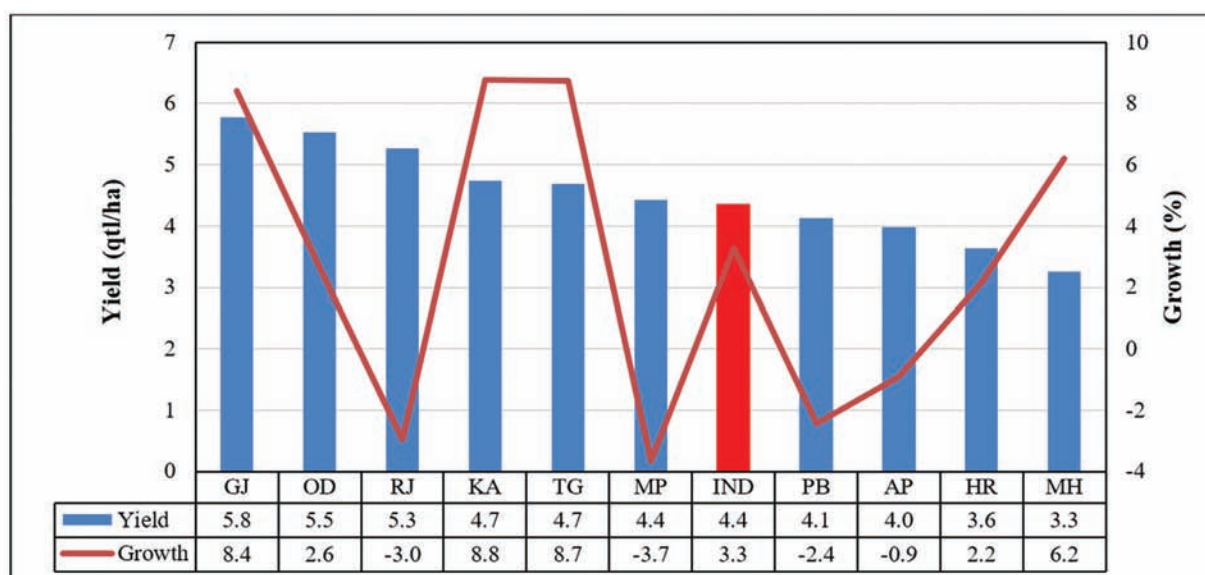
Box 3.1: Special Project on Cotton for Improving Cotton Yield

A special project on cotton under the National Food Security Mission (NFSM), “Targeting technologies for agro-ecological zones—large-scale demonstration of best practices for enhancing cotton productivity” is being implemented in 65 districts in 8 States. Under this project, three technologies i.e. High Density Planting System (HDPS), Closer Spacing (CS) and Extra Long Staple are being implemented in selected districts of the selected States. HDPS in shallow soil and CS in medium soil were implemented in PPP mode during 2023-24 kharif season, which had positive impact on cotton yield. The average yield increase in the HDPS demonstration plot was 30.4 percent and in the CS demonstration plot was 39.1 percent during 2023-24.

Source: ICAR-Central Institute for Cotton Research and Directorate of Cotton Development



Chart 3.15: Average Yield and Growth in Yield of Cotton in Major Producing States, TE2023-24



Source: Economics, Statistics & Evaluation Division, Ministry of Agriculture and Farmers Welfare

Yield Gap Analysis

- 3.28 The magnitude of yield gaps provides some indication of potential for increasing crop yield. Therefore, yield gaps for selected *kharif* crops at all-India level have been analyzed in the following section. The average yield under Front Line Demonstrations (FLD) and realized yield under farmers' practices provided by various Indian Council of Agricultural Research (ICAR) institutes and actual yield from Economics, Statistics & Evaluation Division have been used for this analysis. Two types of yield gaps considered for the purpose are as follows:
- Yield Gap (A): Difference between potential yield (yield achieved under FLD, where best scientific and management practices are followed) and realized yield under farmers' practices.
 - Yield Gap (B): Gap between State average yield and potential yield achieved under FLD.
- 3.29 Yield Gap (A) is due to various socio-economic constraints like non-availability of inputs and services, risk aversion behaviour of farmers, fragmented land holdings, lack of farm mechanization, weak institutions, lack of knowledge, etc. while, yield gap (B) is caused by both lack of technological and scientific management practices and socio-economic constraints faced by farmers.
- 3.30 Yield Gap (A) and (B) for selected *kharif* crops at all-India level have been displayed in Table 3.3. It is evident from the Table that Yield Gap (A) was found to be relatively in a narrow range from 13 percent in moong to 28.6 percent in jowar (*kharif*), while yield Gap (B) was substantially higher than Yield Gap (A) and varied from 15.8 percent in paddy (*kharif*) to 59.8 percent in jowar (*kharif*). Yield Gap (B) was more than 30 percent for all crops, except paddy and cotton. The size of yield gaps vary across States and were high in States with poor resource endowments. The yield gaps in selected



crops in major States are given in Annex Tables 3.1. As land and other production resources are becoming scarce, increased production will need to rely on increasing yield with special focus on narrowing prevailing large yield gaps. These significant yield gaps clearly indicate considerable scope for improving crop yields. It is evident that exploitable yield gaps are relatively large in millets, pulses, oilseeds and cotton, therefore, closing the exploitable yield gaps would help in increasing production. Adopting appropriate crop management practices and ensuring availability of quality seed and other inputs can enable improve yields and thus close/reduce the gaps.

Table 3.3: Yield Gap in Major Kharif Crops

Crop	Yield Gap (in %)	
	A	B
Paddy (<i>kharif</i>)	13.9	15.8
Maize (<i>kharif</i>)	22.5	48.8
Jowar (<i>kharif</i>)	28.6	59.8
Bajra	16.9	39.7
Ragi	23.8	47.6
Tur	19.5	37.6
Urad	19.4	31.2
Moong	13.0	33.7
Sunflower	15.9	48.1
Soybean	19.2	41.2
Cotton	15.5	24.1

Source: Various ICAR-Institutions

- 3.31 Although various research institutes and Government have devised strategies to bridge these yield gaps, but need to be implemented effectively with active participation of farmers and other stakeholders. For instance, encouraging results have been obtained from a collaborative project undertaken by ICAR-IIMR, Ludhiana, International Maize and Wheat Improvement Centre (CIMMYT), State Agricultural Universities (SAUs) and State Agricultural Departments (Box 3.2). Similar efforts can be made for other crops for improving the crop yields and bridging the yield gap.



Box 3.2: Participatory Innovation Platform on Potential Yield Realization of Maize-Based Cropping Systems in Punjab and Haryana

ICAR-IIMR, Ludhiana in collaboration with International Maize and Wheat Improvement Centre (CIMMYT), State Agricultural Universities (SAUs) and State Agricultural Departments initiated a project namely, “Participatory innovation platform on potential yield realization of maize-based cropping systems in Punjab and Haryana”. Information on new agronomic production techniques and agricultural inputs, including seeds of superior maize varieties and insecticides and herbicides to fight fall armyworm were provided to the participatory farmers. The productivity at the learning sites reached up to 77qtl/ha, which shows that maize can be a potential crop for diversification.

Source: ICAR-Indian Institute of Maize Research, Ludhiana

District Level Productivity Trends

- 3.32 On the basis of average yield of major crops at all-India level during TE2015-16 and TE2022-23, districts have been classified into high and low yield categories. All major districts contributing 90 percent of total production of a crop have been considered for classification of districts under the following two categories:

High: Average yield $\geq$ All-India yield

Low: Average yield $<$ All-India yield

- 3.33 The number of districts and their area share, based on yield performance for selected *kharif* crops, rice, maize, tur, moong, soybean, groundnut and cotton, are displayed in Table 3.4. Average productivity of rice increased from 24 quintals per hectare in TE2015-16 to 27.8 quintals per hectare in TE2022-23, number of districts in high-productivity category increased from 143 to 192, and area share from 46.6 percent to 53.9 percent during the period. In case of maize, the all-India productivity has shown a notable increase from 26.2 quintals per hectare in TE2015-16 to 33.8 quintals per hectare in TE2022-23, but number of districts in ‘high’ category remains unchanged with small reduction in area share.
- 3.34 For tur (pigeon pea), productivity at all-India level increased from 7.3 quintals per hectare to 8.6 quintals per hectare, an increase of 17.8 percent in the current period as compared to previous period. Although number of high-yielding districts remained at 71 during both periods, but area under high productivity category increased from 36.9 percent to 44.3 percent and area under low-yield group decreased from 53.9 percent to 44.2 percent. Moong saw a substantial increase of 32.6 percent in yield during the current period as compared to previous period. However, number of districts under high-yield category decreased from 75 to 53 along with reduction in area share between the given periods. On the other hand, districts under low-yield category declined from 61 to 51, while area share significantly increased from 44.7 percent in TE2015-16 to 65.5 percent in TE2022-23.



Table 3.4: Status of Productivity Level of Selected *Kharif* Crops in Major Producing Districts

Crop \ Period		TE2015-16		TE2022-23	
		High	Low	High	Low
Rice	Yield	24.0 qtl/ha		27.8 qtl/ha	
	No. of District [Area Share]	143 [46.6]	141 [40.3]	192 [53.9]	134 [33.8]
Maize	Yield	26.2 qtl/ha		33.8 qtl/ha	
	No. of District [Area Share]	93 [45.2]	74 [39.1]	93 [42.9]	70 [42.2]
Tur	Yield	7.3 qtl/ha		8.6 qtl/ha	
	No. of District [Area Share]	71 [36.9]	60 [53.9]	71 [44.3]	39 [44.2]
Moong	Yield	4.6 qtl/ha		6.1 qtl/ha	
	No. of District [Area Share]	75 [44.9]	61 [44.7]	53 [23.5]	51 [65.5]
Soybean	Yield	9.0 qtl/ha		10.6 qtl/ha	
	No. of District [Area Share]	25 [29]	34 [61.8]	27 [40]	26 [50.2]
Groundnut	Yield	15.9 qtl/ha		18.5 qtl/ha	
	No. of District [Area Share]	48 [46.9]	31 [40.9]	53 [52.8]	35 [36.1]
Cotton	Yield	4.6 qtl/ha		4.4 qtl/ha	
	No. of District [Area Share]	30 [31.2]	41 [58.2]	47 [35]	38 [53.9]

Note: 1. Major producing districts accounting for 90 percent of the total production of the respective crops have been considered

2. Number in square bracket [] shows the area share of crop during the corresponding season

Source: Economics, Statistics & Evaluation Division, Ministry of Agriculture and Farmers Welfare

- 3.35 Soybean also registered significant growth (17.8%) in productivity as well as increase in number of high-yield districts during the current period as compared to earlier period. Area share of high-yielding region also increased significantly from 29 percent to 40 percent. Productivity of groundnut improved from 15.9 quintals per hectare in TE2015-16 to 18.5 quintals per hectare in TE2022-23, with increase



in number of high-productivity districts from 48 to 53 and area share from 46.9 percent to 52.8 percent.

- 3.36 Cotton experienced a slight decline in productivity, from 4.6 quintals per hectare to 4.4 quintals per hectare between TE2015-16 and TE2022-23, however, number of districts under high-yield category increased from 30 to 47 along with small increase in area share (31.2% to 35%).

International Benchmarking of India's Crop Yield

- 3.37 Table 3.5 presents a comparison of the yield of major *kharif* crops in major producing countries to assess India's position at the international level during TE2023. There is a significant gap between all-India and world average yield for all selected crops, except groundnut. In case of groundnut, all-India average yield was higher than the world average but was significantly lower than the world highest yield in the United State of America.

Table 3.5: International Productivity Comparison for Major Crops (TE2023)

(yield in qtl/ha)

Crop	World Average	World Highest except India	All-India Average	State Highest
Paddy	47.3	USA (85.1) [1.1]	42.6 [25.4]	PB (65.2)
Maize	58.5	USA (110.4) [31]	34.3 [2.9]	TN (67.7)
Jowar/Sorghum	14.4	Egypt (51.4) [1.5]	11.1 [7.3]	AP (35.4)
Tur/Pigeon pea	8.9	Malawi (16.6) [8.4]	8.3 [77.6]	GJ (11.6)
Ground nut	17.3	USA (44.4) [4.9]	20.1 [18.7]	WB (27.2)
Soybean	27.3	USA (34) [32]	10.7 [3.7]	TG (16.1)
Sunflower	19.1	China (29.6) [4.1]	10.2 [0.5]	HR (16.9)

Notes: 1. Figures in parentheses show average yield

2. Figures in Square Brackets [] show share of the country in total world production of the crop.

Sources: 1. FAOSTAT for World Average and World Highest

2. Economics, Statistics & Evaluation Division, Ministry of Agriculture and Farmers Welfare



- 3.38 The all-India average yield of paddy was close to the world average but was half of the world highest yield in the United States of America (USA). Paddy yield in Punjab was higher than the world average but lower than the yield in USA. In the case of maize, all-India average yield was significantly lower than the world average yield and the highest yield in the world. Even in Tamil Nadu, the country's highest-yield State, average yield was 61.3 percent of the world highest yield recorded in the USA. All-India average yield of sorghum was substantially lower than the world highest and average yields. Andhra Pradesh, which has the highest yield in the country, had average yield much higher than the world average but little over two-third of the world highest yield (Egypt). In case of soybean, all-India average yield was less than 40 percent of the world average yield and less than one-third of the world highest yield. Almost a similar trend was observed in sunflower.
- 3.39 The above analysis clearly indicates that there is significant gap between all-India average yield and the world average and highest yield. Therefore, there is a strong need to raise crop yield levels to match world averages and the best yields by adopting modern technologies, including high yielding crop varieties, scientific management practices and ensuring timely availability of quality inputs and services. In addition, agricultural practices and technologies from countries with highest yield, which are suitable for India's climatic conditions, can also be adopted to improve yield levels in the country.

Drivers of Productivity

- 3.40 This section discusses important factors such as high yielding variety seed, balanced use of fertilizers, timely access to affordable and adequate credit, incorporation of modern technologies, farm mechanization, irrigation facilities, etc., which are crucial for achieving an efficient and sustainable productivity growth. Major initiatives undertaken by the Government in this regard have also been highlighted in this section.

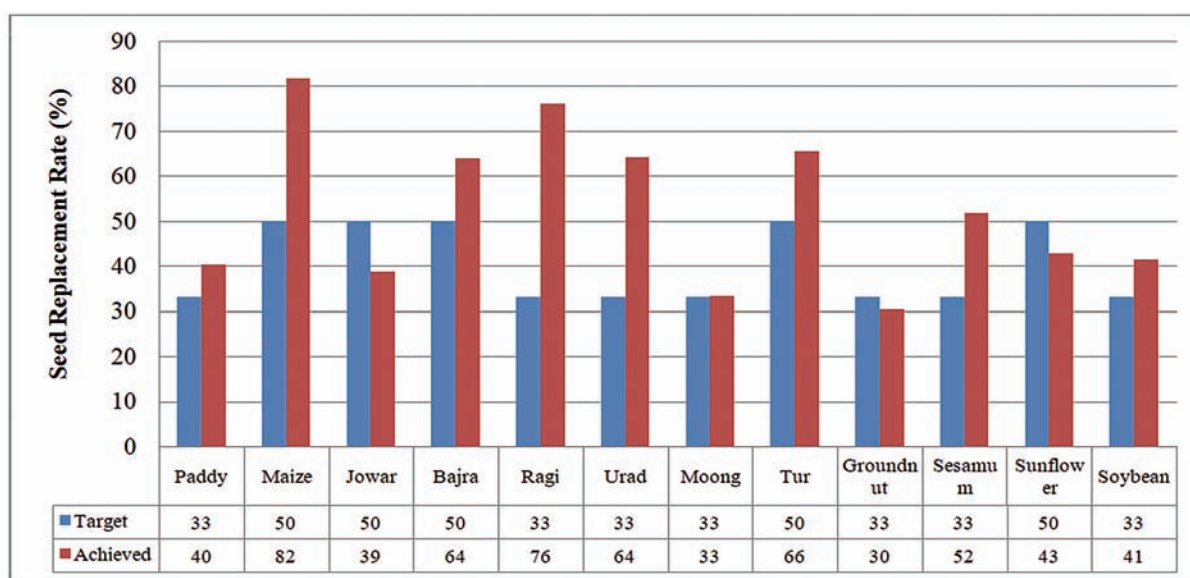
Improved Seed and Variety Replacement

- 3.41 One of the main causes of low productivity of crops is non-availability of quality seed. In many crops, farmers are still using popular old varieties and farm saved seeds either due to high cost of certified seeds or non-availability. Although significant progress has been made in improving quality seed availability and seed/variety replacement rate through various initiatives taken by the Government but more needs to be done as progress is uneven across States and in some crops far from sufficient. The issue was also highlighted by farmers during the meeting of the Commission with farmers and farmers' associations.
- 3.42 The Seed Replacement Rate (SRR) in most of mandated crops was higher than the target during 2023-24, except jowar, groundnut and sunflower (Chart 3.16). As per Annual Report (2023-24) of All India Coordinated Research Project (AICRP), there has been significant enhancement in VRR (Chart 3.17). In rice, share of varieties notified during last 5 years in total breeder seed indent was 24.6 percent and 10 year old varieties was 57.5 percent in 2024-25 as compared to 12.8 percent and 43.2 percent respectively in 2018-19. Among pulses, tur had the share of 70 percent for 5 year old varieties and 91.3 percent for 10 year old varieties in 2024-25 compared to 3.5 percent and 36.5



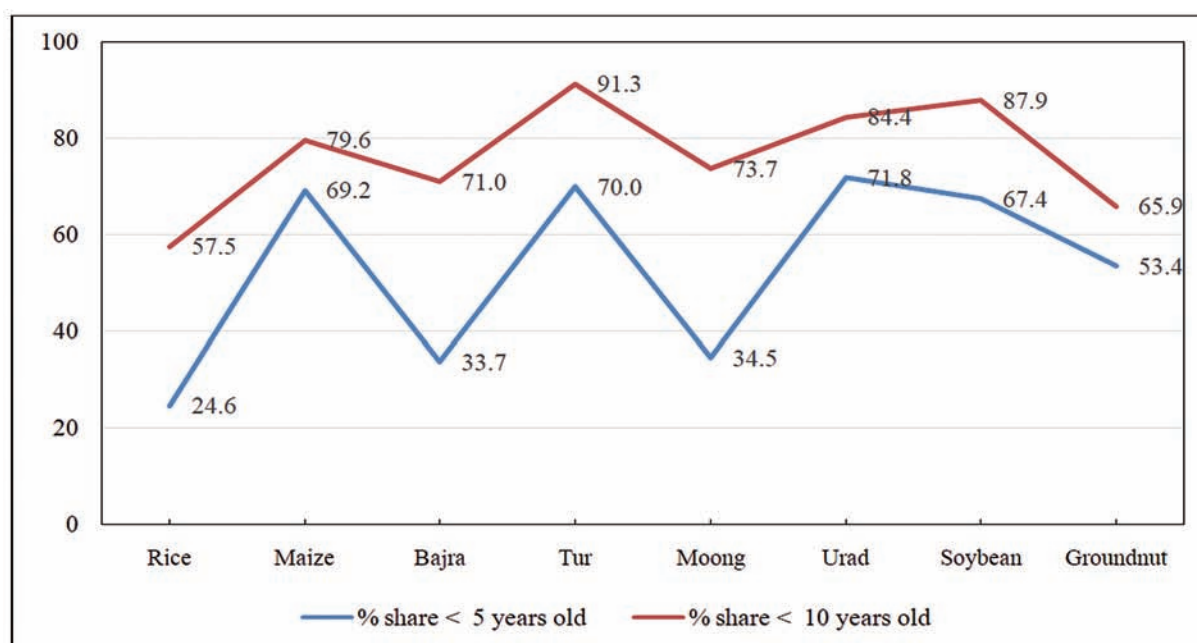
percent in 2018-19, respectively. In case of moong, the share of less than 5 years old varieties was 34.5 percent and share of less than 10 years old varieties was 73.7 percent in 2024-25. The VRR in most crops is steadily increasing through introduction of new crop varieties into seed chain but more efforts are needed to bring newer varieties into seed chain and motivate farmers to use newer varieties and quality seed through more demonstrations and sharing success stories of other farmers. State Government should also make concerted efforts in this regard.

Chart 3.16: Target and Achievement of Seed Replacement Rate in Major *Kharif* Crops, 2023-24



Source: Seed Division, Ministry of Agriculture and Farmers Welfare

Chart 3.17: Varietal Replacement Rate (VRR) in Selected *Kharif* Crops (2024-25)



Source: Annual Report Glimpses 2023-24, ICAR- Indian Institute of Seed Science, Mau



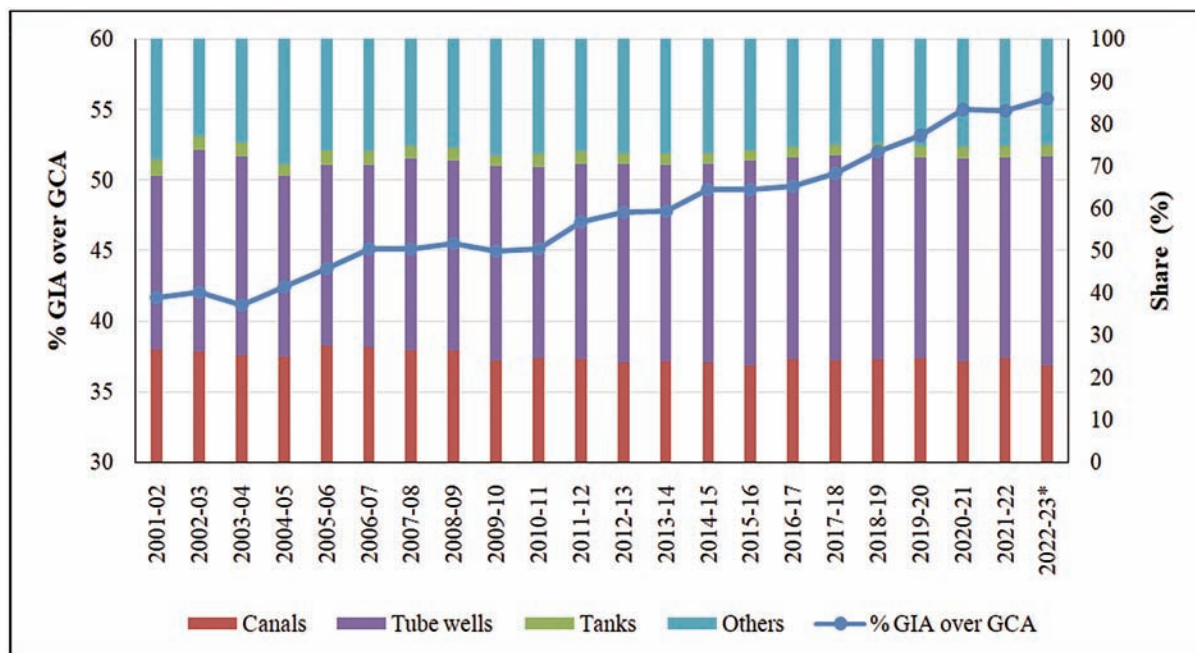
- 3.43 Recently, 109 climate-resilient and nutritionally rich varieties (69 of field crops and 40 of horticulture crops) suitable to different agro-climatic conditions were released by Hon'ble Prime Minister. These varieties will have significant impact on productivity, food and nutritional security but need to be brought in seed chain. The National Mission on High Yielding Varieties/Hybrids seed with an allocation of ₹100 crore announced in the recent budget, will help in strengthening research and development efforts for high yielding, pest & disease resistant and climate resilient varieties and availability of newly released varieties.

Irrigation and Efficiency in Water Management

- 3.44 Irrigation plays an important role in increasing agricultural output through direct production effects and by increasing the productivity of complementary inputs as well as increases resilience to climate change. The Government of India has also accorded high priority to irrigation development and water conservation and management. For this purpose, in order to extend coverage of irrigation and improve water use efficiency “*Pradhan Mantri Krishi Sinchayee Yojana*” (PMKSY) was implemented in 2015-16 with four major components (i) Accelerated Irrigation Benefit Program (AIBP); (ii) *Har Khet Ko Pani* (HKKP); (iii) Watershed Development and (iv) Per Drop More Crop (PDMC).
- 3.45 Chart 3.18 presents the trends in irrigation coverage in the country from 2001-02 to 2022-2023(P). It is evident from the Chart that irrigation coverage has significantly expanded over the years. Irrigation intensity (gross irrigated area as share of gross cropped area) has increased from 41.7 percent in 2001-02 to 55.8 percent in 2022-23(P) but has increased faster in the recent years. However, there are large variations in irrigation coverage across States and crops. The coverage of irrigation varied from less than 15 percent in millets and about 26 percent in total pulses to about 67 percent in rice (Chart 3.19). Similarly, percentage share of gross irrigated area to gross cropped area ranged from less than 15 percent in Assam to nearly 100 percent in Haryana and Punjab among major producing States (Chart 3.20). There is a need to improve irrigation coverage in those States and crops where irrigation facilities are less. It is interesting to note that share of tube-wells in gross irrigated area has increased significantly from 40.8 percent in 2001-02 to 49.3 percent in 2022-23, while share of canal and other sources such as tanks and other wells has declined during the period. High dependence on tube-well irrigation has led to depletion of groundwater resources in many States.



Chart 3.18: Trends in Irrigation Coverage from 2001-02 to 2022-23

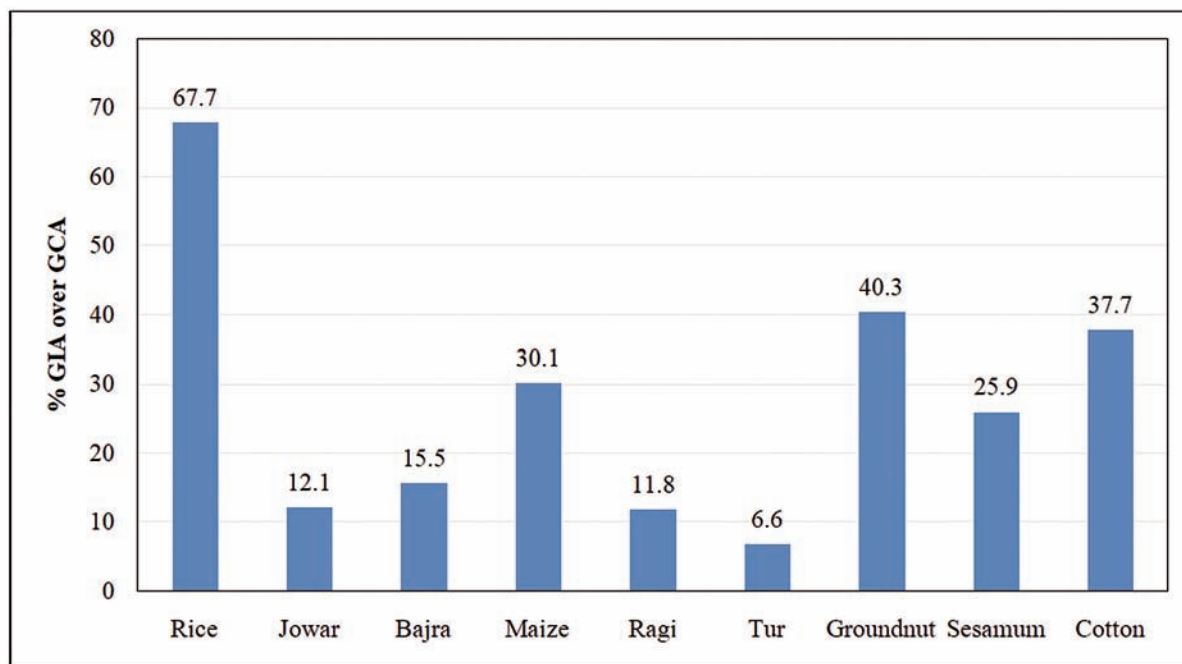


Notes: 1. * Provisional

2. GIA= Gross Irrigated Area, GCA= Gross Cropped Area

Source: Economics, Statistics & Evaluation Division, Ministry of Agriculture and Farmers Welfare

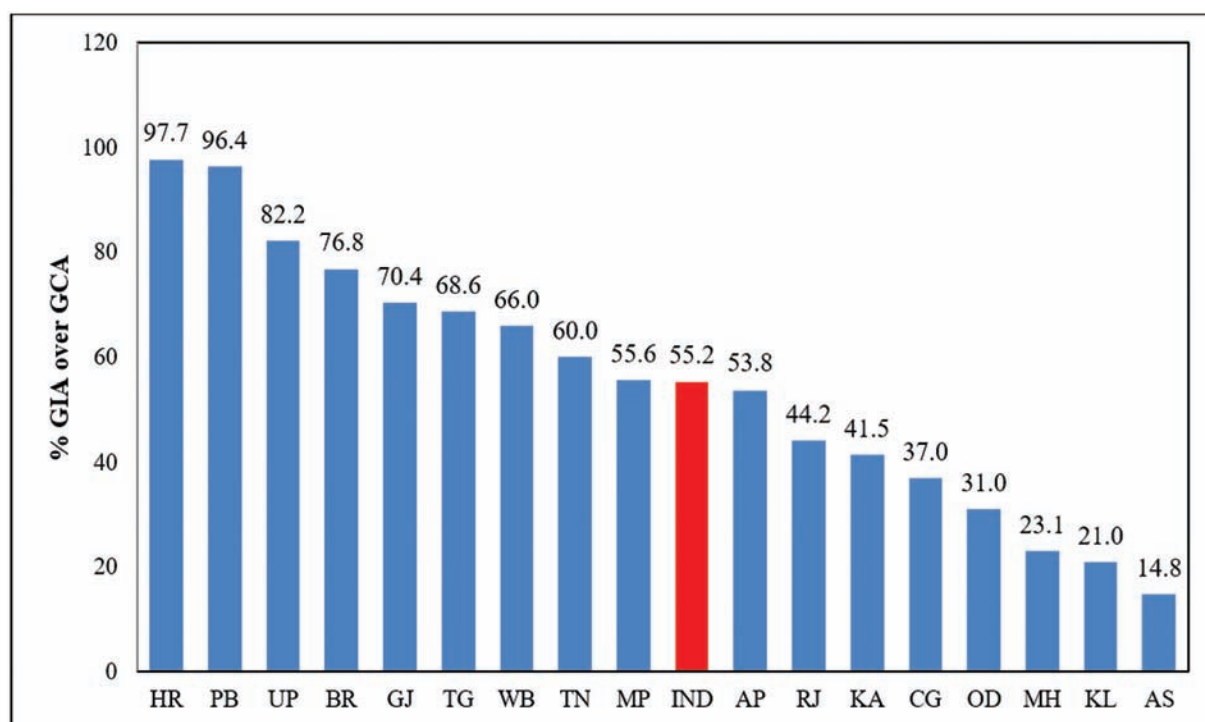
Chart 3.19: Irrigation Coverage in Major Kharif Crops (TE2022-23)



Source: Economics, Statistics & Evaluation Division, Ministry of Agriculture and Farmers Welfare



Chart 3.20: Irrigation Coverage in Major States (TE2022-23)



Note: States having Gross Cropped Area share more than 1% in 2022-23 have been considered for this analysis
Source: Economics, Statistics & Evaluation Division, Ministry of Agriculture and Farmers Welfare

- 3.46 In order to improve water use efficiency, Government is promoting micro-irrigation under PDMC and financial assistance is given to farmers. Additional incentives/subsidies are being given by some States for encouraging more farmers to adopt micro-irrigation. Despite impressive increase in micro-irrigation coverage at all-India level during last 10 years, progress remains low in many States. Around 80 percent coverage of micro-irrigation is in a few States like Karnataka, Gujarat, Tamil Nadu, Maharashtra, Andhra Pradesh and Rajasthan. State-wise details of area covered under micro-irrigation under 'Per Drop More Crop (PDMC)' Scheme are given in Annex Table 3.2. High cost of micro-irrigation systems, poor after sales services and small size of land holdings are some of the constraints in the progress of this Scheme. It has also been observed that farmers lack technical knowledge for installation, operation and maintenance of micro-irrigation systems. Hence, farmers should be properly trained in use and maintenance of these systems for obtaining better results. In addition, special emphasis should be given on repair, renovation and restoration of traditional water bodies including tanks.

Soil Health and Nutrient Management

- 3.47 Growing plant nutrient deficiency, particularly soil organic carbon, in Indian soils has become a critical concern for enhancing agricultural productivity. Soil Health Card Scheme, launched in 2015, has helped in providing information about soil nutrient status and recommendation on optimal fertilizer nutrient use as well as strengthening soil testing facilities in the country. As on 27th February 2025, about 24.8 crore soil health cards have been generated, 8,272 soil testing labs have been set up and 665 village level Soil Testing Labs (STLs) have been established in 17 States. However, some State Governments are



facing various constraints such as shortage of trained manpower, proper laboratory equipments, lack of training, etc. Despite impressive performances of SHC in terms of number of soil health cards, setting up of new STLs and strengthening of existing STLs and related infrastructure, imbalanced and inefficient use of fertilizer nutrients still remains a problem in several parts of the country.

- 3.48 Consumption pattern of NPK between 2018-19 and 2023-24 (P) has been presented in Table 3.6. Despite growing concern over imbalanced use of nutrients, N:P:K use ratio in the country has worsened over the last few years, with marginal improvement in the intervening years. The ratio deteriorated from 6.5:2.8:1 in 2020-21 to 11.8:4.6:1 in 2022-23, however it showed marginal improvement in 2023-24(P), but still far away from the ideal ratio of 4:2:1. Higher subsidy on urea vis-à-vis P&K has resulted in excessive consumption of N, leading to distortion in N:P:K ratio. This distorted ratio is posing a serious threat to soil health, resulting in a persistent decline in fertilizers crop response. Therefore, farmers should be educated and encouraged towards balanced use of fertilizers based on soil health data to have positive impact on crop yield.

Table 3.6: All-India Consumption Ratio of N and P_2O_5 in relation to K_2O

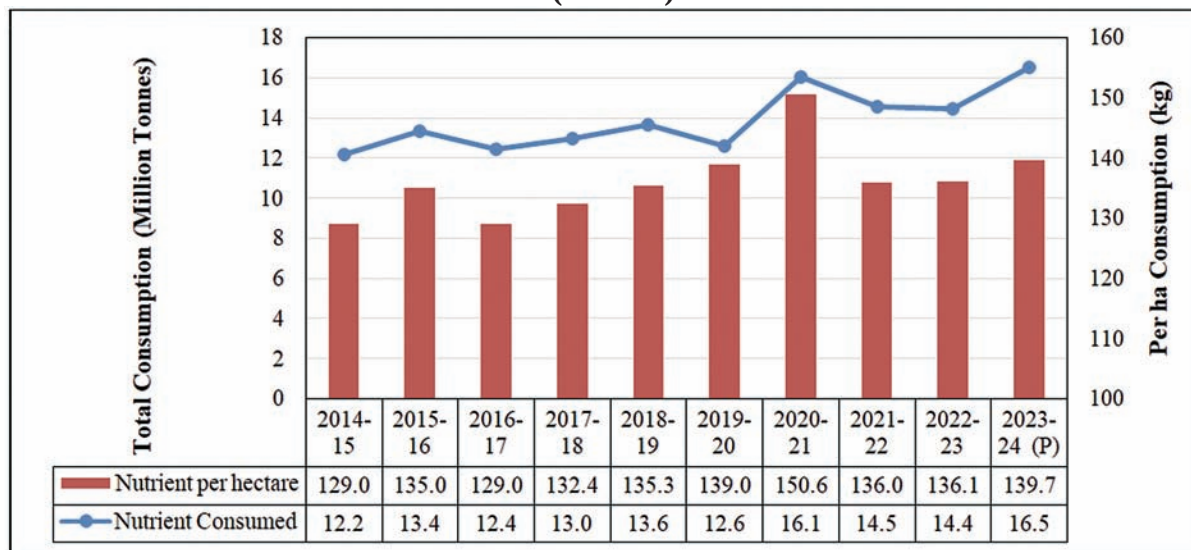
Year	N:P:K Ratio
2018-19	6.6 : 2.6 : 1
2019-20	7.3 : 2.9 : 1
2020-21	6.5 : 2.8 : 1
2021-22	7.7 : 3.1 : 1
2022-23	11.8 : 4.6 : 1
2023-24(P)	10.9 : 4.4 : 1

Source: Fertilizer Association of India

- 3.49 Chart 3.21 present trends in total consumption during *kharif* season and per hectare consumption of fertilizer nutrients in India during last 10 years. A fluctuating trend can be observed in total consumption and per hectare consumption of nutrients during the period. The highest per hectare consumption of nutrients was recorded in 2020-21, which declined in 2021-22 but rose during the last two years. Total nutrients consumption in *kharif* season, which stood at 16.1 million tonnes in 2020-21, declined over the next two years, before reaching the highest level (16.5 MT) in 2023-24(P). However, there are large variations in fertilizers consumption across different regions and States.



Chart 3.21: Trends in Total and Per Hectare Consumption of Fertilizer Nutrients (N+P+K) in India@

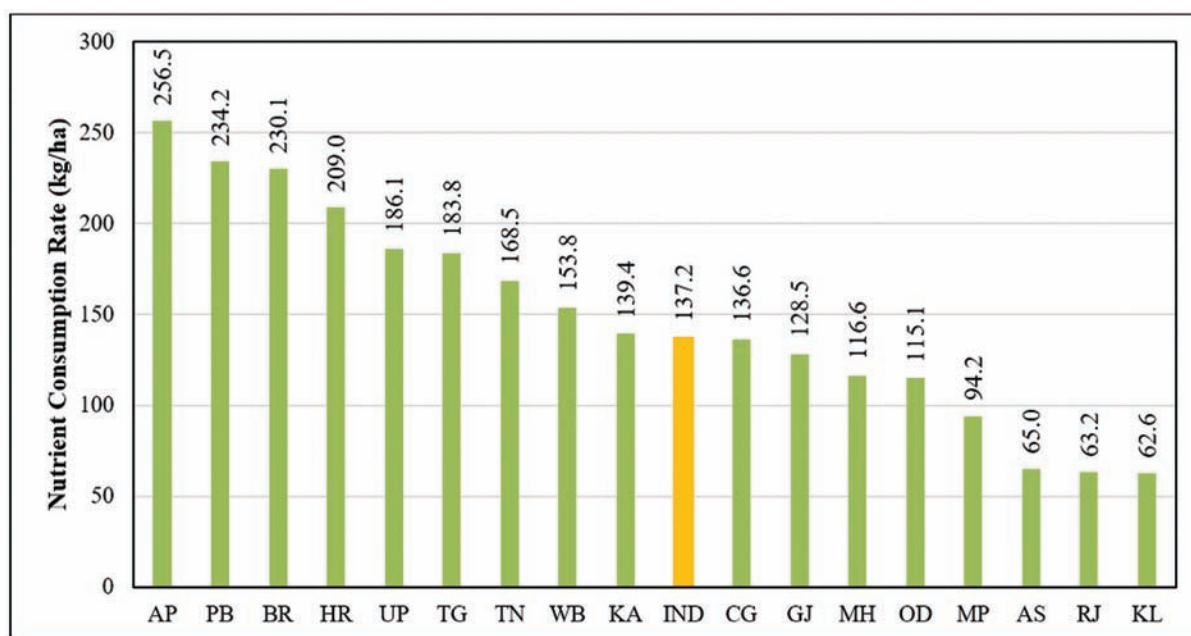


Notes: @- Total consumption is for kharif season and per ha consumption is for both kharif and rabi season
(P)-Provisional

Source: Fertilizer Association of India

- 3.50 Chart 3.22 shows the pattern of per hectare consumption of fertilizer nutrients in major States during TE2023-24. Per hectare nutrient consumption was the highest in Andhra Pradesh, followed by Punjab, Bihar and Haryana, while lowest in Kerala, Rajasthan, Assam and Madhya Pradesh.

Chart 3.22: Per ha of Fertilizer Nutrients (N+P+K) Consumption in Major States in India: TE2023-24



Note: States with more than 1 percent share in Total Cropped Area have been considered

Source: Fertilizer Association of India



- 3.51 In order to understand consumption pattern at district-level, districts in different zones of the country were categorised into different groups based on consumption of fertilizer nutrients (kg/ha) and are given in Table 3.7. It is evident from the Table that share of districts in the region with the highest fertilizer consumption (>200 kg/ha) was highest in North (42.8%), followed by South (32.6%) and the lowest (7.5%) in Western Zone. In the Northern Zone, fertilizer consumption was significantly higher in most districts of Haryana, Punjab, Western Uttar Pradesh, while in Southern Zone, fertilizer consumption was higher in Telangana, Andhra Pradesh and Tamil Nadu. In the Eastern Zone, Bihar had the highest number of districts (60.5%) under the highest fertilizer consumption category. On the other hand, 27.5 percent of districts in East and 20.3 percent districts in Western Zone have low fertilizer consumption, even less than recommended doses of fertilizer nutrients. Therefore, there is need to monitor consumption in high category districts to avoid adverse effects on soil and water but also promote fertilizer use in low-consumption areas, particularly Eastern and Western region.

Table 3.7: Classification of Districts as per Consumption of Fertilizer Nutrients (N+P+K) in Different Zones of India

(No. of Districts)

Zone	>200 kg/ha	150-200 kg/ha	100-150 kg/ha	50-100 kg/ha	<50 kg/ha
East	32	19	22	30	39
West	14	27	64	44	38
North	71	33	22	15	25
South	46	26	22	23	24

Source: Fertilizer Association of India

Farm Mechanization

- 3.52 In view of the increasing problem of shortage of labour for various agricultural operation and rising wages, there is a dire need for promoting mechanization to execute farm operations in a timely and efficient manner. The Government has taken important initiatives for increasing the level of mechanization in the country. In 2014-15, “Sub Mission on Agricultural Mechanization (SMAM)” was implemented with an objective to make farm machines more accessible and affordable, especially for the small and marginal farmers and thereby increasing the overall level of mechanization in the country. Subsequently in 2018, with an objective to promote in-situ management of crop residue by retention and incorporation into the soil through the use of appropriate mechanization, a Scheme on “Promotion of Agricultural Mechanization for in-situ Management of Crop Residue” in the States of Punjab, Haryana, Uttar Pradesh and NCT of Delhi was launched, which has been merged under SMAM and implemented as Centrally Sponsored component under *Rashtriya Krishi Vikas Yojana* (RKVY).



- 3.53 Table 3.8 presents the details of Custom Hiring Centres/Hi-Tech Hubs/Farm Machinery Banks established under SMAM and Crop Residue Management (CRM). State-wise cumulative progress of SMAM and CRM Scheme as on 31st January 2025 has been shown in Annex Table 3.3. 'FARMS' App (Farm Machinery Solutions App), a multi-lingual mobile app based aggregator platform has been developed to connect farmers with Custom Hiring Service Centres in their area. This mobile app incorporates a fair and transparent rental process focusing on quality, dependability and timely delivery of services. As on 31st January 2025, more than 75 thousand CHCs/Service Providers are registered on this mobile app to rent out around 1.76 lakh agricultural machines and about 5.24 crore farmers are also registered.

Table 3.8: Number of Custom Hiring Centres/Hi-Tech Hubs/FMBs Established under SMAM and CRM

Year	Custom Hiring Centres Established	Hi-Tech Hubs Established	Farm Machinery Banks Established at Village Level	CRM (CHCs) (from April 2018)
2014-15	663	14	436	-
2015-16	268	13	218	-
2016-17	1458	13	1877	-
2017-18	1818	39	1942	-
2018-19	3871	106	1481	7382
2019-20	2155	60	1964	8475
2020-21	3979	55	5453	13805
2021-22	4612	103	3420	7528
2022-23	4194	72	3670	1889
2023-24	2533	119	3131	1505
2024-25	1250	108	1078	1384
Total	26801	702	24670	41968

Source: M & T Division, Ministry of Agriculture and Farmers Welfare

- 3.54 Although the level of agricultural mechanization in the country has increased to some extent due to the efforts made in this regard, there is a need to intensify these efforts in view of the growing shortage of labour for agricultural operations. Since most of farmers belong to small and marginal category, therefore, research institutes should focus on developing small and economical agricultural equipments suitable for small farms to increase the level of mechanization. Further, there is a need for more research and

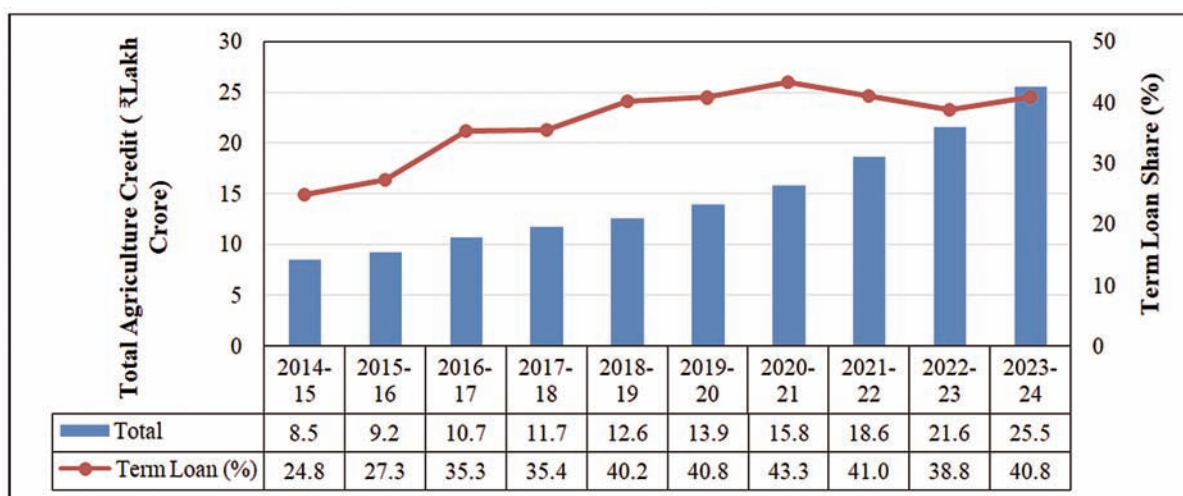


development in the area of agricultural mechanization, including development, testing and standardization of need based small farm equipments and strengthening the role of group ownership through FPOs/ PACS/SHGs, etc.

Increased Availability of Credit to Agriculture

- 3.55 An effective agriculture credit delivery system is imperative for providing timely, adequate, and equitable access to credit for raising agricultural productivity and income. Chart 3.23 presents the trend in agricultural credit during last ten years. It is evident from the Chart that there has been a double digit growth in agricultural credit from ₹8.5 lakh crore in 2014-15 to ₹25.5 lakh crore in 2023-24. It is worth noting that share of term loan, which is essential for capital formation in agriculture and long term growth of the sector, has witnessed a continuous rise from 24.8 percent in 2014-15 to 43.3 percent in 2020-21. The share declined during 2021-22 and 2022-23 but the trend reversed in 2023-24 and increased to 40.8 percent. The increase in the share of term loan is encouraging as it indicates long-term investment in agriculture for better productivity and efficiency.

Chart 3.23: Trend in Institutional Agricultural Credit in Last 10 Years

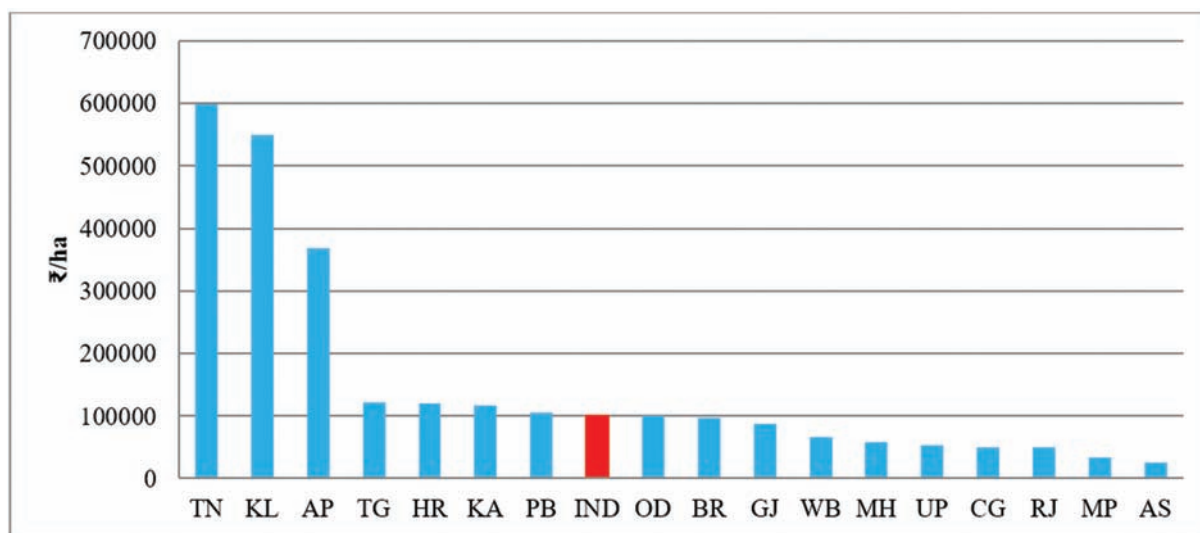


Source: National Bank for Agriculture and Rural Development

- 3.56 There is a significant disparity in flow of agricultural credit availability across regions and States. Chart 3.24 exhibits per hectare credit in major States during TE2023-24. The agricultural credit per hectare was as high as ₹6 lakh in Tamil Nadu, 5.5 lakh in Kerala and 3.5 lakh in Andhra Pradesh, while in States like Assam, Madhya Pradesh, Rajasthan and Chhattisgarh it was extremely low. This indicates wide disparities in agricultural credit distribution among States, which needs to be addressed to achieve the objective of equity in access to credit.



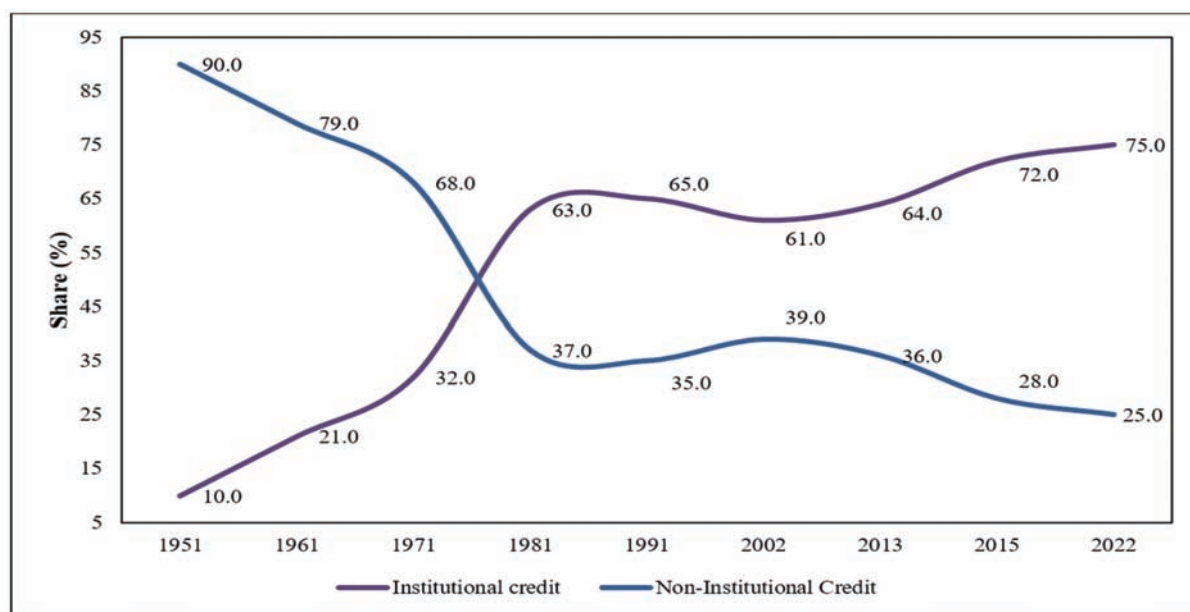
Chart 3.24: Institutional Agricultural Credit per hectare in Major States (TE2023-24)



Source: National Bank for Agriculture and Rural Development

- 3.57 In India, majority of farmers are small and marginal, who are dependent on non-institutional sources of credit to meet expenses for purchase of seeds, fertilizers, pesticides, etc. Therefore, the Government introduced *Kisan Credit Card* (KCC) Scheme with an aim to provide adequate and timely credit support from the banking system under a single window with flexible and simplified procedure to farmers for their cultivation and other needs. As on 31st December 2024, there were 7.72 crore operational KCC accounts with a loan outstanding of ₹10.05 lakh crore. Due to various initiatives of the Government, share of institutional credit has significantly increased during the last 10 years from 64 percent in 2013 to 75 percent in 2022 (Chart 3.25).

Chart 3.25: Share of Institutional and Non-Institutional Credit



Source: Ministry of Agriculture and Farmers Welfare



- 3.58 Under the Modified Interest Subvention Scheme (MISS), short-term agri-loans are provided through KCC for working capital requirements at the concessional interest rate of 7 percent and additional 3 percent subvention is also given to farmers for prompt repayment of loans under Prompt Repayment Incentive (PRI). The Government has launched *Kisan RIN* portal for faster and more efficient capturing and settlement of MISS claims. With an aim to improve access to credit and boost agricultural productivity, *Kisan Credit Card* limit under the Modified Interest Subvention Scheme has been increased from ₹3 lakh to ₹5 lakh in the current budget.

Crop Insurance for Managing Production Risks

- 3.59 ‘*Pradhan Mantri Fasal Bima Yojana*’ (PMFBY), launched in 2016 to provide affordable crop insurance product to farmers for comprehensive risk cover, has made an impressive progress, as is evident from Table 3.9. The number of farmers covered under the scheme has increased from 2 crore in *kharif* 2021 to 3.1 crore in *kharif* 2024. However, it is worth noting that number of non-loanee farmers has increased much faster than loanee farmers during the last four seasons, indicating increasing acceptance of the Scheme among farmers. The share of farmers in gross premium has also declined significantly as in States like Maharashtra, Odisha, Meghalaya, Puducherry and Jharkhand, farmers’ share is paid by the State Government and farmers pay a token premium of ₹1. However, still coverage of scheme is limited and needs to be expanded.
- 3.60 Government has taken various steps to make the scheme more effective. Operational guidelines of the scheme have been revised from time to time with latest revision from *kharif*-2023. Under the revised scheme, latest technologies such as Yield Estimation System Using Technology (YES-TECH), Weather Information and Network Data System (WINDS) for setting up Automatic Weather Stations at block level, automatic Rain Gauges at panchayat level, etc. have been incorporated to ensure transparency and objectivity in implementation of the scheme. For large scale use of technology in implementation of the scheme, Government has created Fund for Innovation & Technology (FIAT) with a corpus of ₹824.77 crore.

Table 3.9: Performance of Crop Insurance under PMFBY during Last Four *Kharif* Seasons

(as on 31.01.2025)

Year	2021	2022	2023	2024*
Farmers Covered (crore)	2.0	2.3	3.0	3.1
Farmers Premium (₹‘000 crore)	1.9	2.1	1.3	1.4
Gross Premium (₹‘000 crore)	15.7	15.9	16.4	17.1
Sum Insured (₹‘000 crore)	89.2	108.7	143.6	153.5
Total Applications (crore)	4.4	6.1	7.9	8.8
Area Insured (crore ha)	2.1	2.2	2.8	2.9
Non-Loanee Farmers Ratio (%)	25.2	22.0	40.8	37.7
Claims Settled (₹‘000 crore)	12.7	10.2	10.4	0.2
Farmers Benefited (crore)	2.0	1.8	2.0	0.1

Note : * Provisional

Source: Credit Division, Ministry of Agriculture and Farmers Welfare



Digitalization of Agriculture

- 3.61 In order to leverage a wide range of technologies like Artificial Intelligence (AI), block chain, Internet of Things (IoT), etc. in Indian agriculture, Government launched Digital Agriculture Mission on 2nd September 2024 with a financial outlay ₹2,817 crore. The key components of the Mission are Digital Public Infrastructure for Agriculture such as Agristack for farmers' registry, village maps and the crop sown registry; *Krishi* Decision Support System for crop and soil maps, drought/flood monitoring, technology-based yield assessment, etc.; Comprehensive soil fertility and profile maps and Digital General Crop Estimation Survey (DGCES), etc. Digitalization in agriculture will help in improving crop planning and estimation, implementation of schemes, transparency, easier access to services and evidence-based decision making and monitoring. The key targets of Agristack include creation of digital identities for 11 crore farmers over three years and launching the Digital Crop Survey nationwide within two years, covering 400 districts in FY 2024-25 and all districts in FY 2025-26. As on 3rd March 2025, a total of 3.8 crore Farmer IDs have been created and Digital Crop Survey (DCS) has been undertaken in 436 districts in *kharif* 2024 and started in 17 States across 462 districts during *rabi* 2024-25.
- 3.62 The mission is expected to create direct and indirect employment in agriculture, providing opportunities for around 2.5 lakh trained local youth and *Krishi Sakhis*. Digitization of records will enable digital authentication for accessing services and benefits thus reducing paperwork and the need for physical visits of the beneficiaries of various Government schemes. It will further enhance efficiency and transparency in Government Schemes, crop insurance, and loan systems through accurate data on crop area and yield. Generation of crop maps will ensure smooth monitoring for better disaster response and efficient settlement of insurance claims. Development of digital infrastructure of this kind will help to optimize value chains and provide tailored advisory services for crop planning, health, pest management, and irrigation.

Recapitulation

- 3.63 Rice and maize registered positive growth in production due to positive growth in both area and yield in TE2023-24. In case of jowar and bajra, moderate growth in yield was recorded, which contributed to positive growth in production despite decline in acreage during the period. Among pulses, urad registered an impressive growth in yield during the current period, however, tur witnessed a significant decline in yield during TE2023-24. As far as oilseeds are concerned, groundnut and nigerseed witnessed a sharp decline in area, resulting in a negative growth in production despite positive growth in yield. On the other hand, soybean, sesamum and sunflower recorded positive growth in production and yield during the period. Decline in area, production and yield of cotton during TE2023-24 presents a depressing picture, which needs to be reversed.
- 3.64 Comparison of India's average yield with the yield of other major producing countries shows that yield in India is substantially lower than world average. Furthermore, wide variations in yield were observed



across the States in the country. The high yield gaps in different crops call for comprehensive strategy at district level to improve efficiency in production for further improvement in yield.

- 3.65 As there is limitation for area expansion, concerted efforts are required for yield improvement through use of quality inputs as well as latest production technologies and best farm practices. There is an immense need to develop high yielding variety seeds and develop seed chains for improving SRR and VRR. Moreover, farmers should be motivated to adopt quality seeds for obtaining better yield. There is also a need to expand the irrigation coverage in States and crops where irrigation facilities are comparatively low. Farmers should also be educated about efficient use of water and fertilizers not only for improving the soil health but also for improving the yield.
- 3.66 There has been significant increase in flow of credit to agriculture and share of term loan has also increased. The limit through *Kisan* Credit Cards under Modified Interest Subvention Scheme has been increased from ₹3 lakh to ₹5 lakh, which will help the small and marginal farmers in fulfilling short term credit needs with regard to agriculture and ensure inclusive growth. However, more efforts are required for reducing the inter-State disparities in the distribution of agricultural credit. In view of the growing scarcity of labour for agricultural operations and rising wages, more intensified efforts are required to promote mechanization and develop cost effective farm implements suitable for small farms in the country. High cost machinery can be provided on rental basis by establishing more CHCs/ machinery banks. As far as crop insurance is concerned, it is expected that with inclusion of latest technologies and remote sensing, claim settlement will be faster and will attract more farmers under the coverage of PMFBY. Digital Agriculture Mission is another remarkable initiative to make Indian agriculture more efficient and competitive.



CHAPTER 4



Trade Competitiveness of Indian Agriculture

- 4.1 Agricultural trade in India has displayed robust resilience in recent years despite several global economic and geopolitical challenges. India's agricultural exports in 2023-24 were about ₹4.2 lakh crore, marginally lower than ₹4.4 lakh crore in 2022-23 due to various trade restrictions on key export commodities combined with erratic weather pattern. However, agri-exports recorded an annual compound growth rate of 8 percent during last 10 years. Indian agri-export basket is dominated by rice, marine products, spices, meat and meat products, accounting for nearly half of the total agri-exports. A focused strategy toward diversification of export basket, integration with global value chains and compliance to sanitary and phytosanitary measures will foster competitiveness of Indian agricultural exports. Agricultural imports have more than doubled during the last 10 years from ₹1.45 lakh crore in 2014-15 to ₹2.91 lakh crore in 2023-24, and edible oils and pulses accounted for more than 50 percent of total imports. Therefore, it is imperative to achieve self-sufficiency in edible oils and pulses production with improvement in productivity and area expansion in order to reduce dependency on highly volatile global markets. This chapter presents the key findings of India's agricultural trade vis-à-vis domestic and international price trends along with recent changes in trade policy of mandated *kharif* crops. In addition, the chapter also provides an overview of global trade outlook for major commodities.

Global Trade Performance

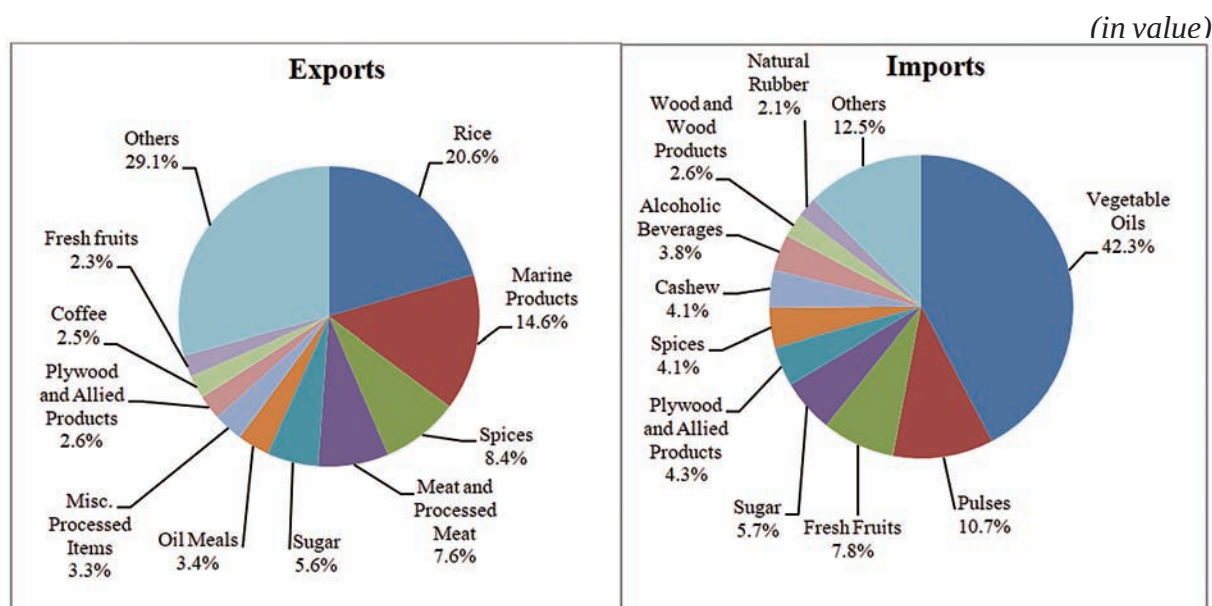
- 4.2 The value of world merchandise trade reached around US\$23.8 trillion in 2023, down by 5 percent over 2022, as per World Trade Organisation's (WTO) Global Trade Outlook and Statistics 2024. This decline was influenced by a combination of factors, which include reduced trade volumes, lower primary commodity prices and exchange rate fluctuations. According to World Bank's Global Economic Prospects, January 2025, world trade in goods experienced a rebound in 2024 and increased by 2.7 percent over previous year (Annex Table 4.1). Both WTO and World Bank project global goods trade to grow at around 3 percent in 2025.
- 4.3 Even though the total value of world merchandise trade decreased in 2023 including agricultural trade, but there was a slight increase of 1 percent in trade of food products over previous year. Despite year-on-year decline in 2023, compared to pre-pandemic level in 2019, value of agricultural exports was significantly up by 32 percent (Annex Table 4.2).



India's Agricultural Trade Performance

- 4.4 India exported agricultural commodities worth ₹418.6 thousand crore, while agricultural imports were ₹290.9 thousand crore in 2023-24, as per Directorate General of Commercial Intelligence and Statistics (DGCIS). Both exports (-5.2%) and imports (-4.3%) of agricultural products declined in 2023-24 over 2022-23. The share of agriculture in total exports was 11.6 percent and agricultural imports accounted for 5.2 percent of the total imports in 2023-24 (Annex Table 4.3).
- 4.5 The composition of India's agricultural exports and imports in 2023-24 is shown in Chart 4.1. With a share of 20.6 percent, rice continues to be the single largest commodity in agricultural exports from India, followed by marine products (14.6%), spices (8.4%) and meat and processed meat (7.6%) (Annex Table 4.6). It is worth noting that export earnings from basmati rice increased by 25.6 percent in 2023-24 over previous year as the Government removed the Minimum Export Price (MEP) for basmati rice in October 2023, while value of exports of non-basmati rice declined by (-)26 percent and sugar exports by (-)49.5 percent due to various export restrictions.
- 4.6 Among the key import items, vegetable oils, with a share of 42.3 percent in total agricultural imports, was the single largest commodity group in 2023-24. Pulses constituted the second largest share of 10.7 percent, followed by fresh fruits (7.8%) and sugar (5.7%) (Annex Table 4.7). It is also pertinent to note that among the key commodities imported by India, import bill for vegetable oils declined by (-)26.4 percent in 2023-24 over previous year, which is attributed to significant decline in global prices of edible oils. However, the share of pulses in total agri-imports increased from 5.2 percent in 2022-23 to 10.7 percent in 2023-24 due to rising demand, and about 7 percent fall in domestic production caused by erratic weather pattern and easing of restrictions on imports of pulses.

Chart 4.1: Composition of India's Agricultural Exports and Imports, 2023-24



Source: Directorate General of Commercial Intelligence and Statistics, Ministry of Commerce and Industry



- 4.7 India's agricultural exports for the period April-December 2024 were ₹325.6 thousand crore, an increase of 7.9 percent over corresponding period last year, while India's agricultural imports increased by 20 percent from ₹218.1 thousand crore to ₹261.8 thousand crore during this period. Increase in exports is attributed to higher exports of rice (both basmati and non-basmati), spices and buffalo meat, while increased imports of pulses and vegetable oils resulted in higher imports.
- 4.8 Indian agricultural exports face various non-tariff barriers such as residue of pesticides and other chemicals, microbiological contamination etc. in agricultural products in various countries, which adversely affect agricultural exports. Therefore, efforts should be made to create awareness amongst farmers about judicious use of safe and permissible pesticides, adoption of Good Agricultural Practices (GAPs) and Good Manufacturing Practices (GMPs), Hazard Analysis Critical Control Points (HACCP), Standard Operating Procedures (SOPs), etc.

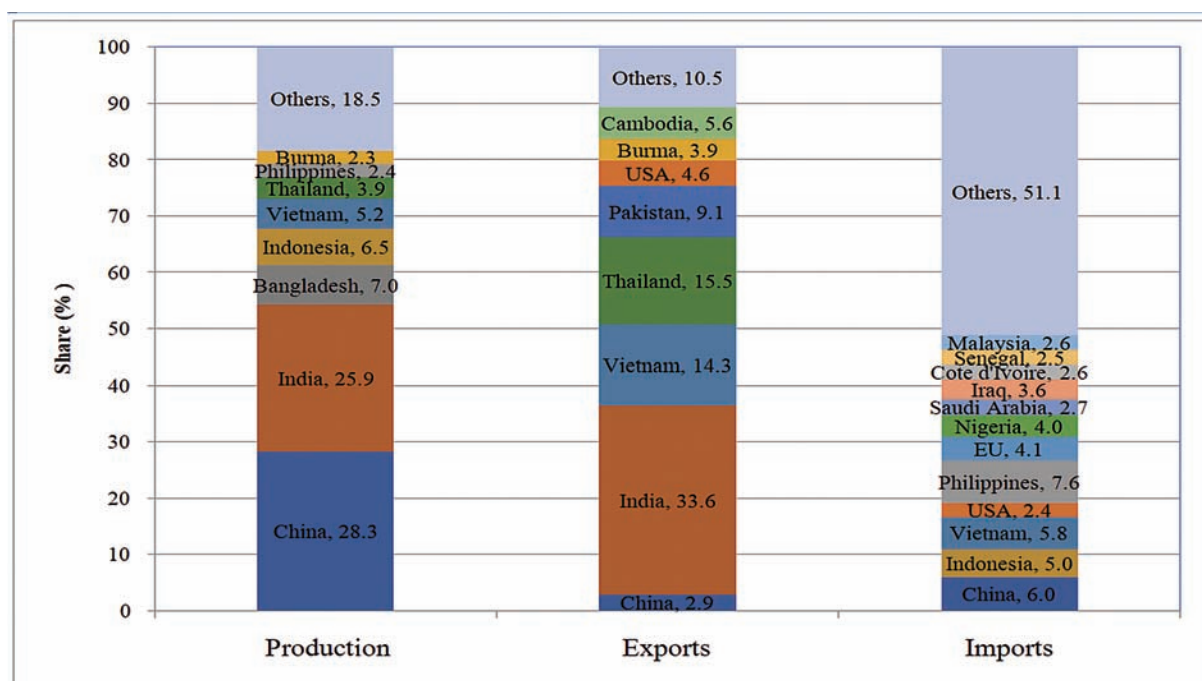
Trade Patterns and Trade Policy of Major Kharif Crops

Rice

Global Production and Trade

- 4.9 Global rice production in TE2023-24 was 517.8 million tonnes, with China and India constituting more than half of the total production, as per United States Department of Agriculture (USDA). Global rice production is forecast to increase to a record high of 532.9 million tonnes in 2024-25 consequent to substantial increase in India and moderate increase in China, Indonesia and Vietnam. Chart 4.2 depicts the share of major rice producers, exporters and importers in TE2023-24. It can be seen that South Asia plays a significant role in the global rice production and exports.

Chart 4.2: Global Players in Rice Market, TE2023-24



Source: United States Department of Agriculture



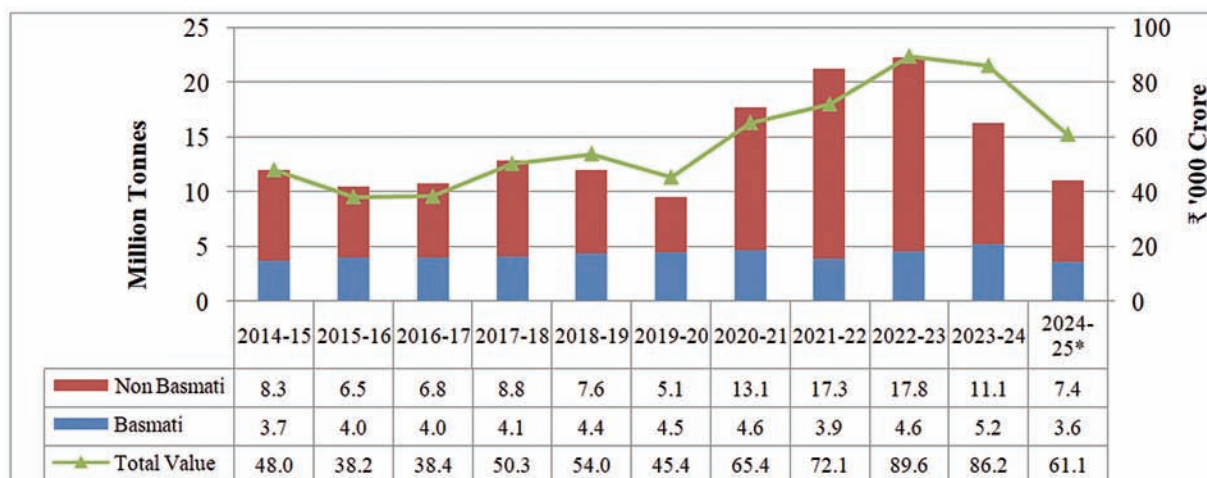
- 4.10 As per USDA, about 11 percent of global rice production was traded in TE2023-24. India has remained the largest exporter of rice in the world since 2011-12 and accounted for more than one-third (33.6%) of global rice exports in TE2023-24, followed by Thailand (15.5%), Vietnam (14.3%) and Pakistan (9.1%) (Chart 4.2). Although China is the largest producer, but it continues to be the second largest importer of rice, with a global share of 6 percent after Philippines (7.6%). Other major importers of rice were Vietnam, Indonesia and EU. It is worth noting that the top five importers accounted for 28.4 percent of the world imports of rice, while share of top five exporters was 78 percent in world exports, which indicates that exports of rice are highly concentrated than imports.
- 4.11 Global rice trade is projected at 58.5 million tonnes in 2025, about 1.2 million tonnes less than in 2024, as per USDA Rice Outlook, February 2025. However, India would continue to be the leading rice exporter with an estimated export of about 22.5 million tonnes in 2025, accounting for 38.4 percent of global rice exports. As per the Outlook, lower exports from Burma, Pakistan, Taiwan, Thailand, Turkey and Vietnam are projected to slightly offset the larger exports from Argentina, Brazil, European Union (EU), India, Paraguay, USA, and Uruguay in 2025. As regards imports, countries like Brazil, Cuba, EU, Guinea, Indonesia, Iraq, Kenya, Malaysia, Sri Lanka, and Vietnam are projected to see a decline in imports in 2025, but an increase is expected in Afghanistan, Angola, Bangladesh, Benin, Canada, China, Cote d'Ivoire, Ethiopia, Gambia, Ghana, Iran, Kazakhstan, Laos, Madagascar, Mozambique, Nepal, North Korea, the Philippines, Somalia, South Africa, Tanzania, Togo, Turkey, the UAE, USA and Venezuela. It is worth noting that India has been the most affordable rice supplier to the world during the last few years.

India's Trade

- 4.12 Rice is the most important staple food in India, constituting over 40 percent of total foodgrains production in the country. Over the years, India has emerged as a major player in the global rice trade due to steady increase in production, rising global demand, supportive Government policies and competitive domestic prices. The trends in India's exports of rice (basmati and non-basmati) from 2014-15 to 2024-25 (till November 2024) are shown in Chart 4.3. It is observed that, India exported an average of 4.3 million tonnes of basmati and 10.2 million tonnes of non-basmati rice in the last 10 years. India's exports of basmati rice witnessed a steady growth during 2014-15 to 2023-24, except a marginal fall in 2021-22, while non-basmati rice exports remained volatile during 2014-15 and 2019-20, but registered significant growth thereafter and reached a record high of 17.8 million tonnes in 2022-23, followed by a decline in 2023-24 due to complete ban on export of non-basmati white rice in July 2023.
- 4.13 During April-November 2024, India exported 7.4 million tonnes of non-basmati rice, slightly lower compared to April-November 2023, while exports of basmati rice marginally increased during the period and were 3.6 million tonnes. In value terms, total rice exports increased by 14.9 percent during April-November 2024 compared with April-November 2023.



Chart 4.3: Trends in India's Exports of Rice, 2014-15 to 2024-25



Note : * For 2024-25 (April-November)

Source: Directorate General of Commercial Intelligence and Statistics, Ministry of Commerce and Industry

- 4.14 In 2023-24, major export destinations of Indian basmati rice were Saudi Arabia (20.9%), Iraq (15.7%) and Iran (12.8%), while for non-basmati rice, major export destinations were Benin (11.5%), Guinea (8.2%), Togo (6.4%) and Vietnam (6.2%). The top ten importers of basmati rice accounted for a share of 78 percent (in volume), whereas the corresponding figure in case of non-basmati rice was about 60 percent.

Trade Policy

- 4.15 In order to ensure sufficient domestic supply as well as price stability, Government takes appropriate trade policy measures for rice from time to time. Government of India banned export of 100 percent broken rice and imposed 20 percent export duty on other types of rice except basmati and parboiled on 9th September, 2022, followed by a complete ban on export of non-basmati white rice from July 2023. Subsequently, Government imposed 20 percent export duty on parboiled non-basmati rice on 25th August, 2023. In order to safeguard against misclassification of exports of non-basmati white rice as basmati rice, on 27th August, 2023, Government imposed MEP for basmati rice exports and made registration mandatory for issue of Registration-cum-Allocation Certificate (RCAC) for any contract of basmati rice consignment with value US\$1200/MT and above from Agricultural and Processed Food Products Export Development Authority (APEDA). However, in September 2024, Government removed the ban on non-basmati white rice exports and set a MEP of \$490 per tonne. In addition, export duty on parboiled rice was reduced from 20 percent to 10 percent and MEP of \$490 per tonne on non-basmati white rice exports was lifted and the export duty on parboiled rice was further reduced from 10 percent to zero in October 2024. Moreover, the export policy of broken rice has been revised from 'Prohibited' to 'Free' w.e.f. 7th March, 2025. The details of changes in rice trade policy since 2011 are given in Annex Table 4.9.

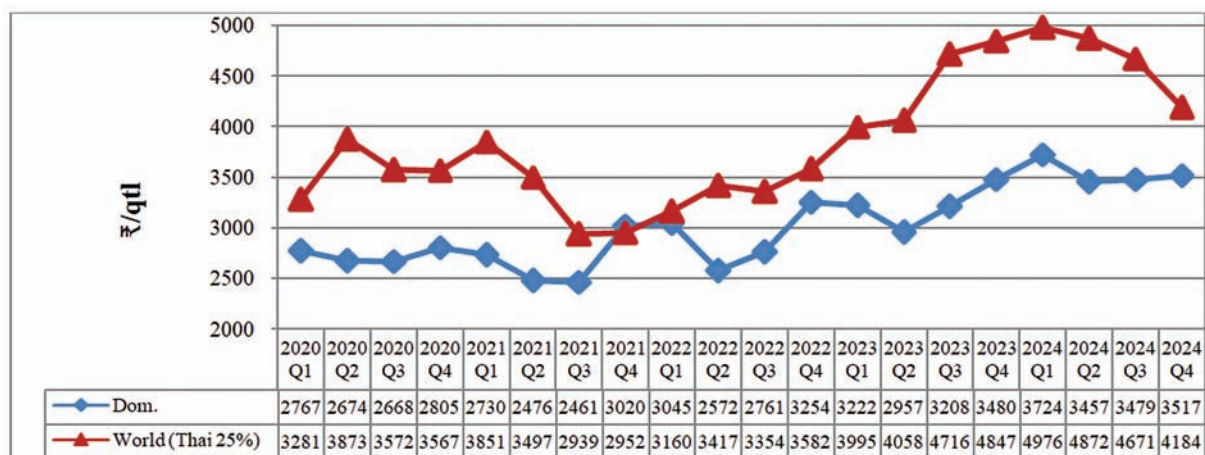


- 4.16 India's rice exports, particularly basmati, have been facing restrictions and bans in some countries due to standards related to food safety, quality and health issues due to residue of pesticides and other chemicals. In order to protect interest of consumer and establish fair practices in trade of basmati rice, the Food Safety and Standards Authority of India (FSSAI) has notified the identity standards for basmati rice *vide*, Food Safety and Standards (Food Products Standards and Food Additives) First Amendment Regulations, 2023 and enforced w.e.f. 1st August, 2023. FSSAI has specified various identity and quality parameters for basmati rice such as average size of grain and their elongation ratio after cooking, maximum limits of moisture, amylose content, uric acid, defective/damaged grains and incidental presence of other non-basmati rice, etc. Ensuring compliance of these measures will not only enhance reputation and trustworthiness of India's basmati rice, but also boost export competitiveness in the world market.

Comparative Trends in Prices

- 4.17 The comparative trends of domestic and international prices of rice for the period 2020 to 2024 are shown in Chart 4.4. It is observed that the international prices of rice remained higher than domestic prices during the entire period, except 2021(Q4), making Indian rice competitive in the world market (Annex Table 4.8). The chart shows that during 2020(Q1) to 2021(Q3), domestic prices of rice were mostly stable and stagnant, but the international prices remained highly volatile due to global supply chain disruptions, caused by COVID-19 pandemic. However, from 2021(Q3) till 2024(Q1), both domestic and international prices followed upward trajectory, with more sharp rise (69.3%) in international prices due to geopolitical conflicts, weather related issues and ban on rice exports by India. However, from 2024(Q1) to 2024(Q4), international prices declined considerably, but domestic prices witnessed a slight increase. It is evident from the Chart that world prices of rice are more volatile than domestic prices.

Chart 4.4: Domestic and International Prices of Rice, 2020 to 2024



Note : 1. Domestic prices in the present report may slightly vary from the earlier reports, as data from UPAg has been used in place of AGMARKNET

2. Rice (Thailand), 25 percent broken, white rice, milled indicative survey price, government standard, f.o.b. Bangkok

3. Domestic prices of paddy converted into rice at the ratio of 1:0.67

Source: 1. Unified Portal for Agricultural Statistics (UPAg), Ministry of Agriculture and Farmers Welfare, for computing weighted average domestic wholesale prices

2. World Bank Pink Sheet for international prices of rice (Thailand)

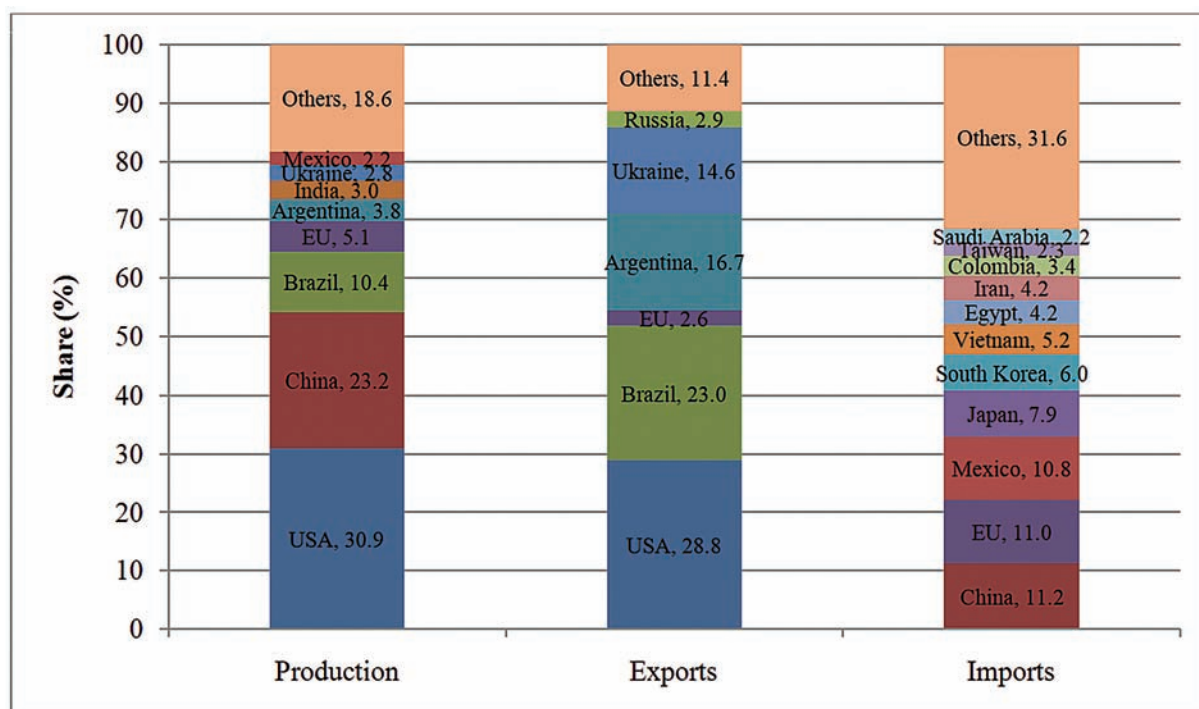


Maize

Global Production and Trade

- 4.18 Global production of maize was 1204 million tonnes in TE2023-24 and USA continues to be the largest producer of maize in the world with a share of 30.9 percent, followed by China (23.2%), Brazil (10.4%), EU (5.1%) and Argentina (3.8%) (Chart 4.5). With a modest share of 3 percent, India ranks sixth in world maize production.
- 4.19 About 16 percent of the global maize production was traded in TE2023-24. It is observed that the exports of maize are highly concentrated, with top three exporters, USA (28.8%), Brazil (23%) and Argentina (16.7%), accounting for more than two-third of total exports. On the contrary, imports of maize are geographically more dispersed and China was the largest importer (11.2%), followed by EU (11%), Mexico (10.8%), Japan (7.9%) and South Korea (6%) in TE2023-24.

Chart 4.5: World Maize Production, Exports and Imports, TE2023-24



Source: United States Department of Agriculture

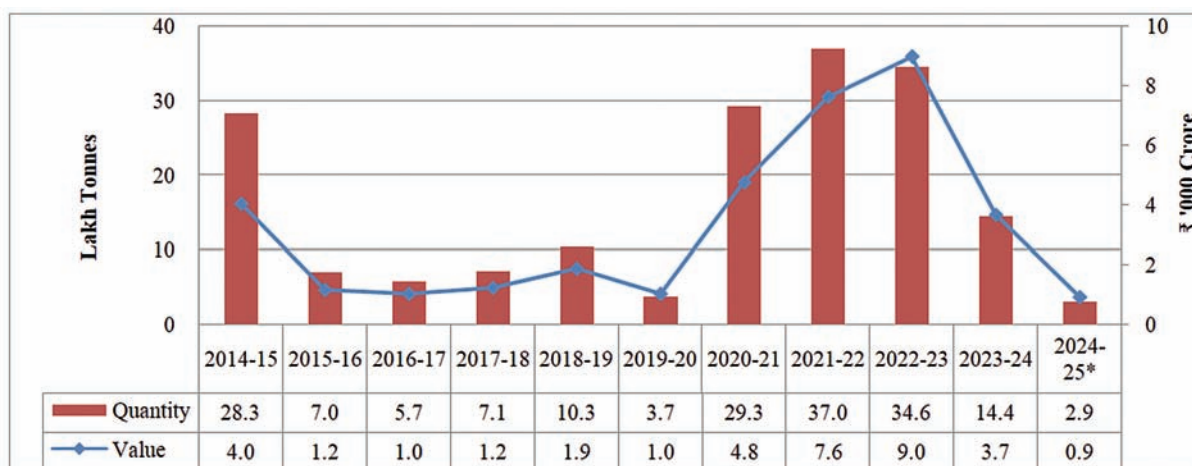
India's Trade

- 4.20 Maize is the third most important cereal crop in India after rice and wheat. The demand for maize is increasing around the world due to its multiple uses as staple food, livestock feed and for industrial uses including ethanol production. India exported 28.3 lakh tonnes of maize in 2014-15, but exports remained subdued between 2015-16 and 2019-20, driven by higher domestic prices compared to international prices. However, exports witnessed a significant growth during 2020-21 and 2022-23 due to lower production of maize in major producing countries like Argentina, Ukraine and USA.



Although, volume of exports marginally declined in 2022-23, but the value of exports increased due to high international prices. A sharp fall in maize exports was recorded during 2023-24 due to rise in domestic demand for industrial uses, particularly ethanol production, marginal fall in domestic production and higher domestic prices. During April-November 2024, maize exports were significantly lower at 2.9 lakh tonnes compared to 12.8 lakh tonnes in the corresponding period last year.

Chart 4.6: India's Exports of Maize, 2014-15 to 2024-25



Note : * For 2024-25 (April- November)

Source: Directorate General of Commercial Intelligence and Statistics, Ministry of Commerce and Industry

Trade Policy

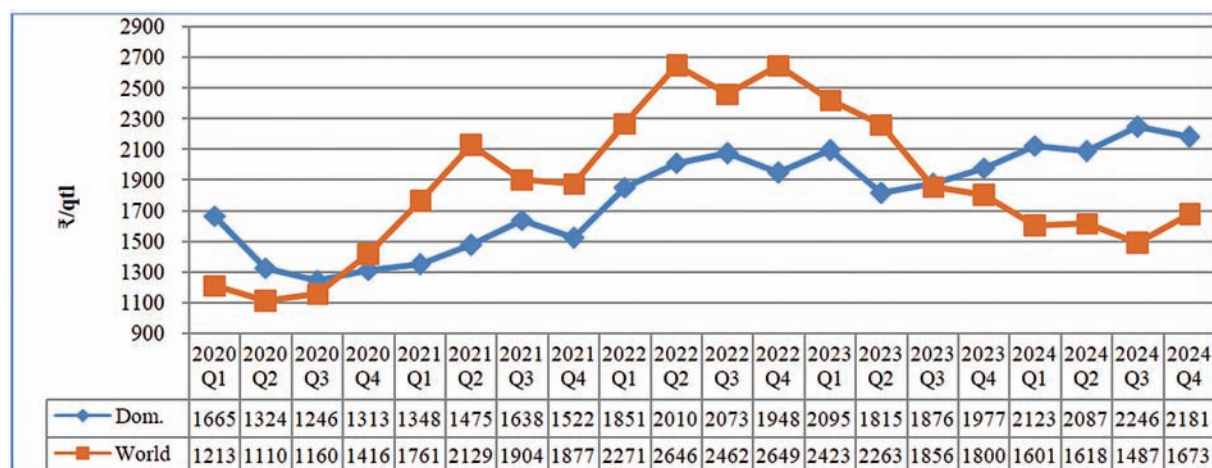
- 4.21 Maize other than seed quality attracts a basic import duty of 50 percent and imports of seed quality maize are “Restricted” in India. In addition, imports of maize, other than seed quality, up to five lakh tonnes in a financial year, are allowed at a concessional import duty of 15 percent under Tariff Rate Quota (TRQ), subject to conditions prescribed vide notification dated 23rd June, 2020. The export of maize from India is under “Free” category.

Comparative Trends in Prices

- 4.22 Chart 4.7 depicts the comparative trends in domestic and international prices of maize during the period 2020 to 2024. It can be seen that the domestic prices were higher than the international prices from 2020(Q1) to 2020(Q3), but the trend reversed from 2020(Q4) and the international prices of maize surpassed domestic prices with notable difference till 2023(Q2), which led to record exports of maize from India during this period. However, domestic maize prices rose above international prices during 2023(Q3) to 2024(Q4) making Indian exports less competitive in world markets.



Chart 4.7: Domestic and International Prices of Maize, 2020 to 2024



Note : 1. Domestic prices in the present report may slightly vary from the earlier reports, as data from UPag has been used in place of AGMARKNET

2. Maize (US), No. 2, yellow, f.o.b. US Gulf ports for international price of maize

Source: 1. UPag, Ministry of Agriculture and Farmers Welfare, for computing weighted average domestic wholesale prices

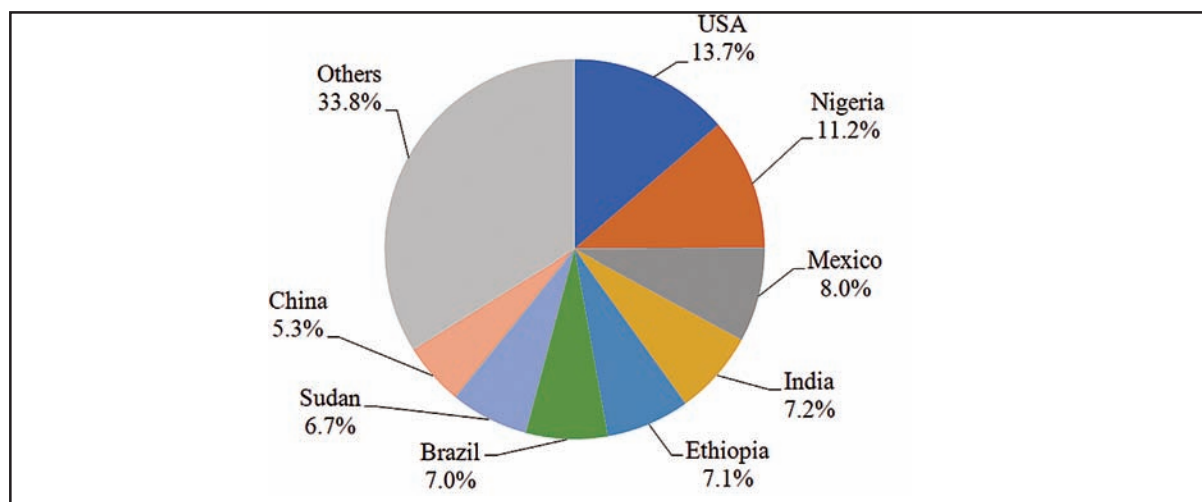
2. World Bank Pink Sheet for international prices

Sorghum/Jowar

Global Production and Trade

- 4.23 Jowar is drought tolerant crop grown in sub-tropical and semi-arid regions of Asia and Africa. USA was the largest producer of jowar with a share of 13.7 percent in TE2023-24, followed by Nigeria (11.2%), Mexico (8%), India (7.2%) and Ethiopia (7.1%) (Chart 4.8). About 16 percent of global production was traded in TE2023-24 and USA was the dominant player with a global share of 58.3 percent, followed by Australia (25.1%) and Argentina (13.1%). With over 85.6 percent share of total imports, China is the largest importer of jowar, followed by Japan (2.2%) and Mexico (2.1%).

Chart 4.8: Share of Major Producers in World Jowar Production, TE2023-24



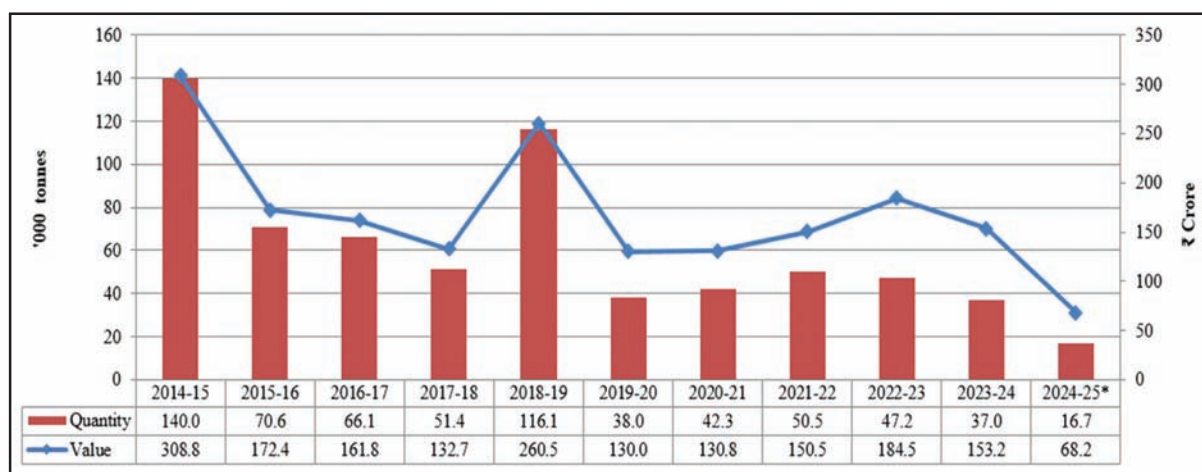
Source: United States Department of Agriculture



India's Trade

- 4.24 Although jowar is the second major millet in terms of production and area in India, but small quantity is exported to neighboring countries like UAE, Saudi Arabia, Bangladesh and Kenya. The trends in exports of jowar from India during the period 2014-15 to 2024-25 (till November 2024) are shown in Chart 4.9. It is observed that exports of jowar from India, which was highest at 1.4 lakh tonnes (₹308.8 crore) in 2014-15 followed a downward trend and reached the lowest at 38 thousand tonnes (₹130 crore) in 2019-20, except a temporary surge in 2018-19. Subsequently, a moderate growth in jowar exports was noticed from 2019-20 to 2021-22, but the trend reversed thereafter. However, it is worth noting that between 2021-22 and 2022-23, while volume of exports declined by 6.4 percent, value of exports significantly increased by 22.6 percent. In 2024-25 (till November 2024), exports of jowar were marginally lower than the corresponding period last year. In view of huge export potential of millets due to nutritional and health benefits, efforts are needed to formulate a robust strategy for promoting exports of value-added products.

Chart 4.9: India's Exports of Jowar, 2014-15 to 2024-25



Note : * For 2024-25 (April-November)

Source: Directorate General of Commercial Intelligence and Statistics, Ministry of Commerce and Industry

Trade Policy

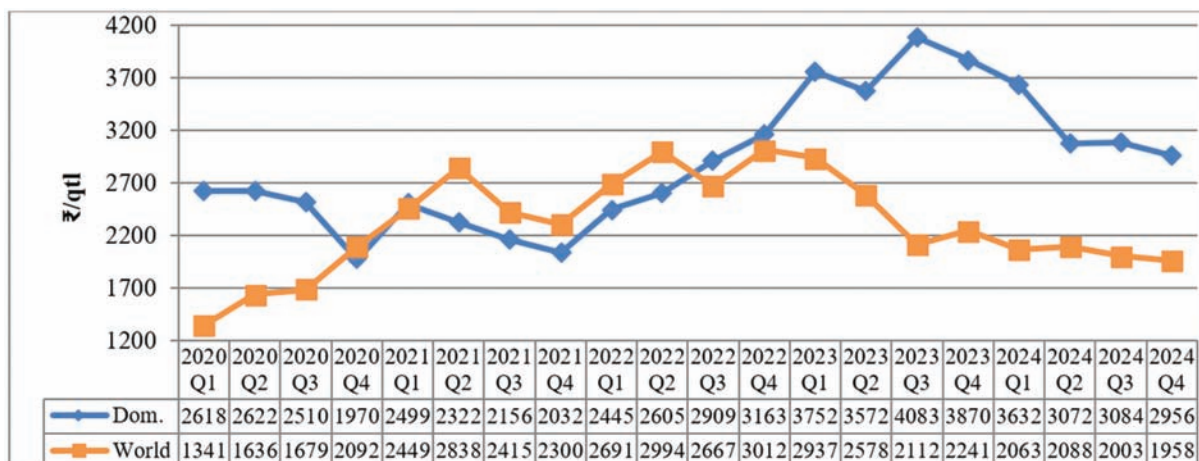
- 4.25 Imports of jowar, other than seed quality are “Free” in India with an import duty of 50 percent from 2017. Exports are also under “Free” category and do not require any license from DGFT.

Comparative Trends in Prices

- 4.26 Chart 4.10 depicts the comparative trends in international and domestic prices of jowar. From 2020(Q1) to 2020(Q3), domestic prices of jowar were higher than the international prices, but remained below the international prices during the period 2020(Q4) to 2022(Q2), except 2021(Q1). From 2022(Q3), domestic prices surpassed international prices due to rise in domestic demand for millet and value-added products, as a result of promotional/awareness activities under National Year of Millets and International Year of Millets.



Chart 4.10: Domestic and International Prices of Jowar, 2020 to 2024



Note : Domestic prices in the present report may slightly vary from the earlier reports, as data from UPAg has been used in place of AGMARKNET

Source: 1. UPAg, Ministry of Agriculture and Farmers Welfare, for computing weighted average domestic wholesale prices

2. International Grain Council (FPMA) for International prices

Pulses

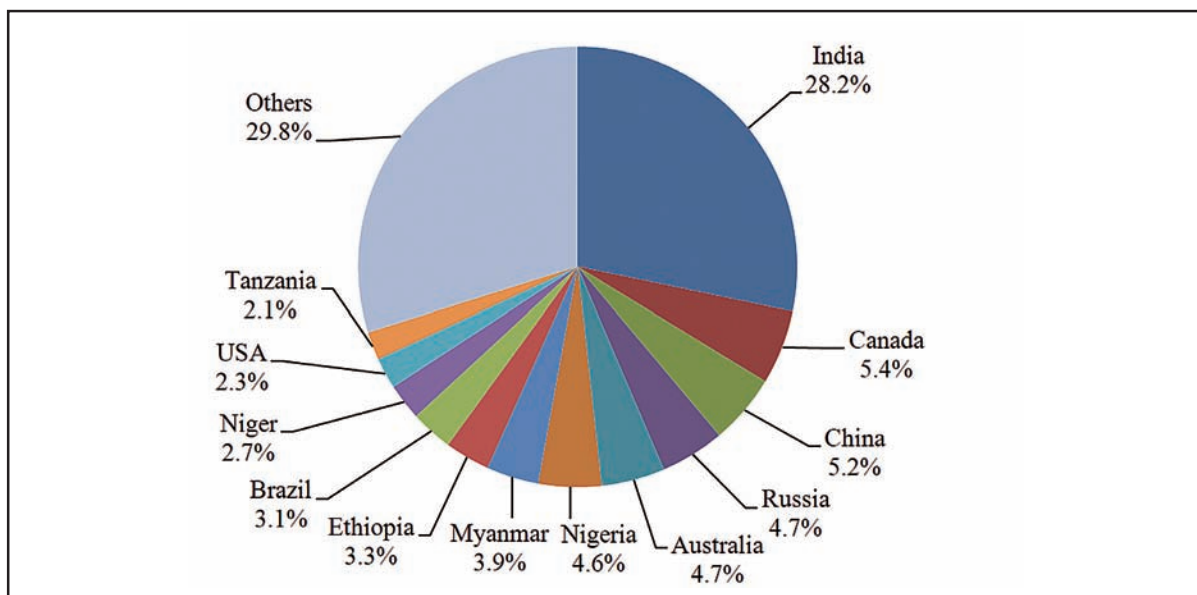
- 4.27 Pulses being main source of protein for the large vegetarian population, India is the largest producer, consumer and importer of pulses in the world. Although India leads in production of certain pulses, but low yields and rising demand have resulted in growing reliance on pulses imports. While noticeable progress has been made in India's efforts towards achieving self-sufficiency in pulses production, but meeting the growing domestic demand is still a big challenge. Therefore, understanding the global pulses dynamics is crucial for a stable and sustainable domestic market in India for the benefits of producers and consumers.

Global Production and Trade

- 4.28 In TE2023, the annual global production of pulses was 93.3 million tonnes, marginally higher than TE2022, as per the estimates of Food and Agricultural Organization (FAO). Of this, production of tur/arhar accounted for 5.5 percent, lentil 6.9 percent and chick pea about 17.9 percent in TE2023. Chart 4.11 shows shares of leading producers of pulses in the world. India, with a share of 28.2 percent, was the largest producer, followed by Canada (5.4%), China (5.2%), Russia (4.7%) and Australia (4.7%).



Chart 4.11: Share of Major Producers in World Production of Pulses, TE2023



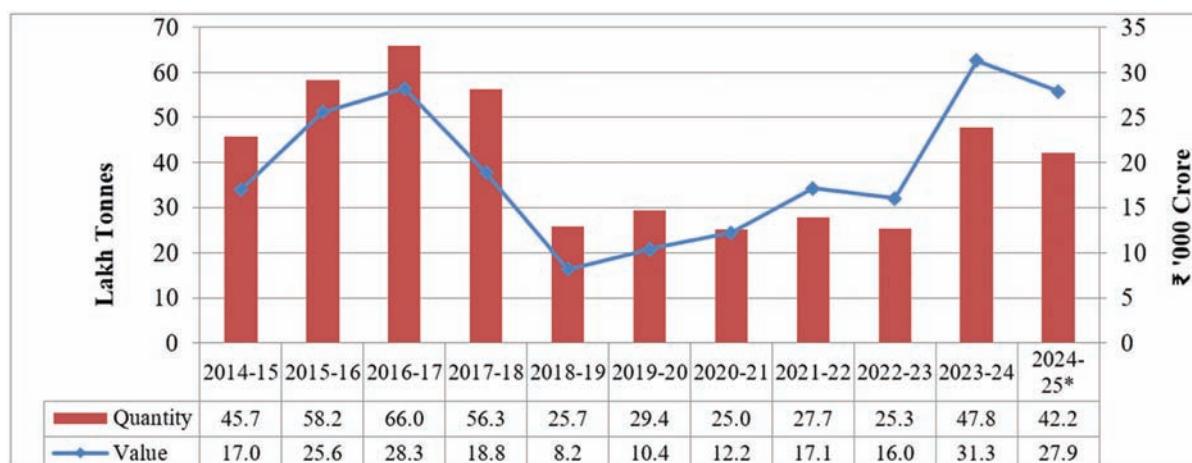
Source: Food and Agricultural Organization

India's Trade

- 4.29 Chart 4.12 depicts the trends in volume and value of imports of pulses in India during the last 10 years. Imports of pulses increased significantly from 45.7 lakh tonnes (₹17 thousand crore) in 2014-15 to a record high of 66 lakh tonnes (₹28.3 thousand crore) in 2016-17. Government launched series of initiatives to increase domestic production, ensure remunerative prices to farmers, higher import duties, quantitative restrictions, ban on imports, removal of export restrictions etc. to reduce dependence on imports. As a result, imports declined to 25.7 lakh tonnes in 2018-19, while registering the lowest import value at ₹8.2 thousand crore. Subsequently, between 2018-19 and 2022-23, although volume of imports remained moderately stable at about 26-27 lakh tonnes, but value of imports nearly doubled due to higher international prices of pulses. However, pulses imports significantly increased to 47.8 lakh tonnes (₹31.3 thousand crore) in 2023-24, almost twice the imports in 2022-23 due to fall in domestic production consequent to adverse weather conditions and rising demand. In order to increase supply and contain surge in domestic prices, Government eased import restrictions and took steps to ensure remunerative prices to farmer. It is worth noting that in 2024-25 (till November 2024), 42.2 lakh tonnes of pulses valued at ₹27.9 thousand crore have been imported as compared to 24.1 lakh tonnes valued at ₹17.5 thousand crore during the corresponding period in 2023-24.



Chart 4.12: India's Imports of Pulses, 2014-15 to 2024-25



Note : * For 2024-25 (April- November)

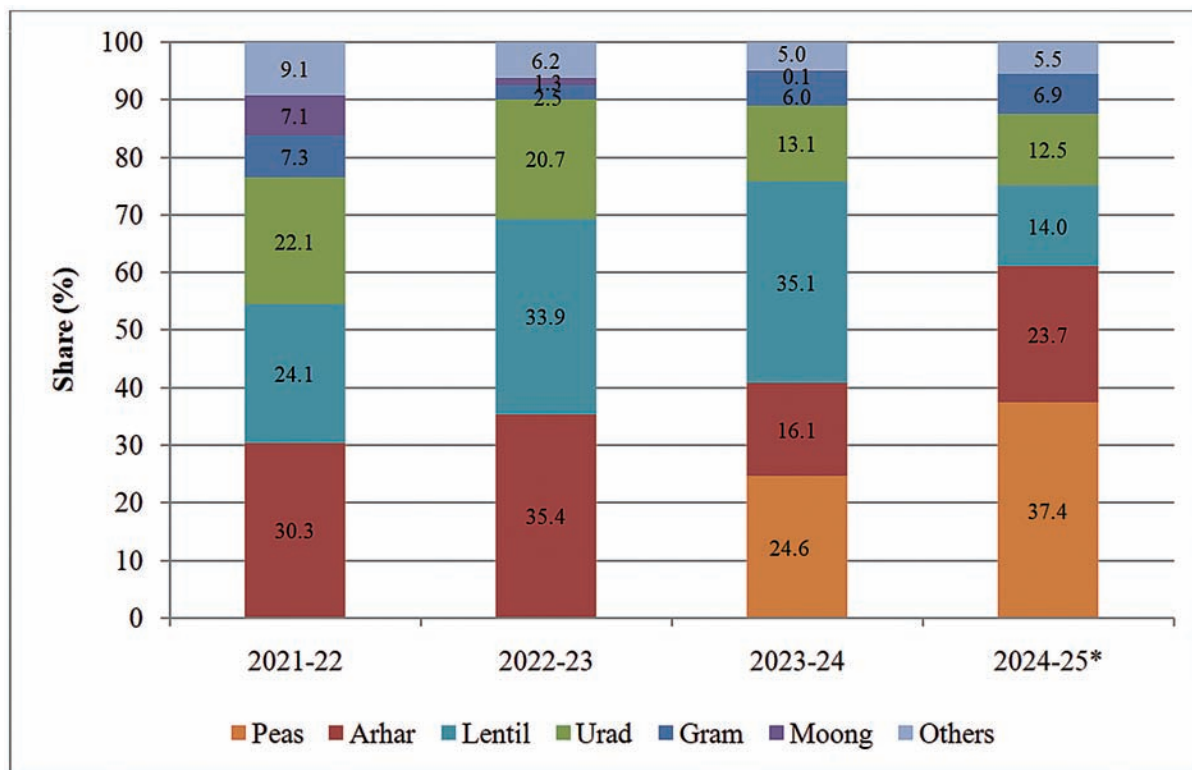
Source: Directorate General of Commercial Intelligence and Statistics, Ministry of Commerce and Industry

- 4.30 India achieved some success in gram and moong production between 2013-14 and 2022-23 due to development of short-duration varieties and concerted efforts by Government to ensure remunerative price to farmers. Imports of yellow/white peas and gram declined substantially from 2017-18 till 2022-23. However, there has been reversal in 2023-24 and imports of pulses, particularly peas and lentil increased significantly. Further, India imported larger quantity of tur/arhar and imports of peas reached seven-year high in 2024-25. Such unrestricted import of cheap pulses will have adverse impact on domestic prices and farmers and calls for a review of trade policy regarding pulses. Further, in order to reduce dependence on imports, India also needs to encourage domestic production of pulses, particularly tur/arhar and urad in *kharif* season at massive scale along with strengthening of procurement mechanism. In the Union Budget 2025-26, Government has announced six-year 'Mission for Aatmanirbharta in Pulses' with special focus on tur/arhar, urad and lentil with an allocation of ₹1,000 crore for development and availability of climate resilient seeds, improving yields and post-harvest warehousing facilities and assuring remunerative prices to farmers through assured procurement.
- 4.31 Chart 4.13 shows the composition of India's pulses imports from 2021-22 to 2024-25 (till November 2024). India is highly dependent on imports for meeting domestic requirement of tur/arhar, lentil and urad. The combined share of tur/arhar, lentil and urad in total pulses imports (in volume) was 74.1 percent in TE2023-24. The share of tur in total pulses imports increased from 30.3 to 35.4 percent in 2022-23 and lentil from 24.1 to 33.9 percent. However, Government allowed duty-free import of yellow peas in December 2023, which resulted in steep rise in imports of peas and accounted for 24.6 percent share in 2023-24, which increased to 37.4 percent in 2024-25 (April-November) due to substantially lower price of peas compared to other pulses. During 2023-24, share of urad and tur declined significantly, but share of lentil and gram increased. In 2024-25 (till November 2024), with 37.4 percent, peas had the largest share, followed by tur (23.7%), lentil (14%) and urad (12.5%). Concerted efforts to boost domestic production of key pulses like lentil, tur and urad will help India to reduce dependence on imports.



Chart 4.13: Changing Composition of India's Pulses Imports, 2021-22 to 2024-25

(in volume)



Note : * For 2024-25 (April- November)

Source: Directorate General of Commercial Intelligence and Statistics, Ministry of Commerce and Industry

- 4.32 India imports lentil mainly from Australia and Canada, tur from Mozambique and Myanmar and urad from Myanmar. Major import origins for various pulses imported by India during 2023-24 and 2024-25 (till November 2024) are given in Annex Table 4.5.

Trade Policy

- 4.33 India's pulses import policy is based on several factors like ensuring availability of pulses to consumers at affordable prices, remunerative prices to the farmers and increasing domestic production to reduce import dependence. Therefore, India calibrates pulses trade policy through tariffs and other restrictions both on imports and exports. Major policy changes in pulses trade policy since 2017 are given in Annex Table 4.10 and changes during the last three years are given below.

- Lentil:** From 12th February, 2022, Agriculture Infrastructure and Development Cess (AIDC) and import duty on lentil was reduced to 'Zero' (import from any country except USA). The duty on imports from U.S.A was reduced from 30 percent to 20 percent and was applicable till 30th September, 2022, which was extended further till 31st March, 2023 and then upto 31st March, 2024. *Vide* notification dated 21st December, 2023, it has now been extended till 31st March, 2025.

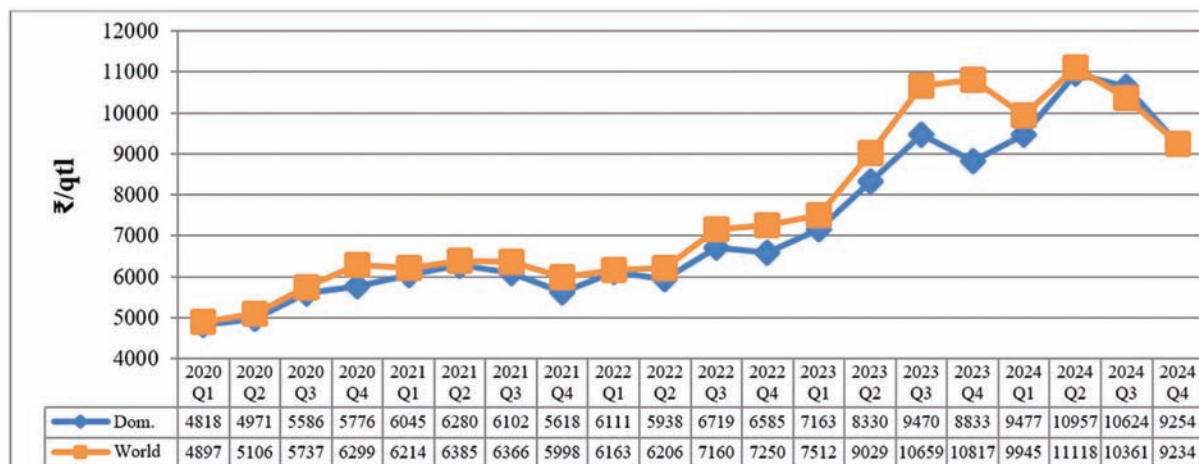


- **Tur and Urad:** The import policy for tur and urad was revised from “Restricted” to “Free” up to 31st October, 2021, *vide* notification dated 15th May, 2021, which was extended to 31st March, 2023, then to 31st March, 2024 and 31st March, 2025. The “Free” import policy of tur has now been extended up to 31st March, 2026 *vide* notification dated 20th January, 2025.
- **Yellow Peas:** *Vide* notification dated 8th December, 2023, import of yellow peas was made “Free” up to 31st March, 2024, which was further extended up to 28th February, 2025.
- **Moong:** *Vide* notification dated 11th February, 2022, import policy of moong was revised from “Free” to “Restricted” and since then, moong imports continue to be “Restricted” in India.

Comparative Trends in Prices

- 4.34 Domestic and international prices of tur during the period 2020 to 2024 are shown in Chart 4.14. Both prices moved in unison during the period, with international prices marginally higher than the domestic prices. Moderate increase was seen in both prices from 2020(Q1) to 2022(Q2), but a significant and steady rise in prices was recorded from 2022(Q2) to 2024(Q2), barring some temporary dips in between. During this period, both domestic and international prices rose significantly, hitting the peak in 2024(Q2), but declined during the last two quarters due to anticipated higher domestic production and imports.

Chart 4.14: Domestic and International Prices of Tur, 2020 to 2024



Note : Domestic prices in the present report may slightly vary from the earlier reports, as data from UPAg has been used in place of AGMARKNET

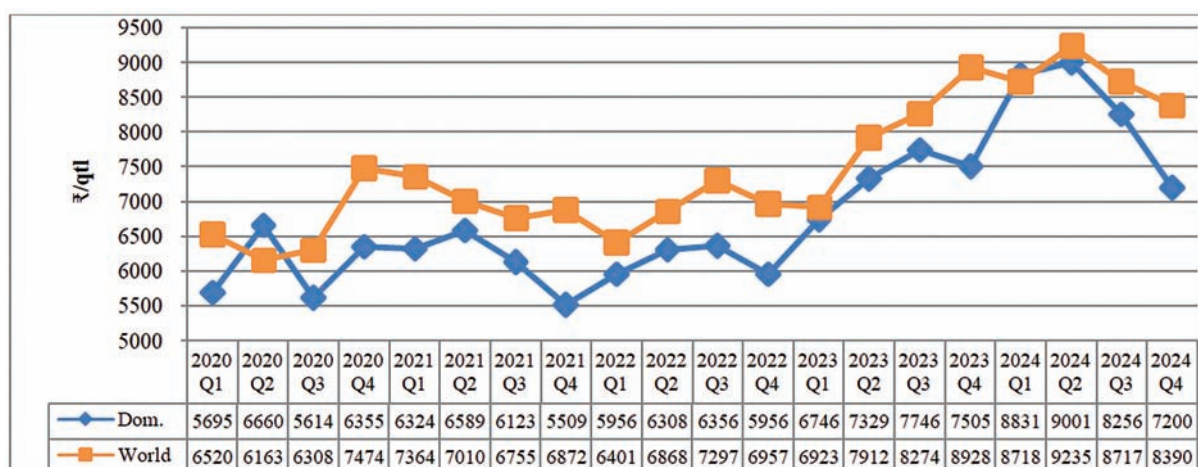
Source: 1. UPAg, Ministry of Agriculture and Farmers Welfare, for computing weighted average domestic wholesale prices
2. Agriwatch for international prices

- 4.35 The trends of domestic and international prices of urad during the period 2020 to 2024 are shown in Chart 4.15. It can be seen that international prices of urad continued to be higher than domestic prices for the entire period, except 2020(Q2) and 2024(Q1). International prices of urad increased significantly (18.5%) in 2020(Q4) over previous quarter, but declined during next three quarters. Significant rise of 33.4 percent was observed in international prices from 2023(Q1) to 2024(Q2), while domestic prices



of urad remained volatile during 2020 to 2022, but recorded sharp rise of 51.1 percent between 2022(Q4) and 2024(Q2). However, both the prices witnessed downward trend during the last two quarters, with more steep fall in domestic prices.

Chart 4.15: Domestic and International Prices of Urad, 2020 to 2024



Note: Domestic prices in the present report may slightly vary from the earlier reports, as data from UPAg has been used in place of AGMARKNET

Source: 1. UPAg, Ministry of Agriculture and Farmers Welfare, for computing weighted average domestic wholesale prices

2. Agriwatch for international prices

Oilseeds and Oilseed Products

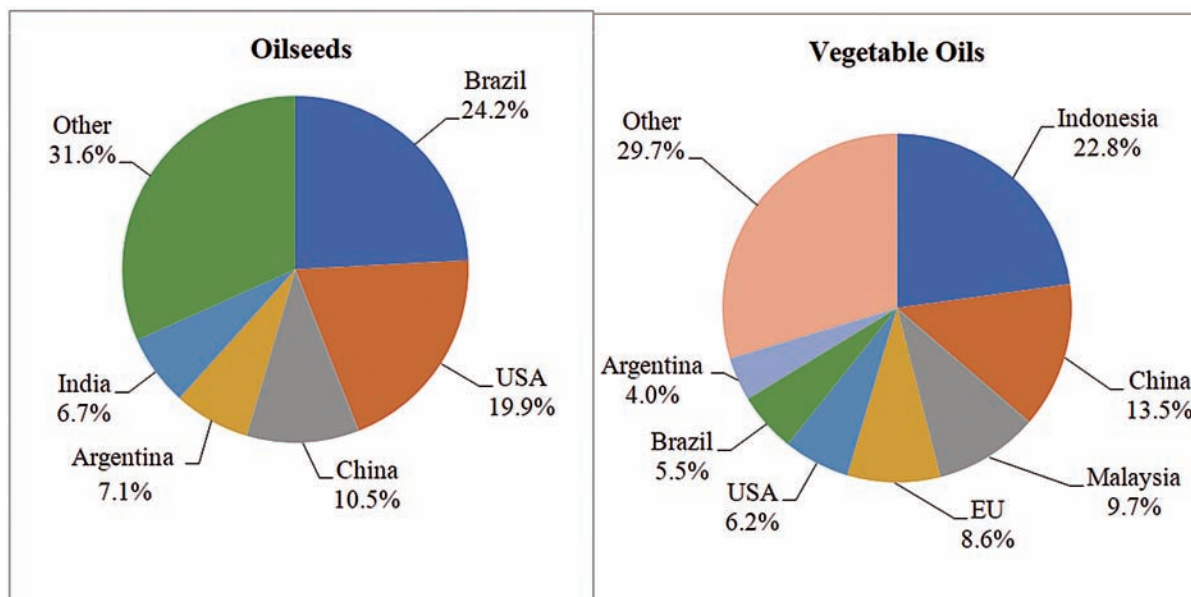
Global Production and Trade

- 4.36 Major oilseeds produced globally are soybean, rapeseed and mustard (R&M), sunflower seed, groundnut and cotton seed. In addition, palm cultivation is a dominant source of edible oils and accounts for more than one-third of world vegetable oil production. World production of major oilseeds in 2023-24 was about 657 million tonnes, 3 percent higher than 2022-23 and 31.2 percent of this was traded. With a dominant share of 59.4 percent, soybean was the largest produced oilseed in the world, followed by R&M (13.4%), sunflower (8.7%), groundnut (7.9%) and cotton (6.5%) in TE2023-24.
- 4.37 The share of major producers of oilseeds in TE2023-24 is shown in Chart 4.16. Brazil accounted for 24.2 percent of total production of oilseeds, followed by USA (19.9%), China (10.5%), Argentina (7.1%) and India (6.7%). Top three producers, Brazil, USA and China, accounted for more than half of the global production of oilseeds, while top two exporters, Brazil and USA, accounted for 75.3 percent of global exports and share of top two importers, China (55.8%) and EU (11.3%) was over 67 percent in TE2023-24.
- 4.38 Global production of major vegetable oils stood at 216 million tonnes in TE2023-24, of which 38.2 percent was traded. Palm oil had the largest share (34.9%) in total vegetable oils production, followed by soybean oil (28.2%), rapeseed oil (15%) and sunflower oil (9.8%). Indonesia was the largest producer of vegetable oils with a share of 22.8 percent, followed by China (13.5%), Malaysia (9.7%),



EU (8.6%) and USA (6.2%) in TE2023-24. It is important to note that with a share of 51.2 percent, Indonesia and Malaysia accounted for more than half of global exports of vegetable oils in TE2023-24, while India continued to be the largest importer with a share of 19.7 percent, followed by EU (11.9%) and China (11.5%) in TE2023-24.

Chart 4.16: Major Producers of Oilseeds and Vegetable Oils, TE2023-24



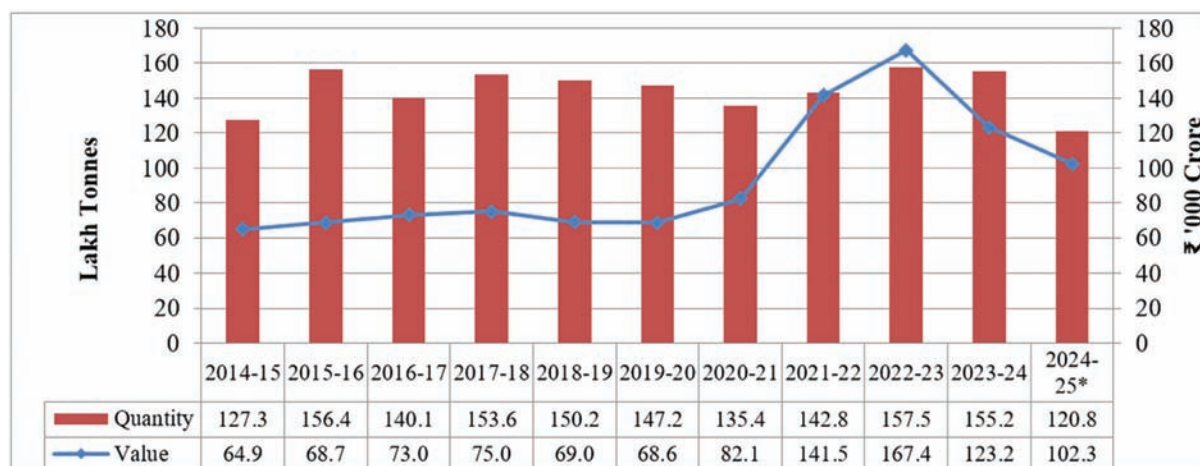
Source: United States Department of Agriculture

India's Trade

- 4.39 Domestic production of edible oils is not sufficient to meet the rising demand in the country and about 60 percent of domestic demand is met through imports. The trends in India's imports of edible oils from 2014-15 to 2024-25 (till November 2024) are shown in Chart 4.17. It may be observed that the volume of imports of edible oils significantly increased by 22.9 percent between 2014-15 and 2015-16, driven by reduced domestic production and cheaper international prices, particularly that of palm oil. It is worth noting that in 2016-17, although volume of imports declined by 10.5 percent, but value of imports surged by 6.4 percent over 2015-16. Subsequently, after registering a rise in 2017-18, volume of imports declined by 11.9 percent between 2017-18 and 2020-21, caused by COVID-19 induced demand shock and supply disruptions, but import bill increased by 9.5 percent during the period. Thereafter, most significant rise in value of imports of edible oils was noticed during 2020-21 to 2022-23, when volume of imports increased by 16.3 percent, but value of imports more than doubled due to steep rise in world prices. On the contrary, between 2022-23 and 2023-24, there was marginal fall in volume of imports (-1.4%), but import bill reduced significantly (-26.4%) due to lower global prices. However, in 2024-25 (till November 2024), India has already imported 120.8 lakh tonnes of edible oils with an import bill of ₹102.3 thousand crore, which is significantly higher than 97.1 lakh tonnes with an import bill of ₹78.7 thousand crore in 2023-24 (till November 2023).



Chart 4.17: Trends in Imports of Edible Oils in India, 2014-15 to 2024-25



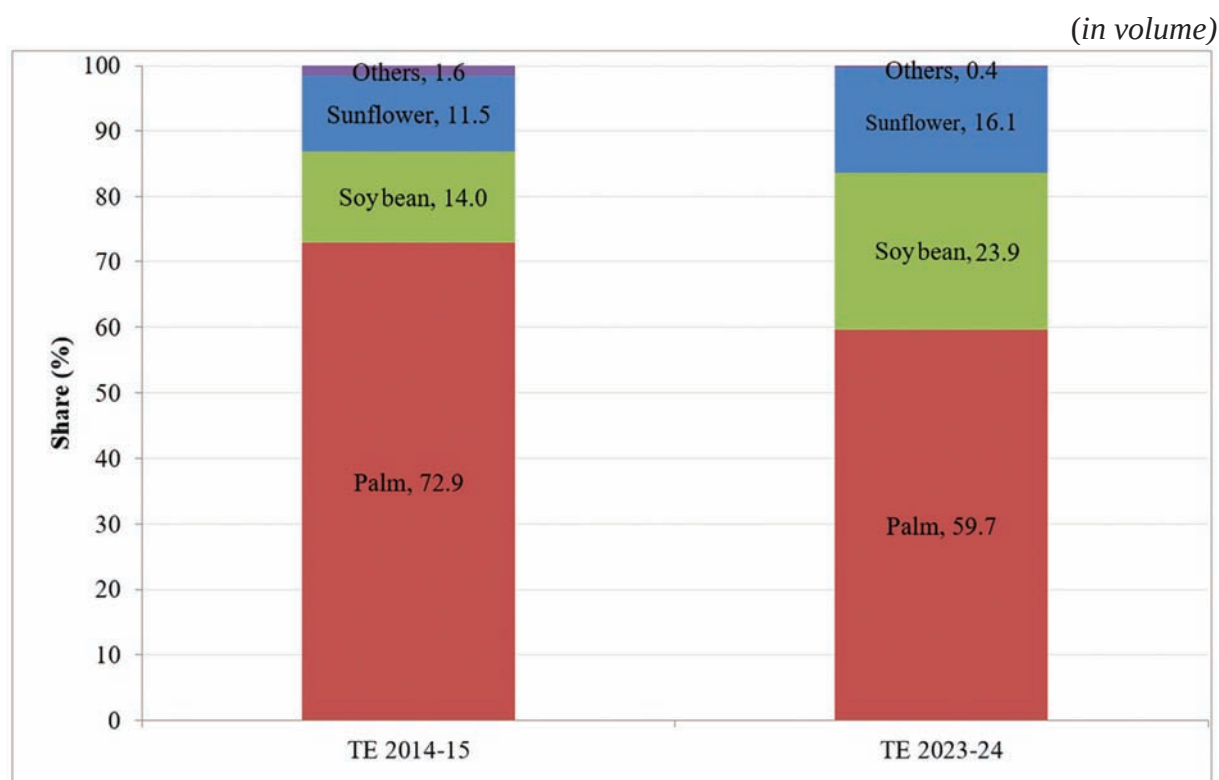
Note : * For 2024-25 (April- November)

Source: Directorate General of Commercial Intelligence and Statistics, Ministry of Commerce and Industry

- 4.40 During past few years, demand for various edible oils in India has changed due to technological, economic and policy changes across countries as well as shift in consumer preferences. Chart 4.18 depicts the changing share of major edible oils in India's imports of edible oils in TE2014-15 and TE2023-24. Although the share of palm oil has declined from 72.9 percent in TE2014-15 to 59.7 percent in TE2023-24, it still continues to be the largest imported edible oil in the country due to lower price compared to other vegetable oils and its expanding uses beyond traditional food like personal care, cosmetics, pharmaceutical products, etc. Among the other competing oils, share of soybean oil has significantly increased from 14 percent to 23.9 percent, while share of sunflower oil has increased from 11.5 percent to 16.1 percent. It is observed that the share of soft oils in total imports has also increased sharply during the last decade.



**Chart 4.18: Changing Composition of India's Import of Major Edible Oils,
TE2014-15 and TE2023-24**



Source: Directorate General of Commercial Intelligence and Statistics, Ministry of Commerce and Industry

- 4.41 In order to increase the domestic oilseeds production and achieve self-reliance in edible oils, Government approved the National Mission on Edible Oils–Oilseeds (NMEO–Oilseeds) on 3rd October, 2024, with a financial outlay of ₹10,103 crore. The Mission will be implemented over a seven-year period i.e. from 2024-25 to 2030-31 with a target to increase oilseed production from 39 million tonnes in 2022-23 to 69.7 million tonnes by 2030-31. The key focus of the Mission will be to enhance the production of primary oilseed crops such as R&M, groundnut, soybean, sunflower, and sesamum, as well as increasing collection and extraction efficiency from secondary sources like cottonseed, rice bran, and tree borne oils. In addition, Government has also undertaken a series of measures to enhance domestic production of palm oil under a Centrally Sponsored Scheme named National Mission on Edible Oils–Oil Palm (NMEO–OP) in 2021, with an outlay of ₹11,040 crore. The objectives of NMEO–OP are to enhance area under oil palm cultivation and to increase palm oil production in the country.

Soybean Complex

Global Production and Trade

Soybean

- 4.42 During 2023-24, global soybean production was 395 million tonnes, 4.4 percent higher than the previous year, as per USDA. Soybean production is projected to rise in major producing nations like



Brazil, USA and Argentina in 2024-25. With 39.3 percent of global share, Brazil was the largest producer of soybean in TE2023-24, followed by USA (31%), Argentina (10.3%), China (5.1%) and India (3.2%). Globally, about 45 percent of soybean production was traded in 2023-24 and Brazil and USA were the largest exporters of soybean with a combined share of 86.8 percent of world exports. China, with a share of 61.2 percent of global imports, was the largest importer of soybean.

Soybean Oil

- 4.43 World production of soybean oil was estimated at 62.8 million tonnes in 2023-24, of which 17.9 percent was traded globally. China, the largest importer of soybean seed, was the largest producer of soybean oil, accounting for 28 percent of global production in TE2023-24, followed by USA (19.8%), Brazil (17.4%) and Argentina (11.5%). In TE2023-24, India's share in global production of soybean oil was only 3 percent, but was the largest importer with a share of 35 percent, followed by Bangladesh (5.9%) and EU (5.1%).

Soybean Meal

- 4.44 World production of soybean meal was 252.1 million tonnes in TE2023-24, of which 27.1 percent was traded, as per USDA. China has been consistently the largest producer of soybean meal, with a global share of 29.9 percent, followed by USA (19%), Brazil (16.2%) and Argentina (10.9%). Among the major exporters of soybean meal, Argentina (34.3%), Brazil (30.5%) and USA (19.1%) together accounted for 83.9 percent of global exports, while EU was the largest importer of soybean meal with 24.5 percent share, followed by Vietnam (8.1%) and Indonesia (8%).

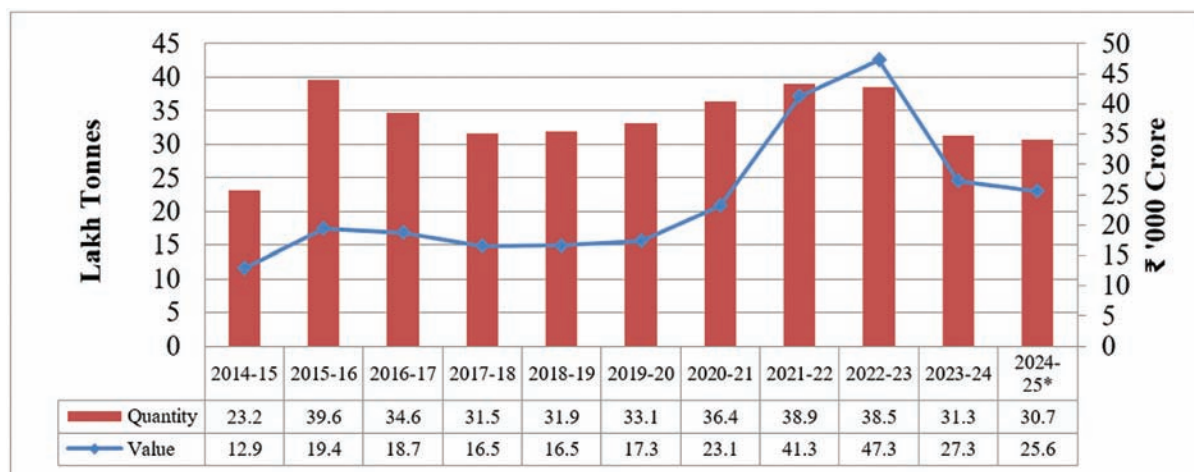
India's Trade

Soybean Oil

- 4.45 Over the years, India has been a leading importer of soybean oil in the world and the trends in import volume and value of soybean oil during the period 2014-15 to 2024-25 (till November 2024) are shown in Chart 4.19. It is observed that soybean oil imports, which were 23.2 lakh tonnes in 2014-15 reached record high at 39.6 lakh tonnes in 2015-16. Although, imports subsequently moderated in 2016-17 and 2017-18, but rebounded steadily thereafter and were recorded at 38.9 lakh tonnes in 2021-22, second highest during the last 10 years. It is pertinent to note that in 2022-23, volume of imports slightly declined (1%) over 2021-22, but value of imports increased substantially (14.5%) due to increase in world prices. The world prices of soybean oil increased from ¹ 12,609 per quintal in 2022(Q1) to ₹14,562 per quintal in 2022(Q2) (Chart 4.22). Imports declined both in volume (-18.6%) and value (-42.3%) term in 2023-24. However, in 2024-25 (till November 2024), India has already imported 30.7 lakh tonnes (valued at ₹25.6 thousand crore) of soybean oil, significantly higher than 23.9 lakh tonnes (valued at ₹21.2 thousand crore) during the corresponding period in previous year.



Chart 4.19: India's Import of Soybean Oil, 2014-15 to 2024-25



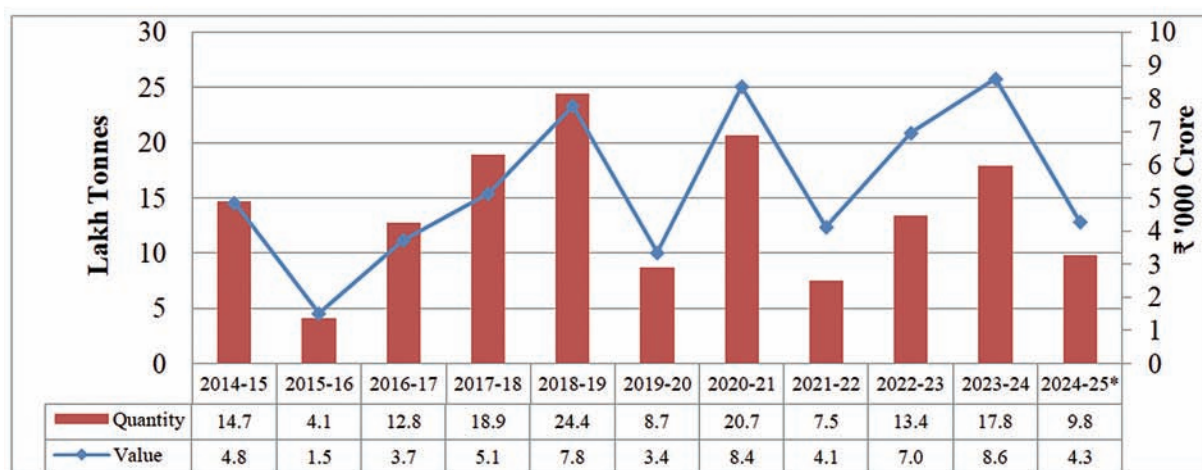
Note : * For 2024-25 (April- November)

Source: Directorate General of Commercial Intelligence and Statistics, Ministry of Commerce and Industry

Soybean Meal

- 4.46 In the recent years, India's growing domestic demand for soybean meal as plant based affordable protein food as well as animal feed and high domestic prices have reduced the potential for exports of soybean meal. Chart 4.20 depicts exports of soybean meal from India. It can be seen that soybean meal exports from India significantly declined from 14.7 lakh tonnes in 2014-15 to 4.1 lakh tonnes in 2015-16 due to lower soybean production in the country. Thereafter, volume of exports increased nearly six times from 2015-16 to 2018-19 and reached at 24.4 lakh tonnes after opening up of Iranian market for Indian soybean meal. Although severe volatility was noticed between 2019-20 and 2021-22, but exports, both in volume and value term, increased during the last three years and more than doubled during 2021-22 to 2023-24. In 2024-25 (till November 2024), India exported 9.8 lakh tonnes (₹4.3 thousand crore) of soybean meal, compared to 9.3 lakh tonnes (₹4.6 thousand crore) during the corresponding period in the previous year.

Chart 4.20: India's Export of Soybean Meal, 2014-15 to 2024-25



Note : * For 2024-25 (April- November)

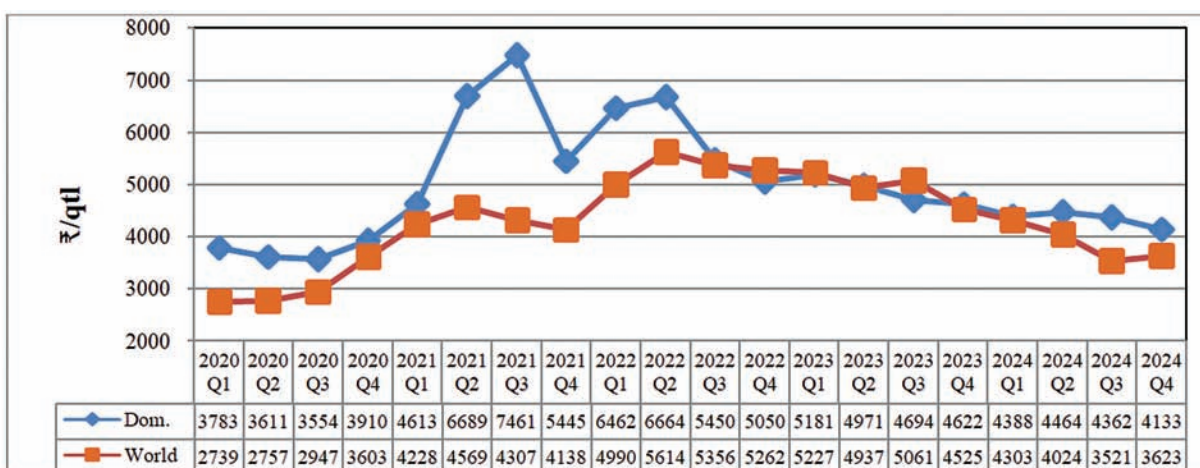
Source: Directorate General of Commercial Intelligence and Statistics, Ministry of Commerce and Industry



Comparative Price Trends

- 4.47 The comparative price trends of domestic and international prices of soybean from 2020 to 2024 are shown in Chart 4.21. It can be seen that domestic prices of soybean remained higher than the international prices during the entire period, except 2022(Q4) and 2023(Q1&Q3). After witnessing a marginal decline between 2020(Q1) and 2020(Q3), domestic prices recorded more than two-fold increase from 2020(Q3) to 2021(Q3). This upward movement in prices was primarily on account of lower output coupled with strong demand for soybean meal from domestic poultry industry. Subsequently, Government allowed import of soybean meal from August 2021, which resulted in sharp decline in domestic prices in 2021(Q4), while international prices increased by 66.8 percent between 2020(Q1) and 2021(Q2). A steady decline in both domestic and global prices was observed after 2022(Q2). Domestic price, which marginally improved in 2024(Q2), declined during last two quarters of 2024, but remained above world prices.

Chart 4.21: Domestic and International Prices of Soybean, 2020 to 2024



Note : Domestic prices for in the present report may slightly vary from the earlier reports, as data from UPAG has been used in place of AGMARKNET

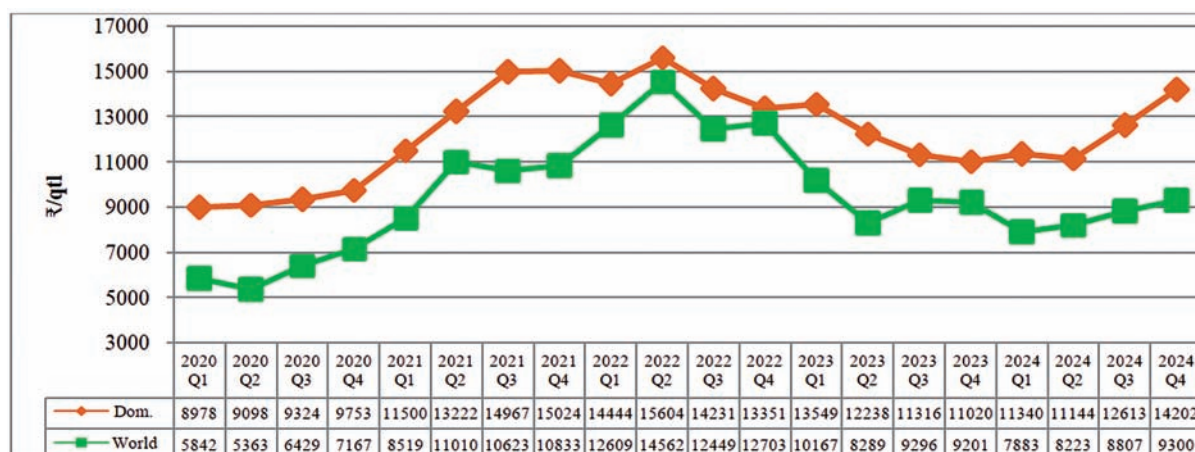
Source: 1. UPAG, Ministry of Agriculture and Farmers Welfare, for computing weighted average domestic wholesale prices

2. World Bank Pink Sheet for international prices

- 4.48 Chart 4.22 depicts the comparative price trends of soybean oil in domestic and world markets for the period 2020 to 2024. It is observed that domestic soybean oil prices consistently remained higher than the international prices for the entire period. With minor fluctuations, 2020(Q1) to 2022(Q2) period was marked by notable rise in both domestic (74%) and international (149%) prices. The upward momentum in prices was attributed to fall in global supply owing to climate, geopolitical and structural factors. On the contrary, soybean oil prices, both in domestic and international markets, witnessed downward trajectory after 2022(Q2), consequent to increase in global supply, subdued demand and reduction in import duty by India. However, from 2024(Q1) to 2024(Q4), both prices witnessed a rising trend due to lower global supply, high transportation costs and increase in import duty in India.



Chart 4.22: Domestic and International Prices of Soybean Oil, 2020 to 2024

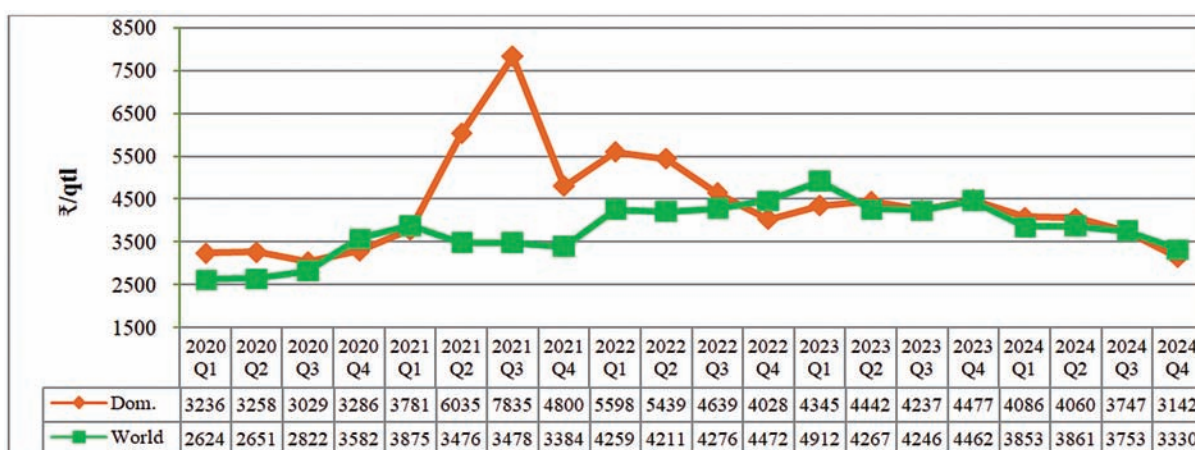


Source: 1. Agricultural Marketing Division, Ministry of Agriculture and Farmers Welfare for domestic wholesale prices

2. World Bank Pink Sheet for international prices

4.49 In the recent years, India's growing poultry and livestock sectors have played a dominant role in influencing domestic soybean meal prices. Chart 4.23 shows trends in prices of soybean meal in domestic and international markets for the period 2020 to 2024. It can be seen that during this period, international soybean meal prices remained below the domestic prices, except a few quarters. The domestic prices remained stable from 2020(Q1) to 2020(Q2), but more than doubled from 2020(Q3) to 2021(Q3) and the prices reached all-time high at ₹7,835/quintal in 2021(Q3), driven by rising domestic demand. However, the price rise could not sustain for long and domestic meal prices fell sharply by 38.7 percent between 2021(Q3) and 2021(Q4), when Government allowed imports of 12 lakh tonnes crushed and de-oiled Genetically Modified (GM) soya cake (only Non-living organism) upto 31st October, 2021. On the other hand, international prices increased moderately from 2020(Q1) to 2021(Q1), but fell slightly thereafter till 2021(Q4). Both the prices increased between 2021(Q4) and 2022(Q1), but followed a moderately falling trend afterward due to higher global production and supplies specifically from Argentina, Brazil and USA.

Chart 4.23: Domestic and International Prices of Soybean Meal, 2020 to 2024



Source: 1 Solvent Extractors Association for domestic wholesale prices

2. World Bank Pink Sheet for international prices



Groundnut

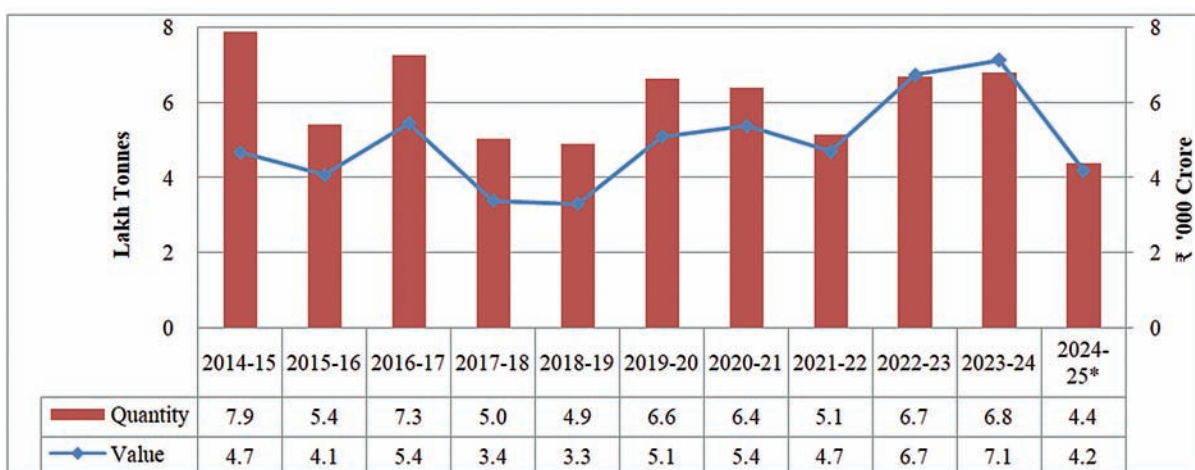
Global Production and Trade

- 4.50 As per USDA, global production of groundnut was estimated at 49.5 million tonnes in 2023-24, out of which 9.2 percent was traded. Due to rising consumption of groundnut as nut for snacking, only 6.1 million tonnes of groundnut oil was produced globally in 2023-24. Due to high domestic consumption of groundnut oil in producing countries, only 6.3 percent groundnut oil was traded globally. In TE2023-24, China was the major producer of groundnut oil in the world with 50.4 percent share, followed by India (19.3%). China was also the largest importer of groundnut oil with 69.8 percent share, followed by EU (14.1%), while India was the largest exporter of groundnut oil (36.9%) in TE2023-24. It is worth noting that high domestic demand for groundnut oil in India has resulted in lower exports in the recent years.

India's Trade

- 4.51 India is one of the leading exporter of groundnut in the world and trends and pattern of groundnut exports from India are shown in Chart 4.24. Groundnut exports kept fluctuating during the last decade due to volatility in production caused by erratic weather patterns. India exported on an average about 6.2 lakh tonnes of groundnut worth ₹5 thousand crore during the period 2014-15 to 2023-24. Groundnut exports recorded significant growth both in volume (32.4%) and value (51.9%) from 2021-22 to 2023-24 due to higher international prices. India's main export destinations in 2023-24 were South-East Asian nations and Gulf countries like Indonesia, Vietnam, Philippines, Malaysia, Thailand and UAE. In 2024-25 (till November 2024), India exported 4.4 lakh tonnes of groundnut worth ₹4.2 thousand crore compared to 3.8 lakh tonnes worth ₹4.2 thousand crore during the corresponding period in previous year. India has ample export opportunity due to growing demand for plant-based proteins, peanut butter and meal.

Chart 4.24: India's Export of Groundnut, 2014-15 to 2024-25



Note : * For 2024-25 (April- November)

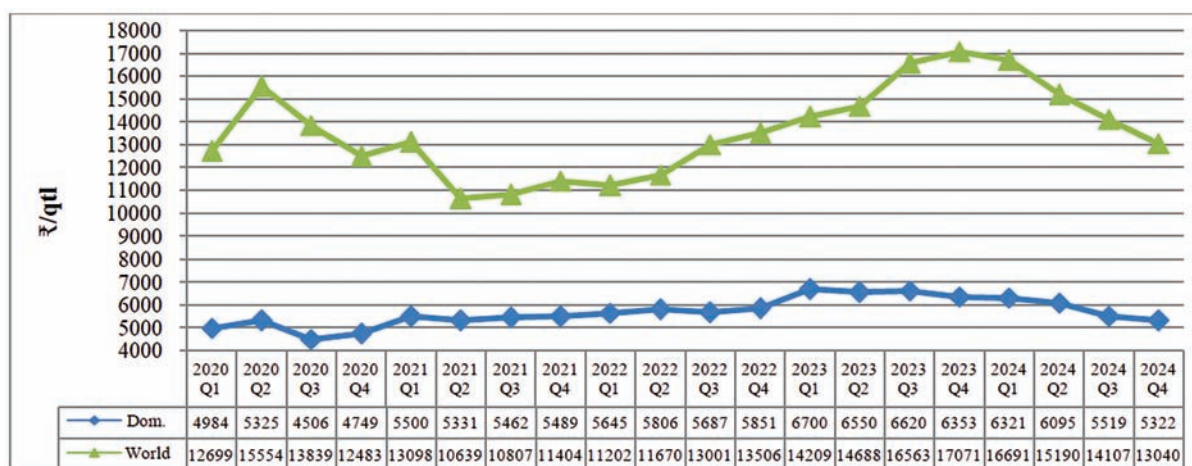
Source: Directorate General of Commercial Intelligence and Statistics, Ministry of Commerce and Industry



Comparative Price Trends

- 4.52 Chart 4.25 shows domestic and international prices of groundnut during the period 2020 to 2024. It is observed that domestic prices of groundnut remained substantially lower than volatile international prices during the entire period. The domestic prices moderately increased from 2020(Q1) to 2023(Q1), but followed a downward trajectory thereafter, with an average price of ₹5,691/quintal during the past five years. On the contrary, international prices increased by 22.5 percent between 2020(Q1) and 2020(Q2) and then steeply declined by 31.6 percent from 2020(Q2) to 2021(Q2). It is pertinent to note that the international prices of groundnut not only increased by 60.5 percent from 2021(Q2) to 2023(Q4), but also hit the historical high at ₹17,071/quintal in 2023(Q4). However, from 2023(Q4) onward both domestic and international prices showed a declining trend, but decline in global prices was much steeper.

Chart 4.25: Domestic and International Prices of Groundnut, 2020 to 2024



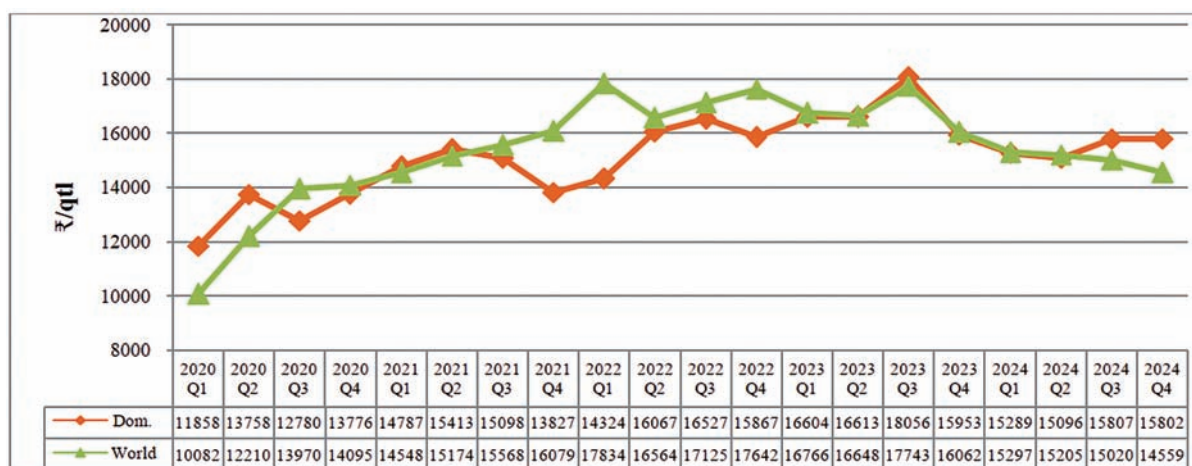
- Note: 1. Domestic prices in the present report may slightly vary from the earlier reports, as data from UPAg has been used in place of AGMARKNET
 2. Groundnuts (U.S.), Runners 40/50, CFR N.W. Europe. Europe beginning January 1999; previously (US), Runners 40/50 shelled basis, c.i.f. Rotterdam
 3. The data for international prices has been taken from World Bank Pink Sheet for the current report for the period 2020(Q1) to 2024(Q4)

Source: 1. UPAg, Ministry of Agriculture and Farmers Welfare, for computing weighted average domestic wholesale prices
 2. World Bank Pink Sheet for international prices

- 4.53 The trends in domestic and international prices of groundnut oil for the period 2020 to 2024 are presented in Chart 4.26. It is observed that except some minor divergences, both domestic and international prices have moved in unison. From 2020(Q1) to 2022(Q1), international prices of groundnut oil increased by 76.9 percent, remained elevated till 2023(Q3), but moderated thereafter. However, domestic prices of groundnut oil increased by 52.3 percent from 2020(Q1) to 2023(Q3), but remained subdued afterwards. It is worth noting that domestic prices have remained below the international prices since 2020(Q3), except 2021(Q1&Q2), 2023(Q3) and 2024(Q3&Q4), indicating competitiveness of Indian groundnut oil in the global market.



Chart 4.26: Domestic and International Prices of Groundnut Oil, 2020 to 2024



Note: Groundnut oil, Dutch Refined Ground Nut Oil A/O, Ex Tank Rotterdam, beginning December 2020; January 1999–November 2020, U.S. crude, FOB South-East; previously any origin, c.i.f. Rotterdam.

Source: 1. Agricultural Marketing Division, Ministry of Agriculture and Farmers Welfare for domestic wholesale prices at Mumbai centre

2. World Bank Pink Sheet for international prices

Sunflower

Global Production and Trade

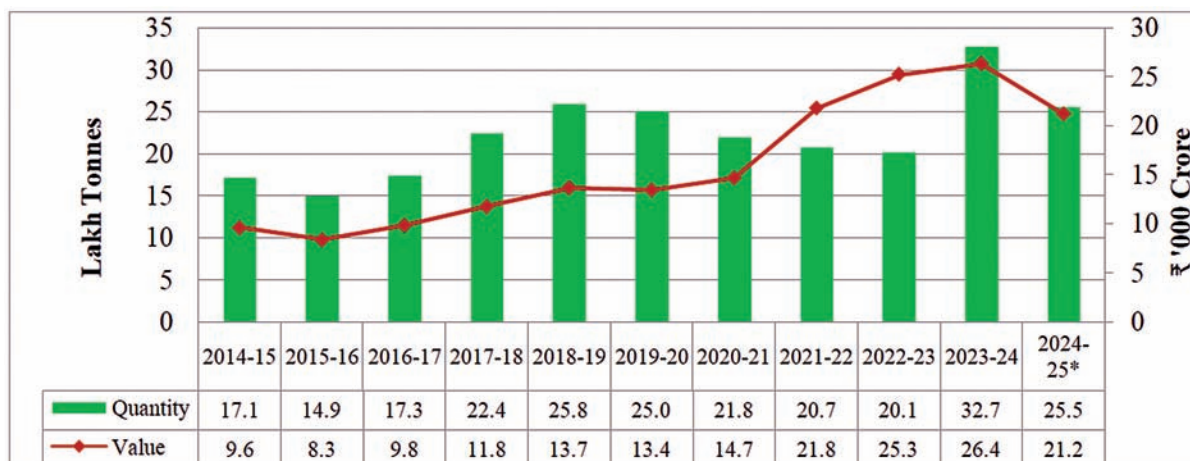
- 4.54 World production of sunflower seed was 55.2 million tonnes in TE2023-24, out of which about 6.3 percent was traded, as per USDA. Russia (29.5%) and Ukraine (27.3%) together accounted for more than half of the world sunflower seed production in TE2023-24. Ukraine (35.3%) was also the largest exporters, followed by EU (13.4%) and Russia (8.5%), while EU (40.2%) and Turkey (19.1%) were top importers of sunflower seed in TE2023-24.
- 4.55 Global production of sunflower oil was 21.2 million tonnes in TE2023-24 and more than 60 percent of which was traded. Ukraine (40.4%) and Russia (28.5%) accounted for nearly 70 percent of the global exports, while EU was the largest importer with 20.1 percent share, followed by Turkey (12.5%).

India's Trade

- 4.56 Chart 4.27 depicts trends in imports of sunflower oil during the period 2014-15 to 2024-25 (till November 2024). It is observed that imports of sunflower oil marginally declined between 2014-15 and 2015-16, but subsequently followed an upward trend and reached at 25.8 lakh tonne (₹13.7 thousand crore) in 2018-19 due to rising domestic demand. Although a marginal fall in volume of imports was noticed between 2018-19 and 2022-23, but import bill continued to surge during this period, driven by escalating global sunflower oil prices owing to Russia-Ukraine conflict and supply chain disruptions. It is worth noting that while most noticeable growth in volume of imports (63%) of sunflower oil was recorded in 2023-24 over previous year, import bill increased only marginally (4.4%) due to fall in sunflower oil prices in the international market. In 2024-25 (till November 2024), India has so far imported 25.5 lakh tonnes (valued at ₹21.2 thousand crore) of sunflower oil compared to 20.6 lakh tonnes (valued at ₹16.9 thousand crore) during the corresponding period in previous year.



Chart 4.27: India's Imports of Sunflower Oil, 2014-15 to 2024-25



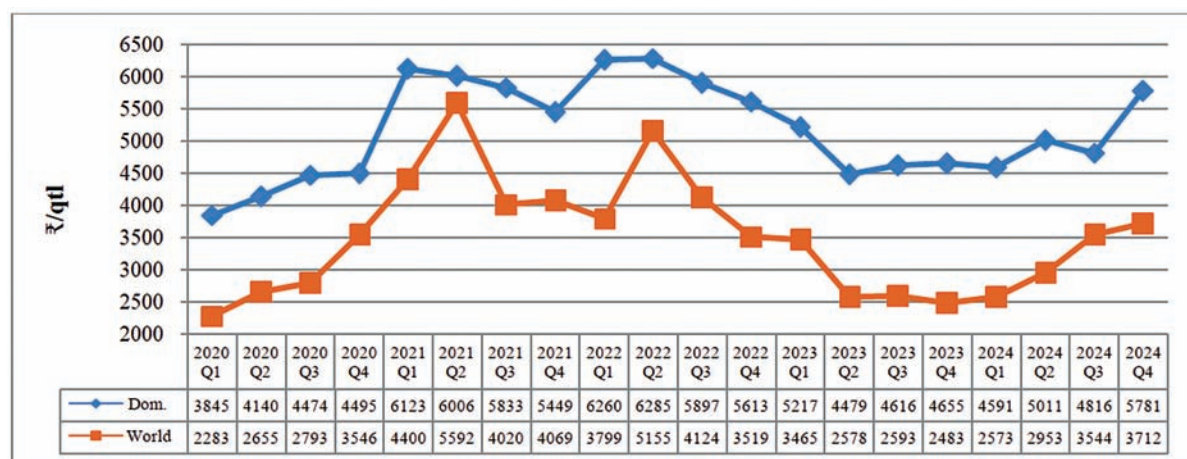
Note : * For 2024-25 (April- November)

Source: Directorate General of Commercial Intelligence and Statistics, Ministry of Commerce and Industry

Comparative Price Trends

- 4.57 Domestic prices of sunflower seed remained higher than the international prices during the period (Chart 4.28). The domestic prices followed an upward trend from 2020(Q1) to 2021(Q1), with a record increase of 36.2 percent in a single quarter between 2020(Q4) and 2021(Q1). Thereafter, the domestic prices remained elevated till 2022(Q2), but declined by 28.7 percent from 2022(Q2) to 2023(Q2). Similarly, the international prices increased sharply by 145 percent from 2020(Q1) to 2021(Q2), then exhibited high volatility till 2022(Q2) and thereafter recorded steep decline of 50 percent till 2023(Q2). The fall in sunflower seed prices was due to lower demand caused by increased competition from other edible oils, particularly palm oil and soybean oil. However, after remaining stable from 2023(Q2) to 2024(Q1), both domestic and international prices have followed steady rising trend.

Chart 4.28: Domestic and International Prices of Sunflower Seed, 2020 to 2024



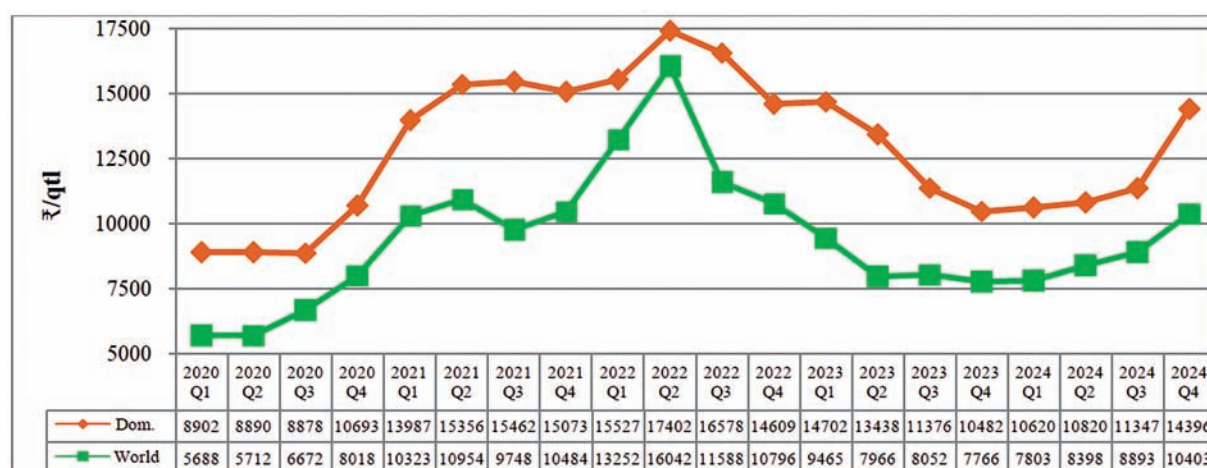
Note : Domestic prices in the present report may slightly vary from the earlier reports, as data from UPAg has been used in place of AGMARKNET

Source: 1. UPAg, Ministry of Agriculture and Farmers Welfare for computing weighted average domestic wholesale prices
2. Agriwatch for international prices (Russian, oil grade FOB Black Sea EUR/mT)



- 4.58 The domestic and international prices of sunflower oil during the period 2020 to 2024 are shown in Chart 4.29. Domestic prices remained higher than international prices during the period, but followed a similar trend. From 2020(Q3), both prices started witnessing significant increase and hit the peak in 2022(Q2). On the contrary, from 2022(Q2) till 2023(Q4), both the prices saw downward trend with more steep decline in international prices (-51.6%) than domestic prices (-39.8%). Afterwards, domestic and global prices have witnessed upward trajectory. The average mark-up of the domestic and international prices, which widened significantly to ₹4,592/quintal in 2021, has narrowed to ₹2,921/quintal in 2024.

Chart 4.29: Domestic and International Prices of Sunflower Oil, 2020 to 2024



Source: 1. Agricultural Marketing Division, Ministry of Agriculture and Farmers Welfare for domestic wholesale prices at Jamnagar centre
2. World Bank Pink Sheet for international prices

Trade Policy of Edible Oils

- 4.59 Adverse weather conditions, supply chain disruptions, complexity of price regulation, growing use of biofuels and availability of alternative oils significantly impacted the supply and price of edible oils in the world market. Government of India periodically reviews trade policy related to oilseeds and edible oils in order to meet high domestic demand, to encourage domestic production, to ensure remunerative prices to the oilseed growers and to stabilize domestic edible oil prices. The changes in trade policy of edible oils since 2017 are given in Annex Table 4.11 and a brief summary of recent changes in import policy of edible oils is given below:

- Due to rise in world and domestic prices of most edible oils in the first-half of 2021, Government of India reduced effective applicable import duty on crude soybean and sunflower oil from 35 percent to 27.5 percent and on refined oils from 45 percent to 37.5 percent in August 2021, which was further reduced to 22.5 percent on crude and 32.5 percent on refined in September 2021 and to 5.5 percent on crude and 17.5 percent on refined in October 2021. Thereafter, in June 2023, the duty on refined soybean and sunflower oil was further reduced from 17.5 percent to 13.75 percent. However, despite a significant fall in the international prices of edible oils from 2023, import duty on both crude and refined oils remained at a very low level, which adversely affected



oilseeds farmers and industry in the country. Therefore, Government increased the duty on crude soybean and sunflower oil from 5.5 percent to 27.5 percent and on refined oils from 13.75 percent to 35.75 percent in September 2024.

- As regards palm oil, in June 2021, effective applicable import duty on crude palm oil was reduced from 32.5 percent to 27.5 percent and on refined palm oil from 45 percent to 37.5 percent. In September 2021, duty on crude palm oil was further reduced to 22.5 percent and on refined to 32.5 percent. Subsequently, duty on both crude and refined variant was reduced to 7.5 and 17.5 percent in October 2021. Thereafter, in December 2021, duty on refined oil was reduced to 12.5 percent and on crude oil to 5.5 percent in February 2022. As the lower import duty adversely affected oilseeds farmers and industry in the country, Government increased the duty on crude palm oil from 5.5 percent to 27.5 percent and on refined palm oil from 12.5 percent to 35.75 percent in September 2024.
- Although, increase in import duty is expected to benefit both farmers and refining industry in India, but large scale duty free imports of edible oils from countries like Nepal and Bangladesh under South Asian Free Trade Area (SAFTA) Agreement is negating the purpose of increase in import duties. Due to duty advantage under SAFTA, imports of refined soybean and sunflower oil from Nepal increased from 158 tonnes in August 2024 to 13,091 tonnes in October 2024 and 22,432 tonnes in November 2024, which will adversely affect domestic oilseed growers, industry as well as edible oils market. Hence, it is imperative to regulate the duty free imports of edible oils under SAFTA under Rules of Origin (ROO) and local value added content under the Free Trade Agreement (FTA) rules as Nepal is not a producer of sunflower and soybean oilseeds.

Table 4.1: Import Duty on Edible Oils w.e.f. 14th September, 2024

(in percent)

Edible Oils	Basic Custom Duty (BCD)	Agriculture Infrastructure and Development Cess (AIDC)	Social Welfare Surcharge at 10 percent of BCD plus AIDC	Effective applicable import duty
Crude Palm Oil	20	5	10	27.50
Crude Soybean Oil	20	5	10	27.50
Crude Sunflower Seed Oil	20	5	10	27.50
RBD Palmolein	32.50	Nil	10	35.75
RBD Palm Oil	32.50	Nil	10	35.75
Refined Soybean Oil	32.50	Nil	10	35.75
Refined Sunflower Oil	32.50	Nil	10	35.75

Note : The effective Most Favored Nation(MFN) import duty on crude rapeseed/cottonseed oil is 38.5 percent and refined rapeseed/cottonseed oil is 49.5 percent

Source: Central Board of Indirect Taxes and Customs (CBIC), Department of Revenue

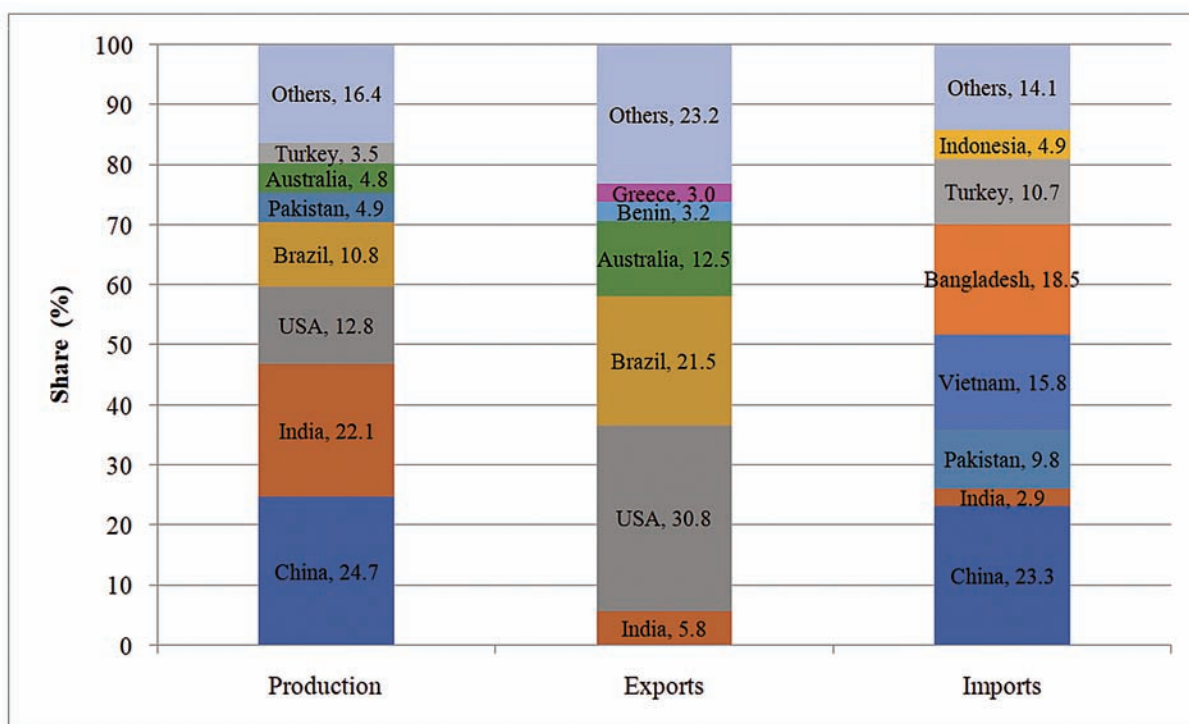


Cotton

Global Production and Trade

- 4.60 Cotton is the most commonly used natural fiber in the world and is a key component of textile industry, which makes it a leading non-food commercial crop. World cotton production increased marginally from 24.9 million tonnes in 2021-22 to 25.3 million tonnes in 2022-23. However, in 2023-24, cotton production marginally declined to 24.6 million tonnes and 39.2 percent of world cotton production was traded. Major producers, exporters and importers of cotton in the world in TE2023-24 are presented in Chart 4.30. With a share of 24.7 percent, China was the largest producer of cotton in TE2023-24, followed by India (22.1%), USA (12.8%) and Brazil (10.8%). USA was the largest exporter of cotton with a share of 30.8 percent, followed by Brazil (21.5%), Australia (12.5%) and India (5.8%). China (23.3%) continued to be the largest importer of cotton, followed by Bangladesh (18.5%) and Vietnam (15.8%).

Chart 4.30: Global Players in Cotton Markets, TE2023-24



Source: United States Department of Agriculture

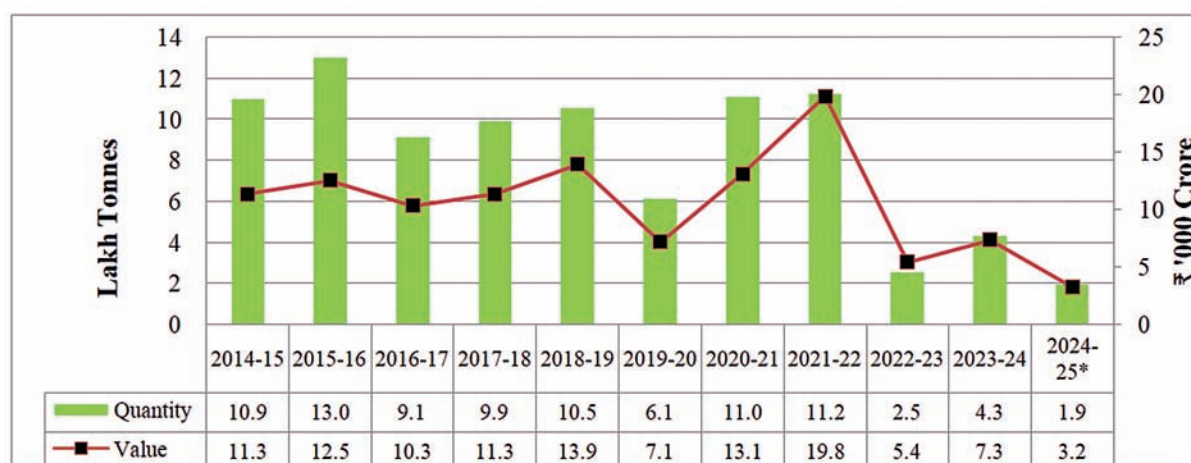
- 4.61 As per USDA, global cotton production in 2024-25 is forecast to increase by 6.6 percent over previous season, largely driven by higher production in USA, Brazil, and China, partially offset by lower production in Pakistan and India. World cotton trade is estimated to decline marginally (-3.5%) in 2024-25, primarily due to a significant decline in imports from China. Brazil is forecast to be the largest exporter for the second consecutive season.



India's Trade

- 4.62 Being one of the largest producers of cotton in the world, India accounts for about 24 percent of the global production. India is also a leading exporter of raw cotton, but Indian exports are volatile. The trends in cotton exports from India, in value and volume terms, during the period 2014-15 to 2024-25 (till November 2024), are shown in Chart 4.31. It is observed that exports of cotton from India, which were 10.9 lakh tonnes in 2014-15 declined to 6.1 lakh tonnes in 2019-20 mainly due to lower production and rise in domestic demand. On the contrary, remarkable growth in cotton exports was recorded from 2019-20 to 2021-22 when India's cotton exports rebounded to 11.2 lakh tonnes, valued at ₹19.8 thousand crore. This was mainly due to higher exports to Bangladesh, China, Vietnam and Indonesia. During 2022-23, cotton exports fell drastically to decadal lowest at 2.5 lakh tonnes, but marginally improved in 2023-24. In 2024-25 (till November 2024), India has exported 1.9 lakh tonnes of cotton worth ₹3.2 thousand crore, almost same as in corresponding period last year.

Chart 4.31: India's Exports of Cotton, 2014-15 to 2024-25



Note : * For 2024-25 (April- November)

Source: Directorate General of Commercial Intelligence and Statistics, Ministry of Commerce and Industry

Trade Policy

- 4.63 Imports of cotton, except seed of cotton, are under “Free” category. Government exempted custom duty on import of cotton from 14th April, 2022 to 30th September, 2022, to control the increase in domestic prices. The effective import duty including Social Welfare Cess (SWC) on short staple cotton is 11 percent w.e.f. 20th February, 2024, but on extra-long staple cotton it is “Nil”. At present, exports of cotton are under “Free” category and registration requirement for exports has been dispensed with from December 2014.

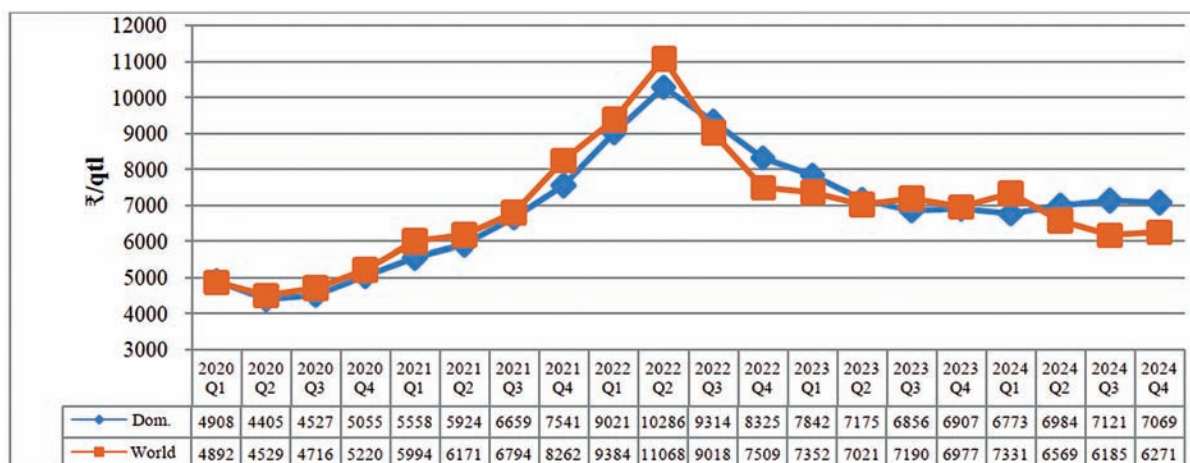
Comparative Price Trends

- 4.64 Chart 4.32 presents comparative trends in domestic and international prices of raw cotton during the period 2020 to 2024. It is evident from the Chart that domestic and international prices of raw cotton have consistently remained aligned to each other during the period, barring 2024 (Q1 to Q4). It is



observed that from 2020(Q1) to 2022(Q2), both domestic and international prices more than doubled due to steep fall in global cotton production and COVID-19 pandemic induced disruptions. However, from 2022(Q2) onward, both the prices exhibited downward trend with more sharp decline in international prices.

Chart 4.32: Domestic and International Prices of Raw Cotton, 2020 to 2024



- Note : 1. Domestic prices of cotton from AGMARKNET are not available hence domestic prices from Agricultural Marketing Division have been used, which may slightly vary from data in the earlier reports
2. Cotton (Cotton Outlook “Cotlook A index”), middling 1-3/32 inch, traded in Far East, C/F beginning 2006; previously Northern Europe, c.i.f.
3. International prices of cotton (lint) converted into Kapas at the ratio of 0.41

Source: 1. Agricultural Marketing Division, Ministry of Agriculture and Farmers Welfare for computing weighted average domestic wholesale prices

2. World Bank Pink Sheet for international prices

Global Outlook

- 4.65 As per International Grains Council (IGC), world rice production is forecast at 534 million tonnes in 2024-25, an increase of 1.9 percent over previous year, while consumption is forecast to rise by 1.7 percent. Trade is projected to remain at the same level at 57 million tonnes. World soybean production and consumption is also forecast to increase by 5.6 and 6.5 percent in 2024-25, respectively. World maize production, consumption and trade are forecast to marginally decline in 2024-25, resulting in lower carryover stocks.
- 4.66 As per USDA estimates, world production of major oilseeds such as soybean, peanut, palm kernel and cotton is forecast to increase, but production of sunflower and rapeseed will decline in 2024-25 over previous year. Exports of all oilseeds, except soybean, will decrease in 2024-25. As regards India, marginal increase in oilseeds production is estimated in 2024-25 over last year. Among major edible oils, world production of soybean, palm & palm kernel and cottonseed oil will increase in 2024-25 and global exports of soybean, rapeseed and palm oil are estimated to increase.
- 4.67 According to OECD-FAO Agricultural Outlook 2024-2033, most of the agricultural commodities have been projected to face slowdown in demand in the next decade due to declining population



growth rate and stabilizing per capita income. Emerging economies like India and Southeast Asia will continue to be the primary drivers of demand growth, but China's dominance will decline. The primary source of growth in agricultural production will be productivity gains. Global agricultural trade is projected to continue expanding albeit at a slower pace than in the past, but real international reference prices for major agricultural commodities are projected to decline slightly over the next decade.

- 4.68 The Outlook projects that the global rice production will reach 587 million tonnes by 2033, driven primarily by yield improvements and growth will be concentrated in Asian countries, with India leading the way and becoming the world's largest rice producer. Global rice consumption is projected to increase steadily, primarily driven by population growth particularly in Africa and Asia. Global rice trade will continue to grow, although at a slower pace than in the previous decade and Asia will remain the dominant rice exporting region, with India, Thailand, and Vietnam as the leading exporters. Africa will see significant growth in rice imports, as domestic production has not kept pace with rising demand, but China's rice imports will decline due to increase in domestic production. The Outlook forecasts a gradual decline in real prices of rice over the next decade, due to expected production recovery, India's export policies, and competition from other rice-exporting countries.
- 4.69 Global pulses supply is estimated to increase by 25 million tonnes in 2033 with 40 percent contribution from Asia, specifically India and consumption is expected to regain importance in diets around the world. Global pulses trade is projected to reach 22 million tonnes by 2033 and Canada will remain the top exporter with Australia and Russia other significant exporters. Real prices of pulses are expected to decline over the coming decade.
- 4.70 Among the oilseeds, soybean production is projected to grow at a slower pace compared to previous decade and about 80 percent of production growth will be predominantly driven by increase in yield. In the recent years, Brazil has been the largest producer of soybeans and its production is expected to grow slightly stronger than USA. Other oilseeds production, including rapeseed, sunflower seed, and groundnuts, is also expected to increase at moderate pace. Global soybean exports are estimated to slow down considerably on account of projected slower import demand from China. Oilseeds prices are expected to decline in real term over the next decade, but uncertainties related to environmental regulations, biofuel policies, and consumer preferences may significantly alter this outlook.
- 4.71 The Outlook anticipates global cotton production to reach 29 million tonnes by 2033 with an annual growth of 1.3 percent, despite challenges such as competition from synthetic fibres and concerns regarding environmental sustainability. This growth is expected to be primarily driven by improved yield with a smaller contribution from the expansion of areas. China is expected to retain its position as the world's largest cotton producer, followed by India. Global trade of raw cotton is projected to increase moderately to 12.4 million tons in 2033, driven by strong demand from Bangladesh and Vietnam. International cotton prices in real terms are projected to tend slightly downwards in the medium term influenced by competition from synthetic fibers and changes in consumer preferences.



Recapitulation

- 4.72 Amidst global trade challenges and domestic policy adjustments, India's agricultural trade has shown strong resilience. Although, agricultural exports marginally declined in 2023-24 as compared to previous year due to restrictions on rice and sugar exports, but this has not affected the positive growth momentum of exports of other agricultural commodities. As regards agricultural imports, while there is a decline in total import bill in 2023-24 over previous year, record high surge in pulses imports and heavy dependence on imports for edible oils and matter of concern.
- 4.73 As per USDA and IGC, global rice production and consumption is forecast to increase in 2024-25, but world rice trade is estimated at the same level as in 2023-24. As per IGC estimates, world maize production and trade are forecast to be lower in 2024-25 than previous year. World soybean output, consumption and trade are forecast to rise in 2024-25.
- 4.74 As per USDA, global outlook for production of major oilseeds such as soybean, peanut, palm kernel and cotton and major edible oils like soybean, palm & palm kernel and cottonseed is positive for 2024-25. Global pulses supply is estimated to increase by 25 million tonnes in next decade, with 40 percent contribution from Asia, specifically by India.
- 4.75 In order to meet the rising demand and to reduce high dependency on imports, India should take proactive steps to achieve self-sufficient in pulses and edible oils by increasing productivity and area under cultivation.



CHAPTER 5



Costs, Returns and Inter-Crop Parity

- 5.1 The Commission considers various factors such as cost of production, demand-supply dynamics, price trends, inter-crop parity, terms of trade, likely impact of price policy on the economy, rational utilization of land, water and other production resources, and a minimum 50 percent margin over cost of production, while recommending Minimum Support Price (MSP) for mandated crops. The Commission holds wide-ranging consultations with farmers and farmer associations/organizations, Central and State Governments, research organizations, industry associations, private food and agribusiness representatives, etc.
- 5.2 The Commission utilizes State-wise cost estimates of mandated crops from the 'Comprehensive Scheme (CS) for Studying the Cost of Cultivation of Principal Crops in India,' provided by the Economics, Statistics & Evaluation Division (ES&E Division) of the Ministry of Agriculture and Farmers Welfare, Government of India. As CS data are available up to the crop season 2023-24, cost projections are made for the crop season 2025-26.
- 5.3 The projected Cost of Cultivation (CoC) for the 2025-26 crop season is calculated from the latest three-year actual cost estimates (2021-22 to 2023-24) for all States. The CoC estimates are not projected for a crop in a State if actual cost data is available only for one year during this period. Additionally, projections are not made if the State's share in all-India production or share of a crop in total production of crop group within the State is negligible, or if the number of sample holdings under the CS for a crop is insufficient.
- 5.4 The projected CoC for crop season 2025-26 captures movement in overall input cost separately for each of the three years viz. 2021-22, 2022-23 and 2023-24 for each of the projected States. An assessment of likely changes in input cost for the crop year 2025-26, with reference to each of the year 2021-22, 2022-23 and 2023-24, is made by constructing Composite Input Price Indices (CIPIs) (base 2011-12=100) for each State for every projected crop. The CIPIs are based on latest prices of different inputs like human labour, bullock labour, machine labour, fertilisers and manures, seeds, pesticides and irrigation as per available data from Labour Bureau, Ministry of Labour and Employment, State Governments and Office of the Economic Adviser, Ministry of Commerce and Industry. Based on these CIPIs, the Commission projected State-wise CoC A_2 , A_2+FL and C_2 for each of the mandated crops.



- 5.5 The State-wise cost of production (CoP) A_2 , A_2 +FL and C_2 projections for the mandated crops are calculated by using respective projected CoC estimates, ratio of Main Product (MP) to Gross Value of Output (GVO) and estimated yield. Subsequently, all-India estimates of CoP A_2 , A_2 +FL and C_2 for a crop are derived as weighted average of State-wise CoPs using their production share in total production as weights. These projected all-India CoP estimates for mandated crops are then considered by the Commission, while formulating price policy recommendations.
- 5.6 The Commission has undertaken cost projection exercise under certain implicit assumptions. First, it is assumed that fixed cost components will not undergo any significant change in the intervening period, for which actual cost estimates are available and cost projections are made. Second, given the inherent variability of yield from year to year due to multiple factors, three-year average yield for latest TE2023-24 has been taken for smoothing fluctuations in yield and consequently in CoP. However, in cases where there are wide fluctuations, Olympic average yield¹ has been used or yield has been estimated based on time series of yield under Comprehensive Scheme. In few cases, yield has been estimated based on yield data published in 'Agricultural Statistics at a Glance 2023' by ES&E Division, Ministry of Agriculture and Farmers Welfare.

Costs and Returns during TE2023-24

- 5.7 The all-India average costs, Gross Value of Output (GVO) and gross returns during TE2023-24 in respect of mandated *kharif* crops have been analysed and are given in Table 5.1. The GVO is estimated at prevailing market prices of main product and by-products during harvest season in village/cluster of villages where the crops are grown and harvested. The details of average cost, GVO and gross returns of mandated *kharif* crops at all-India level for 2021-22, 2022-23 and 2023-24 and State level during TE2023-24, are given in Annex Table 5.1 and 5.2, respectively.

Cost of Cultivation

- 5.8 During TE2023-24, all-India per hectare average CoC A_2 and A_2 +FL was highest for groundnut at ₹63,788 and ₹76,772 and lowest for nigerseed at ₹6,881 and ₹22,406, respectively. Among cereals, per hectare A_2 and A_2 +FL costs were highest for paddy at ₹48,104 and ₹61,314 and lowest for bajra at ₹22,654 and ₹37,123, respectively. In case of pulses, per hectare all-India A_2 and A_2 +FL costs were highest for tur at ₹38,695 and ₹49,080 and lowest for moong at ₹17,055 and ₹28,193, respectively. In case of oilseeds, per hectare all-India A_2 and A_2 +FL costs were highest for groundnut at ₹63,778 and ₹76,772 and lowest for nigerseed at ₹6,881 and ₹22,406, respectively. However, there are significant differences in the cost of cultivation of various crops across States as is evident from Table 5.1(a). The main reason for inter-State variations in CoC is varying expenditure on farm inputs such as labour (human, bullock, machine), seed, fertilisers, manures, insecticides, irrigation etc. due to variations in quantity of inputs used and prices paid by farmers.

¹Olympic average yield is calculated by dropping the highest and the lowest yield from latest five-year yields and calculating the average of the remaining three-year yields.



Gross Value of Output

5.9 All-India average GVO among mandated crops during TE2023-24 was highest for groundnut at ₹1,14,789 per hectare and lowest for nigerseed at ₹23,880 per hectare. Among cereals, per hectare GVO was highest for paddy at ₹91,530 and lowest for bajra at ₹45,780. The GVO from *Shree Anna*/millets was significantly lower than paddy and maize, primarily due to low yields. In case of pulses, GVO was highest for tur (₹73,783/ha), followed by urad (₹39,075/ha) and lowest for moong (₹38,447/ha). Among oilseeds, highest GVO was realised for groundnut (₹1,14,789/ha) and lowest for nigerseed (₹23,880/ha) during TE2023-24. The GVO from nigerseed, sesamum, sunflower and soybean was substantially lower than groundnut due to lower yields. The GVO from cotton was ₹1,12,082 per hectare, second highest among mandated crops.

Table 5.1: All-India Average Costs and Gross Returns over Cost of Cultivation (₹/ha) of Mandated Crops during TE2023-24

Crop	CoC A_2	CoC A_2 +FL	GVO	Gross Returns over	
				A_2 (Col.4 - Col.2)	A_2 +FL (Col.4 - Col.3)
(1)	(2)	(3)	(4)	(5)	(6)
Cereals					
Paddy	48104	61314	91530	43426	30216
Jowar	30795	41209	50216	19421	9007
Bajra	22654	37123	45780	23126	8657
Maize	41988	53388	83544	41556	30156
Ragi	39282	54508	47911	8629	-6596
Pulses					
Arhar (Tur)	38695	49080	73783	35088	24703
Moong	17055	28193	38447	21392	10254
Urad	21465	28444	39075	17610	10631
Oilseeds					
Groundnut	63788	76772	114789	51001	38017
Soybean	37830	43498	63247	25417	19749
Sunflower	34733	40749	54666	19933	13917
Sesamum	18996	31710	45286	26290	13576
Nigerseed	6881	22406	23880	17000	1474
Commercial Crop					
Cotton	61887	76543	112082	50195	35539

Note: All-India CoC, GVO and gross returns of a crop are weighted average of respective CoCs, GVOs and gross returns of projected States

Source: CACP Calculations using CS data



Table 5.1(a): Distribution of States according to Cost of Cultivation (CoC), TE2023-24

Crop	CoC A ₂	
	$\geq$ All-India	$<$ All-India
Paddy	KL, AP, MH, TG, TN, KA, PB, HR, WB, GJ	UP, OD, MP, CG, JH, BR, AS, HP
Jowar	AP, TN, MH	KA, RJ
Bajra	MH, GJ	UP, HR, RJ
Maize	AP, MH, TG, TN, PB	KA, GJ, BR, UP, MP, RJ, HP
Ragi	KA	OD
Tur	MH	TG, KA, GJ, AP, UP, OD
Moong	MH, AP, KA, TN, GJ	RJ, OD, UP
Urad	MH, TN, AP, KA, GJ	MP, RJ, UP, OD
Groundnut	TG, MH, GJ, AP, TN, RJ	WB, KA, OD, UP
Soybean	MH, KA	GJ, RJ, TG, MP
Sunflower	AP	KA
Sesamum	GJ, WB, MP, AP	UP, RJ
Cotton	TN, TG, AP, MH	GJ, PB, MP, KA, OD, RJ, HR
CoC A ₂ +FL		
Paddy	MH, KL, TG, KA, AP, WB, TN, HR	PB, OD, GJ, UP, AS, HP, MP, JH, CG, BR
Jowar	AP, TN, MH	RJ, KA
Bajra	MH, GJ	UP, RJ, HR
Maize	TG, MH, AP, TN, PB, RJ	GJ, KA, BR, UP, HP, MP
Ragi	KA	OD
Tur	MH	GJ, UP, TG, KA, AP, OD
Moong	MH, GJ, AP, KA	RJ, UP, TN, OD
Urad	MH, TN, AP, RJ, GJ, KA	MP, UP, OD
Groundnut	WB, TG, MH, RJ, GJ	TN, AP, KA, OD, UP
Soybean	MH, KA, RJ	GJ, TG, MP
Sunflower	AP	KA
Sesamum	WB, GJ	MP, RJ, AP, UP
Cotton	TN, TG, MH	AP, GJ, RJ, OD, MP, PB, KA, HR

Note: States have been arranged in descending order of CoCs

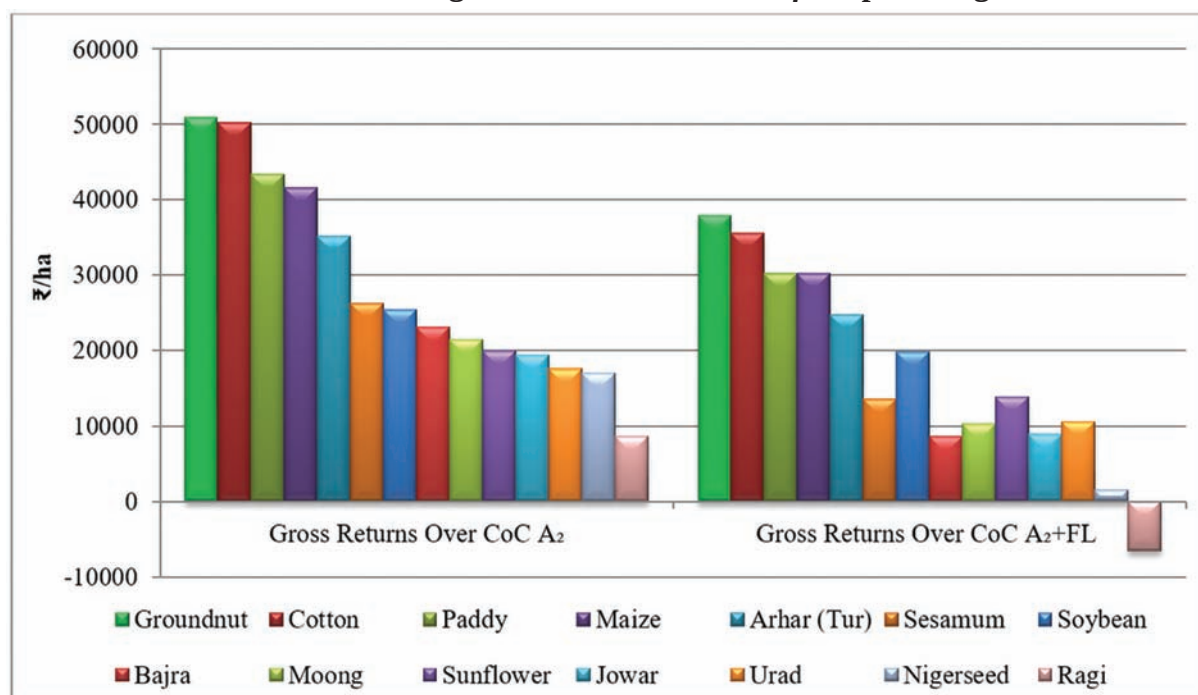
Source: CACP Calculations using CS data



Gross Returns

- 5.10 Chart 5.1 represents all-India average gross returns over actual CoC A_2 and A_2+FL during TE2023-24 for mandated *kharif* crops. The per hectare gross returns over CoC A_2 and A_2+FL were highest for groundnut at ₹51,001 and ₹38,017, respectively, whereas per hectare gross returns over A_2 were lowest for ragi at ₹8,629 and over A_2+FL for nigerseed at ₹1,474. Ragi growers incurred a net loss of ₹6,596 per hectare over A_2+FL due to low yield. All-India average gross returns as percentage of CoC A_2 were highest for nigerseed (247.1%), and lowest for ragi (22%), while average gross returns as percentage of A_2+FL cost were highest for maize (56.5%), followed by groundnut (49.5%), paddy (49.3%) and cotton (46.4%).
- 5.11 During TE2023-24, among cereals, gross returns over CoC A_2 and A_2+FL were highest for paddy mainly due to higher yield, whereas these were lowest for ragi mainly because of higher cost of cultivation and lower yield as compared to other cereal crops.
- 5.12 There are large variations in gross returns of various crops among States (Table 5.1(b)). These variations are largely due to differences in crop yield, cost of cultivation due to quantity of inputs used and input prices paid by farmers and output price.

Chart 5.1: All-India Average Gross Returns of Kharif Crops during TE2023-24



Note: All-India CoC, GVO and gross returns of a crop are weighted average of respective CoCs, GVOs and gross returns of projected States

Source: CACP Calculations using CS data



Table 5.1(b): Distribution of States according to Gross Returns, TE2023-24

Crop	Gross Returns over CoC A ₂	
	$\geq$ All-India	$<$ All-India
Paddy	PB, HR, HP, AP, MP, TG, CG, KA, KL	TN, GJ, WB, AS, OD, BR, UP, JH, MH
Jowar	AP, TN, RJ	KA, MH
Bajra	GJ, UP	HR, RJ, MH
Maize	AP, MH, TG, BR, TN	RJ, KA, HP, MP, GJ, PB, UP
Ragi	OD	KA
Tur	UP, MH, GJ	TG, OD, AP, KA
Moong	GJ, RJ, AP, OD	MH, KA, TN, UP
Urad	AP, MH, TN, OD, RJ	MP, UP, KA, GJ
Groundnut	WB, RJ, GJ	TG, TN, UP, KA, OD, MH, AP
Soybean	KA, MH, RJ	MP, TG, GJ
Sunflower	AP	KA
Sesamum	WB, MP, GJ	RJ, AP, UP
Cotton	PB, RJ, TN, GJ, HR	KA, OD, AP, MP, MH, TG
Gross Returns over CoC A₂ +FL		
Paddy	PB, HR, AP, MP, CG, TG, KA, KL, HP, TN	GJ, BR, WB, UP, AS, OD, JH, MH
Jowar	AP, TN	KA, MH, RJ
Bajra	GJ, UP, HR	RJ, MH
Maize	AP, MH, BR, TG, TN	KA, MP, HP, UP, GJ, RJ, PB
Ragi	-	KA, OD
Tur	UP, MH, GJ	TG, AP, OD, KA
Moong	AP, GJ, OD, RJ	TN, MH, KA, UP
Urad	AP, TN, MH, OD	MP, KA, RJ, UP, GJ
Groundnut	RJ, WB, GJ	TG, TN, UP, KA, OD, AP, MH
Soybean	KA, MH	RJ, MP, TG, GJ
Sunflower	AP	KA
Sesamum	WB, MP, GJ	AP, RJ, UP
Cotton	PB, RJ, GJ, TN, AP, HR, KA	OD, MP, MH, TG

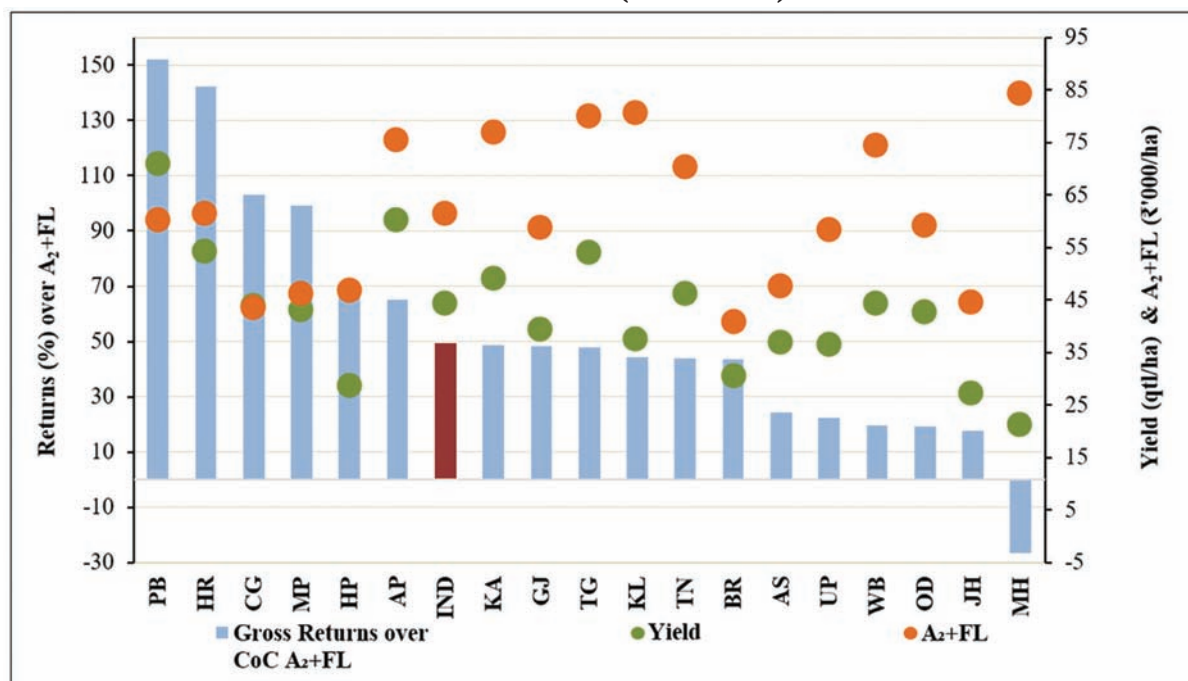
Note: States have been arranged in descending order of returns

Source: CACP Calculations using CS data



- 5.13 Inter-State variations in gross returns from paddy have been shown in Chart 5.2, where gross returns are shown in descending order. The gross returns are significantly higher in Punjab and Haryana due to substantially high yields, while States like West Bengal, Odisha, Jharkhand and Maharashtra have lower returns mainly due to lower yields and high cost of cultivation. Therefore, efforts are needed to improve productivity and reduce cost in high-cost States.

**Chart 5.2: Inter-State Variations in Gross Returns from Paddy:
Yield & Costs (TE2023-24)**



Note: All-India CoC, gross returns and Yield of a crop are weighted average of respective CoCs, gross returns and Yield of projected States

Source: CACP Calculations using CS data

Agricultural Wages and Farm Input Prices

Growth in Daily Wage Rates of Agriculture Labour

- 5.14 Growth in average daily wage rate of agricultural labour during *kharif* season (May to November) in major States and at all-India level, in nominal terms during crop season (May to November) 2022, 2023 and 2024 are given in Table 5.2.
- 5.15 At all-India level, agricultural average daily wage rate increased by 6.9 percent in 2022, 7.4 percent in 2023 and 6.3 percent in 2024, over previous years. The highest growth in average daily wage rate in 2024 over 2023 was recorded in Odisha at 14.1 percent and lowest in Punjab (3.2%). Andhra Pradesh, Assam, Bihar, Gujarat, Haryana, Himachal Pradesh, Karnataka, Maharashtra, Odisha and Rajasthan recorded higher growth in average daily wages as compared to all-India average, while Kerala, Madhya Pradesh, Punjab, Tamil Nadu, Uttar Pradesh and West Bengal registered lower growth than all-India during 2024 over 2023.



Table 5.2: Growth (%) in Average Daily Wage Rates of Agricultural Labour during Kharif Season (2022 to 2024)

State	Growth (%)		
	2022	2023	2024
Andhra Pradesh	14.6	12.8	8.0
Assam	6.5	9.0	8.9
Bihar	7.9	8.8	6.3
Gujarat	8.0	6.5	7.6
Haryana	8.3	6.5	8.7
Himachal Pradesh	-2.1	4.0	8.7
Karnataka	9.0	7.6	7.2
Kerala	3.9	3.8	5.9
Madhya Pradesh	5.9	5.8	3.8
Maharashtra	4.2	5.2	6.5
Odisha	9.3	7.5	14.1
Punjab	6.1	5.3	3.2
Rajasthan	2.2	8.8	9.8
Tamil Nadu	1.8	8.9	3.6
Uttar Pradesh	7.9	5.7	5.1
West Bengal	4.3	3.9	4.0
All-India	6.9	7.4	6.3

Note 1: Average daily wage rates of agriculture labour are related to man only

2: Growth is derived from the average taken for the period from May to November

3: All-India daily wage rate is weighted average of daily wage rates of States

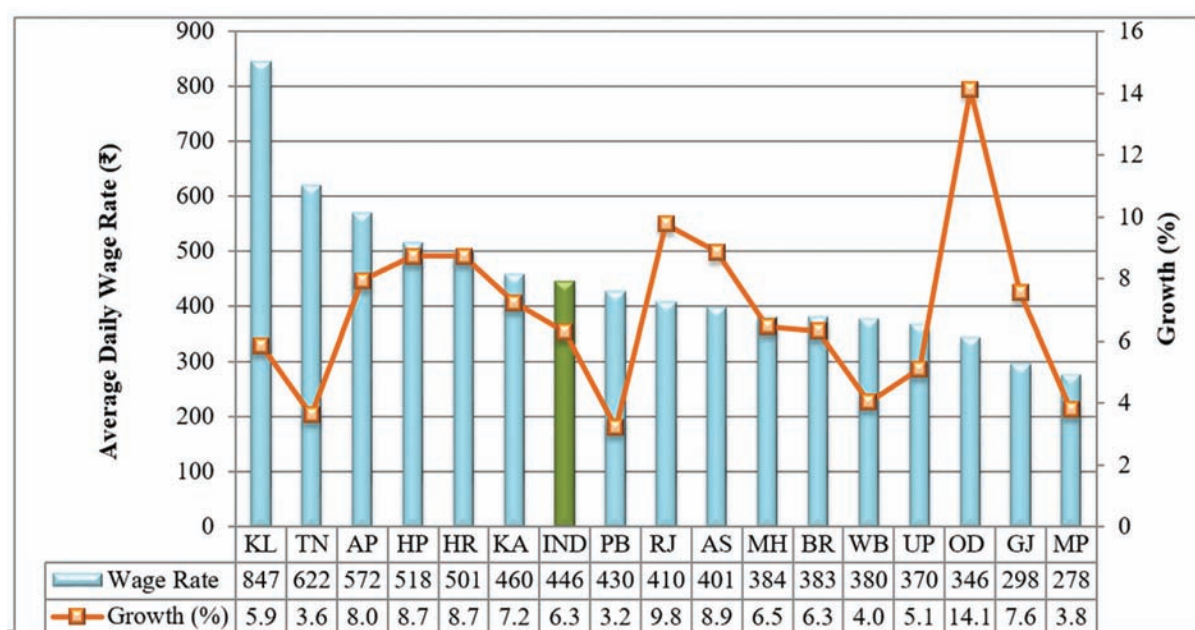
Source: Labour Bureau, Ministry of Labour and Employment and CACP Calculations

Daily Wages of Agriculture Labour

- 5.16 Chart 5.3 depicts State-wise average daily wages of agricultural labour during May to November 2024 and growth in wages during May to November 2024 over same period in the previous year. All-India average daily wage of agricultural labour was ₹446 during 2024, which was highest in Kerala at ₹847, about three times of Madhya Pradesh (₹278), the State having lowest wage.
- 5.17 Wage rates have increased significantly in most of the States during the last season and resulted in significant increase in cost of cultivation/production. Therefore, strategies for promotion of mechanization of farm operations would help in reducing dependence on human labour, leading to lower cultivation costs and enhanced farming profitability. Annex Table 5.3 provides details on the monthly average of daily wage rates for agricultural labour at current prices.



Chart 5.3: Average Daily Wage Rates for *Kharif* season 2024 and Growth over 2023 Season



Note 1: Average daily wage rates of agriculture labour are related to man only
 2: Growth is derived from the average taken for the period from May to November
 3: All-India daily wage rate is weighted average of daily wage rates of States
 Source: Labour Bureau, Ministry of Labour and Employment and CACP Calculations

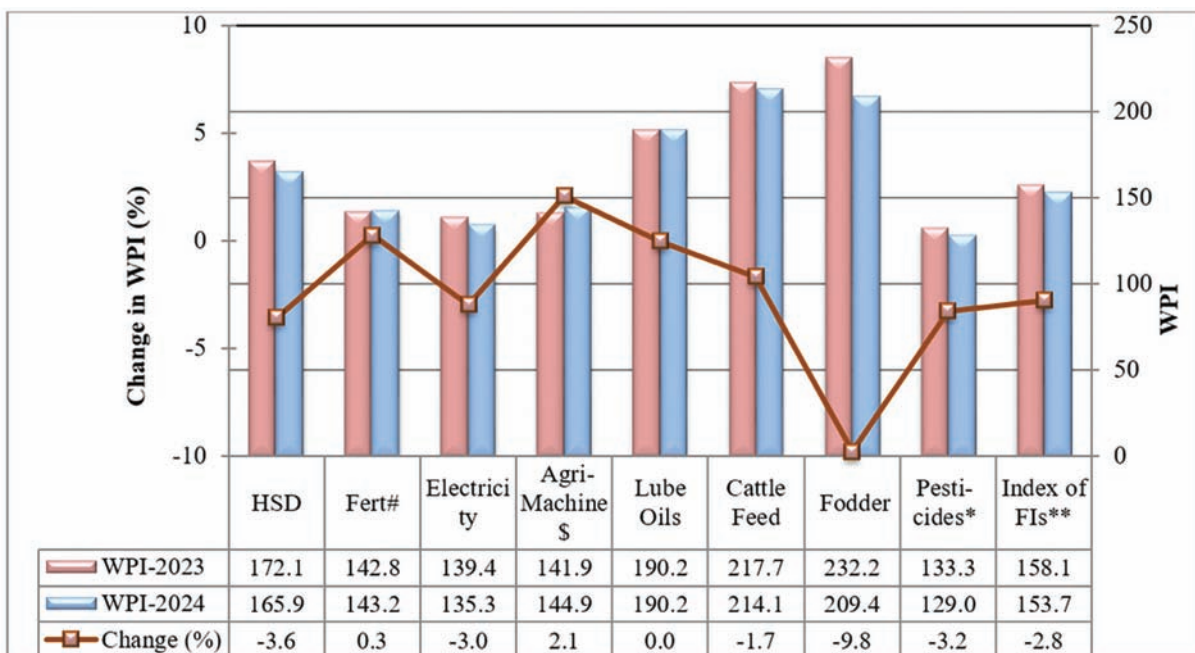
Movement in Farm Input Prices

- 5.18 The Commission analysed price trends of major farm inputs to assess the effect of changes in input prices on cost of cultivation/production. The changes in Wholesale Price Index (WPI) with base 2011-12 of major farm inputs during *kharif* season (May to November) in 2024 over 2023 are given in Chart 5.4. The indices of fodder (-9.8%), high-speed diesel (-3.6%), pesticides and other agrochemical products (-3.2%), electricity (-3%) and cattle feed (-1.7%) declined in 2024 over 2023, whereas the indices of manufacture of agricultural and forestry machinery² (2.1%), and fertilizers and nitrogen compounds (0.3%) increased in 2024 over 2023. The weighted index of above-mentioned selected farm input prices decreased by (-)2.8 percent in 2024. The monthly wholesale price indices of various farm inputs during 2023 and 2024 are given in Annex Table 5.4.

²Manufacture of agricultural and forestry machinery consists of 'agricultural tractors', 'harvesters', 'agriculture implements', 'threshers' and 'soil preparation & cultivation machinery (other than tractors)'.



Chart 5.4: Movement in Wholesale Price Index (WPI) of Farm Inputs



Note 1: WPI of the farm input shown in graph is average of monthly WPIs from May to November

2: Index of Farm Inputs is weighted average of WPIs of farm inputs mentioned in Chart

#: Fertilizers and Nitrogen Compounds

\$: Manufacture of Agricultural and Forestry Machinery

*: Pesticides and other Agrochemical Products

** : Index of Farm Inputs

Source: Office of the Economic Adviser, Ministry of Commerce and Industry and CACP Calculations

Cost Projections for Crop Season 2025-26

Projection of Farm Input Price Index

- 5.19 The Commission has computed all-India weighted 'input weights' for major farm inputs for TE2023-24 and 'Composite Input Price Indices (CIPIs)' from 2022-23 to 2025-26 for mandated *kharif* crops (base 2011-12=100) and are given in Table 5.3. The 'input price indices' and 'CIPIs' for crop season 2024-25 and 2025-26 for a State have been constructed using actual 'weights', 'input price indices' for TE2023-24, and latest farm input prices. Subsequently, based on these State-wise 'actual weights' and 'input price indices', crop-wise all-India weighted average 'input weights' and 'CIPIs' for all inputs, with weights being relative share of States in all-India area under the crop during TE2023-24, have been computed. Further, these crop-wise all-India weighted average 'input weights' and 'CIPIs' have been used to compute input-wise all-India weighted average 'input weights' and 'CIPIs', respectively, for all *kharif* crops, with weights being relative shares of crops in total production of mandated *kharif* crops at all-India level during TE2023-24. Finally, these input-wise all-India weighted average 'input weights' and 'CIPIs' have been used to compute all-India weighted average 'Composite Input Price Index (CIPi)' for all inputs of mandated *kharif* crops.



5.20 Table 5.3 shows that all-India CIPI for *kharif* crops increased by 4.4 percent in 2023-24, 6.7 percent in 2024-25 and 6.9 percent in 2025-26, while CIPI registered the highest increase in machine labour (10.2%), followed by seed (8.3%), bullock labour (7%), human labour (5.7%) and lowest increase in insecticides (1.1%) in 2025-26 over 2024-25.

Table 5.3: Trends in All-India Kharif Crops Farm Input Price Indices (Base 2011-12 = 100)

Inputs	Weights (TE2023-24)	Kharif Crops Farm Input Price Index				Percentage Change in Input Price Index 2025-26 over 2024-25
		2022-23	2023-24	2024-25	2025-26	
Human Labour (HL)	0.49	239.2	254.8	268.8	284.2	5.7
Bullock Labour (BL)	0.04	389.1	408.5	436.0	466.3	7.0
Machine Labour (ML)	0.22	215.2	218.3	241.2	265.9	10.2
Seeds	0.08	223.0	238.9	258.1	279.5	8.3
Fertilizers	0.08	177.9	178.7	186.2	194.4	4.4
Manures	0.02	234.4	239.4	251.0	263.4	4.9
Insecticides	0.04	143.1	139.5	140.9	142.5	1.1
Irrigation Charges	0.03	226.1	220.0	240.9	263.8	9.5
Composite Input Price Index (CIPI)		228.8	238.9	254.9	272.4	6.9
Percentage Change		-	4.4	6.7	6.9	-

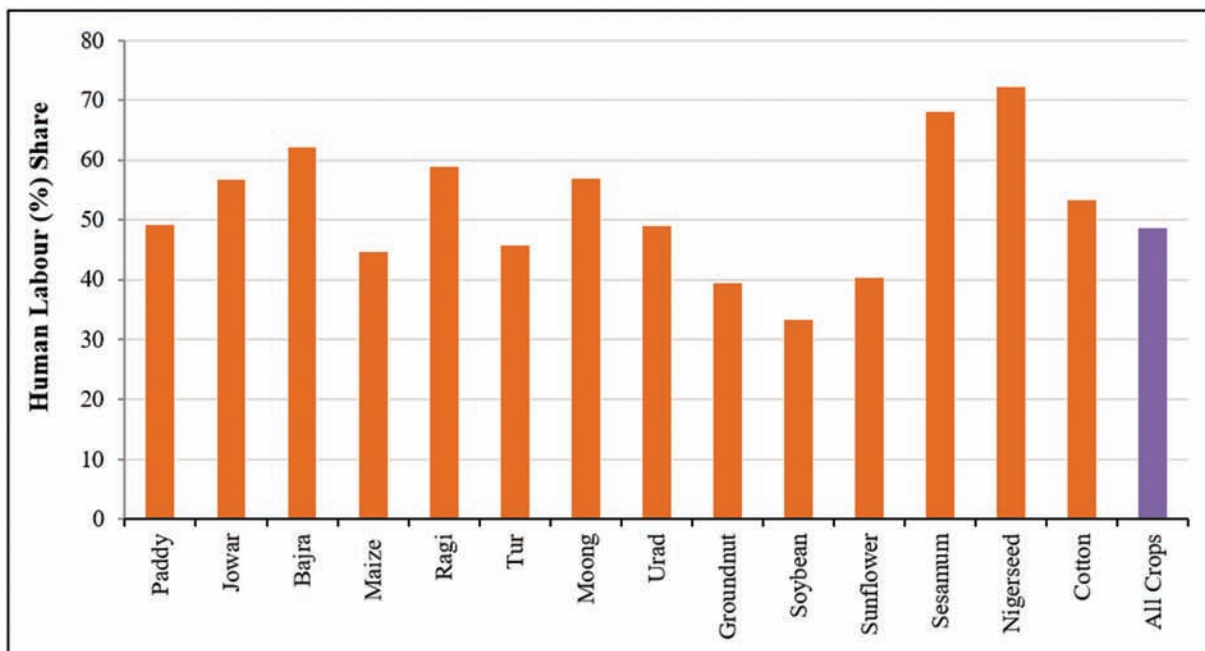
Note: All-India Weights and CIPIs are weighted average of respective weights and CIPIs of projected States

Source: CACP Calculations using CS data

5.21 As is evident from Chart 5.5, the share of human labour in total operational cost for majority of mandated crops is more than 50 percent and the availability of human labour and rising wages have become major constraints. Therefore, there is a need to develop national and State level strategies to promote sustainable agricultural mechanization and encourage farmers to adopt farm mechanization to reduce costs and improve profitability.



Chart 5.5: Crop-Wise Human Labour (%) Share for Kharif Crops, TE2023-24



Note: Human labour share for all crops is weighted average of the human labour share of mandated crops with weights being respective crop area share

Source: CACP Calculations using CS data

Projected Cost of Production

- 5.22 Based on State-wise actual cost estimates during TE2023-24 and projected CIPIs, State-wise estimates of CoC A_2 , A_2 +FL and C_2 for each of the mandated *kharif* crops have been projected for the crop year 2025-26. Using these State-wise estimates of CoC, average ratio of Main Product (MP) to Gross Value of Output (GVO) during TE2023-24, and estimated yield, State-wise CoP A_2 , A_2 +FL and C_2 for each crop for 2025-26 are projected. Subsequently, all-India weighted average projected CoP A_2 , A_2 +FL and C_2 of the mandated crops, with weights being the respective share of the States in all-India production during TE2023-24, have been worked out for 2025-26, and are given in Table 5.4.
- 5.23 Among mandated crops, all-India projected CoP A_2 was highest for sunflower (₹4,379/qlt), while A_2 +FL (₹6,564/qlt) and C_2 (₹8,632/qlt) were highest for sesamum. Projected A_2 CoP was lowest for bajra (₹1,046/qlt), while A_2 +FL (₹1,508/qlt) and C_2 (₹1,952/qlt) were lowest for maize.
- 5.24 Among cereals, per quintal projected CoP A_2 , A_2 +FL and C_2 were highest for ragi at ₹2,337, ₹3,257 and ₹3,976, respectively, while A_2 was lowest for bajra (₹1,046/qlt), and A_2 +FL (₹1,508/qlt) and C_2 (₹1,952/qlt) were lowest for maize. Among pulses, projected CoP A_2 varied from ₹3,510 per quintal in case of moong to ₹3,874 per quintal in urad, A_2 +FL ranged from ₹5,038 in tur to ₹5,845 in moong and C_2 varied from ₹6,829 in urad to ₹7,476 in moong. In case of oilseeds, all-India projected A_2 was lowest for nigerseed (₹1,960/qlt), while A_2 +FL (₹3,552/qlt) and C_2 (₹4,638/qlt) were lowest for



soybean. Projected A_2 was highest for sunflower (₹4,379/ql) and A_2+FL (₹6,564/ql) and C_2 (₹8,632/ql) were highest for sesamum. The all-India A_2 , A_2+FL and C_2 CoP for cotton were projected at ₹4,130, ₹5,140 and ₹6,717, respectively. The highest increase in all-India A_2+FL CoP in 2025-26 was recorded in bajra (14.7%) followed by ragi (13.9%), due to decline in yield and rise in cost of both crops, while paddy (3%) and moong (1%) recorded lowest increase in A_2+FL .

- 5.25 State-wise and all-India cost of production of mandated *kharif* crops for KMS2025-26 and production share during TE2023-24 are given in Annex Table 5.5. Cost of production varied significantly across States and crops. The cost of production in some crops and States was high mainly due to low yield and/or high cost of cultivation, therefore, efforts are needed to improve productivity and rationalise input costs to reduce costs for improving profitability. State-wise break-up of actual CoC estimates of mandated crops from 2021-22 to 2023-24 are given in Annex Tables 5.6(a) to 5.6(n), respectively.

Table 5.4: Projected Cost of Production (CoP) of Mandated *Kharif* Crops, KMS2025-26

Crop	Projected Cost of Production (₹/ql)		
	A_2	A_2+FL	C_2
Paddy	1238	1579	2090
Jowar	1859	2466	3206
Bajra	1046	1703	2209
Maize	1183	1508	1952
Ragi	2337	3257	3976
Arhar (Tur)	3848	5038	6839
Moong	3510	5845	7476
Urad	3874	5114	6829
Groundnut	4065	4842	6047
Soybean	3082	3552	4638
Sunflower	4379	5147	6364
Sesamum	3957	6564	8632
Nigerseed	1960	6358	8025
Cotton	4130	5140	6717

Note: All-India CoP of a crop is weighted average of CoPs of projected States

Source: CACP Calculations using CS data



Production vis-a-vis Cost Projections

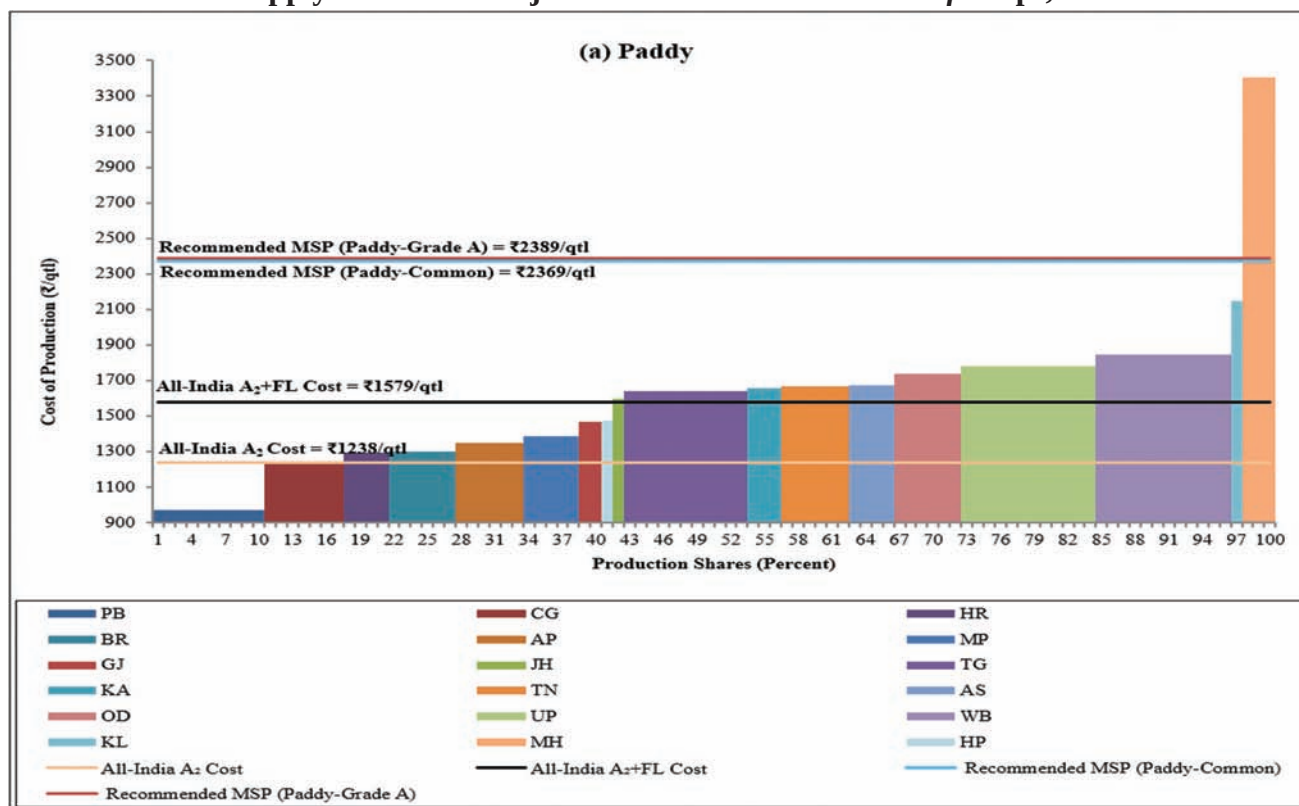
- 5.26 Charts 5.6(a) to 5.6(m) show crop-wise supply curves³ of projected A_2 +FL cost of production by States in ascending order with corresponding share in all-India production. The supply curve presented in Chart 5.6(a) shows that projected A_2 +FL CoP for paddy is lowest in Punjab (₹967/ql) mainly due to higher yield and highest in Maharashtra (₹3,407/ql) due to significantly lower yield and high cost of cultivation, compared to other States. Out of 18 States for which cost projections are made, 10 States have higher cost of production than all-India average.
- 5.27 For jowar, CoP A_2 +FL is lowest in Andhra Pradesh (₹1,459/ql) due to higher yield, and highest in Karnataka (₹2,889/ql) (Chart 5.6(b)). Maharashtra, the largest jowar producing State, has A_2 +FL CoP at ₹2,581 per quintal, which is higher than all-India CoP. Supply curve of bajra presented in Chart 5.6(c) shows that A_2 +FL is lowest in Haryana (₹1,427/ql) due to lowest cost of cultivation and higher yield, and highest in Maharashtra (₹2,668/ql) due to high cost of cultivation. Among top three bajra producing States, projected CoP of Uttar Pradesh (₹1,529/ql) and Gujarat (₹1,545/ql) is lower than Rajasthan (₹1,804/ql), because of significantly higher yield in Uttar Pradesh and Gujarat.
- 5.28 Chart 5.6(d) shows that in case of maize, CoP A_2 +FL is lowest in Andhra Pradesh (₹1,039/ql) mainly due to higher yield and highest in Rajasthan (₹2,313/ql) as crop yield in the State is low. Among top-five maize producers in the country during TE2023-24, Tamil Nadu (₹1,609/ql) has the highest CoP, while Bihar (₹1,121/ql) has the lowest CoP. In case of ragi, the projected CoP A_2 +FL in Odisha (₹4,029/ql) is higher than Karnataka (₹3,225/ql), the largest ragi producing State (Chart 5.6(e)).
- 5.29 In case of pulses, CoP A_2 +FL for tur is lowest for Uttar Pradesh (₹4,275/ql) and highest for Telangana (₹5,718/ql) (Chart 5.6(f)). In top two producers of tur in the country, which account for two-third of production share of projected States, A_2 +FL for Maharashtra (₹5,340/ql) is higher than Karnataka (₹4,859/ql). As presented in Chart 5.6(g), for moong, A_2 +FL is lowest in Andhra Pradesh (₹4,772/ql) and highest in Maharashtra (₹6,783/ql). Projected CoP for Rajasthan (₹5,806/ql), the second largest moong producing State in the country, is marginally lower than all-India CoP.
- 5.30 For urad, A_2 +FL CoP is lowest in Andhra Pradesh (₹3,654/ql) due to higher yield, while Maharashtra (₹7,021/ql) has the highest CoP due to higher cost of cultivation (Chart 5.6(h)). The A_2 +FL of the largest urad producer, Madhya Pradesh, has been projected at ₹4,485 per quintal, lower than all-India average. Six out of nine States have higher A_2 +FL CoP than all-India average, therefore, efforts are required to reduce CoP in these States. Among major pulses producing States, Maharashtra has substantially higher cost of production (A_2 +FL) for all three pulses, namely, tur, moong and urad primarily due to very high costs compared with other States.
- 5.31 In case of *kharif* oilseeds, projected A_2 +FL CoP for groundnut ranged from ₹3,088 per quintal in Rajasthan to ₹8,428 per quintal in Maharashtra (Chart 5.6(i)). Among top three producers, which

³Supply curve for a crop is a graphical presentation of CoP, which represents the quantum of a crop produced at different CoP in various States.

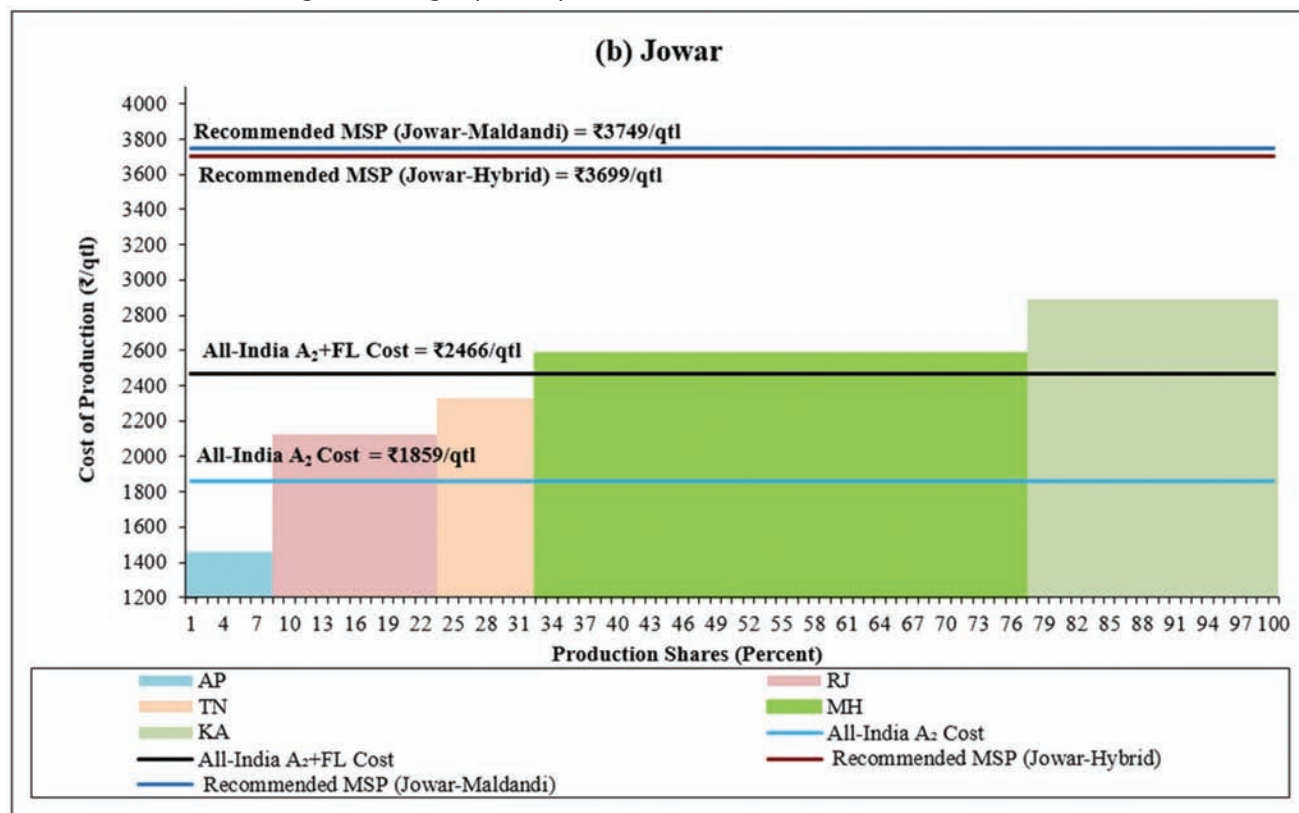


account for 79.3 percent share of the production of projected States, lowest CoP was recorded in Rajasthan, mainly due to higher yield and highest in Tamil Nadu (₹5,669/qlt). CoP for the largest groundnut producing State, Gujarat, has been projected at ₹4,761 per quintal, lower than all-India CoP.

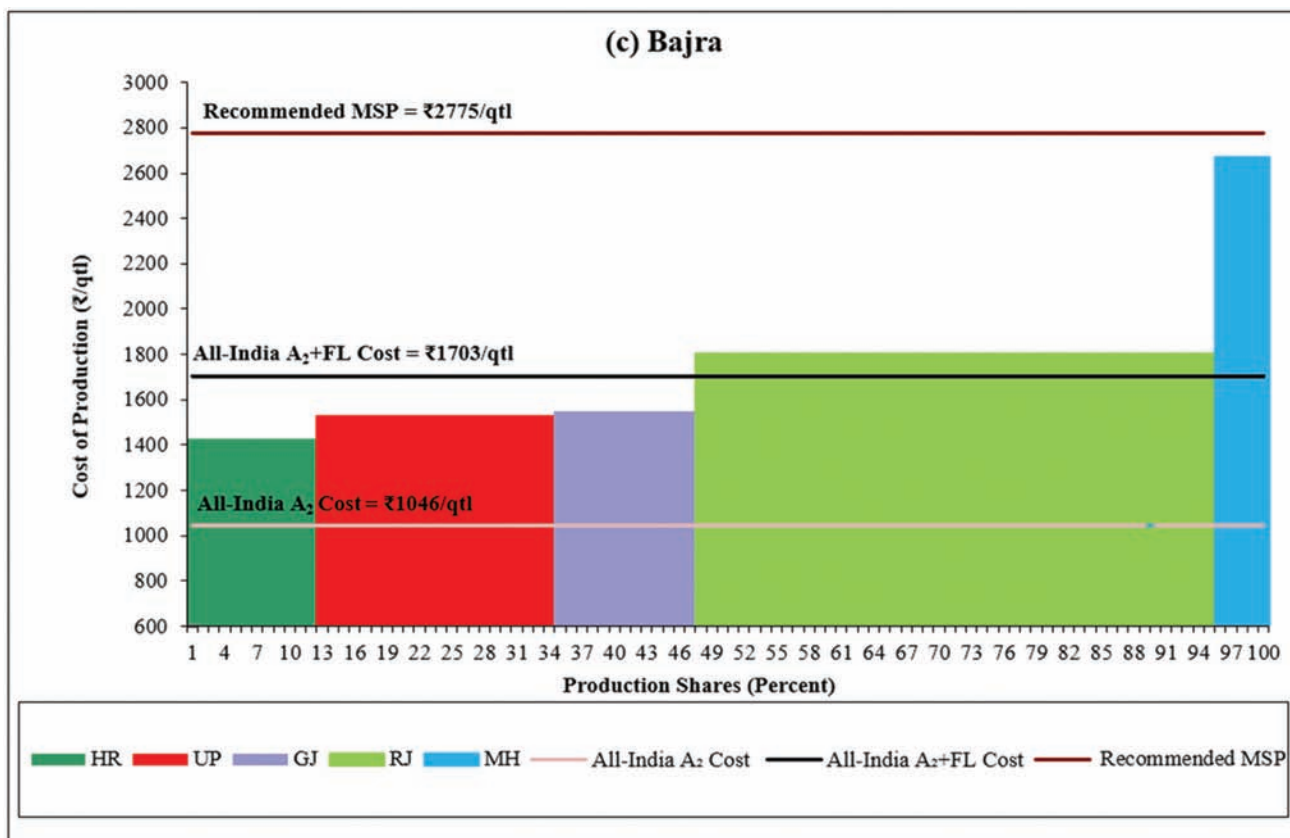
- 5.32 The supply curve of soybean (Chart 5.6(j)) shows that CoP A_2 +FL is lowest (₹3,062/qlt) in Madhya Pradesh, and highest (₹4,206/qlt) in Rajasthan. The CoP for largest soybean producing State, Maharashtra (₹3,925/qlt), is higher than all-India CoP (₹3,552/qlt) as well as for Madhya Pradesh, the second largest soybean producer. For sunflower (Chart 5.6(k)), projected CoP A_2 +FL for Karnataka (₹5,201/qlt) is higher than Andhra Pradesh (₹4,421/qlt). West Bengal has the lowest A_2 +FL CoP (₹5,421/qlt) for sesamum mainly due to higher yield, while Rajasthan (₹8,320/qlt) has the highest CoP (Chart 5.6(l)).
- 5.33 The supply curve of cotton (Chart 5.6(m)) shows that CoP A_2 +FL is lowest in Punjab (₹4,039/qlt) and highest in Tamil Nadu (₹6,513/qlt). Among top three cotton producing States, Gujarat has the lowest CoP (₹4,641/qlt), while Telangana (₹5,785/qlt) has the highest CoP due to higher cost of cultivation.
- 5.34 Among cereals, increase in projected A_2 +FL CoP for KMS2025-26 over 2024-25 is highest in bajra (14.7%) and lowest in paddy (3%). In case of pulses, highest increase has been registered in tur (5.8%), followed by urad (4.7%) and lowest in moong (1%), while in case of oilseeds, highest increase (9.4%) is recorded in nigerseed and lowest (6.1%) in sunflower. In case of cotton, CoP increased by 8.3 percent in 2025-26 over 2024-25 due to rise in costs (Annex Table 5.7).
- 5.35 The share of production covered at the recommended MSP is 96.9 percent in paddy, jowar (81.6%), bajra (87.6%), maize (86.3%), ragi (68.1%), tur (86.1%), moong (50.3%), urad (87.8%), groundnut (91.3%), soybean (99.2%), sunflower (60.7%), nigerseed (49%), sesamum (87.3%), and 99.9 percent for cotton. The MSP margins over all-India projected A_2 +FL cost of production was highest for bajra at 62.9 percent, followed by maize (59.2%), tur (58.8%), urad (52.5%) and 50 percent for other kharif crops.
- 5.36 The highest MSP margins over projected CoP A_2 +FL was in Punjab (145.1%) for paddy, Andhra Pradesh (153.6 %) for jowar, Haryana (94.4%) for bajra, Andhra Pradesh (131%) for maize, Karnataka (51.5%) for ragi, Uttar Pradesh (87.1%) for tur, Andhra Pradesh for moong (83.7%) and urad (113.5%), Rajasthan (135.2%) for groundnut, Madhya Pradesh (74%) for soybean, Andhra Pradesh (74.6%) for sunflower, West Bengal (81.6%) for sesamum, and Punjab (90.9%) for cotton.

Chart 5.6: Supply Curve and Projected CoP for Mandated *Kharif* Crops, KMS2025-26

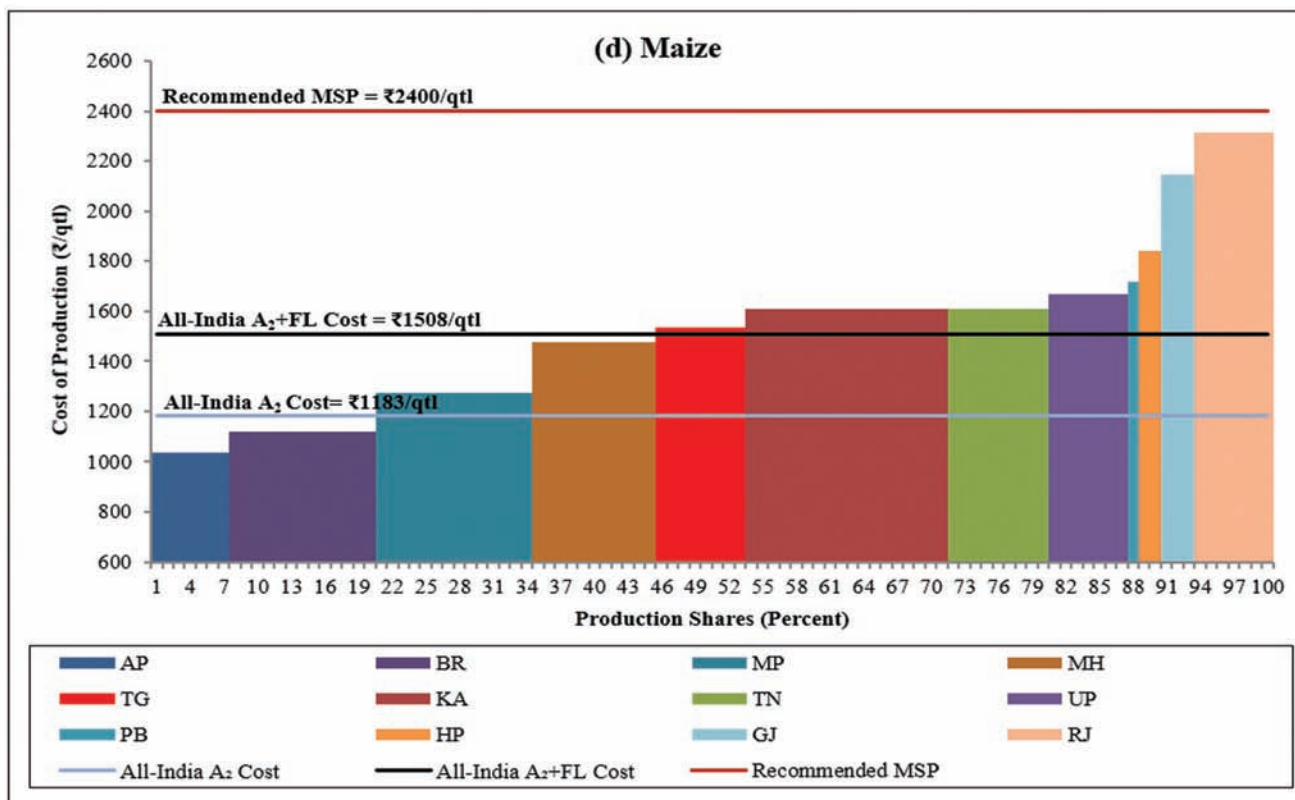
Note: All-India CoP is weighted average of CoP of States mentioned in the Chart



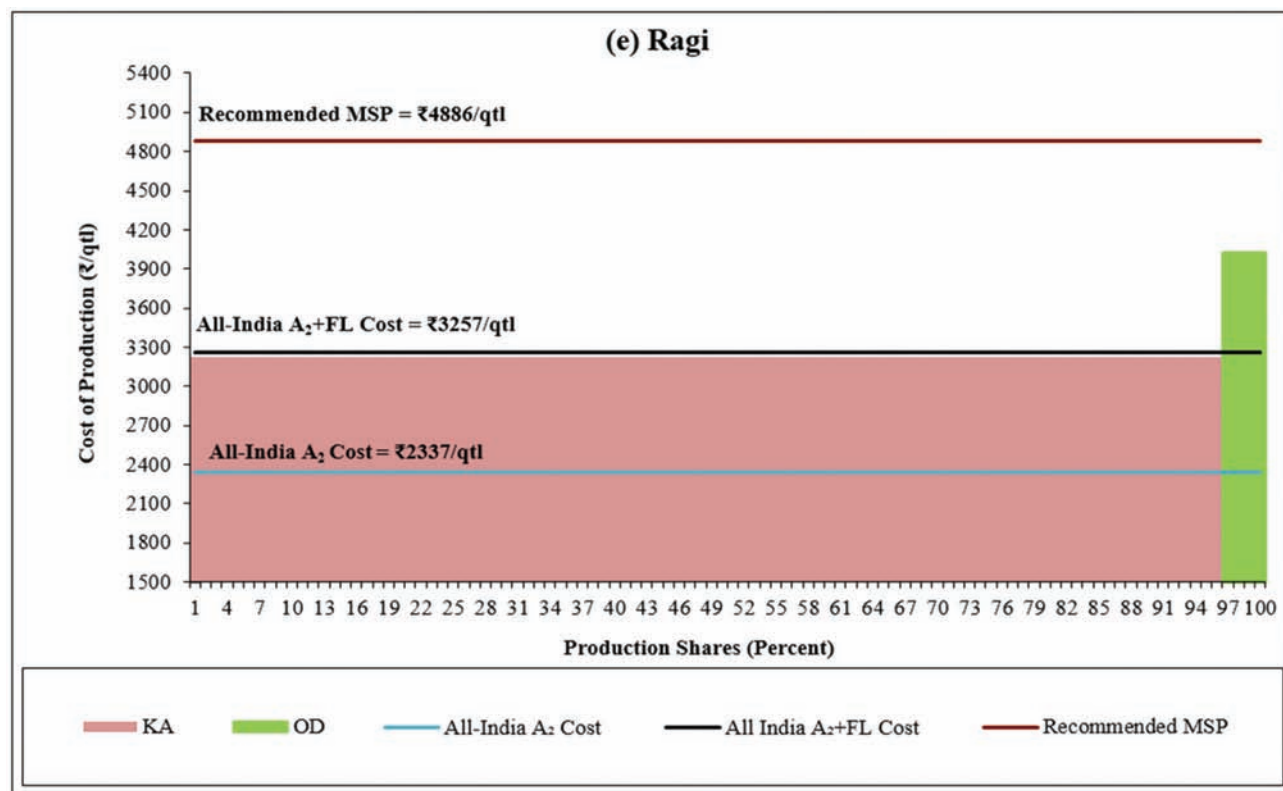
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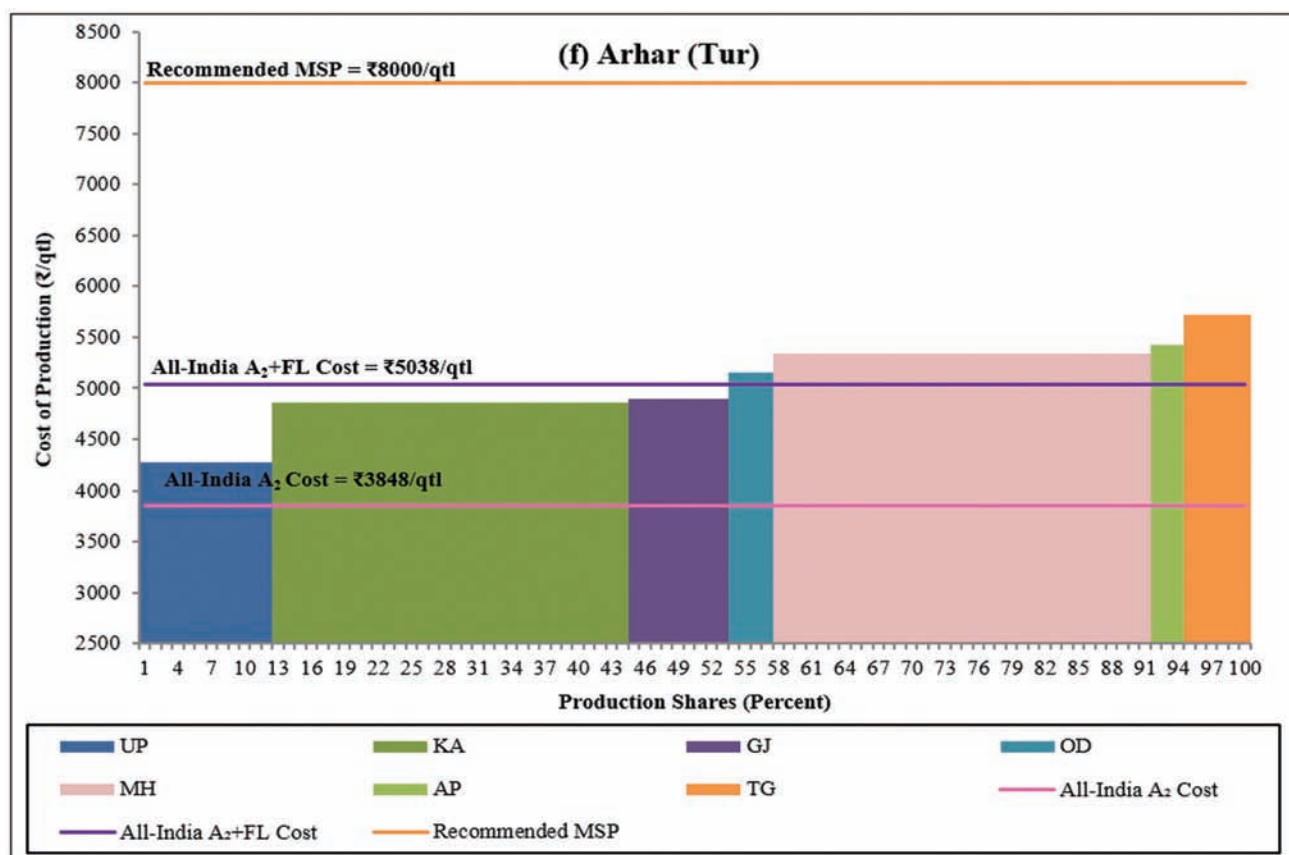
Note: All-India CoP is weighted average of CoP of States mentioned in the Chart



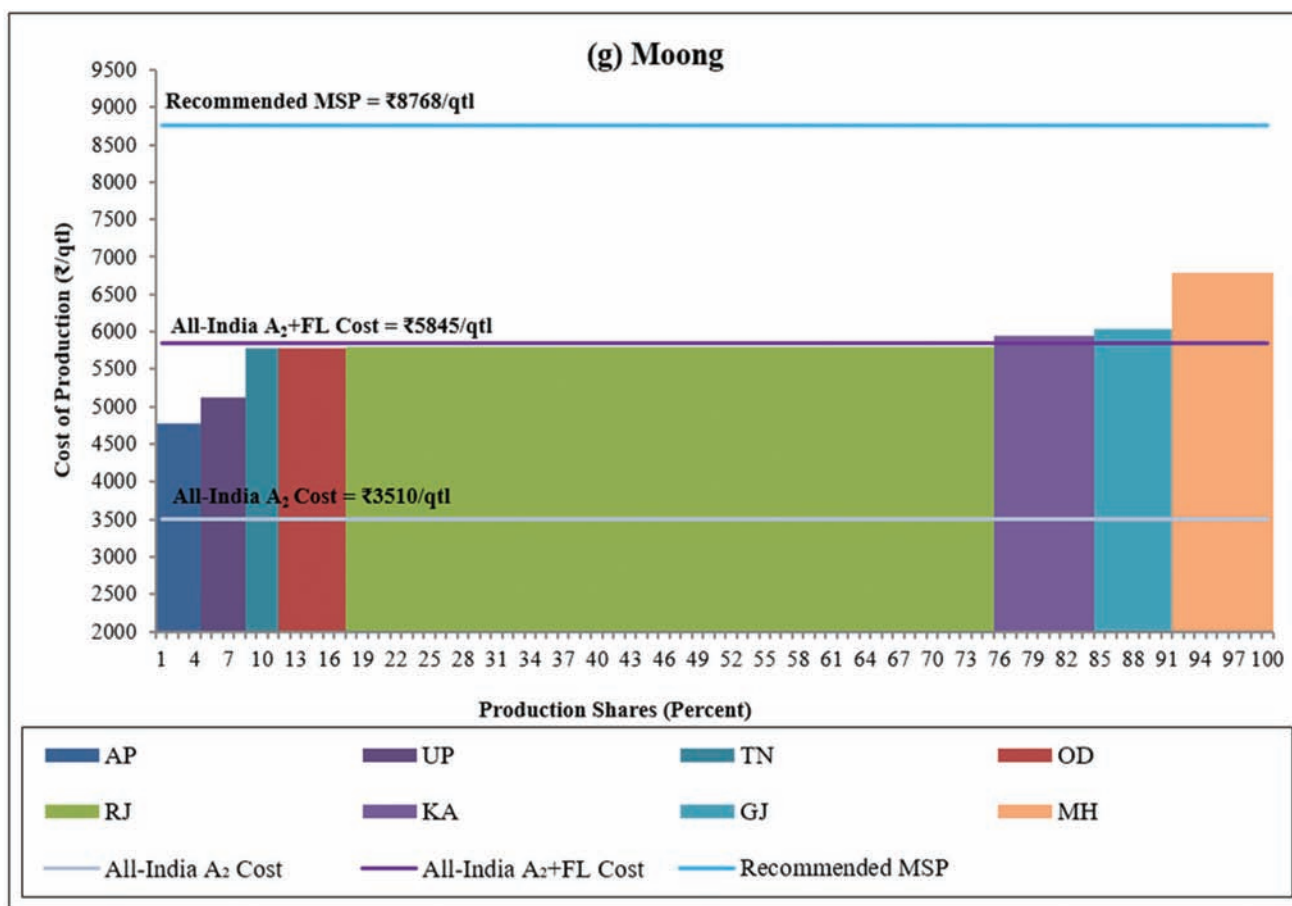
Note: All-India CoP is weighted average of CoP of States mentioned in the Chart



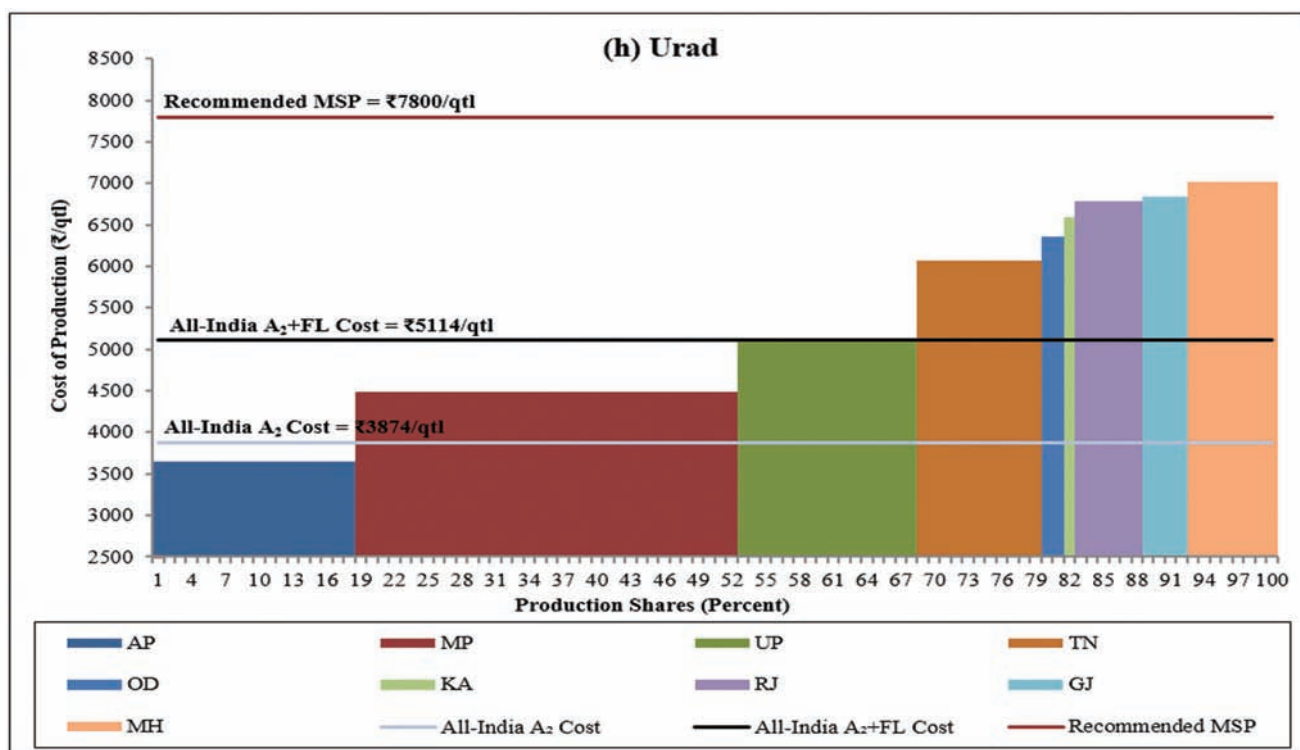
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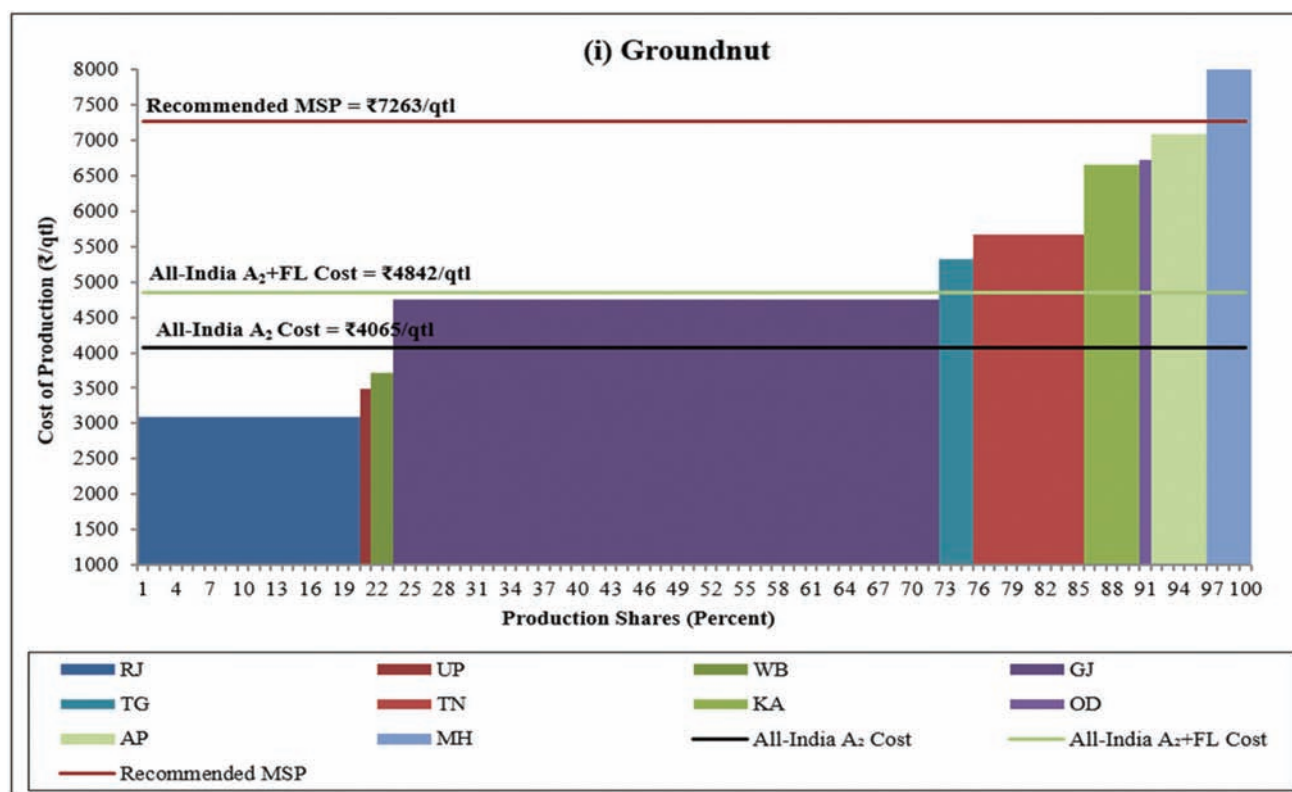
Note: All-India CoP is weighted average of CoP of States mentioned in the Chart



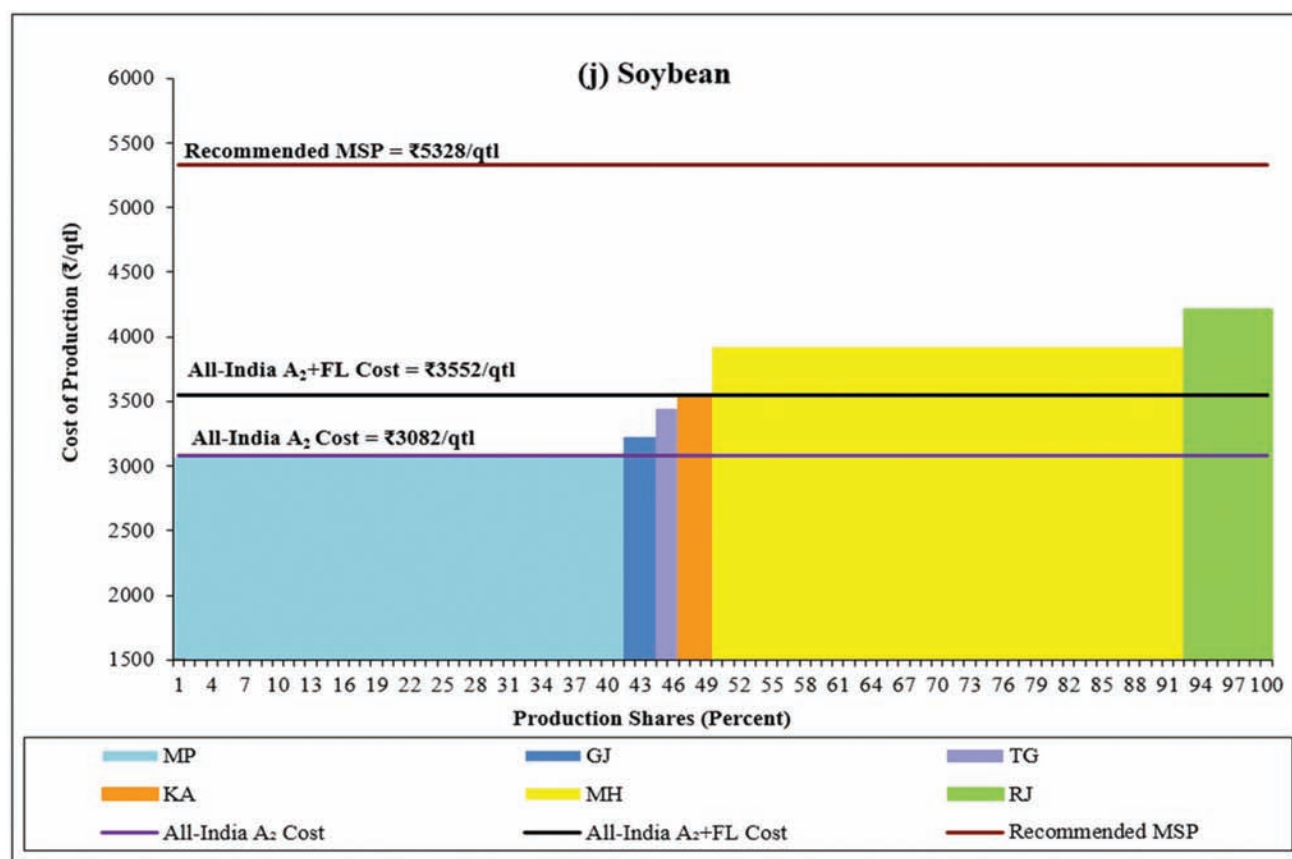
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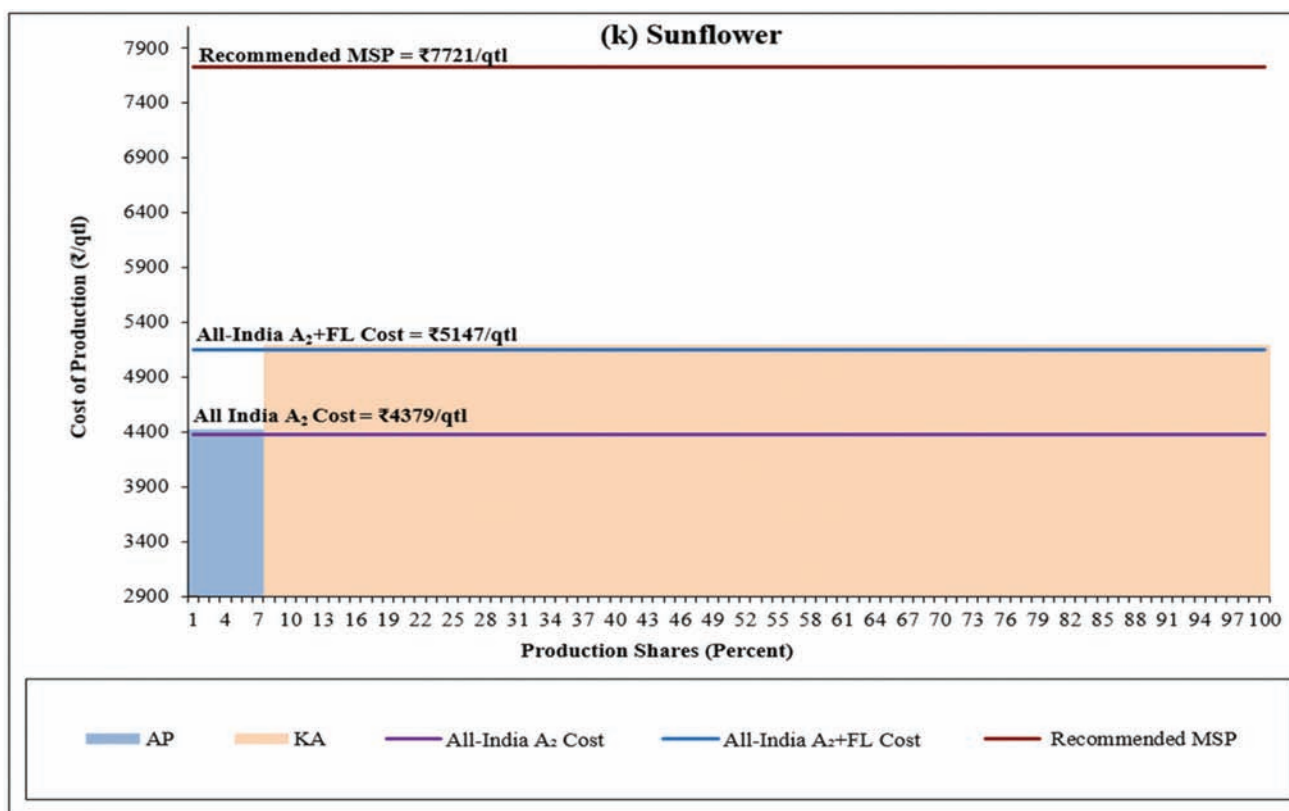
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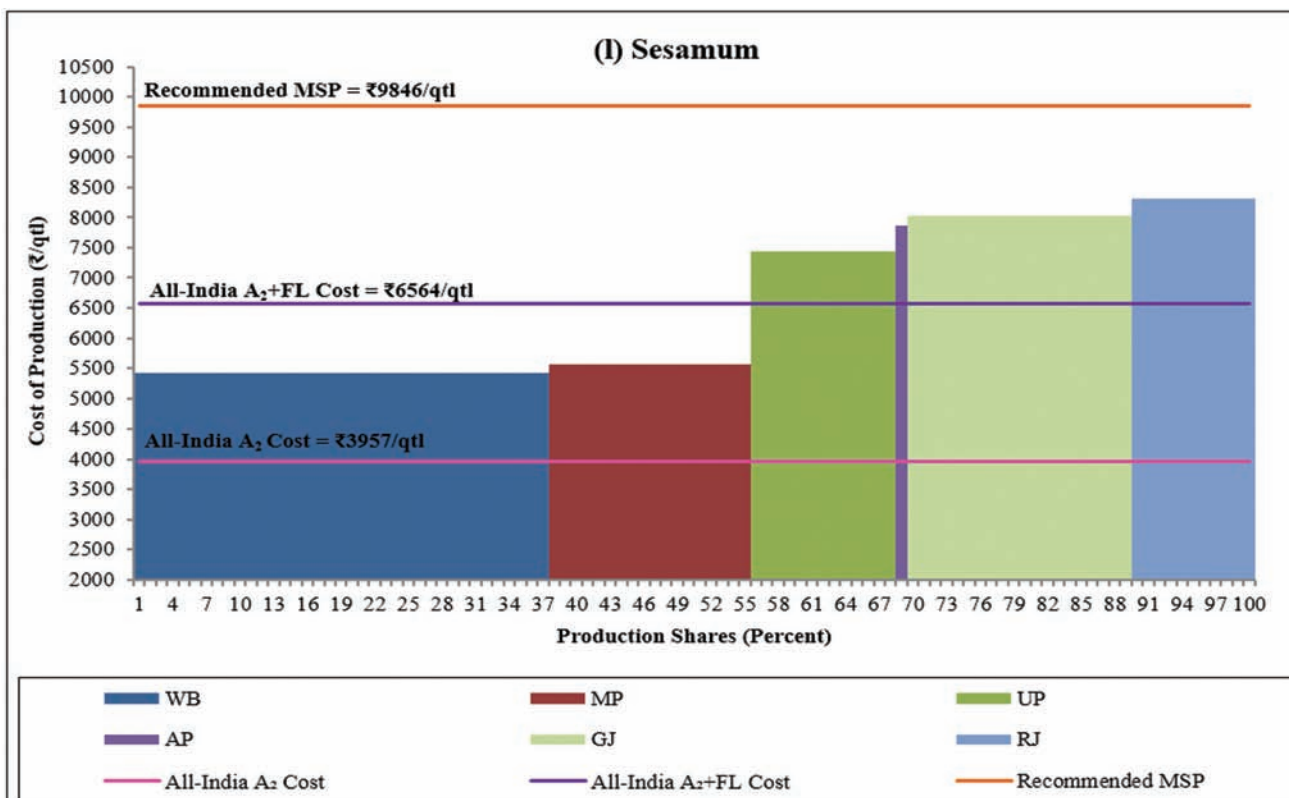
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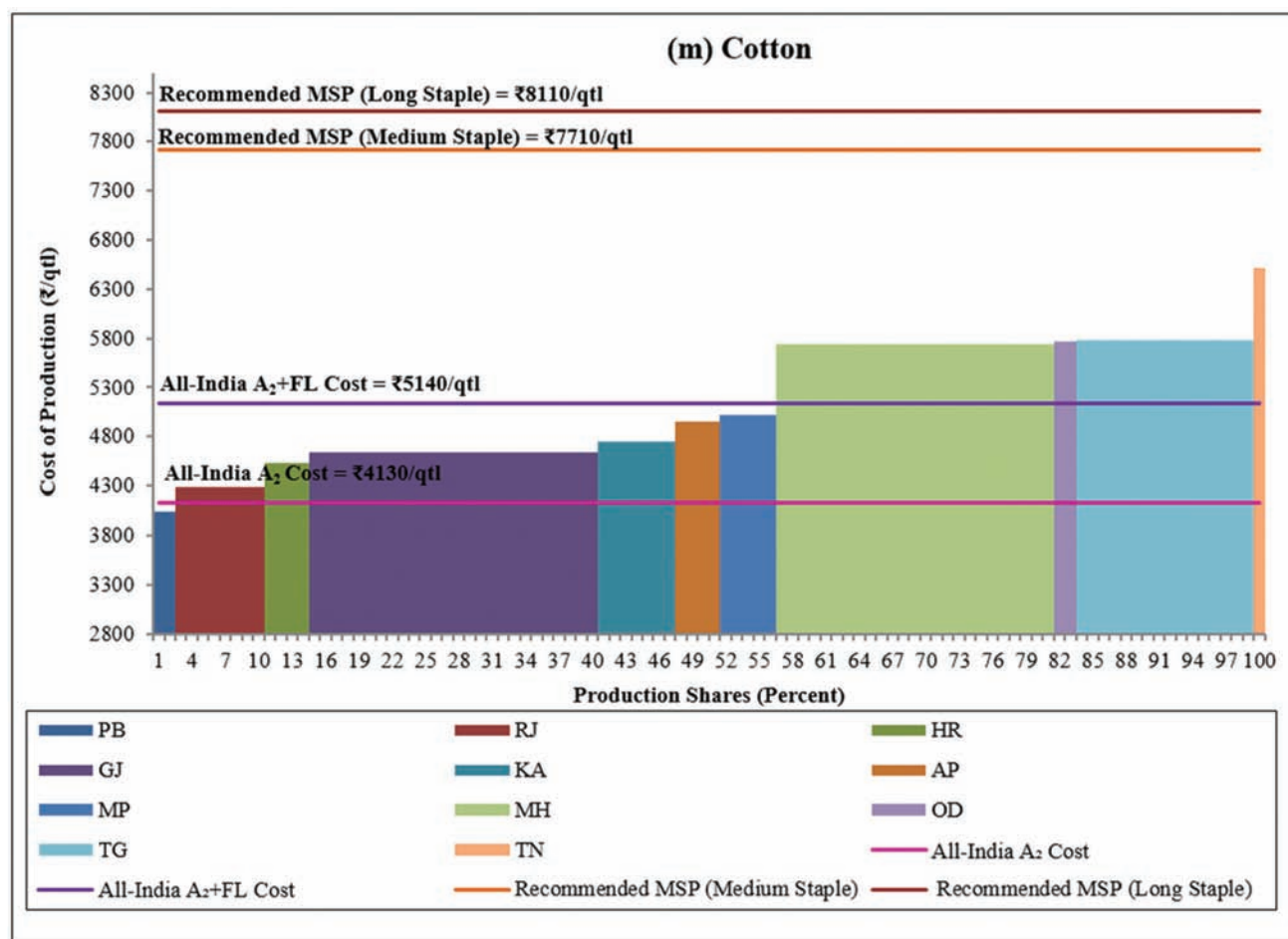
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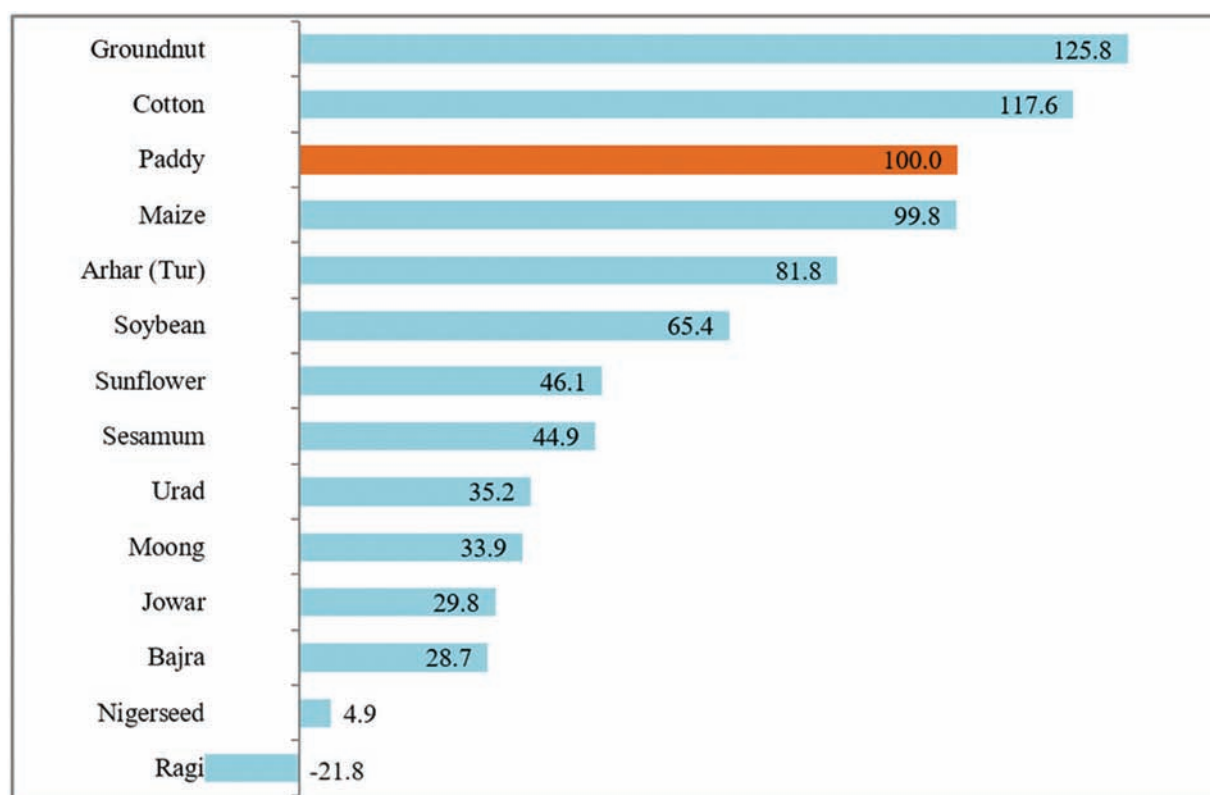
Inter-Crop Parity in Returns of Kharif Crops

- 5.37 Inter-crop parity is an important factor for determination of MSP, therefore, the Commission analysed per hectare relative average gross returns of different *kharif* crops in percentage terms with respect to paddy. Chart 5.7 shows per hectare relative average gross returns over CoC A_2+FL in percentage term for each of the mandated crops with reference to paddy during TE2023-24. It is evident from the Chart that groundnut at 125.8 percent was the most profitable crop during TE2023-24 as compared to other *kharif* crops considered here.
- 5.38 Among cereals, returns from maize were marginally lower than paddy, while in case of nutri-cereals, relative returns were substantially lower compared to paddy and that has resulted in fall in area under nutri-cereals. Among pulses, returns from tur were highest, about 2.4 times of moong and 2.3 times of urad, mainly due to significantly higher yield of tur. Among oilseeds, the returns from groundnut were 1.9 times of soybean, 2.7 times of sunflower, 2.8 times of sesamum and about 21 times higher than nigerseed. Groundnut and cotton have relatively higher returns than paddy but relative returns from nutri-cereals, all three pulses, soybean, sunflower, sesamum and nigerseed were significantly lower compared with paddy.



- 5.39 Relative average gross returns of mandated crops for 2021-22, 2022-23 and 2023-24 with respect to paddy over actual cost of cultivation are given in Annex Table 5.8. Relative returns over A₂+FL with respect to paddy showed that relative profitability of the jowar has increased between 2021-22 and 2023-24, while it has declined in maize, tur, soybean, sunflower and sesamum. For moong and nigerseed, relative returns declined during 2022-23 over 2021-22 but improved during 2023-24 over 2022-23. Relative returns for bajra, ragi, urad, groundnut and cotton increased during 2022-23 over 2021-22 but declined during 2023-24 over 2022-23.

Chart 5.7: Relative Average Gross Returns (%) over CoC A₂+FL with respect to Paddy during TE2023-24



Note: All-India gross returns of a crop are weighted average of gross returns of projected States

Source: CACP Calculations using CS data

- 5.40 Relative average gross returns of different crops in selected States during TE2023-24 with respect to paddy over actual cost of cultivation are given in Annex Table 5.9. Paddy was the most profitable crop in Haryana, Himachal Pradesh, Madhya Pradesh and Punjab, whereas cotton was the most profitable crop in Gujarat, Karnataka, Odisha and Tamil Nadu. Similarly, jowar in Andhra Pradesh and maize in Bihar, Maharashtra and Telangana, tur in Uttar Pradesh and groundnut in Rajasthan and West Bengal were the most profitable crops.
- 5.41 In Andhra Pradesh, jowar and maize had higher profitability than paddy, while groundnut farmers incurred a net loss. Gujarat recorded higher profitability in bajra, tur, and groundnut compared to



paddy. Karnataka farmers incurred net loss in ragi cultivation, while other crops except cotton were less profitable than paddy. In Maharashtra maize, tur and soybean were more profitable, while rice and groundnut farmers incurred net loss. In Odisha, cotton and moong were more profitable than paddy but gross returns from ragi were negative. Paddy was the most profitable crop in Punjab, while returns from maize were significantly lower due to low productivity. Groundnut was the most profitable crop followed by cotton in Rajasthan. In Tamil Nadu, maize and cotton were more profitable than paddy. In Uttar Pradesh, gross returns from moong and sesamum were negative and tur and groundnut were more profitable than paddy. West Bengal witnessed significantly higher profitability in groundnut and sesamum than paddy.

- 5.42 The above analysis indicates that farmers income can be significantly increased by encouraging and incentivizing farmers to shift from less profitable to more profitable crops. Diversification towards jowar and maize in Andhra Pradesh; maize in Bihar; bajra, tur, groundnut, and cotton in Gujarat; cotton in Karnataka; maize, tur, and soybean in Maharashtra; moong and cotton in Odisha; groundnut and cotton in Rajasthan; maize and cotton in Tamil Nadu; maize in Telangana; tur and groundnut in Uttar Pradesh; and groundnut and sesamum in West Bengal would not only enhance farmers income but also align production patterns with evolving market demand for pulses, oilseeds, *Shree Anna* and cotton and achieve *Atmanirbharata* in pulses and edible oils.

Comparison of CACP Cost Estimates with State Estimates

- 5.43 The estimates of projected cost of production/cultivation and yield provided by States and CACP estimates, and suggested Minimum Support Price (MSP) by States for mandated *kharif* crops for KMS2025-26 are given in Annex Table 5.10. It is evident from the Table that there are significant variations in cost estimates provided by the States and CACP cost estimates primarily due to different methodologies and cost concepts used by the States and CACP.
- 5.44 The estimates of projected cost of production for paddy by Andhra Pradesh, Bihar, Gujarat, Karnataka, Kerala, Maharashtra, Punjab, Telangana were higher than CACP projections, while cost estimates provided by Chhattisgarh and West Bengal were lower than CACP estimates. For jowar, Andhra Pradesh, Karnataka and Maharashtra cost estimates were higher than CACP estimates (Annex Table 5.10). In case of ragi, cost estimates of Karnataka were higher than the CACP estimates. In case of maize, cost estimates of Andhra Pradesh, Gujarat, Karnataka, Maharashtra and Telangana were higher than CACP estimates, while Bihar cost was lower than CACP estimates.
- 5.45 In case of pulses, CoP estimates of tur by Karnataka and Maharashtra were higher than CACP projections, while Andhra Pradesh, Gujarat and Telangana CoP was lower. For moong and urad, cost estimates provided by Andhra Pradesh, Karnataka and Maharashtra were higher than CACP cost estimates, while cost estimates of Gujarat were lower than CACP estimates.
- 5.46 For oilseeds, the estimates of cost of production by Karnataka and Maharashtra for groundnut and soybean, Karnataka for sunflower and Telangana for soybean were higher than CACP estimates,



whereas CoP estimates of Andhra Pradesh for groundnut, sunflower and sesamum, Gujarat and Telangana for groundnut, sesamum in case of Gujarat, were lower than CACP cost estimates. The CoP estimates of cotton by Gujarat, Karnataka, Maharashtra, Punjab and Telangana were higher than CACP projections, while estimated CoP in Andhra Pradesh was lower than CACP estimates.

Table 5.5: Comparison of CACP Cost Estimates and CoP Estimates provided by Major States

Crop	States having Cost Estimates	
	$\geq$ CACP estimates	$<$ CACP estimates
Paddy	AP, BR, GJ, KA, KL, MH, PB, TG	CG, WB
Jowar	AP, KA, MH	-
Bajra	GJ, MH	-
Maize	AP, GJ, KA, MH, TG	BR
Ragi	KA	-
Tur	KA, MH	AP, GJ, TG
Moong	AP, KA, MH	GJ
Urad	AP, KA, MH	GJ
Groundnut	KA, MH	AP, GJ, TG
Soybean	KA, MH, TG	-
Sunflower	KA	AP
Sesamum	-	AP, GJ
Cotton	GJ, KA, MH, PB, TG	AP

Source: State Governments and CACP Calculations

Issues Related to Sample Size under Comprehensive Scheme

- 5.47 The Commission has observed that there are some crops in the States whose share in the all-India crop area/production as well as in the area and production of crop group in the State is significant but are not covered under the Comprehensive Scheme. In addition, sample size for certain crops in the State and all-India is inadequate, and may not be a representative sample for cost projections (Annex Table 5.11 and Annex Table 5.12). The Commission recommends that sample size of jowar in Andhra Pradesh and Madhya Pradesh, bajra in Tamil Nadu, maize in Jharkhand and Punjab, tur in Madhya Pradesh, moong in Madhya Pradesh, Haryana, Punjab, Telangana, Uttar Pradesh and West Bengal, urad in Chhattisgarh, Jharkhand, and Karnataka, sunflower in Andhra Pradesh and Bihar and sesamum



in Telangana should be increased. The sample size of ragi, sunflower and nigerseed even at all-India level is thin and can undermine the reliability and representativeness of cost projections, therefore, sample size for these crops needs to be increased.

- 5.48 Annex Table 5.13 provides information about the crops that occupy reasonable share in the all-India area and production or area and production of a crop group in the State, however, CS data for these States are not available. Therefore, inclusion of these States under Comprehensive Scheme would improve representation and reliability of cost projections at all-India level. Therefore, the Commission recommends inclusion of the States for these crops under Comprehensive Scheme and implementation of Comprehensive Scheme in Jammu & Kashmir for paddy and maize, Tripura for paddy, urad and sesamum, and Uttarakhand for paddy, ragi, urad, and sesamum, as these are important crops in these States/UT.

Recapitulation

- 5.49 The average CoC A_2 +FL per hectare during TE2023-24 varied from ₹37,123 for bajra to ₹61,314 for paddy in cereals; ₹28,193 for moong to ₹49,080 for tur in pulses; and ₹22,406 for nigerseed to ₹76,772 for groundnut in oilseeds. Average GVO per hectare ranged from ₹45,780 for bajra to ₹91,530 for paddy in cereals; ₹38,447 for moong to ₹73,783 for tur in pulses; and ₹23,880 for nigerseed to ₹1,14,789 for groundnut in oilseeds. For cotton, average CoC A_2 +FL per hectare was ₹76,543 and GVO was ₹1,12,082 during TE2023-24. Average gross returns over CoC A_2 +FL per hectare during TE2023-24 were ₹8,657 for bajra and ₹30,216 for paddy in cereals, while ragi showed negative returns (-₹6596). In case of pulses, gross returns were lowest from ₹10,254 for moong highest (₹24,703) for tur while in oilseeds, gross returns were in the range of ₹1,474 for nigerseed to ₹38,017 for groundnut. In case of cotton, gross returns over CoC A_2 +FL were ₹35,539 per hectare.
- 5.50 All-India nominal agricultural average daily wage rate increased by 6.3 percent in 2024 over 2023, while weighted index of WPIs of selected farm input price decreased by 2.8 percent. All-India CIPI for mandated *kharif* crops registered an increase of 6.9 percent in 2025-26 over 2024-25. The human labour cost accounted for nearly 50 percent of cost of cultivation and as labour availability and rising wages are becoming major constraints, efforts should be made to promote farm mechanisation to reduce cost and improve profitability.
- 5.51 The all-India projected CoP A_2 +FL per quintal was estimated at ₹1,579 for paddy, ₹2,466 for jowar, ₹1,703 for bajra, ₹1,508 for maize, ₹3,257 for ragi, ₹5,038 for tur, ₹5,845 for moong, ₹5,114 for urad, ₹4,842 for groundnut, ₹3,552 for soybean, ₹5,147 for sunflower, ₹6,564 for sesamum, ₹6,358 for nigerseed, and ₹5,140 for cotton for 2025-26. The increase in projected CoP ranged from one percent for moong and three percent for paddy to 14.7 percent for bajra in 2025-26 over 2024-25. Average relative returns over CoC A_2 +FL with respect to paddy during TE2023-24 were negative for ragi (-21.8%), and ranged from 33.9 percent for moong to 81.8 percent for tur in pulses,



and 4.9 percent for nigerseed to 125.8 percent for groundnut in oilseeds, and were 117.6 percent in cotton.

- 5.52 In States where the cost of production for a crop is higher, efforts should be focused on improving crop yield and promoting the efficient use of inputs to reduce cost of cultivation. The CS data should be collected for States that significantly contribute to the national area and production, as well as within the State's crop group, to ensure an adequate sample size for improving the representativeness and reliability of the cost estimates at the national and State level.
- 5.53 Per quintal MSP recommended at ₹2,369 for paddy, ₹3,699 for jowar, ₹2,775 for bajra, ₹2,400 for maize, ₹4,886 for ragi, ₹8,000 for tur, ₹8,768 for moong, ₹7,800 for urad, ₹7,263 for groundnut, ₹5,328 for soybean, ₹7,721 for sunflower, ₹9,846 for sesamum, ₹9,537 for nigerseed, and ₹7,710 for cotton; would cover projected $A_2 + FL$ CoP for all States for jowar, bajra, maize, ragi, tur, moong, urad, soybean, sunflower, sesamum, nigerseed and cotton; and seventeen out of eighteen States for paddy and nine out of ten States for groundnut.



CHAPTER 6



Considerations and Recommendations for Price Policy

- 6.1 The Commission considers multiple factors, which are part of its broad terms of reference, while recommending the Minimum Support Price (MSP). These include the cost of production, demand-supply situation, domestic and international prices, inter-crop parity, terms of trade between agriculture and non-agriculture sector, rational utilisation of land, water and other production resources, likely impact of the price policy on the economy and a minimum 50 percent margin over the cost of production. The Commission uses cost estimates provided by the Economics, Statistics and Evaluation Division of the Ministry of Agriculture and Farmers Welfare under the “Comprehensive Scheme for Studying the Cost of Cultivation of Principal Crops in India”. In addition, the Commission holds regional meetings with State Governments and wide-ranging stakeholder consultations with representatives of Central Ministries/Departments, farmers and farmers’ associations, research organizations, industry associations, private sector agri-food companies, etc.

Demand-Supply Situation and Outlook

Domestic Scenario

- 6.2 India achieved record production of foodgrains (332.3 million tonnes) and rice (137.8 million tonnes) in 2023-24. As per Second Advance Estimates for 2024-25, *kharif* rice production is estimated to touch a new record (120.7 million tonnes) in 2024-25, while *rabi* production is estimated to increase by 7.9 percent. The total stocks of rice in the Central Pool were about 34 million tonnes on 31st January 2025 (3.5 times the buffer stock norms of 7.6 million tonnes as on 1st January). Despite such high stocks, annual rate of inflation for paddy based on WPI remained high (9.8%) during 2024 and was 6.22 percent in January 2025. Rice exports, which declined during 2023-24 due to export restrictions, were 3.3 percent higher during April-November 2024 than the corresponding period last year. The domestic rice supply in 2024-25 is likely to be at a record high due to higher production and excess stocks.
- 6.3 Maize production, which marginally declined in 2023-24, is set to achieve a new record in 2024-25 as production of *kharif* maize that accounts for more than 60 percent of total production is estimated to increase by 10.3 percent in 2024-25, while *rabi* maize production is estimated to rise by 3.4 percent. The demand for maize is expected to increase due to rising demand from both the feed and ethanol industries.



- 6.4 In case of *Shree Anna*/nutri-cereals, jowar production witnessed a record 24.2 percent increase during 2023-24, while bajra and ragi production significantly declined. *Kharif* jowar production is estimated to further rise by 44.7 percent in 2024-25 but *rabi* production is likely to decline by 3.4 percent. Ragi production is forecast to marginally increase (1.3%) during 2024-25, while *kharif* bajra production is projected to again fall (-)1.8 percent during 2024-25. Decline in production of bajra during 2023-24 and 2024-25 and ragi during 2022-23 and 2023-24 warrants immediate attention as these crops are grown mainly under rainfed conditions and by small and marginal farmers.
- 6.5 Total pulses production, which reached a record level of 27.3 million tonnes in 2021-22, witnessed a continuous decline during the last two years. However, increase in pulses production during both *kharif* (2.5%) and *rabi* (4.5%) seasons in 2024-25 and estimates of higher output of tur (2.8%) and *kharif* moong (30%) is reason for optimism. However, nearly 20 percent decline in production of urad during *kharif* 2024-25 is a real cause of concern as dependence on imports for urad is significantly high. There was a significant increase in imports of pulses, particularly tur/arhar and yellow peas during the first eight months of 2024-25, which adversely affected domestic producer prices.
- 6.6 In case of oilseeds (edible oils), India is highly dependent on imports and a record 15.7 million tonnes (₹1.67 lakh crore) of edible oils were imported in 2022-23, which marginally declined in 2023-24. During the first eight months of 2024-25, India has already imported 12.1 million tonnes (₹1.02 lakh crore), significantly higher than the corresponding period last year. Total oilseeds production, which declined (4.1%) in 2023-24, is expected to rebound in 2024-25 due to significantly higher production of *kharif* oilseeds (14.4%) and 6.2 percent increase in *rabi* oilseeds production compared with 2023-24 season. However, recent increase in import tariff on edible oils and National Mission on Edible Oils (NMEO) are expected to have a positive impact on domestic oilseeds production. During KMS2024-25, market prices of groundnut and soybean were substantially lower than MSP and Government had to procure large quantities of both crops.
- 6.7 Cotton production, which declined by 3.4 percent in 2023-24, is estimated to witness a steeper decline (-9.5%) during 2024-25 as per the Second Advance Estimates. India's lower opening stocks and reduced production are expected to result in higher imports in 2024-25. Both demand and supply of cotton are estimated to decline in 2024-25 but closing stocks are estimated to fall by a whopping 42 percent, indicating tight supply position of cotton during 2025-26.

Global Outlook

- 6.8 Global rice production, supply and utilization are forecast to increase in 2024-25, while rice trade is estimated to remain at the same level as in 2023-24. World stocks are predicted to rise by 3.1 percent and reach a fresh peak in 2024-25 but little change is estimated in Stock to Use Ratio (SUR) due to record consumption (539 million tonnes) in 2024-25.
- 6.9 World maize production and supply are estimated to be lower in 2024-25, while consumption is forecast to rise due to stronger feed and industrial use. International trade in maize for 2024-25 is



projected to drop by roughly six percent and stocks are forecast to fall below the opening level in 2024-25, resulting in lower SUR.

- 6.10 As per USDA estimates, global production, supply and utilization of oilseeds complex (oilseeds, oil meal and vegetable oils) is forecast to increase by about three percent in 2024-25, while trade is estimated to increase by about one percent. World stocks of oilseeds (5.7%) and oil meals (11.4%) are expected to have a sizeable increase but vegetable oil stocks are projected to fall by about eight percent in 2024-25. World production, supply, and utilization of soybean are forecast to expand by 6-8 percent, while trade is predicted to increase marginally (1%) in 2024-25. World stocks and SUR of soybean are estimated to increase substantially in 2024-25.
- 6.11 Both USDA and International Cotton Advisory Committee (ICAC) forecast an increase in world cotton production and decline in trade in 2024-25. USDA forecasts indicate that global stocks and SUR will increase but ICAC forecasts a marginal increase in stocks but a decline in SUR.

Price Trends

- 6.12 Market prices of paddy have generally remained above the MSP over the last five marketing seasons due to timely and significant procurement of paddy by the Government. During KMS2024-25, market prices stayed above MSP but showed a steady decline in December 2024 and January 2025 but again moved up during February and March 2025. The FAO All Rice Price Index (FARPI) showed a continuous decline during the last six months and reached 105.9 points in February 2025, 6.8 percent lower than in January 2025, 24.6 percent lower than in February 2024 and at its lowest level since April 2022 mainly due to limited demand and forecast of higher supplies from Vietnam and India.
- 6.13 Market prices of maize, which generally remained below the MSP during KMS2022-23 and 2023-24, were higher than MSP during the last 3-4 months of KMS2024-25. Maize prices are expected to remain firm in the coming season primarily due to higher demand from poultry feed industry and ethanol producers. World prices of maize witnessed a steady decline since October 2022 but rose during last four months and average price was about US\$203 per metric tonne in December 2024 but about 33 percent lower than in December 2022.
- 6.14 Market prices of tur remained significantly high during KMS2023-24, showed a declining trend from October 2024 and dipped below the MSP in January-February 2025, falling below ₹7,000 per quintal in March 2025. Average market price of urad, in which import dependence is high, remained much above the MSP during 2023-24 but witnessed a declining trend since September 2024 and reached around ₹7,000 per quintal in February-March 2025. Moong prices have remained lower than MSP during last five seasons but prices recorded an increasing trend during January 2022 to April 2024. Market prices witnessed a declining trend and were below the MSP during KMS2024-25, reaching around ₹7,600 per quintal in February 2025. Cheap and unrestricted imports of pulses adversely affected domestic prices, which might discourage farmers from growing these crops.



- 6.15 The market prices of groundnut and soybean, two major *kharif* oilseeds, remained fairly remunerative and higher than MSP during the last 2-3 seasons but ruled significantly below the MSP during KMS2024-25. The worst sufferer was groundnut where market prices were more than 20 percent below the MSP, while in soybean prices were about 15 percent lower than MSP. Inadequate procurement (much lower than sanctioned quantity) by State Government agencies, mainly due to lack of storage and warehousing facilities and quality issues, has further aggravated the problem. It acts as a constraint as well as disincentive for crop diversification and achieving *Aatmanirbharta* in oilseeds. Therefore, it is necessary to create effective procurement mechanism at State level for oilseeds and pulses.
- 6.16 Market prices of cotton, which remained higher than MSP during KMS2021-22 and KMS2022-23, fell below the MSP during KMS2023-24 and 2024-25. The CCI has already procured about 95 lakh bales as on 4th March 2025 during the current season.

Trade Performance

- 6.17 India is an important player in world trade of agricultural products and ranks among top 10 exporters. The country is a leading exporter of many agricultural products and has a significant potential to expand its market access across different countries and products. India's agricultural exports, which reached an all-time high in 2022-23 but declined in 2023-24 due to various export restrictions, are expected to cross new highs in 2024-25 as agricultural exports have increased about eight percent in April-December 2024 over April-December 2023.
- 6.18 Rice continues to be the top export product, followed by marine products and spices. India, which was a net exporter of cotton, has become a net importer due to continuous decline in cotton production. India's agricultural imports, which declined during 2023-24 primarily due to sharp fall in edible oils import bill, are likely to be higher during 2024-25 as imports during April-December 2024 were about 20 percent higher than April-December 2023 due to significant rise in imports of edible oils (17.9%) and pulses (55.5%).

Productivity Growth

- 6.19 During TE2023-24, *kharif* cereals, rice, maize, jowar, bajra and moong recorded a moderate yield growth, while urad yield increased significantly (7%) but yield of tur declined (-3.2%). Cotton yield also witnessed a decline during the recent years. Yield levels of mandated *kharif* crops remain significantly lower than major producing countries and yield gaps between actual and potential yield are substantial. Since yield improvement is an important instrument to contain cost of production, investment in technology including high-yielding climate resilient varieties, irrigation, and R&D is must. Concerted efforts to bridge yield gaps through advanced production technologies, scientific farm management practices, mechanization, and efficient input use should be made.

Food Inflation

- 6.20 The Consumer Food Price Index (CFPI) inflation in 2024 averaged 8.4 percent compared to 6.8



percent in 2023. Similarly, annual rate of inflation based on WPI Food Articles showed an increasing trend, from 5.87 percent in 2023 to 8.17 percent in 2024. The annual rate of inflation for paddy based on WPI increased from 8.6 percent in 2023 to 9.8 percent in 2024, while CPI-based inflation for cereals and products declined from 12.8 percent in 2023 to 7.7 percent in 2024. Inflation for pulses rose from 10.8 percent (WPI) in 2023 to 15 percent in 2024 and from 11.6 percent (CPI) in 2023 to 13.1 percent in 2024. The WPI inflation for oilseeds and CPI based inflation for oils & fats remained in the negative zone during 2024, while WPI inflation for vegetable oils moved from negative zone in 2023 to 4.4 percent in 2024. The overall WPI-based inflation for food articles was 5.88 percent in January 2025, lower than January 2024 (6.91%), while inflation rate based on All India Consumer Food Price Index (CFPI) for January 2025 was 6.02 percent, lower than 8.3 percent in January 2024. Food inflation, which remained high during 2024, is expected to moderate due to improved supply situation of pulses and cereals, however, uncertainties about global developments and weather conditions could influence price trends in the coming season.

Procurement Operations

- 6.21 Rice procurement reached a record high of 60.2 million tonnes in KMS2020-21 but declined during the next three seasons due to high market prices and was 52.5 million tonnes in KMS2023-24. Punjab (23.6%) remained the top contributor, followed by Chhattisgarh (15.8%) and Telangana (12.2%), while share of West Bengal, the largest producer, was 3.2 percent and share of Uttar Pradesh was 6.9 percent in KMS2023-24. More than 93 percent of marketed surplus was procured in Punjab, followed by Chhattisgarh (86.7%), Haryana (74.6%) and Odisha (71.6%), while the share was quite low in West Bengal and Uttar Pradesh, highlighting disparities in procurement operations across States. Rice procurement was 46.2 million tonnes during KMS2024-25 (as on 6th March 2025), 4.9 percent higher than the corresponding period last year.
- 6.22 Due to high market prices, procurement of pulses and maize was negligible during KMS2024-25. Procurement of oilseeds witnessed a sharp increase as prices of groundnut and soybean were substantially lower than the MSP. However, actual procurement of oilseeds was significantly lower than sanctioned quantity due to shortage of storage and warehousing facilities and quality issues.
- 6.23 There was no procurement of cotton during 2021-22 and 2022-23 as market prices were substantially high. Cotton procurement saw significant increase in 2023-24 and 2024-25 due to depressed domestic and global prices. CCI procured 32.8 lakh bales in 2023-24 and has already procured about 95 lakh bales in 2024-25 (as on 3rd March 2025).

Terms of Trade

- 6.24 Farmers' terms of trade index (FToT) remained below 100 during the last decade and was 99.5 in 2023-24, slight improvement over 2022-23. The terms of trade for agriculture (AGRTOT) have remained favourable towards agriculture since 2012-13, declined in 2021-22 and 2022-23 but recorded an improvement in 2023-24.



Cost of Production, Profitability and Inter-Crop Parity

- 6.25 During TE2023-24, the average Cost of Cultivation (CoC) A_2 +FL per hectare varied from ₹37,123 for bajra to ₹61,314 for paddy in cereals and from ₹28,193 in moong to ₹49,080 in tur for pulses. For oilseeds, the CoC ranged from ₹22,406 in nigerseed to ₹76,772 in groundnut, while it was ₹76,543 in cotton. Gross returns over CoC A_2 +FL per hectare ranged from ₹8,657 for bajra to ₹30,216 for paddy in cereals but ragi had negative returns (-₹6,596). In pulses, returns varied from ₹10,254 for moong to ₹24,703 for tur, while in oilseeds, gross returns varied from ₹1,474 for nigerseed to ₹38,017 for groundnut. Cotton was the second most profitable crop with gross returns of ₹35,539 per hectare.
- 6.26 The all-India projected Cost of Production (CoP) A_2 +FL per quintal for 2025-26 was estimated at ₹1,579 for paddy, ₹2,466 for jowar, ₹1,703 for bajra, ₹1,508 for maize, ₹3,257 for ragi, ₹5,038 for tur, ₹5,845 for moong, and ₹5,114 for urad. In case of oilseeds, CoP was ₹4,842 for groundnut, ₹3,552 for soybean, ₹5,147 for sunflower, ₹6,564 for sesamum, and ₹6,358 for nigerseed. For cotton, the projected CoP was ₹5,140 per quintal. The increase in CoP for 2025-26 varied from one percent in moong and three percent in paddy to 14.7 percent for bajra compared with 2024-25.
- 6.27 The analysis of inter-crop parity reveals that paddy and maize are equally profitable over A_2 +FL cost, cotton and groundnut are more profitable than paddy, while all other mandated crops have lower profitability compared to paddy with ragi having negative returns.
- 6.28 The Commission on the basis of detailed analysis of relevant factors discussed above makes the following price and non-price policy recommendations:

Non-Price Recommendations

Promote Balanced Production Pattern

- 6.29 There has been a growing imbalance between domestic production and demand for different agricultural commodities in the country. India has large surpluses of rice and has become the largest exporter of rice in the world but is highly dependent on imports for meeting domestic demand for edible oils and pulses. The continuous cultivation of paddy-wheat monoculture in the northwest Indo-Gangetic Plains over last five decades has led to several sustainability challenges, including declining groundwater table, soil degradation, increased greenhouse gas emissions, etc. as well as decline in area under pulses, oilseeds, *Shree Anna*/nutri-cereals, maize, etc. Therefore, there is a need to have a strategic shift towards alternative crops to achieve *Aatmanirbharta* in pulses and oilseeds as well as address sustainability challenges. It is worth mentioning that the Commission has tried to realign price incentives in favour of pulses, oilseeds and *Shree Anna* to encourage farmers to diversify crop choices and adopt modern technologies but yield levels of these crops are so low that their relative profitability is much lower than the competing crops. The Commission recommends that a comprehensive diversification strategy with a special focus on technological innovations and better price incentive for pulses and oilseeds should be prepared.



Improve Productivity and Bridge Yield Gaps

- 6.30 Although most of the crops have experienced significant yield improvements, yields are stagnating or declining for some crops, which would have profound implications for Indian agriculture. In addition, there are large gaps between actual and the potential yield in many crops due to socio-economic and biophysical factors. Therefore, understanding how changes in management practices including irrigation, nutrient management, disease and pest management, etc. could help in bridging yield gaps is critical to address the issue of stagnating/declining yields. The Commission recommends that investment in agricultural Research and Development (R&D) should be stepped up and effective and timely availability of climate resilient high yielding variety seeds, quality inputs and services, and improvement in Seed Replacement Rate (SRR) and Varietal Replacement Rate (VRR) should be accorded highest priority.

Improve Soil Health and Productivity

- 6.31 Balanced application of fertilizers is key in improving nutrient use efficiency, soil health and crop productivity. However, Indian agriculture faces challenges of both excessive fertilizer use in some regions and imbalanced use of nutrients in most of the regions, primarily due to the heavy subsidy on urea compared to P & K fertilizers, and poor extension services. Government has launched various initiatives to promote sustainable and balanced use of fertilizers as well as alternative fertilizers including organic fertilizers. The Commission recommends two-pronged approach; one, massive awareness programmes and campaigns highlighting the harmful effects of imbalanced use, and two, address the issue of lopsided fertiliser subsidies, which substantially lower the prices of urea relative to P and K and complex fertilizers.

Focus on Irrigation Development and Management

- 6.32 Assured irrigation is essential for development of agriculture especially when variability in rainfall pattern due to climate change is becoming a concern for the agriculture sector. Government has taken various initiatives to improve irrigation facilities and water use efficiency at farm level including the flagship programme '*Pradhan Mantri Krishi Sinchayee Yojana*', which has resulted in significant growth in irrigation development but over exploitation of water resources and low water use efficiency remains a major concern.
- 6.33 The Commission recommends that concerted efforts should be made to increase coverage of area under micro-irrigation through awareness and training programmes for farmers and ensuring supply of good quality equipments along with post installation maintenance services. In addition, special emphasis on repair, renovation and restoration of traditional water bodies including tanks should be given.

Promote Farm Mechanization

- 6.34 Mechanization of agriculture has become a pressing necessity due to rising labour costs and non-availability of labour. Sub-Mission on Agricultural Mechanization (SMAM) has played an important role in promoting mechanization but level of mechanization in some crops and States is low. Given the labour scarcity and Government's initiatives to promote farm mechanization, the Commission



recommends significant increase in allocation under SMAM and expansion of CHC network in underserved areas. The issue of high GST rates on farm machinery and implements requires favourable consideration to encourage mechanization.

Emphasis on Easy and Equitable Access to Credit

- 6.35 Agricultural credit has witnessed a double-digit growth and increased from ₹8.5 lakh crore in 2014-15 to ₹25.5 lakh crore in 2023-24. However, there is a significant disparity in agricultural credit availability across regions and States. In India, majority of farmers are small and marginal, who are dependent on non-institutional sources of credit. The Commission recommends that special efforts should be made to extend agricultural credit to small and marginal farmers and to regions with low credit intensity. More coordinated efforts from all stakeholders including Government and the lending agencies are needed to meet the objective of equitable and universal distribution of agricultural credit.

Reduce Cost of Production

- 6.36 There are large variations in cost of production across the States due to differences in level of productivity, access to technology, inputs and services, and prices of inputs and outputs. In case of paddy, projected cost of production during KMS2025-26 varied from ₹967/ql in Punjab to ₹3,407/ql in Maharashtra, in groundnut from ₹3,088/ql in Rajasthan to ₹8,428/ql in Maharashtra and in cotton from ₹4,039 in Punjab to ₹6,513 in Tamil Nadu. A strategic and prudent response to rising cost of production lies in improving productivity, adoption of new innovative technologies, scientific practices and effective extension services. Additionally, farmers should be encouraged and incentivized to shift from less profitable crops to alternative crops, which are more suitable and profitable.

Address Market Distortions in Agriculture

- 6.37 State interventions in the form of high market fees, taxes and other incidental charges on procurement result in high procurement incidentals and economic cost of grains, and direct or indirect bonus over and above the MSP by some States severely distort markets. Such interventions also significantly restrict participation of private sector, discourage competition and adversely affect global competitiveness. In addition, bonus on paddy, which affects inter-crop parity, encourages farmers to produce more paddy and thereby, adversely affects crop diversification efforts in the States. The Commission recommends that States should rationalise market fee and other charges to create efficient and competitive national market and crop-specific incentive should be discontinued. The farmers should be encouraged and incentivised for diversification, particularly towards pulses, oilseeds and millets.

Strengthen Procurement Operations at State Level

- 6.38 Although procurement of rice is quite significant and 47.7 percent of marketed surplus was procured at all-India level during TE2023-24, the level of procurement is low in some major producing States like West Bengal, Uttar Pradesh, Tamil Nadu, Karnataka, Bihar and Assam. In addition, actual procurement of pulses and oilseeds under the Price Support Scheme (PSS) has always remained less



than sanctioned quantity e.g., during KMS2024-25, about 56 percent of sanctioned quantity in groundnut and 41 percent in case of soybean was procured due to lack of storage, warehousing facilities and procurement centres, delay in start of procurement operations and quality issues. Since PSS is implemented on the request of the State Governments/UTs, the Commission recommends that States should prepare a comprehensive plan for strengthening procurement operations, creating sufficient storage and warehousing facilities and training farmers to meet FAQ norms as well as provide dryers and other facilities.

Underestimation of MSP Procurement by the National Statistical Office (NSO)

- 6.39 The National Sample Survey (NSS) Report “Situation Assessment of Agricultural Households and Land and Livestock Holdings of Households in Rural India, 2019 (NSS 77th Round)” has grossly underestimated paddy output sold under MSP and coverage of farmers under procurement operations.
- 6.40 The actual procurement of paddy (% of production) was significantly higher in all major States compared with NSS estimates during KMS2018-19. For example, in case of Punjab, actual procurement was 88.4 percent of total production compared with 31.1 percent as per NSS estimates. Similarly, in Haryana, actual procurement was 87.3 percent, while NSS estimate was only 27.1 percent. The divergence in these two estimates was quite significant in other States as well. Similarly, percentage of paddy farmers who sold their produce to procurement agency as per NSS estimates was significantly lower than actual number of beneficiary farmers under MSP procurement operations.
- 6.41 Such underestimates create widespread misperceptions about the effectiveness of procurement operations and adversely affect farmers. Therefore, the Commission recommends that the issue of divergence between actual procurement and NSS estimates should be addressed.

Review Import Duty Structure of Pulses and Edible Oils

- 6.42 Ensuring remunerative prices to the farmers is one of the important pathways to achieve *Aatmanirbharta* in pulses and edible oils and pro-farmer trade policy mechanism will help in ensuring better prices to farmers. However, unrestricted and cheap imports of pulses and edible oils have resulted in domestic prices of soybean, groundnut, urad, moong and tur falling below the MSP during KMS2024-25.
- 6.43 Despite, increase in effective import duty on crude edible oils to 27.5 percent and refined oils to 35.75 percent in September 2024, market prices of *kharif* oilseeds remained well below the MSP during this season. However, to make India’s edible oilseeds complex globally competitive, domestic consumption and export of oil meals should be incentivised.
- 6.44 Imports of pulses at low/zero import duty resulted in depressed market prices during both *rabi* and *kharif* seasons in 2024-25. Aligning import duty structure with MSP will ensure remunerative prices to farmers and encourage them to increase area and production of oilseeds and pulses. The Commission recommends that imports of yellow peas should be banned and import duty on other pulses, particularly, tur/arhar, lentil and urad should be increased.



Increase Coverage and Sample Size under the Comprehensive Scheme

- 6.45 There are some crops that have significant share in both all-India and State-level area and production but are not adequately represented in the Comprehensive Scheme. Additionally, sample size for some crops is insufficient, potentially undermining the reliability of cost estimates. To address this, the Commission recommends that sample size for jowar in Andhra Pradesh and Madhya Pradesh, bajra in Tamil Nadu, maize in Jharkhand and Punjab, tur and moong in Madhya Pradesh, and all-India sample size of ragi, sunflower, and nigerseed should be increased. The coverage of Comprehensive Scheme should be extended to Jammu & Kashmir, Tripura and Uttarakhand.

Minimum Support Price (MSP) Recommendations

- 6.46 Considering all the relevant factors, the Commission recommends the following Minimum Support Price (MSP) for the mandated *kharif* crops for KMS2025-26 (Table 6.1):

Table 6.1: MSP Recommended for KMS2025-26

Crops	Projected A_2+FL Cost for KMS2025-26	MSP for KMS2024-25	Recommended MSP for KMS2025-26	(₹/qtl)
				MSP as percent of A_2+FL
Paddy-Common	1579	2300	2369 (3.0)	150
Paddy-Grade A	-	2320	2389 (3.0)	-
Jowar-Hybrid	2466	3371	3699 (9.7)	150
Jowar-Maldandi	-	3421	3749 (9.6)	-
Bajra	1703	2625	2775 (5.7)	163
Ragi	3257	4290	4886 (13.9)	150
Maize	1508	2225	2400 (7.9)	159
Tur/Arhar	5038	7550	8000 (6.0)	159
Moong	5845	8682	8768 (1.0)	150
Urad	5114	7400	7800 (5.4)	153
Groundnut	4842	6783	7263 (7.1)	150
Sunflower Seed	5147	7280	7721 (6.1)	150
Soybean (Yellow)	3552	4892	5328 (8.9)	150
Sesamum	6564	9267	9846 (6.2)	150
Nigerseed	6358	8717	9537 (9.4)	150
Cotton (Medium Staple)	5140	7121	7710 (8.3)	150
Cotton (Long Staple)	-	7521	8110 (7.8)	-

Note: Figures in parenthesis represent increase in MSP over the previous year



- 6.47 The Commission believes that these non-price and price policy recommendations will incentivise and encourage farmers to adopt new technologies and scientific practices, and provide opportunities for diversification of production pattern in the light of emerging market opportunities.

(Vijay Paul Sharma)
Chairman

(Prem Chand)
Member (Official)

(Ratan Lal Daga)
Member (Non-Official)



Annex Table

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Annex Table 1.1: All-India Estimates of Area under Principal Crops

Crop	Season	(million hectares)											
		2013-14	2014-15	2015-16	2016-17	2017-18	2018-19	2019-20	2020-21	2021-22	2022-23	2023-24	2024-25*
Rice	Kharif	39.45	39.83	39.66	39.85	39.35	39.96	39.01	40.36	41.04	40.4	40.7	43.4
	Rabi	4.69	4.28	3.84	4.15	4.42	4.19	4.65	5.41	5.24	4.3	4.1	4.3
	Total	44.14	44.11	43.5	43.99	43.77	44.16	43.66	45.77	46.28	47.8	47.8	47.7
Wheat	Rabi	30.47	31.47	30.42	30.79	29.65	29.32	31.36	31.13	30.46	31.4	31.8	32.6
Jowar	Kharif	2.28	2.27	2.14	2.06	2.06	1.75	1.76	1.64	1.47	1.4	1.3	1.8
	Rabi	3.52	3.89	3.94	3.57	2.96	2.34	3.07	2.74	2.33	2.1	2.7	2.5
	Total	5.79	6.16	6.08	5.62	5.02	4.09	4.82	4.38	3.8	3.5	4.1	4.3
Bajra	Kharif	7.81	7.32	7.13	7.46	7.48	7.11	7.54	7.65	6.84	7.2	7.0	6.8
Maize	Kharif	7.31	7.56	7.18	7.84	7.43	7.33	7.55	7.76	7.78	8.1	8.3	8.4
	Rabi	1.76	1.62	1.63	1.79	1.95	1.7	2.02	2.14	2.17	2.2	2.2	2.3
	Total	9.07	9.19	8.81	9.63	9.38	9.03	9.57	9.89	9.96	10.7	11.2	10.7
Ragi	Kharif	1.19	1.21	1.14	1.02	1.19	0.89	1	1.16	1.22	1.2	1.2	1.2
Barley	Rabi	0.67	0.71	0.59	0.66	0.66	0.58	0.59	0.59	0.45	0.6	0.6	0.6
Nutri/ Coarse Cereals	Kharif	19.27	18.95	18.23	18.99	18.71	17.53	18.31	18.65	17.74	18.2	18.3	18.7
	Rabi	5.95	6.22	6.15	6.01	5.57	4.61	5.67	5.46	4.96	5.0	5.5	5.4
	Total	25.22	25.17	24.39	25.01	24.29	22.15	23.99	24.12	22.7	24.1	24.9	24.1
Cereals	Kharif	58.72	58.78	57.89	58.84	58.06	57.5	57.33	59.01	58.78	58.6	59.0	62.1
	Rabi	41.11	41.97	40.42	40.95	39.65	38.12	41.68	42	40.66	40.7	41.4	42.3
	Total	99.83	100.75	98.31	99.79	97.71	95.62	99.01	101.01	99.44	103.3	104.6	104.4
Tur (Arhar)	Kharif	3.9	3.85	3.96	5.34	4.44	4.55	4.53	4.72	4.9	4.1	4.1	4.2
Gram	Rabi	9.93	8.25	8.4	9.63	10.56	9.54	9.7	10	10.74	10.5	9.6	9.9
Urad	Kharif	2.35	2.49	2.72	3.48	4.35	4.73	3.7	3.21	3.62	3.1	2.7	2.1
	Rabi	0.72	0.76	0.9	1	0.93	0.88	0.83	0.93	1.01	0.6	0.6	0.6
	Total	3.06	3.25	3.62	4.48	5.28	5.6	4.53	4.14	4.63	4.0	3.5	2.7

(Contd.)



Annex Table 1.1: All-India Estimates of Area under Principal Crops

Crop	Season	2013-14	2014-15	2015-16	2016-17	2017-18	2018-19	2019-20	2020-21	2021-22	2022-23	2023-24	2024-25*
Moong	Kharif	2.34	2.03	2.76	3.37	3.26	3.83	3.52	3.82	3.84	3.5	3.2	3.4
	Rabi	1.04	0.99	1.07	0.96	0.98	0.92	1.06	1.31	1.71	0.2	0.1	0.1
	Total	3.38	3.02	3.83	4.33	4.24	4.75	4.58	5.13	5.55	5.6	5.2	3.5
Lentil	Rabi	1.34	1.47	1.28	1.46	1.55	1.36	1.3	1.47	1.41	1.6	1.7	1.7
	Kharif	10.34	10	11.31	14.36	13.93	14.83	13.54	13.43	14.12	12.1	11.6	11.3
	Rabi	14.88	13.55	13.6	15.08	15.88	14.32	14.45	15.35	16.61	14.6	13.7	14.2
Foodgrains	Total	25.22	23.55	24.91	29.45	29.81	29.15	27.99	28.78	30.73	28.9	27.5	25.5
	Kharif	69.06	68.78	69.2	73.2	72	72.33	70.86	72.44	72.9	70.7	70.6	73.3
	Rabi	55.99	55.53	54.01	56.03	55.53	52.45	56.13	57.35	57.27	55.3	55.1	56.6
Groundnut	Total	125.05	124.3	123.22	129.23	127.52	124.77	126.99	129.8	130.17	132.2	132.1	129.9
	Kharif	4.65	4.01	3.84	4.58	4.14	4.13	4.16	5.17	4.91	4.3	4.0	5.0
	Rabi	0.86	0.76	0.76	0.76	0.75	0.6	0.66	0.84	0.79	0.4	0.3	0.4
Sesamum	Total	5.51	4.77	4.6	5.34	4.89	4.73	4.83	6.01	5.7	5.0	4.7	5.4
	Kharif	1.68	1.75	1.95	1.67	1.58	1.42	1.62	1.72	1.63	1.0	1.0	1.0
	Kharif	0.3	0.23	0.25	0.26	0.22	0.16	0.14	0.13	0.1	0.1	0.1	0.1
Soybean	Kharif	11.72	10.91	11.6	11.18	10.33	11.13	12.19	12.92	12.15	13.1	13.3	12.9
	Kharif	0.26	0.22	0.16	0.17	0.14	0.12	0.13	0.1	0.14	0.2	0.1	0.1
	Rabi	0.42	0.37	0.33	0.21	0.15	0.14	0.1	0.13	0.13	0.1	0.1	0.1
Rapeseed & Mustard	Total	0.67	0.59	0.49	0.38	0.28	0.26	0.23	0.23	0.27	0.4	0.2	0.1
	Rabi	6.65	5.8	5.75	6.07	5.98	6.12	6.86	6.7	7.99	8.9	9.2	8.8
	Rabi	0.18	0.17	0.13	0.14	0.08	0.05	0.05	0.06	0.08	0.1	0.1	0.1
Oilseeds	Total Nine	19.66	18.21	18.86	18.67	17.23	17.71	19.28	20.93	19.75	19.7	19.5	19.9
	Rabi	8.39	7.39	7.22	7.51	7.28	7.09	7.86	7.9	9.19	9.8	9.9	9.5
	Total	28.05	25.6	26.09	26.18	24.51	24.79	27.14	28.83	28.95	30.2	30.2	29.4
Cotton		11.96	12.82	12.29	10.83	12.59	12.61	13.48	13.29	12.37	12.9	12.7	11.5

Note: * Second Advance Estimates; ^s Million Bales of 170 kgs

Source: Economics, Statistics & Evaluation Division, Ministry of Agriculture and Farmers Welfare

Annex Table



Annex Table

Annex Table 1.2 : All-India Estimates of Production of Principal Crops

Crop	Season	2013-14	2014-15	2015-16	2016-17	2017-18	2018-19	2019-20	2020-21	2021-22	2022-23	2023-24	2024-25*
Rice	Kharif	91.5	91.39	91.41	96.3	97.14	102.04	102.28	105.21	111	110.5	113.3	120.7
	Rabi	15.15	14.09	13	13.4	15.62	14.44	16.59	19.16	18.47	15.0	14.6	15.8
	Total	106.65	105.48	104.41	109.7	112.76	116.48	118.87	124.37	129.47	135.8	137.8	136.4
Wheat	Rabi	95.85	86.53	92.29	98.51	99.87	103.6	107.86	109.59	107.74	110.6	113.3	115.4
Jowar	Kharif	2.39	2.3	1.82	1.96	2.27	1.74	1.7	1.99	1.6	1.5	1.5	2.2
	Rabi	3.15	3.15	2.42	2.6	2.53	1.74	3.08	2.83	2.55	2.3	3.2	3.1
	Total	5.54	5.45	4.24	4.57	4.8	3.48	4.77	4.81	4.15	3.8	4.7	5.3
Bajra	Kharif	9.25	9.18	8.07	9.73	9.21	8.66	10.36	10.86	9.78	10.4	9.7	9.5
Maize	Kharif	17.15	17.01	16.05	18.92	20.12	19.41	19.43	21.56	22.68	23.7	22.3	24.8
		7.11	7.16	6.51	6.98	8.63	8.3	9.34	10.09	11.05	11.7	12.0	12.4
	Total	24.26	24.17	22.57	25.9	28.75	27.72	28.77	31.65	33.73	38.1	37.7	37.3
Ragi	Kharif	1.98	2.06	1.82	1.39	1.99	1.24	1.76	2	1.7	1.7	1.7	1.7
Barley	Rabi	1.83	1.61	1.44	1.75	1.78	1.63	1.72	1.66	1.37	1.9	1.7	2.0
Nutri/	Kharif	31.2	30.94	28.15	32.44	34.03	31.38	33.61	36.75	36.13	37.6	35.5	38.6
	Rabi	12.09	11.92	10.37	11.33	12.94	11.67	14.13	14.57	14.97	15.9	16.9	17.5
	Total	43.29	42.86	38.52	43.77	46.97	43.06	47.75	51.32	51.1	57.3	56.9	56.0
Cereals	Kharif	122.7	122.34	119.56	128.74	131.16	133.42	135.89	141.96	147.13	148.1	148.8	159.2
	Rabi	123.09	112.53	115.66	123.24	128.44	129.71	138.59	143.32	141.19	141.5	144.8	148.7
	Total	245.79	234.87	235.22	251.98	259.6	263.13	274.48	285.28	288.31	303.6	308.1	307.9
Tur	Kharif	3.17	2.81	2.56	4.87	4.29	3.32	3.89	4.32	4.22	3.3	3.4	3.5
	Rabi	9.53	7.33	7.06	9.38	11.38	9.94	11.08	11.91	13.54	12.3	11.0	11.5
	Kharif	1.15	1.28	1.25	2.18	2.75	2.36	1.33	1.51	1.87	1.8	1.6	1.3
Gram	Rabi	0.55	0.68	0.7	0.66	0.74	0.7	0.75	0.72	0.91	0.6	0.5	0.5
	Total	1.7	1.96	1.95	2.83	3.49	3.06	2.08	2.23	2.78	2.6	2.3	1.8
	Kharif	0.96	0.87	1	1.64	1.43	1.78	1.83	2	1.48	1.7	1.2	1.5
Moong	Rabi	0.65	0.64	0.59	0.52	0.59	0.67	0.68	1.09	1.69	0.1	0.1	0.1
	Total	1.61	1.5	1.59	2.17	2.02	2.46	2.51	3.09	3.17	3.7	3.1	1.6

(Contd.)



Annex Table 1.2 : All-India Estimates of Production of Principal Crops

Crop	Season	(million tonnes)											
		2013-14	2014-15	2015-16	2016-17	2017-18	2018-19	2019-20	2020-21	2021-22	2022-23	2023-24	2024-25*
Lentil	Rabi	1.02	1.04	0.98	1.22	1.62	1.23	1.1	1.49	1.27	1.6	1.8	1.8
Total	Kharif	6	5.73	5.53	9.58	9.31	8.09	7.92	8.62	8.24	7.6	7.0	7.2
Pulses	Rabi	13.26	11.42	10.79	13.55	16.11	13.98	15.1	16.84	19.07	16.4	15.2	15.9
	Total	19.26	17.15	16.32	23.13	25.42	22.08	23.03	25.46	27.3	26.1	24.3	23.0
Total	Kharif	128.69	128.07	125.09	138.33	140.47	141.52	143.81	150.58	155.36	155.7	155.8	166.4
Foodgrains	Rabi	136.35	123.96	126.45	136.78	144.55	143.69	153.69	160.17	160.25	157.8	160.0	164.5
	Total	265.05	252.03	251.54	275.11	285.01	285.21	297.5	310.74	315.62	329.7	332.3	330.9
Groundnut	Kharif	8.06	5.93	5.37	6.05	7.6	5.39	8.39	8.53	8.43	8.6	8.7	10.4
	Rabi	1.66	1.47	1.37	1.41	1.66	1.34	1.56	1.72	1.7	1.1	0.8	0.9
	Total	9.71	7.4	6.73	7.46	9.25	6.73	9.95	10.24	10.13	10.3	10.2	11.3
Sesamum	Kharif	0.71	0.83	0.85	0.75	0.76	0.69	0.66	0.82	0.79	0.4	0.4	0.4
Nigerseed	Kharif	0.1	0.08	0.07	0.09	0.07	0.05	0.04	0.04	0.03	0.0	0.0	0.0
Soybean	Kharif	11.86	10.37	8.57	13.16	10.93	13.27	11.23	12.61	12.99	15.0	13.1	15.1
Sunflower	Kharif	0.17	0.14	0.07	0.1	0.08	0.09	0.09	0.08	0.11	0.2	0.1	0.1
	Rabi	0.34	0.29	0.23	0.15	0.14	0.13	0.12	0.15	0.14	0.1	0.1	0.1
	Total	0.51	0.43	0.3	0.25	0.22	0.22	0.21	0.23	0.25	0.4	0.2	0.2
Rapeseed & Mustard	Rabi	7.88	6.28	6.8	7.92	8.43	9.26	9.12	10.21	11.96	12.6	13.3	12.9
Safflower	Rabi	0.11	0.09	0.05	0.09	0.06	0.02	0.04	0.04	0.06	0.1	0.1	0.1
Total Nine	Kharif	22.63	19.22	16.68	21.51	21.01	20.68	22.25	23.72	23.97	26.2	24.2	27.6
Oilseeds	Rabi	10.13	8.29	8.57	9.76	10.45	10.85	10.97	12.22	13.99	14.2	14.3	14.0
	Total	32.75	27.51	25.25	31.28	31.46	31.52	33.22	35.95	37.96	41.4	39.7	41.7
Cotton ^{\$}		35.9	34.81	30.01	32.58	32.81	28.04	36.07	35.25	31.12	33.7	32.5	29.4

Note: * Second Advance Estimates; ^{\$} Million Bales of 170 kgs

Source: Economics, Statistics & Evaluation Division, Ministry of Agriculture and Farmers Welfare



Annex Table

Annex Table 1.3: All-India Estimates of Yield of Principal Crops

Crop	Season	2013-14	2014-15	2015-16	2016-17	2017-18	2018-19	2019-20	2020-21	2021-22	2022-23	2023-24	2024-25*
		(kg/ha)	(kg/ha)	(kg/ha)	(kg/ha)	(kg/ha)	(kg/ha)	(kg/ha)	(kg/ha)	(kg/ha)	(kg/ha)	(kg/ha)	(kg/ha)
Rice	Kharif	2319	2295	2305	2417	2469	2553	2622	2607	2705	2735	2780	2779
	Rabi	3232	3291	3382	3230	3531	3444	3569	3541	3525	3467	3582	3654
	Total	2416	2391	2400	2494	2576	2638	2722	2717	2798	2838	2882	2859
Wheat	Rabi	3145	2750	3034	3200	3368	3533	3440	3521	3537	3521	3559	3540
	Kharif	1050	1014	850	954	1104	989	967	1210	1090	1071	1170	1197
	Rabi	896	808	615	730	853	744	1002	1033	1093	1086	1165	1244
Jowar	Total	957	884	697	812	956	849	989	1099	1092	1079	1162	1224
	Kharif	1184	1255	1132	1305	1231	1219	1374	1420	1430	1443	1387	1388
	Kharif	2346	2249	2236	2413	2706	2648	2572	2779	2914	2940	2671	2941
Maize	Rabi	4050	4414	4006	3897	4436	4893	4631	4723	5084	5355	5529	5428
	Total	2676	2632	2563	2689	3065	3070	3006	3199	3387	3545	3351	3472
	Kharif	1661	1706	1601	1363	1662	1390	1747	1724	1396	1454	1375	1456
Barley	Rabi	2718	2280	2439	2663	2695	2837	2920	2796	3025	3044	3082	3153
	Kharif	1619	1633	1544	1708	1818	1790	1836	1970	2037	2065	1944	2065
	Rabi	2034	1915	1685	1885	2323	2532	2491	2667	3018	3218	3097	3243
Nutri/ Coarse Cereals	Total	1717	1703	1579	1750	1934	1944	1991	2128	2251	2381	2283	2329
	Kharif	2089	2081	2065	2188	2259	2320	2370	2406	2503	2527	2521	2565
	Rabi	2995	2681	2862	3010	3239	3403	3325	3412	3472	3478	3500	3514
Tur (Arhar)	Total	2462	2331	2393	2525	2657	2752	2772	2824	2899	2939	2945	2949
	Kharif	813	729	646	913	967	729	859	914	861	814	827	836
	Rabi	960	889	840	974	1078	1041	1142	1192	1261	1172	1151	1160
Gram	Kharif	490	516	459	626	632	500	359	469	515	571	598	609
	Rabi	768	892	773	656	798	796	904	778	903	988	828	850
	Total	555	604	537	632	662	546	459	538	599	657	656	663
Moong	Kharif	410	428	363	488	440	466	519	522	385	493	364	440
	Rabi	620	640	554	546	600	727	645	833	987	730	733	795
	Total	475	498	416	500	477	516	548	601	570	663	598	453

(Contd.)



Annex Table 1.3 : All-India Estimates of Yield of Principal Crops

Crop	Season	2013-14	2014-15	2015-16	2016-17	2017-18	2018-19	2019-20	2020-21	2021-22	2022-23	2023-24	2024-25*
Pulses	Rabi	759	705	765	838	1047	901	847	1017	899	952	1028	1045
	Kharif	580	573	489	667	668	546	585	642	583	630	600	635
	Rabi	891	843	794	898	1015	976	1045	1097	1148	1118	1106	1115
	Total	764	728	655	785	853	757	823	885	888	902	881	903
Foodgrains	Kharif	1864	1862	1808	1890	1951	1957	2029	2079	2131	2203	2205	2269
	Rabi	2435	2232	2341	2441	2603	2740	2738	2793	2798	2854	2903	2910
	Total	2120	2028	2041	2129	2235	2286	2343	2394	2425	2494	2515	2548
	Kharif	1735	1478	1399	1321	1834	1304	2016	1649	1717	2009	2142	2088
Groundnut	Rabi	1926	1948	1801	1861	2222	2238	2352	2034	2147	2619	2208	2201
	Total	1764	1552	1465	1398	1893	1422	2063	1703	1777	2075	2163	2097
	Kharif	426	474	436	448	478	485	405	474	485	375	392	402
	Kharif	328	328	295	332	321	290	303	317	337	325	323	284
Soybean	Kharif	1012	951	738	1177	1058	1192	921	976	1069	1145	985	1169
Sunflower	Kharif	668	660	421	567	627	766	731	785	802	933	894	877
	Rabi	808	777	698	737	924	874	1174	1187	1048	1061	1397	1312
	Total	755	734	609	660	782	826	931	1011	923	996	1144	1078
	Rabi	1185	1083	1183	1304	1410	1511	1331	1524	1497	1428	1444	1470
Rapeseed & Mustard													
Safflower	Rabi	638	515	416	651	673	537	843	640	777	826	768	736
Total Nine	Kharif	1151	1056	884	1153	1219	1168	1154	1133	1214	1326	1239	1388
Oilseeds	Rabi	1206	1122	1186	1300	1436	1531	1397	1547	1522	1446	1447	1474
	Total	1168	1075	968	1195	1284	1271	1224	1247	1312	1368	1314	1416
Cotton		510	462	415	512	443	378	455	451	428	443	436	437

Note: * Second Advance Estimates; \$ Million Bales of 170 kgs

Source: Economics, Statistics & Evaluation Division, Ministry of Agriculture and Farmers Welfare

Annex Table 1.4: Share of Major States in All-India Production of Mandated Kharif Crops, TE2023-24

Rice	Jowar		Bajra		Maize		Ragi		Tur		Urad		Moong		Groundnut		Sesamum		Nigerseed		Soybean		Sunflower		Cotton		
	State	Share (%)	State	Share (%)	State	Share (%)	State	Share (%)	State	Share (%)	State	Share (%)	State	Share (%)	State	Share (%)	State	Share (%)	State	Share (%)	State	Share (%)	State	Share (%)	State	Share (%)	
WB	11.9	MH	36.6	RJ	41.6	KA	15.3	KA	65.3	MH	29.3	MP	30.0	MP	39.7	GJ	44.6	WB	32.2	OD	49.0	MH	42.3	KA	56.4	GJ	26.1
UP	11.8	KA	18.5	UP	19.4	MP	12.4	TN	13.4	KA	27.6	AP	15.5	RJ	29.2	RJ	18.5	GJ	17.3	CG	16.3	MP	40.6	HR	9.1	MH	25.3
TG	11.2	RJ	12.2	GJ	11.6	BR	11.7	UK	6.9	UP	10.2	UP	13.9	KA	4.4	TN	9.3	MP	16.0	AS	10.3	RJ	8.0	OD	8.3	TG	16.1
PB	10.0	UP	8.2	HR	10.9	MH	9.1	MH	6.0	GJ	7.7	TN	9.3	MH	4.3	MP	7.8	UP	11.2	MP	9.9	KA	3.4	TG	7.0	RJ	8.1
CG	6.8	TN	7.4	MP	8.8	TN	7.9	OD	2.7	JH	6.5	MH	7.0	BR	3.6	AP	4.7	RJ	9.3	MH	3.9	GJ	2.7	AP	4.3	KA	6.8
OD	6.5	AP	6.9	MH	4.2	TG	7.2	AP	1.9	TG	5.4	RJ	5.7	GJ	3.6	KA	4.3	TN	3.4	AP	3.4	TG	2.1	MH	3.7	MP	4.8
AP	5.7	MP	4.5	KA	1.6	WB	6.8	JH	1.2	MP	4.1	JH	4.1	OD	3.1	MH	3.1	KA	2.1	GJ	2.6	Oth*	0.8	BR	3.4	AP	4.1
BR	5.6	TG	3.7	TN	1.2	AP	5.9	AR	1.1	OD	3.8	GJ	3.8	AP	2.2	TG	2.7	TG	1.9	JH	2.5			TN	2.7	HR	3.9
TN	5.5	GJ	1.3	Oth*	0.8	RJ	5.8	Oth*	1.4	AP	2.2	WB	2.0	UP	1.9	WB	2.1	AS	1.2	WB	1.8			WB	2.1	OD	2.1
MP	4.7	Oth*	0.6			UP	5.7			TN	1.2	AS	2.0	WB	1.7	UP	1.2	AP	1.2	Oth*	0.3			UP	1.4	PB	1.8
HR	3.9					GJ	2.4			Oth*	2.1	TG	1.7	TN	1.7	OD	0.8	CG	0.8					PB	0.8	Oth*	1.0
AS	3.9					HP	1.9					OD	1.4	PB	1.6	Oth*	0.8	Oth*	3.4					Oth*	0.9		
KA	2.9					JK	1.6					KA	1.2	HR	1.2												
MH	2.8					JH	1.5					CG	1.0	TG	0.8												
GJ	1.7					PB	1.1					Oth*	1.4	Oth*	1.1												
JH	1.5					CG	1.0																				
Oth*	3.6					Oth*	2.8																				

Note: * States having less than 1 percent share in total production have been clubbed as others

Source: Economics, Statistics & Evaluation Division, Ministry of Agriculture and Farmers Welfare





Annex Table 1.5: Compound Annual Growth Rate in Value of Output from Agriculture and Allied Sectors, 2015-16 to 2022-23

(Percent)

State	Agriculture and allied Sectors*	Crop						Livestock				Fishing & Aquaculture
		All Crops	Cereals	Pulses	Oilseeds	Fruits & Vegetables	Condiments & Spices	Cotton	Total Livestock	Milk Group	Meat & Eggs Group	
Andhra Pradesh	7.2	2.4	1.4	-0.2	2.1	6.6	-0.6	-1.2	6.6	4.6	10.1	15.0
Assam	1.7	0.6	0.6	0.4	0.9	1.0	-2.2	-	2.4	2.6	2.2	6.2
Bihar	4.0	1.9	1.7	-1.8	1.1	3.7	1.8	-	6.5	6.4	10.1	7.6
Chhattisgarh	4.6	3.9	6.0	-5.5	-5.9	3.5	1.0	-	4.3	6.2	5.5	9.4
Gujarat	3.4	3.3	4.0	22.8	7.1	2.6	3.8	-2.4	3.9	4.8	1.2	1.4
Haryana	2.3	0.1	0.6	3.9	7.0	-1.9	-8.3	-4.4	5.1	5.1	8.0	7.4
Himachal Pradesh	1.4	0.6	-1.2	2.6	4.1	1.1	2.3	-	3.3	3.7	-0.5	5.4
Jammu & Kashmir	3.3	4.0	0.2	3.6	1.9	6.6	-1.5	-	2.2	3.0	-0.9	4.2
Jharkhand	3.8	1.4	-5.2	4.2	11.1	2.7	10.7	-	7.2	6.4	9.8	12.3
Karnataka	8.7	7.5	6.0	5.8	6.2	5.8	18.2	12.8	11.5	11.5	16.5	10.7
Kerala	0.3	0.4	1.8	-10.3	-0.6	-1.2	0.2	-	-0.6	-0.3	-1.0	2.2
Madhya Pradesh	5.5	4.9	4.8	-0.8	1.7	6.8	26.5	-6.3	7.3	7.4	10.1	17.0
Maharashtra	5.0	4.7	4.4	12.7	11.3	4.7	5.9	2.4	6.0	6.0	8.3	-0.3
Odisha	4.5	3.3	4.9	2.1	3.5	1.6	9.6	12.2	4.3	3.7	5.4	11.6

(Contd.)



Annex Table 1.5: Compound Annual Growth Rate in Value of Output from Agriculture and Allied Sectors, 2015-16 to 2022-23

(Percent)

State	Agriculture and allied Sectors*	Crop							Livestock			Fishing & Aquaculture
		All Crops	Cereals	Pulses	Oilseeds	Fruits & Vegetables	Condiments & Spices	Cotton	Total Livestock	Milk Group	Meat & Eggs Group	
Punjab	2.1	1.0	0.5	3.8	6.0	5.9	8.0	-7.7	3.9	4.3	3.0	6.7
Rajasthan	5.0	2.9	3.2	8.2	7.9	7.0	-2.0	12.6	7.5	9.2	6.2	7.5
Tamil Nadu	4.6	3.3	7.7	-1.0	3.3	4.0	2.9	-2.8	6.3	5.5	6.7	1.9
Telangana	8.5	9.3	19.2	7.0	1.9	-2.6	8.0	7.0	7.4	4.2	10.6	10.8
Tripura	2.9	2.1	0.1	2.7	-0.2	2.2	10.5	-12.8	6.0	6.1	10.1	2.9
Uttar Pradesh	3.3	3.3	3.5	8.8	7.2	3.2	3.4	18.9	3.1	4.2	-1.4	7.5
Uttarakhand	0.0	-0.6	0.8	3.4	-0.4	0.4	10.8	NA	0.9	1.7	-5.9	7.7
West Bengal	2.9	1.2	1.5	4.3	3.4	1.6	-5.9	-17.0	7.2	4.6	9.8	2.9
All India	4.4	3.3	3.4	5.4	5.6	3.6	7.9	1.2	5.7	5.8	7.1	8.6

Note: * Agriculture and Allied Sectors excluding Forestry and Logging

Source: CACP calculations using data from Ministry of Statistics and Programme Implementations



Annex Table 2.1: List of DCP and Non-DCP States for Rice

S. No.	States /UTs
	DCP States/UTs
1	Andhra Pradesh
2	Bihar
3	Chhattisgarh
4	Gujarat
5	Himachal Pradesh
6	Jharkhand
7	Karnataka
8	Kerala
9	Madhya Pradesh
10	Maharashtra
11	Odisha
12	Tamil Nadu
13	Telangana
14	Tripura
15	Uttarakhand
16	West Bengal
	Non-DCP States/UTs
17	Assam
18	Chandigarh
19	Haryana
20	J&K
21	Punjab
22	Uttar Pradesh
23	Puducherry

Source: Department of Food and Public Distribution, Ministry of Consumer Affairs, Food and Public Distribution



Annex Table

Annex Table 2.2: Number of Paddy Farmers Benefitted from Procurement

States	2015-16	2016-17	2017-18	2018-19	2019-20	2020-21	2021-22	2022-23	2023-24	2024-25*
Andhra Pradesh	276699	544570	498100	581839	679789	803945	839210	678604	479159	345026
Assam	7288	6960	4332	12921	26537	20401	60025	62423	39577	24339
Bihar	275484	287830	163425	210028	279402	497097	642234	577064	372556	462964
Chhattisgarh	1110163	1327944	1014195	1571414	1838593	2053490	2105972	2047842	2472441	1778738
Haryana	212351	556654	681984	830792	1891622	549466	310083	282725	282103	310100
Jharkhand	53945	39480	42346	34595	53305	104092	139359	31855	10927	31599
Karnataka	14587	-	-	25120	27123	54319	73161	7646	0	0
Kerala	162737	125530	140777	172795	220066	252160	254072	250373	198478	61527
Madhya Pradesh	199984	287759	278898	362089	436399	587223	661756	647699	605553	669271
Maharashtra	111503	149279	116641	269148	526476	624292	595604	516941	375519	307725
Odisha	1078596	1101193	798619	1010390	1161796	1394647	1514727	1710773	1532237	1446077
Punjab	1206216	940560	1142621	1143713	1125238	1057674	933263	903942	914803	904471
Tamil Nadu	850640	73367	413076	498097	585241	852152	400949	599912	393887	287482
Telangana	535007	1088312	1077667	1474828	1988630	2164354	1734675	2016798	1500655	861247
Uttar Pradesh	433556	435320	492913	684013	706549	1022286	947325	940226	692604	662854
Uttarakhand	51772	79470	7618	62564	83910	78129	78798	58038	41009	33779
West Bengal	721273	634696	350181	733357	805186	949362	1318238	1102759	704417	833117
Others	6615	6244	7466	15938	23492	47193	70199	61457	42308	28408
All India	7308416	7685168	7230859	9693641	12459354	13112282	12679650	12497077	10658233	9048724

Note: *upto 02.03.2025

Source: 1. Food Corporation of India

2. Central Food Grains Procurement Portal for year 2024-25



Annex Table 2.3: Procurement of Nutri-Cereals and Maize in Major Producing States during KMS2023-24 and KMS2024-25

(in tonnes)

KMS	Commodity	AP	GJ	HR	KA	MP	MH	UP	UK	TN	TOTAL
2023-24	Jowar (Kharif)	24466	0	0	85402	0	462	13340	0	0	123670
	Jowar (Rabi)	4903	2946	0	105019	0	86625	0	0	0	199493
	Bajra (Kharif)	0	33359	231334	0	0	0	355039	0	0	619732
	Bajra (Rabi)	0	76726		0	0	0	0	0	0	76726
	Maize (Kharif)	0	0	0	0	0	25	4451	0	0	4476
	Maize (Rabi)	0	0	0	0	0	56	0	0	0	56
	Ragi	376	1	0	226576	0	189	0.7	1889	1889	230921
	Total	29745	113032	231334	416997	0	87357	372830.7	1889	1889	1255074
2024-25 *	Jowar (Kharif)	0	27	0	0	908	196	46,942	0	0	48073
	Jowar (Rabi)	0	0	0	0	0	0	0	0	0	0
	Bajra (Kharif)	0	3818	156000	0	0	0	101311	0	0	261129
	Bajra (Rabi)	0	0	0	0	0	0	0	0	0	0
	Maize (Kharif)	0	0	0	0	0	1	0	0	0	1
	Maize (Rabi)	0	0	0	0	0	0	0	0	0	0
	Ragi	160	5	0	0	0	0	0	3139	222	3526
	Total	160	3850	156000	0	908	197	148253	3139	222	312729

Note: * Figures reported as on 03.02.2025

Source: Food Corporation of India



Annex Table 2.4: State-wise Procurement of Pulses (KMS2022-23 to KMS2024-25)

Crop	State	Procurement in '000 Tonnes			Percentage (%)**		
		2022-23	2023-24	2024-25*	2022-23	2023-24	2024-25*
Tur	Andhra Pradesh	-	-	2.2	-	-	2.0
	Gujarat	-	-	-	-	-	-
	Karnataka	-	-	0.1	-	-	0.01
	Maharashtra	-	-	-	-	-	-
	Tamil Nadu	-	-	-	-	-	-
	Telangana	-	-	0.05	-	-	0.03
	Total	-	-	2.3	-	-	0.1
Moong	Andhra Pradesh	-	-	-	-	-	-
	Gujarat	0.03	-	-	0.05	-	-
	Haryana	0.8	0.7	-	5.3	4.3	-
	Karnataka	25.3	-	26.5	13.2	-	-
	Maharashtra	0.2	-	0.0	0.1	-	-
	Rajasthan	93.7	53.0	90.3	8.0	6.5	9.0
	Telangana	0.1	-	1.0	-	-	-
	Total	120.1	53.8	117.7	7.1	4.8	7.8
Urad	Gujarat	-	-	-	-	-	-
	Maharashtra	0.04	-	-	0.02	-	-
	Rajasthan	0.001	-	0.04	0.0005	-	-
	Total	0.04	-	0.04	0.01	-	-

Note: 1. * Figures reported as on 03.02.2025

2. ** Procurement as a percentage of total production

Source: National Agricultural Cooperative Marketing Federation of India



**Annex Table 2.5: State-wise Procurement of Groundnut and Soybean
(KMS2022-23 to KMS2024-25)**

Crop	State	Procurement in '000 Tonnes			Percentage (%)**		
		2022-23	2023-24	2024-25*	2022-23	2023-24	2024-25*
Groundnut	Gujarat	-	1.2	190.1	-	0.03	3.8
	Rajasthan	1.0	27.1	44.2	0.1	1.3	1.9
	Uttar Pradesh	6.1	46.5	13.7	5.3	31.5	3.2
	Karnataka	-	-	0.6	-	-	0.2
	Total	7.1	74.8	248.7	0.1	1.1	3.0
Soybean	Gujarat	-	0.1	47.3	-	0.03	9.5
	Karnataka	-	-	18.3	-	-	3.9
	Madhya Pradesh	-	-	388.9	-	-	7.3
	Maharashtra	-	0.8	752.5	-	0.01	10.3
	Rajasthan	-	-	96.7	-	-	8.2
	Telangana	-	6.1	83.1	-	2.3	29.7
	Total	-	7.0	1386.8	-	0.1	9.2

Note: 1. * Figures reported as on 03.02.2025

2. **Procurement as a percentage of total production

Source: National Agricultural Cooperative Marketing Federation of India



Annex Table 2.6: Break-up of Economic Cost of Rice

(₹/qtl)

Particulars	2022-23	2023-24	2024-25 (RE)
A) Pooled Cost of Grain	2854.9	3081.6	3238.2
B) Procurement Incidentals (a+b+c+d)	475.3	477.3	517.0
a) Statutory /Obligatory Costs	312.9	285.8	290.5
Mandi /Statutory Charges	63.5	55.8	60.8
Artiya Commission	55.3	52.3	55.7
Gunny Cost	155.5	147.8	143.6
Milling Charges & Driage Allowance	38.6	29.9	30.4
b) Labour & Transport Charges	66.5	69.8	74.1
Mandi Labour	23.3	22.7	25.8
Forwarding Charges	-	0.0	0.0
Internal Movement	43.2	47.1	48.3
c) Storage & Interest Charges paid to State Agencies	15.2	36.5	49.7
Storage Charges	3.3	3.5	4.0
Interest	51.8	54.0	45.7
Previous year Arrears Expenditure	-40.0	-21.0	0.0
d) Others Charges	80.8	85.3	102.7
Others		-0.2	0.0
Administrative Charges to State /Agencies [D]	33.4	28.0	29.7
Upgradation Charges/Fortifying Cost	47.4	57.5	73.0
C) Distribution Cost	392.2	325.0	287.0
Freight	157.6	128.8	112.7
Handling Charges	60.6	66.6	64.6
Storage Charges	55.5	48.2	45.5
Interest	62.2	24.8	0.0
Shortages	19.0	7.4	7.1
Administrative Overheads	37.3	49.1	57.1
Economic Cost (A+B+C)	3722.4	3883.8	4042.2

Source: Food Corporation of India

Annex Table 3.1: Yield Gap in Major *Kharif* Crops in Selected States

States	Yield Gap (in %)	
	A	B
<i>Paddy (Kharif)</i>		
Andhra Pradesh	8.2	0.2
Assam	12.4	32.9
Bihar	20.7	25.5
Chhattisgarh	18.0	27.1
Jharkhand	20.4	51.2
Madhya Pradesh	4.0	14.7
Maharashtra	9.7	37.1
Odisha	16.9	45.2
West Bengal	10.6	8.9
<i>Maize (Kharif)</i>		
Andhra Pradesh	7.1	38.8
Bihar	16.8	61.5
Gujarat	16.4	42.7
Jammu and Kashmir	33.5	61.5
Karnataka	13.6	51.8
Madhya Pradesh	30.4	46.9
Maharashtra	36.9	56.5
Punjab	4.4	10.4
Rajasthan	23.9	27.5
Tamil Nadu	12.3	-1.7
Uttar Pradesh	31.8	55.0
<i>Bajra</i>		
Gujarat	8.7	27.0
Haryana	13.4	13.0
Karnataka	11.8	42.5
Madhya Pradesh	16.3	2.3
Maharashtra	17.3	55.8
Rajasthan	24.2	42.3
Tamil Nadu	39.7	16.3
Uttar Pradesh	14.8	27.5

Annex Table 3.1: Yield Gap in Major *Kharif* Crops in Selected States

States	Yield Gap (in %)	
	A	B
<i>Ragi</i>		
Andhra Pradesh	22.7	54.5
Jharkhand	33.9	54.0
Karnataka	22.2	50.7
Maharashtra	32.6	34.4
Tamil Nadu	23.6	6.0
Uttarakhand	34.7	25.2
<i>Tur</i>		
Andhra Pradesh	28.5	73.8
Gujarat	16.1	25.2
Jharkhand	23.8	28.8
Karnataka	19.2	32.7
Madhya Pradesh	21.3	39.0
Maharashtra	16.2	38.0
Odisha	22.2	16.7
Tamil Nadu	21.3	8.3
Telangana	20.0	45.3
Uttar Pradesh	24.3	26.4
<i>Moong</i>		
Andhra Pradesh	10.1	12.8
Gujarat	19.4	-9.7
Karnataka	13.8	39.7
Maharashtra	15.0	20.0
Odisha	21.6	50.0
Rajasthan	11.4	61.9
Uttar Pradesh	15.2	38.4
<i>Urad</i>		
Andhra Pradesh	11.8	4.2
Assam	23.0	47.6
Chhattisgarh	20.7	57.5
Gujarat	14.7	-10.3
Jharkhand	23.5	9.2

Annex Table 3.1: Yield Gap in Major *Kharif* Crops in Selected States

States	Yield Gap (in %)	
	A	B
Karnataka	15.9	42.7
Madhya Pradesh	23.8	26.2
Maharashtra	17.1	34.2
Odisha	21.4	54.8
Rajasthan	19.5	50.6
Tamil Nadu	24.5	37.8
Uttar Pradesh	17.5	44.3
Soybean		
Gujarat	22.4	-9.8
Karnataka	16.7	39.2
Madhya Pradesh	21.3	56.0
Maharashtra	16.9	49.8
Rajasthan	13.5	38.7
Telangana	12.1	30.6
Sunflower		
Karnataka	15.9	48.1
Cotton		
Andhra Pradesh	16.2	41.2
Gujarat	8.8	-1.8
Haryana	14.8	33.3
Karnataka	12.1	19.0
Madhya Pradesh	18.2	20.0
Maharashtra	9.8	45.9
Odisha	19.6	-7.8
Punjab	14.3	26.8
Rajasthan	5.4	5.4
Telangana	19.7	23.0

Source: Various ICAR-Institutions



Annex Table

Annex Table 3.2: Area Covered under Micro Irrigation through Per Drop More Crop (PDMC) Scheme

(Area in ha)

State	2015-16	2016-17	2017-18	2018-19	2019-20	2020-21	2021-22	2022-23	2023-24	2024-25 (P)
Andhra Pradesh	94104	141098	186441	200269	122080	-	14654	90775	81970	48104
Bihar	5155	4228	3143	924	3967	1275	2087	4252	5650	1896
Chhattisgarh	8056	19227	13087	18929	27688	18634	18640	23885	-	7740
Goa	92	123	236	0	285	132	0	7	92	34
Gujarat	142681	165948	143134	140778	107649	101329	78790	106109	130505	91549
Haryana	3117	5701	10751	10469	22652	12446	43045	36786	33626	2779
Himachal Pradesh	3306	937	1197	422	2573	2051	0	406	335	542
Jharkhand	4528	5810	1544	3978	5989	1909	4868	6049	5958	2461
Jammu & Kashmir	0	0	0	0	70	1020	14	0	-	40
Karnataka	64220	139405	236107	234853	250591	321178	336053	166841	256046	101699
Kerala	561	310	358	965	583	389	385	1758	139	321
Madhya Pradesh	75224	54323	39761	35195	14597	17717	69890	46568	53326	19318
Maharashtra	35242	106172	132829	159959	171097	49878	113479	168389	88667	607
Odisha	2907	4611	3036	10081	8498	13592	30331	22419	31343	8842
Punjab	1799	1950	600	507	942	742	3719	3039	2864	556
Rajasthan	56346	47650	48205	53982	58573	69191	78232	191004	219764	123355
Tamil Nadu	32288	44778	105695	172445	263494	214147	102329	70856	129021	60320
Telangana	39864	61980	89474	40381	4548	11308	35538	40711	4604	10695
Uttarakhand	721	3199	2182	4256.41	6790	4128	3700	7231	833	448
Uttar Pradesh	1598	32511	28235	55086	56953	58104	33931	65226	61561	71122
West Bengal	0	0	2137	13370	21824	14821	14365	25120	13773	7730
Arunachal Pradesh	0	0	0	0	0	6898	3009	1798	1594	1491
Assam	0	0	782	70	10873	6731	12849	7960	5902	4043
Manipur	0	0	0	1600	2554	4455	2894	3409	350	2379
Meghalaya	0	0	0	0	0	-	0	0	795	52
Mizoram	398	0	0	0	2348	519	427	826	511	-
Nagaland	0	0	0	0	2830	1520	5518	7626	7666	-
Sikkim	773	0	0	0	2524	3000	2914	2772	4179	1216
Tripura	0	0	0	0	0	-	3413	732	366	-
Ladakh	-	-	-	-	-	-	-	3	-	-
Grand Total	572980	839961	1048934	1158519	1172572	937114	1015073	1102556	1141440	569339

Note: (P) Provisional

Source: Ministry of Agriculture and Farmers Welfare



Annex Table 3.3: State-wise Progress of SMAM and CRM Schemes since Inception (as on 31.01.2025)

State	Funds Released (₹ in Crores)		Machines Distributed under Subsidy to Individual Farmers (Nos.)		CHCs/Hi-tech Hubs/Farm Machinery Banks Established (Nos.)	
	SMAM	CRM	SMAM	CRM	SMAM	CRM
Andhra Pradesh	855	-	251514	-	10598	-
Arunachal Pradesh	66	-	40824	-	13	-
Assam	124	-	3163	-	787	-
Bihar	157	-	39535	-	2005	-
Chhattisgarh	321	-	102736	-	2978	-
Gujarat	93	-	29733	-	247	-
Haryana	267	1082	24870	100822	2313	6775
Himachal Pradesh	174	-	43637	-	53	-
Jammu & Kashmir	77	-	19822	-	387	-
Jharkhand	34	-	-	-	527	-
Karnataka	1099	-	306785	-	790	-
Kerala	298	-	118105	-	1687	-
Madhya Pradesh	512	-	279300	-	1778	-
Maharashtra	603	-	103525	-	1430	-
Manipur	113	-	20283	-	820	-
Meghalaya	25	-	2457	-	8	-
Mizoram	44	-	5730	-	339	-
Nagaland	232	-	21373	-	609	-
Odisha	347	-	81831	-	1778	-
Punjab	109	1756	13648	147668	1267	25917
Rajasthan	145	-	32793	-	1854	-
Sikkim	51	-	7128	-	52	-
Tamil Nadu	804	-	80758	-	4153	-
Telangana	56	-	28954	-	195	-
Tripura	206	-	54915	-	727	-
Uttar Pradesh	672	764	182346	74548	11591	9276
Uttarakhand	343	-	42294	-	2400	-
West Bengal	102	-	10997	-	774	-
NCT of Delhi	0	6	0	247	0	-
Dadra & Nagar Haveli	1	-	89	-	0	-
Puducherry	10	-	621	-	13	-
Ladakh	1	-	1314	-	0	-
Total	7939	3608	1951080	323285	52173	41968

Source: M&T Division, Department of Agriculture & Farmers Welfare, Ministry of Agriculture and Farmers Welfare



Annex Table 4.1: Leading Exporters and Importers in World Merchandise Trade, 2023

(US\$ Billion and percentage)

Rank	Exporters	Value	Share	Annual Percentage Change	Rank	Importers	Value	Share	Annual Percentage Change
1	China	3380	14.2	-5	1	United States of America	3173	13.1	-6
2	United States of America	2020	8.5	-2	2	China	2557	10.6	-6
3	Germany	1688	7.1	1	3	Germany	1463	6.0	-8
4	Netherlands	935	3.9	-3	4	Netherlands	842	3.5	-6
5	Japan	717	3.0	-4	5	United Kingdom	791	3.3	-4
6	Italy	677	2.8	3	6	France	786	3.2	-5
7	France	648	2.7	5	7	Japan	786	3.2	-12
8	Korea, Republic of	632	2.7	-8	8	India	673	2.8	-7
9	Mexico	593	2.5	3	9	Hong Kong, China	654	2.7	-2
						Retained imports ¹	184	0.8	12
10	Hong Kong, China	574	2.4	-6	10	Korea, Republic of	643	2.7	-12
	Domestic exports	21	0.1	30					
	Re-exports	553	2.3	-7					
11	Canada	569	2.4	-5	11	Italy	640	2.6	-8
12	Belgium	562	2.4	-12	12	Mexico	621	2.6	-1
13	United Kingdom	521	2.2	-2	13	Canada	570	2.4	-2
14	United Arab Emirates ¹	488	2.1	-5	14	Belgium	547	2.3	-12
15	Singapore	476	2.0	-8	15	Spain	470	1.9	-5
	Domestic exports	213	0.9	-11					
	Re-exports	263	1.1	-5					
16	Chinese Taipei	432	1.8	-10	16	United Arab Emirates ¹	449	1.9	7
17	India	432	1.8	-5	17	Singapore	423	1.7	-11
						Retained imports ¹	160	0.7	-20
18	Russian Federation	424	1.8	-28	18	Poland	370	1.5	-3
19	Spain	423	1.8	2	19	Switzerland	364	1.5	2
20	Switzerland	420	1.8	5	20	Türkiye	362	1.5	-1
21	Poland	382	1.6	6	21	Chinese Taipei	359	1.5	-18
22	Australia	371	1.6	-10	22	Viet Nam	326	1.3	-9

(Contd.)



Annex Table 4.1: Leading Exporters and Importers in World Merchandise Trade, 2023

(US\$ Billion and percentage)

Rank	Exporters	Value	Share	Annual Percentage Change	Rank	Importers	Value	Share	Annual Percentage Change
23	Viet Nam	354	1.5	-5	23	Russian Federation ²	304	1.3	10
24	Brazil	340	1.4	2	24	Thailand	290	1.2	-4
25	Saudi Arabia, Kingdom of	322	1.4	-22	25	Australia	288	1.2	-7
26	Malaysia	313	1.3	-11	26	Malaysia	266	1.1	-10
27	Thailand	285	1.2	-1	27	Brazil	253	1.0	-14
28	Indonesia	259	1.1	-11	28	Czech Republic	231	1.0	-3
29	Türkiye	256	1.1	1	29	Austria	225	0.9	-4
30	Czech Republic	255	1.1	6	30	Indonesia	222	0.9	-7
	Total of above ³	19748	83.0	-		Total of above ³	19944	82.3	-
	World³	23784	100.0	-5		World³	24235	100.0	-6

Note: (1) Secretariat estimates.

(2) Imports are valued f.o.b.

(3) Includes significant re-exports or imports for re-export.

Source: Global Trade Outlook and Statistics April 2024 (WTO-UNCTAD)



Annex Table 4.2: Top 10 Exporters and Importers of Agricultural Products, 2022

(US\$ Billion and percentage)

	Value	Share in world exports/imports				Annual percentage change			
	2022	2000	2005	2010	2022	2010-22	2020	2021	2022
Exporters									
European Union	799	38.9	41.9	37.4	34.3	4	2	16	5
Extra-EU exports	267	12.6	13.0	11.9	11.5	4	3	13	2
United States of America	222	13.0	9.8	10.5	9.6	4	3	18	10
Brazil	148	2.8	4.1	5.1	6.4	7	5	19	33
China	103	3.0	3.4	3.8	4.4	6	-4	13	17
Canada	91	6.3	4.9	3.8	3.9	5	7	25	5
Indonesia	69	1.4	1.7	2.7	3.0	6	8	39	6
Australia	56	3.0	2.5	2.0	2.4	6	-11	49	20
Argentina	55	2.2	2.3	2.6	2.4	4	-7	18	29
India	55	1.1	1.2	1.7	2.4	7	4	30	8
Mexico	53	1.7	1.5	1.4	2.3	9	4	17	12
Above 10	1651	73.3	73.2	70.9	71.0	-	-	-	-
Importers									
European Union	749	36.3	39.2	35.7	31.7	3	1	16	9
Extra-EU imports	229	13.0	12.5	11.1	9.7	3	-2	13	15
China	291	3.3	5.0	7.8	12.3	9	8	26	7
United States of America	260	11.6	10.7	8.4	11.0	7	3	22	14
Japan	94	10.5	7.3	5.6	4.0	2	-7	11	10
United Kingdom	83	5.8	5.9	4.6	3.5	2	2	4	12
Canada (1)	52	2.6	2.4	2.3	2.2	4	1	14	12
Korea, Republic of	52	2.2	1.9	1.9	2.2	6	-2	22	16
India	46	0.7	0.8	1.3	1.9	8	-6	46	20
Mexico (1)	45	1.9	1.8	1.7	1.9	6	-6	36	16
Türkiye	32	0.7	0.7	0.9	1.4	8	0	24	32
Above 10	1705	75.5	75.8	70.0	72.1	-	-	-	-

(1) Imports are valued f.o.b.

Source: World Trade Statistical Review 2023 (latest publication, 2024 is not available)



Annex Table 4.3 : India's Total Exports and Imports vis-a-vis Agricultural Exports and Imports, 2014-15 to 2023-24

(₹ '000 crore)

Year	Total Exports	Agri-Exports	Total Imports	Agri-Imports
2014-15	1896.3	245.5	2737.1	144.8
2015-16	1716.4	222.5	2490.3	163.3
2016-17	1849.4	233.6	2577.7	185.3
2017-18	1956.5	258.7	3001.0	175.8
2018-19	2307.7	283.5	3594.7	161.9
2019-20	2219.9	262.0	3361.0	169.7
2020-21	2159.0	321.4	2916.0	172.8
2021-22	3147.0	390.8	4572.8	257.0
2022-23	3621.5	441.4	5749.8	304.1
2023-24	3619.0	418.6	5616.0	290.9

Source: Directorate General of Commercial Intelligence and Statistics, Ministry of Commerce and Industry

Annex Table

Annex Table 4.4 : Major Export Destinations of Indian Rice, 2020-21 to 2023-24

Country	Non-Basmati Rice					Basmati Rice				
	2020-2021	2021-2022	2022-2023	2023-2024	Country	2020-2021	2021-2022	2022-2023	2023-2024	
Benin	12.3	15.3	15.6	12.8	Saudi Arabia	10.4	6.7	9.5	11.0	
Guinea	6.1	6.7	9.1	9.1	Iraq	6.5	4.9	3.6	8.2	
Togo	7.8	8.4	9.4	7.1	Iran	7.5	10.0	10.0	6.7	
Vietnam	2.9	7.1	6.4	6.9	UAE	2.3	2.6	3.2	3.1	
Senegal	10.4	10.9	13.3	6.4	Yemen	3.4	2.1	2.9	3.1	
Cote D' Ivoire	7.3	9.3	12.1	6.2	USA	1.8	1.6	2.0	2.3	
Kenya	1.7	1.6	5.9	5.7	UK	1.7	1.3	1.4	1.9	
Somalia	3.7	4.6	4.8	5.1	Kuwait	1.8	1.5	1.6	1.8	
Nepal	13.3	14.0	7.7	3.7	Oman	1.0	0.8	1.1	1.6	
Mozambique	2.0	2.5	3.6	3.4	Qatar	1.1	0.7	0.9	1.2	
Others	63.9	92.5	90.0	44.7	Others	9.0	7.3	9.3	11.5	
Total	131.5	172.9	177.9	111.2	Total	46.3	39.4	45.6	52.4	

Source: Directorate General of Commercial Intelligence and Statistics (DGCIIS), Ministry of Commerce and Industry





Annex Table 4.5: India's Top Import Origins of Pulses

(in tonnes)

Country	Peas			Country	Chana / Gram			Moong		
	2023-2024	2024-2025 (Apr-Nov)	Share in 2023-2024 (%)		2023-2024	2024-2025 (Apr-Nov)	Share in 2023-2024 (%)	2023-2024	2024-2025 (Apr-Nov)	Share in 2023-2024 (%)
Canada	612054	713087	52.0	Tanzania	140640	58571	49.4	2865		66.9
Russia	355790	507440	30.3	Sudan	67587	32780	23.7	770	480	18.0
Turkey	140212	90068	11.9	Australia	50324	122742	17.7	250	250	5.8
Others	68057	266072	5.8	Others	26126	77487	9.2	399	1034	9.3
Total	1176114	1576667	100.0	Total	284677	291580	100.0	4284	1764	100.0

Country	Urad			Country	Lentil			Tur/Arhar		
	2023-2024	2024-2025 (Apr-Nov)	Share in 2023-2024 (%)		2023-2024	2024-2025 (Apr-Nov)	Share in 2023-2024 (%)	2023-2024	2024-2025 (Apr-Nov)	Share in 2023-2024 (%)
Myanmar	607490	479358	97.3	Australia	822531	284731	49.1	264414	295801	34.3
Singapore	11374	8250	1.8	Canada	760746	256819	45.4	208651	194720	27.1
Brazil	3999	39139	0.6	Russia	45472	7443	2.7	178527	243159	23.2
Others	1359	1304	0.2	Others	47345	41592	2.8	119434	266003	15.5
Total	624222	528051	100.0	Total	1676095	590585	100.0	771025	999683	100.0

Source: Directorate General of Commercial Intelligence & Statistics, Ministry of Commerce and Industry



Annex Table 4.6 : India's Agricultural Exports of Major commodities

(₹'000 crore)

Commodity	Apr-Nov 2023	Apr-Nov 2024(P)	Percent increase/decrease over previous year	Share in Total Agri Exports in Apr-Nov 2024(P)
Rice	53.2	61.1	14.8	21.6
Marine Products	43.2	41.5	-4.0	14.7
Spices	21.0	23.4	11.1	8.3
Meat and Processed Meat	20.3	21.6	6.6	7.7
Sugar	13.9	11.3	-18.9	4.0
Coffee	6.6	9.6	44.6	3.4
Misc Processed Items	9.0	9.6	6.5	3.4
Tobacco Unmanufactured	6.1	8.7	44.2	3.1
Plywood And Allied Products	7.2	7.4	3.3	2.6
Oil Meals	8.6	7.3	-15.0	2.6
Castor Oil	5.7	6.4	13.1	2.3
Processed Fruits And Juices	5.0	5.3	5.8	1.9
Total Agri Exports	265.3	282.4	6.4	100

Note: (P) Provisional

Source: Directorate General of Commercial Intelligence and Statistics, Ministry of Commerce and Industry



Annex Table 4.7 : India's Agricultural Imports of Major commodities

(₹'000 crore)

Commodity	Apr-Nov 2023	Apr-Nov 2024(P)	Percent increase/ decrease over previous year	Share in Total Agri Imports in Apr-Nov 2024(P)
Vegetable Oils	87.0	102.2	17.5	44.4
Pulses	17.3	27.4	58.4	11.9
Fresh Fruits	14.4	16.1	12.0	7.0
Cashew	9.2	9.9	8.1	4.3
Plywood And Allied Products	8.2	8.8	7.4	3.8
Spices	8.4	8.7	3.8	3.8
Sugar	9.3	8.0	-14.0	3.5
Natural Rubber	4.0	6.8	68.0	3.0
Cotton Raw Includ. Waste	3.9	6.5	68.0	2.8
Wood And Wood Products	5.3	6.3	18.9	2.7
Alcoholic Beverages	6.7	6.0	-10.3	2.6
Total Agri Imports	192.9	230.3	19.4	100.0

Note: (P) Provisional

Source: Directorate General of Commercial Intelligence and Statistics, Ministry of Commerce and Industry

Annex Table

Annex Table 4.8: Quarterly Domestic and International Prices of Kharif Crops

Year and Quarter	Rice			Maize		Jowar		Tur/Arhar		Urad		Moong		Cotton*	
	D	I (Thailand 25 % Broken)	I (India 25 % Broken)	D	I	D	I	D	I	D	I	D	I	D	I
2020 Q1	2767	3281	2584	1665	1213	2618	1341	4818	4897	5695	6520	7404	7662	4908	4892
2020 Q2	2674	3873	2873	1324	1110	2622	1636	4971	5106	6660	6163	7412	8158	4405	4529
2020 Q3	2668	3572	2712	1246	1160	2510	1679	5586	5737	5614	6308	5584	7979	4527	4716
2020 Q4	2805	3567	2594	1313	1416	1970	2092	5776	6299	6355	7474	6393	8578	5055	5220
2021 Q1	2730	3851	2744	1,348	1761	2499	2449	6045	6214	6324	7364	6598	8260	5558	5994
2021 Q2	2476	3497	2727	1,475	2129	2322	2838	6280	6385	6589	7010	6486	7108	5924	6171
2021 Q3	2461	2939	2671	1638	1904	2156	2415	6102	6366	6123	6755	6017	7163	6659	6794
2021 Q4	3020	2952	2523	1522	1877	2032	2300	5618	5998	5509	6872	6053	7150	7541	8262
2022 Q1	3045	3160	2522	1851	2271	2445	2691	6111	6163	5956	6401	6322	**	9021	9384
2022 Q2	2572	3417	2549	2010	2646	2605	2994	5938	6206	6308	6868	6525	**	10286	11068
2022 Q3	2761	3354	2748	2073	2462	2909	2667	6719	7160	6356	7297	6243	**	9314	9018
2022 Q4	3254	3582	3041	1948	2649	3163	3012	6585	7250	5956	6957	6765	**	8325	7509
2023 Q1	3222	3995	3352	2095	2423	3752	2937	7163	7512	6746	6923	7502	**	7842	7352
2023 Q2	2957	4058	3467	1815	2263	3572	2578	8330	9029	7329	7912	7495	**	7175	7021
2023 Q3	3208	4716	#	1876	1856	4083	2112	9470	10659	7746	8274	7951	**	6856	7190
2023 Q4	3480	4847	#	1977	1800	3870	2241	8833	10817	7505	8928	7899	**	6907	6977
2024 Q1	3724	4976	#	2123	1601	3632	2063	9477	9945	8831	8718	8427	**	6773	7331
2024 Q2	3457	4872	#	2087	1618	3072	2088	10957	11118	9001	9235	8363	**	6984	6569
2024 Q3	3479	4671	#	2246	1487	3084	2003	10624	10361	8256	8717	7621	**	7121	6185
2024 Q4	3517	4184	#	2181	1673	2956	1958	9254	9234	7200	8390	6919	**	7069	6271

(Contd.)





Annex Table 4.8 : Quarterly Domestic and International Prices of Kharif Crops

(₹/qtl)

Year and Quarter	Soybean		Soybean Oil		Soybean Meal		Groundnut		Groundnut Oil		Sunflower Seed		Sunflower Oil	
	D	I	D	I	D	I	D	I	D	I	D	I	D	I
2020 Q1	3783	2739	8978	5842	3236	2624	4984	12699	11858	10082	3845	2283	8902	5688
2020 Q2	3611	2757	9098	5363	3258	2651	5325	15554	13758	12210	4140	2655	8890	5712
2020 Q3	3554	2947	9324	6429	3029	2822	4506	13839	12780	13970	4474	2793	8878	6672
2020 Q4	3910	3603	9753	7167	3286	3582	4749	12483	13776	14095	4495	3546	10693	8018
2021 Q1	4613	4228	11500	8519	3781	3875	5500	13098	14787	14548	6123	4400	13987	10323
2021 Q2	6689	4569	13222	11010	6035	3476	5331	10639	15413	15174	6006	5592	15356	10954
2021 Q3	7461	4307	14967	10623	7835	3478	5462	10807	15098	15568	5833	4020	15462	9748
2021 Q4	5445	4138	15024	10833	4800	3384	5489	11404	13827	16079	5449	4069	15073	10484
2022 Q1	6462	4990	14444	12609	5598	4259	5645	11202	14324	17834	6260	3799	15527	13252
2022 Q2	6664	5614	15604	14562	5439	4211	5806	11670	16067	16564	6285	5155	17402	16042
2022 Q3	5450	5356	14231	12449	4639	4276	5687	13001	16527	17125	5897	4124	16578	11588
2022 Q4	5050	5262	13351	12703	4028	4472	5851	13506	15867	17642	5613	3519	14609	10796
2023 Q1	5181	5227	13549	10167	4345	4912	6700	14209	16604	16766	5217	3465	14702	9465
2023 Q2	4971	4937	12238	8289	4442	4267	6550	14688	16613	16648	4479	2578	13438	7966
2023 Q3	4694	5061	11316	9296	4237	4246	6620	16563	18056	17743	4616	2593	11376	8052
2023 Q4	4622	4525	11020	9201	4477	4462	6353	17071	15953	16062	4655	2483	10482	7766
2024 Q1	4388	4303	11340	7883	4086	3853	6321	16691	15289	15297	4591	2573	10620	7803
2024 Q2	4464	4024	11144	8223	4060	3861	6095	15190	15096	15205	5011	2953	10820	8398
2024 Q3	4362	3521	12613	8807	3747	3753	5519	14107	15807	15020	4816	3544	11347	8893
2024 Q4	4133	3623	14202	9300	3142	3330	5322	13040	15802	14559	5781	3712	14396	10403

Note : # International prices of rice (India), 25 percent broken in export market are not available in FAO report

* International Prices of Cotton (lint) converted into Kapas at the ratio of 0.41

** International prices for moong not available for 2022, 2023 and 2024 as imports for moong is restricted since February 2022

D: Domestic, I: International

Source: 1. UPAG, Department of Agriculture and Farmers Welfare, for computing domestic prices for Paddy, Maize, Jowar, Arhar, Moong, Cotton, Soybean, Groundnut and Sunflower = seed.

2. Agricultural Marketing Division, Department of Agriculture and Farmers Welfare, Ministry of Agriculture and Farmers Welfare, All India Average Monthly Wholesale prices for domestic prices for Soybean Oil, Groundnut Oil, and Sunflower Oil.

3. Solvent Extractors Association for domestic prices of Paddy, Maize, Cotton

4. World Bank Pink Sheet for international prices of Paddy, Maize, Cotton, Soybean, Soybean Oil, Groundnut, Groundnut oil and Sunflower oil

5. International Grain Council (IFPMA) for International prices for Jowar

6. Agriwatch for International Prices of Sunflower seed, Pulses viz. Arhar, Urad & Moong (International prices for moong not available for 2022, 2023 and 2024 as imports for moong is restricted since February 2022).



Annex Table

Annex Table 4.9: Trade policy of Rice since 2011

Month	Trade Policy	Product	Description
Export Policy			
September 2011	Export under OGL	Non-basmati rice	Export of non-basmati rice by private parties out of privately held stocks have been allowed under the Open General License (OGL).
October 2015	“Free” to “Restricted”	Rice of seed quality and other rice in husk (paddy)	Export of rice of seed quality and other rice in husk (paddy) was placed from “Free” to “Restricted” category.
December 2017	Maximum Residue Limits (MRLs) reduced by EU	Basmati rice	MRLs for fungicide (Tricyclazole) was reduced from 1PPM (Parts Per Million) to 0.01 PPM, which adversely affected the basmati rice exports from India. The MRL on other pesticides set by EU is 6.0 mg/kg for Isoprothiolane and 0.01 mg/kg for Buprofezin. Middle-East countries like Oman, Egypt, Jordan, Saudi Arabia and UAE are also adopting similar pesticide residue norms for imported rice at par with the EU.
July 2021	Mandatory Certificate of Inspection from Export Inspection Council/Export Inspection Agency & registered with APEDA	Basmati rice	For export of rice to EU and other European countries, a Certificate of Inspection from Export Inspection Council/Export Inspection Agency is mandatory, vide DGFT notification dated 1 st July 2021. Additionally, all shipments of exports of basmati rice from India have to be mandatorily registered with APEDA.
September 2022	Export duty increased and amended from “Free” to “Prohibited”	Rice	W.e.f. 9 th September 2022, export duty on rice in the husk (paddy or rough), husked (brown) rice, semi-milled or wholly-milled rice whether or not polished or glazed (other than Parboiled rice and Basmati rice) has been imposed at the rate of 20 percent and export policy of broken rice, has been amended from “Free” to “Prohibited” category. These changes in policy will have adverse impact on rice exports.
August 2023	Export duty on rice	Par boiled non-basmati rice	Government imposed 20 percent export duty on par boiled non-basmati rice on 25 th August, 2023.
August 2023	Minimum Export Price (MEP)	Basmati rice	In order to safeguard against misclassification of exports of non-basmati white rice as basmati rice, Government imposed Minimum Export Price (MEP) for basmati rice exports. In this regard, any contract to basmati rice consignment with value exceeding US\$1200/mt has to be mandatorily registered for issue of Registration-cum-Allocation Certificate (RCAC) from Agricultural and Processed Food Products Export Development Authority (APEDA).

(Contd.)



Annex Table 4.9: Trade policy of Rice since 2011

Month	Trade Policy	Product	Description
September 2023	Export of rice through NCEL	Non-basmati rice	W.e.f. 25 th September 2023 export of 75,000 MT non-basmati white rice to UAE is permitted through National Cooperative Exports Limited (NCEL).
October 2023	Export of fixed allocated quantity of rice	Non-basmati rice	W.e.f. 18 th October 2023, export of fixed allocated quantity of non-basmati white rice permitted to Nepal, Cameroon, Cote d' Ivore, Republic of Guinea, Malaysia, Philippines and Seychelles.
November 2023	Export of specified quantity of rice	Non-basmati rice	W.e.f. 30 th November 2023, export of specified quantity of white grain & broken rice permitted through NCEL Bhutan, Mali, Senegal, Gambia & Indonesia.
December 2023	Export of fixed allocated quantity of rice	Non-basmati rice	W.e.f. 7 th December 2023, export of 20 MT non-basmati white rice permitted to Rice Exporters Federation for export to Nepal for earthquake victim.
December 2023	Export of specified quantity of rice	Non-basmati rice	W.e.f. 7 th December 2023, exports of specified quantity of non-basmati white rice permitted through NCEL to Comoros, Madagascar, Equatorial Guinea, Egypt and Kenya.
December 2023	Certificate of Inspection for Export	Rice	W.e.f. 12 th December 2023, export of rice (basmati & non-basmati) to UK, Iceland, Liechtenstein, Norway and Switzerland will only require Certificate of Inspection from Export Inspection Agency/ Export Inspection Council. Exports to other EU nations will not require certificate for a period of six months.
September 2024	"Prohibited" to "Free"	Non-basmati rice	W.e.f. 28 th September 2024, the export policy for non-basmati white rice amended from "Prohibited" to "Free" subject to Minimum Export Price (MEP) of US\$490/tonne until further order.
September 2024	Export duty on rice	Par-boiled non-basmati rice	Government reduced the export duty on parboiled non-basmati rice from 20 percent to 10 percent on 27 th September 2024.
October 2024	Minimum Export Price lifted	Non-basmati rice	W.e.f. 23 rd October 2024, the requirement of Minimum Export Price (MEP) of US\$490/tonne was lifted.
October 2024	Export duty on rice	Par-boiled non-basmati rice	Government reduced the export duty on parboiled non-basmati rice from 10 percent to zero percent on 22 nd October 2024.
March 2025	"Prohibited" to "Free"	Broken rice	W.e.f. 7 th March, 2025, the export policy for broken rice has been amended from "Prohibited" to "Free".

(Contd.)

Annex Table

Annex Table

Annex Table 4.9: Trade policy of Rice since 2011

Month	Trade Policy	Product	Description
March 2008 to April 2009	Zero import duty	Rice	In view of lower domestic supply, import of milled and semi-milled rice was allowed at zero percent import duty during 1 st March 2008 to 1 st April 2009.
February 2021	Mandatory import of non-Genetically Modified rice	Rice	Import of rice into India is permitted only if the importer furnishes a certificate from competent Government authority of the country of export that the rice exported is non-Genetically Modified (GM) w.e.f. 1 st March 2021 vide notification dated 8 th February 2021 issued by Food Safety and Standards Authority of India (FSSAI).

Source: 1. Directorate General of Foreign Trade, Ministry of Commerce and Industry
2. Central Board of Indirect Taxes & Customs, Department of Revenue





Annex Table 4.10: Trade policy of Pulses since 2017

Month	Trade Policy	Product	Import Policy	Description
December 2017	Import duty & Quantitative Restrictions (QR) imposed	Pulses	Government imposed import duties on imports of pulses like tur, chickpeas, lentil and gram in order to check the steep decline in domestic prices owing to increased domestic production. Further, to protect the pulses growers, the imports of tur, urad and moong were brought under QR.	
March 2018	Increase in import duty	Chickpeas, lentil and gram		In March 2018, the import duties on chickpeas, lentil and gram were raised to 60 percent due to sluggish prices in domestic market.
April 2018	“Free” to “Restricted” & QR	Yellow peas		The import policy of yellow peas was also revised from “Free” to “Restricted” in April 2018 and quantitative restriction of 1 lakh tonne was imposed on its imports for the period 1 st April 2018 to 30 th September 2018.
December 2018	Restriction on import	Yellow peas, green peas, dun peas and kaspas peas		Import of peas (including yellow peas, green peas, dun peas and kaspas peas) was “Restricted” till 31st March 2019 vide notification dated 28 th December 2018.
December 2019	Annual quota & MIP	Peas, urad, moong and tur		The import of peas was subject to an annual quota of 1.5 lakh tonnes and a Minimum Import Price (MIP) in CIF of ₹ 200/kg w.e.f. 18 th December 2019. Additionally, on 18th December 2019 till 31 st March 2020, an annual quota of 4 lakh tonnes on import of urad was also imposed. Quantitative restrictions on peas, moong and tur were later extended till 31 st March 2021. For green peas and other categories, import quota was fixed at 75,000 tonnes each, while for yellow peas, imports were prohibited altogether. For moong, import quota was fixed at 1.5 lakh tonnes and for tur, at 4 lakh tonnes.
February 2021	Import duty	Tur		Import policy on tur was kept unchanged till February 2021.
February 2021	Increase in import duty	Urad, moong and tur		Subsequently, import duty was increased to 30 percent for urad, moong and tur w.e.f. 2 nd February 2021.
March 2021	Continuation of annual quota	Urad		Government notified continuation of an annual quota of 4 lakh tonnes for import of urad for fiscal year 2021-22 vide notification dated 30 th March 2021.
May 2021	“Restricted” to “Free”	Tur and urad		The import policy for tur and urad was revised from “Restricted” to “Free” for the period up to 31 st October 2021, vide notification dated 15 th May 2021.
May 2021	“Free” import policy	Moong		As regards moong, the imports were made “Free” till 31st October 2021, vide notification dated 15 th May 2021.

(Contd.)

Annex Table



Annex Table

Annex Table 4.10: Trade policy of Pulses since 2017

Month	Trade Policy	Product	Description
December 2021	“Free” import policy	Moong	This policy was extended up to 31 st March 2022, again vide notification dated 20 th December 2021.
February 2022	“Free” to “Restricted”	Moong	Vide notification dated 11 th February 2022, import policy of “moong” was revised from “Free” to “Restricted” and since then, moong imports continue to be “Restricted” in India.
February 2022	Change in duty rates	Lentil	Vide notification dated 12 th February 2022, Agriculture Infrastructure and Development Cess (AIDC) on lentil was declared as ‘Zero’. The import duty on lentil was notified as ‘Zero’ (import from any country) and reduced from 30 percent to 20 percent (import from U.S.A.). This became applicable till 30 th September 2022.
March 2022	Extension of ‘Free’ import policy	Tur, Urad	Vide notification dated 29 th March 2022, ‘Free’ import policy of tur and urad was extended up to 31 st March 2023.
July 2022	Extension of “Nil” AIDC	Lentil	Vide notification dated 23 rd July 2022, AIDC on Lentil was ‘Zero’ and became applicable upto and inclusive of 31 st March 2023.
December 2022	Extension of ‘Free’ import policy	Tur, Urad	Vide notification dated 28 th December 2022, the “Free” import policy for tur and urad was extended upto 31 st March 2024.
December 2022	Extension of “Nil” AIDC	Lentil	Vide notification dated 29 th December 2022, AIDC on lentil was ‘Zero’ and became applicable upto and inclusive of 31 st March 2024.
March, 2023	Nil import duty	Tur whole	Further, vide notification dated 3 rd March, 2023, Government has also reduced the import duty on ‘tur whole’ to ‘Nil’ w.e.f. 4 th March, 2023.
December 2023	“Restricted” to “Free”	Yellow Peas	Vide notification dated 8 th December 2023, import of yellow peas is “Free” upto 31 st March 2024.
December 2023	Extension of “Nil” AIDC	Lentil	Vide notification dated 21 st December 2023, AIDC on lentil was ‘Zero’ and became applicable upto and inclusive of 31 st March 2025.
December 2023	Extension of ‘Free’ import policy	Tur, Urad	Vide notification dated 28 th December 2023, the “Free” import policy for tur and urad was extended upto 31 st March 2025.
February 2024	Extension of “Free” import policy	Yellow Peas	Vide notification dated 23 rd February 2024, the “Free” import policy for yellow peas was extended upto 30 th April 2024.
April 2024	Extension of “Free” import policy	Yellow Peas	Vide notification dated 5 th April 2024, the “Free” import policy for yellow peas was extended upto 30 th June 2024.

(Contd.)



Annex Table 4.10: Trade policy of Pulses since 2017

Month	Trade Policy	Product	Description
May 2024	Extension of “Free” import policy	Yellow Peas	Vide notification dated 8 th May 2024, the “Free” import policy for yellow peas was extended upto 31 st October 2024.
September 2024	Extension of “Free” import policy	Yellow Peas	Vide notification dated 13 th September 2024, the “Free” import policy for yellow peas was extended upto 31 st December 2024.
December 2024	Extension of “Free” import policy	Yellow Peas	Vide notification dated 24 th December 2024, the “Free” import policy for yellow peas was extended upto 28 th February 2025.
January 2025	Extension of “Free” import policy	Tur	Vide notification dated 20 th January 2025, the “Free” import policy for tur and urad was extended upto 31 st March 2026.
			Export Policy
September 2017	Lifted ban on export	Tur, urad and moong	In September 2017, Government lifted ban on export of tur, urad and moong, however, permission from APEDA was needed before exports.
November 2017	Removed prohibition on export	All types of pulses	Government removed prohibition on export of all types of pulses placing them under “Free” category, subject to exports routed through Customs Electronic Data Interchange (EDI) Ports only. Exports through Land Custom Stations (LCS), Indo-Bangladesh and Indo-Nepal border were also allowed but subject to registration of quantity with DGFT.

Source: 1. Directorate General of Foreign Trade, Ministry of Commerce and Industry

2. Central Board of Indirect Taxes & Customs, Department of Revenue

Annex Table

Annex Table 4.11(a): Trade policy of Edible Oils since 2017 (Soybean, Sunflower and Cotton seed Oil)

Month	Trade Policy	Product	Description
Import Policy			
November 2017	Increase in import duty	Crude soybean oil	The import duty on crude soybean oil was increased from 17.5 percent to 30 percent in November 2017.
March 2018	Increase in import duty	Refined soybean oil	The import duty on refined soybean oil was raised from 20 percent to 35 percent in March 2018.
March 2018	Increase in import duty	Crude sunflower oil	Import duty on crude sunflower oil was increased from 12.5 percent to 25 percent in March 2018.
March 2018	Increase in import duty	Refined sunflower oil	Import duty on refined sunflower oil, the import duty was increased from 20 to 35 percent in March 2018.
June 2018	Increase in import duty	Crude soybean oil	The import duty on crude soybean oil was increased from 30 percent to 35 percent in June 2018.
June 2018	Increase in import duty	Refined soybean oil	The import duty on refined soybean oil was raised from 35 percent to 45 percent in June 2018.
June 2018	Increase in import duty	Crude sunflower oil	Import duty on crude sunflower oil was increased from 25 percent to 35 percent in June 2018.
June 2018	Increase in import duty	Refined sunflower oil	Import duty on refined sunflower oil, the import duty was increased 35 to 45 percent in June 2018.
June 2018	Increase in import duty	Crude cotton seed oil	Import duty on crude cotton seed oil was raised from 30 percent to 35 percent in June 2018.
June 2018	Increase in import duty	Refined cotton seed oil	Import duty on refined cotton seed oil increased from 35 percent to 45 percent in June 2018.
Union Budget 2021-22	Import duty reduced & AIDC imposed	Crude soybean oil and crude sunflower oil	In the Union Budget 2021-22, the basic customs duty on imports of crude soybean oil and crude sunflower oil was cut from 35 percent to 15 percent and additionally, a 20 percent Agriculture Infrastructure and Development Cess (AIDC) was imposed on these two oils. Thus, the effective import duties on crude soybean oil and crude sunflower oil remained unchanged at 35 percent.

(Contd.)





Annex Table 4.11(a): Trade policy of Edible Oils since 2017 (Soybean, Sunflower and Cotton seed Oil)

Month	Trade Policy	Product	Description
August 2021	Import duty reduced	Crude soybean oil and crude sunflower oil	The basic import duty on crude soybean oil and crude sunflower oil was rationalized from 15 percent to 7.5 percent and thereafter, it was reduced to 2.5 percent w.e.f. 11 th September 2021.
August 2021	Import duty reduced	Refined soybean oil and sunflower oil	The basic import duty on refined variants of soybean and sunflower oil was reduced from 45 percent to 37.5 percent w.e.f. 20 th August 2021.
September 2021	Import duty reduced	Refined soybean oil and sunflower oil	The basic import duty on refined variants of soybean and sunflower oil was reduced to 32.5 percent w.e.f. 11 th September 2021.
October 2021	AIDC reduced	Crude soybean oil and crude sunflower oil	The AIDC on crude soybean oil and crude sunflower oil was reduced from 20 percent to 5 percent along with elimination of the basic import duty, earlier set at 2.5 percent. The AIDC of 5 percent on crude soybean oil and crude sunflower oil was imposed till 31 st March 2022.
October 2021	Import duty reduced	Refined soybean oil and sunflower oil	The basic import duty on refined variants of soybean and sunflower oil was reduced to 17.5 percent w.e.f. 14 th October 2021.
August 2022	Import duty & AIDC	Crude soybean oil and sunflower oil, refined soybean oil and sunflower oil	At present, the import duty on crude soybean oil and sunflower oil is zero. However, after taking into account 5 percent AIDC and 10 percent social welfare cess thereupon, the effective duty on these edible oils is 5.5 percent. For refined soybean and sunflower oil, the basic customs duty is 17.5 percent and taking into account 10 percent social welfare cess, the effective duty comes to 19.25 percent. These concessional customs duties on edible oils import have been extended till 31 st March 2023 vide Central Board of Indirect Taxes and Customs (CBIC) notification dated 31 st August 2022.
March 2023	TRQ on Import	Crude sunflower seed oil	A Tariff Rate Quota (TRQ) for import of crude sunflower seed oil was fixed at 20 lakh Metric Tonne (MT)/year during financial year 2022-23. Vide notification dated 1 st March 2023, it was further clarified that no TRQ shall be allocated for imports of crude sunflower seed oil in financial year 2023-24.
June 2023	Import duty reduced	Refined soybean oil and sunflower oil	The Basic Import Duty on Refined Soybean Oil and Refined Sunflower Oil was reduced from 17.5 percent to 12.5 percent w.e.f. 15 th June 2023, vide notification dated 14 th June 2023. This duty has been extended upto 31 st March, 2024.

(Contd.)

Annex Table

Annex Table

Annex Table 4.11(a): Trade policy of Edible Oils since 2017 (Soybean, Sunflower and Cotton seed Oil)

Month	Trade Policy	Product	Description
December 2023	AIDC period extended	Crude soybean oil and crude sunflower seed oil	AIDC of 5 percent imposed on Crude soybean oil, crude sunflower seed oil was extended till 31 st March 2022 extended till 31 st March 2025.
December 2023	Import Duty and AIDC	Crude soybean oil and crude sunflower seed oil	At present, the import duty on crude soybean oil and sunflower oil is zero. However, after taking into account 5 percent AIDC and 10 percent social welfare cess thereupon, the effective duty on these edible oils is 5.5 percent. For refined soybean and sunflower oil, the basic customs duty is 12.5 percent and taking into account 10 percent social welfare cess, the effective duty comes to 13.75 percent. These concessional customs duties on edible oils import have been extended till 31 st March 2025 vide CBIC notification dated 21 st December 2023.
September 2024	Import Duty Increased	Crude soybean oil, crude sunflower oil and refined soybean oil and refined sunflower oil	W.e.f. 14 th September 2024, the basic customs duty on crude soybean oil and crude sunflower oil increased from 0 percent to 20 percent, making the effective duty on these crude oils as 27.5 percent. Additionally, the basic customs duty on refined sunflower oil and refined soybean oil increased from 12.5 percent to 32.5 percent making the effective duty on these refined oils as 35.75 percent.

Source: 1. Directorate General of Foreign Trade, Ministry of Commerce and Industry

2. Central Board of Indirect Taxes & Customs, Department of Revenue





Annex Table 4.11(b): Trade policy of Edible Oils since 2017 (Palm Oil)

Month	Trade Policy	Product	Description
Import Policy			
November 2017	Increase in import duty	Crude Palm Oil (CPO)	The import duty CPO of edible grade was raised from 15 percent to 30 percent in November 2017.
November 2017	Increase in import duty	RBD palmolein	Import duty on RBD palmolein was increased from 25 percent to 40 percent in November 2017.
March 2018	Increase in import duty	Crude Palm Oil (CPO)	The import duty CPO of edible grade was raised to 44 percent in March 2018.
March 2018	Increase in import duty	RBD palmolein	Import duty on RBD palmolein was increased to 54 percent in March 2018.
January 2019	Import duty reduced	Crude Palm Oil (CPO)	The import duty CPO of edible grade was reduced to 40 percent in January 2019.
January 2019	Import duty reduced	RBD palmolein	However, Import duty on RBD palmolein was reduced to 45 percent for imports from Malaysia and 50 percent for shipments from Indonesia in January 2019.
September 2019	Safeguard duty	RBD palmolein	In September 2019, a safeguard duty of 5 percent was imposed on imports of RBD palmolein from Malaysia under Malaysia-India Comprehensive Economic Cooperation Agreement (MICECA) although imports under ASEAN agreement did not attract this safeguard duty.
January 2020	Import duty reduced	Refined & crude palm oil	Import duty on refined palm oil was lowered to 45 percent while that on crude palm oil was reduced from 45 percent to 37.5 percent w.e.f. 1 st January 2020 under ASEAN and MICECA agreement.
January 2020	“Free” to “Restricted”	Refined palm oil	In January 2020, the import policy of refined palm oil was amended from “Free” to “Restricted” category.
November 2020	Import duty reduced	Crude Palm Oil (CPO)	W.e.f. 27 th November 2020, the basic import duty on CPO was reduced from 37.5 percent to 27.5 percent in order to control the rise in price of palm oil and allow greater availability of oil in the market for consumption.
Union Budget 2021-22	Import duty reduced & AIDC imposed	Crude Palm Oil (CPO)	In the Union Budget 2021-22, the basic import duty on CPO was cut from 27.5 percent to 15 percent. In addition, 17.5 percent AIDC was also imposed on the imports of CPO, which narrowed the duty gap between palm oil and other edible oils.

(Contd.)

Annex Table



Annex Table 4.11(b): Trade policy of Edible Oils since 2017 (Palm Oil)

Month	Trade Policy	Product	Description
June 2021	Import duty reduced	Crude & refined palm oil	The basic import duty on crude palm oil was reduced from 15 to 10 percent and that on refined variants of palm oil was reduced to 37.5 percent in the month of June 2021.
September 2021	Import duty reduced & AIDC imposed	Crude & refined palm oil	The basic import duty on crude palm oil was further reduced from 10 percent to 2.5 percent and that on refined palm oils from 37.5 percent to 32.5 percent in September 2021. Simultaneously, Agriculture Infrastructure and Development Cess (AIDC) on crude palm oil was increased to 20 percent.
October 2021	Import duty reduced	Crude Palm Oil (CPO)	Government slashed the basic import duty on CPO from 2.5 percent to 'nil' and AIDC on CPO from 20 percent to 7.5 percent w.e.f. 14 th October 2021.
December 2021	"Restricted" to "Free"	Refined palm oils	The import of refined palm oils was revised from "Restricted" to "Free" category till 31 st December 2022 with the exception on imports of these oils through any port in Kerala.
February 2022	AIDC reduced	Crude and refined edible oils	The AIDC on crude palm oil was further reduced from 7.5 to 5 percent w.e.f. 12 th February 2022 and the duty cuts on all the above crude and refined edible oils were made applicable till 30 th September 2022.
March 2022	Import duty reduced	Refined palm oil	The basic import duty on refined palm oil was reduced from 32.5 percent to 17.5 percent, applicable till 31 st March 2022.
August 2022	Concessional import duty extended	Palm oil	The concessional import duty on palm oil import has been extended for another six months from 30 th September 2022 to 31 st March 2023 vide Central Board of Indirect Taxes and Customs (CBIC) notification dated 31 st August 2022.
December 2023	Import duty and AIDC	Crude Palm Oil (CPO)	At present, the import duty on CPO is zero. However, after taking into account 5 percent AIDC and 10 percent social welfare cess thereupon, the effective duty on CPO is 5.5 percent. This has been extended till 31 st March 2025 vide notification dated 21 st December 2023.
September 2024	Import Duty Increased	Crude and refined palm oil	W.e.f. 14 th September 2024, the basic customs duty on crude palm oil increased from 0 percent to 20 percent, making the effective duty on crude palm oil as 27.5 percent. Additionally, the basic customs duty on refined palm oil increased from 12.5 percent to 32.5 percent making the effective duty on refined palm oil as 35.75 percent.

Source: 1. Directorate General of Foreign Trade, Ministry of Commerce and Industry

2. Central Board of Indirect Taxes & Customs, Department of Revenue



Annex Table 4.11(c): Export Policy for Edible Oils

Month	Trade Policy	Product	Description
April 2018	“Free” export without any quantitative ceiling	Edible oils (except mustard oil)	Except mustard oil, exports of all edible oils have been made “Free” without any quantitative ceiling, pack size etc. w.e.f. 6 th April 2018. However, export of mustard oil is permitted in packs of up to 5 kg with a Minimum Export Price (MEP) of US\$900/tonne.

Source: 1. Directorate General of Foreign Trade, Ministry of Commerce and Industry
2. Central Board of Indirect Taxes & Customs, Department of Revenue



Annex Table 5.1: Actual Costs and Gross Returns over Actual Cost of Cultivation (₹/ha) of Kharif crops for 2021-22, 2022-23 and 2023-24

Year	CoC A ₂	CoC A ₂ +FL	GVO	Gross Returns over CoC	
				A ₂ (Col.4 - Col.2)	A ₂ +FL (Col.4 - Col.3)
(1)	(2)	(3)	(4)	(5)	(6)
Paddy					
2021-22	45848	59160	81326	35477	22166
2022-23	48650	62055	91751	43100	29696
2023-24	49813	62727	101514	51701	38787
TE2023-24	48104	61314	91530	43426	30216
Jowar					
2021-22	28890	39449	43070	14180	3620
2022-23	31120	41253	51303	20182	10049
2023-24	32376	42926	56277	23901	13351
TE2023-24	30795	41209	50216	19421	9007
Bajra					
2021-22	21809	36150	40269	18460	4119
2022-23	22013	35975	47405	25391	11430
2023-24	24140	39244	49667	25528	10423
TE2023-24	22654	37123	45780	23126	8657
Maize					
2021-22	39601	50984	78838	39237	27853
2022-23	42020	53348	84762	42742	31414
2023-24	44344	55832	87033	42689	31202
TE2023-24	41988	53388	83544	41556	30156
Ragi					
2021-22	37327	53174	44697	7369	-8477
2022-23	39141	53757	49540	10399	-4217
2023-24	41377	56592	49497	8120	-7095
TE2023-24	39282	54508	47911	8629	-6596
Arhar (Tur)					
2021-22	35947	45457	70639	34692	25182
2022-23	37884	48023	75315	37431	27292
2023-24	42252	53761	75394	33142	21633
TE2023-24	38695	49080	73783	35088	24703
Moong					
2021-22	16965	26863	34486	17521	7623
2022-23	16658	28024	37197	20538	9173
2023-24	17541	29691	43658	26117	13967
TE2023-24	17055	28193	38447	21392	10254
Urad					
2021-22	21237	27323	35749	14512	8425
2022-23	21602	28573	40670	19068	12097
2023-24	21557	29436	40807	19249	11371
TE2023-24	21465	28444	39075	17610	10631

(Contd...)



**Annex Table 5.1: Actual Costs and Gross Returns over Actual Cost of Cultivation (₹/ha)
of Kharif crops for 2021-22, 2022-23 and 2023-24**

Year	CoC A ₂	CoC A ₂ +FL	GVO	Gross Returns over CoC	
				A ₂ (Col.4 - Col.2)	A ₂ +FL (Col.4 - Col.3)
(1)	(2)	(3)	(4)	(5)	(6)
Groundnut					
2021-22	58388	70238	96393	38005	26156
2022-23	63111	75908	114069	50957	38160
2023-24	69865	84171	133906	64041	49735
TE2023-24	63788	76772	114789	51001	38017
Soybean					
2021-22	35476	40622	59582	24106	18959
2022-23	38271	43977	62845	24574	18868
2023-24	39742	45895	67315	27572	21420
TE2023-24	37830	43498	63247	25417	19749
Sunflower					
2021-22	33044	39116	54968	21924	15852
2022-23	32208	39942	58017	25810	18075
2023-24	38946	43188	51012	12065	7823
TE2023-24	34733	40749	54666	19933	13917
Sesamum					
2021-22	17362	29742	40272	22910	10530
2022-23	19293	31937	45871	26577	13933
2023-24	20332	33450	49716	29384	16266
TE2023-24	18996	31710	45286	26290	13576
Nigerseed					
2021-22	7123	21456	22509	15387	1053
2022-23	7892	23552	24294	16401	742
2023-24	5628	22210	24838	19211	2629
TE2023-24	6881	22406	23880	17000	1474
Cotton					
2021-22	58217	72019	99787	41570	27768
2022-23	61651	76252	120011	58360	43759
2023-24	65792	81357	116447	50655	35090
TE2023-24	61887	76543	112082	50195	35539

Note: All-India CoC, GVO and gross returns of a crop are weighted average of respective CoCs, GVOs and gross returns of projected States

Source: CACP Calculations using CS data



Annex Table 5.2: Average Gross Returns over Actual Cost of Cultivation (₹/ha) of Kharif Crops in Selected States during TE2023-24

State/Crop	CoC A ₂	CoC A ₂ +FL	GVO	Gross Returns over CoC	
				A ₂ (Col.4 - Col.2)	A ₂ +FL (Col.4 - Col.3)
(1)	(2)	(3)	(4)	(5)	(6)
Paddy					
Andhra Pradesh	67684	75427	124530	56846	49103
Assam	28095	47402	59003	30908	11601
Bihar	30774	40838	58584	27810	17747
Chhattisgarh	35334	43249	87779	52444	44530
Gujarat	49368	58695	86997	37628	28301
Haryana	52489	61409	148822	96334	87413
Himachal Pradesh	21462	46753	79759	58297	33006
Jharkhand	32662	44591	52536	19874	7944
Karnataka	61751	76827	114081	52330	37254
Kerala	68970	80768	116573	47603	35805
Madhya Pradesh	36031	46063	91789	55758	45726
Maharashtra	67512	84498	62132	-5380	-22367
Odisha	40660	58888	70182	29522	11294
Punjab	54296	60331	152161	97865	91830
Tamil Nadu	63134	70216	101147	38013	30931
Telangana	64559	79982	118375	53816	38393
Uttar Pradesh	46557	58481	71484	24927	13003
West Bengal	51502	74461	88994	37492	14533
All-India	48104	61314	91530	43426	30216
Jowar					
Andhra Pradesh	58340	60077	150856	92516	90778
Karnataka	25865	35239	43912	18047	8673
Maharashtra	34183	41944	46906	12722	4962
Rajasthan	16472	36921	39315	22843	2394
Tamil Nadu	41344	50729	70925	29581	20196
All-India	30795	41209	50216	19421	9007
Bajra					
Gujarat	42060	55665	95300	53240	39635
Haryana	21256	31775	42406	21150	10631
Maharashtra	45189	60870	64681	19492	3811
Rajasthan	18028	33430	37985	19957	4555
Uttar Pradesh	21767	34085	45948	24181	11863
All-India	22654	37123	45780	23126	8657
Maize					
Andhra Pradesh	63298	69632	135789	72492	66157
Bihar	39673	48154	92349	52676	44195

(Contd...)



Annex Table 5.2: Average Gross Returns over Actual Cost of Cultivation (₹/ha) of Kharif Crops in Selected States during TE2023-24

State/Crop	CoC A ₂	CoC A ₂ +FL	GVO	Gross Returns over CoC	
				A ₂ (Col.4 - Col.2)	A ₂ +FL (Col.4 - Col.3)
(1)	(2)	(3)	(4)	(5)	(6)
Gujarat	40983	52596	63076	22094	10481
Himachal Pradesh	17825	40030	52758	34933	12728
Karnataka	41159	50168	78797	37638	28629
Madhya Pradesh	32205	38516	63203	30999	24687
Maharashtra	60723	72045	123810	63087	51765
Punjab	47517	59193	68538	21020	9345
Rajasthan	27412	55592	65277	37865	9685
Tamil Nadu	57565	66116	105963	48398	39847
Telangana	58086	73255	115353	57267	42098
Uttar Pradesh	32664	40559	51781	19117	11221
All-India	41988	53388	83544	41556	30156
Ragi					
Karnataka	41015	55784	49174	8159	-6610
Odisha	13235	35355	28651	15416	-6704
All-India	39282	54508	47911	8629	-6596
Arhar (Tur)					
Andhra Pradesh	27664	30831	45404	17739	14572
Gujarat	29569	38524	75181	45612	36658
Karnataka	30020	35950	46405	16385	10455
Maharashtra	64144	79729	118125	53981	38396
Odisha	5892	16487	27066	21175	10579
Telangana	30554	36919	57239	26685	20320
Uttar Pradesh	18936	38485	80894	61958	42409
All-India	38695	49080	73783	35088	24703
Moong					
Andhra Pradesh	27912	29231	50892	22980	21662
Gujarat	17619	32014	48819	31200	16805
Karnataka	23304	28759	36724	13421	7965
Maharashtra	35132	44759	53425	18293	8666
Odisha	10212	18532	31718	21506	13186
Rajasthan	14412	27325	38064	23652	10740
Tamil Nadu	18918	23237	31910	12992	8673
Uttar Pradesh	9836	26940	17429	7593	-9511
All-India	17055	28193	38447	21392	10254
Urad					
Andhra Pradesh	30550	33652	76004	45454	42352
Gujarat	23841	32736	33366	9525	631

(Contd...)



Annex Table 5.2: Average Gross Returns over Actual Cost of Cultivation (₹/ha) of Kharif Crops in Selected States during TE2023-24

State/Crop	CoC A ₂	CoC A ₂ +FL	GVO	Gross Returns over CoC	
				A ₂ (Col.4 - Col.2)	A ₂ +FL (Col.4 - Col.3)
(1)	(2)	(3)	(4)	(5)	(6)
Karnataka	25134	29045	35961	10827	6917
Madhya Pradesh	20033	23927	31688	11655	7761
Maharashtra	32357	40552	54577	22219	14025
Odisha	9084	19049	29985	20901	10936
Rajasthan	16226	33467	35540	19314	2072
Tamil Nadu	32121	36574	53453	21331	16879
Uttar Pradesh	11278	20844	22201	10924	1358
All-India	21465	28444	39075	17610	10631
Groundnut					
Andhra Pradesh	66191	74318	72291	6101	-2027
Gujarat	68041	76849	120876	52835	44027
Karnataka	45624	58719	76043	30418	17324
Maharashtra	69255	99385	88768	19513	-10616
Odisha	33722	54089	56468	22747	2379
Rajasthan	64233	80844	162926	98693	82082
Tamil Nadu	64292	75707	104495	40202	28788
Telangana	90847	104949	140968	50120	36019
Uttar Pradesh	25732	38542	64128	38396	25587
West Bengal	58648	107266	180269	121621	73003
All-India	63788	76772	114789	51001	38017
Soybean					
Gujarat	37035	42534	41469	4434	-1064
Karnataka	40453	45915	73535	33082	27620
Madhya Pradesh	26872	31603	47639	20767	16036
Maharashtra	51985	57342	83078	31093	25736
Rajasthan	33894	45696	62141	28248	16445
Telangana	30994	37483	45257	14264	7774
All-India	37830	43498	63247	25417	19749
Sunflower					
Andhra Pradesh	36995	42868	74516	37520	31648
Karnataka	34510	40560	53122	18611	12562
All-India	34733	40749	54666	19933	13917
Sesamum					
Andhra Pradesh	22526	24834	32546	10020	7712
Gujarat	30426	42681	56723	26297	14042
Madhya Pradesh	22595	28239	54485	31890	26246
Rajasthan	8504	24852	27674	19170	2822

(Contd...)



Annex Table 5.2: Average Gross Returns over Actual Cost of Cultivation (₹/ha) of Kharif Crops in Selected States during TE2023-24

State/Crop	CoC A_2	CoC A_2+FL	GVO	Gross Returns over CoC	
				A_2 (Col.4 - Col.2)	A_2+FL (Col.4 - Col.3)
(1)	(2)	(3)	(4)	(5)	(6)
Uttar Pradesh	10061	19563	19355	9294	-208
West Bengal	28171	51084	79913	51742	28829
All-India	18996	31710	45286	26290	13576
Nigerseed					
Odisha	6881	22406	23880	17000	1474
All-India	6881	22406	23880	17000	1474
Cotton					
Andhra Pradesh	70209	76391	117808	47599	41417
Gujarat	57388	71954	133173	75786	61220
Haryana	38249	54020	93052	54802	39031
Karnataka	51049	61547	100266	49217	38719
Madhya Pradesh	52558	65497	89394	36836	23897
Maharashtra	67986	82047	101538	33552	19491
Odisha	46606	68559	94734	48128	26176
Punjab	56026	64033	149803	93777	85770
Rajasthan	43968	71545	136072	92104	64526
Tamil Nadu	80268	102981	159262	78994	56281
Telangana	75409	89829	108675	33267	18847
All-India	61887	76543	112082	50195	35539

Note: All-India CoC, GVO and gross returns of a crop are weighted average of respective CoCs, GVOs and gross returns of projected States

Source: CACP Calculations using CS data



Annex Table

Annex Table 5.3: Monthly Average Daily Wage Rates (₹/day) for Agricultural Labour (Man) during Kharif season 2021 to 2024

Year/Month	AP	AS	BR	GJ	HR	HP	KA	KL	MP	MH	OD	PB	RJ	TN	UP	WB	IND
2021																	
May	405	311	303	236	392	453	360	729	238	327	251	371	330	541	304	334	361
June	402	311	304	239	396	463	361	729	236	329	248	374	331	540	306	334	361
July	404	312	306	242	398	484	360	729	239	325	265	374	327	538	311	337	363
August	400	316	307	242	399	494	362	730	241	328	264	372	337	542	309	338	364
September	410	323	308	242	401	463	371	758	241	333	261	372	335	541	310	338	368
October	421	323	309	242	405	460	374	758	240	331	259	373	343	544	309	338	370
November	429	326	311	242	406	460	374	758	238	328	259	373	346	543	310	338	371
December	430	329	313	244	410	463	378	758	241	329	271	372	346	543	315	339	373
2022																	
January	424	330	316	245	410	460	379	761	246	331	273	373	337	544	319	340	374
February	430	338	317	247	412	458	387	761	248	332	267	383	341	550	320	339	376
March	443	333	320	248	413	458	389	762	248	333	267	385	334	550	323	340	379
April	453	333	321	252	421	458	389	762	250	334	272	387	334	550	327	342	382
May	457	334	323	253	422	458	388	763	250	340	275	393	334	551	328	343	384
June	460	335	327	256	422	458	391	763	251	341	276	393	339	554	328	346	386
July	464	338	331	260	427	459	391	768	252	341	280	395	341	549	330	352	388
August	468	338	332	262	435	459	396	773	253	343	282	395	344	550	335	354	391
September	472	338	332	262	437	458	402	773	255	343	287	396	347	550	335	354	392
October	480	340	334	263	442	458	407	776	256	344	286	397	345	551	337	354	395
November	487	342	341	264	442	458	416	776	257	345	289	399	351	553	337	354	398
December	489	342	341	266	449	458	417	776	257	347	291	399	352	555	338	355	399
2023																	
January	489	343	342	268	452	460	419	780	259	349	293	400	355	558	341	355	401
February	496	349	344	269	452	465	419	784	260	350	294	402	357	560	342	360	403
March	500	352	346	271	452	465	421	786	259	351	295	402	359	561	345	362	405
April	506	354	349	271	452	466	423	788	260	351	298	402	359	572	346	363	408

(Contd...)



Annex Table 5.3: Monthly Average Daily Wage Rates (₹/day) for Agricultural Labour (Man) during Kharif season 2021 to 2024

Year/Month	AP	AS	BR	GJ	HR	HP	KA	KL	MP	MH	OD	PB	RJ	TN	UP	WB	IND
May	515	359	353	272	460	466	423	790	265	352	298	407	365	583	347	364	412
June	520	364	356	274	461	467	424	791	265	355	298	413	367	590	349	364	414
July	527	367	357	277	461	464	427	798	268	357	300	417	372	600	350	365	418
August	532	368	361	277	461	476	430	799	269	362	303	417	374	603	352	365	421
September	536	369	365	278	461	476	431	800	269	364	305	417	376	605	354	365	422
October	538	372	366	279	461	505	431	802	269	366	308	417	379	610	355	365	424
November	541	379	367	282	461	485	435	819	271	368	312	426	380	610	356	366	427
December	545	384	368	282	462	485	438	819	271	370	317	427	383	611	356	368	429
2024																	
January	550	386	368	283	466	485	442	825	272	371	320	427	390	611	357	371	431
February	552	386	372	283	472	485	447	828	272	371	323	428	396	611	358	371	433
March	555	387	373	282	474	485	451	830	272	372	325	428	398	613	359	371	434
April	558	389	374	283	481	492	451	834	273	372	329	428	400	613	363	371	436
May	561	391	377	285	487	492	453	840	273	374	334	428	402	613	365	372	438
June	567	391	380	291	491	513	457	842	274	376	337	428	404	613	365	374	440
July	571	394	382	297	496	517	458	844	276	377	342	428	406	618	368	377	443
August	575	400	386	299	502	522	458	848	279	386	349	430	409	621	369	381	447
September	575	406	386	302	507	526	463	849	280	389	352	430	412	626	372	383	450
October	576	410	386	302	512	526	465	850	281	391	352	432	415	630	374	384	451
November	578	415	388	308	512	533	465	852	284	392	357	432	419	632	376	386	453

Note: 1: Daily Wage rate - Average of agricultural operations, such as, ploughing/tilling; sowing (including planting/transplanting/weeding); harvesting/ winnowing/threshing; watering & irrigation and plant protection (applying pesticides, treating seeds, etc.)

2: All-India daily wage rate is weighted average of daily wage rates of States mentioned in the Table

Source: Labour Bureau, Ministry of Labour and Employment and CACP Calculations



Annex Table 5.4: Monthly Wholesale Price Index (Base: 2011-12) of Major Farm Inputs during Kharif season 2023 and 2024

Year/Month	High Speed Diesel	Fertilizers and Nitrogen Compounds	Electricity	Manufacture of Agricultural and Forestry Machinery	Lube Oils	Cattle Feed	Fodder	Pesticides and other Agrochemical Products	Index of Farm Inputs
2023									
May	169.5	144.5	138.4	141	190.2	214.2	227.5	135.8	156.9
June	169	144.3	135.2	141.4	190.2	215.7	222.8	134.0	155.5
July	170.2	143.6	134.2	141.8	190.2	216.8	232.1	132.1	155.9
August	171.9	141.4	135.1	141.2	190.2	219.7	231.5	132.9	156.4
September	174.4	140.6	137	142.2	190.2	220.2	237.9	132.8	158.1
October	175.1	142.2	142.5	142.5	190.2	218.6	242.5	133.0	160.5
November	174.3	143	153.3	143.2	190.2	218.9	230.9	132.8	163.1
December	172	142.5	158.4	143.9	190.2	219.1	225.6	132.3	163.7
2024									
January	171.8	141.4	157.2	144.1	190.2	218.3	222.7	131.2	162.9
February	172.1	142.0	157.0	144.1	190.2	215.9	220.5	130.0	162.8
March	170.3	142.6	144.6	143.9	190.2	216.3	219.3	130.5	158.4
April	168.5	143.3	141.1	145.3	190.2	214.7	207.2	127.1	156.2
May	167.7	143.3	135.7	145.2	190.2	214.5	204.4	127.8	154.1
June	166.0	143.3	132.0	145.0	190.2	214.2	204.9	128.0	152.5
July	167.4	143.4	132.9	143.5	190.2	214.1	205.6	128.9	153.2
August	166.7	143.2	134.8	144.3	190.2	214.8	208.8	129.8	153.8
September	165.1	142.7	136.7	145.2	190.2	214.9	210.2	128.8	153.9
October	164.2	143.2	137.4	145.3	190.2	214.1	213.2	130.3	154.1
November	164.4	143.2	137.3	145.6	190.2	212.3	219.0	129.5	154.4

Note : Index of Farm Inputs is weighted average of WPIs of farm inputs

Source : Office of the Economic Adviser, Ministry of Commerce and Industry and CACP Calculations



Annex Table 5.5: Projected Cost of Production (A_2 , A_2+FL & C_2) and Yield for Kharif Crops for KMS2025-26 and Production Share during TE2023-24

State/Crop	Yield (qtl/ha)	Cost of Production (₹/qtl)			Share in Production(%)
		A_2	A_2+FL	C_2	
Paddy					
Andhra Pradesh	60.18	1207	1346	1926	5.72
Assam	36.90	992	1669	2038	3.86
Bihar	30.32	980	1300	1738	5.62
Chhattisgarh	42.72	1005	1230	1647	6.83
Gujarat	40.22	1235	1467	1815	1.71
Haryana	53.38	1105	1293	2128	3.90
Himachal Pradesh	28.38	676	1473	2025	0.12
Jharkhand	27.05	1169	1598	2050	1.47
Karnataka	49.02	1332	1657	2157	2.91
Kerala	37.41	1832	2146	2713	0.39
Madhya Pradesh	43.12	1085	1386	1814	4.73
Maharashtra	21.07	2722	3407	3918	2.83
Odisha	42.60	1197	1734	2128	6.45
Punjab	71.10	870	967	1593	9.98
Tamil Nadu	46.20	1496	1663	2134	5.52
Telangana	53.83	1325	1641	2219	11.20
Uttar Pradesh	36.50	1415	1777	2329	11.76
West Bengal	44.26	1270	1843	2294	11.88
All-India	44.18	1238	1579	2090	96.90
Jowar					
Andhra Pradesh	49.42	1416	1459	2266	6.86
Karnataka	11.11	2123	2889	3802	18.55
Maharashtra	11.88	2104	2581	3274	36.58
Rajasthan	10.57	950	2130	2914	12.24
Tamil Nadu	22.30	1898	2329	2726	7.41
All-India	13.60	1859	2466	3206	81.64
Bajra					
Gujarat	30.35	1167	1545	1994	11.55
Haryana	22.48	955	1427	2274	10.90
Maharashtra	24.64	1986	2668	3186	4.17
Rajasthan	11.53	973	1804	2192	41.58
Uttar Pradesh	19.68	977	1529	2127	19.39
All-India	15.92	1046	1703	2209	87.60

(Contd...)



Annex Table 5.5: Projected Cost of Production (A_2 , A_2+FL & C_2) and Yield for *Kharif* Crops for KMS2025-26 and Production Share during TE2023-24

State/Crop	Yield (qtl/ha)	Cost of Production (₹/qtl)			Share in Production(%)
		A_2	A_2+FL	C_2	
Maize					
Andhra Pradesh	74.13	945	1039	1525	5.91
Bihar	45.40	924	1121	1510	11.65
Gujarat	22.21	1672	2146	2647	2.36
Himachal Pradesh	19.01	821	1843	2374	1.89
Karnataka	34.82	1319	1608	2141	15.31
Madhya Pradesh	32.44	1063	1272	1610	12.39
Maharashtra	57.52	1245	1476	1838	9.09
Punjab	35.16	1379	1718	2237	1.08
Rajasthan	23.19	1136	2313	2740	5.84
Tamil Nadu	47.28	1401	1609	2021	7.91
Telangana	57.36	1216	1534	2053	7.19
Uttar Pradesh	26.66	1344	1669	2202	5.70
All-India	38.78	1183	1508	1952	86.32
Ragi					
Karnataka	14.96	2372	3225	3940	65.38
Odisha	9.36	1506	4029	4837	2.71
All-India	14.62	2337	3257	3976	68.08
Arhar (Tur)					
Andhra Pradesh	6.60	4873	5431	7558	2.21
Gujarat	9.41	3762	4899	6730	7.66
Karnataka	7.20	4058	4859	6562	27.58
Maharashtra	18.15	4298	5340	6669	29.31
Odisha	4.14	1840	5148	6958	3.79
Telangana	7.16	4730	5718	8186	5.38
Uttar Pradesh	9.43	2105	4275	7252	10.17
All India	10.83	3848	5038	6839	86.11
Moong					
Andhra Pradesh	7.52	4558	4772	6619	2.15
Gujarat	6.05	3327	6037	7631	3.61
Karnataka	5.26	4800	5938	7686	4.38
Maharashtra	7.96	5332	6783	8189	4.31
Odisha	4.10	3182	5781	7747	3.10
Rajasthan	4.96	3066	5806	7338	29.19

(Contd...)



Annex Table 5.5: Projected Cost of Production (A_2 , A_2+FL & C_2) and Yield for Kharif Crops for KMS2025-26 and Production Share during TE2023-24

State/Crop	Yield (qtl/ha)	Cost of Production (₹/qtl)			Share in Production(%)
		A_2	A_2+FL	C_2	
Tamil Nadu	4.10	4702	5775	7818	1.65
Uttar Pradesh	6.06	1870	5125	7418	1.89
All India	5.24	3510	5845	7476	50.30
Urad					
Andhra Pradesh	10.78	3318	3654	5599	15.55
Gujarat	4.72	4982	6842	8374	3.83
Karnataka	5.10	5707	6593	8376	1.22
Madhya Pradesh	6.49	3756	4485	5600	29.97
Maharashtra	7.06	5605	7021	8430	6.99
Odisha	3.68	3026	6351	8437	1.37
Rajasthan	5.19	3294	6791	8071	5.70
Tamil Nadu	6.84	5334	6071	8144	9.26
Uttar Pradesh	5.44	2763	5091	7942	13.93
All India	6.52	3874	5114	6829	87.83
Groundnut					
Andhra Pradesh	11.58	6309	7086	8928	4.70
Gujarat	17.69	4215	4761	5962	44.64
Karnataka	10.96	5172	6649	8058	4.28
Maharashtra	11.90	5874	8428	9962	3.11
Odisha	8.42	4195	6728	8429	0.84
Rajasthan	25.78	2453	3088	3851	18.47
Tamil Nadu	15.82	4815	5669	7129	9.27
Telangana	19.13	4607	5323	6674	2.70
Uttar Pradesh	12.06	2334	3496	4666	1.19
West Bengal	33.83	2049	3715	5069	2.11
All India	17.33	4065	4842	6047	91.33
Soybean					
Gujarat	15.66	2808	3226	4212	2.70
Karnataka	14.53	3121	3544	4669	3.40
Madhya Pradesh	11.24	2604	3062	4175	40.58
Maharashtra	15.86	3560	3925	4957	42.34
Rajasthan	10.82	3126	4206	5309	8.05
Telangana	16.09	2835	3442	5090	2.11
All India	13.24	3082	3552	4638	99.17

(Contd...)



Annex Table 5.5: Projected Cost of Production (A_2 , A_2+FL & C_2) and Yield for Kharif Crops for KMS2025-26 and Production Share during TE2023-24

State/Crop	Yield (qtl/ha)	Cost of Production (₹/qtl)			Share in Production(%)
		A_2	A_2+FL	C_2	
Sunflower					
Andhra Pradesh	11.35	3815	4421	6463	4.25
Karnataka	9.21	4422	5201	6356	56.45
All India	9.39	4379	5147	6364	61
Sesamum					
Andra Pradesh	3.51	7143	7874	9979	1.17
Gujarat	5.28	5721	8025	10201	17.28
Madhya Pradesh	5.58	4444	5553	7549	16.01
Rajasthan	3.22	2853	8320	10385	9.31
Uttar Pradesh	2.52	3832	7448	10229	11.25
West Bengal	11.41	3015	5421	7216	32.24
All India	5.40	3957	6564	8632	87.27
Nigerseed					
Odisha	3.72	1960	6358	8025	49.03
All-India	3.72	1960	6358	8025	49.03
Cotton					
Andhra Pradesh	16.29	4553	4953	6729	4.10
Gujarat	16.68	3701	4641	6218	26.07
Haryana	12.26	3214	4537	6703	3.93
Karnataka	13.11	3943	4754	6573	6.76
Madhya Pradesh	15.96	4020	5021	6680	4.78
Maharashtra	14.20	4755	5739	7140	25.29
Odisha	14.34	3925	5770	7373	2.09
Punjab	18.48	3534	4039	6260	1.77
Rajasthan	18.00	2641	4294	5683	8.10
Tamil Nadu	19.12	5078	6513	7897	0.90
Telangana	16.91	4857	5785	7348	16.14
All-India	15.51	4130	5140	6717	99.92

Note: 1. Production share are related to production of projected States

2. All-India CoP of a crop is weighted average of CoPs of projected States

Source: CACP Calculations using CS data



Annex Table 5.6(a): Paddy : Break-up of Cost of Cultivation

(₹/ha)

Cost Items	Andhra Pradesh					Assam				
	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23
Operational Cost	70347	74253	77008	5.6	3.7	44169	46309	49599	4.8	7.1
Human Labour										
<i>Casual</i>	26210	26775	27249	2.2	1.8	8584	8641	8751	0.7	1.3
<i>Attached</i>	103	110	87	6.4	-21.1	10	7	13	-28.8	85.0
<i>Family</i>	7801	7506	7921	-3.8	5.5	17326	18715	21882	8.0	16.9
Total	34115	34390	35256	0.8	2.5	25920	27362	30646	5.6	12.0
Bullock Labour										
<i>Hired</i>	128	150	157	17.5	4.3	97	103	26	5.9	-74.6
<i>Owned</i>	19	15	24	-22.6	61.8	4414	4146	5181	-6.1	24.9
Total	147	165	181	12.3	9.5	4511	4249	5207	-5.8	22.5
Machine Labour										
<i>Hired</i>	13880	15795	16444	13.8	4.1	8701	9403	8041	8.1	-14.5
<i>Owned</i>	668	901	1412	34.9	56.8	789	788	987	-0.1	25.2
Total	14548	16696	17857	14.8	7.0	9490	10191	9028	7.4	-11.4
Seed	2256	2411	2535	6.8	5.1	1309	1408	1563	7.6	11.0
Fertilisers and Manure										
<i>Fertilisers</i>	9279	9845	9870	6.1	0.3	1056	1099	1004	4.0	-8.6
<i>Manure</i>	969	789	293	-18.6	-62.8	835	906	1046	8.6	15.5
Total	10248	10634	10164	3.8	-4.4	1891	2005	2050	6.0	2.3
Other Inputs										
Insecticides	6053	6608	7523	9.2	13.8	24	28	29	16.1	5.6
Irrigation charges	590	661	833	12.0	26.1	195	212	220	8.8	3.4
Crop Insurance	0	0	0	-	-	0	0	0	-	-
Interest on working capital	1895	2023	2094	6.7	3.5	813	836	840	2.8	0.4
Miscellaneous	495	667	567	34.8	-15.0	16	17	17	8.8	0.2
Fixed Cost	33241	36960	44673	11.2	20.9	13479	15350	17968	13.9	17.1
Rental value of owned land	30374	33864	43007	11.5	27.0	11786	13628	15891	15.6	16.6
Rent paid for leased-in land	1823	2015	408	10.6	-79.8	12	2	53	-80.1	2206.5
Land revenue, cesses & taxes	0	0	8	-	-	170	172	174	0.9	1.1
Depreciation on implements & Farm buildings	128	136	154	5.9	13.4	500	509	538	1.8	5.6
Interest on fixed capital	916	944	1097	3.1	16.1	1012	1039	1312	2.7	26.3
Yield(Qtl/ha)	57.09	59.41	64.03	4.1	7.8	36.84	36.43	37.43	-1.1	2.7

Source: ES&E Division, Ministry of Agriculture & Farmers Welfare



Annex Table 5.6(a): Paddy : Break-up of Cost of Cultivation

(₹/ha)

Cost Items	Bihar					Chhattisgarh				
	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23
Operational Cost	37312	40780	42009	9.3	3.0	41975	42501	43048	1.3	1.3
Human Labour										
<i>Casual</i>	12594	14585	15439	15.8	5.9	7635	7913	8308	3.6	5.0
<i>Attached</i>	392	367	9	-6.3	-97.4	3	3	1	0.7	-60.2
<i>Family</i>	9865	10129	10197	2.7	0.7	7923	7762	8058	-2.0	3.8
Total	22850	25081	25645	9.8	2.2	15561	15678	16367	0.7	4.4
Bullock Labour										
<i>Hired</i>	0	0	0	-	-	177	187	182	5.5	-2.4
<i>Owned</i>	0	0	0	-	-	4238	4483	4283	5.8	-4.5
Total	0	0	0	-	-	4415	4669	4465	5.8	-4.4
Machine Labour										
<i>Hired</i>	5495	5581	5708	1.6	2.3	10686	10794	10956	1.0	1.5
<i>Owned</i>	129	239	252	84.3	5.8	256	273	279	6.9	2.1
Total	5624	5820	5960	3.5	2.4	10942	11067	11235	1.1	1.5
Seed	2389	2571	2705	7.6	5.2	2980	2972	3074	-0.3	3.4
Fertilisers and Manure										
<i>Fertilisers</i>	3681	3940	4194	7.0	6.4	3787	3802	4001	0.4	5.2
<i>Manure</i>	272	282	285	3.8	0.9	843	892	921	5.7	3.3
Total	3953	4222	4479	6.8	6.1	4631	4694	4922	1.4	4.9
Other Inputs										
Insecticides	167	162	154	-3.4	-4.7	1568	1524	1584	-2.8	4.0
Irrigation charges	1491	1991	2098	33.5	5.4	245	241	251	-1.7	3.8
Crop Insurance	0	0	0	-	-	514	511	0	-0.6	-
Interest on working capital	832	929	964	11.7	3.8	1032	1053	1060	2.0	0.7
Miscellaneous	6	5	5	-8.9	-2.4	86	92	90	6.5	-1.9
Fixed Cost	16998	17773	17944	4.6	1.0	19527	20122	21900	3.0	8.8
Rental value of owned land	14096	14964	15030	6.2	0.4	17200	17780	19472	3.4	9.5
Rent paid for leased-in land	171	179	186	4.6	4.0	0	0	0	-	-
Land revenue, cesses & taxes	160	175	159	9.2	-8.7	7	6	6	-5.6	-1.6
Depreciation on implements & Farm buildings	474	442	466	-6.8	5.4	694	737	772	6.3	4.8
Interest on fixed capital	2097	2014	2102	-4.0	4.4	1627	1598	1649	-1.8	3.2
Yield(Qtl/ha)	28.58	30.17	32.22	5.6	6.8	41.86	42.52	46.67	1.6	9.8

Source: ES&E Division, Ministry of Agriculture & Farmers Welfare



Annex Table 5.6(a): Paddy : Break-up of Cost of Cultivation

(₹/ha)

Cost Items	Gujarat					Haryana				
	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23
Operational Cost	53165	59916	60825	12.7	1.5	57838	59184	60407	2.3	2.1
Human Labour										
<i>Casual</i>	16697	16726	14711	0.2	-12.1	13974	14107	14915	1.0	5.7
<i>Attached</i>	230	111	0	-51.8	-	726	727	686	0.1	-5.6
<i>Family</i>	7822	10209	9950	30.5	-2.5	8591	8684	9486	1.1	9.2
Total	24749	27046	24661	9.3	-8.8	23291	23517	25086	1.0	6.7
Bullock Labour										
<i>Hired</i>	7	28	20	305.4	-30.3	0	0	0	-	-
<i>Owned</i>	44	35	45	-20.1	26.9	0	0	0	-	-
Total	51	63	64	24.1	1.6	0	0	0	-	-
Machine Labour										
<i>Hired</i>	5864	6986	9795	19.1	40.2	5899	6073	6292	3.0	3.6
<i>Owned</i>	2327	3462	3453	48.8	-0.3	3115	3169	3286	1.7	3.7
Total	8190	10448	13248	27.6	26.8	9014	9242	9578	2.5	3.6
Seed	6012	6554	6199	9.0	-5.4	1503	1591	1657	5.8	4.2
Fertilisers and Manure										
<i>Fertilisers</i>	5309	5891	6041	11.0	2.5	5665	6024	6116	6.3	1.5
<i>Manure</i>	3911	4613	4990	17.9	8.2	67	70	73	4.6	5.0
Total	9220	10503	11030	13.9	5.0	5732	6094	6189	6.3	1.6
Other Inputs										
Insecticides	1107	1149	1145	3.7	-0.3	6960	7044	6668	1.2	-5.3
Irrigation charges	2051	2270	2625	10.7	15.6	9623	9944	9453	3.3	-4.9
Crop Insurance	0	0	0	-	-	74	76	79	2.5	3.9
Interest on working capital	1374	1506	1542	9.6	2.3	1492	1530	1543	2.5	0.8
Miscellaneous	412	377	312	-8.3	-17.3	150	146	154	-2.2	5.3
Fixed Cost	16241	18567	19808	14.3	6.7	44291	43677	55049	-1.4	26.0
Rental value of owned land	12478	14159	15242	13.5	7.6	36701	35658	47286	-2.8	32.6
Rent paid for leased-in land	362	399	404	10.2	1.3	1275	1334	1421	4.6	6.5
Land revenue, cesses & taxes	10	11	9	10.4	-11.6	0	0	0	-	-
Depreciation on implements & Farm buildings	305	320	359	4.9	12.2	910	955	904	4.9	-5.4
Interest on fixed capital	3086	3678	3794	19.2	3.2	5405	5730	5438	6.0	-5.1
Yield(Qtl/ha)	37.00	41.24	40.18	11.5	-2.6	52.87	53.22	56.18	0.7	5.6

Source: ES&E Division, Ministry of Agriculture & Farmers Welfare



Annex Table 5.6(a): Paddy : Break-up of Cost of Cultivation

(₹/ha)

Cost Items	Himachal Pradesh					Jharkhand				
	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23
Operational Cost	44320	46168	48463	4.2	5.0	44836	43012	43803	-4.1	1.8
Human Labour										
<i>Casual</i>	3303	4076	2902	23.4	-28.8	11372	12541	12405	10.3	-1.1
<i>Attached</i>	20	20	26	-2.7	30.4	0	0	0	-	-
<i>Family</i>	24121	23961	27792	-0.7	16.0	13518	10860	11410	-19.7	5.1
Total	27444	28057	30720	2.2	9.5	24890	23401	23815	-6.0	1.8
Bullock Labour										
<i>Hired</i>	64	180	31	183.4	-83.0	10	3	4	-67.8	4.5
<i>Owned</i>	1147	849	1178	-25.9	38.7	4243	3560	3484	-16.1	-2.1
Total	1210	1030	1209	-14.9	17.4	4253	3563	3488	-16.2	-2.1
Machine Labour										
<i>Hired</i>	5463	6574	5270	20.3	-19.8	3927	4649	4891	18.4	5.2
<i>Owned</i>	2393	2410	3878	0.7	60.9	65	48	47	-26.3	-1.9
Total	7856	8984	9148	14.4	1.8	3992	4697	4938	17.7	5.1
Seed	3317	3633	3858	9.5	6.2	2885	3167	3378	9.8	6.6
Fertilisers and Manure										
<i>Fertilisers</i>	1204	1286	914	6.8	-28.9	4088	3991	4026	-2.4	0.9
<i>Manure</i>	919	610	693	-33.6	13.5	1175	785	843	-33.2	7.4
Total	2122	1896	1607	-10.7	-15.2	5263	4776	4869	-9.3	2.0
Other Inputs										
Insecticides	1216	1189	603	-2.2	-49.3	0	0	0	-	-
Irrigation charges	516	664	648	28.5	-2.4	2604	2433	2334	-6.6	-4.1
Crop Insurance	0	0	0	-	-	0	0	0	-	-
Interest on working capital	612	673	626	9.9	-6.9	949	974	982	2.7	0.7
Miscellaneous	26	43	44	67.4	1.5	0	0	0	-	-
Fixed Cost	23534	23739	18781	0.9	-20.9	17580	19355	13328	10.1	-31.1
Rental value of owned land	19661	20065	14911	2.1	-25.7	14766	16643	10640	12.7	-36.1
Rent paid for leased-in land	111	60	26	-46.4	-56.1	0	0	198	-	-
Land revenue, cesses & taxes	9	9	8	1.3	-12.6	135	126	108	-6.8	-14.2
Depreciation on implements & Farm buildings	340	362	383	6.4	5.7	514	537	506	4.5	-5.9
Interest on fixed capital	3413	3244	3452	-4.9	6.4	2165	2049	1876	-5.4	-8.4
Yield(Qtl/ha)	30.92	29.94	24.29	-3.2	-18.9	27.35	27.74	26.05	1.4	-6.1

Source: ES&E Division, Ministry of Agriculture & Farmers Welfare



Annex Table 5.6(a): Paddy : Break-up of Cost of Cultivation

(₹/ha)

Cost Items	Karnataka					Kerala				
	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23
Operational Cost	70806	79020	79261	11.6	0.3	78127	81905	78959	4.8	-3.6
Human Labour										
<i>Casual</i>	13299	16069	16701	20.8	3.9	30953	29909	28969	-3.4	-3.1
<i>Attached</i>	216	100	0	-53.7	-	0	0	122	-	-
<i>Family</i>	12260	17178	15791	40.1	-8.1	11095	12892	11408	16.2	-11.5
Total	25776	33347	32492	29.4	-2.6	42047	42801	40499	1.8	-5.4
Bullock Labour										
<i>Hired</i>	60	78	246	29.7	213.4	19	20	0	4.5	-
<i>Owned</i>	3507	3504	6966	-0.1	98.8	0	0	0	-	-
Total	3568	3582	7212	0.4	101.3	19	20	0	4.5	-
Machine Labour										
<i>Hired</i>	17125	15916	15493	-7.1	-2.7	17499	18245	18336	4.3	0.5
<i>Owned</i>	1693	637	1388	-62.4	118.1	113	83	60	-26.3	-27.8
Total	18818	16553	16881	-12.0	2.0	17611	18328	18396	4.1	0.4
Seed	2404	2508	2712	4.3	8.1	3812	3869	3866	1.5	-0.1
Fertilisers and Manure										
<i>Fertilisers</i>	14883	17284	12406	16.1	-28.2	7063	8385	7687	18.7	-8.3
<i>Manure</i>	175	191	686	8.8	259.5	1342	1674	1576	24.7	-5.8
Total	15058	17475	13092	16.0	-25.1	8406	10059	9264	19.7	-7.9
Other Inputs										
Insecticides	2399	2676	3603	11.5	34.7	2005	2258	2698	12.6	19.5
Irrigation charges	429	411	811	-4.2	97.2	204	228	226	11.7	-1.1
Crop Insurance	7	0	0	-	-	71	90	88	27.1	-3.2
Interest on working capital	1774	1874	1923	5.6	2.6	2031	2091	2047	3.0	-2.1
Miscellaneous	573	594	536	3.7	-9.8	1919	2160	1876	12.6	-13.2
Fixed Cost	23254	27372	29661	17.7	8.4	22783	25814	24564	13.3	-4.8
Rental value of owned land	21501	25829	28101	20.1	8.8	21200	24514	23098	15.6	-5.8
Rent paid for leased-in land	0	0	368	-	-	420	101	5	-75.8	-95.4
Land revenue, cesses & taxes	14	17	20	18.7	16.7	514	578	773	12.6	33.7
Depreciation on implements & Farm buildings	366	330	281	-9.9	-14.9	288	292	342	1.4	17.3
Interest on fixed capital	1373	1197	891	-12.8	-25.5	361	329	346	-9.0	5.1
Yield(Qtl/ha)	48.00	48.00	51.05	0.0	6.4	36.23	39.65	36.35	9.4	-8.3

Source: ES&E Division, Ministry of Agriculture & Farmers Welfare



Annex Table 5.6(a): Paddy : Break-up of Cost of Cultivation

(₹/ha)

Cost Items	Madhya Pradesh					Maharashtra				
	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23
Operational Cost	44323	45623	46314	2.9	1.5	81228	83995	86704	3.4	3.2
Human Labour										
<i>Casual</i>	8021	8151	8135	1.6	-0.2	25126	27789	25279	10.6	-9.0
<i>Attached</i>	102	104	36	2.0	-65.4	0	0	103	-	-
<i>Family</i>	9742	10153	10201	4.2	0.5	14147	15414	21398	9.0	38.8
Total	17865	18408	18372	3.0	-0.2	39273	43204	46780	10.0	8.3
Bullock Labour										
<i>Hired</i>	260	268	280	3.1	4.4	482	1525	3084	216.6	102.2
<i>Owned</i>	1848	1854	1865	0.3	0.6	6731	5600	3501	-16.8	-37.5
Total	2108	2122	2145	0.7	1.1	7213	7125	6584	-1.2	-7.6
Machine Labour										
<i>Hired</i>	9586	9713	10110	1.3	4.1	13187	10918	13670	-17.2	25.2
<i>Owned</i>	603	629	643	4.4	2.2	2360	2561	1945	8.5	-24.1
Total	10189	10342	10754	1.5	4.0	15547	13479	15614	-13.3	15.8
Seed	3083	3151	3299	2.2	4.7	3760	4033	4032	7.3	0.0
Fertilisers and Manure										
<i>Fertilisers</i>	3819	4069	4186	6.5	2.9	8028	8730	6791	8.7	-22.2
<i>Manure</i>	1936	1969	1944	1.7	-1.2	3018	3105	2647	2.9	-14.8
Total	5755	6037	6130	4.9	1.5	11046	11835	9438	7.1	-20.3
<i>Other Inputs</i>										
Insecticides	2282	2383	2482	4.4	4.1	1158	1295	1166	11.8	-9.9
Irrigation charges	1468	1571	1507	7.0	-4.0	163	160	289	-1.7	80.3
Crop Insurance	261	266	258	1.8	-3.1	170	0	0	-	-
Interest on working capital	1048	1075	1094	2.6	1.8	2033	2078	1979	2.2	-4.8
Miscellaneous	263	267	273	1.7	2.0	866	786	822	-9.3	4.6
Fixed Cost	19385	20430	21093	5.4	3.2	12121	12272	15294	1.2	24.6
Rental value of owned land	16028	17004	17491	6.1	2.9	9277	9669	12126	4.2	25.4
Rent paid for leased-in land	0	0	0	-	-	0	0	0	-	-
Land revenue, cesses & taxes	3	3	3	6.9	-4.5	34	32	29	-3.9	-9.8
Depreciation on implements & Farm buildings	617	632	671	2.3	6.3	452	472	549	4.6	16.2
Interest on fixed capital	2737	2792	2928	2.0	4.9	2358	2098	2590	-11.1	23.5
Yield(Qtl/ha)	44.87	44.11	40.37	-1.7	-8.5	20.55	21.73	20.92	5.8	-3.7

Source: ES&E Division, Ministry of Agriculture & Farmers Welfare



Annex Table 5.6(a): Paddy : Break-up of Cost of Cultivation

(₹/ha)

Cost Items	Odisha					Punjab				
	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23
Operational Cost	54635	59339	59520	8.6	0.3	48255	49562	50667	2.7	2.2
Human Labour										
<i>Casual</i>	14342	15204	15869	6.0	4.4	9347	10070	10464	7.7	3.9
<i>Attached</i>	9	9	15	-0.2	57.8	1624	1653	1664	1.8	0.7
<i>Family</i>	17425	18373	18887	5.4	2.8	5840	6162	6104	5.5	-0.9
Total	31776	33586	34770	5.7	3.5	16811	17885	18232	6.4	1.9
Bullock Labour										
<i>Hired</i>	348	346	286	-0.6	-17.1	0	0	0	-	-
<i>Owned</i>	2222	1891	1911	-14.9	1.1	4	5	5	1.1	4.2
Total	2569	2236	2198	-13.0	-1.7	5	5	5	-7.4	4.2
Machine Labour										
<i>Hired</i>	8636	9713	9340	12.5	-3.8	5858	6024	6233	2.8	3.5
<i>Owned</i>	299	479	976	60.0	103.8	5604	5789	6073	3.3	4.9
Total	8935	10192	10316	14.1	1.2	11462	11814	12306	3.1	4.2
Seed	1687	1801	1881	6.7	4.5	1755	1830	1869	4.3	2.2
Fertilisers and Manure										
<i>Fertilisers</i>	4983	6387	5432	28.2	-15.0	4052	4081	4085	0.7	0.1
<i>Manure</i>	2287	2454	2547	7.3	3.8	399	413	407	3.7	-1.5
Total	7270	8841	7979	21.6	-9.7	4451	4494	4492	1.0	0.0
Other Inputs										
Insecticides	1026	1175	870	14.5	-26.0	5841	5914	5985	1.2	1.2
Irrigation charges	167	194	192	15.9	-1.1	6479	6138	6264	-5.3	2.1
Crop Insurance	8	7	6	-11.0	-13.1	0	0	0	-	-
Interest on working capital	1128	1241	1231	10.1	-0.8	1285	1315	1350	2.3	2.7
Miscellaneous	69	66	76	-4.4	15.7	166	169	164	1.4	-2.9
Fixed Cost	16656	19329	20710	16.0	7.1	53089	56230	57100	5.9	1.5
Rental value of owned land	14784	17499	18706	18.4	6.9	37673	40571	40456	7.7	-0.3
Rent paid for leased-in land	618	748	638	21.1	-14.8	8962	9043	9632	0.9	6.5
Land revenue, cesses & taxes	21	20	21	-2.3	3.2	0	0	0	-	-
Depreciation on implements & Farm buildings	355	341	407	-3.9	19.2	1566	1612	1695	2.9	5.2
Interest on fixed capital	877	720	938	-17.9	30.3	4889	5004	5316	2.4	6.2
Yield(Qtl/ha)	40.48	44.86	42.47	10.8	-5.3	69.80	70.60	72.91	1.1	3.3

Source: ES&E Division, Ministry of Agriculture & Farmers Welfare



Annex Table 5.6(a): Paddy : Break-up of Cost of Cultivation

(₹/ha)

Cost Items	Tamil Nadu					Telangana				
	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23
Operational Cost	66178	68912	72731	4.1	5.5	77384	76822	75258	-0.7	-2.0
Human Labour										
<i>Casual</i>	14866	16570	17453	11.5	5.3	17923	17514	16275	-2.3	-7.1
<i>Attached</i>	58	52	42	-10.7	-19.2	332	368	249	10.6	-32.4
<i>Family</i>	7128	6627	7492	-7.0	13.1	16646	14496	15127	-12.9	4.4
Total	22052	23249	24987	5.4	7.5	34901	32377	31650	-7.2	-2.2
Bullock Labour										
<i>Hired</i>	86	84	120	-1.9	41.7	173	120	145	-30.4	20.2
<i>Owned</i>	2	0	5	-	-	207	137	196	-33.9	43.2
Total	88	84	124	-4.2	47.1	380	258	341	-32.3	32.4
Machine Labour										
<i>Hired</i>	16308	16223	16867	-0.5	4.0	19348	19389	20883	0.2	7.7
<i>Owned</i>	1943	1813	2387	-6.7	31.6	2177	2622	1647	20.4	-37.2
Total	18252	18037	19254	-1.2	6.8	21525	22011	22531	2.3	2.4
Seed	8225	8471	8216	3.0	-3.0	2702	2765	3096	2.4	11.9
Fertilisers and Manure										
<i>Fertilisers</i>	7746	8651	9640	11.7	11.4	8817	9667	8336	9.6	-13.8
<i>Manure</i>	508	328	454	-35.5	38.7	81	103	109	26.1	6.1
Total	8254	8978	10094	8.8	12.4	8898	9770	8445	9.8	-13.6
<i>Other Inputs</i>										
Insecticides	2165	2304	2835	6.4	23.0	5248	5461	4937	4.1	-9.6
Irrigation charges	5016	5572	4931	11.1	-11.5	1694	2112	2317	24.6	9.7
Crop Insurance	58	69	27	19.5	-60.8	0	0	0	-	-
Interest on working capital	1789	1887	1977	5.5	4.7	1841	1889	1822	2.6	-3.5
Miscellaneous	280	260	286	-7.0	9.8	195	180	120	-7.8	-33.1
Fixed Cost	22202	24889	24752	12.1	-0.5	35883	34009	38247	-5.2	12.5
Rental value of owned land	14745	17538	18443	18.9	5.2	31880	29329	30464	-8.0	3.9
Rent paid for leased-in land	163	149	261	-8.2	74.9	1747	2475	5320	41.7	114.9
Land revenue, cesses & taxes	7	7	7	0.5	0.0	0	0	0	-	-
Depreciation on implements & Farm buildings	717	771	746	7.5	-3.2	281	306	351	8.9	14.7
Interest on fixed capital	6571	6424	5295	-2.2	-17.6	1974	1898	2112	-3.8	11.3
Yield(Qtl/ha)	45.25	47.73	45.63	5.5	-4.4	54.56	52.15	54.77	-4.4	5.0

Source: ES&E Division, Ministry of Agriculture & Farmers Welfare



Annex Table 5.6(a): Paddy : Break-up of Cost of Cultivation

(₹/ha)

Cost Items	Uttar Pradesh					West Bengal				
	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23
Operational Cost	54670	57107	60056	4.5	5.2	71099	76818	73839	8.0	-3.9
Human Labour										
<i>Casual</i>	13588	13666	15466	0.6	13.2	19939	20811	23963	4.4	15.1
<i>Attached</i>	9	3	10	-64.0	201.8	19	1	0	-92.7	-
<i>Family</i>	11731	12116	11927	3.3	-1.6	25582	26177	17118	2.3	-34.6
Total	25329	25785	27403	1.8	6.3	45539	46989	41081	3.2	-12.6
Bullock Labour										
<i>Hired</i>	0	0	0	-	-	518	410	70	-20.9	-82.9
<i>Owned</i>	14	15	16	6.2	11.7	921	1101	518	19.6	-53.0
Total	14	15	16	6.2	11.7	1439	1511	588	5.0	-61.1
Machine Labour										
<i>Hired</i>	8132	8345	8883	2.6	6.5	8450	10382	12411	22.9	19.5
<i>Owned</i>	819	914	1532	11.6	67.6	38	200	25	423.4	-87.3
Total	8950	9258	10415	3.4	12.5	8489	10582	12436	24.7	17.5
Seed	4842	5498	5709	13.6	3.8	2641	2985	3321	13.0	11.3
Fertilisers and Manure										
<i>Fertilisers</i>	4597	4366	5157	-5.0	18.1	6124	6792	7188	10.9	5.8
<i>Manure</i>	125	203	200	61.8	-1.3	1619	1436	1335	-11.3	-7.0
Total	4722	4569	5357	-3.3	17.3	7743	8228	8523	6.3	3.6
Other Inputs										
Insecticides	972	970	803	-0.2	-17.3	2026	2179	3650	7.5	67.5
Irrigation charges	8471	9584	8823	13.1	-7.9	1776	2712	2430	52.6	-10.4
Crop Insurance	0	0	0	-	-	0	0	0	-	-
Interest on working capital	1301	1363	1458	4.8	7.0	1379	1535	1719	11.3	12.0
Miscellaneous	69	64	71	-7.8	10.5	66	97	93	48.0	-4.9
Fixed Cost	19987	20848	27194	4.3	30.4	19348	23821	25885	23.1	8.7
Rental value of owned land	15780	16451	22105	4.3	34.4	18302	22752	24632	24.3	8.3
Rent paid for leased-in land	438	415	445	-5.3	7.2	119	43	336	-64.3	686.4
Land revenue, cesses & taxes	2	1	2	-5.8	60.0	0	0	0	-	-
Depreciation on implements & Farm buildings	700	765	842	9.4	10.1	392	414	324	5.6	-21.8
Interest on fixed capital	3068	3215	3799	4.8	18.2	536	612	593	14.3	-3.1
Yield(Qtl/ha)	36.15	32.91	40.45	-9.0	22.9	41.86	45.02	45.89	7.5	1.9

Source: ES&E Division, Ministry of Agriculture & Farmers Welfare



Annex Table 5.6(b): Jowar : Break-up of Cost of Cultivation

(₹/ha)

Cost Items	Andhra Pradesh					Karnataka				
	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23
Operational Cost	55491	61713	56512	11.2	-8.4	31312	35605	37619	13.7	5.7
Human Labour										
<i>Casual</i>	22341	26252	22633	17.5	-13.8	8453	9914	9248	17.3	-6.7
<i>Attached</i>	0	0	0	-	-	123	145	35	17.1	-76.0
<i>Family</i>	2456	1192	1565	-51.5	31.2	8158	9282	10682	13.8	15.1
Total	24797	27445	24198	10.7	-11.8	16734	19341	19965	15.6	3.2
Bullock Labour										
<i>Hired</i>	1238	1233	1064	-0.4	-13.7	1925	1185	2242	-38.4	89.2
<i>Owned</i>	0	0	123	-	-	1536	2606	1793	69.6	-31.2
Total	1238	1233	1187	-0.4	-3.7	3461	3791	4035	9.5	6.4
Machine Labour										
<i>Hired</i>	10705	11406	9228	6.5	-19.1	4714	5773	6549	22.5	13.4
<i>Owned</i>	83	246	2501	196.5	917.6	1184	1308	1509	10.5	15.3
Total	10788	11652	11729	8.0	0.7	5897	7082	8057	20.1	13.8
Seed	3178	3504	3282	10.3	-6.3	770	852	800	10.7	-6.1
Fertilisers and Manure										
<i>Fertilisers</i>	6013	6674	7278	11.0	9.0	2931	2942	2747	0.4	-6.6
<i>Manure</i>	0	0	0	-	-	39	0	137	-	-
Total	6013	6674	7278	11.0	9.0	2970	2942	2884	-1.0	-2.0
<i>Other Inputs</i>										
Insecticides	5687	7043	4973	23.9	-29.4	453	441	426	-2.6	-3.5
Irrigation charges	1982	2066	2044	4.3	-1.1	235	262	503	11.7	91.6
Crop Insurance	0	0	0	-	-	0	0	2	-	-
Interest on working capital	1607	1834	1665	14.1	-9.2	702	798	816	13.7	2.3
Miscellaneous	200	262	155	30.8	-41.0	90	95	131	5.9	37.4
Fixed Cost	42908	50236	34814	17.1	-30.7	10603	12352	14714	16.5	19.1
Rental value of owned land	42156	48889	28221	16.0	-42.3	8203	10193	12687	24.3	24.5
Rent paid for leased-in land	0	580	5691	-	880.8	0	0	156	-	-
Land revenue, cesses & taxes	0	0	0	-	-	5	5	6	-4.9	20.3
Depreciation on implements & Farm buildings	76	80	89	5.2	10.9	314	319	374	1.7	17.3
Interest on fixed capital	676	687	813	1.6	18.4	2081	1835	1490	-11.8	-18.8
Yield(Qtl/ha)	47.27	52.16	48.83	10.3	-6.4	11.08	11.42	10.62	3.1	-7.0

Source: ES&E Division, Ministry of Agriculture & Farmers Welfare



Annex Table 5.6(b): Jowar : Break-up of Cost of Cultivation

(₹/ha)

Cost Items	Maharashtra					Rajasthan				
	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23
Operational Cost	40159	41148	43170	2.5	4.9	37256	37080	34981	-0.5	-5.7
Human Labour										
<i>Casual</i>	10744	13045	13425	21.4	2.9	4506	5506	3277	22.2	-40.5
<i>Attached</i>	71	235	512	230.2	117.8	0	0	0	-	-
<i>Family</i>	7525	7031	8726	-6.6	24.1	22695	19903	18751	-12.3	-5.8
Total	18340	20311	22663	10.7	11.6	27201	25409	22029	-6.6	-13.3
Bullock Labour										
<i>Hired</i>	1468	630	821	-57.1	30.3	80	0	0	-	-
<i>Owned</i>	709	1951	2135	175.1	9.4	152	220	0	45.0	-
Total	2178	2581	2956	18.5	14.5	232	220	0	-4.9	-
Machine Labour										
<i>Hired</i>	12731	9920	7746	-22.1	-21.9	4424	3646	7963	-17.6	118.4
<i>Owned</i>	578	2473	3352	327.8	35.5	2052	4164	1470	103.0	-64.7
Total	13309	12393	11098	-6.9	-10.4	6476	7810	9433	20.6	20.8
Seed	1455	1303	1251	-10.5	-4.0	1525	1692	1799	11.0	6.3
Fertilisers and Manure										
<i>Fertilisers</i>	1582	1789	2228	13.0	24.5	1351	1345	1202	-0.5	-10.6
<i>Manure</i>	27	28	33	5.3	16.8	13	15	0	17.9	-
Total	1609	1817	2260	12.9	24.4	1364	1360	1202	-0.3	-11.6
Other Inputs										
Insecticides	15	16	16	4.4	2.1	0	41	0	-	-
Irrigation charges	2186	1658	1843	-24.1	11.2	0	1	0	-	-
Crop Insurance	47	0	0	-	-	0	0	0	-	-
Interest on working capital	989	1034	1044	4.5	1.0	441	521	492	18.0	-5.5
Miscellaneous	32	36	39	14.4	6.6	17	26	27	52.2	6.4
Fixed Cost	11378	11626	12520	2.2	7.7	13843	15470	14702	11.8	-5.0
Rental value of owned land	7170	7926	8361	10.5	5.5	5793	6604	7264	14.0	10.0
Rent paid for leased-in land	0	0	0	-	-	0	0	0	-	-
Land revenue, cesses & taxes	32	29	25	-9.9	-12.8	0	0	0	-	-
Depreciation on implements & Farm buildings	392	408	470	4.0	15.0	507	464	476	-8.4	2.5
Interest on fixed capital	3784	3263	3664	-13.8	12.3	7543	8402	6961	11.4	-17.1
Yield(Qtl/ha)	10.76	11.63	11.28	8.1	-3.0	11.29	10.94	9.49	-3.1	-13.3

Source: ES&E Division, Ministry of Agriculture and Farmers Welfare



Annex Table 5.6(b): Jowar : Break-up of Cost of Cultivation

(₹/ha)

Cost Items	Tamil Nadu				
	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23
Operational Cost	46051	50247	54932	9.1	9.3
Human Labour					
<i>Casual</i>	22232	24522	28072	10.3	14.5
<i>Attached</i>	0	0	0	-	-
<i>Family</i>	9503	8623	10029	-9.3	16.3
Total	31735	33146	38102	4.4	15.0
Bullock Labour					
<i>Hired</i>	0	0	0	-	-
<i>Owned</i>	0	0	0	-	-
Total	0	0	0	-	-
Machine Labour					
<i>Hired</i>	5923	6331	6388	6.9	0.9
<i>Owned</i>	0	0	638	-	-
Total	5923	6331	7026	6.9	11.0
Seed	1081	983	1766	-9.0	79.6
Fertilisers and Manure					
<i>Fertilisers</i>	1936	2523	1737	30.3	-31.1
<i>Manure</i>	3596	5179	3887	44.0	-24.9
Total	5531	7702	5624	39.2	-27.0
Other Inputs					
Insecticides	169	259	376	53.7	45.0
Irrigation charges	420	493	597	17.5	21.0
Crop Insurance	17	0	0	-	-
Interest on working capital	1108	1261	1361	13.9	7.9
Miscellaneous	66	71	80	8.2	12.6
Fixed Cost	8868	6834	16192	-22.9	136.9
Rental value of owned land	7706	5628	14855	-27.0	164.0
Rent paid for leased-in land	0	0	0	-	-
Land revenue, cesses & taxes	11	12	11	3.1	-5.6
Depreciation on implements & Farm buildings	293	297	334	1.3	12.5
Interest on fixed capital	857	898	992	4.7	10.4
Yield(Qtl/ha)	11.72	27.96	27.23	138.6	-2.6

Source: ES&E Division, Ministry of Agriculture and Farmers Welfare



Annex Table 5.6(c): Bajra : Break-up of Cost of Cultivation

(₹/ha)

Cost Items	Gujarat					Haryana				
	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23
Operational Cost	51553	56138	58464	8.9	4.1	30606	31374	31939	2.5	1.8
Human Labour										
<i>Casual</i>	14360	17372	18873	21.0	8.6	5646	5487	5904	-2.8	7.6
<i>Attached</i>	6	4	0	-21.7	-	0	0	5	-	-
<i>Family</i>	13594	13670	13553	0.6	-0.9	10518	10777	10262	2.5	-4.8
Total	27959	31046	32426	11.0	4.4	16163	16264	16170	0.6	-0.6
Bullock Labour										
<i>Hired</i>	608	410	406	-32.6	-0.9	0	0	0	-	-
<i>Owned</i>	120	39	55	-67.7	42.9	4	0	0	-	-
Total	728	449	462	-38.3	2.8	4	0	0	-	-
Machine Labour										
<i>Hired</i>	7280	7289	7166	0.1	-1.7	7172	7519	7857	4.8	4.5
<i>Owned</i>	2264	3880	4727	71.4	21.8	1719	1832	1971	6.6	7.6
Total	9544	11169	11892	17.0	6.5	8890	9350	9827	5.2	5.1
Seed	2613	2567	2805	-1.8	9.2	996	1041	1107	4.4	6.3
Fertilisers and Manure										
<i>Fertilisers</i>	3852	3991	3351	3.6	-16.0	2433	2560	2701	5.2	5.5
<i>Manure</i>	720	640	662	-11.2	3.5	3	0	0	-	-
Total	4572	4631	4013	1.3	-13.3	2436	2560	2701	5.1	5.5
Other Inputs										
Insecticides	177	170	257	-3.9	51.3	177	184	195	4.0	5.6
Irrigation charges	4623	4639	5103	0.3	10.0	1310	1329	1259	1.5	-5.2
Crop Insurance	0	0	0	-	-	0	1	0	-	-
Interest on working capital	1150	1287	1361	11.9	5.8	609	624	657	2.5	5.2
Miscellaneous	186	180	146	-3.2	-19.3	20	21	24	2.2	12.3
Fixed Cost	16744	19364	21624	15.6	11.7	12945	13693	17653	5.8	28.9
Rental value of owned land	13662	15908	17872	16.4	12.3	10152	10893	14603	7.3	34.1
Rent paid for leased-in land	0	0	39	-	-	161	0	124	-	-
Land revenue, cesses & taxes	4	3	4	-1.1	9.2	0	0	0	-	-
Depreciation on implements & Farm buildings	231	269	291	16.2	8.3	356	375	389	5.3	3.7
Interest on fixed capital	2847	3184	3418	11.8	7.4	2276	2425	2537	6.5	4.6
Yield(Qtl/ha)	27.73	30.50	32.81	10.0	7.6	14.34	14.62	15.34	2.0	4.9

Source: ES&E Division, Ministry of Agriculture and Farmers Welfare



Annex Table 5.6(c): Bajra : Break-up of Cost of Cultivation

(₹/ha)

Cost Items	Maharashtra					Rajasthan				
	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23
Operational Cost	58926	60485	61931	2.6	2.4	31618	31801	35468	0.6	11.5
Human Labour										
<i>Casual</i>	13856	15253	13907	10.1	-8.8	4739	5591	6147	18.0	9.9
<i>Attached</i>	874	146	82	-83.3	-44.1	3	4	24	33.7	519.7
<i>Family</i>	16118	14797	16128	-8.2	9.0	15667	14460	16078	-7.7	11.2
Total	30848	30196	30117	-2.1	-0.3	20409	20055	22248	-1.7	10.9
Bullock Labour										
<i>Hired</i>	835	890	712	6.5	-20.0	0	0	0	-	-
<i>Owned</i>	889	562	762	-36.7	35.5	9	6	0	-31.3	-
Total	1724	1452	1474	-15.8	1.5	9	6	0	-31.3	-
Machine Labour										
<i>Hired</i>	14151	17160	14502	21.3	-15.5	6274	5923	6141	-5.6	3.7
<i>Owned</i>	2909	3425	7206	17.7	110.4	1205	1565	2815	29.9	79.9
Total	17060	20585	21707	20.7	5.5	7479	7488	8956	0.1	19.6
Seed	1313	1426	1556	8.6	9.1	1199	1706	1722	42.3	0.9
Fertilisers and Manure										
<i>Fertilisers</i>	3969	2687	3245	-32.3	20.8	733	922	938	25.9	1.6
<i>Manure</i>	355	450	444	26.9	-1.4	352	744	750	111.7	0.7
Total	4324	3137	3689	-27.4	17.6	1084	1667	1687	53.7	1.2
Other Inputs										
Insecticides	35	37	39	6.9	6.0	62	93	54	48.1	-42.2
Irrigation charges	2278	2213	1884	-2.8	-14.9	863	233	182	-73.0	-21.6
Crop Insurance	0	0	21	-	-	0	0	0	-	-
Interest on working capital	1297	1385	1388	6.7	0.3	483	525	588	8.7	11.8
Miscellaneous	48	54	56	12.2	3.9	28	27	31	-1.6	11.9
Fixed Cost	14000	15059	15525	7.6	3.1	7926	8988	9080	13.4	1.0
Rental value of owned land	9995	11140	11212	11.5	0.6	5260	6757	6767	28.5	0.2
Rent paid for leased-in land	0	0	0	-	-	51	187	0	265.5	-
Land revenue, cesses & taxes	29	33	30	15.4	-8.2	1	0	0	-	-
Depreciation on implements & Farm buildings	372	379	425	2.1	11.9	394	365	405	-7.4	11.1
Interest on fixed capital	3605	3506	3859	-2.7	10.1	2221	1680	1908	-24.4	13.6
Yield(Qtl/ha)	26.34	24.06	21.82	-8.7	-9.3	10.86	12.62	11.12	16.2	-11.9

Source: ES&E Division, Ministry of Agriculture and Farmers Welfare



Annex Table 5.6(c): Bajra : Break-up of Cost of Cultivation

(₹/ha)

Cost Items	Uttar Pradesh				
	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23
Operational Cost	29008	32287	35741	11.3	10.7
Human Labour					
<i>Casual</i>	8071	7845	8944	-2.8	14.0
<i>Attached</i>	0	153	0	-	-
<i>Family</i>	9964	13098	13890	31.4	6.0
Total	18035	21096	22834	17.0	8.2
Bullock Labour					
<i>Hired</i>	0	0	0	-	-
<i>Owned</i>	20	0	0	-	-
Total	20	0	0	-	-
Machine Labour					
<i>Hired</i>	6620	5904	7192	-10.8	21.8
<i>Owned</i>	366	663	639	81.1	-3.6
Total	6986	6567	7832	-6.0	19.3
Seed	1641	1686	1764	2.8	4.6
Fertilisers and Manure					
<i>Fertilisers</i>	956	1225	1625	28.2	32.6
<i>Manure</i>	0	0	0	-	-
Total	956	1225	1625	28.2	32.6
<i>Other Inputs</i>					
Insecticides	54	78	126	45.0	62.0
Irrigation charges	718	1031	884	43.6	-14.3
Crop Insurance	0	0	0	-	-
Interest on working capital	577	581	662	0.8	13.9
Miscellaneous	21	22	14	3.7	-34.8
Fixed Cost	14679	16587	19614	13.0	18.3
Rental value of owned land	11330	13651	15611	20.5	14.4
Rent paid for leased-in land	1557	906	1610	-41.8	77.6
Land revenue, cesses & taxes	3	5	5	47.7	8.8
Depreciation on implements & Farm buildings	353	364	416	3.1	14.4
Interest on fixed capital	1436	1662	1973	15.7	18.7
Yield(Qtl/ha)	18.91	18.84	21.30	-0.4	13.1

Source: ES&E Division, Ministry of Agriculture and Farmers Welfare



Annex Table 5.6(d): Maize : Break-up of Cost of Cultivation

(₹/ha)

Cost Items	Andhra Pradesh					Bihar				
	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23
Operational Cost	63363	67149	66314	6.0	-1.2	44961	48304	49133	7.4	1.7
Human Labour										
<i>Casual</i>	13906	15807	15079	13.7	-4.6	10619	11376	11933	7.1	4.9
<i>Attached</i>	0	0	0	-	-	0	0	0	-	-
<i>Family</i>	6499	5564	6940	-14.4	24.7	8093	8722	8629	7.8	-1.1
Total	20405	21371	22019	4.7	3.0	18712	20098	20561	7.4	2.3
Bullock Labour										
<i>Hired</i>	333	261	345	-21.5	32.2	0	0	0	-	-
<i>Owned</i>	723	118	84	-83.7	-29.0	0	0	0	-	-
Total	1056	379	429	-64.1	13.2	0	0	0	-	-
Machine Labour										
<i>Hired</i>	10535	14318	10273	35.9	-28.3	7829	8095	8318	3.4	2.8
<i>Owned</i>	2302	279	3670	-87.9	1215.6	141	152	158	8.4	3.5
Total	12837	14597	13942	13.7	-4.5	7970	8247	8475	3.5	2.8
Seed	8177	7762	8625	-5.1	11.1	4038	4701	4795	16.4	2.0
Fertilisers and Manure										
<i>Fertilisers</i>	10852	11736	10708	8.1	-8.8	4916	5361	5505	9.0	2.7
<i>Manure</i>	9	8	9	-7.5	14.3	297	159	154	-46.3	-3.5
Total	10860	11744	10717	8.1	-8.7	5213	5520	5659	5.9	2.5
Other Inputs										
Insecticides	5593	6648	5904	18.9	-11.2	8	0	94	-	-
Irrigation charges	2347	2411	2576	2.7	6.9	7903	8538	8321	8.0	-2.5
Crop Insurance	0	0	0	-	-	0	0	0	-	-
Interest on working capital	1723	1866	1799	8.3	-3.6	1117	1199	1227	7.4	2.3
Miscellaneous	365	370	301	1.3	-18.5	0	0	0	-	-
Fixed Cost	39486	41553	39296	5.2	-5.4	19114	22305	19535	16.7	-12.4
Rental value of owned land	37893	40053	26363	5.7	-34.2	17130	20232	17117	18.1	-15.4
Rent paid for leased-in land	0	56	11340	-	20317.8	0	0	271	-	-
Land revenue, cesses & taxes	0	0	0	-	-	189	196	192	4.0	-2.3
Depreciation on implements & Farm buildings	237	213	224	-10.3	5.5	390	402	425	3.2	5.7
Interest on fixed capital	1355	1231	1369	-9.1	11.2	1406	1475	1530	4.9	3.8
Yield(Qtl/ha)	71.90	74.90	70.15	4.2	-6.3	46.07	46.45	51.65	0.8	11.2

Source: ES&E Division, Ministry of Agriculture and Farmers Welfare



Annex Table 5.6(d): Maize : Break-up of Cost of Cultivation

(₹/ha)

Cost Items	Gujarat					Himachal Pradesh				
	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23
Operational Cost	51793	52192	52648	0.8	0.9	40061	40954	37652	2.2	-8.1
Human Labour										
<i>Casual</i>	14257	15154	15207	6.3	0.3	742	827	686	11.4	-17.1
<i>Attached</i>	213	288	0	34.8	-	40	13	12	-68.8	-4.3
<i>Family</i>	11955	11117	11767	-7.0	5.8	22483	22890	21242	1.8	-7.2
Total	26426	26559	26974	0.5	1.6	23265	23730	21940	2.0	-7.5
Bullock Labour										
<i>Hired</i>	1933	1791	1608	-7.3	-10.2	307	397	176	29.4	-55.6
<i>Owned</i>	885	1034	1185	16.9	14.6	3220	4004	1345	24.3	-66.4
Total	2818	2825	2794	0.3	-1.1	3527	4400	1522	24.8	-65.4
Machine Labour										
<i>Hired</i>	7156	5295	6192	-26.0	16.9	5329	4509	4864	-15.4	7.9
<i>Owned</i>	1014	2648	604	161.2	-77.2	417	815	1226	95.6	50.4
Total	8170	7943	6796	-2.8	-14.4	5746	5324	6090	-7.3	14.4
Seed	4266	4547	4651	6.6	2.3	1905	2060	2627	8.1	27.5
Fertilisers and Manure										
<i>Fertilisers</i>	4847	4122	3755	-15.0	-8.9	988	957	1235	-3.2	29.0
<i>Manure</i>	1206	2027	2588	68.0	27.7	3844	3676	3456	-4.4	-6.0
Total	6053	6148	6343	1.6	3.2	4832	4633	4691	-4.1	1.2
Other Inputs										
Insecticides	843	900	1034	6.7	14.9	212	232	250	9.4	7.7
Irrigation charges	1862	1866	2649	0.2	42.0	4	5	5	31.9	3.7
Crop Insurance	0	0	0	-	-	0	0	0	-	-
Interest on working capital	1207	1245	1239	3.1	-0.5	533	547	497	2.8	-9.2
Miscellaneous	148	159	168	7.4	6.1	37	22	30	-39.9	35.2
Fixed Cost	14433	14870	16085	3.0	8.2	14090	15181	15940	7.7	5.0
Rental value of owned land	10218	10482	10717	2.6	2.2	11615	12385	12720	6.6	2.7
Rent paid for leased-in land	105	33	12	-68.3	-63.5	74	0	5	-	-
Land revenue, cesses & taxes	9	8	7	-5.4	-10.9	7	8	8	14.1	12.5
Depreciation on implements & Farm buildings	284	318	379	12.0	19.3	369	439	513	18.9	16.9
Interest on fixed capital	3818	4030	4969	5.5	23.3	2025	2349	2693	16.0	14.6
Yield(Qtl/ha)	22.48	21.66	22.48	-3.7	3.8	19.32	18.60	19.10	-3.7	2.7

Source: ES&E Division, Ministry of Agriculture and Farmers Welfare



Annex Table 5.6(d): Maize : Break-up of Cost of Cultivation

(₹/ha)

Cost Items	Karnataka					Madhya Pradesh				
	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23
Operational Cost	47599	47405	54528	-0.4	15.0	37359	38059	38506	1.9	1.2
Human Labour										
<i>Casual</i>	9668	10587	11673	9.5	10.3	7643	7983	8118	4.4	1.7
<i>Attached</i>	0	0	0	-	-	32	34	34	5.5	1.4
<i>Family</i>	9101	7985	9940	-12.3	24.5	6194	6304	6438	1.8	2.1
Total	18770	18572	21613	-1.1	16.4	13869	14320	14590	3.3	1.9
Bullock Labour										
<i>Hired</i>	1296	1310	1563	1.0	19.3	170	179	185	5.3	3.7
<i>Owned</i>	2427	2370	2308	-2.3	-2.6	4600	4494	4275	-2.3	-4.9
Total	3723	3680	3871	-1.2	5.2	4769	4673	4460	-2.0	-4.6
Machine Labour										
<i>Hired</i>	8703	9058	11352	4.1	25.3	7224	7371	7550	2.0	2.4
<i>Owned</i>	1712	1838	1797	7.4	-2.3	499	496	514	-0.6	3.7
Total	10415	10896	13149	4.6	20.7	7722	7867	8064	1.9	2.5
Seed	4418	4284	4857	-3.0	13.4	3812	3827	3766	0.4	-1.6
Fertilisers and Manure										
<i>Fertilisers</i>	7374	6607	6440	-10.4	-2.5	3104	3201	3385	3.1	5.7
<i>Manure</i>	179	201	432	12.3	115.0	487	514	530	5.6	3.1
Total	7553	6807	6872	-9.9	1.0	3591	3715	3915	3.5	5.4
Other Inputs										
Insecticides	837	1142	1758	36.5	53.9	2236	2285	2342	2.2	2.5
Irrigation charges	325	413	786	27.1	90.4	8	0	2	-	-
Crop Insurance	50	52	0	2.8	-	218	228	217	4.8	-4.9
Interest on working capital	1167	1195	1351	2.4	13.1	944	962	972	1.9	1.0
Miscellaneous	342	365	273	6.5	-25.2	188	181	178	-4.0	-1.6
Fixed Cost	18474	21191	22385	14.7	5.6	13019	13915	14939	6.9	7.4
Rental value of owned land	15835	18298	19224	15.6	5.1	11245	12159	13123	8.1	7.9
Rent paid for leased-in land	0	0	0	-	-	0	0	0	-	-
Land revenue, cesses & taxes	8	10	10	19.7	-2.0	3	3	3	13.2	2.4
Depreciation on implements & Farm buildings	246	281	417	14.3	48.6	528	538	551	1.7	2.6
Interest on fixed capital	2385	2602	2734	9.1	5.1	1243	1216	1262	-2.2	3.8
Yield(Qtl/ha)	35.46	35.77	33.22	0.9	-7.1	31.14	31.86	34.33	2.3	7.8

Source: ES&E Division, Ministry of Agriculture and Farmers Welfare



Annex Table 5.6(d): Maize : Break-up of Cost of Cultivation

(₹/ha)

Cost Items	Maharashtra					Punjab				
	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23
Operational Cost	66159	72258	75115	9.2	4.0	57955	56095	55742	-3.2	-0.6
Human Labour										
<i>Casual</i>	17487	16518	16679	-5.5	1.0	9688	10112	10022	4.4	-0.9
<i>Attached</i>	552	1710	194	210.1	-88.7	1295	1305	0	0.8	-
<i>Family</i>	9687	11546	12734	19.2	10.3	11854	11377	11795	-4.0	3.7
Total	27726	29774	29607	7.4	-0.6	22837	22794	21817	-0.2	-4.3
Bullock Labour										
<i>Hired</i>	1403	1730	1265	23.3	-26.9	44	42	0	-3.5	-
<i>Owned</i>	2399	1706	2043	-28.9	19.7	50	0	44	-	-
Total	3802	3436	3308	-9.6	-3.7	94	42	44	-55.2	3.6
Machine Labour										
<i>Hired</i>	11223	11401	11362	1.6	-0.3	9609	8911	9106	-7.3	2.2
<i>Owned</i>	3228	5683	7641	76.1	34.5	5290	4916	5032	-7.1	2.4
Total	14451	17084	19004	18.2	11.2	14899	13827	14138	-7.2	2.2
Seed	5436	5711	6066	5.1	6.2	7996	7840	7972	-1.9	1.7
Fertilisers and Manure										
<i>Fertilisers</i>	7424	7785	8177	4.9	5.0	4107	3906	4027	-4.9	3.1
<i>Manure</i>	1620	2054	1736	26.7	-15.5	3182	3097	3196	-2.7	3.2
Total	9045	9839	9913	8.8	0.8	7289	7003	7223	-3.9	3.1
Other Inputs										
Insecticides	1702	1939	2195	14.0	13.2	2200	2009	1980	-8.7	-1.4
Irrigation charges	2071	2409	2867	16.4	19.0	1096	1085	1102	-1.0	1.5
Crop Insurance	0	0	24	-	-	0	0	0	-	-
Interest on working capital	1711	1840	1890	7.5	2.7	1397	1355	1332	-3.0	-1.7
Miscellaneous	215	226	242	5.3	7.1	147	140	136	-4.9	-2.6
Fixed Cost	26185	27708	25318	5.8	-8.6	20117	24689	24086	22.7	-2.4
Rental value of owned land	20768	21389	19761	3.0	-7.6	13260	21200	20339	59.9	-4.1
Rent paid for leased-in land	0	0	0	-	-	4186	1026	1094	-75.5	6.5
Land revenue, cesses & taxes	18	22	22	25.6	-2.6	0	0	0	-	-
Depreciation on implements & Farm buildings	793	958	791	20.9	-17.4	494	471	514	-4.7	9.0
Interest on fixed capital	4607	5338	4744	15.9	-11.1	2177	1991	2140	-8.5	7.5
Yield(Qtl/ha)	57.00	57.90	57.66	1.6	-0.4	37.51	34.35	33.63	-8.4	-2.1

Source: ES&E Division, Ministry of Agriculture and Farmers Welfare



Annex Table 5.6(d): Maize : Break-up of Cost of Cultivation

(₹/ha)

Cost Items	Rajasthan					Tamil Nadu				
	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23
Operational Cost	50070	55838	58138	11.5	4.1	62758	63774	69285	1.6	8.6
Human Labour										
<i>Casual</i>	3052	3261	5706	6.9	75.0	17588	18896	20496	7.4	8.5
<i>Attached</i>	262	390	0	48.9	-	20	19	18	-7.9	-3.7
<i>Family</i>	27492	29762	27285	8.3	-8.3	10751	7313	7590	-32.0	3.8
Total	30806	33412	32992	8.5	-1.3	28359	26227	28104	-7.5	7.2
Bullock Labour										
<i>Hired</i>	99	183	606	83.8	231.8	157	5	28	-96.5	419.7
<i>Owned</i>	763	684	301	-10.4	-56.0	0	1	0	-	-
Total	862	866	907	0.5	4.7	157	7	28	-95.7	316.3
Machine Labour										
<i>Hired</i>	9597	10931	12434	13.9	13.7	12319	13983	14715	13.5	5.2
<i>Owned</i>	835	844	1014	1.1	20.2	1048	1566	2654	49.4	69.5
Total	10432	11775	13448	12.9	14.2	13368	15549	17370	16.3	11.7
Seed	3464	3912	4815	12.9	23.1	6986	7012	7198	0.4	2.7
Fertilisers and Manure										
<i>Fertilisers</i>	2094	2597	2249	24.0	-13.4	6209	7235	8232	16.5	13.8
<i>Manure</i>	1107	1858	2013	67.9	8.4	1844	1572	1555	-14.7	-1.1
Total	3201	4455	4263	39.2	-4.3	8053	8807	9786	9.4	11.1
Other Inputs										
Insecticides	194	216	248	11.3	15.0	2105	2200	2767	4.5	25.8
Irrigation charges	416	403	520	-3.2	29.0	1836	1997	1906	8.7	-4.5
Crop Insurance	0	0	0	-	-	189	152	0	-19.6	-
Interest on working capital	684	790	935	15.5	18.3	1576	1711	1870	8.6	9.3
Miscellaneous	11	8	10	-27.2	32.6	129	112	256	-13.4	128.8
Fixed Cost	13377	12654	15418	-5.4	21.8	19859	20795	22808	4.7	9.7
Rental value of owned land	9281	10239	12817	10.3	25.2	13090	15340	17268	17.2	12.6
Rent paid for leased-in land	628	0	0	-	-	491	505	0	3.0	-
Land revenue, cesses & taxes	0	0	0	-	-	10	9	9	-4.3	-8.2
Depreciation on implements & Farm buildings	807	635	659	-21.4	3.9	509	475	524	-6.6	10.3
Interest on fixed capital	2661	1780	1941	-33.1	9.1	5760	4465	5007	-22.5	12.1
Yield(Qtl/ha)	22.61	23.42	23.55	3.6	0.6	42.14	49.97	49.72	18.6	-0.5

Source: ES&E Division, Ministry of Agriculture and Farmers Welfare



Annex Table 5.6(d): Maize : Break-up of Cost of Cultivation

(₹/ha)

Cost Items	Telangana					Uttar Pradesh				
	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23
Operational Cost	64118	71065	71085	10.8	0.0	39196	39156	42259	-0.1	7.9
Human Labour										
<i>Casual</i>	12761	13967	13113	9.5	-6.1	11264	10636	11958	-5.6	12.4
<i>Attached</i>	1003	1519	249	51.4	-83.6	0	0	0	-	-
<i>Family</i>	14338	15388	15780	7.3	2.5	7886	7580	8222	-3.9	8.5
Total	28102	30874	29142	9.9	-5.6	19150	18216	20179	-4.9	10.8
Bullock Labour										
<i>Hired</i>	902	1002	1222	11.1	21.9	0	0	0	-	-
<i>Owned</i>	1528	1086	989	-28.9	-9.0	0	0	184	-	-
Total	2430	2088	2211	-14.1	5.9	0	0	184	-	-
Machine Labour										
<i>Hired</i>	10864	11500	12327	5.9	7.2	8181	7930	6501	-3.1	-18.0
<i>Owned</i>	2179	4359	5330	100.0	22.3	430	288	1642	-33.1	471.0
Total	13044	15859	17657	21.6	11.3	8611	8218	8143	-4.6	-0.9
Seed	6954	7764	7912	11.7	1.9	5617	6029	6828	7.3	13.3
Fertilisers and Manure										
<i>Fertilisers</i>	7692	7509	7262	-2.4	-3.3	3111	3270	3395	5.1	3.8
<i>Manure</i>	0	22	48	-	116.3	0	150	144	-	-4.3
Total	7692	7531	7310	-2.1	-2.9	3111	3420	3539	9.9	3.5
Other Inputs										
Insecticides	3500	4336	4251	23.9	-2.0	174	328	257	88.1	-21.7
Irrigation charges	767	830	809	8.3	-2.6	1577	1981	2089	25.6	5.5
Crop Insurance	0	0	0	-	-	0	0	0	-	-
Interest on working capital	1508	1687	1676	11.8	-0.7	949	957	1031	0.8	7.8
Miscellaneous	121	94	118	-22.1	25.4	8	8	9	7.6	13.9
Fixed Cost	31932	35503	36006	11.2	1.4	15197	15999	19916	5.3	24.5
Rental value of owned land	26435	29507	25849	11.6	-12.4	12882	13864	17352	7.6	25.2
Rent paid for leased-in land	2344	2391	6694	2.0	180.0	0	0	0	-	-
Land revenue, cesses & taxes	0	0	0	-	-	4	4	5	-4.9	22.9
Depreciation on implements & Farm buildings	579	759	731	31.2	-3.7	336	350	367	4.3	4.9
Interest on fixed capital	2574	2845	2732	10.6	-4.0	1974	1780	2192	-9.8	23.1
Yield(Qtl/ha)	53.97	60.03	58.08	11.2	-3.2	27.08	20.70	24.27	-23.6	17.2

Source: ES&E Division, Ministry of Agriculture & Farmers Welfare



Annex Table 5.6(e): Ragi : Break-up of Cost of Cultivation

(₹/ha)

Cost Items	Karnataka					Odisha				
	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23
Operational Cost	54196	54438	57314	0.4	5.3	33986	36336	34998	6.9	-3.7
Human Labour										
<i>Casual</i>	15432	17645	16185	14.3	-8.3	2626	4559	1786	73.6	-60.8
<i>Attached</i>	0	0	0	-	-	0	0	0	-	-
<i>Family</i>	15447	14258	14601	-7.7	2.4	21993	21408	22958	-2.7	7.2
Total	30879	31903	30786	3.3	-3.5	24619	25967	24744	5.5	-4.7
Bullock Labour										
<i>Hired</i>	596	734	2090	23.2	184.8	312	273	115	-12.5	-58.0
<i>Owned</i>	1821	1961	961	7.7	-51.0	3135	3176	4458	1.3	40.4
Total	2417	2695	3051	11.5	13.2	3447	3449	4572	0.1	32.6
Machine Labour										
<i>Hired</i>	9799	6041	8662	-38.3	43.4	1854	2115	1532	14.0	-27.6
<i>Owned</i>	4222	5674	4429	34.4	-21.9	99	552	406	458.8	-26.4
Total	14021	11715	13091	-16.4	11.7	1953	2667	1938	36.5	-27.3
Seed	885	1007	878	13.8	-12.9	423	459	467	8.6	1.8
Fertilisers and Manure										
<i>Fertilisers</i>	3128	4488	6707	43.5	49.4	1018	1471	1127	44.5	-23.4
<i>Manure</i>	1055	797	535	-24.5	-32.9	2125	1826	1769	-14.1	-3.1
Total	4183	5285	7242	26.3	37.0	3143	3296	2895	4.9	-12.2
<i>Other Inputs</i>										
Insecticides	0	0	9	-	-	0	0	0	-	-
Irrigation charges	302	316	698	4.6	120.7	24	31	0	32.2	-
Crop Insurance	0	0	0	-	-	0	0	0	-	-
Interest on working capital	1174	1218	1294	3.7	6.3	363	452	365	24.5	-19.4
Miscellaneous	334	299	265	-10.4	-11.3	14	14	16	5.0	9.0
Fixed Cost	11081	11937	15454	7.7	29.5	7307	7995	8213	9.4	2.7
Rental value of owned land	10344	11136	11653	7.7	4.6	6651	7358	7479	10.6	1.6
Rent paid for leased-in land	0	0	178	-	-	0	0	0	-	-
Land revenue, cesses & taxes	16	18	25	12.7	38.4	21	17	14	-17.2	-16.8
Depreciation on implements & Farm buildings	193	206	765	6.8	270.8	212	229	250	8.2	9.3
Interest on fixed capital	527	576	2834	9.4	391.7	423	391	469	-7.5	19.8
Yield(Qtl/ha)	13.42	12.94	11.64	-3.6	-10.0	8.70	9.54	9.85	9.7	3.2

Source: ES&E Division, Ministry of Agriculture and Farmers Welfare



Annex Table 5.6(f): Arhar (Tur) : Break-up of Cost of Cultivation

(₹/ha)

Cost Items	Andhra Pradesh					Gujarat				
	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23
Operational Cost	27966	29823	30808	6.6	3.3	36703	37422	38539	2.0	3.0
Human Labour										
<i>Casual</i>	5432	5867	6955	8.0	18.5	7915	8130	9256	2.7	13.8
<i>Attached</i>	1	0	0	-	-	167	119	0	-28.6	-
<i>Family</i>	2566	3453	3481	34.6	0.8	8997	8296	9570	-7.8	15.4
Total	7999	9321	10436	16.5	12.0	17079	16546	18826	-3.1	13.8
Bullock Labour										
<i>Hired</i>	1279	1835	1649	43.4	-10.1	263	282	347	7.4	23.0
<i>Owned</i>	748	967	1352	29.4	39.8	496	506	742	2.0	46.6
Total	2027	2802	3001	38.3	7.1	759	788	1089	3.9	38.1
Machine Labour										
<i>Hired</i>	7465	8136	8436	9.0	3.7	5298	7646	5723	44.3	-25.2
<i>Owned</i>	783	144	755	-81.6	425.1	5322	3555	3333	-33.2	-6.3
Total	8248	8280	9192	0.4	11.0	10620	11202	9056	5.5	-19.2
Seed	1598	1758	1855	10.0	5.5	1175	1262	1342	7.3	6.4
Fertilisers and Manure										
<i>Fertilisers</i>	3659	4138	3043	13.1	-26.5	1374	1543	1828	12.3	18.5
<i>Manure</i>	911	839	415	-7.9	-50.5	466	209	233	-55.1	11.2
Total	4570	4977	3458	8.9	-30.5	1841	1752	2061	-4.8	17.6
Other Inputs										
Insecticides	2662	1793	1928	-32.6	7.6	2282	2621	2837	14.8	8.3
Irrigation charges	20	0	12	-	-	2021	2298	2365	13.7	2.9
Crop Insurance	0	0	0	-	-	0	0	0	-	-
Interest on working capital	770	799	828	3.8	3.6	840	883	878	5.1	-0.5
Miscellaneous	72	94	99	30.0	4.8	86	70	86	-18.5	22.2
Fixed Cost	14598	14219	17551	-2.6	23.4	16184	19835	20920	22.6	5.5
Rental value of owned land	12511	12585	12828	0.6	1.9	10724	12971	13811	20.9	6.5
Rent paid for leased-in land	38	58	2864	53.3	4815.8	648	538	0	-17.0	-
Land revenue, cesses & taxes	0	0	46	-	-	19	17	14	-12.0	-15.9
Depreciation on implements & Farm buildings	336	272	281	-19.2	3.5	476	544	651	14.4	19.6
Interest on fixed capital	1713	1304	1531	-23.9	17.5	4317	5765	6444	33.6	11.8
Yield(Qtl/ha)	6.37	5.97	6.02	-6.3	0.8	9.52	9.84	8.87	3.3	-9.9

Source: ES&E Division, Ministry of Agriculture and Farmers Welfare



Annex Table 5.6(f): Arhar (Tur) : Break-up of Cost of Cultivation

(₹/ha)

Cost Items	Karnataka					Maharashtra				
	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23
Operational Cost	31162	33788	42094	8.4	24.6	73882	76147	86631	3.1	13.8
Human Labour										
<i>Casual</i>	5660	6198	8802	9.5	42.0	21018	17470	20851	-16.9	19.4
<i>Attached</i>	115	94	557	-18.4	491.3	606	952	324	57.2	-65.9
<i>Family</i>	4747	6077	6966	28.0	14.6	14654	14393	17709	-1.8	23.0
Total	10522	12369	16325	17.6	32.0	36277	32815	38885	-9.5	18.5
Bullock Labour										
<i>Hired</i>	2636	2938	3171	11.4	7.9	1900	1656	2123	-12.9	28.3
<i>Owned</i>	2061	1487	1603	-27.8	7.8	3967	4486	5009	13.1	11.7
Total	4697	4425	4773	-5.8	7.9	5868	6141	7132	4.7	16.1
Machine Labour										
<i>Hired</i>	8077	8187	11789	1.4	44.0	13661	16735	17743	22.5	6.0
<i>Owned</i>	693	846	730	22.1	-13.7	1181	1192	1871	1.0	56.9
Total	8770	9033	12519	3.0	38.6	14842	17928	19614	20.8	9.4
Seed	983	1176	1308	19.6	11.2	2602	2953	3419	13.5	15.8
Fertilisers and Manure										
<i>Fertilisers</i>	2743	3108	3010	13.3	-3.2	5595	7181	8188	28.3	14.0
<i>Manure</i>	15	15	18	-0.1	18.6	763	636	631	-16.7	-0.7
Total	2758	3123	3027	13.2	-3.1	6358	7817	8820	22.9	12.8
Other Inputs										
Insecticides	2427	2591	2677	6.8	3.3	5235	5675	5578	8.4	-1.7
Irrigation charges	72	81	158	12.7	94.3	689	709	843	2.9	18.8
Crop Insurance	9	10	22	9.3	127.7	87	99	107	13.4	8.3
Interest on working capital	800	840	1065	4.9	26.8	1795	1871	2089	4.3	11.6
Miscellaneous	125	141	221	12.9	57.1	129	138	145	7.0	5.0
Fixed Cost	12298	13435	13015	9.2	-3.1	25913	24792	26008	-4.3	4.9
Rental value of owned land	10179	11482	11779	12.8	2.6	20167	19220	19688	-4.7	2.4
Rent paid for leased-in land	0	0	0	-	-	0	0	0	-	-
Land revenue, cesses & taxes	8	8	7	2.9	-9.3	35	34	39	-1.4	13.7
Depreciation on implements & Farm buildings	214	240	327	12.5	36.3	753	775	891	2.8	15.0
Interest on fixed capital	1898	1704	901	-10.2	-47.1	4958	4763	5390	-3.9	13.2
Yield(Qtl/ha)	6.67	5.71	5.23	-14.4	-8.4	19.91	16.22	16.64	-18.5	2.6

Source: ES&E Division, Ministry of Agriculture and Farmers Welfare



Annex Table 5.6(f): Arhar (Tur) : Break-up of Cost of Cultivation

(₹/ha)

Cost Items	Odisha					Telangana				
	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23
Operational Cost	15882	15986	16409	0.7	2.6	35240	36290	37686	3.0	3.8
Human Labour										
<i>Casual</i>	1595	2163	1849	35.6	-14.5	5549	10077	10576	81.6	5.0
<i>Attached</i>	0	0	0	19.5	-59.2	62	146	305	134.6	109.1
<i>Family</i>	10706	10175	10906	-5.0	7.2	7887	5497	5713	-30.3	3.9
Total	12301	12338	12756	0.3	3.4	13498	15719	16594	16.5	5.6
Bullock Labour										
<i>Hired</i>	3	88	61	3163.5	-31.2	1508	2096	1558	38.9	-25.7
<i>Owned</i>	1465	1555	1056	6.1	-32.1	1532	125	620	-91.9	397.4
Total	1468	1643	1116	12.0	-32.1	3040	2220	2178	-27.0	-1.9
Machine Labour										
<i>Hired</i>	579	616	618	6.5	0.3	6566	7470	9573	13.8	28.2
<i>Owned</i>	60	69	108	14.8	56.4	3035	2032	214	-33.1	-89.5
Total	639	685	726	7.3	6.0	9602	9502	9787	-1.0	3.0
Seed	1192	1140	1322	-4.4	16.0	1479	1483	1638	0.3	10.5
Fertilisers and Manure										
<i>Fertilisers</i>	48	0	183	-	-	2856	2469	2918	-13.5	18.2
<i>Manure</i>	72	0	0	-	-	0	0	0	-	-
Total	120	0	183	-	-	2856	2469	2918	-13.5	18.2
Other Inputs										
Insecticides	3	0	121	-	-	3793	3804	3429	0.3	-9.8
Irrigation charges	0	0	13	-	-	50	56	62	12.9	10.0
Crop Insurance	0	0	0	-	-	0	0	0	-	-
Interest on working capital	157	176	167	12.3	-5.3	829	933	969	12.6	3.8
Miscellaneous	3	3	5	5.0	53.2	94	104	111	10.7	6.6
Fixed Cost	7718	7675	8559	-0.6	11.5	14935	18734	21061	25.4	12.4
Rental value of owned land	6490	6588	7153	1.5	8.6	12611	16444	18626	30.4	13.3
Rent paid for leased-in land	12	0	0	-	-	30	83	0	175.9	-
Land revenue, cesses & taxes	16	16	16	1.3	0.6	0	0	0	-	-
Depreciation on implements & Farm buildings	353	349	422	-1.2	20.9	504	449	476	-10.9	6.2
Interest on fixed capital	847	722	968	-14.7	34.0	1791	1759	1958	-1.8	11.3
Yield(Qtl/ha)	4.09	3.97	3.52	-2.9	-11.3	7.01	7.66	6.80	9.3	-11.2

Source: ES&E Division, Ministry of Agriculture and Farmers Welfare



Annex Table 5.6(f): Arhar (Tur) : Break-up of Cost of Cultivation

(₹/ha)

Cost Items	Uttar Pradesh				
	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23
Operational Cost	37840	36917	36840	-2.4	-0.2
Human Labour					
<i>Casual</i>	4890	6854	8193	40.2	19.5
<i>Attached</i>	0	31	32	-	3.7
<i>Family</i>	21753	18933	17959	-13.0	-5.1
Total	26643	25818	26184	-3.1	1.4
Bullock Labour					
<i>Hired</i>	0	0	0	-	-
<i>Owned</i>	0	0	0	-	-
Total	0	0	0	-	-
Machine Labour					
<i>Hired</i>	4656	4175	4313	-10.3	3.3
<i>Owned</i>	1269	1527	878	20.3	-42.5
Total	5925	5702	5191	-3.8	-9.0
Seed	3407	3364	3264	-1.3	-3.0
Fertilisers and Manure					
<i>Fertilisers</i>	186	44	50	-76.5	13.8
<i>Manure</i>	0	0	0	-	-
Total	186	44	50	-76.5	13.8
<i>Other Inputs</i>					
Insecticides	0	0	155	-	-
Irrigation charges	1170	1444	1424	23.4	-1.4
Crop Insurance	0	0	0	-	-
Interest on working capital	487	545	572	11.8	5.0
Miscellaneous	21	0	0	-	-
Fixed Cost	32477	34245	30975	5.4	-9.5
Rental value of owned land	26279	28113	24887	7.0	-11.5
Rent paid for leased-in land	0	0	662	-	-
Land revenue, cesses & taxes	8	8	7	-3.9	-9.7
Depreciation on implements & Farm buildings	1108	1066	997	-3.8	-6.4
Interest on fixed capital	5082	5058	4422	-0.5	-12.6
Yield(Qtl/ha)	9.32	10.03	8.95	7.6	-10.8

Source: ES&E Division, Ministry of Agriculture and Farmers Welfare



Annex Table 5.6(g): Moong : Break-up of Cost of Cultivation

(₹/ha)

Cost Items	Andhra Pradesh					Gujarat				
	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23
Operational Cost	29103	27478	26174	-5.6	-4.7	29394	30361	33861	3.3	11.5
Human Labour										
<i>Casual</i>	7404	9997	10340	35.0	3.4	6462	1954	2643	-69.8	35.3
<i>Attached</i>	97	0	0	-	-	0	0	0	-	-
<i>Family</i>	2171	783	1003	-64.0	28.2	10423	15508	17252	48.8	11.2
Total	9671	10780	11343	11.5	5.2	16886	17462	19895	3.4	13.9
Bullock Labour										
<i>Hired</i>	99	0	11	-	-	0	0	6	-	-
<i>Owned</i>	323	0	88	-	-	0	0	85	-	-
Total	423	0	98	-	-	0	0	91	-	-
Machine Labour										
<i>Hired</i>	6284	6575	4456	4.6	-32.2	4098	5665	5701	38.2	0.6
<i>Owned</i>	305	381	380	25.1	-0.3	2740	1612	1474	-41.2	-8.6
Total	6588	6956	4836	5.6	-30.5	6838	7277	7175	6.4	-1.4
Seed	3052	3372	3539	10.5	4.9	2048	1982	2144	-3.3	8.2
Fertilisers and Manure										
<i>Fertilisers</i>	257	0	28	-	-	536	548	674	2.3	22.9
<i>Manure</i>	0	0	0	-	-	0	0	95	-	-
Total	257	0	28	-	-	536	548	769	2.3	40.2
Other Inputs										
Insecticides	7840	5201	5203	-33.7	0.0	918	1018	1118	10.9	9.7
Irrigation charges	406	314	315	-22.7	0.2	1503	1530	2061	1.8	34.7
Crop Insurance	0	0	0	-	-	0	0	0	-	-
Interest on working capital	816	809	763	-0.9	-5.7	575	450	503	-21.7	11.8
Miscellaneous	49	47	49	-4.5	5.6	89	94	105	6.0	11.2
Fixed Cost	12751	17582	16392	37.9	-6.8	10987	9363	12193	-14.8	30.2
Rental value of owned land	9881	15356	16107	55.4	4.9	6608	7198	9781	8.9	35.9
Rent paid for leased-in land	2641	1956	0	-25.9	-	1769	0	0	-	-
Land revenue, cesses & taxes	0	0	0	-	-	4	5	4	14.9	-6.9
Depreciation on implements & Farm buildings	92	121	127	31.4	4.8	246	189	210	-23.2	11.1
Interest on fixed capital	137	149	158	8.4	6.4	2360	1971	2198	-16.5	11.5
Yield(Qtl/ha)	7.01	7.68	7.87	9.5	2.5	6.61	5.80	5.73	-12.3	-1.2

Source: ES&E Division, Ministry of Agriculture and Farmers Welfare



Annex Table 5.6(g): Moong : Break-up of Cost of Cultivation

(₹/ha)

Cost Items	Karnataka					Maharashtra				
	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23
Operational Cost	27974	27997	28514	0.1	1.8	42374	44419	46441	4.8	4.6
Human Labour										
<i>Casual</i>	5143	5120	6269	-0.4	22.4	12608	13594	11417	7.8	-16.0
<i>Attached</i>	0	13	135	-	943.3	55	199	1024	265.5	413.7
<i>Family</i>	6695	4597	5074	-31.3	10.4	7840	8627	12414	10.0	43.9
Total	11838	9730	11478	-17.8	18.0	20503	22420	24854	9.4	10.9
Bullock Labour										
<i>Hired</i>	878	417	1058	-52.5	153.4	1765	1559	77	-11.7	-95.0
<i>Owned</i>	1837	998	860	-45.7	-13.9	829	1096	3023	32.3	175.7
Total	2715	1416	1918	-47.9	35.5	2594	2655	3100	2.4	16.7
Machine Labour										
<i>Hired</i>	6265	8685	5228	38.6	-39.8	9770	10024	7477	2.6	-25.4
<i>Owned</i>	1789	2379	4150	33.0	74.4	1211	2975	4415	145.7	48.4
Total	8054	11064	9378	37.4	-15.2	10981	12999	11892	18.4	-8.5
Seed	1484	1824	1563	22.9	-14.3	2360	2335	2477	-1.1	6.1
Fertilisers and Manure										
<i>Fertilisers</i>	2252	1966	1815	-12.7	-7.7	1770	1860	1977	5.1	6.3
<i>Manure</i>	0	99	84	-	-15.6	2002	0	0	-	-
Total	2252	2065	1898	-8.3	-8.1	3772	1860	1977	-50.7	6.3
<i>Other Inputs</i>										
Insecticides	891	1081	1231	21.3	13.9	873	796	793	-8.8	-0.4
Irrigation charges	1	1	220	2.5	18085.1	211	230	274	9.4	19.1
Crop Insurance	12	0	0	-	-	1	1	2	16.8	22.0
Interest on working capital	645	709	710	10.0	0.2	1046	1085	1031	3.6	-4.9
Miscellaneous	82	107	118	30.5	10.7	34	38	40	12.4	4.7
Fixed Cost	9138	10264	10855	12.3	5.8	11325	10981	12650	-3.0	15.2
Rental value of owned land	7499	8634	8159	15.1	-5.5	8225	8262	10230	0.4	23.8
Rent paid for leased-in land	0	0	802	-	-	0	0	0	-	-
Land revenue, cesses & taxes	7	8	6	11.0	-25.8	16	18	16	10.5	-11.3
Depreciation on implements & Farm buildings	278	328	362	17.8	10.3	308	331	353	7.5	6.7
Interest on fixed capital	1354	1294	1526	-4.4	18.0	2776	2370	2051	-14.6	-13.5
Yield(Qtl/ha)	4.60	4.95	4.18	7.6	-15.6	8.09	7.81	7.99	-3.4	2.3

Source: ES&E Division, Ministry of Agriculture and Farmers Welfare



Annex Table 5.6(g): Moong : Break-up of Cost of Cultivation

(₹/ha)

Cost Items	Odisha					Rajasthan				
	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23
Operational Cost	16909	18522	19378	9.5	4.6	24350	26937	29296	10.6	8.8
Human Labour										
<i>Casual</i>	3298	2785	4386	-15.6	57.5	2995	2178	2550	-27.3	17.1
<i>Attached</i>	5	6	3	12.0	-45.6	0	0	0	-	-
<i>Family</i>	7600	9330	8029	22.8	-13.9	11329	13659	13749	20.6	0.7
Total	10903	12121	12418	11.2	2.5	14324	15838	16299	10.6	2.9
Bullock Labour										
<i>Hired</i>	15	19	35	25.9	87.3	0	0	0	-	-
<i>Owned</i>	1211	1059	567	-12.6	-46.4	0	0	226	-	-
Total	1226	1078	602	-12.1	-44.1	0	0	226	-	-
Machine Labour										
<i>Hired</i>	1968	2394	3196	21.7	33.5	4528	5154	6228	13.8	20.8
<i>Owned</i>	55	111	146	102.8	31.8	1080	1517	1936	40.5	27.6
Total	2023	2505	3342	23.9	33.4	5607	6671	8164	19.0	22.4
Seed	2063	2164	2320	4.9	7.2	2143	2012	2293	-6.1	14.0
Fertilisers and Manure										
<i>Fertilisers</i>	170	134	170	-21.5	27.0	868	964	720	11.1	-25.3
<i>Manure</i>	55	0	0	-	-	539	548	646	1.7	17.8
Total	225	134	170	-40.7	27.0	1407	1512	1366	7.5	-9.7
Other Inputs										
Insecticides	113	145	119	29.3	-18.3	422	448	422	6.2	-5.8
Irrigation charges	72	94	61	31.9	-35.5	13	6	10	-57.1	80.9
Crop Insurance	0	0	0	-	-	0	0	0	-	-
Interest on working capital	282	279	344	-1.3	23.5	395	402	471	2.0	17.1
Miscellaneous	3	3	3	-6.8	4.7	39	48	45	25.2	-7.6
Fixed Cost	8494	8384	9034	-1.3	7.8	8170	8398	9960	2.8	18.6
Rental value of owned land	7718	7714	8325	0.0	7.9	5361	6199	7476	15.6	20.6
Rent paid for leased-in land	13	21	8	60.6	-59.6	0	0	0	-	-
Land revenue, cesses & taxes	11	11	12	1.2	1.8	0	0	0	-	-
Depreciation on implements & Farm buildings	264	219	226	-17.0	2.8	500	415	476	-17.0	14.7
Interest on fixed capital	488	418	463	-14.4	10.8	2308	1784	2008	-22.7	12.6
Yield(Qtl/ha)	4.07	4.05	4.17	-0.4	3.0	4.39	5.41	5.07	23.4	-6.3

Source: ES&E Division, Ministry of Agriculture and Farmers Welfare



Annex Table 5.6(g): Moong : Break-up of Cost of Cultivation

(₹/ha)

Cost Items	Tamil Nadu					Uttar Pradesh				
	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23
Operational Cost	21210	22307	24222	5.2	8.6	27744	24998	27347	-9.9	9.4
Human Labour										
<i>Casual</i>	7860	8247	10299	4.9	24.9	2046	546	1012	-73.3	85.3
<i>Attached</i>	0	0	8	-	-	0	0	0	-	-
<i>Family</i>	4028	4643	4286	15.3	-7.7	18154	16479	16679	-9.2	1.2
Total	11888	12890	14592	8.4	13.2	20199	17025	17691	-15.7	3.9
Bullock Labour										
<i>Hired</i>	0	0	0	-	-	0	0	0	-	-
<i>Owned</i>	0	0	0	-	-	0	0	0	-	-
Total	0	0	0	-	-	0	0	0	-	-
Machine Labour										
<i>Hired</i>	3673	4123	4033	12.3	-2.2	4374	4766	6139	9.0	28.8
<i>Owned</i>	755	80	123	-89.3	53.2	97	188	0	94.1	-
Total	4428	4204	4156	-5.1	-1.1	4471	4954	6139	10.8	23.9
Seed	1898	1990	2381	4.8	19.7	1977	2055	2276	4.0	10.8
Fertilisers and Manure										
<i>Fertilisers</i>	522	453	466	-13.2	2.9	155	0	189	-	-
<i>Manure</i>	406	401	420	-1.1	4.8	0	0	0	-	-
Total	927	854	886	-7.9	3.8	155	0	189	-	-
<i>Other Inputs</i>										
Insecticides	1349	1644	1389	21.9	-15.5	93	144	155	54.1	7.5
Irrigation charges	94	84	97	-10.5	14.3	548	551	560	0.5	1.7
Crop Insurance	12	0	4	-	-	0	0	0	-	-
Interest on working capital	521	535	604	2.8	12.9	291	258	323	-11.2	25.2
Miscellaneous	93	106	112	14.0	5.6	11	12	13	8.1	13.4
Fixed Cost	8108	7473	11687	-7.8	56.4	6274	5526	6261	-11.9	13.3
Rental value of owned land	5681	6597	10709	16.1	62.3	5517	4672	5099	-15.3	9.1
Rent paid for leased-in land	427	0	0	-	-	0	6	0	-	-
Land revenue, cesses & taxes	4	4	5	-8.6	30.0	2	2	2	-13.3	4.0
Depreciation on implements & Farm buildings	698	404	429	-42.1	6.2	195	189	332	-3.1	75.4
Interest on fixed capital	1298	467	544	-64.0	16.3	559	656	828	17.3	26.2
Yield(Qtl/ha)	4.21	4.10	3.99	-2.6	-2.7	2.98	2.20	2.24	-26.2	1.8

Source: ES&E Division, Ministry of Agriculture and Farmers Welfare



Annex Table 5.6(h): Urad : Break-up of Cost of Cultivation

(₹/ha)

Cost Items	Andhra Pradesh					Gujarat				
	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23
Operational Cost	32068	32690	33073	1.9	1.2	29668	31918	35559	7.6	11.4
Human Labour										
<i>Casual</i>	12335	12568	11604	1.9	-7.7	8435	7200	8636	-14.6	19.9
<i>Attached</i>	215	169	0	-21.4	-	5	0	0	-	-
<i>Family</i>	2583	2957	3765	14.5	27.3	7522	9133	10028	21.4	9.8
Total	15133	15694	15369	3.7	-2.1	15962	16333	18664	2.3	14.3
Bullock Labour										
<i>Hired</i>	423	255	0	-39.8	-	162	0	165	-	-
<i>Owned</i>	0	0	267	-	-	102	269	134	163.9	-50.0
Total	423	255	267	-39.8	4.7	264	269	300	2.0	11.6
Machine Labour										
<i>Hired</i>	5798	6206	6175	7.0	-0.5	6518	7173	7870	10.1	9.7
<i>Owned</i>	751	532	816	-29.1	53.4	1305	1658	1741	27.0	5.0
Total	6549	6738	6991	2.9	3.8	7823	8831	9612	12.9	8.8
Seed	3809	3770	4029	-1.0	6.9	1755	2321	2369	32.2	2.1
Fertilisers and Manure										
<i>Fertilisers</i>	1337	1010	1019	-24.5	0.9	808	813	865	0.6	6.4
<i>Manure</i>	25	0	0	-	-	1454	1434	1635	-1.4	14.0
Total	1362	1010	1019	-25.9	0.9	2262	2246	2499	-0.7	11.3
Other Inputs										
Insecticides	3584	3987	4178	11.3	4.8	710	983	1083	38.5	10.1
Irrigation charges	234	302	293	29.4	-3.1	116	127	133	9.6	4.3
Crop Insurance	0	0	0	-	-	0	0	0	-	-
Interest on working capital	893	901	888	0.8	-1.4	671	690	774	2.9	12.1
Miscellaneous	82	34	39	-59.0	16.5	105	117	133	11.0	13.5
Fixed Cost	21266	24157	20615	13.6	-14.7	7105	7788	10131	9.6	30.1
Rental value of owned land	19837	23045	19012	16.2	-17.5	4348	5020	7211	15.5	43.6
Rent paid for leased-in land	950	753	1223	-20.7	62.4	217	37	0	-82.9	-
Land revenue, cesses & taxes	0	0	0	-	-	7	6	6	-14.3	2.6
Depreciation on implements & Farm buildings	75	59	63	-21.9	7.3	243	262	283	8.0	8.1
Interest on fixed capital	403	301	317	-25.5	5.4	2292	2463	2631	7.5	6.8
Yield(Qtl/ha)	10.89	10.76	10.69	-1.2	-0.7	4.14	4.46	5.24	7.7	17.5

Source: ES&E Division, Ministry of Agriculture and Farmers Welfare



Annex Table 5.6(h): Urad : Break-up of Cost of Cultivation

(₹/ha)

Cost Items	Karnataka					Madhya Pradesh				
	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23
Operational Cost	26399	29576	30615	12.0	3.5	22853	23583	24493	3.2	3.9
Human Labour										
<i>Casual</i>	8245	9035	7514	9.6	-16.8	4055	4199	4519	3.6	7.6
<i>Attached</i>	214	257	0	20.2	-	18	11	0	-39.7	-
<i>Family</i>	4021	4089	3620	1.7	-11.5	3719	3848	4114	3.5	6.9
Total	12480	13380	11134	7.2	-16.8	7791	8058	8633	3.4	7.1
Bullock Labour										
<i>Hired</i>	924	1763	2840	90.9	61.1	0	3	4	-	33.4
<i>Owned</i>	1357	828	0	-39.0	-	0	0	0	-	-
Total	2280	2591	2840	13.6	9.6	0	3	4	-	33.4
Machine Labour										
<i>Hired</i>	5285	6269	9003	18.6	43.6	8519	8733	8111	2.5	-7.1
<i>Owned</i>	598	852	1091	42.3	28.1	170	167	439	-1.9	163.5
Total	5884	7121	10094	21.0	41.8	8689	8899	8550	2.4	-3.9
Seed	1312	1237	1181	-5.7	-4.5	2360	2347	2419	-0.6	3.1
Fertilisers and Manure										
<i>Fertilisers</i>	2440	2668	2217	9.3	-16.9	1945	2001	2019	2.8	0.9
<i>Manure</i>	0	315	0	-	-	231	329	584	42.7	77.5
Total	2440	2983	2217	22.2	-25.7	2176	2330	2603	7.1	11.7
Other Inputs										
Insecticides	1205	1358	2214	12.7	63.0	1164	1251	1521	7.4	21.6
Irrigation charges	0	7	0	-	-	0	0	0	-	-
Crop Insurance	0	0	0	-	-	19	20	25	3.9	26.7
Interest on working capital	678	772	818	13.9	5.9	580	598	618	3.1	3.3
Miscellaneous	120	126	116	5.5	-7.9	74	78	120	4.4	54.3
Fixed Cost	9164	10383	8554	13.3	-17.6	6978	7529	9030	7.9	19.9
Rental value of owned land	8409	9562	7640	13.7	-20.1	6164	6670	8105	8.2	21.5
Rent paid for leased-in land	0	0	0	-	-	0	0	0	-	-
Land revenue, cesses & taxes	5	4	5	-18.6	29.4	3	2	2	-15.0	-2.2
Depreciation on implements & Farm buildings	125	149	255	19.6	70.6	259	278	307	7.3	10.4
Interest on fixed capital	626	667	654	6.5	-1.9	552	578	616	4.8	6.4
Yield(Qtl/ha)	5.39	5.18	4.74	-3.9	-8.5	5.84	6.58	7.06	12.7	7.3

Source: ES&E Division, Ministry of Agriculture and Farmers Welfare



Annex Table 5.6(h): Urad : Break-up of Cost of Cultivation

(₹/ha)

Cost Items	Maharashtra					Odisha				
	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23
Operational Cost	38528	39908	42430	3.6	6.3	17327	19101	19949	10.2	4.4
Human Labour										
<i>Casual</i>	10929	11374	11888	4.1	4.5	1941	1931	3897	-0.5	101.8
<i>Attached</i>	620	66	0	-89.4	-	2	0	0	-	-
<i>Family</i>	7607	7782	9194	2.3	18.1	9249	11066	9578	19.6	-13.4
Total	19157	19222	21082	0.3	9.7	11192	12997	13475	16.1	3.7
Bullock Labour										
<i>Hired</i>	2042	1888	348	-7.6	-81.6	105	190	218	80.0	15.2
<i>Owned</i>	3634	2997	4387	-17.5	46.4	1205	931	776	-22.7	-16.7
Total	5677	4885	4734	-13.9	-3.1	1311	1121	994	-14.5	-11.3
Machine Labour										
<i>Hired</i>	5255	7446	8109	41.7	8.9	1591	1727	2120	8.6	22.7
<i>Owned</i>	633	205	237	-67.6	15.4	160	135	215	-15.8	59.4
Total	5888	7650	8346	29.9	9.1	1751	1862	2335	6.4	25.4
Seed	2195	2409	2307	9.7	-4.2	2568	2674	2602	4.1	-2.7
Fertilisers and Manure										
<i>Fertilisers</i>	2567	3142	3422	22.4	8.9	173	157	183	-9.6	17.0
<i>Manure</i>	235	300	0	27.7	-	56	11	13	-80.4	14.6
Total	2803	3442	3422	22.8	-0.6	229	167	196	-26.9	16.8
Other Inputs										
Insecticides	1196	1226	1293	2.5	5.4	25	33	30	32.0	-10.7
Irrigation charges	592	0	128	-	-	0	0	0	-	-
Crop Insurance	10	11	10	15.1	-7.2	0	0	0	-	-
Interest on working capital	937	974	1007	3.9	3.5	245	243	314	-0.5	29.1
Miscellaneous	75	89	100	19.0	12.3	6	3	3	-56.9	6.9
Fixed Cost	9625	10640	10633	10.5	-0.1	7333	7902	9329	7.8	18.1
Rental value of owned land	8536	9487	9271	11.1	-2.3	6690	7257	8495	8.5	17.1
Rent paid for leased-in land	0	0	0	-	-	0	0	81	-	-
Land revenue, cesses & taxes	19	16	15	-17.4	-6.4	12	13	15	1.7	19.9
Depreciation on implements & Farm buildings	206	247	286	20.2	15.7	202	210	237	4.0	13.0
Interest on fixed capital	864	890	1061	2.9	19.2	428	422	500	-1.4	18.5
Yield(Qtl/ha)	8.06	6.86	6.25	-14.9	-8.9	3.50	3.65	3.88	4.3	6.3

Source: ES&E Division, Ministry of Agriculture and Farmers Welfare



Annex Table 5.6(h): Urad : Break-up of Cost of Cultivation

(₹/ha)

Cost Items	Rajasthan					Tamil Nadu				
	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23
Operational Cost	31398	32249	35894	2.7	11.3	33447	37768	37436	12.9	-0.9
Human Labour										
<i>Casual</i>	4736	2596	4107	-45.2	58.2	11408	12238	12703	7.3	3.8
<i>Attached</i>	485	178	0	-63.3	-	18	7	0	-58.5	-
<i>Family</i>	16201	18496	17027	14.2	-7.9	3434	4912	5011	43.0	2.0
Total	21421	21270	21135	-0.7	-0.6	14860	17157	17714	15.5	3.2
Bullock Labour										
<i>Hired</i>	0	1	0	-	-	0	23	0	-	-
<i>Owned</i>	138	158	172	14.0	8.8	0	0	0	-	-
Total	138	159	172	14.8	8.1	0	23	0	-	-
Machine Labour										
<i>Hired</i>	5923	6531	8077	10.3	23.7	8770	9759	8893	11.3	-8.9
<i>Owned</i>	1289	1090	1581	-15.4	45.0	718	708	828	-1.5	16.9
Total	7211	7621	9658	5.7	26.7	9489	10467	9721	10.3	-7.1
Seed	1752	2051	2229	17.1	8.7	2630	3037	3428	15.5	12.9
Fertilisers and Manure										
<i>Fertilisers</i>	204	489	569	139.4	16.4	2339	2546	2149	8.9	-15.6
<i>Manure</i>	0	0	1321	-	-	733	1189	1279	62.3	7.5
Total	204	489	1890	139.4	286.7	3072	3736	3427	21.6	-8.2
Other Inputs										
Insecticides	193	233	86	20.8	-63.1	1927	1770	1501	-8.1	-15.2
Irrigation charges	0	0	143	-	-	495	506	553	2.1	9.3
Crop Insurance	0	0	0	-	-	0	0	1	-	-
Interest on working capital	461	417	572	-9.5	37.2	909	996	983	9.5	-1.3
Miscellaneous	18	9	9	-47.0	-1.7	66	78	108	18.7	38.8
Fixed Cost	6301	7814	8333	24.0	6.6	13747	13125	17275	-4.5	31.6
Rental value of owned land	4849	6240	6685	28.7	7.1	11235	10949	15017	-2.5	37.2
Rent paid for leased-in land	0	0	0	-	-	6	4	2	-36.4	-39.9
Land revenue, cesses & taxes	2	0	0	-	-	7	5	6	-18.1	9.3
Depreciation on implements & Farm buildings	277	287	296	3.7	3.0	297	348	394	16.8	13.4
Interest on fixed capital	1173	1288	1352	9.7	5.0	2202	1819	1856	-17.4	2.1
Yield(Qtl/ha)	4.85	5.66	5.05	16.7	-10.8	6.31	7.88	6.32	24.8	-19.8

Source: ES&E Division, Ministry of Agriculture and Farmers Welfare



Annex Table 5.6(h): Urad : Break-up of Cost of Cultivation

(₹/ha)

Cost Items	Uttar Pradesh				
	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23
Operational Cost	18617	20841	21558	11.9	3.4
Human Labour					
<i>Casual</i>	3770	2909	2395	-22.8	-17.7
<i>Attached</i>	0	0	0	-	-
<i>Family</i>	7812	10082	10803	29.0	7.2
Total	11582	12991	13198	12.2	1.6
Bullock Labour					
<i>Hired</i>	0	0	0	-	-
<i>Owned</i>	0	0	0	-	-
Total	0	0	0	-	-
Machine Labour					
<i>Hired</i>	3915	4880	4824	24.6	-1.1
<i>Owned</i>	603	130	456	-78.4	249.8
Total	4518	5010	5279	10.9	5.4
Seed	1991	1913	2160	-3.9	12.9
Fertilisers and Manure					
<i>Fertilisers</i>	1	1	1	14.2	10.1
<i>Manure</i>	0	0	0	-	-
Total	1	1	1	14.2	10.1
<i>Other Inputs</i>					
Insecticides	116	372	334	221.0	-10.3
Irrigation charges	78	224	256	187.9	14.1
Crop Insurance	0	0	0	-	-
Interest on working capital	327	326	326	-0.4	0.0
Miscellaneous	4	3	3	-7.1	0.0
Fixed Cost	9042	7606	10068	-15.9	32.4
Rental value of owned land	6990	5913	7611	-15.4	28.7
Rent paid for leased-in land	284	0	0	-	-
Land revenue, cesses & taxes	4	4	3	-11.6	-24.4
Depreciation on implements & Farm buildings	380	364	476	-4.0	30.7
Interest on fixed capital	1384	1324	1978	-4.3	49.4
Yield(Qtl/ha)	3.02	2.67	2.81	-11.5	5.2

Source: ES&E Division, Ministry of Agriculture and Farmers Welfare



Annex Table 5.6(i): Groundnut : Break-up of Cost of Cultivation

(₹/ha)

Cost Items	Andhra Pradesh					Gujarat				
	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23
Operational Cost	64105	72967	81550	13.8	11.8	70949	73966	83025	4.3	12.2
Human Labour										
<i>Casual</i>	18021	20236	25405	12.3	25.5	16023	17089	19898	6.7	16.4
<i>Attached</i>	10	15	9	57.6	-40.6	0	0	0	-	-
<i>Family</i>	7769	8762	7852	12.8	-10.4	8505	8519	9400	0.2	10.3
Total	25800	29013	33265	12.5	14.7	24528	25608	29298	4.4	14.4
Bullock Labour										
<i>Hired</i>	667	1118	1313	67.5	17.5	461	321	355	-30.4	10.5
<i>Owned</i>	1724	1545	1522	-10.4	-1.5	1160	1347	1055	16.1	-21.7
Total	2391	2663	2835	11.4	6.5	1621	1668	1410	2.9	-15.5
Machine Labour										
<i>Hired</i>	7533	7810	6742	3.7	-13.7	9351	9799	11210	4.8	14.4
<i>Owned</i>	215	237	1382	10.1	483.9	4665	4742	4896	1.7	3.2
Total	7748	8047	8124	3.9	1.0	14016	14541	16106	3.7	10.8
Seed	16728	20361	21573	21.7	6.0	16110	16709	18895	3.7	13.1
Fertilisers and Manure										
<i>Fertilisers</i>	6693	7204	9538	7.6	32.4	3289	3531	4072	7.3	15.3
<i>Manure</i>	335	351	125	4.9	-64.3	4011	3919	4297	-2.3	9.6
Total	7028	7556	9664	7.5	27.9	7300	7450	8369	2.1	12.3
Other Inputs										
Insecticides	1896	2468	2769	30.2	12.2	4177	4581	4990	9.7	8.9
Irrigation charges	559	623	775	11.5	24.4	1131	1186	1461	4.8	23.2
Crop Insurance	0	0	0	-	-	0	0	0	-	-
Interest on working capital	1707	1946	2233	14.0	14.8	1892	1983	2231	4.8	12.5
Miscellaneous	248	291	312	17.3	7.1	174	239	266	37.3	11.3
Fixed Cost	19339	23147	28732	19.7	24.1	20698	23882	29213	15.4	22.3
Rental value of owned land	14776	20647	25607	39.7	24.0	16153	18803	23888	16.4	27.0
Rent paid for leased-in land	2348	242	736	-89.7	203.6	459	550	195	19.9	-64.5
Land revenue, cesses & taxes	0	0	0	-	-	5	5	5	7.8	-0.8
Depreciation on implements & Farm buildings	334	335	338	0.2	0.9	401	459	524	14.6	14.3
Interest on fixed capital	1881	1923	2051	2.2	6.7	3680	4064	4601	10.5	13.2
Yield(Qtl/ha)	9.76	11.73	13.25	20.2	13.0	16.50	16.84	19.73	2.1	17.2

Note: Total cost may not match due to rounding off the figures

Source: ES&E Division, Ministry of Agriculture and Farmers Welfare



Annex Table 5.6(i): Groundnut : Break-up of Cost of Cultivation

(₹/ha)

Cost Items	Karnataka					Maharashtra				
	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23
Operational Cost	55151	59809	60464	8.4	1.1	96819	105314	94764	8.8	-10.0
Human Labour										
<i>Casual</i>	13445	13268	13729	-1.3	3.5	18986	25153	20414	32.5	-18.8
<i>Attached</i>	0	0	0	-	-	1543	248	147	-83.9	-40.8
<i>Family</i>	11103	13772	14409	24.0	4.6	28991	31442	29957	8.5	-4.7
Total	24547	27040	28138	10.2	4.1	49519	56843	50518	14.8	-11.1
Bullock Labour										
<i>Hired</i>	3306	3334	1438	0.9	-56.9	0	125	151	-	20.4
<i>Owned</i>	1273	1187	4126	-6.7	247.4	2145	1605	1728	-25.2	7.7
Total	4578	4521	5564	-1.3	23.1	2145	1731	1879	-19.3	8.6
Machine Labour										
<i>Hired</i>	7697	7703	6282	0.1	-18.5	4803	13392	7856	178.9	-41.3
<i>Owned</i>	1309	1470	1756	12.3	19.4	11193	4790	6558	-57.2	36.9
Total	9006	9174	8037	1.9	-12.4	15996	18182	14414	13.7	-20.7
Seed	10999	12328	11957	12.1	-3.0	12909	11659	11536	-9.7	-1.1
Fertilisers and Manure										
<i>Fertilisers</i>	3161	3934	3759	24.5	-4.4	3714	3926	4976	5.7	26.8
<i>Manure</i>	124	106	184	-15.1	74.0	4947	5588	5146	13.0	-7.9
Total	3285	4039	3942	23.0	-2.4	8661	9514	10122	9.9	6.4
Other Inputs										
Insecticides	368	423	535	15.0	26.4	297	329	335	10.7	1.8
Irrigation charges	896	728	681	-18.8	-6.5	4957	4514	3643	-8.9	-19.3
Crop Insurance	0	0	0	-	-	0	0	0	-	-
Interest on working capital	1335	1395	1396	4.5	0.0	2055	2239	1964	8.9	-12.3
Miscellaneous	136	159	213	17.2	33.7	280	303	351	8.0	16.0
Fixed Cost	16030	18997	15528	18.5	-18.3	19299	19498	20414	1.0	4.7
Rental value of owned land	14923	17817	14222	19.4	-20.2	14404	13942	16047	-3.2	15.1
Rent paid for leased-in land	0	0	0	-	-	0	0	0	-	-
Land revenue, cesses & taxes	6	5	5	-12.1	6.0	22	25	21	10.0	-15.9
Depreciation on implements & Farm buildings	204	235	278	15.4	18.0	312	397	482	27.2	21.4
Interest on fixed capital	897	940	1023	4.7	8.8	4561	5134	3864	12.6	-24.7
Yield(Qtl/ha)	12.23	11.79	8.08	-3.6	-31.5	11.74	11.86	12.09	1.0	1.9

Source: ES&E Division, Ministry of Agriculture and Farmers Welfare



Annex Table 5.6(i): Groundnut : Break-up of Cost of Cultivation

(₹/ha)

Cost Items	Odisha					Rajasthan				
	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23
Operational Cost	49757	55899	55926	12.3	0.0	66625	77683	91746	16.6	18.1
Human Labour										
<i>Casual</i>	11848	15731	12481	32.8	-20.7	2415	4237	6468	75.5	52.7
<i>Attached</i>	4	1	0	-75.1	-	51	31	0	-38.6	-
<i>Family</i>	19106	19909	22089	4.2	11.0	14303	16464	19066	15.1	15.8
Total	30957	35641	34570	15.1	-3.0	16769	20732	25535	23.6	23.2
Bullock Labour										
<i>Hired</i>	1170	1248	2207	6.7	76.8	4	72	63	1612.4	-12.3
<i>Owned</i>	1663	1856	1140	11.6	-38.5	191	96	80	-49.9	-17.0
Total	2833	3104	3347	9.6	7.8	196	168	143	-14.1	-14.9
Machine Labour										
<i>Hired</i>	2696	2903	3275	7.7	12.8	14512	17190	15420	18.5	-10.3
<i>Owned</i>	174	218	175	24.9	-19.7	1468	1986	4459	35.3	124.5
Total	2871	3121	3450	8.7	10.5	15979	19176	19878	20.0	3.7
Seed	9766	10215	10174	4.6	-0.4	17848	20236	24770	13.4	22.4
Fertilisers and Manure										
<i>Fertilisers</i>	2168	2474	3050	14.1	23.3	1489	2449	2855	64.5	16.6
<i>Manure</i>	132	130	131	-1.9	1.0	1113	1250	1316	12.3	5.3
Total	2300	2604	3181	13.2	22.2	2602	3699	4171	42.2	12.7
Other Inputs										
Insecticides	0	0	53	-	-	1414	1326	1868	-6.3	40.9
Irrigation charges	68	93	92	36.0	-0.5	10146	10382	13030	2.3	25.5
Crop Insurance	0	0	0	-	-	0	0	0	-	-
Interest on working capital	929	1091	1025	17.4	-6.0	1585	1855	2202	17.0	18.7
Miscellaneous	33	31	33	-6.2	7.1	86	109	150	27.6	37.0
Fixed Cost	13568	14864	16048	9.5	8.0	24263	21806	24542	-10.1	12.5
Rental value of owned land	12781	14229	15341	11.3	7.8	20044	18303	18966	-8.7	3.6
Rent paid for leased-in land	0	0	0	-	-	995	1326	2955	33.2	122.8
Land revenue, cesses & taxes	16	14	14	-7.5	-5.1	0	0	0	-	-
Depreciation on implements & Farm buildings	212	201	230	-5.4	14.3	376	398	428	5.8	7.7
Interest on fixed capital	560	420	463	-25.0	10.4	2847	1779	2193	-37.5	23.3
Yield(Qtl/ha)	7.81	8.56	8.90	9.6	4.0	25.62	26.26	25.45	2.5	-3.1

Source: ES&E Division, Ministry of Agriculture and Farmers Welfare



Annex Table 5.6(i): Groundnut : Break-up of Cost of Cultivation

(₹/ha)

Cost Items	Tamil Nadu					Telangana				
	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23
Operational Cost	67412	75838	82014	12.5	8.1	94879	104036	104582	9.7	0.5
Human Labour										
<i>Casual</i>	22432	25673	27715	14.4	8.0	23505	27710	26333	17.9	-5.0
<i>Attached</i>	26	85	0	232.2	-	1050	1149	0	9.4	-
<i>Family</i>	9428	11802	13014	25.2	10.3	15169	13559	13577	-10.6	0.1
Total	31887	37560	40729	17.8	8.4	39724	42418	39910	6.8	-5.9
Bullock Labour										
<i>Hired</i>	503	290	292	-42.3	0.5	1425	3088	51	116.6	-98.4
<i>Owned</i>	8	0	55	-	-	1895	389	0	-79.5	-
Total	511	290	346	-43.1	19.3	3320	3477	51	4.7	-98.5
Machine Labour										
<i>Hired</i>	7884	9301	10788	18.0	16.0	12217	14320	15251	17.2	6.5
<i>Owned</i>	979	332	261	-66.1	-21.3	1278	2159	6817	68.9	215.7
Total	8863	9633	11049	8.7	14.7	13495	16479	22068	22.1	33.9
Seed	14368	14321	14699	-0.3	2.6	24155	26092	28287	8.0	8.4
Fertilisers and Manure										
<i>Fertilisers</i>	4369	5800	6389	32.8	10.2	6644	7328	5717	10.3	-22.0
<i>Manure</i>	2174	2405	2401	10.6	-0.2	56	0	0	-	-
Total	6543	8205	8790	25.4	7.1	6700	7328	5717	9.4	-22.0
Other Inputs										
Insecticides	813	850	1085	4.5	27.6	3971	4362	4639	9.8	6.4
Irrigation charges	2480	2781	2960	12.1	6.4	947	986	1042	4.1	5.7
Crop Insurance	0	0	0	-	-	0	0	0	-	-
Interest on working capital	1757	1941	2091	10.4	7.8	2415	2742	2758	13.5	0.6
Miscellaneous	190	258	265	35.2	2.8	151	154	110	2.1	-28.4
Fixed Cost	21103	24960	27965	18.3	12.0	39343	39289	14399	-0.1	-63.4
Rental value of owned land	15647	19289	22927	23.3	18.9	37912	37174	2743	-1.9	-92.6
Rent paid for leased-in land	122	14	0	-88.4	-	338	766	9762	126.5	1175.1
Land revenue, cesses & taxes	6	7	7	10.0	7.6	0	0	0	-	-
Depreciation on implements & Farm buildings	550	551	600	0.2	9.0	118	165	202	39.6	22.5
Interest on fixed capital	4778	5099	4431	6.7	-13.1	975	1184	1692	21.5	42.8
Yield(Qtl/ha)	14.76	15.70	16.99	6.4	8.2	17.84	19.42	20.14	8.8	3.7

Source: ES&E Division, Ministry of Agriculture and Farmers Welfare



Annex Table 5.6(i): Groundnut : Break-up of Cost of Cultivation

(₹/ha)

Cost Items	Uttar Pradesh					West Bengal				
	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23
Operational Cost	35060	39398	39661	12.4	0.7	102495	108358	109351	5.7	0.9
Human Labour										
<i>Casual</i>	3353	5577	5965	66.3	7.0	16711	23329	25287	39.6	8.4
<i>Attached</i>	0	0	0	-	-	0	0	0	-	-
<i>Family</i>	13159	12165	13105	-7.6	7.7	51515	46876	47464	-9.0	1.3
Total	16512	17742	19071	7.4	7.5	68226	70205	72751	2.9	3.6
Bullock Labour										
<i>Hired</i>	0	0	0	-	-	0	0	0	-	-
<i>Owned</i>	0	0	0	-	-	1634	1734	0	6.1	-
Total	0	0	0	-	-	1634	1734	0	6.1	-
Machine Labour										
<i>Hired</i>	7813	8682	6877	11.1	-20.8	1667	2788	3651	67.3	30.9
<i>Owned</i>	99	186	204	87.6	9.3	1974	1886	765	-4.5	-59.4
Total	7912	8868	7080	12.1	-20.2	3642	4674	4416	28.4	-5.5
Seed	7624	9099	9412	19.3	3.4	17154	19255	19619	12.3	1.9
Fertilisers and Manure										
<i>Fertilisers</i>	1646	1554	1812	-5.6	16.6	3872	3954	4517	2.1	14.3
<i>Manure</i>	0	0	0	-	-	0	321	0	-	-
Total	1646	1554	1812	-5.6	16.6	3872	4275	4517	10.4	5.7
Other Inputs										
Insecticides	156	470	530	201.4	12.7	3146	2702	3072	-14.1	13.7
Irrigation charges	540	820	929	51.8	13.2	3265	3637	3088	11.4	-15.1
Crop Insurance	0	0	0	-	-	0	0	0	-	-
Interest on working capital	664	825	805	24.3	-2.5	1545	1863	1875	20.6	0.7
Miscellaneous	5	19	22	263.4	15.0	11	13	11	14.8	-12.1
Fixed Cost	12967	13372	18480	3.1	38.2	39885	49056	51418	23.0	4.8
Rental value of owned land	11155	11276	16569	1.1	46.9	38160	47007	49355	23.2	5.0
Rent paid for leased-in land	3	275	0	7929.4	-	117	191	0	62.6	-
Land revenue, cesses & taxes	1	1	2	-4.9	25.9	0	0	0	-	-
Depreciation on implements & Farm buildings	440	381	403	-13.4	5.9	382	419	486	9.5	16.1
Interest on fixed capital	1368	1439	1506	5.2	4.6	1226	1440	1578	17.5	9.5
Yield(Qtl/ha)	10.44	12.60	13.13	20.7	4.2	32.59	33.96	34.93	4.2	2.9

Source: ES&E Division, Ministry of Agriculture and Farmers Welfare



Annex Table 5.6(j): Soybean : Break-up of Cost of Cultivation

(₹/ha)

Cost Items	Gujarat					Karnataka				
	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23
Operational Cost	40624	42952	42830	5.7	-0.3	38855	45456	53031	17.0	16.7
Human Labour										
<i>Casual</i>	13014	13230	12051	1.7	-8.9	9206	10170	14276	10.5	40.4
<i>Attached</i>	0	0	0	-	-	0	0	1257	-	-
<i>Family</i>	6770	4571	5154	-32.5	12.8	4792	5561	6035	16.0	8.5
Total	19785	17801	17205	-10.0	-3.3	13998	15730	21568	12.4	37.1
Bullock Labour										
<i>Hired</i>	894	1388	1201	55.2	-13.5	552	990	2041	79.5	106.1
<i>Owned</i>	804	240	394	-70.1	63.9	1740	1267	1388	-27.2	9.6
Total	1698	1628	1594	-4.1	-2.1	2291	2256	3429	-1.5	52.0
Machine Labour										
<i>Hired</i>	5104	6994	6409	37.0	-8.4	12591	14434	12413	14.6	-14.0
<i>Owned</i>	1283	1910	2936	48.8	53.7	137	162	235	18.4	44.6
Total	6388	8904	9345	39.4	5.0	12728	14596	12648	14.7	-13.3
Seed	5544	5955	6511	7.4	9.3	3694	4174	5085	13.0	21.8
Fertilisers and Manure										
<i>Fertilisers</i>	1600	2437	2169	52.2	-11.0	3012	3733	3540	24.0	-5.2
<i>Manure</i>	3450	2689	2778	-22.1	3.3	0	1518	2536	-	67.1
Total	5051	5125	4947	1.5	-3.5	3012	5251	6076	74.3	15.7
Other Inputs										
Insecticides	901	1993	1593	121.1	-20.1	1698	1872	2448	10.3	30.8
Irrigation charges	128	309	379	142.0	22.7	192	153	120	-20.5	-21.3
Crop Insurance	0	0	0	-	-	19	21	23	11.7	10.8
Interest on working capital	1026	1163	1142	13.4	-1.8	1032	1209	1424	17.1	17.8
Miscellaneous	105	73	114	-30.3	55.3	191	193	210	1.1	8.9
Fixed Cost	6704	9184	10560	37.0	15.0	15790	16645	18135	5.4	9.0
Rental value of owned land	4876	7162	8297	46.9	15.9	14958	15917	17559	6.4	10.3
Rent paid for leased-in land	0	133	257	-	93.6	0	0	0	-	-
Land revenue, cesses & taxes	6	6	7	-3.0	8.5	8	7	8	-8.0	10.2
Depreciation on implements & Farm buildings	247	259	281	5.2	8.1	128	124	129	-3.1	3.8
Interest on fixed capital	1575	1624	1719	3.1	5.8	696	597	438	-14.3	-26.6
Yield(Qtl/ha)	6.30	8.40	9.36	33.3	11.4	13.55	15.53	14.52	14.6	-6.5

Source: ES&E Division, Ministry of Agriculture and Farmers Welfare



Annex Table 5.6(j): Soybean : Break-up of Cost of Cultivation

(₹/ha)

Cost Items	Madhya Pradesh					Maharashtra				
	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23
Operational Cost	28737	29969	33596	4.3	12.1	53147	59285	57651	11.5	-2.8
Human Labour										
<i>Casual</i>	3470	3660	3837	5.5	4.8	12897	13881	12566	7.6	-9.5
<i>Attached</i>	74	78	80	5.7	2.5	649	560	360	-13.8	-35.6
<i>Family</i>	4623	4734	4834	2.4	2.1	4587	5590	5896	21.9	5.5
Total	8167	8472	8752	3.7	3.3	18133	20031	18822	10.5	-6.0
Bullock Labour										
<i>Hired</i>	8	8	4	-0.8	-46.8	1022	851	1289	-16.7	51.4
<i>Owned</i>	336	342	413	1.6	20.9	3322	3608	3558	8.6	-1.4
Total	345	350	418	1.6	19.3	4343	4459	4847	2.7	8.7
Machine Labour										
<i>Hired</i>	7034	7338	8043	4.3	9.6	11384	12553	11734	10.3	-6.5
<i>Owned</i>	647	618	970	-4.4	56.9	1742	1525	3028	-12.5	98.6
Total	7680	7956	9013	3.6	13.3	13126	14077	14762	7.2	4.9
Seed	5670	5918	7573	4.4	28.0	7546	9064	7908	20.1	-12.8
Fertilisers and Manure										
<i>Fertilisers</i>	2128	2325	2439	9.3	4.9	4070	5037	4450	23.8	-11.7
<i>Manure</i>	945	979	1087	3.6	11.1	909	835	732	-8.2	-12.3
Total	3073	3304	3526	7.5	6.7	4979	5872	5181	17.9	-11.8
Other Inputs										
Insecticides	2519	2639	2831	4.8	7.3	3026	3552	3897	17.4	9.7
Irrigation charges	6	0	1	-	-	166	161	264	-3.1	63.5
Crop Insurance	472	488	513	3.4	5.1	252	331	272	31.3	-17.8
Interest on working capital	731	765	872	4.6	14.0	1472	1627	1568	10.6	-3.6
Miscellaneous	76	78	98	3.0	25.3	103	111	130	7.6	17.5
Fixed Cost	12631	13835	14935	9.5	8.0	16758	17111	18035	2.1	5.4
Rental value of owned land	10079	11168	12039	10.8	7.8	13643	13772	14132	0.9	2.6
Rent paid for leased-in land	0	0	0	-	-	0	0	0	-	-
Land revenue, cesses & taxes	4	4	4	8.0	-6.4	24	23	21	-5.3	-7.7
Depreciation on implements & Farm buildings	771	797	928	3.4	16.5	546	624	707	14.3	13.3
Interest on fixed capital	1777	1866	1964	5.0	5.2	2546	2693	3175	5.8	17.9
Yield(Qtl/ha)	10.74	10.89	12.10	1.4	11.1	16.08	15.97	15.54	-0.7	-2.7

Source: ES&E Division, Ministry of Agriculture and Farmers Welfare



Annex Table 5.6(j): Soybean : Break-up of Cost of Cultivation

(₹/ha)

Cost Items	Rajasthan					Telangana				
	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23
Operational Cost	38568	44205	48397	14.6	9.5	35758	35827	37870	0.2	5.7
Human Labour										
<i>Casual</i>	6564	4986	6506	-24.0	30.5	7323	8305	8794	13.4	5.9
<i>Attached</i>	627	608	88	-3.1	-85.6	559	475	0	-15.1	-
<i>Family</i>	9328	11236	14842	20.5	32.1	7714	6558	5196	-15.0	-20.8
Total	16519	16831	21436	1.9	27.4	15595	15338	13990	-1.7	-8.8
Bullock Labour										
<i>Hired</i>	66	85	0	29.1	-	196	99	36	-49.6	-63.9
<i>Owned</i>	113	91	206	-18.9	125.3	1197	413	510	-65.5	23.5
Total	179	176	206	-1.2	16.6	1393	512	546	-63.3	6.6
Machine Labour										
<i>Hired</i>	6924	8558	7352	23.6	-14.1	8184	6807	11003	-16.8	61.6
<i>Owned</i>	2128	2254	3148	5.9	39.6	1261	2709	82	114.8	-97.0
Total	9052	10812	10500	19.4	-2.9	9445	9516	11085	0.8	16.5
Seed	8910	12055	11085	35.3	-8.0	3822	4936	5723	29.1	15.9
Fertilisers and Manure										
<i>Fertilisers</i>	575	849	1251	47.7	47.4	2625	2137	2857	-18.6	33.7
<i>Manure</i>	0	0	0	-	-	0	0	0	-	-
Total	575	849	1251	47.7	47.4	2625	2137	2857	-18.6	33.7
Other Inputs										
Insecticides	2306	2328	2670	1.0	14.7	1782	2258	2595	26.7	14.9
Irrigation charges	136	143	218	5.0	52.8	161	170	0	6.0	-
Crop Insurance	0	0	0	-	-	0	0	0	-	-
Interest on working capital	886	999	1017	12.8	1.8	850	887	990	4.4	11.6
Miscellaneous	5	13	15	134.8	20.3	85	73	84	-13.8	15.2
Fixed Cost	15367	13767	14371	-10.4	4.4	14303	16273	17957	13.8	10.4
Rental value of owned land	8718	10811	10891	24.0	0.7	12462	14155	13662	13.6	-3.5
Rent paid for leased-in land	3176	34	0	-98.9	-	0	0	2134	-	-
Land revenue, cesses & taxes	1	0	0	-	-	0	0	0	-	-
Depreciation on implements & Farm buildings	922	835	950	-9.4	13.8	255	298	307	17.2	2.9
Interest on fixed capital	2549	2088	2530	-18.1	21.2	1586	1819	1854	14.6	2.0
Yield(Qtl/ha)	9.52	12.21	10.74	28.2	-12.0	8.35	9.87	9.29	18.2	-5.9

Source: ES&E Division, Ministry of Agriculture and Farmers Welfare



Annex Table 5.6(k): Sunflower : Break-up of Cost of Cultivation

(₹/ha)

Cost Items	Andhra Pradesh					Karnataka				
	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23
Operational Cost	40412	43501	42557	7.6	-2.2	38650	39283	36351	1.6	-7.5
Human Labour										
<i>Casual</i>	9336	11868	11300	27.1	-4.8	9234	9280	8057	0.5	-13.2
<i>Attached</i>	0	105	0	-	-	0	0	11	-	-
<i>Family</i>	6100	5021	6497	-17.7	29.4	6069	7966	4113	31.3	-48.4
Total	15436	16994	17796	10.1	4.7	15303	17246	12180	12.7	-29.4
Bullock Labour										
<i>Hired</i>	955	986	1071	3.2	8.6	1600	1221	2369	-23.7	94.0
<i>Owned</i>	421	188	154	-55.3	-18.0	309	778	1093	151.7	40.5
Total	1376	1174	1226	-14.7	4.4	1909	1998	3461	4.7	73.2
Machine Labour										
<i>Hired</i>	8394	8564	8260	2.0	-3.5	7103	5393	8202	-24.1	52.1
<i>Owned</i>	1008	1053	51	4.5	-95.2	3053	3340	479	9.4	-85.7
Total	9402	9617	8311	2.3	-13.6	10155	8732	8681	-14.0	-0.6
Seed	3228	3786	4194	17.3	10.8	3336	3658	4694	9.7	28.3
Fertilisers and Manure										
<i>Fertilisers</i>	6464	7034	5967	8.8	-15.2	4067	3701	4226	-9.0	14.2
<i>Manure</i>	0	0	0	-	-	1786	1911	1121	7.0	-41.4
Total	6464	7034	5967	8.8	-15.2	5853	5612	5346	-4.1	-4.7
Other Inputs										
Insecticides	2619	2803	3071	7.0	9.6	558	502	439	-10.0	-12.6
Irrigation charges	661	732	703	10.6	-3.9	444	500	491	12.6	-1.7
Crop Insurance	0	0	0	-	-	7	0	0	-	-
Interest on working capital	1040	1166	1093	12.1	-6.3	987	949	977	-3.9	2.9
Miscellaneous	186	196	197	5.3	0.2	99	86	80	-13.0	-7.7
Fixed Cost	21137	25066	25470	18.6	1.6	13969	15279	10671	9.4	-30.2
Rental value of owned land	19577	23244	22217	18.7	-4.4	11624	12647	1052	8.8	-91.7
Rent paid for leased-in land	0	0	1558	-	-	0	0	5934	-	-
Land revenue, cesses & taxes	0	0	0	-	-	9	8	6	-7.1	-25.3
Depreciation on implements & Farm buildings	176	197	202	12.2	2.5	275	330	833	19.9	152.3
Interest on fixed capital	1384	1624	1492	17.4	-8.1	2061	2293	2845	11.3	24.0
Yield(Qtl/ha)	10.24	11.88	11.94	16.0	0.5	7.87	8.08	7.57	2.7	-6.3

Source: ES&E Division, Ministry of Agriculture and Farmers Welfare



Annex Table 5.6(I): Sesamum : Break-up of Cost of Cultivation

(₹/ha)

Cost Items	Andhra Pradesh					Gujarat				
	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23
Operational Cost	22901	22598	23235	-1.3	2.8	38045	43011	45790	13.1	6.5
Human Labour										
<i>Casual</i>	10098	9980	10228	-1.2	2.5	7141	10158	11457	42.2	12.8
<i>Attached</i>	0	59	0	-	-	0	0	0	-	-
<i>Family</i>	1875	2367	2682	26.3	13.3	12179	12314	12271	1.1	-0.4
Total	11973	12406	12910	3.6	4.1	19320	22472	23727	16.3	5.6
Bullock Labour										
<i>Hired</i>	766	645	0	-15.8	-	143	0	17	-	-
<i>Owned</i>	0	269	477	-	77.4	65	144	158	121.3	9.1
Total	766	914	477	19.3	-47.8	208	144	175	-30.7	21.2
Machine Labour										
<i>Hired</i>	4010	3630	5113	-9.5	40.9	4084	5871	6430	43.8	9.5
<i>Owned</i>	1817	1270	633	-30.1	-50.1	3356	3076	3672	-8.3	19.4
Total	5827	4900	5746	-15.9	17.3	7440	8948	10102	20.3	12.9
Seed	2065	2438	2620	18.1	7.5	1428	1548	1719	8.4	11.0
Fertilisers and Manure										
<i>Fertilisers</i>	438	496	540	13.4	8.8	4178	3736	3858	-10.6	3.3
<i>Manure</i>	17	18	0	1.7	-	333	380	413	14.3	8.5
Total	455	514	540	13.0	5.1	4511	4116	4271	-8.7	3.8
Other Inputs										
Insecticides	413	510	0	23.4	-	2010	2361	2425	17.5	2.7
Irrigation charges	754	289	304	-61.6	5.1	2249	2398	2244	6.6	-6.4
Crop Insurance	0	0	0	-	-	0	0	0	-	-
Interest on working capital	637	613	623	-3.8	1.6	784	930	1016	18.7	9.2
Miscellaneous	11	14	15	25.5	9.7	95	94	112	-1.2	18.9
Fixed Cost	9470	9839	8606	3.9	-12.5	10177	11831	14020	16.3	18.5
Rental value of owned land	8767	7869	3009	-10.2	-61.8	7763	8770	10905	13.0	24.3
Rent paid for leased-in land	0	1006	4526	-	350.0	48	485	59	908.0	-87.8
Land revenue, cesses & taxes	0	0	0	-	-	5	2	3	-57.3	25.0
Depreciation on implements & Farm buildings	65	80	91	22.8	13.0	185	193	218	4.3	13.4
Interest on fixed capital	637	884	980	38.7	10.9	2176	2382	2835	9.5	19.0
Yield(Qtl/ha)	3.61	3.31	3.60	-8.4	8.8	3.21	5.52	5.93	71.7	7.4

Source: ES&E Division, Ministry of Agriculture and Farmers Welfare



Annex Table 5.6(l): Sesamum : Break-up of Cost of Cultivation

(₹/ha)

Cost Items	Madhya Pradesh					Rajasthan				
	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23
Operational Cost	25664	26935	30839	5.0	14.5	22891	24237	26574	5.9	9.6
Human Labour										
<i>Casual</i>	8896	9182	8913	3.2	-2.9	3400	2483	1977	-27.0	-20.4
<i>Attached</i>	0	0	0	-	-	0	0	0	-	-
<i>Family</i>	5343	5598	5991	4.8	7.0	14646	16641	17756	13.6	6.7
Total	14239	14780	14904	3.8	0.8	18046	19124	19733	6.0	3.2
Bullock Labour										
<i>Hired</i>	0	0	0	-	-	0	0	0	-	-
<i>Owned</i>	0	506	3267	-	545.7	0	463	0	-	-
Total	0	506	3267	-	545.7	0	463	0	-	-
Machine Labour										
<i>Hired</i>	6033	6157	6736	2.1	9.4	2043	1338	3499	-34.5	161.6
<i>Owned</i>	75	76	165	2.5	115.8	1564	2093	1604	33.8	-23.4
Total	6107	6234	6901	2.1	10.7	3607	3430	5102	-4.9	48.7
Seed	1491	1575	1450	5.6	-7.9	510	588	785	15.3	33.5
Fertilisers and Manure										
<i>Fertilisers</i>	1676	1698	1176	1.3	-30.7	458	380	644	-17.2	69.5
<i>Manure</i>	663	617	1537	-6.9	149.0	0	0	0	-	-
Total	2339	2316	2713	-1.0	17.2	458	380	644	-17.2	69.5
Other Inputs										
Insecticides	667	675	661	1.1	-2.1	0	1	6	-	328.2
Irrigation charges	0	0	0	-	-	0	0	15	-	-
Crop Insurance	131	133	96	1.2	-27.5	0	0	0	-	-
Interest on working capital	616	647	753	5.0	16.5	250	230	267	-7.9	16.1
Miscellaneous	75	71	94	-4.7	31.4	20	21	22	2.6	5.8
Fixed Cost	10993	11366	13526	3.4	19.0	6059	7156	8040	18.1	12.4
Rental value of owned land	9788	10111	12502	3.3	23.6	4165	4672	5002	12.2	7.1
Rent paid for leased-in land	0	0	0	-	-	0	0	0	-	-
Land revenue, cesses & taxes	3	2	2	-23.6	7.7	0	0	0	-	-
Depreciation on implements & Farm buildings	454	476	341	4.9	-28.3	220	280	353	27.1	25.9
Interest on fixed capital	747	777	680	3.9	-12.4	1674	2203	2685	31.7	21.9
Yield(Qtl/ha)	5.77	6.18	4.79	7.1	-22.5	2.22	2.39	2.75	7.6	15.1

Source: ES&E Division, Ministry of Agriculture and Farmers Welfare



Annex Table 5.6(I): Sesamum : Break-up of Cost of Cultivation

(₹/ha)

Cost Items	Uttar Pradesh					West Bengal				
	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23
Operational Cost	17758	19091	20715	7.5	8.5	49156	52578	50062	7.0	-4.8
Human Labour										
<i>Casual</i>	3239	4319	5503	33.3	27.4	16518	16556	13624	0.2	-17.7
<i>Attached</i>	0	0	10	-	-	0	0	0	-	-
<i>Family</i>	9570	9366	9571	-2.1	2.2	22896	23330	22513	1.9	-3.5
Total	12808	13684	15084	6.8	10.2	39414	39885	36137	1.2	-9.4
Bullock Labour										
<i>Hired</i>	0	0	0	-	-	0	0	123	-	-
<i>Owned</i>	0	0	0	-	-	0	118	0	-	-
Total	0	0	0	-	-	0	118	123	-	3.9
Machine Labour										
<i>Hired</i>	3198	3917	4080	22.5	4.2	2376	4533	5089	90.8	12.3
<i>Owned</i>	474	154	128	-67.5	-16.8	83	214	129	155.9	-39.6
Total	3672	4071	4209	10.9	3.4	2460	4747	5218	93.0	9.9
Seed	694	618	678	-10.9	9.6	882	873	1020	-1.1	16.9
Fertilisers and Manure										
<i>Fertilisers</i>	11	0	22	-	-	3124	3522	3677	12.7	4.4
<i>Manure</i>	0	0	0	-	-	0	21	20	-	-5.1
Total	11	0	22	-	-	3124	3543	3697	13.4	4.3
Other Inputs										
Insecticides	0	9	0	-	-	951	1011	1126	6.4	11.4
Irrigation charges	323	411	385	27.2	-6.3	1529	1512	1905	-1.1	26.0
Crop Insurance	0	0	0	-	-	0	0	0	-	-
Interest on working capital	248	295	338	18.8	14.6	796	886	835	11.4	-5.8
Miscellaneous	2	2	0	11.1	-	0	2	2	2011.1	-10.5
Fixed Cost	7123	7866	8129	10.4	3.3	17424	20996	25794	20.5	22.9
Rental value of owned land	5698	6314	6422	10.8	1.7	16126	19631	23654	21.7	20.5
Rent paid for leased-in land	7	12	0	65.9	-	48	0	745	-	-
Land revenue, cesses & taxes	3	3	2	2.6	-15.0	0	0	0	-	-
Depreciation on implements & Farm buildings	342	352	404	3.0	14.7	200	221	241	10.4	9.2
Interest on fixed capital	1074	1185	1300	10.3	9.8	1050	1144	1153	9.0	0.8
Yield(Qtl/ha)	1.77	2.18	2.35	23.3	7.8	10.18	11.70	12.36	15.0	5.6

Source: ES&E Division, Ministry of Agriculture and Farmers Welfare



Annex Table 5.6(m): Nigerseed : Break-up of Cost of Cultivation

(₹/ha)

Cost Items	Odisha				
	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23
Operational Cost	21198	23276	21949	9.8	-5.7
Human Labour					
<i>Casual</i>	215	209	526	-2.5	151.3
<i>Attached</i>	0	0	0	-	-
<i>Family</i>	14333	15660	16582	9.3	5.9
Total	14548	15869	17108	9.1	7.8
Bullock Labour					
<i>Hired</i>	40	36	107	-10.8	197.3
<i>Owned</i>	5674	6364	3635	12.1	-42.9
Total	5715	6400	3742	12.0	-41.5
Machine Labour					
<i>Hired</i>	58	35	123	-39.7	253.2
<i>Owned</i>	0	0	18	-	-
Total	58	35	141	-39.7	305.3
Seed	670	742	790	10.9	6.5
Fertilisers and Manure					
<i>Fertilisers</i>	0	0	0	-	-
<i>Manure</i>	0	0	6	-	-
Total	0	0	6	-	-
<i>Other Inputs</i>					
Insecticides	0	0	0	-	-
Irrigation charges	0	0	0	-	-
Crop Insurance	0	0	0	-	-
Interest on working capital	208	231	163	11.0	-29.5
Miscellaneous	0	0	0	-	-
Fixed Cost	6114	6561	6742	7.3	2.8
Rental value of owned land	5627	6073	6210	7.9	2.2
Rent paid for leased-in land	0	0	0	-	-
Land revenue, cesses & taxes	16	16	15	-0.4	-3.7
Depreciation on implements & Farm buildings	243	260	246	7.3	-5.7
Interest on fixed capital	228	211	272	-7.3	28.5
Yield(Qtl/ha)	3.28	3.58	3.22	9.1	-10.1

Source: ES&E Division, Ministry of Agriculture and Farmers Welfare



Annex Table 5.6(n): Cotton : Break-up of Cost of Cultivation

(₹/ha)

Cost Items	Andhra Pradesh					Gujarat				
	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23
Operational Cost	66141	71221	74393	7.7	4.5	64901	68252	77687	5.2	13.8
Human Labour										
<i>Casual</i>	28091	28397	28614	1.1	0.8	19908	20785	24039	4.4	15.7
<i>Attached</i>	0	0	60	-	-	41	33	0	-18.0	-
<i>Family</i>	4636	6118	7791	32.0	27.4	13348	14631	15720	9.6	7.4
Total	32727	34515	36465	5.5	5.6	33296	35449	39760	6.5	12.2
Bullock Labour										
<i>Hired</i>	2650	2376	1916	-10.3	-19.4	839	430	365	-48.8	-15.1
<i>Owned</i>	1513	1553	1937	2.7	24.8	548	866	945	58.1	9.2
Total	4163	3929	3853	-5.6	-1.9	1387	1295	1310	-6.6	1.1
Machine Labour										
<i>Hired</i>	5769	6719	7106	16.5	5.8	7022	7576	9044	7.9	19.4
<i>Owned</i>	1349	1344	1526	-0.4	13.5	5006	4745	5718	-5.2	20.5
Total	7118	8062	8631	13.3	7.1	12029	12322	14762	2.4	19.8
Seed	5140	5655	6169	10.0	9.1	3301	3567	3765	8.1	5.6
Fertilisers and Manure										
<i>Fertilisers</i>	7392	8765	8433	18.6	-3.8	4549	4880	5733	7.3	17.5
<i>Manure</i>	1247	1240	1263	-0.6	1.9	2416	2150	2366	-11.0	10.0
Total	8640	10005	9696	15.8	-3.1	6965	7030	8098	0.9	15.2
Other Inputs										
Insecticides	6194	6748	7224	8.9	7.1	3703	4177	4993	12.8	19.5
Irrigation charges	179	209	206	16.9	-1.4	2461	2624	2979	6.6	13.5
Crop Insurance	0	0	0	-	-	0	0	0	-	-
Interest on working capital	1864	1973	2018	5.8	2.3	1562	1625	1878	4.0	15.6
Miscellaneous	117	125	130	6.7	4.2	197	163	142	-17.5	-12.5
Fixed Cost	37570	33460	33198	-10.9	-0.8	25392	29832	29431	17.5	-1.3
Rental value of owned land	28357	25353	24896	-10.6	-1.8	18519	22074	21429	19.2	-2.9
Rent paid for leased-in land	6272	5038	4913	-19.7	-2.5	898	1173	1216	30.6	3.7
Land revenue, cesses & taxes	0	0	98	-	-	8	9	9	6.1	0.7
Depreciation on implements & Farm buildings	351	361	385	2.8	6.5	536	551	621	2.8	12.7
Interest on fixed capital	2590	2707	2906	4.5	7.4	5431	6025	6156	11.0	2.2
Yield(Qtl/ha)	16.61	15.09	16.12	-9.1	6.8	14.82	16.82	15.82	13.5	-5.9

Source: ES&E Division, Ministry of Agriculture and Farmers Welfare



Annex Table 5.6(n): Cotton : Break-up of Cost of Cultivation

(₹/ha)

Cost Items	Haryana					Karnataka				
	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23
Operational Cost	53096	53974	52557	1.7	-2.6	57766	58433	66880	1.2	14.5
Human Labour										
<i>Casual</i>	11028	11372	10735	3.1	-5.6	20097	19162	21146	-4.7	10.3
<i>Attached</i>	0	0	0	-	-	29	31	108	7.7	250.1
<i>Family</i>	16371	16114	14828	-1.6	-8.0	10238	11167	10087	9.1	-9.7
Total	27399	27486	25563	0.3	-7.0	30364	30360	31341	0.0	3.2
Bullock Labour										
<i>Hired</i>	0	12	0	-	-	2915	3110	3378	6.7	8.6
<i>Owned</i>	147	141	0	-4.2	-	1612	1707	3476	5.9	103.7
Total	147	153	0	3.8	-	4527	4817	6854	6.4	42.3
Machine Labour										
<i>Hired</i>	5462	5600	6091	2.5	8.8	6885	5836	7616	-15.2	30.5
<i>Owned</i>	3997	4020	4215	0.6	4.9	1401	1754	1811	25.2	3.2
Total	9458	9620	10306	1.7	7.1	8287	7590	9427	-8.4	24.2
Seed	3759	4039	4212	7.4	4.3	4094	3970	4604	-3.0	16.0
Fertilisers and Manure										
<i>Fertilisers</i>	3899	3905	3954	0.1	1.3	5018	5636	7078	12.3	25.6
<i>Manure</i>	10	0	0	-	-	0	55	53	-	-3.0
Total	3910	3905	3954	-0.1	1.3	5018	5691	7131	13.4	25.3
Other Inputs										
Insecticides	3316	3421	3262	3.2	-4.6	3699	4191	5135	13.3	22.5
Irrigation charges	3634	3821	3714	5.1	-2.8	132	152	397	15.1	160.7
Crop Insurance	299	321	339	7.6	5.5	0	0	0	-	-
Interest on working capital	1113	1147	1143	3.1	-0.3	1440	1432	1721	-0.5	20.2
Miscellaneous	61	62	64	2.0	2.9	205	230	270	12.0	17.7
Fixed Cost	24755	25838	34744	4.4	34.5	22553	26131	25629	15.9	-1.9
Rental value of owned land	19910	20887	29519	4.9	41.3	20584	23919	22331	16.2	-6.6
Rent paid for leased-in land	55	0	0	-	-	0	0	0	-	-
Land revenue, cesses & taxes	0	0	0	-	-	10	9	10	-10.6	9.1
Depreciation on implements & Farm buildings	800	822	757	2.7	-7.9	330	372	829	12.7	122.8
Interest on fixed capital	3990	4130	4469	3.5	8.2	1629	1832	2459	12.4	34.2
Yield(Qtl/ha)	10.89	9.79	13.96	-10.1	42.6	13.29	12.66	13.38	-4.8	5.7

Source: ES&E Division, Ministry of Agriculture and Farmers Welfare



Annex Table 5.6(n): Cotton : Break-up of Cost of Cultivation

(₹/ha)

Cost Items	Madhya Pradesh					Maharashtra				
	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23
Operational Cost	58134	61327	70777	5.5	15.4	76303	81288	86689	6.5	6.6
Human Labour										
<i>Casual</i>	16865	17197	17985	2.0	4.6	24186	26151	25914	8.1	-0.9
<i>Attached</i>	481	453	0	-5.8	-	727	981	1092	34.9	11.3
<i>Family</i>	12949	12609	13260	-2.6	5.2	12315	13947	15923	13.2	14.2
Total	30295	30259	31246	-0.1	3.3	37228	41078	42929	10.3	4.5
Bullock Labour										
<i>Hired</i>	219	204	0	-7.1	-	2549	2796	2714	9.7	-2.9
<i>Owned</i>	6908	7221	13813	4.5	91.3	6619	7063	7343	6.7	4.0
Total	7127	7425	13813	4.2	86.0	9168	9859	10056	7.5	2.0
Machine Labour										
<i>Hired</i>	2881	3095	3050	7.4	-1.5	6982	7929	7044	13.6	-11.2
<i>Owned</i>	1011	1071	2003	5.9	87.1	804	819	2229	1.8	172.1
Total	3892	4166	5053	7.0	21.3	7787	8748	9273	12.3	6.0
Seed	2001	2059	2086	2.9	1.3	3616	3960	4559	9.5	15.1
Fertilisers and Manure										
<i>Fertilisers</i>	7334	7876	9991	7.4	26.9	9692	9142	10282	-5.7	12.5
<i>Manure</i>	2254	2283	0	1.3	-	785	612	703	-22.0	14.8
Total	9588	10159	9991	6.0	-1.7	10477	9755	10986	-6.9	12.6
Other Inputs										
Insecticides	2315	2425	4974	4.8	105.1	4478	3897	4442	-13.0	14.0
Irrigation charges	1396	1461	1512	4.6	3.5	1454	1745	2096	20.0	20.2
Crop Insurance	0	1740	189	-	-89.2	67	107	93	58.1	-12.8
Interest on working capital	1369	1476	1743	7.8	18.1	1939	2041	2144	5.2	5.1
Miscellaneous	150	157	172	4.7	9.0	90	99	110	10.4	10.4
Fixed Cost	28074	28556	34482	1.7	20.8	18407	21901	21847	19.0	-0.2
Rental value of owned land	18244	18370	21653	0.7	17.9	14551	18461	17767	26.9	-3.8
Rent paid for leased-in land	0	0	0	-	-	0	0	0	-	-
Land revenue, cesses & taxes	7	7	5	-3.6	-19.2	29	27	28	-4.8	2.1
Depreciation on implements & Farm buildings	1558	1616	3059	3.7	89.3	564	520	694	-7.9	33.5
Interest on fixed capital	8265	8563	9764	3.6	14.0	3263	2893	3358	-11.4	16.1
Yield(Qtl/ha)	15.05	16.67	16.17	10.8	-3.0	12.73	14.13	15.40	11.0	9.0

Source: ES&E Division, Ministry of Agriculture and Farmers Welfare



Annex Table 5.6(n): Cotton : Break-up of Cost of Cultivation

(₹/ha)

Cost Items	Odisha					Punjab				
	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23
Operational Cost	61466	71403	69375	16.2	-2.8	57768	58633	59291	1.5	1.1
Human Labour										
<i>Casual</i>	18663	20380	16901	9.2	-17.1	16690	15970	16987	-4.3	6.4
<i>Attached</i>	4	16	40	329.5	156.0	1508	1588	1489	5.3	-6.3
<i>Family</i>	19718	22391	23748	13.6	6.1	8006	8326	7689	4.0	-7.7
Total	38385	42787	40689	11.5	-4.9	26204	25885	26165	-1.2	1.1
Bullock Labour										
<i>Hired</i>	469	635	864	35.3	36.1	0	0	0	-	-
<i>Owned</i>	661	718	558	8.7	-22.4	17	17	22	3.4	28.8
Total	1130	1353	1421	19.7	5.0	17	17	22	3.4	28.8
Machine Labour										
<i>Hired</i>	3778	4568	2599	20.9	-43.1	1629	1642	1750	0.8	6.5
<i>Owned</i>	398	216	1379	-45.7	538.5	9755	9798	10282	0.4	4.9
Total	4176	4784	3978	14.6	-16.8	11384	11440	12031	0.5	5.2
Seed	2568	2745	3188	6.9	16.1	5487	5767	5644	5.1	-2.1
Fertilisers and Manure										
<i>Fertilisers</i>	9367	12413	11944	32.5	-3.8	5345	5632	5450	5.4	-3.2
<i>Manure</i>	2622	3512	3548	34.0	1.0	146	146	0	0.3	-
Total	11988	15925	15492	32.8	-2.7	5491	5778	5450	5.2	-5.7
Other Inputs										
Insecticides	1905	2273	3091	19.3	36.0	6753	7279	7463	7.8	2.5
Irrigation charges	0	0	3	-	-	636	658	679	3.3	3.2
Crop Insurance	0	0	0	-	-	0	0	0	-	-
Interest on working capital	1265	1485	1383	17.4	-6.9	1508	1524	1564	1.1	2.6
Miscellaneous	49	51	131	4.6	155.4	288	286	273	-0.8	-4.4
Fixed Cost	21785	26221	24384	20.4	-7.0	48653	50216	44342	3.2	-11.7
Rental value of owned land	19817	24122	23367	21.7	-3.1	39209	41014	34388	4.6	-16.2
Rent paid for leased-in land	1183	1204	0	1.7	-	4778	4651	5048	-2.7	8.5
Land revenue, cesses & taxes	10	10	0	4.0	-	0	0	0	-	-
Depreciation on implements & Farm buildings	313	345	358	10.2	3.7	645	620	663	-3.9	7.0
Interest on fixed capital	461	540	651	17.0	20.5	4021	3932	4242	-2.2	7.9
Yield(Qtl/ha)	13.86	14.42	14.75	4.0	2.3	22.05	17.59	15.80	-20.2	-10.2

Source: ES&E Division, Ministry of Agriculture and Farmers Welfare



Annex Table 5.6(n): Cotton : Break-up of Cost of Cultivation

(₹/ha)

Cost Items	Rajasthan					Tamil Nadu				
	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23
Operational Cost	72274	74107	64970	2.5	-12.3	100701	101930	101695	1.2	-0.2
Human Labour										
<i>Casual</i>	10739	18227	13043	69.7	-28.4	43163	45201	42334	4.7	-6.3
<i>Attached</i>	0	152	0	-	-	490	166	0	-66.2	-
<i>Family</i>	33963	28221	20549	-16.9	-27.2	23282	21182	23674	-9.0	11.8
Total	44702	46600	33592	4.2	-27.9	66935	66549	66009	-0.6	-0.8
Bullock Labour										
<i>Hired</i>	33	57	64	69.5	12.8	0	160	0	-	-
<i>Owned</i>	220	146	140	-33.7	-4.3	0	0	0	-	-
Total	254	203	204	-20.1	0.5	0	160	0	-	-
Machine Labour										
<i>Hired</i>	4755	4779	5550	0.5	16.1	7764	8383	8213	8.0	-2.0
<i>Owned</i>	3083	4340	5705	40.8	31.4	1530	921	1441	-39.8	56.4
Total	7838	9119	11256	16.3	23.4	9294	9304	9654	0.1	3.8
Seed	5426	5175	6257	-4.6	20.9	3136	3418	3493	9.0	2.2
Fertilisers and Manure										
<i>Fertilisers</i>	2653	3472	4696	30.9	35.3	9024	10307	11073	14.2	7.4
<i>Manure</i>	2639	2340	2289	-11.3	-2.2	2755	2708	2364	-1.7	-12.7
Total	5292	5812	6985	9.8	20.2	11779	13015	13437	10.5	3.2
Other Inputs										
Insecticides	3642	2513	2001	-31.0	-20.4	3066	3121	2753	1.8	-11.8
Irrigation charges	3851	3178	3188	-17.5	0.3	3633	3511	3769	-3.3	7.3
Crop Insurance	0	0	0	-	-	329	220	0	-33.2	-
Interest on working capital	1161	1390	1346	19.8	-3.2	2346	2447	2364	4.3	-3.4
Miscellaneous	108	116	142	7.9	22.0	182	186	216	1.9	16.3
Fixed Cost	26943	31227	22454	15.9	-28.1	23792	29060	31512	22.1	8.4
Rental value of owned land	22231	26235	16388	18.0	-37.5	13114	19640	23977	49.8	22.1
Rent paid for leased-in land	116	585	879	406.4	50.2	473	123	0	-73.9	-
Land revenue, cesses & taxes	2	0	0	-	-	7	8	8	7.2	0.8
Depreciation on implements & Farm buildings	607	520	578	-14.3	11.1	1381	1454	1161	5.3	-20.2
Interest on fixed capital	3988	3887	4609	-2.5	18.6	8817	7835	6367	-11.1	-18.7
Yield(Qtl/ha)	19.31	19.61	15.07	1.5	-23.2	21.38	19.67	16.32	-8.0	-17.0

Source: ES&E Division, Ministry of Agriculture and Farmers Welfare



Annex Table 5.6(n): Cotton : Break-up of Cost of Cultivation

(₹/ha)

Cost Items	Telangana				
	2021-22	2022-23	2023-24	% change in 2022-23 over 2021-22	% change in 2023-24 over 2022-23
Operational Cost	80315	89560	87107	11.5	-2.7
Human Labour					
<i>Casual</i>	32191	35425	29732	10.0	-16.1
<i>Attached</i>	705	596	368	-15.5	-38.3
<i>Family</i>	12530	14528	16202	15.9	11.5
Total	45426	50549	46301	11.3	-8.4
Bullock Labour					
<i>Hired</i>	2139	1798	2059	-16.0	14.5
<i>Owned</i>	4449	3366	4036	-24.3	19.9
Total	6588	5164	6095	-21.6	18.0
Machine Labour					
<i>Hired</i>	7767	8967	8180	15.5	-8.8
<i>Owned</i>	1127	2318	4067	105.6	75.5
Total	8894	11285	12247	26.9	8.5
Seed	4280	4515	5083	5.5	12.6
Fertilisers and Manure					
<i>Fertilisers</i>	7728	9346	8185	20.9	-12.4
<i>Manure</i>	331	231	254	-30.3	10.1
Total	8060	9577	8440	18.8	-11.9
<i>Other Inputs</i>					
Insecticides	4801	5986	6615	24.7	10.5
Irrigation charges	45	45	49	-1.1	8.1
Crop Insurance	0	0	0	-	-
Interest on working capital	2054	2274	2149	10.7	-5.5
Miscellaneous	166	164	129	-0.9	-21.7
Fixed Cost	25443	32248	34106	26.7	5.8
Rental value of owned land	23443	29616	22177	26.3	-25.1
Rent paid for leased-in land	705	885	9908	25.6	1020.0
Land revenue, cesses & taxes	0	0	0	-	-
Depreciation on implements & Farm buildings	267	334	405	25.1	21.2
Interest on fixed capital	1028	1414	1615	37.5	14.2
Yield(Qtl/ha)	15.98	16.18	18.57	1.3	14.8

Source: ES&E Division, Ministry of Agriculture and Farmers Welfare



Annex Table 5.7: All-India Projected Costs of Production of Kharif Crops for KMS2024-25 and KMS2025-26

Crop	Cost of Production (₹/qtl)						Percentage Change (2025-26 over 2024-25)		
	2024-25			2025-26			A_2	A_2+FL	C_2
	A_2	A_2+FL	C_2	A_2	A_2+FL	C_2			
Paddy	1189	1533	2008	1238	1579	2090	4.1	3.0	4.1
Jowar	1679	2247	2958	1859	2466	3206	10.7	9.7	8.4
Bajra	903	1485	1936	1046	1703	2209	15.8	14.7	14.1
Maize	1135	1447	1863	1183	1508	1952	4.2	4.2	4.8
Ragi	2062	2860	3465	2337	3257	3976	13.3	13.9	14.7
Arhar (Tur)	3677	4761	6504	3848	5038	6839	4.7	5.8	5.2
Moong	3536	5788	7304	3510	5845	7476	-0.7	1.0	2.4
Urad	3781	4883	6496	3874	5114	6829	2.5	4.7	5.1
Groundnut	3793	4522	5664	4065	4842	6047	7.2	7.1	6.8
Soybean	2829	3261	4291	3082	3552	4638	8.9	8.9	8.1
Sunflower	4036	4853	6594	4379	5147	6364	8.5	6.1	-3.5
Sesamum	3611	6178	8152	3957	6564	8632	9.6	6.2	5.9
Nigerseed	1888	5811	7342	1960	6358	8025	3.8	9.4	9.3
Cotton	3845	4747	6230	4130	5140	6717	7.4	8.3	7.8

Source: CACP Calculations using CS data



Annex Table 5.8: Relative Average Gross Returns of Kharif Crops with respect to Paddy at All-India Level for 2021-22, 2022-23 and 2023-24

Year/Crop	Average Gross Returns Over Actual CoC (₹/ha)		Relative Average Gross Returns Over Actual CoC(%)	
	A_2	$A_2 + FL$	A_2	$A_2 + FL$
Paddy				
2021-22	35477	22166	100.0	100.0
2022-23	43100	29696	100.0	100.0
2023-24	51701	38787	100.0	100.0
TE2023-24	43426	30216	100.0	100.0
Jowar				
2021-22	14180	3620	40.0	16.3
2022-23	20182	10049	46.8	33.8
2023-24	23901	13351	46.2	34.4
TE2023-24	19421	9007	44.7	29.8
Bajra				
2021-22	18460	4119	52.0	18.6
2022-23	25391	11430	58.9	38.5
2023-24	25528	10423	49.4	26.9
TE2023-24	23126	8657	53.3	28.7
Maize				
2021-22	39237	27853	110.6	125.7
2022-23	42742	31414	99.2	105.8
2023-24	42689	31202	82.6	80.4
TE2023-24	41556	30156	95.7	99.8
Ragi				
2021-22	7369	-8477	20.8	-38.2
2022-23	10399	-4217	24.1	-14.2
2023-24	8120	-7095	15.7	-18.3
TE2023-24	8629	-6596	19.9	-21.8
Arhar (Tur)				
2021-22	34692	25182	97.8	113.6
2022-23	37431	27292	86.8	91.9
2023-24	33142	21633	64.1	55.8
TE2023-24	35088	24703	80.8	81.8
Moong				
2021-22	17521	7623	49.4	34.4
2022-23	20538	9173	47.7	30.9
2023-24	26117	13967	50.5	36.0
TE2023-24	21392	10254	49.3	33.9

(Contd...)



Annex Table 5.8: Relative Average Gross Returns of Kharif Crops with respect to Paddy at All-India Level for 2021-22, 2022-23 and 2023-24

Year/Crop	Average Gross Returns Over Actual CoC (₹/ha)		Relative Average Gross Returns Over Actual CoC(%)	
	A_2	$A_2 + FL$	A_2	$A_2 + FL$
Urad				
2021-22	14512	8425	40.9	38.0
2022-23	19068	12097	44.2	40.7
2023-24	19249	11371	37.2	29.3
TE2023-24	17610	10631	40.6	35.2
Groundnut				
2021-22	38005	26156	107.1	118.0
2022-23	50957	38160	118.2	128.5
2023-24	64041	49735	123.9	128.2
TE2023-24	51001	38017	117.4	125.8
Soybean				
2021-22	24106	18959	67.9	85.5
2022-23	24574	18868	57.0	63.5
2023-24	27572	21420	53.3	55.2
TE2023-24	25417	19749	58.5	65.4
Sunflower				
2021-22	21924	15852	61.8	71.5
2022-23	25810	18075	59.9	60.9
2023-24	12065	7823	23.3	20.2
TE2023-24	19933	13917	45.9	46.1
Sesamum				
2021-22	22910	10530	64.6	47.5
2022-23	26577	13933	61.7	46.9
2023-24	29384	16266	56.8	41.9
TE2023-24	26290	13576	60.5	44.9
Nigerseed				
2021-22	15387	1053	43.4	4.8
2022-23	16401	742	38.1	2.5
2023-24	19211	2629	37.2	6.8
TE2023-24	17000	1474	39.1	4.9
Cotton				
2021-22	41570	27768	117.2	125.3
2022-23	58360	43759	135.4	147.4
2023-24	50655	35090	98.0	90.5
TE2023-24	50195	35539	115.6	117.6

Note: All-India gross returns of a crop are weighted average of respective gross returns of projected States

Source: CACP Calculations using CS data



Annex Table 5.9: Relative Average Gross Returns of Kharif Crops with respect to Paddy in Selected States during TE2023-24

Crop/State	Gross Returns Over (₹/ha)		Relative Average Gross Returns Over(%)	
	A_2	$A_2 + FL$	A_2	$A_2 + FL$
Andhra Pradesh				
Paddy	56846	49103	100.0	100.0
Jowar	92516	90778	162.7	184.9
Maize	72492	66157	127.5	134.7
Arhar (Tur)	17739	14572	31.2	29.7
Moong	22980	21662	40.4	44.1
Urad	45454	42352	80.0	86.3
Groundnut	6101	-2027	10.7	-4.1
Sunflower	37520	31648	66.0	64.5
Sesamum	10020	7712	17.6	15.7
Cotton	47599	41417	83.7	84.3
Assam				
Paddy	30908	11601	100.0	100.0
Bihar				
Paddy	27810	17747	100.0	100.0
Maize	52676	44195	189.4	249.0
Chhattisgarh				
Paddy	52444	44530	100.0	100.0
Gujarat				
Paddy	37628	28301	100.0	100.0
Bajra	53240	39635	141.5	140.0
Maize	22094	10481	58.7	37.0
Arhar (Tur)	45612	36658	121.2	129.5
Moong	31200	16805	82.9	59.4
Urad	9525	631	25.3	2.2
Groundnut	52835	44027	140.4	155.6
Soybean	4434	-1064	11.8	-3.8
Sesamum	26297	14042	69.9	49.6
Cotton	75786	61220	201.4	216.3
Haryana				
Paddy	96334	87413	100.0	100.0
Bajra	21150	10631	22.0	12.2
Cotton	54802	39031	56.9	44.7
Himachal Pradesh				
Paddy	58297	33006	100.0	100.0
Maize	34933	12728	59.9	38.6

(Contd...)



Annex Table 5.9: Relative Average Gross Returns of Kharif Crops with respect to Paddy in Selected States during TE2023-24

Crop/State	Gross Returns Over (₹/ha)		Relative Average Gross Returns Over(%)	
	A_2	$A_2 + FL$	A_2	$A_2 + FL$
Jharkhand				
Paddy	19874	7944	100.0	100.0
Karnataka				
Paddy	52330	37254	100.0	100.0
Jowar	18047	8673	34.5	23.3
Maize	37638	28629	71.9	76.8
Ragi	8159	-6610	15.6	-17.7
Arhar (Tur)	16385	10455	31.3	28.1
Moong	13421	7965	25.6	21.4
Urad	10827	6917	20.7	18.6
Groundnut	30418	17324	58.1	46.5
Soybean	33082	27620	63.2	74.1
Sunflower	18611	12562	35.6	33.7
Cotton	49217	38719	94.1	103.9
Kerala				
Paddy	47603	35805	100.0	100.0
Madhya Pradesh				
Paddy	55758	45726	100.0	100.0
Maize	30999	24687	55.6	54.0
Urad	11655	7761	20.9	17.0
Soybean	20767	16036	37.2	35.1
Sesamum	31890	26246	57.2	57.4
Cotton	36836	23897	66.1	52.3
Maharashtra				
Paddy	-5380	-22367	-100.0	-100.0
Jowar	12722	4962	236.5	22.2
Bajra	19492	3811	362.3	17.0
Maize	63087	51765	1172.6	231.4
Arhar (Tur)	53981	38396	1003.4	171.7
Moong	18293	8666	340.0	38.7
Urad	22219	14025	413.0	62.7
Groundnut	19513	-10616	362.7	-47.5
Soybean	31093	25736	577.9	115.1
Cotton	33552	19491	623.6	87.1
Odisha				
Paddy	29522	11294	100.0	100.0
Ragi	15416	-6704	52.2	-59.4

(Contd...)



Annex Table 5.9: Relative Average Gross Returns of Kharif Crops with respect to Paddy in Selected States during TE2023-24

Crop/State	Gross Returns Over (₹/ha)		Relative Average Gross Returns Over(%)	
	A_2	$A_2 + FL$	A_2	$A_2 + FL$
Arhar (Tur)	21175	10579	71.7	93.7
Moong	21506	13186	72.8	116.8
Urad	20901	10936	70.8	96.8
Groundnut	22747	2379	77.1	21.1
Nigerseed	17000	1474	57.6	13.1
Cotton	48128	26176	163.0	231.8
Punjab				
Paddy	97865	91830	100.0	100.0
Maize	21020	9345	21.5	10.2
Cotton	93777	85770	95.8	93.4
Rajasthan				
Paddy*	55758	45726	100.0	100.0
Jowar	22843	2394	41.0	5.2
Bajra	19957	4555	35.8	10.0
Maize	37865	9685	67.9	21.2
Moong	23652	10740	42.4	23.5
Urad	19314	2072	34.6	4.5
Groundnut	98693	82082	177.0	179.5
Soybean	28248	16445	50.7	36.0
Sesamum	19170	2822	34.4	6.2
Cotton	92104	64526	165.2	141.1
Tamil Nadu				
Paddy	38013	30931	100.0	100.0
Jowar	29581	20196	77.8	65.3
Maize	48398	39847	127.3	128.8
Moong	12992	8673	34.2	28.0
Urad	21331	16879	56.1	54.6
Groundnut	40202	28788	105.8	93.1
Cotton	78994	56281	207.8	182.0
Telangana				
Paddy	53816	38393	100.0	100.0
Maize	57267	42098	106.4	109.6
Arhar (Tur)	26685	20320	49.6	52.9
Groundnut	50120	36019	93.1	93.8
Soybean	14264	7774	26.5	20.2
Cotton	33267	18847	61.8	49.1

(Contd...)



Annex Table 5.9: Relative Average Gross Returns of *Kharif* Crops with respect to Paddy in Selected States during TE2023-24

Crop/State	Gross Returns Over (₹/ha)		Relative Average Gross Returns Over(%)	
	A_2	$A_2 + FL$	A_2	$A_2 + FL$
Uttar Pradesh				
Paddy	24927	13003	100.0	100.0
Bajra	24181	11863	97.0	91.2
Maize	19117	11221	76.7	86.3
Arhar (Tur)	61958	42409	248.6	326.2
Moong	7593	-9511	30.5	-73.1
Urad	10924	1358	43.8	10.4
Groundnut	38396	25587	154.0	196.8
Sesamum	9294	-208	37.3	-1.6
West Bengal				
Paddy	37492	14533	100.0	100.0
Groundnut	121621	73003	324.4	502.3
Sesamum	51742	28829	138.0	198.4

*Average gross returns of Madhya Pradesh have been taken, as CS data for paddy are not collected from Rajasthan

Note: All-India gross returns of a crop are weighted average of respective gross returns of projected States

Source: CACP Calculations using CS data



Annex Table 5.10: Comparison of CACP and State Projected Cost of Production (C_2), and Yield and Suggested Minimum Support Price by State for *Kharif* Crops for KMS2025-26

State/Crop	CACP Projections on the basis of CS data		State Projections		Suggested Minimum Support Price by State (₹/qtl)
	Yield (qtl/ha)	Cost of Production (₹/qtl)	Yield (qtl/ha)	Cost of Production (₹/qtl)	
Paddy					
Andhra Pradesh	60.18	1926	59.00	2407	3465
Assam	36.90	2038	Cost Estimates are not provided		No Suggestion
Bihar	30.32	1738	31.80	2119	3260
Chhattisgarh	42.72	1647	42.55	1492	3100
Gujarat	40.22	1815	46.97	1858	3000
Haryana	53.38	2128	Cost Estimates are not provided		No Suggestion
Himachal Pradesh	28.38	2025	Cost Estimates are not provided		No Suggestion
Jharkhand	27.05	2050	Cost of Production (₹/qtl) is not provided		2234
Karnataka	49.02	2157	72.10	2488	3732
Kerala	37.41	2713	29.58	3146	3758
Madhya Pradesh	43.12	1814	Cost Estimates are not provided		No Suggestion
Maharashtra	21.07	3918	30.18	4159	4783
Odisha	42.60	2128	Cost Estimates are not provided		No Suggestion
Punjab	71.10	1593	64.70	2787	4281
Tamil Nadu	46.20	2134	Cost of Production (₹/qtl) is not provided		3040
Telangana	53.83	2219	49.00	3673	5510
Uttar Pradesh	36.50	2329	Cost Estimates are not provided		2315
West Bengal	44.26	2294	46.43	2104	2800
Jowar					
Andhra Pradesh	49.42	2266	16.00	2804	4046
Karnataka	11.11	3802	15.34	5232	7848
Maharashtra	11.88	3274	15.62	4163	4788
Rajasthan	10.57	2914	Cost Estimates are not provided		No Suggestion
Tamil Nadu	22.30	2726	Cost of Production (₹/qtl) is not provided		4090
Bajra					
Gujarat	30.35	1994	21.76	2212	3500
Haryana	22.48	2274	Cost Estimates are not provided		No Suggestion
Maharashtra	24.64	3186	17.21	4050	4658
Rajasthan	11.53	2192	Cost Estimates are not provided		No Suggestion
Uttar Pradesh	19.68	2127	Cost Estimates are not provided		1780
Maize					
Andhra Pradesh	74.13	1525	48.00	2320	3340
Bihar	45.40	1510	51.50	1293	3220
Gujarat	22.21	2647	20.22	2991	4550
Himachal Pradesh	19.01	2374	Cost Estimates are not provided		No Suggestion
Karnataka	34.82	2141	36.78	2868	4302
Madhya Pradesh	32.44	1610	Cost Estimates are not provided		No Suggestion
Maharashtra	57.52	1838	40.63	2556	2939
Punjab	35.16	2237	Cost Estimates are not provided		3132

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Annex Table 5.10: Comparison of CACP and State Projected Cost of Production (C_2), and Yield and Suggested Minimum Support Price by State for Kharif Crops for KMS2025-26

State/Crop	CACP Projections on the basis of CS data		State Projections		Suggested Minimum Support Price by State (₹/qtl)
	Yield (qtl/ha)	Cost of Production (₹/qtl)	Yield (qtl/ha)	Cost of Production (₹/qtl)	
Rajasthan	23.19	2740	Cost Estimates are not provided		No Suggestion
Tamil Nadu	47.28	2021	Cost of Production (₹/qtl) is not provided		3000
Telangana	57.36	2053	45.00	2191	3286
Uttar Pradesh	26.66	2202	Cost Estimates are not provided		1840
Ragi					
Karnataka	14.96	3940	16.45	6387	9581
Odisha	9.36	4837	Cost Estimates are not provided		No Suggestion
Tur (Arhar)					
Andhra Pradesh	6.60	7558	6.00	7317	10823
Gujarat	9.41	6730	15.17	5304	9100
Karnataka	7.20	6562	7.95	11032	16548
Maharashtra	18.15	6669	13.81	7230	8315
Odisha	4.14	6958	Cost Estimates are not provided		No Suggestion
Telangana	7.16	8186	7.00	7969	11953
Uttar Pradesh	9.43	7252	Cost Estimates are not provided		5205
Moong					
Andhra Pradesh	7.52	6619	7.00	7327	10787
Gujarat	6.05	7631	7.20	6641	9700
Karnataka	5.26	7686	6.87	10622	15933
Maharashtra	7.96	8189	7.34	10561	12146
Odisha	4.10	7747	Cost Estimates are not provided		No Suggestion
Rajasthan	4.96	7338	Cost Estimates are not provided		No Suggestion
Tamil Nadu	4.10	7818	Cost of Production (₹/qtl) is not provided		12280
Uttar Pradesh	6.06	7418	Cost Estimates are not provided		5620
Urad					
Andhra Pradesh	10.78	5599	8.00	6485	9476
Gujarat	4.72	8374	6.18	7075	9500
Karnataka	5.10	8376	6.87	13040	19560
Madhya Pradesh	6.49	5600	Cost Estimates are not provided		No Suggestion
Maharashtra	7.06	8430	8.27	10220	11753
Odisha	3.68	8437	Cost Estimates are not provided		No Suggestion
Rajasthan	5.19	8071	Cost Estimates are not provided		No Suggestion
Tamil Nadu	6.84	8144	Cost of Production (₹/qtl) is not provided		9200
Uttar Pradesh	5.44	7942	Cost Estimates are not provided		4920
Groundnut					
Andhra Pradesh	11.58	8928	10.00	6665	9819
Gujarat	17.69	5962	25.00	4258	8600
Karnataka	10.96	8058	8.20	12387	18581
Maharashtra	11.90	9962	9.06	10276	11817
Odisha	8.42	8429	Cost Estimates are not provided		No Suggestion

(Contd...)



Annex Table 5.10: Comparison of CACP and State Projected Cost of Production (C_2), and Yield and Suggested Minimum Support Price by State for *Kharif* Crops for KMS2025-26

State/Crop	CACP Projections on the basis of CS data		State Projections		Suggested Minimum Support Price by State (₹/qtl)
	Yield (qtl/ha)	Cost of Production (₹/qtl)	Yield (qtl/ha)	Cost of Production (₹/qtl)	
Rajasthan	25.78	3851	Cost Estimates are not provided		No Suggestion
Tamil Nadu	15.82	7129	Cost of Production (₹/qtl) is not provided		9340
Telangana	19.13	6674	15.00	5805	8708
Uttar Pradesh	12.06	4666	Cost Estimates are not provided		5400
West Bengal	33.83	5069	Cost Estimates are not provided		No Suggestion
Soybean					
Gujarat	15.66	4212	Cost Estimates are not provided		No Suggestion
Karnataka	14.53	4669	13.04	7422	11133
Madhya Pradesh	11.24	4175	Cost Estimates are not provided		No Suggestion
Maharashtra	15.86	4957	15.15	6154	7077
Rajasthan	10.82	5309	Cost Estimates are not provided		No Suggestion
Telangana	16.09	5090	15.00	5155	7733
Sunflower					
Andhra Pradesh	11.35	6463	8.00	6295	9502
Karnataka	9.21	6356	8.55	10759	16139
Sesamum					
Andhra Pradesh	3.51	9979	5.00	7332	10998
Gujarat	5.28	10201	5.14	8232	12000
Madhya Pradesh	5.58	7549	Cost Estimates are not provided		No Suggestion
Rajasthan	3.22	10385	Cost Estimates are not provided		No Suggestion
Uttar Pradesh	2.52	10229	Cost Estimates are not provided		5605
West Bengal	11.41	7216	Cost Estimates are not provided		No Suggestion
Nigerseed					
Odisha	3.72	8025	Cost Estimates are not provided		No Suggestion
Cotton					
Andhra Pradesh	16.29	6729	20.00	6089	9133
Gujarat	16.68	6218	18.05	6676	10500
Haryana	12.26	6703	Cost Estimates are not provided		No Suggestion
Karnataka	13.11	6573	17.34	9615	14423
Madhya Pradesh	15.96	6680	Cost Estimates are not provided		No Suggestion
Maharashtra	14.20	7140	13.66	9199	10579
Odisha	14.34	7373	Cost Estimates are not provided		No Suggestion
Punjab	18.48	6260	16.72	8134	12201
Rajasthan	18.00	5683	Cost Estimates are not provided		No Suggestion
Tamil Nadu	19.12	7897	Cost of Production (₹/qtl) is not provided		9420
Telangana	16.91	7348	13.00	11613	17420

Source: State Governments and CACP Calculations



Annex Table 5.11: Crop-wise Projection of State having small Area/Production/Sample of Kharif Crops for Marketing Season 2025-26

Crop	State	Reasons for Inclusion in Projection	Action and Recommendation
Paddy	Himachal Pradesh	Although State area and production share of paddy at national level are very low, yet area and production share of paddy in Kharif cereals at State level are 10.2% and 14.7%, respectively.	Projected Projection is made as State having reason able area and production share of paddy in State Kharif cereals.
	Kerala	Although State area and production share of paddy at national level are low, yet area and production share of Paddy in Kharif cereals at State level are 99.6% and 99.9%, respectively.	Projected Projection is made as State having reason able area and production share of paddy in State Kharif cereals.
Groundnut	Odisha	Although State area and production of groundnut at national level are small, yet area and production share at Kharif oilseeds at State level are 28.5% and 60.3%, respectively.	Projected Projection is made as State having reason able area and production share of ground nut in State kharif oilseeds.
Soybean	Gujarat	Although State area and production shares of soybean in State Kharif pulses are small, yet State area and production shares in all-India are 1.85% and 2.70%, respectively.	Projected Projection is made as State having reason able area and production of soybean at national level.
Sunflower	Andhra Pradesh	Although State area and production share of sunflower in State Kharif oilseeds are low; yet State area and production share at national level are 5.76% and 4.25%, respectively.	Projected Projection is made as State having reason able area and production share of sunflower at national level.
Cotton	Tamil Nadu	Although area and production share of cotton in State Kharif crops at State level are low, yet State area and production share at national level are 1.19% and 0.90%, respectively.	Projected Projection is made as State having reason able area and production share of cotton at national level.

Note: Percentage area and production share presented in Table have been derived on average area and production during TE2023-24

Source 1: ES&E Division, Ministry of Agriculture and Farmers Welfare

2: CACP Calculations

Annex Table



Annex Table

Annex Table 5.12: Crop-wise States not-included in Projection of Kharif Crops for Marketing Season 2025-26

Crop	State	Reasons for not-inclusion in Projection	Action and Recommendation
Jowar	Madhya Pradesh	Although State area and production share in State Kharif cereals are low, yet area and production share in all-India are 2.7% and 4.5% respectively. But CS data is available only for single year during TE2023-24 with thin sample.	Not Projected. CS data should be collected every year with increased sample size.
	Karnataka	Although area and production shares in all-India are 1.9% and 1.6% respectively. But CS data is available only for one year during TE2023-24	Not Projected. CS data should be collected every year.
Bajra	Madhya Pradesh	Although area and production shares in all-India are 5.0% and 8.8% respectively. But CS data is available only for one year during TE2023-24	Not Projected. CS data should be collected every year.
	Tamil Nadu	State Area and Production shares of bajra in all-India are reasonable. Besides, State area and production of bajra are adequate. But CS data is available only with one year during TE2023-24 that too with thin sample size	Not Projected. CS data should be collected every year with increased sample size.
Maize	Chhattisgarh	Although area and production shares in all-India are 1.1% and 1.0% respectively. But CS data is available only for one year during TE2023-24	Not Projected. CS data should be collected every year.
	Jharkhand	Although area and production shares in all-India are 2.5% and 1.5% respectively. But CS data is available with thin sample for 2021-22 and 2022-23 during TE2023-24	Not Projected. CS data should be collected every year with increased sample size.
	Odisha	State area and production share in all-India as well as area and production share in State Kharif cereals are small.	Not Projected.
	Punjab	Although area and production shares in all-India are 0.9% and 1.1% respectively. But CS data is available with thin sample during TE2023-24.	Not Projected. CS data should be collected every year with increased sample size.
	West Bengal	Although area and production shares in all-India are 3.5% and 6.8% respectively. But CS data is available only for one year during TE2023-24	Not Projected. CS data should be collected every year.

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Annex Table 5.12: Crop-wise States not-included in Projection of Kharif Crops for Marketing Season 2025-26

Crop	State	Reasons for not-inclusion in Projection	Action and Recommendation
Ragi	Tamil Nadu	Although area and production shares in all-India are 6.0% and 13.4% respectively. But CS data is available only for one year during TE2023-24	Not Projected. CS data should be collected every year.
Tur	Jharkhand	Although area and production shares in all-India are 4.8% and 6.5% respectively. But CS data is available only for one year during TE2023-24	Not Projected. CS data should be collected every year.
	Madhya Pradesh	Although area and production share of tur in Kharif pulses at national level are 4.17% and 4.14% respectively. But CS data are available with thin sample during TE2023-24.	Not Projected. CS data should be collected every year with increased sample size.
Moong	Bihar	Although area and production shares in all-India are 3.0% and 3.6% respectively. But CS data is available only for one year during TE2023-24.	Not Projected. CS data should be collected every year.
	Haryana	Although area and production shares in all-India are 1.1% and 1.24% respectively. But CS data is available only for two year with thin sample during TE2023-24.	Not Projected. CS data should be collected every year with increased sample size.
	Madhya Pradesh	Although State area and production shares in all-India are significant, and State area and production shares of Moong in State Kharif pulses are also reasonable. But CS data is available for only one year during TE2023-24 that too with thin sample size.	Not Projected. CS data should be collected every year with increased sample size.
	Punjab	Although area and production share of moong in Kharif pulses at state level are 60.52% and 65.28% respectively. But CS data is available for only two year with thin sample size during TE2023-24.	Not Projected. CS data should be collected every year with increased sample size.
	Telangana	Although area and production share of moong in Kharif pulses at state level are 8.12% and 5.60% respectively. But CS sample is of thin size for TE2023-24.	Not Projected. CS data should be collected every year with increased sample size.
	West Bengal	Although area and production share of moong at national level as well as in Kharif pulses at state level are adequate. But CS data is available with thin sample size for TE2023-24.	Not Projected. CS data should be collected every year with increased sample size.

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Annex Table



Annex Table

Annex Table 5.12: Crop-wise States not-included in Projection of Kharif Crops for Marketing Season 2025-26

Crop	State	Reasons for not-inclusion in Projection	Action and Recommendation
Urad	Chhattisgarh	Although area and production shares in all-India are reasonable; and State area and production shares of urad in State Kharif pulses are significant. But CS data are available only for one year during TE2023-24 with thin sample size.	Not Projected. CS data should be collected every year with increased sample size.
	Himachal Pradesh	Although area and production share of urad in Kharif pulses at state level are adequate but the share as well as quantum of the area and production in all-India is very low.	Not Projected.
	Jharkhand	Although area and production shares in all-India are 2.9% and 4.1% respectively. But CS data is available only for one year during TE2023-24	Not Projected. CS data should be collected every year with increased sample size.
	West Bengal	Area and production shares in all-India are 1.8% and 2.0% respectively. But CS data is available with very thin sample for 2021-22 and 2022-23 during TE2023-24	Not Projected. CS data should be collected every year with increased sample size.
Groundnut	Chhattisgarh	Although State area and production shares of groundnut in State Kharif oilseeds are 15.97% and 37.22% respectively. But CS data is available only for one year during TE2023-24	Not Projected. CS data should be collected every year.
	Kerala	Although area and production share of groundnut in Kharif oilseeds at state level are adequate but the share as well as quantum of the area and production in all-India is very low.	Not Projected.
Soybean	Chhattisgarh	Although State area and production share at national level is small, yet area and production share of soybean in Kharif oilseeds at state level are 25.85% and 30.11% respectively. But CS data are available only for one year, that too with small sample size for TE2023-24.	Not Projected. CS data should be collected every year with increased sample size.
	Himachal Pradesh	Although area and production share of soybean in Kharif oilseeds at state level are adequate but the share as well as quantum of the area and production in all-India is very low.	Not Projected.

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Annex Table 5.12: Crop-wise States not-included in Projection of Kharif Crops for Marketing Season 2025-26

Crop	State	Reasons for not-inclusion in Projection	Action and Recommendation
Sunflower	Bihar	Although State area and production share of Sunflower in all-India as well as State Kharif oilseeds are reasonable, but CS sample is of thin size for TE2023-24.	Not Projected. CS data should be collected every year with increased sample size.
Sesamum	Himachal Pradesh	Although area and production share of sesamum in Kharif oilseeds at state level are adequate but the share as well as quantum of the area and production in all-India is very low.	Not Projected.
	Kerala	Although area and production share of sesamum in Kharif oilseeds at state level are adequate but the share as well as quantum of the area and production in all-India is very low.	Not Projected.
	Odisha	Although area and production share of sesamum in Kharif oilseeds at state level are adequate but the share as well as quantum of the area and production in all-India is very low.	Not Projected.
	Telangana	Although area and production share of sesamum at national level is more than 1%, moreover, State area and production share in State Kharif oilseeds are 6.07% and 2.57% respectively, but CS data are available for two year with thin size for TE2023-24.	Not Projected. CS data should be collected every year with increased sample.

Note: Percentage area and production share presented in Table have been derived on average area and production during TE2023-24

Source 1: ES&E Division, Ministry of Agriculture and Farmers Welfare

2: CACP Calculations



Annex Table

Annex Table 5.13: Crop-wise Inclusion of States under Comprehensive Scheme

Crop	State	Reasons for inclusion under Comprehensive Scheme	Action and Recommendation
Paddy	Jammu & Kashmir	Although State area and production shares of paddy at national level are 0.56% and 0.43%, respectively, yet area and production shares of paddy in Kharif cereals at State level are 31.07% and 42.6%, respectively. But, there are no CS data for the State.	Comprehensive Scheme should be implemented within the State.
	Rajasthan	Although State area and production shares of paddy at national level are 0.49% and 0.45%, respectively, yet quantum of area and production of paddy at State level are adequate. But, there are no CS data for the State.	CS data should be collected for the crop.
	Tripura	Although State area and production shares of paddy at national level are 0.52% and 0.59%, respectively, yet quantum of area and production of paddy at State level are adequate. But, there are no CS data for the State.	Comprehensive Scheme should be implemented within the State.
	Uttarakhand	Although State area and production shares of paddy at national level are 0.53% and 0.49%, respectively, yet quantum of area and production of paddy at State level are adequate. But, there are no CS data for the State.	CS data should be collected for the crop.
Jowar	Gujarat	Although area and production shares of jowar in Kharif cereals at State level are low, yet State area and production shares of jowar at national level are 1.03% and 1.34%, respectively. But, there are no CS data for the State.	CS data should be collected for the crop.
	Telangana	State area and production shares of jowar at national level are 2.16% and 3.74%, respectively, moreover, State area and production are reasonable. But, there are no CS data for the State.	CS data should be collected for the crop.
	Uttar Pradesh	Although State area and production shares of jowar at national level are 5.77% and 8.25%, respectively, moreover, State area and production are reasonable. But, there are no CS data for the State.	CS data should be collected for the crop.
Maize	Jammu & Kashmir	Although State area and production shares of maize at national level are 2.68% and 1.57%, respectively, and area and production shares of maize in Kharif cereals at State level are 33.23% and 28.04%, respectively. But, there are no CS data for the State.	Comprehensive Scheme should be implemented within the State.
	Chhattisgarh	State area and production shares of maize at national level are 1.13% and 1.03%, respectively, moreover, State area and production are reasonable. But, there are no CS data for the State.	CS data should be collected for the crop.

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Annex Table 5.13: Crop-wise Inclusion of States under Comprehensive Scheme

Crop	State	Reasons for inclusion under Comprehensive Scheme	Action and Recommendation
Ragi	Andhra Pradesh	State area and production shares of ragi at national level are 2.34% and 1.92%, respectively, moreover, State area and production are reasonable. But, there are no CS data for the State.	CS data should be collected for the crop.
	Jharkhand	Although area and production shares of ragi in Kharif cereals at State level are low, yet State area and production shares of ragi at national level are 1.86% and 1.19%, respectively. But, there are no CS data for the State.	CS data should be collected for the crop.
	Maharashtra	State area and production shares of ragi at national level are 6.08% and 6.05%, respectively, moreover, State area and production are reasonable. But, there are no CS data for the State.	CS data should be collected for the crop.
	Uttarakhand	State area and production shares of ragi at national level are 6.42% and 6.86%, respectively, moreover, State area and production are reasonable. But, there are no CS data for the State.	CS data should be collected for the crop.
Tur	Assam	Although State area and production shares of tur at national level are low, yet tur is the only pulse grown at State level. But, there are no CS data for the State.	CS data should be collected for the crop.
	Tamil Nadu	State area and production shares of tur at national level are 1.00% and 1.18%, respectively. But, there are no CS data for the State.	CS data should be collected for the crop.
	Tripura	Although State area and production shares of tur at national level are low, yet area and production shares of tur in Kharif pulses at State level are 21.19% and 20.75%, respectively. But, there are no CS data for the State.	Comprehensive Scheme should be implemented within the State.
Urad	Assam	State area and production shares of urad at national level are 1.93% and 2.01%, respectively, moreover, area and production shares of urad in Kharif pulses at State level are 32.91% and 27.98%, respectively, yet there are no CS data for the State.	CS data should be collected for the crop.
	Telangana	State area and production shares of urad at national level are 0.90% and 1.66%, respectively, moreover, area and production shares of urad in Kharif pulses at State level are 7.97% and 8.93%, respectively, yet there are no CS data for the State.	CS data should be collected for the crop.

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Annex Table



Annex Table

Annex Table 5.13: Crop-wise Inclusion of States under Comprehensive Scheme

Crop	State	Reasons for inclusion under Comprehensive Scheme	Action and Recommendation
	Tripura	Although State area and production shares of urad at national level are low, yet area and production shares of urad in Kharif pulses at State level are 29.29% and 27.65%, respectively. But, there are no CS data for the State.	Comprehensive Scheme should be implemented within the State.
	Uttarakhand	Although State area and production shares of urad at national level are low, yet area and production shares of urad in Kharif pulses at State level are 20.48% and 22.43%, respectively. But, there are no CS data for the State.	CS data should be collected for the crop.
Groundnut	Madhya Pradesh	State area and production shares of groundnut at national level are 8.18% and 7.83%, respectively, moreover, area and production shares of groundnut in Kharif oilseeds at State level are 5.37% and 9.67%, respectively, yet there are no CS data for the State.	CS data should be collected for the crop.
Soybean	Bihar	Although State area and production shares of soybean at national level are low, yet area and production shares of soybean in Kharif oilseeds at State level are 16.62% and 11.74%, respectively. But, there are no CS data for the State.	CS data should be collected for the crop.
	Uttarakhand	Although State area and production shares of soybean at national level are low, yet area and production shares of soybean in Kharif oilseeds at State level are 24.19% and 31.14%, respectively. But, there are no CS data for the State.	CS data should be collected for the crop.
Sunflower	Haryana	State area and production shares of sunflower at national level are 5.33% and 9.10%, respectively. But, there are no CS data for the State.	CS data should be collected for the crop.
	Maharashtra	Although area and production shares of sunflower in Kharif oilseeds at State level are low, yet, State area and production shares of sunflower at national level are 7.12% and 3.69%, respectively. But, there are no CS data for the State.	CS data should be collected for the crop.
	Odisha	State area and production shares of sunflower at national level are 6.58% and 8.25%, respectively, Moreover, area and production shares of sunflower in Kharif oilseeds at State level are 10.97% and 15.10%, respectively, yet there are no CS data for the State.	CS data should be collected for the crop.

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Annex Table 5.13: Crop-wise Inclusion of States under Comprehensive Scheme

Crop	State	Reasons for inclusion under Comprehensive Scheme	Action and Recommendation
	Tamil Nadu	Although State area and production shares of sunflower in Kharif oilseeds at State level are very low, yet area and production shares of sunflower at national level are 2.79% and 2.74%, respectively. But, there are no CS data for the State.	CS data should be collected for the crop.
	Telangana	Although State area and production shares in State Kharif oilseeds are low, yet area and production shares in all-India are 4.45% and 6.96% respectively, But, there are no CS data for the State.	CS data should be collected for the crop.
	Uttar Pradesh	Although area and production shares of sunflower in Kharif oilseeds at State level are low, yet State area and production shares of sunflower at national level are 1.02% and 1.38%, respectively. But, there are no CS data for the State.	CS data should be collected for the crop.
	West Bengal	Although area and production shares of sunflower in Kharif oilseeds at State level are low, yet State area and production shares of sunflower at national level are 1.72% and 2.07%, respectively. But, there are no CS data for the State.	CS data should be collected for the State.
	Assam	Although area and production shares of sesamum at national level are 0.89% and 1.18%, respectively. But, there are no CS data for the State.	CS data should be collected for the crop.
Sesamum	Karnataka	State area and production shares of sesamum in kharif oilseeds at state level are 2.07% and 1.52%, respectively. Moreover, area and production shares of sesamum at national level are 1.47% and 2.07%, respectively. But, there are no CS data for the State.	CS data should be collected for the crop.
	Maharashtra	Although area and production shares of sesamum in kharif oilseeds at State level are low, yet State area and production shares of sesamum at national level are 1.41% and 0.59%, respectively. But, there are no CS data for the State.	CS data should be collected for the crop.
	Tripura	Although area and production shares of sesamum at national level are low, yet area and production shares of sesamum in Kharif oilseeds at State level are 39.88% and 31.36%, respectively. But, there are no CS data for the State.	Comprehensive Scheme should be implemented within the State.
Nigerseed	Andhra Pradesh	Although area and production shares of nigerseed in Kharif oilseeds at State level are low, yet State area and production shares of nigerseed at national level are 2.98% and 3.36%, respectively. But, there are no CS data for the State.	CS data should be collected for the State.

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Annex Table

Annex Table 5.13: Crop-wise Inclusion of States under Comprehensive Scheme

Crop	State	Reasons for inclusion under Comprehensive Scheme	Action and Recommendation
	Assam	State area and production shares of nigerseed at national level are 5.84% and 10.30%, respectively. But, there are no CS data for the State.	CS data should be collected for the crop.
	Gujarat	Although area and production shares of nigerseed in Kharif oilseeds at State level are low, yet State area and production shares of nigerseed at national level are 2.02% and 2.63%, respectively. But, there are no CS data for the State.	CS data should be collected for the crop.
	Jharkhand	State area and production shares of nigerseed at national level are 2.16% and 2.49%, respectively. But, there are no CS data for the State.	CS data should be collected for the crop.
	Madhya Pradesh	Although area and production shares of nigerseed in Kharif oilseeds at State level are low, yet State area and production shares of nigerseed at national level are 10.42% and 9.94%, respectively. But, there are no CS data for the State.	CS data should be collected for the crop.
	Maharashtra	Although area and production shares of nigerseed in Kharif oilseeds at State level are low, yet State area and production shares of nigerseed at national level are 5.85% and 3.87%, respectively. But, there are no CS data for the State.	CS data should be collected for the crop.
	West Bengal	Although area and production shares of nigerseed in Kharif oilseeds at State level are low, yet State area and production shares of nigerseed at national level are 1.13% and 1.80%, respectively. But, there are no CS data for the State.	CS data should be collected for the crop.

Note 1: Union Territory of Jammu and Kashmir also Includes Union Territory of Ladakh

2: Percentage area and production share presented in Table have been derived on average area and production during TE2023-24

Source 1: ES&E Division, Ministry of Agriculture and Farmers Welfare

2: CACP Calculations

Commission for Agricultural Costs & Prices

List of Officers

Advisers

Dr. R Sathish
Dr. Seema
Mr. Vivek Shukla

Joint Directors

Mrs. Divya Sharma
Mrs. Sonal Ramesh

Deputy Directors

Mr. Baidyanath Guru
Mrs. Shivani
Mrs. Shraddha Pal

Assistant Directors

Mr. Deepak Peeceeyen S
Mr. Vipin Singh
Ms. Parul Singh

Sr. Statistical Officers

Mr. B.K. Barnawal
Mr. Ravi Kumar Sharma
Mr. Ashok Kumar
Mrs. Savitri Ramani

Economic Officers

Md. Abdul Aleem
Mr. Mintu Kumar
Mr. Chandra Kumar

Jr. Statistical Officers

Mr. Mahender Kumar Gupta
Mr. Mohammad Shoeb Malik

Sr. Translation Officer

Mr. Manoj Kumar Shaw

Administrative Staff

Mr. Pardeep Kumar
Mr. Pankaj Kumar
Mr. Anil Singh
Mrs. Pooja Budhiraja
Ms. Kamla Manral



Commission for Agricultural Costs & Prices

Department of Agriculture & Farmers Welfare

Ministry of Agriculture & Farmers Welfare

Government of India, New Delhi

March 2025